

Merged research report for the Surreal project

Polish Models of Omnific Arithmetic

Polish Presburger models for Glazer’s Question 2, Borel presentations and support barriers, continuous Presburger arithmetic, local compactness, Polish series fields, Baire-category rigidity, nonsplit models, a claimed ATR_0 bound for Question 1, locally compact cones, and elementary embeddings of lexicographic models

Built from thirteen manuscripts dated October 3 and 4, 2026

Sources: manuscripts 05 (the base text of Part I), 04 (Part I, marked [04]) and 03 (Part II) of batch 86, arrival commit 31fdc6571, placement commit 3d2177df4; pinned to repository commits a11efab09 (05), 3ad5f878c (04) and c5b4881fa (03). Manuscripts 07 (the base text of Part III), 06 (Part III, marked [06]), 08 (Part IV) and 09 (Part V) of batch 86, arrival commit fb8414869, placement commit 0bd0e5527; pinned to 0f95145cb (07), fa2f3e419 (06), 883e0b3b2 (08) and 4af956fe0 (09). Batch-87 manuscripts 03, 01, 04–06 (sources 12, 10, 13–15): arrival fa0a0576e, placement d151b39ca; pins 0f084afa9 (12, 15), 6fef5383b (10, 14), eab47e330 (13). Batch-89 manuscript 02 (source 16, Part X): arrival 7c0f2d9f9, placement 23adb85f9, pin 0f084afa9. Their titles and contributions are in [section 1.2](#); the provenance of every section is in [section E](#), and the non-claims of every source are collected in [section F](#).

Abstract

Elliot Glazer's *A Topological Tennenbaum Theorem* shows that an uncountable Polish model of $Q + IOpen$ cannot have continuous addition and Borel multiplication, and asks (Question 2) whether some uncountable Polish space supports a model of Presburger arithmetic with continuous addition. Part I answers that question, as printed, affirmatively. The nonnegative cone $M_{\mathbb{R}} = (\mathbb{R}_{\geq 0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N})$ of the lexicographic group $\mathbb{R} \overrightarrow{\times} \mathbb{Z}$, with the usual topology on the real coordinate and the discrete one on the integer coordinate, is a perfect, locally compact, uncountable Polish model of the full theory $\text{Th}(\mathbb{N}; 0, 1, +, <)$ whose addition is continuous. Replacing $\mathbb{R}$ by the lexicographically ordered group $\mathbb{Q}^{\mathbb{N}}$ with discrete coordinates gives a model homeomorphic to Baire space, with Type-2 computable addition and division by standard integers. Every definable relation is Δ_2^0 , and this is optimal for the order; the truncated subtraction and the predecessor are discontinuous exactly on explicit loci; no unital semiring multiplication exists on either model; the self-embeddings are classified (positive dilations in the real case, row-finite matrices with increasing positive pivots in the Baire case), and both models embed additively in Conway's omnific integers $\mathbf{Oz}$ as $r\omega + n$ and $\sum_j a_j \omega^{1/(j+1)} + n$. Part I also isolates the coset-invariance hypothesis needed to carry the repository's integer Cooper lemmas to nonstandard groups. Glazer himself speculated that the answer to Question 2 would be negative; the affirmative answer is given for the question as printed, and its priority has not been established.

Part II is the other side of the same boundary. For a countable linear order L , its well-ordered subsets form a Borel set exactly when L is scattered, and a Π_1^1 -complete set otherwise; the same condition decides whether a full real Hahn field $\mathbb{R}((t^\Gamma))$ has a standard Borel parametrization with Borel coefficient maps. The finite Puiseux principal-part ring $\mathcal{A}_{\text{fin}}$, of which $M_{\mathbb{R}}$ is one stratum, satisfies open induction and has a Polish presentation on which addition and multiplication are continuous off explicit closed nowhere dense cancellation loci; by Glazer's Corollary 1 no Borel-compatible Polish recoding makes its addition continuous everywhere.

Part III merges two further manuscripts on continuous Presburger arithmetic. A Borel invariant order on a Polish abelian group has Δ_2^0 cones, so every divisible Polish group with Borel positive cone yields a Polish model; the cones $(\mathbb{Q}^\alpha \overrightarrow{\times} \mathbb{Z})_{\geq 0}$, α countably infinite, give $\aleph_1$ pairwise nonisomorphic models on Baire space, other families give exactly $2^{\aleph_0}$, and one order formula has unbounded oscillation ranks. The cone of $\mathbb{Z} + t\mathbb{R}[t]$ (t infinite) carries a locally compact Polish topology with continuous addition and multiplication and fails open induction at $x^2 < t$; and in every Polish topological cone of a discretely ordered ring with both operations continuous, 0 is isolated, which is Glazer's Theorem 1 with PA weakened to PA^- . Part IV shows that a discretely ordered ring whose additive group is Hausdorff, locally compact and second countable with Borel order is countable and discrete, whatever its multiplication, and classifies the locally compact Borel $\mathbb{Z}$ -groups as $\mathbb{R}^d \times E$. Part V shows that a full Hahn field $k((t^\Gamma))$ admits a Polish additive topology with Baire order and Borel scalar maps exactly when k is countable and $\Gamma \cong \mathbb{Z}$; the Puiseux field over a countable field admits none, its Levi-Civita completion exactly one; and every integer part of such a field is countable and closed discrete.

Parts VI–X add six later manuscripts: a discretely ordered ring whose additive group has a Hausdorff Baire ccc topology and whose positive cone has the Baire property is countable and discrete, whatever its multiplication (VI); nonsplit Polish Presburger models exist on every countable core, with computable discontinuity sets and optimal revision numbers (VII); a manuscript claims, without independent review, that Glazer's Theorem 2 is provable in ATR_0 (VIII); the locally compact cones of Borel ordered Polish groups are the countable sums of lines and closed rays (IX); and all self-embeddings of $(\mathbb{Q}^\alpha \overrightarrow{\times} \mathbb{Z})_{\geq 0}$, α countable, are continuous exactly when $\alpha \leq \omega$ and have closed images exactly when $\alpha < \omega \cdot 2$ (X; the counterexamples use choice).

The report merges thirteen independently written manuscripts; it states which source proved each result, prints shared results once or as marked second routes, keeps every source's limitations, and cites the collection's earlier results instead of reprinting them as new. It is an unrefereed AI-assisted draft, and nothing in it has been formalized.

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1 Scope, sources and conventions

This report collects thirteen manuscripts, dated October 3 and 4, 2026, on one subject: where Elliot Glazer’s topological Tennenbaum programme meets the repository’s omnific arithmetic. Their common spine is the question which arithmetic operations of a Polish or standard Borel presentation of a substructure of $\mathbf{Oz}$ can be continuous. Three of them (03, 04, 05) were written first; the four later ones (06, 07, 08, 09) arrived together, and 08 and 09 cite 05 and 03 as earlier companion manuscripts. [Part I](#) (label prefix `pma:pr:`) constructs Polish models of Presburger arithmetic with continuous addition and studies them; its base text is manuscript 05, and the further results of manuscript 04 are printed as marked sections. [Part II](#) (label prefix `pma:bor:`) is manuscript 03: Borel presentations of Hahn fields, support barriers, and the finite principal-part ring on which Glazer’s theorem forbids continuous addition. Parts I and II meet in [section 26](#). [Part III](#) (label prefix `pma:cp:`) merges manuscripts 07 (base) and 06 (marked [06]) on continuous Presburger arithmetic: Borel orders on Polish groups, a general realization theorem, ordinal and continuum families on Baire space, a Polish semiring cone, and the isolated-zero theorem; it meets Parts I and II in [section 42](#). [Part IV](#) (prefix `pma:lc:`) is manuscript 08: local compactness forces arithmetic rigidity. [Part V](#) (prefix `pma:ser:`) is manuscript 09, which continues Part II: Polish topologies on Hahn, Puiseux and Levi-Civita fields and their integer parts.

Five further manuscripts (sources 10 and 12–15, batch 87) continue the report. [Part VI](#) (prefix `pma:br:`) merges sources 12 (base) and 10 (marked [10]), which prove that a Baire-category regularity of the order forces arithmetic rigidity without local compactness. [Part VII](#) (prefix `pma:nsp:`) is source 13: nonsplit Polish Presburger models, exact continuity and optimal computation from streams. [Part VIII](#) (prefix `pma:atr:`) is source 14, which claims an ATR_0 proof of Glazer’s Theorem 2. [Part IX](#) (prefix `pma:lcc:`) is source 15: locally compact cones and the limits of multiplication.

A thirteenth manuscript, source 16 of batch 89, continues Parts I and III. [Part X](#) (prefix `pma:emb:`) is source 16: elementary embeddings and Polish elementary submodels of the ordinal-indexed models $(\mathbb{Q}^\alpha \vec{\times} \mathbb{Z})_{\geq 0}$, with two sharp ordinal thresholds.

1.1 Glazer’s paper and its two questions

Glazer’s paper [1] is an eight-page preprint, dated November 20, 2023 on its first page and submitted to arXiv on November 22, 2023 as arXiv:2311.13699. On October 3, 2026 the arXiv record listed one version (v1) and no journal reference. It answers negatively a question of Enayat, Hamkins and Wcisłó [2]: no model of Peano arithmetic on a perfect Polish space has continuous addition and multiplication (its Theorem 1). Its Theorem 2 says that if $(M, +, \cdot) \models Q + I\Delta_0 + \text{Exp}$ lives on an uncountable Polish space with Borel addition and multiplication, then neither operation is continuous. Glazer notes that the axiom Exp is used only to obtain the function $x \mapsto 2^x$, and only for the discontinuity of multiplication; this gives his Corollary 1: *there is no model of $Q + \text{IOpen}$ on an uncountable Polish space in which addition is continuous and multiplication is Borel*. The paper ends with two questions. Question 1 asks whether Theorem 2 is provable in ATR_0 . Question 2 is quoted in [section 2](#): does some uncountable Polish space support a model of Presburger arithmetic with continuous addition?

[write] (Batch 87, 3 October 2026.) Two different questions of Glazer carry the number 1 in the repository. His earlier paper *A choiceless box game paradox* (arXiv:2211.10474, 2022) ends its Section 5 with another question: does ZF prove that the box game $G([3], \mathbb{R}, \{0, 1\})$ is winnable, that is, can three players facing continuum many boxes of bits guarantee that at

most one of them guesses wrong? Batch 87's manuscript 02 studies that question in the research-report collection's `SetTheory/Cardinals/docs/reports/ordinals-and-order-types/measurable-box-games`, and answers it only for blind outputs measurable for the cylinder σ -algebra. There it is Research question Q1. It is *not* Question 1 of this report, which asks whether Theorem 2 of the Topological Tennenbaum paper is provable in ATR_0 ; the two reports share no theorem and no notation.

Glazer's speculation. Immediately after Question 2, Glazer writes that he speculates that an affirmative answer to Question 1 “would lead to a negative answer to Question 2” [1, p. 8]. His reason is an analogy: removing the exponential function from the proof that multiplication is discontinuous would parallel removing multiplication from a proof that addition is discontinuous. Neither source of [Part I](#) mentions this sentence; it is recorded here because [Part I](#) goes the other way.

How this report reads it [write]. [Part I](#) answers Question 2 *as printed* affirmatively ([theorem 3.1](#)). The question asks only for an uncountable Polish space, a model of Presburger arithmetic and continuous addition; it does not ask that the topology be the order topology, and it mentions no multiplication. The speculation is a heuristic about proofs, not a stated implication, and nothing here bears on Question 1, which stays open; no conclusion about Question 1 is drawn from the affirmative answer. Three facts locate the difference. First, the affirmative models are not an artefact of a missing definability condition: every definable relation in them is Δ_2^0 ([theorem 7.1](#)), so the order and every other definable relation are Borel, as in the Borel models of Glazer's theorems. Second, what fails in them is multiplication: no unital semiring multiplication exists on them at all ([theorem 10.4](#)), so Corollary 1 does not apply. Third, on a ring that does carry Borel multiplication and open induction, Corollary 1 does forbid continuous addition under every Borel-compatible Polish recoding ([theorem 25.2](#)), and [section 26](#) shows that the real model of [Part I](#) sits inside that ring. We do not know whether Glazer intended Question 2 with a further hypothesis, for instance an expansion to a model of a stronger theory, under which his speculation could hold; that would be a different question.

[write] (Batch 87.) “Question 1, which stays open” was written for Parts I and II; see the paragraph on Parts VI–IX below, where a manuscript's affirmative answer, not independently reviewed, is recorded.

Parts III–V and the speculation [write]. None of sources 06–09 quotes the sentence either. Source 07 comes closest: after restating Question 1 it observes that, because the affirmative Presburger construction stands, a route from a weaker metatheory to a *negative* Presburger conclusion cannot yield that conclusion as stated ([section 43](#)). Read literally, the outcome the speculation predicts is therefore excluded for Question 2 as printed, whatever the answer to Question 1; only a variant of Question 2 with further hypotheses could carry it. Source 06's isolated-zero theorem ([theorem 39.1](#)) removes a hypothesis from one of Glazer's proofs, in the spirit of his analogy, but from his Theorem 1: with Glazer's subset-sum and zero-one method, and without his injectivity claim, it weakens Peano arithmetic to PA^- in “no perfect Polish space carries a model with continuous addition and multiplication” ([corollary 39.2](#)). It does not bear on Question 1, which concerns his Theorem 2; Glazer himself remarks [1, p. 7] that Theorem 1 and Corollary 1 can be proven in ATR_0 , and source 06's project 6 asks for the strength of the isolated-zero principle. Sources 08 and 09 concern rings, where multiplication is present, and both state that they answer neither of Glazer's questions. Priority of source 06's strengthening is not established.

Parts VI–IX, Question 1 and the speculation [write]. (Batch 87.) Source 14 ([Part VIII](#)) claims an affirmative answer to Question 1: Glazer’s Theorem 2 is provable in ATR_0 , under a stated convention for coding Polish presentations and Borel data ([corollary 111.2](#)), with a stronger, cofinal conclusion ([theorem 104.1](#)). This report prints that answer as a claim. It is unrefereed and AI-assisted; its ATR_0 bookkeeping rests partly on inputs it declares rather than proves, and its author did not see the Gaifman–Dimitracopoulos chapter whose internal MRDP theorem it uses. It has not been independently reviewed, either here or in the repository’s Hilbert’s-tenth programme, whose review of the archive read its internal-MRDP input and states that it does not certify the claimed resolution ([section E](#)). The statements of this report that Question 1 stays open were made for Parts I–V and remain true in the sense that no reviewed answer is known; they are qualified by this paragraph. If source 14’s claim is correct, the outcome Glazer speculated, a negative answer to Question 2 following from an affirmative answer to Question 1, does not occur for Question 2 as printed, which Part I answers affirmatively; as noted above, that outcome was already excluded for Question 2 as printed, whatever the answer to Question 1. Source 14 does not mention the speculation. Sources 10, 12 and 15 state that they do not answer Question 1, and source 13 that its results do not resolve it; their theorems concern signed group topologies (10, 12), inherited cones (15) and the positive models of Part I (13).

Priority. Priority of the affirmative answer is **not established**. Source 05 searched for the exact question and combinations of its terms and found no earlier answer, and says that this does not prove that none exists; source 04 says that historical priority has not been established. The intake re-checked only that the arXiv record has one version and no journal reference (October 3, 2026). The answer is an elementary construction; an earlier publication of it, or a published variant of the question, may exist.

1.2 Thirteen manuscripts, one report

Source	Manuscript (delivered pages)	Pin	Printed as
05	<i>Polish Presburger Arithmetic inside the Omnific Integers: An affirmative construction for Glazer’s Question 2, effective Baire-space models, and structural obstructions</i> (24, A4); archive <code>polish_presburger_glazer</code>	<code>a11efab09</code>	base text of Part I ; worked examples in section A
04	<i>Polish Presburger Arithmetic: A positive construction for Glazer’s question, an effective Baire-space model, and exact regularity boundaries</i> (25, letter); archive <code>Polish_Presburger_Glazer_ProveIt</code>	<code>3ad5f878c</code>	Part I , every statement marked [04]; second proofs; sections B and C
03	<i>Borel Presentations and Support Barriers in Omnific Arithmetic: Scattered exponent orders, Hahn fields, and the exact discontinuities of finite normal forms</i> (23, letter); archive <code>glazer_proveit_research</code>	<code>c5b4881fa</code>	Part II and section D
07	<i>Continuous Presburger Arithmetic on Polish Spaces: An affirmative construction for Glazer’s question, Baire-space models, and omnific realizations</i> (27, letter); archive <code>polish_presburger_research</code>	<code>0f95145cb</code>	base text of Part III ; sections G and H
06	<i>Continuous Presburger Arithmetic on Polish Spaces: Lexicographic models, an isolated-zero obstruction, and a boundary between additive and semiring arithmetic</i> (21, letter); archive <code>polish_presburger_article</code>	<code>fa2f3e419</code>	Part III , every passage marked [06]; sections I and J
08	<i>Local Compactness Forces Arithmetic Rigidity: Borel ordered groups, Presburger models, and multiplication without regularity assumptions</i> (23, A4); archive <code>glazer_proveit_local_compactness</code>	<code>883e0b3b2</code>	Part IV and section K
09	<i>Polishability at the Hahn–Puisseux–Levi–Civita Boundary: Automatic order bounds, local Polish hulls, and countable omnific integer parts</i> (30, letter); archive <code>Polish_Hahn_Puisseux_Levi_Civita_Research</code>	<code>4af956fe0</code>	Part V and sections L and M

The numbers are the manuscript numbers of batch 86 of the repository’s intake (manuscripts 01 and 02 of that batch are unrelated OEIS reports; manuscript 10, on the standard cut in one infinite interval, was written into the report `Algebra/SurrealNumbers/docs/surreal/discrete-initial-subgroups-and-omnific-normalization`). Manuscripts 03–05 arrived in commit `31fdc6571` and were placed in `3d2177df4`; manuscripts 06–09 arrived in `fb8414869` and were placed in `0bd0e5527`. Every pin is an ancestor of its placement commit, and the repository files the sources inspected are unchanged between their pins and the placements.

Why one report. 04 and 05 prove the same main theorem independently: they share 0.6% of their word 8-grams, essentially only Glazer’s quoted question and his bibliography entry, and have different pins, notation and supplementary theorems. 03 answers the same brief from the other side: its finite principal-part ring contains the real model of 04 and 05 as one stratum (section 26), and its Theorem 10.2 (theorem 25.2) is Glazer’s obstruction on that ring. No earlier report of the collection names Glazer, his questions, Polish models of Presburger arithmetic, or Borel presentations of Hahn fields.

Why Parts III–V are here. Sources 06 and 07 answer Glazer’s Question 2 again and go on from Part I’s models; neither is an edition of the other or of 04 and 05 (at most 0.74% shared word 8-grams among manuscripts 03–10). Source 08 names 05 and 03 as its prior work and asks what local compactness forces on every Borel ordered presentation; source 09 names them too and asks the Polish-group question that follows 03’s Borel classification. No other report of the collection treats Polish topologies on Puiseux or Levi-Civita fields. Source 07 is the base of Part III because its realization theorem has the weakest hypotheses (Part III).

Why base 05. On the shared theorems 05 has the weaker hypotheses: its Polish criterion (theorem 5.2) is an equivalence for every Polish group topology on the coefficient group, where 04’s transfer theorem (theorem 5.4) is a sufficient condition under the stronger hypothesis that the positive cone be open and division continuous; its Δ_2^0 theorem holds under the same weaker hypothesis; and it treats the models explicitly as fragments of **Oz**. Results of 04 not in 05 are printed in full and marked [04]: continuity of fixed division under its hypothesis, the general closed-order obstruction and the invariant-metric obstruction, the reduction of all unital additive maps to the divisible part, the matrix classification of the self-embeddings of the Baire model with automatic continuity, and the Laurent-series model with its growth-order obstruction. Where 04 proves a result of 05 by a different argument, its proof is printed as a marked second proof.

Sources of Parts VI–IX [write]. Five manuscripts of batch 87 arrived in commit fa0a0576e and were placed in d151b39ca. They are numbered here by their file prefixes, which continue the batch-86 numbers 03–09: batch-87 manuscript 01 is source 10, manuscript 03 is source 12, 04 is 13, 05 is 14 and 06 is 15. Batch-87 manuscript 02, on Glazer’s box-game paper, became the report SetTheory/Cardinals/docs/reports/ordinals-and-order-types/measurable-box-games; prefix 11 is unused. All five pins are ancestors of the placement commit; at the pins of sources 10, 12, 14 and 15 this report consisted of the delivered base text of Part I, and source 13’s pin is one commit before the report was placed. Sources 10, 12, 13 and 14 describe themselves as prepared with ChatGPT, source 15 as prepared with AI assistance.

Source	Manuscript (delivered pages)	Pin	Printed as
12	<i>Baire-Category Rigidity for Discretely Ordered Rings: Beyond local compactness, without measurable multiplication</i> (23, A4); archive Baire_Category_Arithmetic_Rigidity; batch 87, manuscript 03	0f084afa9	base text of Part VI ; sections N and O
10	<i>Baire Order Forces Arithmetic Rigidity: Discretely ordered rings with Polish addition and completely unrestricted multiplication</i> (23, A4); archive Glazer_ProveIt_Baire_Rigidity; manuscript 01	6fef5383b	Part VI , every passage marked [10]; sections P and Q
13	<i>Polish Presburger Arithmetic Beyond the Split Construction: Nonsplit models, exact continuity, and optimal computation from streams</i> (39, A4); archive polish_presburger_frontiers; manuscript 04	eab47e330	Part VII ; sections R and S
14	<i>Diophantine Definability and Topological Arithmetic: An ATR_0 proof of Glazer's theorem, cofinal unary discontinuities, and nonsplit Presburger models</i> (26, letter); archive glazer_atr0_research; manuscript 05	6fef5383b	Part VIII ; sections T and U
15	<i>Polish Arithmetic: Multiplication Obstructions and Local Compactness</i> (32, A4); archive polish_arithmetic_package; manuscript 06	0f084afa9	Part IX ; section V

Why Parts VI–IX are here. Each of the five answers or continues a question or result of this report. Sources 12 and 10 answer Part IV's [question 57.1](#); source 13 answers four questions of Part I and one of Part III, wholly or in part; source 14 claims to answer Glazer's Question 1, which Part III prints as [question 43.1](#); source 15 classifies the locally compact cones of Part IV without assuming a locally compact ambient group. Sources 12 and 10 prove the same principal theorem by different proofs and are merged as one part, with base 12 for its weaker hypotheses ([Part VI](#)); sources 13, 14 and 15 are parts of their own. One construction, $\mathbb{R} \overrightarrow{\times} K$ with a countable nonsplit $\mathbb{Z}$ -group K , occurs in sources 13, 14 and 15; it is printed once, in [Part VII](#), and flagged in Parts VIII and IX. No two of the five are editions of one another: the placement measured under 0.8% shared word 8-grams between any two of them.

Source of Part X [write]. (Batch 89.) Source 16 is manuscript 02 of batch 89. It arrived in commit 7c0f2d9f9 with three other manuscripts, a fifth arriving in 8ea27d6c0, and all five were placed in 23adb85f9; the other four went to surreal-well-orders, birthday-cut-offs-and-hereditary-sets and the research-report collection's naming-elementary-embeddings. It is numbered by its file prefix 16 (the batch-87 numbers end at 15). Its pin 0f084afa9 is an ancestor of the placement; at the pin this report was the delivered base text of Part I. It names this report's questions on endomorphisms and on elementary-submodel lattices and answers them for its family, so it is a part of this report, not a report of its own. It is not an

edition of any source: the placement measured 1.24% shared word 8-grams with this report, shared Presburger phrasing. It describes itself as prepared for Vladimir Reshetnikov; its delivery README calls it AI-assisted.

Source	Manuscript (delivered pages)	Pin	Printed as
16	<i>Elementary Embeddings and Polish Submodels of Lexicographic Presburger Arithmetic: Matrix normal forms, closed images, and two sharp ordinal thresholds</i> (23, A4); archive Glazer_ProveIt_Embeddings_and_Thresholds; batch 89, manuscript 02	0f084afa9	Part X; sections W and X

Reading conventions. In [Part I](#), unmarked text is source 05's; text from source 04 is marked [04] (statements in their titles, inserted passages in the sentence that introduces them). In [Part II](#) all unmarked text is source 03's; in [Part III](#) unmarked text is source 07's and text from source 06 is marked [06]; in [Parts IV](#) and [V](#) all unmarked text is that of source 08 and source 09 respectively. In [Part VI](#) unmarked text is source 12's and text from source 10 is marked [10]; in [Parts VII](#) to [IX](#) all unmarked text is that of sources 13, 14 and 15 respectively, and in [Part X](#) that of source 16. Text written for this report is marked [merge] (a mathematical observation combining the sources, with proof) or [write] (editorial notes: provenance, citations of the collection, corrections). Phrases such as "this manuscript" or "this article" inside a source's text refer to that source. References to the collection's other reports give their labels in typewriter type, for example `odg:thm:floor`; these labels are not hyperlinks.

Labels. Every label carries the prefix `pma:`. Source 05's labels are `pma:pr:<label>`, source 03's are `pma:bor:<label>`. Source 04's labels are `pma:pr:<label>`, except the ten that coincide with labels of 05 (`eq:cone`, `eq:division`, `sec:algebra`, `sec:baire`, `sec:embeddings`, `sec:formalization`, `sec:multiplication`, `sec:questions`, `thm:baire`, `thm:qe`), which take the sub-prefix `pma:pr:s04:`. Of these, `eq:cone` and `eq:division` denote displays identical to 05's and are printed once, under 05's labels; source 04's displays `eq:divisionformula`, `eq:monuslocus`, `eq:surrealR` and `eq:surrealQ` are likewise printed once as 05's (4.2), (8.2), (9.1) and (9.2), and its Definition 2.1 (`def:split`) is 05's setup in [section 3](#). Where 04 states a theorem that 05 also states, 04's label is a second label of 05's statement (for example `pma:pr:thm:monuslocus` is [theorem 8.3](#)). Labels written for the merge have the bare prefix `pma:` or a part prefix. Sources 07 and 06 have labels `pma:cp:<label>`, except six labels of 06 that coincide with labels of 07, which take `pma:cp:s06:` ([Part III](#)); source 08 has `pma:lc:`, source 09 `pma:ser:`. A few labels were added to unlabelled statements of these sources so that notes can cite them. Parts VI–IX use `pma:br:` (sources 12 and 10; seven labels of source 10 that coincide with labels of source 12 take `pma:br:s10:`), `pma:nsp:` (13), `pma:atr:` (14) and `pma:lcc:` (15); their introductions list the labels they add. The batch-87 write also labelled fifteen questions of Parts I, III and IV so that notes can cite them: `pma:pr:q:zerodim`, `pma:pr:q:mindchange`, `pma:pr:q:beyondsplit`, `pma:pr:q:rigidity`, `pma:pr:q:discprofiles`, `pma:pr:q:effectivetruth`, `pma:pr:q:metrics`, `pma:cp:q:glazerone`, `pma:cp:q:spaces`, `pma:cp:q:nonsplit`, `pma:cp:q:connected`, `pma:lc:q:weakenborel`, `pma:lc:q:sizespectra`, `pma:lc:q:strength` and `pma:lc:q:certified`. [Part X](#) uses `pma:emb:` (source 16; its introduction lists the labels added), and the batch-89 write labelled four further questions of [Part I](#): `pma:pr:q:coefficientgroups`, `pma:pr:q:lattices`, `pma:pr:q:surrealcutoffs` and `pma:pr:q:embeddingcategories`.

1.3 Notation

The sources use clashing symbols. The table fixes the convention of this report; renamed symbols are marked $\rightarrow$. Its second block, for Parts III–V, was added with them. The third block, for Parts VI–IX, was added with those, and the fourth, for Part X, with it.

Notion	This report	In the sources
Presburger arithmetic	$\text{Th}(\mathbb{N}; 0, 1, +, <)$	05: the same; 04: $\text{Pr}_{\mathbb{N}} = \text{Th}(\mathbb{N}; 0, 1, +)$, kept in 04's text. The two are definitionally equivalent: $x \leq y$ is $\exists z (x + z = y)$.
Lexicographic group, coefficient group D dominant	$D \overrightarrow{\times} \mathbb{Z} = G_D$	05: the same; 04: $D \oplus_{\text{lex}} \mathbb{Z} = G(D) \rightarrow G_D$
Nonnegative cone	$M_D; M_E$ for a subgroup E	04: $M(D), M(E) \rightarrow M_D, M_E$
Baire coefficient group	$D_B = \mathbb{Q}^{\mathbb{N}}$, lexicographic, each $\mathbb{Q}$ discrete	05: the same; 04: $D_{\omega} \rightarrow D_B$ (the same group, order and topology)
Baire model	$M_B = (D_B \overrightarrow{\times} \mathbb{Z})_{\geq 0}$	04: $M_{\omega} \rightarrow M_B$ (the same model)
Laurent model (04)	$M_L = M_{D_L}, D_L = \mathbb{Q}((t))$	04 only
Baire space	$\mathbb{N}^{\mathbb{N}}$	04's macro $\backslash B = \mathbb{N}^{\mathbb{N}}$; 03's $\backslash B = \mathcal{B}$ is defined but never used
Homeomorphisms with Baire space	05: $h : \mathbb{N}^{\mathbb{N}} \rightarrow M_B$ (tag code); 04: $\Psi : M_B \rightarrow \mathbb{N}^{\mathbb{N}}$, charts $\Psi_{\geq}, \Psi_{>}$, integer code $\zeta : \mathbb{Z} \rightarrow \mathbb{N}$	04: $H, H_{\geq}, H_{>}, h \rightarrow \Psi, \Psi_{\geq}, \Psi_{>}, \zeta$. The two codings are different maps.
Lexicographic symbol	$\overrightarrow{\times} (05); \mathbb{Z}_{\text{lex}}^d$ for lexicographic powers (03)	04's subscript lex in $\oplus_{\text{lex}} \rightarrow \overrightarrow{\times}$
Division by a standard m	$q_m(x) \in M,$ $r_m(x) \in \{0, \dots, m-1\}$ (05); $Q_m(x) \in M,$ $R_m(x) = (0, z \bmod m) \in M$ (04)	kept; R_m is r_m viewed as an element of the model
Congruence predicates	$D_m(t) \iff t \in mG$ (05); $u \equiv_m v \iff u - v \in mG$ (04)	kept; $u \equiv_m v \iff D_m(u - v)$
Coset invariance	" mG -invariant" (05) = " m -residue invariant" (04)	the same notion
Cofinal infinite element	"cofinal infinite additive unit" (05) = "arithmetically infinite order unit" (04)	the same notion; not a multiplicative unit
Truncated subtraction	$T(x, y)$ (05), $x \dot{-} y$ (04)	the same operation; $\text{Disc}(f)$ is the discontinuity set of f
Pivots, filtration (04)	$p(k), F_k$ in theorem 11.4 ; F_n in proposition 10.6	local; p elsewhere is the predecessor; 03's closed pieces are $F_{Z,k}$
Growth classes (04)	$\mathcal{A}(M)$	local to proposition 10.7 ; not the ring $\mathcal{A}_{\text{fin}}$
Finite principal-part ring (03)	$\mathcal{A}_{\text{fin}} = \mathbb{Z} \oplus \bigoplus_{q \in \mathbb{Q}_{>0}} \mathbb{R}X^q$, cone $\mathcal{A}_{\text{fin}, \geq 0}$	03: $\mathcal{A}, \mathcal{A}_{\geq 0} \rightarrow \mathcal{A}_{\text{fin}}, \mathcal{A}_{\text{fin}, \geq 0}$. Not the collection's $\mathbf{Oz}$, which the base of the omnific Diophantine report once called A .
Rational Hahn ring (03)	$\mathcal{H}_{\mathbb{Q}}$, (17.1)	03: $H \rightarrow \mathcal{H}_{\mathbb{Q}}$
Kleene–Brouwer embedding (03)	$\mu_{\text{KB}} : \mathbb{N}^{<\mathbb{N}} \rightarrow \mathbb{Q} \cap (0, 1)$	03: $h \rightarrow \mu_{\text{KB}}$. Not the dyadic coding κ or the midpoint coding μ into $(1, 2)$ of the omnific-notations report.

Notion	This report	In the sources
Growth variable (03)	$X = t^{-1}$, read as ω	03 §2.1: increasing- t convention in sections 18 and 19 , reverse X -convention from section 22 on; $q \mapsto -q$ converts
Well-ordered subsets (03)	$\text{WO}(L) \subseteq 2^L$	not the well-orderings $\text{WO}_\mu(X)$ of type μ of the surreal-well-orders report
Local sets (03)	W (proof of theorem 23.2); $\mathcal{Z}$ (lemma 24.2); D_+ , $D_\times$ (theorem 24.3)	03: $D \rightarrow W$ and $\mathcal{Z}$; D_+ , $D_\times$ kept. In Part I , D is always the divisible coefficient group.
L	a countable linear order (03)	in 05's and 04's elimination proofs, a lower bound or a common multiple (local)
e_j, e	unit vectors of $\mathbb{Q}^{\mathbb{N}}$ (05, 04); computable enumerations $e, e_+, e_{\geq 0}$ (05, section 6)	as in 05
Omnific integers	$\mathbf{Oz} = \mathbb{Z} \oplus \Pi$, growth notation ω^γ	as in the collection's docs/NOTATION.md
<i>Parts III–V (sources 07, 06, 08, 09)</i>		
The variable t	Part III: t positive <i>infinite</i> (06's $\mathbb{Z} + t\mathbb{R}[t]$, $t \mapsto \omega$); Part V: t positive <i>infinitesimal</i> (09's Hahn convention, $t^q \mapsto \omega^{-q}$)	kept in each part; tempting false reading: 06's t and 09's t are reciprocal, not equal. 03 uses $X = t^{-1}$ (infinite); 08 writes X for 06's t
Polynomial ring $\mathbb{Z} + t\mathbb{R}[t]$	A , cone $P = A_{\geq 0}$, topology τ_P by degree and constant term (06)	= 08's $A_* = \mathbb{Z} + X\mathbb{R}[X]$ = the ring $\backslash\text{Pmod}[X]$ of the omnific Diophantine report (odg:frac:sub:models) and, at $t = \omega$, $\mathbb{Z} + \omega\mathbb{R}[\omega]$; under $t \mapsto X$ the integer-exponent subring of 03's $\mathcal{A}_{\text{fin}}$. Not 03's cone $\mathcal{A}_{\text{fin}, \geq 0}$
Presburger arithmetic	$\text{Th}(\mathbb{N}; 0, 1, +, <)$; 06: Pres	07's macro for $\text{Th}(\mathbb{N}; 0, 1, +, <)$ is expanded; 06's Pres = $\text{Th}(\mathbb{N}, 0, 1, +, <)$ kept. 08: signed language $\mathcal{L}_{\text{Pres}}$
Lexicographic product	$\vec{\times}$	07's $A \times_{\text{lex}} \mathbb{Z}$, 06's $D \text{ lex } \mathbb{Z}$ and 08's $\mathbb{R}^d \times_{\text{lex}} \mathbb{Z} \rightarrow \vec{\times}$ (same order: left factor dominant)
Real model	$M_{\mathbb{R}} = M_{\mathbb{R}}$	the same set and topology in 06, 07, 08; 07's second topology $M_{\mathbb{R}}^{\sharp}$ is a different space on the same algebra
Model families (07)	M_A, G_A (general A); $M_\alpha = (\mathbb{Q}^\alpha \vec{\times} \mathbb{Z})_{\geq 0}$; M_s with $B_s = \mathbb{Q} + \mathbb{Q}s$, $s \in (1, 2)$ irrational	M_ω is Part I's M_B ; 07's H_s is a convex subgroup
Model families (06)	$M(D), G(D)$; $M_\omega (= M_B)$; M_α (as 07); M'_s, G'_s, H'_s for $s \in 2^{\mathbb{N}}$	06: $M_s, G_s, H_s \rightarrow M'_s, G'_s, H'_s$ (07's M_s is another family); 06's $H = \mathbb{Q}^{\mathbb{N}}$ in section 35.5 is local
Models of Part IV	$G_d = \mathbb{R}^d \vec{\times} \mathbb{Z}$, M_d (so $M_1 = M_{\mathbb{R}}$); $H_d \subseteq \mathbf{Oz}$; $\mathbb{R}^d \times E$	08's D in theorem 47.2 is a countable discrete group, not Part I's divisible D
Series fields of Part V	$\mathcal{P}_k$ (Puisseux), $\mathcal{L}_k$ (Levi-Civita), $\mathcal{H}_k = k((t^{\mathbb{Q}}))$, integer part I_k ; hulls F_A, J_C	09's macros $\backslash\text{PP}$, $\backslash\text{LL}$, $\backslash\text{HH}$, $\backslash\text{II}$ renamed in the source only; same symbols printed. $\mathcal{P}_k$ is not 03's cone $\mathcal{A}_{\text{fin}, \geq 0}$; 03's $\mathcal{P}$ is the real-coefficient Puisseux field

Notion	This report	In the sources
Well-founded trees Typography	WF 1, $\dot{-}$, IOpen, ProveIt	09's WF $\rightarrow$ 03's WF (same set) 07's $\backslash\text{ind}$, $\backslash\text{dotminus}$; 08's IOpen; 07's ProveIt $\rightarrow$ these (same meaning)
Local letters	A (a Polish group in 07, a ring in 06 and 08); P (a cone); H, D, B, J	local to the statement or proof where they are introduced
<i>Parts VI–IX (sources 12, 10, 13, 14, 15)</i>		
Ordinary-integer-bounded part (10)	$\text{Fin}(A) = \{x : x \leq \underline{n} \text{ for some ordinary } n\}$	10's macro $\backslash\text{Fin}$ renamed in the source only; printed Fin in roman type. Not Part II's Fin (sans-serif), the set of finite subsets
Principal convex hull (12, 10)	$H_a = \{x : x \leq na \text{ for some ordinary } n\}$	the same in both sources; local to Part VI. Not source 08's finite surreal scales $H_d \subseteq \mathbf{Oz}$, 03's ring $\mathcal{H}_{\mathbb{Q}}$, or source 15's $H = G^0$
Unit-image spectrum (12, 10)	$\mathcal{U}(G)$	10: $\mathcal{S}(G, e) \rightarrow \mathcal{U}(G)$ (the same set)
Cardinal functions (12, 10)	c, d (12); cell, dens, ext, L (10)	kept in each source's text; c = cell is cellularity, d = dens density. c is the continuum, written $\mathfrak{c}$ in Part III
Continuous scalars (10) The ring $\mathbb{Z} + t\mathbb{R}[t]$ (12, 10, 15)	$\text{Sc}_{\text{cont}}(A)$ 12: $A = \mathbb{Z} + t\mathbb{R}[t]$; 10: $A = \mathbb{Z} + X\mathbb{R}[X]$; 15: R_0	not source 08's Borel scalars $\mathcal{B}(A)$ the ring of Part III and source 08's A_* ; t and 10's X are positive <i>infinite</i> , as in Part III, not Part V's infinitesimal t
Lexicographic products (12–15) Baire model (13)	$\overrightarrow{\times}$ $M_\omega, D_\omega = \mathbb{Q}^\omega$	12's and 13's $\oplus_{\text{lex}}$, 13's, 14's and 15's $\text{lex} \rightarrow \overrightarrow{\times}$ (left factor dominant) kept; Part I's $M_B, D_B, \mathbb{Q}^{\mathbb{N}}$ (the same objects)
Countable core (13, 14, 15)	K (13, general); $K_* = G_{\zeta_*}$ (14); $C = N - N$ (15)	14: $D \rightarrow K_*$ in its Section 10 and $D \rightarrow K$ in the commentary on its question on nonsplit topology, because D is the divisible coefficient group in Parts I and VII. 13's explicit core is $\text{isg's } G_{\varepsilon_2}$
Divisible kernel (13)	$D = \ker r = \bigcap_m mG$ in section 95	local; r is the canonical residue map, not Part I's remainder r_m
Revision numbers (13) Exponential and unit (14)	$\mu(f), \mu_{\text{eff}}(f)$ $e(x) = 2^x$, internal; e_0 the unit of K_*	not Part II's μ_{KB} 14's e is not a unit; Part I's e_j are unit vectors
Root scales (14)	$t_n, q_n, X_n, h_{k,n}$	local to section 108 ; unrelated to the variable t of Parts III and V
Cantor space (14) Surreal embeddings (14, 15)	$2^{\mathbb{N}}$ $\Phi, \Phi_\omega = \omega\Phi$ (14); Φ_L (15)	14's $\backslash C = 2^\omega \rightarrow 2^{\mathbb{N}}$ 14's $\Psi \rightarrow \Phi_\omega$, because Ψ is Part I's Baire homeomorphism; 10's Ψ in theorem 79.14 is local, like source 08's in theorem 46.2
Sets of Cantor space (14) Components (15)	D, D^*, E_n $H = G^0$ and its infinitesimal subgroup C	local to sections 110 and 111 source 08's C and B_C , letters exchanged; tempting false reading: C is the identity component, as in Part IV
Models on a countable order (15)	M_L, G_L, D_L, V_L (L italic)	not Part I's Laurent model $M_L = M_L$

Notion	This report	In the sources
Locally compact carrier (15)	X_{lc}	15 only
Typography	$\text{IOpen}, \text{Pres}, \text{Provelt}, \div$	12's roman IOpen , Pres, 13's $\backslash\text{dotminus}$ and 14's $\text{ProveIt} \rightarrow$ these (same meaning)
<i>Part X (source 16)</i>		
Coefficient group (16)	$H_\alpha = \mathbb{Q}^\alpha$ (full product, discrete coordinates, lexicographic), $G_\alpha = H_\alpha \times \mathbb{Z}$ ordered as $H_\alpha \xrightarrow{\gamma} \mathbb{Z}$, $M_\alpha = (G_\alpha)_{\geq 0}$	kept: H_α is Part III's A_α (source 07), not renamed because source 16's A is a matrix; H_ω is Part I's D_B . Tempting false reading: H_α is not Part VI's principal hull H_a , source 07's convex subgroup H_s , source 08's $H_d \subseteq \mathbf{Oz}$ or source 15's $H = G^0$
Finite-support vectors (16)	$D_B = \mathbb{Q}^\omega$, $D_B^{\text{fin}} = \mathbb{Q}^{(\omega)}$; dual $H_\alpha^\circ = \mathbb{Q}^{(\alpha)}$, pairing $\langle c, h \rangle$, annihilator $V^\perp$	16: $D, D_{\text{fin}} \rightarrow D_B, D_B^{\text{fin}}$ (the same group as Part I's); D is the generic divisible group in Parts I and VII
Valuation, pivots (16)	$v(h)$, the first nonzero index; pivot map p ; pivot subset $S \subseteq \alpha$; $\text{otp}(S)$	v and p as in source 04's theorem 11.4 ; p elsewhere is the predecessor; S is not source 04's shift S (example 11.6)
Matrices and maps (16)	$A = (a_{ji})$, left inverse L, U_V , E_S , quotient map B , projection π	local to Part X ; A is not Part III's group A or A_α , B not the Baire coefficient group
Wild functional, shear (16)	φ (exists by choice); Sh_φ ; graph space W_φ	16: $S_\varphi \rightarrow \text{Sh}_\varphi$, because S is the pivot subset; not Part I's shift σ
Terminal tail (16)	$\alpha = \lambda + n$, $K_\alpha \cong \mathbb{Q}^n$	16's local K in the proof of theorem 134.1 $\rightarrow K_\alpha$ (the same set); not source 13's core K or source 14's K_*
Restriction, lexicographic sum (16)	$x \upharpoonright \omega$; $\overrightarrow{\times}$	16's macro $\backslash\text{res} (\upharpoonright) \rightarrow \backslash\text{rstr}$, because Part I's $\backslash\text{res}$ is res ; 16's roman $\text{lex} \rightarrow \overrightarrow{\times}$
Omnific map (16)	ι_α , exponents $\gamma_i = (i+1)^{-1}$	= Part III's $J_\alpha, \varepsilon_\beta$ (theorem 40.1); Part I's ι_B for $\alpha = \omega$
Local letters (16)	$N, \Gamma = N - N$ in the proof of theorem 131.1	16: $S, K \rightarrow N, \Gamma$, because S is the pivot subset

1.4 Relation to the repository

The formal project `Logic/PresburgerArithmetic`. Both sources of [Part I](#) inspected this project at their pins, 05 its Lean file `Lean/PresburgerArithmetic/Cooper.lean` and 04 its Rocq file `Coq/Cooper.v`. The project decides Presburger sentences over the ordinary integers. Its periodicity lemmas assume a predicate on $\mathbb{Z}$ invariant under one standard shift; [example 4.4](#) and [proposition 4.6](#) show that this hypothesis cannot be carried verbatim to nonstandard $\mathbb{Z}$ -groups, and [lemma 4.7](#) and [theorem 4.8](#) give the model-uniform replacement. This is a porting requirement, not a defect of the integer theorems. [Section 13.5](#) names the declarations. **Nothing in this report is formalized**, and its placement in the Surreal collection, with a pointer from the formal project, confers no formal status.

The Surreal collection. Results already in the collection are cited, not reprinted as new:

- `exr:prop:division` and the paragraph after it in the `exponential-relations-over-omnific-integers` report: $\mathbf{Oz}$ is a $\mathbb{Z}$ -group, hence a model of Presburger arithmetic in its ordered-group presentation. [Proposition 4.2](#) and [theorem 4.9](#) prove the classical facts again for the split groups G_D .
- `isg:cf:main:presburger` in the `discrete-initial-subgroups-and-omnific-normalization` report: a set-sized $\mathbb{Z}$ -group has an initial realization in $\mathbf{Oz}$ if and only if it splits as $G^{\text{dv}} \xrightarrow{\sim} \mathbb{Z}$ with $G^{\text{dv}} = \bigcap_n nG$. The groups G_D of [Part I](#) are split by construction, with $\bigcap_m mG_D = D \times \{0\}$ ([corollary 4.3](#)), so that theorem applies to them [merge]; whether 05's maps $\iota_{\mathbb{R}}, \iota_{\mathbb{B}}$ are initial realizations in its sense is not checked here.
- `onot:neg:lem:kbtree`, `onot:neg:thm:pionesone` and `onot:neg:rem:secondroute` in the `omnific-notations` report: the lightface Π_1^1 -completeness of support validity. 03's [lemma 18.2](#) and [theorem 22.1](#) are their boldface counterparts (notes after those statements).
- `odg:thm:floor` in the `omnific-diophantine-geometry` report, proved in Lean for every surreal: 03's [lemma 23.1](#) is its restriction to the Puiseux field (note after it).
- `dsn:cor:openinduction` in the `definable-surreals-and-omnific-integers` report, and the Shepherdson paragraph of the `omnific-diophantine-geometry` report: the route of 03's [theorem 23.2](#) (note after it).

All these labels exist at the repository's current state and already existed at the pins of sources 03–05. Of them only `odg:thm:floor` is formalized (`docs/FORMALIZATION.md`, row “Full `odg:thm:floor`”); the others are recorded there as pending.

Parts III–V. Sources 06 and 07 cite the formal Presburger project (its `Lean Syntax.lean`, `Decision.lean`, `Audit.lean`) and source 06 also `Logic/Interpretability/PAHF/Lean/PAHF/PASyntax.lean`; their descriptions are correct at the current state ([section 41](#)). Source 06 cites `isg:thm:suspension`, `isg:cf:main:presburger` and `isg:cf:thm:PA`, and the omnific Diophantine report's `odg:frac:sub:models`, `odg:frac:prop:intersection` and `odg:frac:ex:workspace` for the ring $\mathbb{Z} + t\mathbb{R}[t]$; the notes of Parts III–V also cite `odg:thm:induction`, `odg:rem:ringinduction`, `odg:thm:floor`, `dsn:cor:openinduction`, `onot:neg:lem:kbtree` and `onot:neg:thm:pionesone`. All exist at the current state; of them only `odg:thm:floor` is formalized. Placement beside the formal Presburger project confers no formal status on Parts III–V either.

Parts VI–IX. Sources 10 and 12 read `Logic/PresburgerArithmetic/README.md` and `Algebra/SurrealNumbers/README.md`; source 13 also the Lean files `Syntax.lean` and `Decision.lean` of the Presburger project, the `discrete-initial-subgroups-and-omnific-normalization` report and the Hilbert's-tenth programme's scope note; source 14 `Logic/PeanoArithmetic/ListCoding/Lean/PAListCoding/ExponentiationDiophantine.lean` (`naturalPowGraph_diophantine`, `power_polynomial_exists`, proved from `Mathlib's Dioph.pow_dioph`) and its Coq sibling `ExponentiationDiophantine.v`; source 15 the two Presburger Lean files and `OmnificIntegers.lean`, `OmnificPurelyInfiniteModule.lean` and `OmnificAlgebraDiscreteness.lean` in `Algebra/SurrealNumbers/Surreal/Foundations`. Their descriptions are correct at the current state; the two READMEs have changed since the pins only by the catalogue's pointers to this report (2b7b388ba), and the other files are unchanged. Those Lean and Coq files are formal developments, but about the ordinary

integers, the standard natural numbers and the omnific coefficient construction; none states a result of Parts VI–IX, and source 14 stresses that the standard exponentiation theorem is not the internal statement it uses. The notes of Parts VII and VIII cite `isg:cf:prop:family`, `isg:cf:thm:familytest`, `isg:cf:lem:noninteger`, `isg:cf:eq:explicitparameter`, `isg:cf:main:presburger` and `isg:cf:cor:unit`; all exist at the current state, and `docs/FORMALIZATION.md` records the results among them as pending. Nothing in Parts VI–IX is formalized, and placement beside these projects confers no formal status.

Part X. Source 16 read this report at its pin (source 05’s delivered text), the batch-86 archives then in `docs/incoming` (now Parts III–V and source 03 of `discrete-initial-subgroups-and-omnific-normalization`), `Logic/PresburgerArithmetic/README.md` and `Algebra/SurrealNumbers/README.md`. Its descriptions are correct at the pin; the two READMEs have changed since only by the catalogue’s counts and pointers. Its formalization route ([section 137](#)) proposes modules that do not exist; the Presburger project’s Lean and Rocq files are about the ordinary integers. Nothing in Part X is formalized, and `docs/FORMALIZATION.md` maps none of its labels.

1.5 The status statements of the sources

The thirteen manuscripts state their status as follows; these statements apply to the parts of this report taken from them.

Source 05: status and scope

The mathematical assertions designated as theorems are accompanied by proofs. The construction answers the cited question *as written*; independent bibliographic priority has not been established. Classical Presburger elimination and elementary topological-group facts are not claimed as new. This AI-assisted manuscript is not independently refereed or Lean-verified. The accompanying executable tests are finite checks, not a replacement for the proofs.

Source 04: from its abstract

The construction gives an affirmative answer to Question 2 as printed in Elliot Glazer’s 2023 paper *A Topological Tennenbaum Theorem*. The historical priority of this answer has not been established. The new results are supported here by conventional proofs and executable exact tests, not by a claimed Lean or Rocq kernel check.

Source 03: status and provenance

Status and provenance. Complete mathematical arguments are provided for the constructions and deductions, with classical inputs identified. ProveIt’s existing coanalytic support lower bound and omnific floor formula are credited, not claimed as new. The main contribution of this report is the presentation dichotomy and its algebraic consequences, together with the exact finite-support cancellation analysis. Global novelty and independent correctness review remain open; no established famous conjecture or Lean verification is claimed.

Source 07: status

Status. The main constructions and their consequences are proved below. Classical model theory and standard descriptive set theory are identified as inputs. No new result in this report has been checked by Lean or Rocq, and no claim of publication priority or external peer review is made.

Source 06: from its abstract

The article connects these constructions to ProveIt's existing Presburger elimination development and its polynomial omnific subrings. All new assertions are proved in ordinary mathematics here; no machine-checked certification or unrestricted publication-priority claim is made.

Source 08: status

Status. The central results below have complete mathematical proofs, with classical inputs identified. Their bibliographic priority has not been established. This is an AI-assisted, unrefereed research manuscript, not a claim of an independently certified breakthrough or a Lean-checked proof. The included exact-arithmetic tests are supplementary checks only.

Source 09: status

Status. The central deductions are proved in this manuscript. Classical category theorems, Hahn and Puiseux constructions, real-closedness inputs, and previous ProveIt support and floor results are explicitly credited. Historical novelty has not been established by an exhaustive literature search. This is an unrefereed research contribution, not a claimed solution of a famous conjecture and not a Lean- or Rocq-verified development.

Source 12: status and scope

The results designated as theorems have mathematical proofs below. Historical priority has not been established. This is an AI-assisted, unrefereed manuscript, not an independently certified breakthrough and not a Lean- or Rocq-checked development. The accompanying exact-arithmetic checks are supplementary, not proofs of the topological theorems.

Source 10: status

Complete mathematical proofs are supplied for the main claims. The proposed contributions have not been independently refereed, and bibliographic priority has not been established. This is not a claim of a Lean- or Rocq-checked theorem. Classical inputs and prior project results are distinguished from the deductions made here.

Source 13: from its abstract

The contributions are conventional proofs, with explicit classical inputs and limited exact computational diagnostics. The earlier existence construction is credited, and no first-publication priority, refereeing, or new proof-assistant verification is claimed.

Source 14: status

Complete mathematical proofs of the stated contributions are supplied. The classical MRDP and coded descriptive-set-theoretic inputs are identified explicitly. The manuscript is not a proof-assistant formalization. Priority beyond the inspected literature and repository materials has not been established.

Source 15: from its abstract

We give full mathematical proofs, an exact comparison with the inspected ProveIt sources, and a research agenda. Historical priority and a machine-checked formalization of these extensions are not claimed.

Source 16: status (from its title page)

The article supplies mathematical proofs and explicitly credits the existing Presburger constructions, Hahn-group theory, and topological linear algebra. It resolves specified structural questions for this family of models; it does not claim a literature-certified breakthrough or a new solution of Glazer’s already-addressed Question 2. The accompanying computations check finite rational examples, not the infinite theorems. No new proof-assistant formalization is asserted.

[write] Source 14’s claimed answer to Glazer’s Question 1 has not been independently reviewed ([Part VIII](#)).

[write] Source 16 did not see source 04’s results on the Baire model, which it proves again for $\alpha = \omega$ ([Part X](#)).

Every limitation and non-claim of the thirteen sources is also listed, source by source, in [section F](#).

Part I**Polish Presburger models**

This part answers Glazer’s Question 2 as printed. Its base text is source 05; source 04’s further results and second proofs are marked [04]. Repository pins: 05 inspected commit `a11efab09`, 04 commit `3ad5f878c`; the files they name are unchanged at the placement.

2 The research overlap and the precise target

The relevant intersection is *topological and effective model theory of weak arithmetic*. Glazer’s work supplies a sharply formulated question; ProveIt supplies two concrete mathematical interfaces: an executable Cooper quantifier-elimination development and a normal-form construction of the omnific integers. This makes it possible to connect an external Polish topology, internal induction, and exact surreal coordinates without appealing to the repository’s unreviewed research claims.

In the final section of [1], Glazer asks:

Does some uncountable Polish space support a model of Presburger arithmetic

with continuous addition?

The source is Question 2 on page 8 of arXiv:2311.13699v1. We use precisely that formulation. The topology need not be the arithmetic order topology, and no multiplication is required. These two distinctions are essential.

Glazer’s obstruction for stronger arithmetic includes the impossibility of an uncountable Polish model of $Q + I$ Open with continuous addition and Borel multiplication [1, Corollary 1]. The earlier study of Enayat, Hamkins, and Wcisło [2] motivates the broader problem of finding topological presentations of arithmetic. Our construction does not contradict those results: it supports the additive language, and its additive structure prevents a semiring expansion.

2.1 Repository interfaces actually inspected

The repository snapshot used here is

a11efab09a2d08f692a09829cb6d3440ddd1795a.

The Presburger README [14] describes a complete Lean integer quantifier eliminator and an independent Rocq/Coq proof of the normalized one-variable step. The inspected file PresburgerArithmetic/Cooper.lean [15] states `periodic_interval` and `periodic_has_residue` for predicates on the ordinary type `Int`. Their hypotheses are entirely appropriate there. They must not be transferred unchanged to a nonstandard ordered group; section 4.2 gives an explicit counterexample and the correct replacement.

The surreal README [16] records an omnific ring whose normal forms have nonnegative exponents and integer constant term. Only this normal-form interface and the standard arithmetic of its additive group are needed below. We do not assume that the present topology or any of this article’s theorems has already been formalized in that repository.

2.2 What is established, and what is not

The principal result is an explicit affirmative construction, rather than a claim that a difficult negative theorem has been overturned. The lexicographic algebra is classical. The contribution developed here is the precise topology and its application to the stated question, together with an effective Baire presentation and a collection of sharp structural consequences.

The literature search verified the question in the source PDF and checked the arXiv record; it did not establish the absence of an earlier solution. Consequently, “an affirmative answer to the stated question” is the appropriate conclusion, not an independently certified first solution. No answer to Glazer’s separate reverse-mathematical Question 1 is claimed.

All constructions and proofs take place in ordinary ZFC with set-sized carriers. The notation **Oz** is used only to identify where explicit set-sized images lie. No topology on an unrestricted proper class, and no set of all surreal numbers, is used.

2.3 Source 04’s framing

Source 04 opens with the same question and the same two repository connections (its Section 1). It adds three points, in its own words; “this article” and “this package” there mean source 04.

The interpretation of the signature matters. Throughout this article, Presburger arithmetic is

$$\text{Pr}_{\mathbb{N}} = \text{Th}(\mathbb{N}; 0, 1, +),$$

with its definable order, and with no multiplication of two variables. Continuity of addition does *not* mean continuity of every definable function. Indeed, the difference between these requirements is central to our result. Adding order and fixed congruence predicates to the language does not change which first-order sets are definable, but the predicates need not be open or closed in the topology.

The characterization of Presburger groups as discretely ordered abelian groups with divisible quotient by the standard integers, and their quantifier elimination after naming congruences, are classical. A convenient modern reference is Jeřábek's preliminaries [4, Section 2]. We include proofs of the facts needed here so the construction can be audited independently of that characterization.

The specific contribution is the topological application to the printed question, strengthened by an effective model on literal Baire space and an analysis of definable operations and self-embeddings. The finite-residue argument also identifies an exact hypothesis needed when transporting a standard-integer proof to arbitrary models. It is not a claim that the repository's integer proof is defective.

This is an unrefereed research manuscript. The statements called theorems below have proofs in the manuscript; the Python program is an independent finite diagnostic and a usable stream implementation. No new theorem in this package is represented as machine-checked. The literature check verified the wording of Glazer's question and the relevant background; it did not establish that the affirmative construction has never appeared elsewhere. Thus an answer to the specified question is claimed at the mathematical level, but first-publication priority is not claimed.

3 The construction in one theorem

We include 0 in $\mathbb{N}$. For an ordered abelian group D , let

$$G_D = D \overrightarrow{\times} \mathbb{Z}$$

mean the direct-product additive group with the D coordinate more significant in the order. Thus

$$(d, n) < (e, m) \iff d < e \text{ or } (d = e \text{ and } n < m). \quad (3.1)$$

Write $\mathbf{1} = (0, 1)$ and let

$$M_D = (G_D)_{\geq 0} = (D_{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}). \quad (3.2)$$

The ambient product topology, whenever specified, is separate from (3.2)'s arithmetic order.

Theorem 3.1 (Polish and effective Presburger models). *The following assertions hold.*

- (1) *With $D = \mathbb{R}$ in its usual topology, $M_{\mathbb{R}}$ is an uncountable, locally compact Polish space without isolated points. Coordinatewise addition is continuous, and*

$$(M_{\mathbb{R}}; 0, \mathbf{1}, +, <) \models \text{Th}(\mathbb{N}; 0, 1, +, <).$$

- (2) *With $D = \mathbb{Q}^{\mathbb{N}}$, ordered by the first differing coordinate and with each rational coordinate discrete, the space $M_{\mathbb{B}} = M_D$ is homeomorphic to $\mathbb{N}^{\mathbb{N}}$. There is an explicit such homeomorphism under which addition, successor, and division by each positive standard integer are Type-2 computable.*

- (3) In both models, every first-order definable relation, with parameters, is Δ_2^0 . The order is neither open nor closed. In the Baire presentation, each formula has a computable bound on the number of changes in a convergent truth approximation, relative to its parameter codes.
- (4) Both models embed as ordered additive submonoids of $\mathbf{Oz}$, fixing 0 and 1. Neither admits any unital semiring multiplication with its given addition, zero, and unit.

The proofs occupy [sections 4 to 7, 9 and 10](#). The first assertion alone answers the source question. The later assertions explain how much regularity is possible and where it necessarily stops.

For orientation, an element $(r, -17)$ with $r > 0$ is a *positive* element of $M_{\mathbb{R}}$. Indeed it exceeds every standard numeral $(0, k)$. Nevertheless,

$$(1/j, 0) \longrightarrow (0, 0) \quad (j \longrightarrow \infty). \quad (3.3)$$

in the chosen topology. Arithmetic size and topological proximity are different notions. The construction would fail if one silently demanded that this topology be the order topology.

Source 04's statement [04]. Source 04 (its Section 1.2) sets up the same real model:

Let $\mathbb{Z}$ have the discrete topology and put

$$M_{\mathbb{R}} = (\mathbb{R}_{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}). \quad (3.4)$$

Set $0 = (0, 0)$, $1 = (0, 1)$, and

$$(r, z) + (s, w) = (r + s, z + w).$$

Order the ambient group $\mathbb{R} \times \mathbb{Z}$ lexicographically, with the real coordinate dominant. Then $M_{\mathbb{R}}$ is exactly its nonnegative cone.

Remark 3.2 (Source 04's statement of [theorem 3.1](#) [04]). The structure $(M_{\mathbb{R}}; 0, 1, +)$ is a model of $\text{Pr}_{\mathbb{N}}$. With the topology inherited from $\mathbb{R} \times \mathbb{Z}_{\text{discrete}}$, its universe is an uncountable perfect locally compact Polish space, and addition is jointly continuous. There is also a model of $\text{Pr}_{\mathbb{N}}$ on $\mathbb{N}^{\mathbb{N}}$ with jointly continuous addition. In both models, quotient and remainder by each fixed positive standard integer are continuous, and every first-order definable relation with parameters is Δ_2^0 .

The algebraic assertion is proved in [section 4](#) and [section 4.3](#); the topology and the explicit Baire-space realization are proved in [sections 5.2, 5.4 and 6.3](#). The proof uses familiar $\mathbb{Z}$ -groups, not an alteration of what Presburger arithmetic means.

[merge] The two statements agree. Source 04's adds continuity of quotient and remainder by each fixed standard integer, which it proves for every admissible coefficient group ([theorem 5.4](#)) and which 05 proves for its two models ([proposition 8.1](#)); source 05's adds the optimal order rank, the finite-mind-change truth approximation and the omnific realization. Both prove the no-semiring and self-embedding results.

4 Full Presburger arithmetic, not merely a few axioms

4.1 Discretely ordered groups with standard residues

Definition 4.1. A $\mathbb{Z}$ -group in this article is a linearly ordered abelian group G with a distinguished least positive element 1 , such that for every positive standard integer m and

every $x \in G$ there exist $q \in G$ and $r \in \{0, \dots, m-1\}$ satisfying

$$x = mq + r\mathbf{1}. \quad (4.1)$$

Here mq denotes repeated addition by a *standard* integer, not a new multiplication operation on G .

Ordered abelian groups are torsion-free. The quotient and remainder in (4.1) are unique: if $m(q - q') = (r' - r)\mathbf{1}$ and $q > q'$, then $q - q' \geq \mathbf{1}$, whereas $r' - r < m$; the other inequality is similar. We suppress $\mathbf{1}$ in standard integer constants when no confusion is possible.

Proposition 4.2 (The split lexicographic group). *If D is a divisible ordered abelian group, then $D \overrightarrow{\times} \mathbb{Z}$ is a $\mathbb{Z}$ -group. Its standard division algorithm is*

$$(d, n) = m(d/m, \lfloor n/m \rfloor) + (0, n \bmod m). \quad (4.2)$$

If $(d, n) \geq 0$, the displayed quotient is nonnegative.

Proof. Coordinatewise addition preserves the lexicographic order. A positive (d, n) either has $d > 0$, in which case it exceeds $(0, 1)$, or has $d = 0$ and $n \geq 1$. Thus $(0, 1)$ is the least positive element. Divisibility and torsion-freeness of D give the unique element d/m . Ordinary Euclidean division of n now proves (4.2). If $d > 0$, then $d/m > 0$, regardless of the integer quotient. If $d = 0$ and $n \geq 0$, the integer quotient is nonnegative. $\square$

Source 04 proves the same statement (its Lemma 2.2) with the same explicit quotient and the same nonnegativity clause; its uniqueness argument compares coordinates instead of using least positivity.

Second proof (source 04). If $(d, z) > 0$ and $d > 0$, then $(d, z) > (0, 1)$. If $d = 0$, positivity says $z \geq 1$. This proves discreteness. Existence in (4.1) follows from divisibility of D and ordinary integer Euclidean division, including for negative z .

For uniqueness, equality of $mq + r\mathbf{1}$ and $mq' + r'\mathbf{1}$ first gives equality of the D -coordinates of q, q' . Equality of the integer coordinates then gives ordinary uniqueness of quotient and remainder. Finally, if $d > 0$, then $d/m > 0$. If $d = 0$ and $z \geq 0$, then $\lfloor z/m \rfloor \geq 0$. $\square$

Corollary 4.3 ([04]). *For $m \geq 1$,*

$$mG_D = D \times m\mathbb{Z}, \quad (d, z) \equiv (e, w) \pmod{mG_D} \iff z \equiv w \pmod{m}.$$

Moreover,

$$\bigcap_{m \geq 1} mG_D = D \times \{0\}. \quad (4.3)$$

Proof. Every D -coordinate is an m -multiple. The only ordinary integer divisible by every positive integer is zero. $\square$

The standard subgroup $\mathbb{Z}\mathbf{1}$ is convex, and $G_D/\mathbb{Z}\mathbf{1}$ is the ordered divisible group D . This gives the usual quick recognition of G_D as a Presburger group. For the precise model-theoretic conclusion, we next give a direct elimination argument.

The implication from these axioms to *full* Presburger arithmetic is important enough to prove. We use the classical elimination architecture of Presburger and Cooper; [3] supplies historical and algorithmic context. The proof below makes explicit which algebraic hypotheses are needed in a nonstandard group.

4.2 Coset invariance versus standard-shift periodicity

For $m \geq 1$, write $mG = \{mh : h \in G\}$. A predicate P is mG -invariant if

$$y - z \in mG \implies (P(y) \leftrightarrow P(z)). \quad (4.4)$$

This is stronger than invariance under the single standard shift $m\mathbf{1}$.

Example 4.4 (A false generalization of an integer lemma). In $G = \mathbb{R} \overrightarrow{\times} \mathbb{Z}$, let $P(d, n)$ mean $d > 0$. For every positive standard m ,

$$P(x + m\mathbf{1}) \leftrightarrow P(x)$$

holds. Yet P has witnesses and is false at every standard residue $0, 1, \dots, m-1$. Thus

$$(\exists x P(x)) \not\iff \bigvee_{0 \leq r < m} P(r\mathbf{1}).$$

It also fails the naive finite-interval reduction with lower bound 0 and upper bound $(1, 0)$. No flaw in the repository's Int theorem is involved: its domain is the ordinary integers, where every multiple is obtained by a standard finite iteration.

The general counterexample [04]. Source 04 phrases coset invariance as a definition and gives the counterexample for every nonzero divisible D , with the predicate $d \neq 0$ in place of $d > 0$. In its text G is a linearly ordered abelian group with a least positive element $\mathbf{1}$ and unique division by positive standard integers as in (4.1), that is, a $\mathbb{Z}$ -group.

Definition 4.5 ([04]). A predicate $P \subseteq G$ is m -residue invariant if

$$u - v \in mG \implies (P(u) \iff P(v)).$$

No definability hypothesis is imposed in this definition.

The condition is stronger than invariance under the single translation $x \mapsto x + m\mathbf{1}$. This distinction is invisible over the ordinary group $\mathbb{Z}$, but is essential over nonstandard groups.

Proposition 4.6 (One-step periodicity is insufficient [04]). If $D \neq 0$, the external predicate

$$P(d, z) \iff d \neq 0$$

on G_D satisfies $P(x + m\mathbf{1}) \leftrightarrow P(x)$ for every standard $m \geq 1$. Nevertheless P is nonempty and is false at every standard residue $0, 1, \dots, m-1$.

Proof. A standard integer translation does not change the D -coordinate. A nonzero D -coordinate exists, whereas every standard integer has D -coordinate zero. In particular P is not m -residue invariant: $(d, 0) - (0, 0) \in mG_D$ for every d . $\square$

Lemma 4.7 (The coset-invariant finite search). Let G be a $\mathbb{Z}$ -group, $m \geq 1$ standard, and P an mG -invariant predicate. For $L, U \in G$,

$$\begin{aligned} \exists y (L \leq y \leq U \wedge P(y)) \\ \iff \bigvee_{0 \leq r < m} (L + r \leq U \wedge P(L + r)). \end{aligned} \quad (4.5)$$

With only an upper bound, candidates $U - r$ suffice; with only a lower bound, candidates $L + r$ suffice. With neither bound, the candidates are $0, \dots, m-1$.

Proof. Given a witness $y \geq L$, write $y - L = mh + r$ with $0 \leq r < m$. One has $h \geq 0$: otherwise $h \leq -1$, and then $mh + r \leq -m + (m - 1) = -1$, contradicting $y - L \geq 0$. Consequently

$$L \leq L + r \leq y \leq U.$$

The difference $y - (L + r) = mh$ lies in mG , so $P(L + r)$ holds. Conversely every displayed candidate is already a witness. The same argument applied to $U - y$ proves the upper-bound version. Without bounds, use the standard remainder of y itself. $\square$

Congruence predicates have exactly the stronger invariance (4.4). If M is a common multiple of finitely many standard moduli, any Boolean combination of congruences with those moduli is MG -invariant. This remains true with arbitrary nonstandard parameters: adding Mh to a variable changes an affine congruence term by an element of each relevant modulus subgroup.

4.3 Second route [04]: finite witnesses at a boundary

Source 04's form of lemma 4.7 works with strict bounds and an explicit finite candidate set. It is the same theorem with a different normalization of the bounds, and it is the step that 04 uses in its elimination proof below. Here G is a $\mathbb{Z}$ -group, and a standard integer r also denotes $r1$.

Theorem 4.8 (Finite boundary criterion [04]). *Let A, B be finite lists of elements of G , let $m \geq 1$, and let P be m -residue invariant. Put*

$$\Phi(x) := P(x) \wedge \bigwedge_{a \in A} (a < x) \wedge \bigwedge_{b \in B} (x < b).$$

Define a finite candidate set C by

$$C = \begin{cases} \{a_* + 1, \dots, a_* + m\}, & A \neq \emptyset, \quad a_* = \max A, \\ \{b_* - 1, \dots, b_* - m\}, & A = \emptyset, B \neq \emptyset, \quad b_* = \min B, \\ \{0, \dots, m - 1\}, & A = B = \emptyset. \end{cases} \quad (4.6)$$

Then

$$(\exists x \in G) \Phi(x) \iff \bigvee_{c \in C} \Phi(c). \quad (4.7)$$

Proof. The reverse direction is immediate. Suppose first that $A \neq \emptyset$ and x is a witness. Divide $x - a_* - 1$ by m to obtain

$$x - a_* = mq + s, \quad 1 \leq s \leq m.$$

Here q is an element of G , not necessarily a standard integer. If $q < 0$, discreteness gives $q \leq -1$, whence $mq + s \leq 0$, contradicting $x > a_*$. Thus $q \geq 0$ and $c = a_* + s \leq x$. This c is larger than every lower bound and smaller than every upper bound because $c \leq x < b$. Also $x - c = mq \in mG$, so $P(c)$ holds.

If A is empty and B is nonempty, divide $b_* - x - 1$ instead. The result is $b_* - x = mq + s$ with $q \geq 0$ and $1 \leq s \leq m$. The candidate $c = b_* - s$ satisfies $x \leq c < b_* \leq b$ for every upper bound and is in the same residue class as x . Finally, when both lists are empty, take the unique standard representative of x modulo mG . $\square$

This proof uses no induction through a nonstandard interval. Its only finite searches have standard lengths fixed by the formula being eliminated. Weak bounds can be converted to strict ones using discreteness: $a \leq x$ is $a - 1 < x$, and $x \leq b$ is $x < b + 1$.

4.4 A self-contained elimination argument

Expand the ordered-group language by predicates

$$D_m(t) \iff t \in mG \quad (m \geq 2),$$

and regard D_1 as always true. All of these are definable by existential formulas in the group language.

Theorem 4.9 (Classical Presburger elimination over $\mathbb{Z}$ -groups). *In the expanded language*

$$\{0, 1, +, -, \leq, (D_m)_{m \geq 2}\},$$

the axioms of $\mathbb{Z}$ -groups eliminate quantifiers and form a complete theory. In particular every $\mathbb{Z}$ -group is elementarily equivalent to $(\mathbb{Z}; 0, 1, +, -, \leq, (m \mid \cdot)_m)$.

Proof. It suffices to eliminate one existential quantifier from a quantifier-free formula. Rewrite the formula in disjunctive normal form. Negated inequalities are replaced using discreteness, for example $\neg(s \leq t) \leftrightarrow t + 1 \leq s$. An equality is the conjunction of two inequalities; a disequality is a disjunction of strict inequalities. Finally, standard residue exhaustiveness gives

$$\neg D_m(t) \iff \bigvee_{r=1}^{m-1} D_m(t - r).$$

Thus it is enough to eliminate $\exists x$ from a conjunction of affine inequalities and positive divisibility atoms, along with conjuncts not involving x .

Let A be a positive common multiple of every nonzero absolute coefficient of x occurring in that conjunction; take $A = 1$ if there are none. Set $y = Ax$ and add the condition $D_A(y)$. An atom

$$ax + t \leq 0, \quad a \neq 0,$$

can be multiplied by the positive standard integer $k = A/|a|$, giving

$$\text{sgn}(a)y + kt \leq 0.$$

It is now an upper or lower bound on y . For a congruence, torsion-free cancellation gives the equivalence

$$D_m(ax + t) \iff D_{km}(\text{sgn}(a)y + kt). \quad (4.8)$$

Indeed $kz \in kmG$ if and only if $z \in mG$: one direction multiplies a witness, and the other cancels k from $kz = kmh$. The additional atom $D_A(y)$ is exactly what ensures that every candidate y comes from a unique x .

After normalizing signs, we have finitely many lower bounds $L_i \leq y$, finitely many upper bounds $y \leq U_j$, and a conjunction $P(y)$ of congruence conditions. Let M be a positive common multiple of all their moduli, including A . Then P is MG -invariant.

If lower bounds exist, choose their maximum L . By [lemma 4.7](#), a witness exists exactly when one of $L, L + 1, \dots, L + M - 1$ satisfies all the normalized conjuncts. The maximum need not be introduced as a new function symbol: take a finite disjunction over the indices i , require $L_h \leq L_i$ for every h , and substitute $L_i + r$ for y . This produces a quantifier-free formula in the parameters. If there are upper bounds but no lower bounds, choose a minimal U_j and substitute $U_j - r$. If no bounds exist, substitute the standard residues r . Conjuncts independent of x are retained throughout.

This eliminates one existential quantifier. Induction on formula complexity, replacing universal quantification by negated existential quantification, eliminates all quantifiers.

A quantifier-free sentence uses only standard integer multiples of 1. Their order and their residues are the same in every $\mathbb{Z}$ -group, by least positivity and uniqueness of standard remainders. Therefore all $\mathbb{Z}$ -groups satisfy the same sentences. The ordinary integers are one such group, proving completeness and the final assertion. $\square$

4.5 Second route [04]: normalization and elimination

Source 04 proves [theorem 4.9](#) by a differently organized argument: it keeps negated congruences instead of rewriting them through residues, substitutes away equalities in the eliminated variable, and uses strict bounds with [theorem 4.8](#). Its statement is uniform over all groups G of [section 4.3](#).

Use the group language

$$L_{Pr} = \{0, 1, +, -, <, (\equiv_m)_{m \geq 2}\},$$

where $u \equiv_m v$ means $u - v \in mG$. Affine terms have integer coefficients. The congruence predicates are definable by existential quantification in the underlying additive language.

Theorem 4.10 (Quantifier elimination, with a uniform proof [04]). *Every L_{Pr} formula is equivalent, uniformly in all groups G satisfying the hypotheses above, to a quantifier-free Boolean combination of affine inequalities, affine equalities, and congruences. Consequently every such G is elementarily equivalent to $(\mathbb{Z}; 0, 1, +, -, <)$ in the reduct language.*

Proof. It is enough to eliminate one existential quantifier from a conjunction of literals, after writing the quantifier-free part in disjunctive normal form. Negated equality is split into two strict inequalities, and negated inequalities are converted by discreteness. Negated congruences can be kept as negated congruences.

Let x be the variable to eliminate. If it occurs with no nonzero coefficient, the quantifier is vacuous. Otherwise let L be the least common multiple of the positive absolute values of its nonzero coefficients. Introduce $y = Lx$ and impose $y \in LG$. There is a unique x for every such y because G is torsion-free.

For an atom with term $ax + t$, $a \neq 0$, put $B = L/|a|$. Multiplying an inequality or equality by the positive standard integer B replaces its x -coefficient by $\text{sgn}(a)$ on y . For divisibility use

$$ax + t \in nG \iff \text{sgn}(a)y + Bt \in nBG. \quad (4.9)$$

Indeed the forward implication multiplies a witness by B , and the reverse implication cancels B in the torsion-free group. Atoms with zero x -coefficient remain independent conditions.

The resulting conjunction has only coefficient 1 or -1 on y in its inequalities, equalities, and congruences. An equality involving y permits substitution. In the absence of such an equality, the inequalities form finite strict lower and upper lists. The remaining congruences and negated congruences form a predicate $P(y)$ invariant modulo mG , where m is the least common multiple of their moduli, including L . For example, if $n \mid m$, replacing y by $y + mg$ cannot change membership of $\pm y + t$ in nG .

Apply [theorem 4.8](#). The maximum or minimum appearing there is not an extra function symbol: take a finite disjunction over choices of a largest lower bound, or a smallest upper bound, and add the finitely many inequalities that express that choice. Substitution of

the finite candidates leaves a quantifier-free formula in the remaining variables. Negation handles universal quantifiers, so induction on the external formula tree proves elimination.

Finally a closed quantifier-free formula has the same truth value in every such G as in $\mathbb{Z}$. Its terms are standard numerals. Their equality and order are fixed, and their congruences have the usual residues by division. Eliminating all quantifiers therefore proves elementary equivalence. $\square$

Corollary 4.11 (The entire induction scheme). *For every divisible ordered abelian group D ,*

$$(M_D; 0, 1, +, <) \models \text{Th}(\mathbb{N}; 0, 1, +, <).$$

In particular it satisfies induction for every first-order formula of this additive language, with arbitrary parameters from M_D .

Proof. By [proposition 4.2](#) and [theorem 4.9](#), $G_D \equiv \mathbb{Z}$. The nonnegative cone is uniformly definable by $x \geq 0$. Relativizing every quantifier to that cone translates any sentence about M_D into a sentence about G_D . The same translation in $\mathbb{Z}$ gives the ordinary nonnegative integers. Each universally closed induction instance is therefore true in M_D . $\square$

Proposition 4.12 (Definable least elements). *Every nonempty definable subset of M_D , allowing parameters, has a least element.*

Proof. Eliminate quantifiers and take a finite disjunction of conjunctions as in [theorem 4.9](#). Include $x \geq 0$ explicitly. For each conjunction, the substitution $y = Ax$ introduces a lower bound $y \geq 0$. Hence a nonempty conjunction has a least solution among the finitely many valid candidates at its maximum lower bound; the residue argument shows that one of these candidates lies below every witness. Dividing by A is order-preserving. Finally take the minimum of the least solutions of the nonempty disjuncts. $\square$

This also gives a direct induction check after elimination: a nonempty set of counterexamples to an induction instance would have a least member. It cannot be 0, and its predecessor would contradict the induction step. The external standard part $\{(0, n) : n \in \mathbb{N}\}$ is a proper inductive subset when $D \neq 0$, but is not definable, even with parameters. Induction quantifies over definable sets, not over every external subset.

Source 04 proves [corollary 4.11](#) (its Corollary 3.5, for $\text{Pr}_{\mathbb{N}}$) by the same relativization, and adds [04]:

The same argument shows that every definable subset of M_D^k is a Boolean combination of restrictions of group affine inequalities, equalities, and standard congruences. This observation will control its Borel complexity without an induction over projections of Borel sets.

5 Polishness and a locally compact example

A Polish space is a separable space whose topology is induced by some complete metric. A particular convenient metric may be incomplete without the space ceasing to be Polish. This distinction matters for open rays.

5.1 A useful exact criterion

We first record the standard subspace criterion with a proof, so that the completeness issue is visible.

Lemma 5.1 (Polish subspaces). *A subspace A of a Polish space X is Polish if and only if A is a G_δ subset of X .*

Proof. Fix a compatible complete metric d on X . If $A = \bigcap_j U_j$ with U_j open, consider on A the continuous functions $f_j(x) = 1/d(x, X \setminus U_j)$; when $U_j = X$, set $f_j = 0$. The graph of $x \mapsto (f_j(x))_j$ is closed in $X \times \mathbb{R}^\mathbb{N}$. Indeed a limit outside some U_j would force f_j to tend to infinity, not to a real number. Projection identifies this closed graph homeomorphically with A . Countable products and closed subspaces of complete separable metric spaces are complete and separable.

Conversely, let δ be a complete metric inducing the subspace topology on A . For each n , let W_n be the union of all ambient open sets U that meet A and satisfy

$$\text{diam}_d(U) < 2^{-n}, \quad \text{diam}_\delta(U \cap A) < 2^{-n}.$$

These unions cover A . If $x \in \bigcap_n W_n$, choose $x \in U_n$ and $a_n \in U_n \cap A$. Then $a_n \rightarrow x$ in d . For each fixed n , all sufficiently late a_k belong to U_n , so (a_k) is δ -Cauchy. Its δ -limit lies in A , and agreement of the two topologies identifies that limit with x . Thus $A = \bigcap_n W_n$. $\square$

Theorem 5.2 (The positive-cone criterion). *Let D be a divisible ordered abelian group equipped with a Polish topological-group topology, not necessarily its order topology. Give $\mathbb{Z}$ the discrete topology. Then the subspace $M_D \subseteq D \times \mathbb{Z}$ is Polish if and only if $D_{>0}$ is G_δ in D . When this holds, $D_{>0}$ is in fact Δ_2^0 , and addition on M_D is continuous.*

Proof. If $D_{>0}$ is G_δ , so is its union with the closed singleton $\{0\}$. On each clopen integer slice, the carrier (3.2) is either $D_{>0}$ or $D_{\geq 0}$. A countable topological sum of G_δ subspaces is G_δ in the corresponding sum: its complement is a countable union of F_σ subsets of the clopen slices. Apply lemma 5.1.

Conversely, the slice with integer coordinate -1 is closed in M_D and homeomorphic to $D_{>0}$. If M_D is Polish, so is that slice, and lemma 5.1 makes $D_{>0}$ a G_δ subset of D . Negation is a homeomorphism on D , so $D_{<0}$ is G_δ as well. The complement of $D_{>0}$ is $D_{<0} \cup \{0\}$, also G_δ . Thus $D_{>0}$ is both F_σ and G_δ .

Finally the cone is closed under coordinatewise addition. Restricting continuous addition on $D \times \mathbb{Z}$ to a subspace with its image in that subspace gives a continuous map $M_D^2 \rightarrow M_D$. $\square$

5.2 Admissible coefficient groups and continuous division [04]

Source 04 proves a sufficient version of theorem 5.2 under a stronger hypothesis, and obtains in return the continuity of division by standard integers for every such D ; source 05 proves that continuity only for its two explicit models (proposition 8.1). The Polish clause of theorem 5.4 is the case of theorem 5.2 in which $D_{>0}$ is open.

Definition 5.3 ([04]). *An admissible topological divisible group is a divisible linearly ordered abelian group D with a Polish group topology such that $D_{>0}$ is open and each map $d \mapsto d/m$, $m \geq 1$, is continuous. The specified topology is additional data; it is not silently identified with the order topology of G_D or M_D .*

Theorem 5.4 (Transfer [04]). *Let D be an admissible topological divisible group, and give G_D the product topology with discrete $\mathbb{Z}$. Then M_D , with its subspace topology, is Polish. Addition on M_D is jointly continuous. The functions*

$$Q_m(d, z) = (d/m, \lfloor z/m \rfloor), \quad R_m(d, z) = (0, z \bmod m)$$

are continuous maps from M_D to itself for each fixed $m \geq 1$.

Proof. The ambient group is Polish. Its complement of M_D is

$$(D_{<0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{Z}_{<0}).$$

The first set is open and the second is closed. In a metrizable space every open set is F_σ , so this complement is F_σ and M_D is G_δ . A G_δ subspace of a Polish space is Polish. Equivalently, the integer-coordinate slices are a countable disjoint sum of copies of the Polish spaces $D_{\geq 0}$ and $D_{>0}$.

The set M_D is closed under addition; restriction of the continuous ambient addition is therefore continuous as a map into the subspace. The displayed formulas for Q_m, R_m are continuous in the ambient coordinates: division on D is continuous, and every function on the discrete coordinate is continuous. Their images belong to M_D by [proposition 4.2](#). $\square$

5.3 The real-coordinate model, including a complete metric

For $D = \mathbb{R}$, the positive cone is open. More explicitly, the slice

$$X_n = \{r : (r, n) \in M_{\mathbb{R}}\}$$

is $[0, \infty)$ when $n \geq 0$ and $(0, \infty)$ when $n < 0$. Hence

$$M_{\mathbb{R}} \cong \bigsqcup_{n \geq 0} [0, \infty) \sqcup \bigsqcup_{n < 0} (0, \infty). \quad (5.1)$$

This proves separability, local compactness, and absence of isolated points directly.

For a fully explicit compatible complete metric, define

$$d_n(r, s) = \begin{cases} \min\{1, |r - s|\}, & n \geq 0, \\ \min\{1, |\log r - \log s|\}, & n < 0, \end{cases}$$

and set

$$d((r, n), (s, m)) = \begin{cases} d_n(r, s), & n = m, \\ 2, & n \neq m. \end{cases} \quad (5.2)$$

Every Cauchy sequence is eventually contained in one slice. The closed ray with the first metric is complete; the open ray with the logarithmic metric is isometric, before truncation, to $\mathbb{R}$. Thus (5.2) is complete. It induces exactly the topology in (5.1), which is the original product-subspace topology. Continuity of addition follows from that topology, not from a claim of uniform continuity in this metric.

The map $r \mapsto (r, 0)$ on $r > 0$ establishes uncountability. Together with [corollary 4.11](#), this completes the simplest proof of the affirmative answer.

Remark 5.5 (Why the arithmetic order topology is not used). The least positive element is **1**. In G_D , the order interval $(x - 1, x + 1)$ is the singleton $\{x\}$. The positive cone has the same discreteness at its minimum. Consequently its arithmetic order topology is discrete, and an uncountable such topology cannot be Polish. The topology of (5.1) is different, as (5.2) and the convergence in (3.3) explicitly demonstrate.

5.4 The real-coordinate model, source 04's account [04]

Source 04 gives the same slice decomposition and the same complete metric as (5.2), except that distinct slices are at distance 1; it adds two arithmetic illustrations and a remark.

Take $D = \mathbb{R}$ with the usual order and topology. It is admissible, so [corollary 4.11](#) and [theorems 5.4](#) and [7.2](#) apply to (3.4).

Topology, perfection, and a complete metric. For $z \geq 0$, the slice of integer coordinate z is a copy of $[0, \infty)$; for $z < 0$, it is a copy of $(0, \infty)$. Each slice is clopen in $M_{\mathbb{R}}$. Thus $M_{\mathbb{R}}$ is uncountable, locally compact, and has no isolated points. In particular every standard numeral $(0, n)$ is nonisolated.

It is important not to assert that the inherited Euclidean metric is complete: $(0, \infty)$ with that metric is not. Here is a compatible complete metric, with different integer slices distance 1 apart:

$$\rho((r, z), (s, w)) = \begin{cases} 1, & z \neq w, \\ \min\{1, |r - s|\}, & z = w \geq 0, \\ \min\{1, |\log r - \log s|\}, & z = w < 0. \end{cases} \quad (5.3)$$

The last line only occurs for positive r, s . A Cauchy sequence is eventually in one slice. On a nonnegative slice use completeness of $[0, \infty)$; on a negative slice use the homeomorphism $r \mapsto \log r$ onto $\mathbb{R}$. The truncated metrics are metrics, and the inter-slice distance preserves the triangle inequality. Countable dense subsets in the slices give a countable dense subset of the whole space.

Arithmetically huge elements converging to zero. Put $a_j = (1/j, 0)$ for $j \geq 1$. Then

$$a_j > n\mathbf{1} \quad \text{for every standard } n, \quad a_j \longrightarrow 0 \quad \text{topologically.}$$

There is no contradiction. The arithmetic order and the chosen topology are distinct structures. The arithmetic order topology on M_D would be discrete, since every nonzero point has a predecessor and a successor, and 0 is isolated by its immediate successor. That discrete topology is *not* the topology used in the construction.

For a concrete division example, in $M_{\mathbb{R}}$ one has

$$(2, -7) = 3(2/3, -3) + (0, 2).$$

The quotient $(2/3, -3)$ is positive because its real coordinate is positive. This illustrates why restricting the integer coordinate to nonnegative values at all positive real coordinates would destroy the desired arithmetic: the negative integer sheets are necessary.

Remark 5.6 (Why a Cartesian product of two positive cones is not used [04]). The universe is not $\mathbb{R}_{\geq 0} \times \mathbb{N}$. The latter does not have the required predecessor property at points $(r, 0)$ with $r > 0$. By contrast, in $M_{\mathbb{R}}$ every positive (r, z) has predecessor $(r, z - 1)$, except that the ordinary zero remains the least element as required.

6 An explicit Baire-space realization

Let

$$D_B = \mathbb{Q}^{\mathbb{N}}$$

with coordinatewise addition. Each factor $\mathbb{Q}$ is discrete, *not* given its Euclidean subspace topology. Two distinct sequences are compared at the least index at which they differ. This lexicographic order makes D_B a divisible ordered abelian group. Its product topology is Polish and zero-dimensional; for example,

$$d_D(a, b) = 2^{-\min\{j: a_j \neq b_j\}} \quad (a \neq b), \quad d_D(a, a) = 0$$

is a compatible complete metric.

The sets

$$C_{k,q} = \{a : a_0 = \dots = a_{k-1} = 0, a_k = q\}, \quad k \in \mathbb{N}, q \in \mathbb{Q}_{>0},$$

are clopen cylinders, and

$$(D_B)_{>0} = \bigsqcup_{k,q>0} C_{k,q}. \quad (6.1)$$

No restriction is placed on later coordinates. Each cylinder is homeomorphic to $\mathbb{Q}^{\mathbb{N}}$. The nonnegative cone $(D_B)_{\geq 0}$ is closed, since the negative cone is open. [theorem 5.2](#) already proves that

$$M_B = (D_B \overrightarrow{\times} \mathbb{Z})_{\geq 0}$$

is Polish. We now identify its homeomorphism type without relying on a space-classification theorem.

6.1 The zero branch and its explicit tree code

Fix computable bijections

$$e : \mathbb{N} \rightarrow \mathbb{Q}, \quad e_+ : \mathbb{N} \rightarrow \mathbb{Q}_{>0}, \quad e_{\geq 0} : \mathbb{N} \rightarrow \mathbb{Q}_{\geq 0},$$

with computable inverses. A code $s \in \mathbb{N}^{\mathbb{N}}$ determines a sequence $a \in (D_B)_{\geq 0}$ as follows. As long as all previously chosen coefficients are zero, set $a_j = e_{\geq 0}(s_j)$. Once the first positive coefficient has appeared, use $a_j = e(s_j)$ at all later stages. This is a bijection: the inverse uses the corresponding enumeration at each step. A finite output prefix depends on exactly a finite input prefix, in both directions. Thus it is a homeomorphism

$$\mathbb{N}^{\mathbb{N}} \cong (D_B)_{\geq 0}. \quad (6.2)$$

The all-zero rational sequence is not omitted; it is the branch that continues choosing zero indefinitely.

Theorem 6.1 (Baire-space carrier). *There is an explicit homeomorphism $h : \mathbb{N}^{\mathbb{N}} \rightarrow M_B$. In particular M_B is uncountable, has no isolated points, and is nowhere locally compact.*

Proof. Partition M_B into the following countably many clopen blocks. For each $n \geq 0$, take the entire slice $(D_B)_{\geq 0} \times \{n\}$. For $n < 0$, partition that slice further into $C_{k,q} \times \{n\}$ using (6.1). By (6.2), every nonnegative slice is homeomorphic to Baire space; every negative-slice block is so by coding its unrestricted rational tail. A countable disjoint sum of copies of $\mathbb{N}^{\mathbb{N}}$ is homeomorphic to $\mathbb{N}^{\mathbb{N}}$: use the first natural coordinate to choose a block and the remaining coordinates to choose a point in it.

For completeness, no nonempty basic cylinder of $\mathbb{N}^{\mathbb{N}}$ is relatively compact. Its projection to the next unrestricted discrete coordinate has infinite image, whereas the continuous image of a compact set in a discrete space is finite. Every neighborhood contains such a cylinder. The other two assertions follow from the elementary properties of the product $\mathbb{N}^{\mathbb{N}}$. $\square$

6.2 A total code for every sequence of natural numbers

Here is a fully specified block indexing, also used in the accompanying program. Let $\langle a, b \rangle$ be Cantor's pairing function. The first digit of a Baire code is a tag. An even tag $2n$ specifies the nonnegative integer coordinate n and the tree code (6.2). An odd tag

$$2\langle a, \langle k, b \rangle \rangle + 1 \quad (6.3)$$

specifies integer coordinate $-a - 1$, first positive rational coordinate k , and value $e_+(b)$ there. The remaining digits code the unrestricted rational tail using e .

Every natural tag is of exactly one of these forms. Every infinite natural sequence therefore codes a valid element; no hidden domain test is required. Conversely, a point with negative integer coordinate has a first positive rational coefficient, so its tag and tail are uniquely recoverable. This describes h and h^{-1} explicitly.

Theorem 6.2 (Computable continuous arithmetic). *Under this presentation, the operations*

$$x + y, \quad x + \mathbf{1}, \quad q_m(x) = \lfloor x/m \rfloor, \quad r_m(x) = x \bmod m$$

are Type-2 computable for every positive standard integer m . Here $q_m(x)$ is the unique quotient in the model's Euclidean division by m , not a real-valued floor operation. The remainders have their finite discrete topology.

Proof. A Type-2 computation reads finitely many input digits to produce each requested output digit. Decode the integer coordinates n, z of the two inputs. Each rational coefficient $a_j + b_j$ is computable from finite prefixes. If $n + z \geq 0$, output tag $2(n + z)$ and encode the coefficient sequence by the nonnegative-cone tree. The tree decisions only inspect finite prefixes.

If $n + z < 0$, at least one input has a negative integer coordinate and hence a strictly positive coefficient sequence. Its tag supplies an index k of a positive leading coefficient. Since both coefficient sequences are lex-nonnegative, their sum is strictly positive and its first nonzero coordinate occurs at or before k . Search these finitely many coordinates, output the negative tag (6.3), and then encode the tail. Thus the tag itself is computable; the algorithm does not attempt to decide whether an arbitrary sequence is identically zero.

For division use

$$q_m(a, n) = (a/m, \lfloor n/m \rfloor), \quad r_m(a, n) = n \bmod m.$$

If the quotient's integer coordinate is negative, the input integer coordinate was negative as well, and the same known leading index can be used. Otherwise the nonnegative tree code applies. Constants and successor are special cases of these finite-prefix procedures. $\square$

The adjective “computable” here concerns operations on infinite sequence codes. It does not assert that every element is computable, or that equality and order of arbitrary streams can be decided in finite time. The distinction will be quantified next.

6.3 Second route [04]: explicit charts

Source 04 builds the same space: its M_ω is M_B , with the same group $\mathbb{Q}^{\mathbb{N}}$, the same discrete coordinates and the same lexicographic order. Its charts differ from the tag coding of [theorem 6.1](#): they go from the model to Baire space and code the nonnegative and the negative integer slices separately. Source 04's homeomorphism H , its charts $H_{\geq}$, $H_{>}$ and its integer coding h are written Ψ , $\Psi_{\geq}$, $\Psi_{>}$ and ζ here ([section 1.3](#)).

6.3.1 The lexicographic rational product

Let

$$D_B = \mathbb{Q}^{\mathbb{N}}$$

with coordinatewise addition, the product topology in which *each copy of $\mathbb{Q}$ is discrete*, and lexicographic order by the first unequal coordinate. A nonzero sequence has a least nonzero index, so the order is defined on all sequences, not merely on finite-support ones. It is a linear order compatible with addition. Division by standard integers is coordinatewise.

The metric

$$d(u, v) = \begin{cases} 0, & u = v, \\ 2^{-k}, & k = \min\{i : u_i \neq v_i\}, \end{cases}$$

is a complete compatible ultrametric. Finite-support rational sequences form a countable dense set. The positive cone is

$$(D_B)_{>0} = \bigcup_{k \in \mathbb{N}} \bigcup_{q \in \mathbb{Q}_{>0}} \{d : d_0 = \dots = d_{k-1} = 0, d_k = q\},$$

an open union of clopen cylinders. Hence D_B is admissible. Write

$$M_B = M_{D_B}.$$

It is already an uncountable Polish Presburger model with continuous addition. We now identify its space explicitly with $\mathbb{N}^{\mathbb{N}}$.

6.3.2 Two elementary coding homeomorphisms

Fix bijections

$$c : \mathbb{Q} \longrightarrow \mathbb{N}, \quad b : \mathbb{Q}_{>0} \longrightarrow \mathbb{N}_{>0}, \quad a : \mathbb{N} \times \mathbb{Q}_{>0} \longrightarrow \mathbb{N}, \quad \zeta : \mathbb{Z} \longrightarrow \mathbb{N}.$$

Since their domains and codomains are discrete, these are homeomorphisms between discrete spaces.

For $d \in (D_B)_{\geq 0}$ define $\Psi_{\geq}(d) \in \mathbb{N}^{\mathbb{N}}$ as follows. Send the zero sequence to the zero sequence. If k is the first nonzero index of d , then $d_k > 0$, and set

$$\Psi_{\geq}(d)_i = \begin{cases} 0, & i < k, \\ b(d_k), & i = k, \\ c(d_i), & i > k. \end{cases} \quad (6.4)$$

Every nonzero element of $\mathbb{N}^{\mathbb{N}}$ has a first positive entry, so this coding has an evident inverse. Both it and its inverse preserve information in finite initial segments: to determine the first n coordinates, only the first n coordinates are needed. It is therefore a homeomorphism, including at the zero sequence.

For $d \in (D_B)_{>0}$, let k again be its first nonzero index and put

$$\Psi_{>}(d)_0 = a(k, d_k), \quad \Psi_{>}(d)_{j+1} = c(d_{k+1+j}) \quad (j \geq 0). \quad (6.5)$$

The pair (k, d_k) identifies a clopen positive cylinder; within that cylinder the tail is an arbitrary rational sequence. This is another homeomorphism onto $\mathbb{N}^{\mathbb{N}}$.

Theorem 6.3 (A concrete Baire homeomorphism [04]). *The map*

$$\Psi : M_B \longrightarrow \mathbb{N}^{\mathbb{N}}, \quad \Psi(d, z) = \begin{cases} \zeta(z) \frown \Psi_{\geq}(d), & z \geq 0, \\ \zeta(z) \frown \Psi_{>}(d), & z < 0, \end{cases} \quad (6.6)$$

is a homeomorphism. Here $\zeta(z) \frown u$ means prepend the one entry $\zeta(z)$ to the sequence u .

Proof. The integer slices of M_B are clopen. Their images are exactly the clopen cylinders of $\mathbb{N}^{\mathbb{N}}$ specified by the first coordinate. On each slice the map is one of the two homeomorphisms just proved. These slice maps assemble to a bijection whose map and inverse are continuous. $\square$

Transporting addition by Ψ gives a binary operation on the *literal* set $\mathbb{N}^{\mathbb{N}}$:

$$u \oplus v = \Psi(\Psi^{-1}(u) + \Psi^{-1}(v)). \quad (6.7)$$

This proves the Baire-space assertion of [remark 3.2](#); it does not merely produce a Borel-isomorphic universe. The space is perfect and nowhere locally compact. For the latter assertion, every nonempty basic cylinder in $\mathbb{N}^{\mathbb{N}}$ has infinitely many disjoint next-coordinate cylinders covering it, with no finite subcover. If a compact neighborhood existed, a closed basic cylinder inside it would be compact, a contradiction.

6.3.3 Effective operations on infinite streams

A map on Baire space is *Type-2 computable* here if an algorithm, given oracle access to the input sequences, returns every requested output coordinate after reading finitely many input coordinates. The number of queries may depend on the input. This is a statement about computations with infinite input oracles, not an enumeration of an uncountable model by ordinary natural-number indices.

Choose the bijections above effectively. For example, index positive rationals by the binary tree starting at $1/1$, with children

$$\frac{p}{q} \mapsto \frac{p}{p+q}, \quad \frac{p+q}{q}.$$

Every positive rational in lowest terms has a unique finite path: repeatedly subtract the smaller numerator or denominator from the larger. The sum strictly decreases until $(1, 1)$ is reached. Recording whether the larger entry was the numerator or denominator and reversing the record recovers the path. This proves both bijectivity and effective invertibility. Enumerate $\mathbb{Q}$ by interleaving positive and negative rationals and adding zero. Combine this enumeration with an effective pairing of two natural numbers to obtain a and with a signed-integer coding to obtain ζ .

Theorem 6.4 (Effective strengthening [04]). *The homeomorphism Ψ and its inverse can be chosen Type-2 computable. With this choice, the addition $\oplus$ in (6.7), and quotient and remainder by each fixed positive integer, are Type-2 computable on $\mathbb{N}^{\mathbb{N}}$.*

Proof. For $\Psi_{\geq}$, a requested coordinate i requires only the first $i + 1$ coefficients: either all are zero, the first nonzero is at i , or it occurred earlier. These are precisely the three cases in (6.4). The same finite-prefix procedure computes its inverse. No decision whether an entire infinite sequence is zero is made.

For $\Psi_{>}$, search for the first nonzero coefficient. The search terminates because this chart has a positive, not merely nonnegative, domain. Once found, all output coordinates are

computable; its inverse reads the encoded pair in the first coordinate and then the tail. Reading the integer coordinate chooses the appropriate chart of Ψ and Ψ^{-1} .

Coordinatewise rational addition and division are computable, as are integer addition, quotient, and remainder. Compose these algorithms with Ψ and Ψ^{-1} . Whenever the output has negative integer coordinate its D_B -coordinate is strictly positive, so the positive-chart search is guaranteed to terminate. For addition, for instance, a negative sum of integer coordinates requires at least one negative input integer coordinate; that input has strictly positive D_B -coordinate, and the other has nonnegative D_B -coordinate. Their sum is strictly positive. $\square$

The companion `verification.py` implements these maps as memoized indexed streams. A caller supplies a function returning the i th natural number of a sequence. The module exports `baire_add`, `baire_quotient`, and `baire_remainder`. Its round-trip checks include long initial zero blocks, which exercise the boundary chart at the zero coefficient sequence.

7 Definability, effective truth, and the optimal order rank

A subset of a metrizable space is Δ_2^0 when it is both F_σ and G_δ . Finite Boolean combinations of such sets are again Δ_2^0 . Closed sets, open sets, and clopen sets belong to this class.

Theorem 7.1 (All definable relations have low Borel rank). *Under the hypotheses of [theorem 5.2](#), every first-order definable subset of M_D^k , with parameters, is Δ_2^0 .*

Proof. In $G_D = D \times \mathbb{Z}$, the strict positive cone is

$$(D_{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{Z}_{>0}).$$

It is Δ_2^0 , as is its complement. Affine group terms are continuous maps to G_D , so their inequalities have Δ_2^0 truth sets. Equalities have closed truth sets because the ambient group is Hausdorff.

Divisibility is especially simple. Since D is divisible,

$$mG_D = D \times m\mathbb{Z}. \quad (7.1)$$

This is clopen. Thus congruence atoms have clopen truth sets, including when their affine terms contain parameters. Translate a cone formula into the group and apply [theorem 4.9](#). The result is a finite Boolean combination of these atoms. Finally restrict to M_D^k . $\square$

Second route [04]. Source 04 proves [theorem 7.1](#) for its admissible groups ([section 5.2](#)), a special case of the hypotheses of [theorem 5.2](#), by an explicit description of the order relation:

Theorem 7.2 (Low complexity of all definable relations [04]). *For every admissible D , every first-order definable relation with parameters on M_D , in every finite arity, is simultaneously F_σ and G_δ .*

Proof. It suffices by [theorem 4.10](#) to consider affine atoms. Affine group maps are continuous, so an affine equality defines a closed set. A congruence condition depends only on finitely many discrete integer coordinates and is clopen, by [corollary 4.3](#).

For the order, write two group elements as $(d, z), (e, w)$. The strict comparison is

$$(d, z) < (e, w) \iff e - d > 0 \quad \text{or} \quad (e = d \text{ and } z < w). \quad (7.2)$$

The first condition is open; the second is closed because equality of D -coordinates is closed and the integer comparison is clopen. Thus the order relation is F_σ . Its complement has the identical form with the opposite strict comparison and weak integer comparison, hence is also F_σ . Pulling back by affine maps preserves both classes. Finite Boolean combinations of Δ_2^0 sets remain Δ_2^0 , which proves the claim. $\square$

This is stronger than merely obtaining a Borel presentation. The proof also explains why quantification does not raise the complexity in this particular presentation: its apparent projections are replaced by finite residue searches and finite Boolean operations.

7.1 A formula-by-formula bound on changes of mind

Theorem 7.3 (Effective finite-mind-change truth). *For each first-order Presburger formula $\varphi(\bar{x}; \bar{p})$, one can compute a natural number K_φ and an algorithm producing*

$$A_s(\bar{x}; \bar{p}) \in \{0, 1\} \quad (s \in \mathbb{N})$$

from finite prefixes of the Baire codes, such that

$$\lim_s A_s(\bar{x}; \bar{p}) = \mathbf{1}_{M_B \models \varphi(\bar{x}; \bar{p})}.$$

For every input tuple the displayed sequence changes value at most K_φ times. Parameter codes are supplied as oracles; they are not assumed computable.

Proof. Compute a quantifier-free group formula equivalent to the relativized cone formula, by [theorem 4.9](#). An affine group term has the form (c, n) , where n is an exactly computable integer and each rational coefficient c_j is computable from finite prefixes of the inputs.

To approximate the sign of (c, n) , first use the sign of n . At stage s , inspect $c_0, \dots, c_{s-1}$. If a nonzero coefficient has been found, use the sign of the first such coefficient instead. If $c = 0$, the initial answer was correct. Otherwise the answer becomes correct when the first nonzero coefficient is reached and never changes again. An equality atom can be handled in the same way: it starts with the answer determined by n and becomes permanently false if a nonzero coefficient is found. A congruence atom is decided immediately from n by [\(7.1\)](#).

Evaluate the fixed Boolean combination of these approximations at each stage. Its value can change only when an equality or inequality atom changes. The number of those atom occurrences is a computable bound K_φ . Each stage reads only finitely many digits. The truth set is effectively F_σ because it is

$$\bigcup_N \bigcap_{s \geq N} \{\bar{x} : A_s(\bar{x}; \bar{p}) = 1\},$$

where the stage sets are effectively clopen relative to the parameter codes. The same expression with 0 proves effective G_δ status. $\square$

No effective stopping time is asserted: a later rational coordinate may be the first nonzero one. Deciding sentences and deciding arbitrary parameter instances are different problems. The former is ordinary Presburger decidability; the latter is represented here by a convergent, uniformly bounded-change process.

7.2 Why rank one is impossible for the order

Theorem 7.4 (Open or closed arithmetic order forces discreteness). *Let X carry a model of Presburger arithmetic and a topology in which successor $x \mapsto x + 1$ is continuous. If either of the relations $<$ or $\leq$ is open or closed in X^2 , then X is discrete. Consequently, in any uncountable Polish model with continuous addition, the arithmetic order is neither open nor closed.*

Proof. If $\leq$ is open, then for fixed a both sets $\{x : x \leq a\}$ and $\{x : a \leq x\}$ are open. Their intersection is $\{a\}$, so every point is isolated.

If $<$ is closed, taking the complement and reversing coordinates makes $\leq$ open. If $<$ is open, the discrete-order equivalence

$$x \leq y \iff x < y + 1$$

and continuity of successor make $\leq$ open. Finally, if $\leq$ is closed, then

$$x < y \iff x + 1 \leq y$$

makes $<$ closed, reducing to the previous case. An uncountable discrete space is not separable, hence not Polish. $\square$

Together, [theorems 7.1](#) and [7.4](#) make the Δ_2^0 order bound optimal. This is a statement about the order relation, not about every individual definable set: many congruence sets are clopen. It also rules out a continuous Boolean-valued order test on the Baire model, despite its computable addition.

7.3 A general obstruction from closed order [04]

In a Presburger model M , define truncated subtraction by

$$x \dot{-} y = \begin{cases} z, & y + z = x \text{ for some } z \in M, \\ 0, & x < y. \end{cases}$$

The witness in the first case is unique. Equivalently this is $\max\{x - y, 0\}$ in the ordered group completion.

Theorem 7.5 (Closed order forces discreteness [04]). *Let a model of $\text{Pr}_{\mathbb{N}}$ carry a Hausdorff topology. If its weak order relation is closed in M^2 , then every point of M is isolated. If the topology is also separable, then M is countable.*

Proof. By totality, the complement of the converse weak order is strict order. Thus $<$ is open in M^2 . Fixing either argument gives open order rays. For $x > 0$, let $x - 1$ be its predecessor. Discreteness of the arithmetic order gives

$$\{x\} = \{y : x - 1 < y < x + 1\}.$$

At zero one has $\{0\} = \{y : y < 1\}$. Each singleton is open, so the topology is discrete. A dense subset of a discrete space must contain every point; separability therefore implies countability. $\square$

Corollary 7.6 (No continuous monus on an uncountable separable model [04]). *If truncated subtraction on a Hausdorff Presburger model is jointly continuous, the topology is discrete. In particular no uncountable separable Hausdorff Presburger model has continuous truncated subtraction.*

Proof. The relation $x \leq y$ is exactly $x \dot{-} y = 0$. A singleton is closed in a Hausdorff space, so continuity would make weak order closed. Apply [theorem 7.5](#). $\square$

No continuity hypothesis on addition was needed for this obstruction. Consequently, extending the signature of our examples by their definable monus operation destroys the assertion that all named operations are continuous. Definitional equivalence of first-order theories does not by itself preserve this topological property.

Remark 7.7 (Two incomparable order obstructions [merge]). [Theorem 7.4](#) (source 05) and [theorem 7.5](#) (source 04) both show that the order of a nondiscrete model cannot be too simple, under incomparable hypotheses. 05 assumes that successor is continuous, assumes no separation axiom, and treats four cases ($<$ or $\leq$, open or closed). 04 assumes only that the weak order $\leq$ is closed and the topology Hausdorff, with no continuity at all. Neither implies the other: 04 covers a closed $\leq$ in a topology where successor is discontinuous, 05 covers open orders and topologies that are not Hausdorff. Both are therefore printed. For the metrizable models of this part, where successor is continuous, both give the same conclusion, and so does [corollary 7.6](#) for truncated subtraction, whose exact discontinuity locus is [theorem 8.3](#).

8 Which definable operations are continuous?

The coordinate topology makes several arithmetic operations exceptionally regular, but not every definable operation. This section identifies exact boundaries in the two explicit models.

Proposition 8.1 (Euclidean self-similarity). *In $M_{\mathbb{R}}$ and $M_{\mathbb{B}}$, for each positive standard integer m the map*

$$x \mapsto (q_m(x), r_m(x)) \quad (8.1)$$

is a homeomorphism from the model onto $M \times \{0, \dots, m-1\}$, where the finite factor is discrete. Its inverse is $(q, r) \mapsto mq + r\mathbf{1}$.

Proof. Uniqueness and existence are [proposition 4.2](#). In both chosen coefficient topologies, division of d by m is continuous. The integer maps $n \mapsto \lfloor n/m \rfloor$ and $n \mapsto n \bmod m$ are continuous on a discrete space. Hence (8.1) is continuous into the product. Repeated addition and addition of a fixed standard numeral make its inverse continuous. $\square$

The same formula also explains why all standard congruences ignore the entire coefficient sequence. Two elements have the same remainder modulo every positive standard integer if and only if their integer coordinates agree. The forward direction uses the elementary fact that an ordinary integer divisible by every positive integer is zero. Their infinite coordinates can still be arbitrary and different.

Proposition 8.2 (The total predecessor has exactly one discontinuity). *Define*

$$p(0) = 0, \quad p(x) = x - \mathbf{1} \quad (x > 0).$$

In either explicit model, p is continuous exactly at the points other than 0.

Proof. Away from 0, this is a restriction of translation by $-\mathbf{1}$ in the ambient topological group. Since $\{0\}$ is closed, the nonzero domain is open, and translation has its image in the cone.

In the real model let $t_j = 1/j$; in the Baire model let $t_j = e_j$, the rational sequence with 1 in coordinate j and zeros elsewhere. In either case $t_j > 0$ and $t_j \rightarrow 0$ in D . Then

$$(t_j, 0) \rightarrow 0, \quad p(t_j, 0) = (t_j, -1).$$

The latter sequence does not converge to $p(0) = 0$, because its integer coordinate is always -1 and integer coordinates are discrete. $\square$

Source 04 proves [proposition 8.2](#) (its Proposition 7.4), treating the numerals $(0, n)$, $n \geq 1$, separately:

Second proof (source 04). At points with positive D -coordinate, $p(d, z) = (d, z - 1)$ locally. At an ordinary numeral $(0, n)$ with $n \geq 1$, the same formula holds on its integer-coordinate slice and maps into M_D , so it is continuous there. At zero, the sequence $(s_j, 0) \rightarrow 0$ used above has $p(s_j, 0) = (s_j, -1)$, which cannot converge to zero. $\square$

Theorem 8.3 (Exact discontinuity locus of truncated subtraction). *For $M = M_{\mathbb{R}}$ or $M_{\mathbb{B}}$, let*

$$T(x, y) = \max\{x - y, 0\}$$

using the arithmetic order in the ambient group. Its discontinuity locus is exactly

$$\{((d, n), (d, z)) \in M^2 : n \neq z\}. \quad (8.2)$$

Proof. Both positive and negative cones in the coefficient group D are open for our two examples. If the coefficient difference $d - e$ is nonzero, its sign is locally fixed. Thus T locally agrees either with the continuous ambient difference or with the constant zero map.

Suppose instead that $x = y$. Along inputs sufficiently near this pair, the integer difference is zero. The output is either their ambient difference or zero. For any neighborhood V of zero in the ambient group, both possible outputs lie in V when the inputs are sufficiently near (x, x) . This proves continuity there.

It remains to consider equal coefficient coordinates and $k = n - z \neq 0$. Use the positive null sequence t_j from [proposition 8.2](#). If $k > 0$, replace y by $y + (t_j, 0)$. These inputs converge to (x, y) , but their truncated differences are zero, whereas $T(x, y) = (0, k)$. If $k < 0$, replace x by $x + (t_j, 0)$. The outputs are (t_j, k) , which cannot converge to the original value 0 because their integer coordinate is the nonzero integer k . All perturbed inputs remain in the positive cone. This proves exactly (8.2). $\square$

Source 04 states [theorem 8.3](#) as its Theorem 7.3, for the same two models and with the same locus, written $\text{Disc}(\dot{-}) = \{((d, z), (e, w)) : d = e \text{ and } z \neq w\}$, and adds [04]: *this locus is a closed nowhere dense subset of M^2 , and truncated subtraction is continuous on its dense open complement*. Its proof covers the addition.

Second proof (source 04), with the nowhere-density clause. Suppose first that $d \neq e$. The sign of $d - e$ is locally constant because both strict cones of D are open. On a neighborhood the monus operation is therefore either the continuous ambient difference or the constant zero map. Its values lie in the positive cone, so it is continuous as a map to that subspace.

Next let $d = e$ and $z = w$. In a sufficiently small neighborhood the two integer coordinates stay equal to these fixed integers. Monus has integer coordinate zero and D -coordinate $\max\{u - v, 0\}$. This positive-part map is continuous at zero in both groups under consideration. Over $\mathbb{R}$ this follows from the usual absolute-value estimate. Over $D_{\mathbb{B}}$, if $u - v$ has its first n coordinates zero, so does its positive part: it is either $u - v$ or zero. These finite-prefix neighborhoods prove continuity at the point.

For the converse, choose $s_j > 0$ in D with $s_j \rightarrow 0$: use $1/j$ in $\mathbb{R}$ or the j th unit vector in $D_{\mathbb{B}}$. Let $x = (d, z)$, $y = (d, w)$ and $z \neq w$. If $z < w$, replace x by $x_j = (d + s_j, z) \in M_D$. Then $x_j \rightarrow x$ but

$$x_j \dot{-} y = (s_j, z - w), \quad x \dot{-} y = 0.$$

The output integer coordinate is constantly negative and cannot converge to the zero integer coordinate. If $z > w$, instead replace y by $y_j = (d + s_j, w)$. Then $x \dot{-} y_j = 0$ for every j , whereas $x \dot{-} y = (0, z - w) \neq 0$. This proves discontinuity in both cases, including at the boundary $d = 0$.

The set in (8.2) is closed: equality of the D -coordinates is closed and inequality of the discrete integer coordinates is clopen. It has empty interior. At any of its points, perturbing the first D -coordinate upward by a sufficiently small positive element stays in M_D^2 , lies in any prescribed neighborhood, and makes the two D -coordinates unequal. Thus the closed set is nowhere dense. $\square$

The standard numerals form a closed, relatively discrete subset of both models, yet zero is not isolated in the whole space. This causes no conflict with [proposition 8.2](#). Nor does continuity of standard division imply continuity of every definable piecewise operation: the branch boundary is decisive.

Source 04 continues [04]:

The failure is not an indiscriminate loss of regularity: the graph of monus is still Δ_2^0 by [theorem 7.2](#), because monus is definable. Its ordinary predecessor specialization has a smaller exceptional set.

8.1 Polish completeness need not be translation-invariant [04]

Proposition 8.4 (A complete-metric obstruction [04]). *If zero is nonisolated in a metrizable Presburger model, there is no compatible complete metric invariant under all additive translations. More generally, for any compatible complete metric, the successor map cannot be a uniform homeomorphism onto its image.*

Proof. Successor is a bijection from the model onto $M \setminus \{0\}$. If it were a uniform homeomorphism onto this subspace, every Cauchy sequence in the image would pull back to a Cauchy sequence in the complete domain and would therefore converge in the image. Thus the image would be a complete subspace of a metric space, hence closed. Since zero is nonisolated, $M \setminus \{0\}$ is not closed, a contradiction. A translation-invariant metric would make successor an isometry, which is a special case. $\square$

This applies to both main examples. Their Polishness and continuous addition therefore cannot be replaced by a claim of completeness in the uniform structure of an invariant additive metric.

Structure or operation	Real-coordinate model	Baire-coordinate model
Underlying space	Perfect, locally compact, Polish	Homeomorphic to $\mathbb{N}^{\mathbb{N}}$
Addition	Continuous	Type-2 computable
Fixed quotient/remainder	Continuous	Type-2 computable
All definable relations	Δ_2^0	Δ_2^0
Monus	Discontinuous exactly on (8.2)	Same exact locus
Total predecessor	Discontinuous only at 0	Discontinuous only at 0
Arithmetic order topology	Discrete; not used	Discrete; not used

9 Explicit set-sized fragments of the omnific integers

We use Conway normal-form notation

$$x = \sum_{\gamma \in S} r_\gamma \omega^\gamma,$$

where S is reverse well-ordered and all coefficients are real. In the omnific ring the occurring exponents are nonnegative, and the coefficient at exponent zero is an ordinary integer. This is the interface recorded in the repository's surreal development [16]. Addition is coefficientwise, and a nonzero normal form has the sign of its leading coefficient.

Theorem 9.1 (One-scale and countable-scale realizations). *There are injective ordered additive maps, fixing 0 and 1,*

$$\iota_{\mathbb{R}} : \mathbb{R} \overrightarrow{\times} \mathbb{Z} \longrightarrow \mathbf{Oz}, \quad \iota_{\mathbb{R}}(r, n) = r\omega + n, \quad (9.1)$$

$$\iota_{\mathbb{B}} : \mathbb{Q}^{\mathbb{N}} \overrightarrow{\times} \mathbb{Z} \longrightarrow \mathbf{Oz}, \quad \iota_{\mathbb{B}}(a, n) = \sum_{j=0}^{\infty} a_j \omega^{1/(j+1)} + n. \quad (9.2)$$

Their positive images, equipped with the transported coefficient topologies, are respectively the Polish models $M_{\mathbb{R}}$ and $M_{\mathbb{B}}$.

Proof. The first formula is a finite normal form. For the second, the fixed exponents

$$1, \frac{1}{2}, \frac{1}{3}, \dots$$

strictly decrease and are positive. Every nonempty subset of these exponents has a largest element, namely the one with least index. Thus every possible support in (9.2) is reverse well-ordered, and adjoining exponent zero preserves that property. The expression is a legitimate normal form, even when infinitely many coefficients are nonzero. All positive-exponent coefficients are real, and the constant term is the integer n , so the value is omnific.

Uniqueness of normal forms gives injectivity. Addition of two such expressions is coordinatewise on a common fixed support, giving additivity. The leading nonzero coefficient is the first nonzero a_j ; if all are zero, the sign is that of n . Therefore the map preserves and reflects the lexicographic order. The same reasoning proves the one-scale case. Transporting a topology through a bijection onto its image preserves all the asserted topological properties. $\square$

9.1 What this realization does and does not preserve

These are ordered additive embeddings, *not* ring embeddings. Both images contain ω , but neither contains ω^2 . Their multiplicative nonclosure is therefore immediate from the normal forms. The stronger theorem in [section 10](#) will show that replacing surreal multiplication by some other multiplication cannot repair the problem while retaining the same carrier, addition, and unit.

The topologies in [theorem 9.1](#) are precisely specified coefficient topologies. The real coordinate is Euclidean in (9.1); the rational coordinates are discrete and combined by a product topology in (9.2). Neither is the discrete arithmetic order topology. No assertion that they are a canonical topology on all omnific integers is intended.

The ordinary constant-term map becomes simply

$$\text{ct}(\iota(d, n)) = n.$$

It is continuous into the discrete integers for these models. Standard congruence data see only this coordinate, matching (7.1). The standard part itself remains externally specified and nondefinable in the Presburger structure.

9.2 Why no proper-class difficulty arises

The source sets $\mathbb{R} \times \mathbb{Z}$ and $\mathbb{Q}^{\mathbb{N}} \times \mathbb{Z}$ are ordinary sets. The normal-form constructions above are definable assignments on those sets. Replacement therefore gives set-sized images. The set of birthdays of their elements has an ordinal supremum; choosing a larger ordinal places both images inside a set-sized birthday cutoff of **No**. No explicit optimal cutoff is needed for any theorem here.

Accordingly, one can conduct the entire argument in the product-group models and use [theorem 9.1](#) only as a final realization theorem. This is also the most economical formalization strategy. It avoids making a class-sized object the carrier of a Polish space or a first-order structure in ordinary set-based semantics.

Source 04's account [04]. Source 04 proves [theorem 9.1](#) as its Proposition 10.1 with the same proof; its second map is written ι_{ω} there and ι_B here. It adds the following remarks on the normal-form rule and on the transported topologies.

The construction is purely set-sized, but it has a natural expression in the surreal normal-form language. We use only the classical normal-form rule: a reverse-well-ordered set of surreal exponents with real coefficients defines a surreal sum; addition is coefficientwise, and order is decided by the greatest exponent with nonzero coefficient. Omnific integers have positive growth exponents and an ordinary integer constant term. This is the normal-form description relevant to the ordered-group study in [5] and to the repository's omnific development [19].

We transfer the topologies from the coefficient models to these images. This is not a claim that the whole proper class **Oz** has a Polish topology, nor that a canonical global surreal topology is being restricted. For example, ω/j converges to zero in the transported real-coefficient topology on the first image, despite each term being arithmetically infinite. Likewise $\omega^{1/(k+1)}$ converges to zero in the second coefficient topology.

These are additive subgroups, not subrings. Squaring ω yields ω^2 , which belongs to neither displayed image. The absence of inherited multiplicative closure is consistent with the stronger nonexistence result [corollary 10.5](#) for *any* compatible semiring multiplication on these positive cones.

For foundations, the domains $\mathbb{R} \times \mathbb{Z}$ and $\mathbb{Q}^{\mathbb{N}} \times \mathbb{Z}$ are sets. One may work entirely with their normal-form codes in ZFC and interpret the notation inside a sufficiently large set-sized portion of the surreal construction. Alternatively one may use a class theory for the ambient surreal field while retaining these set-sized domains. None of the proofs quantifies over a set of all surreals, a set of all class maps, or a Polish space whose points form a proper class.

10 An absolute obstruction to multiplication

A Borel obstruction would not rule out a wildly discontinuous multiplication. For the models constructed here, the obstruction is stronger and purely algebraic.

Definition 10.1. In a Presburger model M , a positive element u is a *cofinal infinite additive unit* if

$$u > n\mathbf{1} \quad \text{for every standard } n,$$

and for every $x \in M$ there is a positive standard integer N with $x \leq Nu$. The word “unit” here refers only to cofinality under repeated addition; it is not a multiplicative unit.

Source 04 splits the hypothesis of [theorem 10.4](#) into two notions and proves a positivity lemma first:

Definition 10.2 ([04]). An element h in an ordered additive monoid is an *order unit* if for every x there is a positive standard integer N with $x \leq Nh$. It is *arithmetically infinite* if $h > n\mathbf{1}$ for every standard n .

Lemma 10.3 (Positivity is automatic in a semiring expansion [04]). *Suppose a Presburger monoid is expanded to a unital semiring with the same addition, zero, and one. Then the product of two positive elements is positive, and multiplication by a positive element is strictly increasing on either side.*

Proof. If $x, y > 0$, write $x = 1 + a$, $y = 1 + b$ with $a, b \geq 0$. Distributivity and the unit laws give $xy = 1 + a + b + ab \geq 1$. If $x > y$, write $x = y + c$ with $c > 0$. For $z > 0$, one has $xz = yz + cz > yz$ and similarly on the other side. Here the order is the additive natural order of the Presburger model, so no separate order axiom for the multiplication has been assumed. $\square$

Theorem 10.4 (Cofinal-infinite-element obstruction). *A Presburger model with a cofinal infinite additive unit cannot be expanded to a unital semiring with the same addition, zero, and multiplicative unit $\mathbf{1}$. No continuity, measurability, commutativity of multiplication, or induction involving multiplication is assumed.*

Proof. Suppose such a multiplication exists. Cofinality applied to u^2 gives a positive standard N such that $u^2 \leq Nu$. Since u exceeds every standard numeral, it has a decomposition

$$u = (N + 1)\mathbf{1} + v \quad (v \in M).$$

Distributivity and the unit law give

$$u^2 = (N + 1)u + vu \geq (N + 1)u > Nu,$$

a contradiction. The first inequality uses only that vu belongs to the nonnegative carrier. The last one follows from $u > 0$ and additive cancellation. In any semiring expansion of this additive monoid, its natural order $x \leq y \leftrightarrow \exists z (x + z = y)$ is already the Presburger order, so no additional order-compatibility axiom was hidden in the argument. $\square$

Corollary 10.5. *Neither $M_{\mathbb{R}}$ nor $M_{\mathbb{B}}$ admits a unital semiring multiplication with the displayed addition and unit.*

Proof. In $M_{\mathbb{R}}$ take $u = (1, 0)$. It exceeds every standard numeral, and an integer $N > r$ gives $(r, n) < Nu$. In $M_{\mathbb{B}}$ take $u = (e_0, 0)$, where e_0 has first coefficient 1 and all later coefficients zero. For any (a, n) choose an integer $N > a_0$; comparison at coordinate zero gives $(a, n) < Nu$. Apply [theorem 10.4](#). $\square$

In their omnific realizations, this cofinal element is ω . The missing larger scale ω^2 is not merely a defect of one chosen multiplication: every possible semiring multiplication would have to produce an element growing faster than every standard multiple of ω , which the additive carrier cannot contain.

For a different uncountable Polish Presburger model that does allow a semiring expansion algebraically, Glazer's cited Borel obstruction for $Q + I\text{Open}$ would still matter. Our argument is not a replacement for that general theorem; it identifies a stronger obstruction in these particular additive structures.

An arithmetically infinite order unit in 04's sense is exactly a cofinal infinite additive unit in 05's sense, so [theorem 10.4](#) is 04's Theorem 9.3 and [corollary 10.5](#) its Corollary 9.4. Source 04's proof routes the strict inequality through [lemma 10.3](#):

Second proof of theorem 10.4 (source 04). Suppose such a multiplication exists and let h be the order unit. Choose a standard $N \geq 1$ with $h^2 \leq Nh$. Since $h > N1$, [lemma 10.3](#) gives $h^2 > Nh$, a contradiction. $\square$

This algebraic argument is specific to the supplied models and does not replace the general topological theorem about stronger arithmetic. Presburger models need not be additive reducts of models of Peano arithmetic.

10.1 A variation with no largest growth class [04]

Let $D_L = \mathbb{Q}((t))$ be the additive group of formal Laurent series

$$f = \sum_{k \geq k_0} a_k t^k, \quad k_0 \in \mathbb{Z}, \quad a_k \in \mathbb{Q},$$

ordered by the sign of the coefficient at the least nonzero exponent. Use the t -adic topology, with rational coefficients discrete. Its basic neighborhoods at zero are

$$F_n = \{f : v(f) \geq n\}, \quad n \in \mathbb{N},$$

where $v(0) = +\infty$. A bounded compatible metric is $\min\{1, 2^{-v(f-g)}\}$ for $f \neq g$.

Proposition 10.6 (A Laurent-series Polish construction [04]). *The group D_L is admissible, and $M_L = M_{D_L}$ is a Polish Presburger model homeomorphic to Baire space, with continuous addition and fixed quotient and remainder. Unlike the two main models, M_L has no arithmetically infinite order unit.*

Proof. A Cauchy sequence of Laurent series eventually has the same entire negative part, because its pairwise differences eventually lie in F_0 . That common negative part is finite. Every subsequent coefficient stabilizes in turn, giving a Laurent-series limit. Thus the metric is complete. Laurent polynomials with rational coefficients form a countable dense set. The leading sign is determined on a basic neighborhood, so the strict positive cone is open. Addition and division by fixed positive integers are coefficientwise continuous.

For the asserted homeomorphism, decompose D_L into the clopen cosets $p + \mathbb{Q}[[t]]$, where p is a finite negative-power rational polynomial. There are countably many such cosets, and each is a rational product. If $p \neq 0$, the sign on its coset is constant, determined by its leading coefficient. The coset $p = 0$ contributes respectively $(D_B)_{>0}$ or $(D_B)_{\geq 0}$ to the positive or nonnegative cone. Thus both Laurent cones are countable disjoint sums of spaces explicitly homeomorphic to $\mathbb{N}^{\mathbb{N}}$, by the codings in [section 6.3](#). Prepending an index codes such a sum by $\mathbb{N}^{\mathbb{N}}$. The integer slices of M_L are another such sum, proving the claim.

Finally, for $f > 0$ with least exponent k , the element $(t^{k-1}, 0)$ is larger than every standard multiple of (f, z) : its leading exponent is strictly earlier. Standard finite elements also fail to be order units. $\square$

The absence of an order unit does not establish the existence of a semiring expansion. In this example another simple obstruction decides that issue negatively.

10.1.1 A growth-order obstruction

For positive arithmetically infinite elements, define

$$x \asymp y \iff (\exists m, n \in \mathbb{N}_{>0})(x \leq my \wedge y \leq nx).$$

Let $\mathcal{A}(M)$ be the set of these Archimedean growth classes, ordered so that $[x] < [y]$ means $nx < y$ for every positive standard n . These are external set-sized classes of elements of our set-sized model; no first-order definability is being asserted.

Proposition 10.7 (Growth classes must embed into their strict final segments [04]). *If a Presburger monoid M has a unital semiring expansion, then for every arithmetically infinite h , the map*

$$[x] \mapsto [hx]$$

is an order embedding of $\mathcal{A}(M)$ into its strict final segment above $[h]$.

Proof. Multiplication by $h > 0$ preserves and reflects every comparison $x \leq ny$. Preservation follows from distributivity and positivity. For reflection, if $x > ny$ then strict monotonicity gives $hx > n(hy)$. It follows that multiplication induces an order embedding on growth classes. Because x is infinite, $x > n\mathbf{1}$ for every standard n , so $hx > nh$ for every n . Thus its class lies strictly above $[h]$. $\square$

Corollary 10.8 ([04]). *The Laurent-series model M_L also has no unital semiring expansion.*

Proof. Its infinite growth classes are determined exactly by the least Laurent exponent, with increasing growth corresponding to decreasing exponent. Hence $\mathcal{A}(M_L)$ has order type $\mathbb{Z}$. A strict final segment of $\mathbb{Z}$ is well-ordered, while $\mathbb{Z}$ contains an infinite strictly decreasing sequence. There is no order embedding of $\mathbb{Z}$ into such a final segment. Apply [proposition 10.7](#). $\square$

The obstruction is not a statement that every split Presburger group forbids multiplication. For example, take $D = X\mathbb{Q}[X]$ ordered by its highest-degree nonzero coefficient, and let the constant term be an ordinary integer. The ordered ring

$$\mathbb{Z} + X\mathbb{Q}[X]$$

has this split additive group, least positive element 1, and a positive cone closed under its polynomial multiplication. Its growth classes have order type $\mathbb{N}_{>0}$, which does embed in each

of its strict final segments. This countable comparison makes the scope of the obstruction explicit.

11 Endomorphisms, elementary submodels, and a contrast

The two models have markedly different self-embedding behavior. All homomorphisms in the next theorem preserve addition, zero, and **1**; continuity and injectivity are *not* assumed.

11.1 The real model: endomorphisms and elementary submodels

Theorem 11.1 (Rigidity of the real-coordinate model). *Every unital additive endomorphism $f : M_{\mathbb{R}} \rightarrow M_{\mathbb{R}}$ has the form*

$$f(r, n) = (cr, n) \quad (c \in \mathbb{R}_{>0}). \quad (11.1)$$

Conversely each such formula is an endomorphism. All of them are homeomorphisms and Presburger automorphisms. In particular $M_{\mathbb{R}}$ has no proper elementary self-embedding.

Proof. The group completion of the cancellative monoid $M_{\mathbb{R}}$ is $G = \mathbb{R} \overrightarrow{\times} \mathbb{Z}$. Thus f extends uniquely to an additive group map $F : G \rightarrow G$ fixing **1**. Inside G ,

$$\bigcap_{m \geq 1} mG = \mathbb{R} \times \{0\}.$$

One inclusion follows from real divisibility; the other follows because the integer coordinate of an element of the intersection is divisible by every positive integer. Every additive group map preserves this intersection. Hence

$$F(r, 0) = (h(r), 0), \quad F(0, n) = (0, n)$$

for some additive map $h : \mathbb{R} \rightarrow \mathbb{R}$.

Since f maps the positive cone into itself, $h(r) \geq 0$ whenever $r \geq 0$. Thus h is monotone. If $c = h(1)$, rational comparisons and additivity give $h(r) = cr$ for every real r : sandwich r between rational numbers and use the corresponding inequalities for h . Here $c \geq 0$. But $c = 0$ would send the positive element $(1, -1)$ to $(0, -1)$, outside the cone. Thus $c > 0$.

Formula (11.1) and its inverse with scalar $1/c$ preserve the cone, addition, order, and unit, and are continuous in the product coordinates. This proves all assertions. $\square$

Theorem 11.2 (All elementary submodels of the real model). *The elementary submodels of $M_{\mathbb{R}}$ are exactly*

$$M_E = (E \overrightarrow{\times} \mathbb{Z})_{\geq 0}, \quad E \text{ a } \mathbb{Q}\text{-linear subspace of } \mathbb{R}. \quad (11.2)$$

Among them, the only ones Polish in the inherited topology are the standard copy $\mathbb{N}$ and the whole model $M_{\mathbb{R}}$. Every nonzero proper choice of E gives a dense, non-Polish elementary submodel.

Proof. If E is a $\mathbb{Q}$ -linear subspace, then $E \overrightarrow{\times} \mathbb{Z}$ is a $\mathbb{Z}$ -group. Its inclusion in $\mathbb{R} \overrightarrow{\times} \mathbb{Z}$ preserves and reflects the expanded quantifier-free language, including all congruences, since a standard modulus depends only on the integer coordinate. Quantifier elimination makes this inclusion elementary. Relativizing to positive cones proves $M_E \preccurlyeq M_{\mathbb{R}}$.

Conversely, let $S \preccurlyeq M_{\mathbb{R}}$, and let $H = S - S$ be its group completion inside G . If $a, b \in S$ and $a - b \geq 0$, the unique solution of $b + x = a$ lies in S by elementarity. Therefore

$S = H \cap M_{\mathbb{R}}$. Since H contains every standard integer, subtracting the integer coordinate from any $(r, n) \in H$ shows that $H = E \times \mathbb{Z}$ for an additive subgroup $E \leq \mathbb{R}$. For $r \in E_{\geq 0}$, the model's unique division of $(r, 0)$ by a positive standard m has remainder zero and quotient $(r/m, 0)$. Elementarity puts that quotient in S . Applying the same observation to negatives in H shows E is divisible, hence $\mathbb{Q}$ -linear.

If $E \neq 0$, it contains all rational multiples of a nonzero real and is dense in $\mathbb{R}$. Thus M_E is dense in every slice of $M_{\mathbb{R}}$. Suppose M_E is also Polish in the inherited topology. Its closed slice of integer coordinate zero is homeomorphic to $E \cap [0, \infty)$. By [lemma 5.1](#), this is G_δ in $\mathbb{R}$. Its negative is also G_δ , so their union E is G_δ in $\mathbb{R}$. For any $r \in \mathbb{R}$, the dense G_δ sets E and $r + E$ intersect by the Baire category theorem. Hence $r \in E - E = E$, giving $E = \mathbb{R}$. Finally $E = 0$ gives the relatively discrete standard copy $\mathbb{N}$, which is Polish, and $E = \mathbb{R}$ gives the whole model. $\square$

For example, $E = \mathbb{Q}$ gives a countable dense elementary submodel. Its inherited topology is not Polish; there is no inconsistency between its density and the complete metric on the ambient model.

11.2 Reduction to the divisible part [04]

This section considers embeddings in the arithmetic language containing $0, 1, +$. Naming 1 is explicit. Order is already definable, so its preservation is not an extra independent assumption.

Proposition 11.3 (All unital additive maps [04]). *Let D, E be divisible linearly ordered abelian groups. Every map $f : M_D \rightarrow M_E$ preserving $0, 1, +$ is injective and has the unique form*

$$f(d, z) = (T(d), z), \quad (11.3)$$

where $T : D \rightarrow E$ is a $\mathbb{Q}$ -linear order embedding. Conversely every such T defines a map of this kind. Every one of these maps is an elementary embedding of Presburger models.

Proof. If $x > y$, then $x = y + 1 + u$ for some $u \in M_D$. Thus $f(x) = f(y) + 1 + f(u) > f(y)$, so f strictly preserves order and is injective. Every group element is a difference of nonnegative elements. The rule

$$\bar{f}(x - y) = f(x) - f(y)$$

therefore extends f uniquely to an additive order embedding of group completions. It is well defined because $x - y = x' - y'$ implies $x + y' = x' + y$, and f preserves that equality.

For each $d \in D$ and every standard $m \geq 1$, the element $(d, 0)$ is an m -multiple. Its image is consequently in every mG_E , whose intersection is $E \times \{0\}$ by (4.3). Hence $\bar{f}(d, 0) = (T(d), 0)$. Since $f(1) = 1$, the map fixes every standard integer, giving (11.3). Divisibility and torsion-freeness make T $\mathbb{Q}$ -linear; order preservation and injectivity are inherited. The converse follows directly from the formula.

Such a group map preserves and reflects order, equality, and all congruences, the last because integer coordinates are unchanged. Quantifier elimination in [theorem 4.10](#) makes it elementary on the groups. Relativizing to the nonnegative cones makes f elementary as well. $\square$

Source 04 proves [theorem 11.1](#) (its Theorem 8.2) from [proposition 11.3](#):

Second proof (source 04). By [proposition 11.3](#), it remains to classify increasing additive maps $T : \mathbb{R} \rightarrow \mathbb{R}$. Put $c = T(1) > 0$. For rational q one has $T(q) = cq$. For a real r , compare r with rational numbers from below and above and use monotonicity. The resulting inequalities force $T(r) = cr$. The inverse dilation is $f_{1/c}$, and both dilations are continuous in the specified product topology. $\square$

Here 1 in $T(1)$ denotes the real vector $1 \in D$, not the arithmetic unit $(0, 1)$. The distinction prevents a mistaken conclusion that c must equal one. The model is co-Hopfian for unital additive maps but is not rigid.

11.3 The Baire model: all self-embeddings [04] and clopen self-copies

Source 04 classifies every unital additive self-map of M_B ; source 05's shift theorem follows.

For $d \in D_B \setminus \{0\}$ define

$$v(d) = \min\{k : d_k \neq 0\}.$$

Let e_k be the rational sequence with a single 1 in coordinate k , and put

$$F_k = \{d : d_0 = \dots = d_{k-1} = 0\}.$$

Thus $F_0 = D_B$; the subgroups F_k form a neighborhood basis at zero.

Theorem 11.4 (Complete matrix classification [04]). *Every $\mathbb{Q}$ -linear order embedding $T : D_B \rightarrow D_B$, without any assumed continuity, has the form*

$$(Td)_i = \sum_{k:p(k) \leq i} a_{ik} d_k, \quad (11.4)$$

where $p : \mathbb{N} \rightarrow \mathbb{N}$ is strictly increasing and

$$a_{ik} = 0 \ (i < p(k)), \quad a_{p(k),k} > 0. \quad (11.5)$$

The coefficients below each pivot, a_{ik} with $i > p(k)$, are arbitrary rationals. Conversely every such matrix defines an order embedding. All the sums in (11.4) are finite. Every such map is continuous and has closed image. It is surjective exactly when $p(k) = k$ for all k .

Proof. Let T first be an arbitrary $\mathbb{Q}$ -linear order embedding. Since $Te_k > 0$, its first nonzero coefficient is positive. Define $p(k) = v(Te_k)$. The inequalities

$$ne_{k+1} < e_k \quad (n = 1, 2, \dots)$$

remain true after applying T . If $p(k+1) < p(k)$, the image on the left already dominates the right for $n = 1$. If the pivots were equal, a large enough standard n would make the positive leading rational coefficient on the left larger than the one on the right. Both are impossible. Thus $p(k+1) > p(k)$, and in particular $p(k) \geq k$.

Every $d \in F_k$ satisfies $|d| < Ne_k$ for some standard integer $N > 0$: choose $N > |d_k|$, with a further unit of margin. Order preservation implies $|Td| < NTe_k$, hence

$$T(F_k) \subseteq F_{p(k)} \subseteq F_k. \quad (11.6)$$

This already proves continuity at zero, and therefore everywhere by additivity. It also excludes noncontinuous linear maps arising from a choice of a vector-space basis.

Let $a_{ik} = (Te_k)_i$. To compute coordinate i of Td , subtract the finite initial sum $\sum_{k=0}^i d_k e_k$. The remainder belongs to F_{i+1} , whose image has zero coordinates through i by (11.6). Thus $(Td)_i = \sum_{k=0}^i a_{ik} d_k$. Terms with $p(k) > i$ vanish, yielding (11.4) and (11.5).

Conversely, a strictly increasing p satisfies $p(k) \geq k$, so each row in (11.4) has finite support. The formula defines a linear continuous map on all rational sequences. If $d \neq 0$ has first nonzero index k , then all rows before $p(k)$ vanish and

$$(Td)_{p(k)} = a_{p(k),k} d_k.$$

Its sign is the sign of d_k . This proves order preservation and injectivity, including comparison of arbitrary pairs by their difference.

To describe the range, for any proposed target sequence y recursively recover

$$d_k = \frac{y_{p(k)} - \sum_{j < k} a_{p(k),j} d_j}{a_{p(k),k}}. \quad (11.7)$$

Each recovered coordinate is a continuous function of finitely many coordinates of y . The target belongs to the range exactly when, for every nonpivot row i ,

$$y_i = \sum_{k:p(k) < i} a_{ik} d_k. \quad (11.8)$$

These are closed, indeed clopen, finite-coordinate conditions. Their intersection is closed. If every row is a pivot, the recovery solves the system for every target. This happens exactly when $p(k) = k$. If a row j is omitted, no image has its first nonzero coordinate equal to j , so surjectivity fails. $\square$

Corollary 11.5 (Automatic continuity and the topology of self-images [04]). *Every unital additive self-map of M_B is elementary, continuous, and a homeomorphism onto a closed image. Its image is clopen when the complement of $p(\mathbb{N})$ is finite, and closed nowhere dense when that complement is infinite. The automorphisms are exactly the infinite lower triangular rational matrices with strictly positive diagonal, acting on the divisible coordinate and fixing the integer coordinate.*

Proof. Combine proposition 11.3 and theorem 11.4. The inverse on the image is continuous by the coordinate-recovery formula. The group image is $T(D_B) \times \mathbb{Z}$, and its intersection with M_B is exactly the monoid image, because T reflects order.

When finitely many rows are not pivots, (11.8) gives a finite intersection of clopen conditions, so the group and monoid images are clopen. When infinitely many rows are not pivots, they occur arbitrarily far out. At an image point whose D_B -coordinate is positive, keep an initial segment containing its leading coordinate and change a later nonpivot coordinate so that its constraint fails. The result remains positive and lies in any sufficiently specified cylinder. At a point $(0, z)$ in the image, necessarily $z \geq 0$. In any given zero-prefix cylinder first insert a positive pivot far beyond that prefix, using a small-in-topology image vector Te_k . Then violate a still later nonpivot constraint. Positivity is unchanged and the resulting point is outside the image. Thus the image has empty interior even at the boundary. It is closed, hence nowhere dense. $\square$

Example 11.6 (Two contrasting proper elementary copies [04]). The shift

$$S(d)_0 = 0, \quad S(d)_{k+1} = d_k$$

has pivots $p(k) = k + 1$ and produces a proper clopen elementary self-copy of M_B . The insertion map

$$E(d)_{2k} = d_k, \quad E(d)_{2k+1} = 0$$

has pivots $p(k) = 2k$ and produces a closed nowhere dense elementary self-copy. Both are homeomorphisms onto their images. Their order-theoretic and model-theoretic behavior is identical at the level of elementary embedding, while their image interiors are different.

Theorem 11.7 (Clopen elementary self-copies of the Baire model). *Let*

$$\sigma((a_0, a_1, a_2, \dots), n) = ((0, a_0, a_1, a_2, \dots), n).$$

Then $\sigma : M_B \rightarrow M_B$ is a proper elementary embedding and a homeomorphism onto a clopen subspace. Its iterates satisfy

$$\bigcap_{k \geq 0} \sigma^k(M_B) = \{(0, n) : n \in \mathbb{N}\}.$$

Thus a descending sequence of proper clopen elementary self-copies has intersection exactly the standard elementary submodel.

Proof. Inserting an initial zero preserves and reflects the lexicographic order, addition, and every standard congruence. It maps the divisible coefficient group into the subgroup with first coordinate zero. Quantifier elimination gives elementarity, first for the groups and then for their cones. The image is exactly the subset with $a_0 = 0$, which is clopen because the first rational coordinate is discrete. Deleting that zero is a continuous inverse on the image. It is proper, since elements with $a_0 > 0$ are omitted. The k -th image has its first k rational coordinates zero. Membership in all images therefore means that every rational coordinate is zero; positivity then forces $n \geq 0$. $\square$

These clopen submodels are external subsets, not Presburger-definable subsets, even with parameters: a proper definable subset containing zero and closed under successor would violate induction. Topological simplicity and internal definability are different properties.

[merge] The map σ of [theorem 11.7](#) (source 05) is the shift S of [example 11.6](#) (source 04), with pivots $p(k) = k + 1$. Source 05 adds the intersection of the iterated images; source 04 adds the insertion map E , whose image is closed and nowhere dense.

[write] (Batch 89.) Source 16 ([Part X](#)) proves [proposition 11.3](#), [theorem 11.4](#), and [corollary 11.5](#) again, without having seen them, and extends them and [theorem 11.7](#) to the cones $M_\alpha = (\mathbb{Q}^\alpha \vec{\times} \mathbb{Z})_{\geq 0}$ for every countable ordinal α ([theorems 129.1](#), [130.5](#), [130.7](#) and [135.1](#)). Automatic continuity of all self-embeddings fails from $\alpha = \omega + 1$ on, and closedness of all their images from $\alpha = \omega \cdot 2$ on ([theorems 133.3](#) and [134.2](#); both counterexamples use choice).

12 A compactness obstruction independent of multiplication

The positive answer on Baire space does not extend to every familiar uncountable Polish space. In particular compact carriers are impossible. The underlying fact is the standard compact-cancellative-monoid argument, which we include in its commutative form.

Lemma 12.1 (Compact cancellation produces inverses). *Every compact Hausdorff commutative cancellative topological monoid is a group algebraically.*

Proof. Write the identity as 0 and fix a . Let S be the closure of $\{na : n \geq 1\}$. Continuity of addition makes S a nonempty compact subsemigroup. By the compact intersection property and Zorn's lemma, S has a minimal nonempty compact subsemigroup K .

For $x \in K$, the set $x + K$ is a nonempty compact subset of K and is a subsemigroup: for $u, v \in K$,

$$(x + u) + (x + v) = x + (x + u + v) \in x + K.$$

Minimality gives $x + K = K$. Since $x \in K$, there is $y \in K$ with $x + y = x = x + 0$, and cancellation gives $y = 0$. Thus $0 \in K \subseteq S$. But $a + X$ is compact and hence closed, and contains every positive multiple of a . Therefore it contains S and in particular 0. So $a + b = 0$ for some $b \in X$. $\square$

Corollary 12.2 (No compact positive-cone model). *No model of Presburger arithmetic has a compact Hausdorff topology with continuous addition. In particular Cantor space cannot be a carrier, although Baire space can.*

Proof. Its addition is a commutative cancellative monoid. The lemma would give an additive inverse of $\mathbf{1}$, but no nonnegative element can sum with $\mathbf{1}$ to zero. $\square$

This obstruction uses neither Borel definability nor multiplication. It is separate from [theorem 10.4](#): the real model is locally compact but not compact, and the Baire model is nowhere locally compact.

13 Formalization and reproducibility

13.1 A minimal theorem dependency graph

The construction naturally splits into independently checkable layers. The core algebra does not require real analysis or surreal numbers. First define a discretely ordered abelian group with standard residue representatives. Prove the coset-invariant finite-search lemma. Then prove coefficient normalization, elimination, completeness, and the positive-cone interpretation. Only afterward supply the real or rational-sequence carrier and its topology. The omnific realization can be verified last, as an independent bridge.

A practical Lean development can use the following *proposed* modules. These names are a plan, not existing checked files:

Proposed module	Required mathematical content
<code>ZGroup.Basic</code>	Least positive element, standard residue witnesses, uniqueness, and order properties of the quotient.
<code>ZGroup.CosetCooper</code>	Full mG -invariance and the finite-search criterion; the counterexample to weaker periodicity.
<code>ZGroup.Elimination</code>	Affine-term semantics over an arbitrary $\mathbb{Z}$ -group, coefficient scaling, quantifier elimination.
<code>PolishPresburger.Real</code>	The positive cone of $\mathbb{R} \overrightarrow{\times} \mathbb{Z}$ and its complete slice metric.
<code>PolishPresburger.Baire</code>	The rational-sequence cone and the explicit block/tree homeomorphism.
<code>PolishPresburger.Structure</code>	Order-rank obstruction, absence of semiring expansions, and self-embedding theorems.
<code>PolishPresburger.Omnific</code>	The fixed-support normal-form maps and their additive/order laws.

13.2 The nonstandard Cooper boundary must be explicit

The inspected repository theorem `periodic_interval` has an ordinary-integer predicate and a one-step periodicity hypothesis [15]. Its proof can iterate by any ordinary integer quotient. Over a general $\mathbb{Z}$ -group, the quotient in $x - L = mq + r$ is an element of that group, not necessarily a standard numeral. A proof by ordinary natural-number induction cannot supply invariance under that shift for an arbitrary external predicate.

The correct interface is therefore a predicate on the quotient G/mG , or equivalently the hypothesis (4.4). Affine congruence atoms satisfy it by algebra, and Boolean combinations preserve it. This makes the generalized arithmetic lemma reusable and prevents an incorrect formal transfer from being hidden by notation. The original integer theorem should be retained as an instance; it does not need to be weakened or retracted.

The existing decision procedure's semantics are over integers. Reusing its syntax and transformations is plausible, but proving soundness over all $\mathbb{Z}$ -groups is an additional obligation. The repository's existence of an executable integer eliminator, by itself, does not certify [corollary 4.11](#) for a new carrier.

13.3 Executable material included with this manuscript

`polish_presburger.py` implements exact finite-rank rational arithmetic and the Baire codec of [section 6](#). The codec uses a computable positive-rational enumeration and Cantor pairing, supplies addition and standard division on infinite-sequence oracles, and gives a one-change comparison approximation. Every Baire input code is valid. The lower-level encoding routine takes a positive-cone promise, documented in its interface; the public arithmetic operations preserve that promise.

`verify.py` performs deterministic tests using exact `Fraction` arithmetic and a fixed seed. The executed run produced **17,930 successful checks**. They cover pairing and rational-enumeration round trips, additive and order identities, Euclidean division, cone closure, coset-invariant finite candidates, Baire-code prefix round trips, arithmetic output prefixes, the comparison change bound, and the predecessor boundary witness. The deliberately false weak-periodicity generalization is tested as a counterexample, not asserted as a theorem.

Reproduction

With Python 3.10 or later, run `python verify.py`. No external Python package is needed. The script writes `verification_results.json`. Compile the article twice with `pdflatex -interaction=nonstopmode -halt-on-error polish_presburger_glazer.tex` to resolve references.

Tests on finite-rank elements do not quantify over arbitrary elements of $\mathbb{Q}^{\mathbb{N}}$, and finite prefixes do not decide equality of infinite streams. Polishness, induction, and universal semantic statements are established by the proofs, not by the test count. No new Lean or Rocq formalization is included in this archive.

[write] The box above gives source 05's delivery names. In this report 05's programs are `code/05-omnific-presburger-polish_presburger.py` and `code/05-omnific-presburger-verify.py`, its recorded run is `data/05-omnific-presburger-verification_results.json`, and its manuscript is merged into `article.tex`. The runner imports `polish_presburger` by its delivery name and writes its record beside itself, so it is rerun

on a copy with the delivery names restored (the report’s README gives the commands). The intake reran it on such a copy with Python 3.14.4: 17,930 checks passed, and the record differed from the delivered one only in the interpreter version and line endings.

13.4 The formalization interface, source 04’s audit [04]

Source 04 inspected the Rocq file of the formal project rather than the Lean file, at its own pin.

13.4.1 A pinned, limited source audit

The repository files used in this manuscript were checked at commit

3ad5f878c26d2d82c092190af2f3edc50d6feb06.

The relevant Presburger source is `Logic/PresburgerArithmetic/Coq/Cooper.v`; the project description is `Logic/PresburgerArithmetic/README.md`. The latter describes the Lean pipeline through syntax, normal form, one-variable elimination, and sentence decision [17]. The present manuscript does not claim to have rebuilt that repository or audited its unrelated mathematical projects.

In the inspected Coq source, the predicate hypothesis is defined over *ordinary* integers by one-step translation invariance. The lemma `periodic_shift` then proves invariance under arbitrary integer multiples, using induction on ordinary positive and negative integer representations. This supports `periodic_has_residue` and the interval criterion [18]. That is sound for its stated standard-integer domain.

For a model-uniform version, replacing the type of integers by an arbitrary Presburger group while leaving the predicate hypothesis unchanged would be invalid: [proposition 4.6](#) is a counterexample. The quotient in $x = mq + r$ may have a nonzero divisible coordinate, and iteration through standard integer shifts does not reach mq . The repair is conceptual, not a missing induction tactic: use [definition 4.5](#). Congruence literals satisfy that stronger property directly, so [theorem 4.8](#) supplies the needed uniform lemma.

13.4.2 A concrete implementation sequence

A formalization can be divided into independently reviewable stages. The following are proposed new components, not assertions about existing repository declarations.

Abstract arithmetic semantics. Define a class of linearly ordered abelian groups with a named least positive unit and a standard finite quotient–remainder principle. Interpret the existing affine syntax in an arbitrary group of this class. Keep the syntactic integers as *coefficients*; they must not be confused with the semantic carrier.

Finite boundary theorem. Formalize m -residue invariance as invariance under differences in mG . Prove [theorem 4.8](#), separating existence of a finite candidate from any decidability hypothesis on its predicate. Then specialize to a Boolean evaluator when the residue conditions are decidable. This keeps the mathematical statement valid even when the carrier has undecidable equality or order in a computational representation.

Model-uniform elimination. Reuse the finite syntax transformations, with a fresh semantic correctness proof for coefficient normalization, cancellation in (4.9), bound selection, and finite substitution. The correctness endpoint should quantify over all groups satisfying the abstract assumptions. Specialization to $\mathbb{Z}$ must recover the current integer semantics rather than replace it.

Split models and topological structure. Construct G_D and its nonnegative cone, prove explicit division, and obtain the complete Presburger theory by relativization. Next formalize the G_δ argument, continuity of restricted addition, and the coordinate formulas for fixed quotient and remainder. For D_B , construct the two coding homeomorphisms rather than invoking a black-box classification of zero-dimensional Polish spaces.

Self-embeddings. First prove extension from the positive cone to the group completion and preservation of the divisible subgroup. For D_B , prove the filtration lemma (11.6) before introducing matrices. This exposes the reason for automatic continuity and avoids an unproved assumption that an arbitrary linear map is determined by its values on the coordinate vectors. Those vectors are *not* a vector-space basis of the full product; the finite-row conclusion requires the order argument.

13.4.3 Trust and proof-status ledger

Component	Status in this package	Dependence
Presburger characterization and QE	Classical; self-contained argument supplied	Ordered groups and finite syntax
Affirmative Polish construction	Conventional proof supplied	Explicit split group and topology
Baire homeomorphism and stream operations	Proof and executable implementation	Effective rational coding
Monus and embedding theorems	Conventional proofs supplied	Earlier lemmas in this manuscript
Finite arithmetic diagnostics	Executed successfully	Python standard library, exact fractions
New Lean/Rocq kernel certification	Not supplied	Proposed work above
Historical priority of the answer	Not established	Further literature and expert review

The distinction is especially important here because the repository already contains machine-checked material. Reusing its mathematical strategy does not automatically confer its certification status on the new semantic carrier or on the topology.

13.5 Relation to the formal project `Logic/PresburgerArithmetic` [write]

At the placement, the project's Lean development (namespace `PresburgerArithmetic`) and its Rocq file are unchanged from the two pins. The declarations the sources discuss are:

- Lean, `Lean/PresburgerArithmetic/Cooper.lean`: `periodic_shift_of_mod`, `periodic_interval`, `periodic_has_residue`, `periodic_unbounded_above`, `period`

`ic_unbounded_below` and `cooper_finite_criterion`, all for a predicate P on Int with the one-step hypothesis $\forall x (P(x + m) \leftrightarrow P(x))$;

- Lean, Lean/PresburgerArithmetic/Decision.lean: the decision procedure `PresburgerArithmetic.Formula.presburgerArithmetic_decidable` and its correctness theorem `Formula.decideSentence_eq_true_iff`;
- Rocq, Coq/Cooper.v: `periodic`, `periodic_shift`, `periodic_has_residue`, `periodic_interval`, `periodic_unbounded_above`, `periodic_unbounded_below`, `cooper_finite_criterion` and `cooper_step_decidable`, all over $\mathbb{Z}$.

These statements are about the ordinary integers, where they are correct. [Example 4.4](#) and [proposition 4.6](#) show that their periodicity hypothesis cannot be read over a nonstandard $\mathbb{Z}$ -group, and [lemma 4.7](#) and [theorem 4.8](#) state the coset-invariant replacement that a model-uniform version would need. Neither source changed the project, and neither reports a defect in it. **No statement of this report is formalized:** the proposed modules of [section 13](#) and the stages of [section 13.4](#) are plans, and the collection's `docs/FORMALIZATION.md` maps no label of this report.

14 Further research questions

Sources 05 and 04 each end with a list of questions (ten and eleven). Four questions occur in both lists and are merged in [section 14.1](#); the remaining six of 05 and seven of 04 follow, so that this part has seventeen questions. Source 05 introduces its list as follows, and source 04 its own list as in the second paragraph. None of these questions is answered in this report; [Part II](#) bears on [question 14.4](#) and on the last questions of both lists ([section 26](#)).

[write] (Batch 87.) The sentence “None of these questions is answered in this report” was true of Parts I–V. Source 13 ([Part VII](#)) answers [question 14.8](#) for parameter-free formulas, the existence half of [question 14.11](#), [question 14.13](#), and [question 14.16](#) for joint uniform continuity; sources 13 and 15 settle [question 14.3](#) and [question 14.5](#) for topologies inherited from a Polish ordered group ([Part IX](#)). Notes after these questions give the details; the other questions, and the remaining parts of these, stay open.

[write] (Batch 89.) Source 16 ([Part X](#)) answers [questions 14.9](#) and [14.14](#) for the cones $(\mathbb{Q}^\alpha \vec{\times} \mathbb{Z})_{\geq 0}$, α a countable ordinal, and the clause of [question 14.6](#) on continuous order-preserving endomorphisms for the groups $\mathbb{Q}^\alpha$; it asks [question 14.3](#) and the first sentence of [question 14.10](#) again. Notes after these questions say what stays open.

The following questions are proposed directions arising from the proved results. Except where explicitly linked to the source question, their status is not presented as a certification that the literature has no answer.

The construction answers the printed existence question, but it leaves several sharper problems. The questions below distinguish implementation work, structural problems, and possible new obstructions. No solution to them is asserted merely by listing them.

14.1 Questions asked by both sources

Question 14.1 (A formal certificate for the affirmative answer [05, 04]). *Source 05:* Can the one-scale model, the full $\mathbb{Z}$ -group elimination theorem, and its Polish topology be formalized with a compact audit boundary in ProveIt? The key initial milestone is the generic coset-invariant Cooper lemma, rather than merely a finite test of arithmetic identities. *Source 04:*

Can the complete proof of [remark 3.2](#), including the Baire homeomorphism and the full induction scheme via model-uniform elimination, be added to Provelt with no new axioms? The first milestone should be the finite boundary lemma with residue invariance, followed by the semantic specialization checks described in [section 13.4](#).

Question 14.2 (Weak metatheory [05, 04]). *Source 05*: What weak foundational theory suffices to certify the affirmative construction, including the chosen representation of Polishness? The real model has an explicit complete metric, and the Baire model has an explicit sequence code, so a general model-existence theorem is unnecessary. This is distinct from Glazer’s Question 1 concerning the strength needed for his stronger-arithmetic obstruction. *Source 04*: What is the weakest useful subsystem of second-order arithmetic in which the Baire-space construction, its stream algorithms, and its formula-by-formula model correctness can be coded and proved? Distinguish coding all points of Baire space from asserting a set of all such points. This question concerns the positive construction, not a resolution of the separate reverse-mathematical problem posed in Glazer’s Question 1 [1].

Question 14.3 (Connected carriers [05, 04]). *Source 05*: What additional obstruction prevents or permits a continuous-addition model on a connected Polish space? Our locally compact model has countably many ray components, and the arithmetic order cannot be open or closed. A proof for a prescribed connected carrier must use more than the order-topology observation alone. *Source 04*: Can a connected uncountable Polish space support a Presburger model with continuous addition? Our examples do not decide this: their integer coordinate is a continuous map onto a countable discrete space. Requiring all fixed remainders to be continuous would already force nontrivial clopen partitions, so the connected problem must separate continuity of addition from continuity of these definable functions.

[write] (Batch 87.) Negative for topologies inherited from a Polish ordered group: source 13 shows that every cone with a topology inherited from a Polish topological $\mathbb{Z}$ -group has continuous residues and is therefore disconnected (note after [theorem 95.4](#)), and source 15’s [theorem 121.1](#) gives the locally compact case with its exact carrier. For a topology on the cone alone the question stays open ([question 103.4](#)).

[write] (Batch 89.) Source 16 asks it again ([question 138.10](#)); none of its carriers is connected.

Question 14.4 (Multiplication and growth orders [05, 04]). *Source 05*: Can one give an equally explicit Polish continuous-addition Presburger model whose underlying additive structure admits a unital semiring expansion? Any candidate must avoid a cofinal infinite additive unit. If one also requires $Q + IOpen$, the multiplication cannot be Borel by the cited obstruction. Separating algebraic expandability from Borel expandability is the central issue. *Source 04*: Find admissible uncountable topological groups D whose infinite growth order embeds into each of its strict final segments, thereby passing [proposition 10.7](#). For their positive cones, determine whether any semiring multiplication exists, and, separately, which descriptive regularity it can have. Glazer’s obstruction to Borel multiplication under stronger arithmetic remains relevant, but it does not by itself classify bare semiring expansions.

[write] (Batch 86C.) The first sentence of source 05’s part is answered affirmatively in [Part III](#): source 06’s cone of $\mathbb{Z} + t\mathbb{R}[t]$ ([theorem 38.1](#)) is an explicit uncountable, locally compact Polish model of $\text{Th}(\mathbb{N}; 0, 1, +, <)$ with continuous addition whose additive structure carries a unital semiring multiplication, continuous for the same topology. It has no cofinal infinite element and fails open induction ([theorem 38.2](#)). Whether such a model can also satisfy $Q + IOpen$ with non-Borel multiplication, and source 04’s growth-order question, remain open.

[merge] For [question 14.4](#): the real model is the additive monoid of one stratum of 03’s ring $\mathcal{A}_{\text{fin}, \geq 0}$, which does carry a semiring multiplication, Borel and continuous off a closed

nowhere dense set ([theorem 24.3](#)); but $\mathcal{A}_{\text{fin}, \geq 0}$ is not a model of Presburger arithmetic, and on it continuous addition is impossible ([theorem 25.2](#)). 03's [question 28.1](#) asks the multiplicative counterpart on $\mathcal{A}_{\text{fin}, \geq 0}$.

14.2 Further questions of source 05

Question 14.5 (Prescribed zero-dimensional locally compact carriers). Which uncountable zero-dimensional locally compact Polish spaces support continuous-addition Presburger models? Compact spaces are excluded and Baire space is realized, but Baire space is nowhere locally compact. This leaves a meaningful topological region not settled by our examples.

[write] (Batch 87.) None, among cones with a topology inherited from a Borel ordered Polish group: source 15's [corollary 121.2](#) shows that every uncountable locally compact such cone is homeomorphic to the countable sum of lines and closed rays, and its [theorem 121.4](#) shows that 0 is isolated in every zero-dimensional locally compact continuous-addition presentation. Presentations not inherited from a Polish group remain open ([question 126.1](#)).

Question 14.6 (Classifying admissible coefficient groups). Which divisible ordered Polish abelian groups have G_δ positive cone? [Theorem 5.2](#) turns this into an exact criterion for the split construction. Can their possible underlying topologies, order ranks, and continuous order-preserving endomorphisms be classified under useful additional hypotheses?

[write] (Batch 89.) For the groups $\mathbb{Q}^\alpha$, α a countable ordinal (discrete coordinates, lexicographic order), the clause on continuous order-preserving endomorphisms is answered by source 16 for the strictly order-preserving ones, the order embeddings: they are the row-finite rational matrices with strictly increasing pivot map and positive pivots ([theorem 130.5](#)); every order embedding of $\mathbb{Q}^\alpha$ into itself is continuous exactly when $\alpha \leq \omega$ ([theorems 133.1](#) and [133.3](#), the discontinuous ones built with choice); and these groups have F_σ , hence G_δ , positive cones ([\(128.3\)](#); [theorem 35.1](#)). Weakly monotone endomorphisms, the discontinuous order embeddings for $\alpha > \omega$ ([question 138.4](#)), other coefficient groups and the rest of the question stay open.

Question 14.7 (Continuity of a richer definable language). How much of Presburger definable choice can be made continuous in one uncountable Polish presentation? Standard quotients and remainders are continuous here, but total predecessor and truncated subtraction are not everywhere continuous. Is there a natural maximal collection of definable operations admitting simultaneous continuity?

Question 14.8 (Optimal finite-mind-change complexity). For a given Presburger formula, what is the smallest possible uniform change bound in the Baire presentation? The proof gives the number of order/equality atom occurrences after elimination. That bound need not be optimal and may be much larger than necessary. Relating the minimal bound to the finite difference hierarchy or to formula structure would refine the Δ_2^0 result.

[write] (Batch 87.) Answered by source 13 in the rational-stream presentation: for a parameter-free formula the optimum is computable and equals the longest alternating chain of feasible sign patterns, at most the rank of the order and equality forms and at most the arity ([theorems 100.1](#) and [100.4](#)); the arity bound is attained ([theorem 100.2](#)); and no uniform algorithm computes the optimum for a fixed parameter ([theorem 100.5](#)). A bound on revisions is not a bound on running time or on the coordinates read.

Question 14.9 (Polish elementary-submodel lattices). Which lattices of closed or clopen elementary submodels can occur in a Polish continuous-addition Presburger model? The real model has only two Polish elementary submodels with inherited topology, whereas the Baire model has infinitely many proper clopen elementary self-copies with standard intersection. What intermediate behaviors can be realized?

[write] (Batch 89.) Answered by source 16 for the family $M_\alpha = (\mathbb{Q}^\alpha \overrightarrow{\times} \mathbb{Z})_{\geq 0}$, α a countable ordinal, which contains the Baire model $M_\omega = M_B$. Its elementary submodels that are Polish in the inherited topology are the M_V for the closed $\mathbb{Q}$ -subspaces V of $\mathbb{Q}^\alpha$ ([theorem 131.1](#)); they form a complete lattice, with meet the intersection and join the closure of the sum, anti-isomorphic to the lattice of all subspaces of the finite-support space $\mathbb{Q}^{(\alpha)}$ ([theorem 132.4](#)); each is clopen exactly when its pivot subset omits finitely many coordinates, and closed nowhere dense otherwise ([theorem 132.1](#)), and is isomorphic to $M_{\text{otp}(S)}$ for its pivot subset S ([corollary 131.4](#)). The clopen self-copies intersect in the terminal tail, the standard copy exactly when α is 0 or a limit, and the closed submodels that are intersections of clopen self-copies are exactly those containing that tail ([theorems 135.1](#) and [135.2](#)). For $\alpha > \omega$ an automorphism can move a clopen elementary submodel onto a dense non-Polish one ([proposition 133.5](#), built with choice). Which lattices occur in other Polish continuous-addition Presburger models stays open.

Question 14.10 (Canonical surreal cutoffs and topologies). Find useful explicit birthday bounds for the two fixed-support omnific images and formalize the coefficient-topology transport inside those cutoffs. More generally, which support sets admit Polish coefficient topologies with continuous addition and full Presburger induction on the resulting positive cone? Such a classification should keep ring closure separate from additive closure.

[write] (Batch 89.) Source 16 asks the first sentence again for the whole family M_α ([question 138.12](#)); its realization is Part III's [theorem 40.1](#).

14.3 Further questions of source 04

The last question below and the last question of [section 14.2](#) (canonical surreal cutoffs and topologies) overlap in their second halves: both ask which set-sized supports or subgroups of **Oz** carry Polish coefficient topologies with continuous addition. They are printed separately because their first halves differ. [Theorem 25.2](#) gives a negative datum for one ring-closed family ([section 26](#)).

Question 14.11 (Beyond split topological models). Which Polish presentations of Presburger groups admit a continuous integer-coordinate retraction and a topological decomposition $D \overrightarrow{\times} \mathbb{Z}$ of the kind used here? What can occur when the algebraic extension or the topology does not split? Any characterization must separate the standard subgroup, the divisible subgroup, and the full system of residues modulo standard integers.

[write] (Batch 87.) Source 13's [theorem 95.4](#) characterizes, among Polish group presentations, those with a continuous integer-coordinate retraction and a topological decomposition $D \overrightarrow{\times} \mathbb{Z}$: exactly the algebraically split ones, for which the retraction is automatically continuous. Its [theorem 94.1](#) gives nonsplit uncountable perfect Polish models from every countable nonsplit $\mathbb{Z}$ -group, locally compact or homeomorphic to Baire space. What can occur in general is not classified.

[write] (Batch 89.) Source 16's [question 138.9](#) asks which Polish continuous-addition Presburger models have a Polish group completion with a discrete standard integer coordinate; the note after it points here and to [question 57.2](#).

Question 14.12 (Rigidity with continuous addition). Is there an uncountable rigid Presburger model with a Polish topology making addition continuous and order Borel? Rigid models are studied by Jeřábek [4]; our split examples are not rigid, because positive rational scaling of their divisible part gives nontrivial automorphisms. One must therefore move beyond the present easy symmetry.

[write] Source 13 asks this again ([question 103.5](#)); its nonsplit models are not rigid either.

Question 14.13 (Exact discontinuity profiles of definable functions). For each definable function $f : M_D^k \rightarrow M_D$, can its discontinuity set be computed from a quantifier-free description of its graph? For the two main models, determine whether every such set admits a finite description in terms of equality of divisible-coordinate affine expressions and residue conditions. Keep graph complexity separate from target-valued Baire class; they are not interchangeable statements for arbitrary target spaces.

[write] (Batch 87.) Answered by source 13: the discontinuity set is computed from a quantifier-free description of the graph, as a finite union of locally closed sets inside finitely many nowhere dense affine hyperplanes ([theorem 98.3](#) and [corollary 98.4](#)); a description by equalities and residue conditions alone does not suffice, so the proposed fragment is too small ([section 98](#)); and the target-valued Baire class is determined separately ([section 99](#)).

Question 14.14 (Embedding categories for other coefficient groups). Extend [theorem 11.4](#) to products of ordered coefficient groups other than $\mathbb{Q}$, and to countable support systems with more complicated growth orders. Which hypotheses force every order embedding to be continuous? Which image topologies replace the finite-versus-infinite omitted-pivot dichotomy?

[write] (Batch 89.) Answered by source 16 for products $\mathbb{Q}^\alpha$ indexed by countable ordinals, with rational coefficients. The continuous order embeddings $\mathbb{Q}^\alpha \rightarrow \mathbb{Q}^\beta$ are the row-finite matrices with increasing positive pivots ([theorem 130.5](#)); they have continuous left inverses and closed images ([theorem 130.7](#)), and Borel ones are continuous ([theorem 130.3](#)). Every order embedding of $\mathbb{Q}^\alpha$ into itself is continuous exactly when $\alpha \leq \omega$ ([theorems 133.1](#) and [133.3](#)), and has closed image exactly when $\alpha < \omega \cdot 2$ ([theorems 134.1](#) and [134.2](#)); the failures are built with choice ([remark 133.2](#)). The finite-versus-infinite omitted-pivot dichotomy persists for the images of continuous embeddings and for every closed submodel ([theorem 132.1](#)); from $\omega \cdot 2$ on, discontinuous self-embeddings can have dense non-Borel images. Coefficient groups other than $\mathbb{Q}$ and index orders that are not well ordered stay open ([questions 138.7](#) and [138.8](#)).

Question 14.15 (Effective truth versus effective operations). The Baire presentation has computable addition and fixed division, but not a computable test for every definable relation on arbitrary input streams. Give sharp oracle bounds for evaluation of formulas in this presentation, including uniform bounds under parameter coding and quantifier elimination. Distinguish the decidability of closed Presburger sentences from the discontinuous evaluation of an atomic relation on arbitrary infinite names.

Question 14.16 (Metrics compatible with arithmetic translations). What useful uniform continuity properties of addition can be retained by compatible complete metrics after the invariant-metric obstruction in [proposition 8.4](#)? Determine whether addition can be made jointly uniformly continuous, or all translations uniformly continuous in one direction, and quantify the unavoidable distortion of their inverses. This would clarify completion arguments for nonclosed positive cones.

[write] (Batch 87.) Addition cannot be made jointly uniformly continuous for any compatible complete metric on $M_{\mathbb{R}}$, $M_{\mathbb{B}}$ or the countable-core models, and not even translations can be uniformly equicontinuous at the identity (source 13's [corollary 101.3](#)). The other parts of the question stay open.

Question 14.17 (Coefficient topologies inside surreal arithmetic). Classify set-sized omnific subgroups carrying Polish coefficient topologies with continuous addition, and determine how those topologies behave under changes of normal-form support. The natural comparison is between coefficient topology, arithmetic order, and model-theoretic elementary embedding, not between a Polish topology and the whole proper class of surreals.

15 Conclusions of Part I

The affirmative construction can be summarized by one change of viewpoint: use the lexicographic order to interpret arithmetic, and a coefficient topology to interpret continuity. Divisibility of the leading-coordinate group supplies all standard residues; quantifier elimination then supplies full Presburger induction. Neither countability of the standard model nor discreteness of the arithmetic order forces the external presentation to be countable or discrete.

The real model gives a locally compact answer, while the rational-sequence model gives a genuinely effective answer on Baire space. Their order relations have the smallest possible nontrivial Borel rank, yet they exhibit different elementary self-embedding behavior. Their realization inside **Oz** connects these results directly to the repository's normal-form arithmetic. Finally, the cofinal-infinite-element argument shows exactly why these additive examples cannot be mistaken for topological models of full arithmetic.

Source 04's conclusion [04]. The separation needed for the construction is simple: place a divisible ordered group above a discrete integer coordinate algebraically, while retaining the ordinary or product topology on that divisible group. The arithmetic becomes discretely ordered without forcing the topology to be discrete. Unique standard residues give the whole Presburger theory, not just a fragment.

The real-coordinate construction supplies a perfect locally compact example; the rational product supplies literal Baire space and effective stream operations. Their regularity is sharply delimited: every definable relation has Borel rank at most two, whereas monus is discontinuous exactly on an explicitly described geometric boundary. Their self-embeddings range from unavoidable surjective dilations to closed nowhere dense elementary copies, but all are automatically continuous.

For Provolt, the most immediate next step is precise and finite: generalize the Cooper residue lemma with the correct subgroup-invariance hypothesis, then connect the already familiar syntax to these new semantics. That would turn the conventional answer presented here into a kernel-checked one without changing the mathematical construction.

Part II

Borel presentations and support barriers

This part is source 03, pinned to repository commit c5b4881fa. It keeps 03’s definitions, in particular coefficient observability (definition 17.1) and its two exponent conventions: $X = t^{-1}$ is read as ω , and the change $q \mapsto -q$ converts the increasing t -convention into the reverse X -convention (section 17). Four symbols are renamed (section 1.3): 03’s ring $\mathcal{A}$ is $\mathcal{A}_{\text{fin}}$, its rational Hahn ring H is $\mathcal{H}_{\mathbb{Q}}$, its Kleene–Brouwer embedding h is μ_{KB} , and two local sets called D are W and $\mathcal{Z}$. Several of 03’s results restate results already in the collection; they keep their proofs here, because theorem 25.2 needs their exact hypotheses, and a note after each names the collection’s statement. In 03’s text, “this report” means source 03.

16 The intersection and the questions answered

Glazer’s paper *A Topological Tennenbaum Theorem* studies the interaction between arithmetic, Borel definability, and complete separable topologies. The part used here is particularly well matched to omnific arithmetic: an uncountable Polish model of Robinson arithmetic plus open induction cannot have continuous addition and Borel multiplication [1, Corollary 1]. This is weaker than the hypothesis needed in his separate multiplication obstruction; we preserve that distinction throughout.

The relevant ProveIt material is the surreal-number project, especially the reports on omnific notations and computer algebra, and its construction of the omnific ring and floor. The inspected repository guides explicitly distinguish research drafts, finite verification programs, and statements mapped to Lean proofs [20, 21]. That distinction is retained here. The repository already reports a Π_1^1 classification of unrestricted support validity, using Kleene–Brouwer coding. Thus merely repeating a well-founded-tree reduction would not constitute a new advance on this material.

Instead, we ask for an exact boundary and a positive arithmetic counterpart. For which countable exponent orders is support validity Borel? Does the answer also determine whether the whole Hahn field can be given a standard Borel presentation with readable coefficients? Can a concrete uncountable integer part have arithmetic that is continuous almost everywhere in the category sense, even though Glazer rules out global continuity? Precisely where does continuity fail?

16.1 Main theorem map

The report proves the following connected package.

- Support and field-presentation dichotomy.** For countable L , $\text{WO}(L)$ is Borel if and only if L contains no copy of $\mathbb{Q}$. Otherwise it is coanalytic-complete (Theorem 18.4). For countable ordered abelian Γ , the same condition characterizes coefficient-observable standard Borel presentations of $\mathbb{R}((t^\Gamma))$ (Theorem 20.2).
- Exact benchmark complexities and a root-closure obstruction.** The support predicates for $\mathbb{Z}$ and $\mathbb{Z}_{\text{lex}}^d$, $d \geq 2$, are respectively Σ_2^0 -complete and Π_3^0 -complete. A nontrivial countable exponent group in a square-root-closed full real Hahn field cannot lie on the Borel side (Theorem 19.1, Corollary 20.4).

3. **Uniform bounds and localization.** Every Borel coefficient-observable family of valid series has its support order types bounded below one countable ordinal. Arbitrarily large countable support types, and coanalytic-complete validity, already occur in a fixed 0–1 coefficient slice inside $[X, 2X)$ (Theorems 21.2 and 22.1).
4. **A positive model with exact negative boundaries.** The finite principal-part model $\mathcal{A}_{\text{fin}, \geq 0}$ satisfies open induction and has a Polish presentation with explicit closed nowhere dense discontinuity sets for both operations. Yet no Polish topology with the same Borel sets makes addition continuous everywhere (Theorems 23.2, 24.3, and 25.2).

The order-theoretic dichotomy is deduced from classical Hausdorff theory and well-founded-tree completeness. The uniform bound is an application of classical analytic boundedness, for which we include a proof. The integer-part model is of Shepherdson type, not a newly discovered model of arithmetic. The report's claims of contribution concern the explicit combined classification, the observable-field criterion, the benchmark calculations, and the exact topological analysis. These are proved here; bibliographic priority is not asserted.

16.2 A precise source boundary

The source audit used selected repository guides at commit

c5b4881fa603339c7ff90f5e1098ede0651243d8.

The inspected paths are recorded in Appendix D. This was not an exhaustive audit of the repository, nor a rebuild of its Lean development. No new theorem below depends on an unreviewed repository theorem: the required normal-form and integer-part arguments are supplied independently. The repository is used to identify the research overlap and to avoid claiming its previously recorded results as new.

17 Conventions and the meaning of a presentation

17.1 Two opposite exponent conventions

For an ordered abelian group Γ , a Hahn series is written

$$f = \sum_{\gamma \in \Gamma} a_{\gamma} t^{\gamma}, \quad a_{\gamma} \in \mathbb{R},$$

with $\text{supp}(f) = \{\gamma : a_{\gamma} \neq 0\}$ well-ordered in the increasing order of Γ . Its sign is the sign of the coefficient at the least exponent. Thus t is infinitesimal when $\Gamma = \mathbb{Q}$ and $t = t^1$.

For the omnific applications we instead write $X = t^{-1}$, so X is positive infinite. In

$$f = \sum_{q \in \mathbb{Q}} b_q X^q,$$

the support must be *reverse* well-ordered: every nonempty subset has a greatest exponent. The change $q \mapsto -q$ converts one convention into the other. All descriptive-set-theoretic statements in Sections 18 and 19 use the increasing convention. Section 22 explicitly reverses it.

The ring

$$\mathcal{H}_{\mathbb{Q}} = \left\{ n + \sum_{q \in S} b_q X^q : n \in \mathbb{Z}, S \subseteq \mathbb{Q}_{>0} \text{ reverse well-ordered} \right\} \quad (17.1)$$

is a set-sized rational-exponent omnific ring. Replacing X^q by the Conway monomial ω^q identifies it with the corresponding normal-form subring of $\mathbf{Oz}$. Hahn addition and multiplication agree with those normal-form operations. The normal-form viewpoint and its ordered-algebraic context are standard; see [11, 5].

17.2 Borel spaces, completeness, and observability

A Polish space is a separable topological space admitting a compatible complete metric. A standard Borel space is a measurable space isomorphic to a Borel subset of a Polish space. We use the usual boldface pointclasses: Σ_2^0 consists of countable unions of closed sets, Π_3^0 of countable intersections of Σ_2^0 sets, and Π_1^1 denotes coanalytic sets. An analytic set is a projection of a Borel set in a product of Polish spaces. Completeness is with respect to continuous reductions from zero-dimensional Polish spaces. Standard references are [8, 9].

Definition 17.1. Let K be a set of series over a countable exponent alphabet. A *coefficient-observable standard Borel parametrization* of K consists of a standard Borel space Y and a surjection $\theta : Y \rightarrow K$ such that, for every exponent γ , the map

$$y \mapsto [t^\gamma]\theta(y)$$

is Borel as a map to $\mathbb{R}$. Repetitions of names are allowed.

No Borel structure on K is presupposed in this definition. Since the alphabet is countable, coefficient observability says exactly that the composite code map $Y \rightarrow \mathbb{R}^\Gamma$ is Borel. This is stronger than the mere existence of a set-theoretic bijection with the reals, and different from an abstract Borel presentation that is allowed to make every canonical coefficient unobservable. Our negative results explicitly retain the observability hypothesis.

Definition 17.2. For a countable linear order L , put

$$\text{WO}(L) = \{S \in 2^L : (S, <_L) \text{ is well-ordered}\}.$$

The empty support is allowed. The order L is *scattered* if it contains no order-isomorphic copy of $\mathbb{Q}$.

A subset of 2^L is a set of codes, not a new collection of proper-class objects. In particular, asking whether $\text{WO}(L)$ is Borel is an ordinary set-sized descriptive-set-theoretic question.

18 The exact support dichotomy

18.1 A complete obstruction on a dense countable alphabet

Lemma 18.1. *For every countable linear order L , $\text{WO}(L)$ is coanalytic.*

Proof. A support S fails to be well-ordered exactly when there exists $(x_n)_{n \in \mathbb{N}} \in L^{\mathbb{N}}$ with $x_n \in S$ and $x_{n+1} <_L x_n$ for every n . The relation on $(S, (x_n))$ specified by these countably many coordinate conditions is closed, with L discrete. Its projection is analytic. Its complement is $\text{WO}(L)$. $\square$

Let Tr be the closed space of trees $T \subseteq \mathbb{N}^{<\mathbb{N}}$, where trees are closed under initial segments, and let $\text{WF} \subseteq \text{Tr}$ be the well-founded trees. Recall briefly why WF is coanalytic-complete. If A is coanalytic in Baire space, its complement is the projection of the body of a tree R on pairs of finite sequences. For $x \in \mathbb{N}^{\mathbb{N}}$, let

$$T_x = \{s : (x \upharpoonright |s|, s) \in R\}.$$

Membership of a fixed s depends on finitely many coordinates of x , so $x \mapsto T_x$ is continuous, and $x \in A$ exactly when T_x has no branch. This is the standard universal-tree proof [9, Section 16.B].

Lemma 18.2 (Kleene–Brouwer coding). *There is a fixed injection $\mu_{\text{KB}} : \mathbb{N}^{<\mathbb{N}} \rightarrow \mathbb{Q} \cap (0, 1)$ such that*

$$T \in \text{WF} \iff \mu_{\text{KB}}[T] \in \text{WO}(\mathbb{Q}).$$

Consequently $\text{WO}(\mathbb{Q})$ is Π_1^1 -complete.

Proof. Order the finite sequences by the Kleene–Brouwer order: a proper extension precedes its prefix, and otherwise the first differing entry determines the order, with the ordinary order on $\mathbb{N}$. This is a countable linear order. Embed it in $\mathbb{Q} \cap (0, 1)$ as follows. Enumerate its elements. At each finite stage, put the next element at a rational strictly between its nearest already assigned lower and upper neighbors, using 0 and 1 as external bounds when needed. A midpoint works. The resulting μ_{KB} is one fixed order embedding of the entire tree alphabet, not an embedding selected separately for each T .

If T has a branch, its successively longer prefixes form a strictly descending Kleene–Brouwer sequence. If T is well-founded, well-founded induction over its nodes shows that the Kleene–Brouwer order on each descendant subtree is a well-order: it is the increasing $\mathbb{N}$ -indexed sum of the child-subtree orders, followed by the root. Therefore $\text{KB}(T)$ is a well-order exactly when T is well-founded.

The map $T \mapsto \mu_{\text{KB}}[T]$ is continuous. At an exponent in the range of μ_{KB} , its characteristic function reads exactly one tree-membership bit; outside the range it is constantly zero. Thus it reduces WF to $\text{WO}(\mathbb{Q})$. Lemma 18.1 supplies membership in the pointclass. $\square$

[write] **Lemma 18.2** and part (ii) of **theorem 18.4** below are boldface counterparts of results already in the collection’s omnific-notations report: `onot:neg:lem:kbtree` (a tree is well-founded exactly when its Kleene–Brouwer order is a well-order), `onot:neg:thm:pionesone` (Π_1^1 -completeness, for recursive trees, of the validity of decidable dyadic supports in $[1, 2)$ with coefficient one) and `onot:neg:rem:secondroute` (the midpoint codings of that report’s sources 08 and 09, which are the construction of μ_{KB} up to the target interval). Source 03 reproves the reduction to make the continuous (boldface) form explicit, as its contribution ledger says (section D). What is new in section 18 is the dichotomy between scattered and nonscattered orders.

18.2 Why scattered alphabets are different

We use the classical countable Hausdorff characterization: starting with finite linear orders, closing under $\mathbb{Z}$ -indexed sums and finite sums through countable ordinal stages produces precisely the countable scattered orders. A precise formulation is recalled in [10, Definition 1]. Only this characterization, not its more advanced rank consequences, is needed below.

Lemma 18.3. *Suppose $L = \sum_{i \in \mathbb{Z}} L_i$ is a lexicographic sum and each $\text{WO}(L_i)$ is Borel. Then $\text{WO}(L)$ is Borel.*

Proof. For $S \subseteq L$, let $S_i = S \cap L_i$ and $I(S) = \{i : S_i \neq \emptyset\}$. Then

$$S \in \text{WO}(L) \iff (\forall i \in \mathbb{Z} S_i \in \text{WO}(L_i)) \wedge (\exists N \in \mathbb{Z} \forall i < N S_i = \emptyset). \quad (18.1)$$

Necessity of the second condition follows because unboundedly negative occupied indices yield an infinite descending sequence, choosing one element from each of a decreasing sequence of such indices. Conversely, when the condition holds, S is a well-ordered sum of well-ordered fibers over a subset of an integer tail.

Each restriction map $S \mapsto S_i$ is continuous. The fiber-empty conditions are closed. The right-hand side of (18.1) is therefore Borel. For a finite sum the analogous criterion has only finitely many fiber conditions and no additional obstruction. $\square$

Theorem 18.4 (Support dichotomy). *For every countable linear order L :*

- (i) *if L is scattered, $\text{WO}(L)$ is Borel;*
- (ii) *if L is not scattered, $\text{WO}(L)$ is Π_1^1 -complete.*

In particular, Borelness of $\text{WO}(L)$ is equivalent to scatteredness of L .

Proof. For a finite order the support predicate is the whole finite code space. Induction through the Hausdorff construction, using Lemma 18.3 and the finite-sum observation, proves (i).

For (ii), fix an order embedding $j : \mathbb{Q} \rightarrow L$. Extending a support by zero outside $j[\mathbb{Q}]$ gives a continuous map $S \mapsto j[S]$ from $2^{\mathbb{Q}}$ to 2^L , preserving and reflecting well-orderedness. Compose this map with Lemma 18.2, and use Lemma 18.1. A coanalytic-complete set is not Borel, completing the equivalence. $\square$

Remark 18.5. The quantification here is over *all subsets of one fixed countable order*. It is not a statement about the Borel complexity of the class of all scattered orders, nor about finding a Hausdorff decomposition uniformly from an arbitrary code for L . The fixed decomposition used in the positive proof need not be computable.

19 Exact benchmarks: integers, finite lexicographic powers, and rationals

The dichotomy gives a qualitative boundary. On useful discrete alphabets one can calculate the precise Borel level.

Theorem 19.1. *With the product topology on the corresponding binary support spaces:*

- (i) $\text{WO}(\mathbb{Z})$ is Σ_2^0 -complete;
- (ii) for every integer $d \geq 2$, $\text{WO}(\mathbb{Z}_{\text{lex}}^d)$ is Π_3^0 -complete;
- (iii) every well-ordered subset of $\mathbb{Z}_{\text{lex}}^d$ has order type at most ω^d , and this bound is attained.

Proof. A subset of $\mathbb{Z}$ is well-ordered if and only if it is bounded below. Hence

$$\text{WO}(\mathbb{Z}) = \bigcup_{N \in \mathbb{Z}} \{S : \forall n < N (n \notin S)\},$$

a countable union of closed sets. The continuous map $A \subseteq \mathbb{N} \mapsto \{-n - 1 : n \in A\}$ reduces the familiar Σ_2^0 -complete set Fin of finite subsets of $\mathbb{N}$ to $\text{WO}(\mathbb{Z})$. This proves (i).

For completeness of the benchmark being used, Fin is countable and dense in $2^{\mathbb{N}}$. Were it G_δ , it would be comeager, while it is also meager; the Baire theorem rules this out. The usual completeness criterion for Borel sets in zero-dimensional Polish spaces then gives its Σ_2^0 -completeness [9, Section 16.A].

For $d = 2$, view the order as $\sum_{i \in \mathbb{Z}} \mathbb{Z}$. Formula (18.1) is the intersection of a Π_3^0 condition (every fiber is bounded below) and a Σ_2^0 condition (the occupied outer indices are bounded below). The result is Π_3^0 . For $d > 2$, the same formula and induction work: countable intersections of Π_3^0 sets remain Π_3^0 .

For hardness, use

$$\text{Fin}^{\mathbb{N}} = \{a \in 2^{\mathbb{N} \times \mathbb{N}} : \forall m \{n : a(m, n) = 1\} \text{ is finite}\}.$$

This set is Π_3^0 -complete. Indeed, any Π_3^0 subset of a zero-dimensional Polish space is a countable intersection of Σ_2^0 sets; reduce each of these sets to Fin and take the product of the reductions. Map a to the support

$$S_a = \{(m, -n - 1) : a(m, n) = 1\} \subseteq \mathbb{Z}_{\text{lex}}^2. \quad (19.1)$$

The outer indices are automatically bounded below by zero. A fiber is well-ordered exactly when its row is finite. Thus (19.1) is a continuous reduction. Embedding $\mathbb{Z}_{\text{lex}}^2$ into $\mathbb{Z}_{\text{lex}}^d$ gives hardness for all $d \geq 2$.

Finally, a well-ordered subset of $\mathbb{Z}$ has type at most ω . Inductively, a well-ordered support in $\mathbb{Z}^d$ is a sum of at most ω fibers of type at most ω^{d-1} , and so has type at most $\omega^{d-1} \cdot \omega = \omega^d$. The subset $\mathbb{N}^d$, with the same lexicographic convention (first coordinate most significant), has type exactly ω^d . $\square$

Exponent alphabet	Valid-support complexity	Support-type bound
$\mathbb{Z}$	Σ_2^0 -complete	$\leq \omega$
$\mathbb{Z}_{\text{lex}}^d, d \geq 2$	Π_3^0 -complete	$\leq \omega^d$
A fixed countable scattered order	Borel	Some countable bound
$\mathbb{Q}$, or any countable nonscattered order	Π_1^1 -complete	Unbounded in ω_1

The last two support-type assertions follow also from Theorem 21.2 below. An important feature of the table is that the finite lexicographic rank can grow while the exact Borel level stays at Π_3^0 . Support order type and Borel complexity measure different things. Conversely, passing from $\mathbb{Z}$ to its divisible hull $\mathbb{Q}$ is a genuine jump out of the entire Borel hierarchy.

20 An observable-presentation dichotomy for full Hahn fields

Fix a countable ordered abelian group Γ . Let

$$V_\Gamma = \{a \in \mathbb{R}^\Gamma : \text{supp}(a) \in \text{WO}(\Gamma)\}.$$

As a set, V_Γ is the canonical coefficient presentation of the full Hahn field $K_\Gamma = \mathbb{R}((t^\Gamma))$.

Lemma 20.1. *If Γ is scattered, V_Γ is Borel. If Γ is nonscattered, V_Γ is Π_1^1 -complete.*

Proof. The support map

$$\sigma : \mathbb{R}^\Gamma \rightarrow 2^\Gamma, \quad \sigma(a)(\gamma) = 1 \iff a_\gamma \neq 0$$

is Borel. Thus the scattered case follows from Theorem 18.4. In all cases, failure of validity is witnessed by an infinite descending sequence of nonzero coordinates, an analytic condition, so V_Γ is coanalytic. The binary-coefficient inclusion $2^\Gamma \hookrightarrow \mathbb{R}^\Gamma$ is continuous and reduces $\text{WO}(\Gamma)$ to V_Γ , proving completeness in the remaining case. $\square$

Theorem 20.2 (Observable Hahn-field criterion). *For countable ordered abelian Γ , the following are equivalent:*

- (i) Γ is scattered as a linear order;
- (ii) V_Γ is a Borel subset of $\mathbb{R}^\Gamma$;
- (iii) K_Γ admits a coefficient-observable standard Borel parametrization, allowing repeated names;
- (iv) K_Γ admits a standard Borel presentation with Borel coefficient maps, Borel addition and multiplication, Borel inversion on the nonzero elements, and Borel order.

Proof. Lemma 20.1 gives (i) $\Leftrightarrow$ (ii). Assume (ii), and use V_Γ itself as the standard Borel domain. Addition and negation are coordinatewise and therefore Borel. For $\Gamma = \{0\}$ all assertions reduce to those for $\mathbb{R}$. Otherwise, for multiplication enumerate $\Gamma = \{\delta_0, \delta_1, \dots\}$ without repetition. At a fixed exponent γ , the product coefficient is

$$(ab)_\gamma = \lim_{N \rightarrow \infty} \sum_{j < N} a_{\delta_j} b_{\gamma - \delta_j}. \quad (20.1)$$

For valid Hahn supports, only finitely many terms are nonzero. To see the finiteness, infinitely many contributing δ would contain a strictly increasing sequence in the well-ordered support of a ; the corresponding $\gamma - \delta$ would be a strictly decreasing sequence in the support of b . This is impossible. Thus the sums in (20.1) eventually stabilize. They are continuous on the ambient product, and their pointwise limit on the Borel domain is Borel.

The standard Hahn-field theorem supplies closure under multiplication and the existence of multiplicative inverses; its well-ordered-support lemma is the usual Neumann lemma [11]. The graph

$$\{(a, b) \in (V_\Gamma \setminus \{0\}) \times V_\Gamma : ab = 1\}$$

is Borel and has exactly one point in each vertical section. The Lusin–Novikov theorem gives Borel inversion. Equivalently, for an open set U the inverse image under inversion is the projection of this graph intersected with $V_\Gamma \times U$, whose sections have at most one point.

Positivity is Borel because it is expressed by

$$\exists \gamma \in \Gamma (a_\gamma > 0 \wedge \forall \delta < \gamma \ a_\delta = 0).$$

There are only countably many coordinates. Order follows by subtraction. This proves (iv), which immediately implies (iii).

Assume (iii). The coefficient code map from its standard Borel parameter space to $\mathbb{R}^\Gamma$ is Borel, and its image is exactly V_Γ . Consequently V_Γ is analytic. For nonscattered Γ , Lemma 20.1 makes it coanalytic-complete and hence nonanalytic. Thus Γ must be scattered, proving (i). $\square$

Remark 20.3 (What the negative direction does and does not exclude). The theorem excludes even a surjective system of Borel-observable names, not just unique or canonical names. It does not exclude abstract Borel models in which the coefficient functions fail to be Borel. Nor does it exclude useful Borel subfields of K_Γ , including real closed subfields. The word *full* is essential.

20.1 Root closure forces the non-Borel side for full fields

Corollary 20.4. *Suppose $\Gamma \neq \{0\}$ is countable and every positive element of the full Hahn field $\mathbb{R}((t^\Gamma))$ has a square root in that field. Then Γ is nonscattered, and the field admits no coefficient-observable standard Borel parametrization. In particular, this holds whenever the full Hahn field is real closed.*

Proof. For every $\gamma \in \Gamma$, the positive monomial t^γ has a root f with $f^2 = t^\gamma$. Taking least exponents gives $2 \min \text{supp}(f) = \gamma$. Thus Γ is 2-divisible. For $a < b$ in Γ , the element $a + (b - a)/2$ lies strictly between them. A nontrivial ordered group has no endpoints, so Γ is a countable dense linear order without endpoints and is order-isomorphic to $\mathbb{Q}$. Apply Theorem 20.2. $\square$

This is a necessary-condition argument; it does not use a classification of which Hahn fields are real closed. It also explains the source of the jump: arbitrary well-ordered supports over a densely divisible scale are the problem, not the mere availability of algebraic roots. The Puiseux field in Section 23 supplies the contrasting real closed Borel subfield.

21 Every observable family has a countable support bound

We now strengthen the nonexistence statement into a restriction on every individual Borel system of names. The essential input is the classical boundedness theorem for analytic families of well-founded trees [8]. The following proof makes the particular form we need explicit.

Lemma 21.1 (Analytic boundedness, with proof). *If $\mathcal{T} \subseteq \text{WF}$ is analytic, there is a countable ordinal β bounding the root ranks of all trees in $\mathcal{T}$.*

Proof. The empty tree is assigned rank zero and can be discarded. If no trees remain, take $\beta = 1$. Otherwise write the remaining analytic family $\mathcal{T}$ as the projection of a closed set $C \subseteq \mathbb{N}^\mathbb{N} \times \text{Tr}$. Fix a binary coding of trees using an enumeration of $\mathbb{N}^{<\mathbb{N}}$. Construct a countable tree U whose level- k nodes are triples (s, u, v) with $s, u \in \mathbb{N}^k$, $v \in 2^k$, for which some $(y, T) \in C$ satisfies

$$u = y \upharpoonright k, \quad v = \text{code}(T) \upharpoonright k, \quad s \in T.$$

Taking initial segments preserves these conditions, so U is a tree.

If U had an infinite branch, its three coordinates would give $b, y \in \mathbb{N}^\mathbb{N}$ and a binary code z . For each k choose a witness $(y_k, T_k) \in C$ for the level- k node. The witnesses converge to (y, T) , where T has code z : for every fixed number of coordinates the witnesses agree with (y, z) once k is large. Since C and Tr are closed, $(y, T) \in C$. For every j , the node $b \upharpoonright j$ belongs to T_k for all $k \geq j$, and hence belongs to T . Thus $T \in \mathcal{T}$ has a branch, a contradiction. Therefore U is well-founded.

For each $(y, T) \in C$, the assignment

$$s \mapsto (s, y \upharpoonright |s|, \text{code}(T) \upharpoonright |s|)$$

embeds the extension tree T into U , taking children to children. Induction on node ranks gives $\text{rk}(T) \leq \text{rk}(U)$. A well-founded countable tree has countable rank. Taking $\beta = \text{rk}(U) + 1$ gives a strict bound. $\square$

Theorem 21.2 (Uniform support bound). *Let L be countable, Y standard Borel, and $c : Y \rightarrow \mathbb{R}^L$ Borel. Suppose every $\text{supp}(c(y))$ is well-ordered in L . Then there is $\alpha < \omega_1$ such that*

$$\text{otp}(\text{supp}(c(y))) < \alpha \quad \text{for every } y \in Y.$$

The same conclusion holds for reverse well-ordered supports, with the reverse support order used to compute the type.

Proof. For $S \subseteq L$, form the descending-sequence tree D_S . Its root is the empty sequence, and its other nodes are finite strictly decreasing sequences of members of S . The map $S \mapsto D_S$ is continuous on binary support codes; the composite $y \mapsto D_{\text{supp}(c(y))}$ is Borel, since each nonzero coefficient test is Borel. Its image is an analytic family of well-founded trees. Apply Lemma 21.1.

For a well-ordered S of type ξ , the root rank of D_S is exactly ξ . Indeed, the rank of a node ending in $s \in S$ is the order type of the initial segment of S below s ; induction verifies the recursion. The root rank is the supremum of these ranks plus one, namely ξ . Reversing L proves the last assertion. $\square$

Corollary 21.3. *For a countable linear order L , the following are equivalent:*

- (i) L is scattered;
- (ii) the order types of all well-ordered subsets of L have a common countable upper bound;
- (iii) $\text{WO}(L)$ is analytic;
- (iv) $\text{WO}(L)$ is Borel.

Proof. Theorem 18.4 gives (i) $\Leftrightarrow$ (iv) and excludes (iii) in the nonscattered case. If (iv) holds, apply Theorem 21.2 to the identity binary-coefficient coding of $\text{WO}(L)$ to obtain (ii). If L is nonscattered, it contains $\mathbb{Q}$, and every countable ordinal embeds in $\mathbb{Q}$ by the finite-stage rational insertion construction used in Lemma 18.2. Its well-ordered suborders therefore have unbounded countable types, contradicting (ii). $\square$

The bound is uniform over the entire parameter space, even if it is uncountable. It is not asserted to be computable from an arbitrary code for the Borel map. In particular, this is a boldface descriptive theorem, not an unqualified statement that every effective system has a computable ordinal bound.

22 The obstruction already lies in one omnific interval

Return to the X -convention. Let $I = \mathbb{Q} \cap (1/3, 2/3)$ and consider the compact space of candidate coefficient arrays

$$C = \left\{ X + \sum_{q \in I} \varepsilon_q X^q : \varepsilon_q \in \{0, 1\} \right\}. \quad (22.1)$$

Here the displayed sums are only *candidate codes*. A member of $\mathcal{C}$ denotes a Hahn or surreal normal form only when its nonzero support is reverse well-ordered.

Theorem 22.1 (Bounded-interval localization). *The valid-code subset $\mathcal{C}_{\text{valid}}$ of (22.1) is Π_1^1 -complete. Every valid value lies in the omnific interval $[X, 2X)$. The reverse support order types in this slice are unbounded in ω_1 .*

Proof. Take the fixed increasing Kleene–Brouwer embedding μ_{KB} from Lemma 18.2 and put

$$e(s) = \frac{2}{3} - \frac{\mu_{\text{KB}}(s)}{3} \in I.$$

This reverses the Kleene–Brouwer order. Define the candidate code

$$z_T = X + \sum_{s \in T} X^{e(s)}. \quad (22.2)$$

The map $T \mapsto z_T$ is continuous in the coefficient product topology. The support below the leading exponent 1 is reverse well-ordered exactly when $\text{KB}(T)$ is well-ordered, exactly when $T \in \text{WF}$. Appending the single larger exponent 1 changes none of these equivalences. Coanalytic membership follows from the descending-sequence characterization applied to the reverse exponent order.

For every valid code, the lower part is nonnegative because all its nonzero coefficients are positive, while $2X - z_T$ has greatest exponent 1 and coefficient 1. Hence $X \leq z_T < 2X$; the lower equality occurs when the lower support is empty.

Every countable ordinal ξ embeds into I in its reverse order, by the same rational-insertion principle. Taking that binary support gives total reverse support type $1 + \xi$. These types are unbounded below ω_1 . $\square$

Corollary 22.2. *There is no coefficient-observable standard Borel parametrization onto $\mathcal{C}_{\text{valid}}$, even with repeated names. More generally, no such parametrization can cover a family whose support types are unbounded in ω_1 .*

Proof. The general statement is the contrapositive of Theorem 21.2. Apply it to Theorem 22.1. $\square$

This localization removes several possible explanations for the obstruction. It is not due to large coefficients, an unbounded range of leading degrees, or a need to represent the proper class of all surreals. The leading monomial is fixed, the other coefficients are bits, the remaining exponents lie in one fixed rational band, and all actual values lie in one bounded omnific interval.

Remark 22.3 (Relation to ProveIt’s existing lower bound). The use of well-founded trees, and the fact that general support validation is coanalytic-hard, are already recorded in the omnific-notations report [21]. The present localization is included as an explicit corollary with a fixed coefficient cylinder. The more substantial extensions are the scattered/nonscattered classification and its observable-field and uniform-family formulations. None of these descriptions establishes global priority for a result outside the inspected material.

[write] Theorem 22.1 and the collection’s `onot:neg:thm:pionesone` are the same phenomenon in two slices, as the remark above says: there, lightface completeness for recursive trees, with decidable dyadic supports in $[1, 2)$ and coefficient one; here, boldface completeness with the fixed leading monomial X , bits on the band $(1/3, 2/3)$ and values in $[X, 2X)$. The unboundedness of the support types and corollary 22.2 are not in the omnific-notations report.

23 A real closed Borel subfield and its finite omnific integer part

The preceding obstructions do not make exact non-Archimedean algebra impossible. A particularly useful positive model separates finite support, bounded-denominator infinite support, and arbitrary Hahn support.

Let

$$\mathcal{P} = \bigcup_{n \geq 1} \mathbb{R}((t^{1/n})), \quad X = t^{-1}. \quad (23.1)$$

The union is a field: any two denominators have a common multiple. Newton–Puisseux says that the corresponding complex field $\bigcup_n \mathbb{C}((t^{1/n}))$ is algebraically closed [12, 13]. It is exactly $\mathcal{P}(i)$, and $\mathcal{P}$ is ordered by the leading coefficient. Hence $\mathcal{P}$ is real closed.

Define

$$\mathcal{A}_{\text{fin}} = \left\{ n + \sum_{q \in S} a_q X^q : n \in \mathbb{Z}, S \subseteq \mathbb{Q}_{>0} \text{ finite}, a_q \in \mathbb{R} \right\}. \quad (23.2)$$

Zero coefficients may be removed to obtain a unique finite normal form. The notation $\mathbb{Z} \oplus \bigoplus_{q>0} \mathbb{R}X^q$ describes an additive direct sum. It does not mean that $\mathcal{A}_{\text{fin}}$ is an $\mathbb{R}$ -algebra: its constant subring is $\mathbb{Z}$, not $\mathbb{R}$.

Lemma 23.1 (Exact principal-part floor). *The ring $\mathcal{A}_{\text{fin}}$ is an integer part of $\mathcal{P}$. More explicitly, write uniquely*

$$f = p + r + \varepsilon,$$

where p contains the positive powers of X , $r \in \mathbb{R}$ is the constant term, and ε has only negative powers of X . Then p is finite, ε is infinitesimal, and the unique $a \in \mathcal{A}_{\text{fin}}$ with $a \leq f < a + 1$ is

$$a = p + \begin{cases} \lfloor r \rfloor - 1, & r \in \mathbb{Z} \text{ and } \varepsilon < 0, \\ \lfloor r \rfloor, & \text{otherwise.} \end{cases} \quad (23.3)$$

Proof. For $f \in \mathbb{R}((t^{1/n}))$, its exponents are bounded below in the lattice $\frac{1}{n}\mathbb{Z}$. Only finitely many are negative in the t -convention, so the positive- X principal part p is finite. Every remaining nonconstant term has negative X -exponent; if the tail is nonzero, its greatest exponent is negative, so it is smaller in absolute value than every positive real.

When $r \notin \mathbb{Z}$, the fractional part $r - \lfloor r \rfloor$ lies strictly between 0 and 1, and adding an infinitesimal preserves both inequalities. When $r \in \mathbb{Z}$ and $\varepsilon \geq 0$, the remainder is in $[0, 1)$. When $r \in \mathbb{Z}$ and $\varepsilon < 0$, subtracting one from the proposed floor makes the remainder $1 + \varepsilon$, which also lies in $[0, 1)$. This proves the inequalities.

The ring is discretely ordered. A positive element with nonempty positive support exceeds every real, in particular 1; a positive constant is an ordinary positive integer. Thus 1 is its least positive element, and two distinct elements of $\mathcal{A}_{\text{fin}}$ cannot both satisfy the floor inequalities. $\square$

The boundary correction in (23.3) is essential. For example, $\lfloor -X^{-1} \rfloor_{\mathcal{A}_{\text{fin}}} = -1$, not 0. This correction is also explicitly recorded in Provel’s current omnific-floor description [20]; it is not a new correction discovered by this report.

[write] Lemma 23.1 is the collection’s `odg:thm:floor` (omnific-diophantine-geometry report) read in the Puiseux field. Under 03’s identification of X with ω (section 17), $\mathcal{P}$ is a subfield of $\mathbf{No}$; p , r and ε are the positive-growth part, the real constant and the infinitesimal tail of `odg:eq:floor`; and $\mathbf{Oz} \cap \mathcal{P} = \mathcal{A}_{\text{fin}}$, because the positive-growth

part of a Puiseux series is finite. Formula (23.3) is therefore the same formula, and its negative-infinitesimal correction is the case $r \in \mathbb{Z}$, $\varepsilon < 0$ there. `odg:thm:floor` is proved in Lean for every surreal (`Algebra/SurrealNumbers/Surreal/Foundations/OmnificFloor.lean: omnificFloor_spec, existsUnique_omnific_integerPart, omnificFloor_of_integer_negative_tail`; row “Full `odg:thm:floor`” of `docs/FORMALIZATION.md`). The Lean development does not mention $\mathcal{P}$ or $\mathcal{A}_{\text{fin}}$, so the lemma as stated here, with the finiteness of p and the integer-part property of $\mathcal{A}_{\text{fin}}$ in $\mathcal{P}$, is not formalized. Source 03 credits the Surreal README for the correction.

Theorem 23.2. *The nonnegative cone $\mathcal{A}_{\text{fin}, \geq 0}$ is an uncountable model of Robinson arithmetic plus open induction in the ordered semiring language.*

Proof. A discretely ordered ring gives the elementary semiring and predecessor axioms on its nonnegative cone. In particular, if $x > 0$ then $x - 1 \geq 0$. Uncountability follows already from the distinct elements aX , $a > 0$ real.

We prove open induction directly, using the familiar integer-part direction of Shepherdson’s theorem [6, 7]. Let $W \subseteq \mathcal{A}_{\text{fin}, \geq 0}$ be a nonempty set defined by a quantifier-free formula with parameters from $\mathcal{A}_{\text{fin}}$. Interpret the same formula in the real closed field $\mathcal{P}$. Its truth set is a finite union of intervals and points: the finitely many roots of the polynomials appearing in the formula partition the line into regions of constant sign.

Intersect these finitely many cells with $[0, \infty)$ and choose the first one meeting $\mathcal{A}_{\text{fin}}$. A point cell contributes its own element. For an interval cell with lower endpoint $b \in \mathcal{P}$, the integer-part property gives the least integer-part element at or above b , or the least one strictly above b when the endpoint is excluded. This element is nonnegative; if it were above the interval, that interval would contain no element of $\mathcal{A}_{\text{fin}}$ at all. For a cell beginning at 0, the same argument starts with 0 or 1. Thus W has a least element.

If a quantifier-free induction instance failed while its base and successor steps held, the nonempty quantifier-free set of counterexamples would have a least member m . It would satisfy $m > 0$, so $m - 1 \in \mathcal{A}_{\text{fin}, \geq 0}$ would satisfy the formula. The successor step would then force the formula at m , a contradiction. $\square$

[write] The proof of [theorem 23.2](#) is the Shepherdson route of the collection’s `dsn:cor:openinduction (definable-surreals-and-omnific-integers report: open induction for the integer part  $I_A$  of a real closed field  $K_A$ )` and of the paragraph citing Shepherdson’s theorem in the subsection “Sums of squares and induction” of the `omnific-diophantine-geometry report (open induction for the omnific integers of a divisible Hahn workspace)`. Here it is applied to $\mathcal{A}_{\text{fin}}$ inside the real closed field $\mathcal{P}$; uncountability and the semiring axioms are 03’s. `dsn:cor:openinduction` is pending in `docs/FORMALIZATION.md`. As 03 says, $\mathcal{A}_{\text{fin}, \geq 0}$ is a Shepherdson-type model, not a new model of arithmetic.

23.1 Why the Puiseux field itself has Borel codes

In the ambient coefficient space $\mathbb{R}^{\mathbb{Q}}$, the field (23.1) is exactly the union, over $n \geq 1$ and $k \in \mathbb{Z}$, of the closed sets

$$\{a : a_q = 0 \text{ whenever } q \notin \frac{1}{n}\mathbb{Z} \text{ or } q < k/n\}.$$

It is therefore F_σ and standard Borel. The convolution and unique graph arguments of Theorem 20.2 make its field operations Borel. Every such support has increasing order type at most ω .

This gives a useful three-way contrast. Finite principal parts have finite support; the real closed Puiseux field has a uniform support bound ω ; the full rational-exponent Hahn field has supports of unbounded countable type. Real closedness alone does not prevent a Borel coefficient presentation. Full arbitrary-support closure is what makes Corollary 20.4 apply.

24 A Polish finite-normal-form presentation

24.1 The topology and its Borel structure

Write an element of $\mathcal{A}_{\text{fin}, \geq 0}$ uniquely as $(S, n, (a_q)_{q \in S})$, where all $a_q \neq 0$. If $S \neq \emptyset$, positivity means $a_{\max S} > 0$ and the integer n is unrestricted. If $S = \emptyset$, we require $n \geq 0$.

For finite nonempty $S \subseteq \mathbb{Q}_{>0}$ and $n \in \mathbb{Z}$, let

$$U_{S,n} = \{a \in (\mathbb{R} \setminus \{0\})^S : a_{\max S} > 0\}.$$

For $S = \emptyset$, let $U_{\emptyset,n}$ be a singleton when $n \geq 0$. Put on $\mathcal{A}_{\text{fin}, \geq 0}$ the topological disjoint union

$$(\mathcal{A}_{\text{fin}, \geq 0}, \tau_{\text{FS}}) = \bigsqcup_{S,n} U_{S,n}. \quad (24.1)$$

Support and integer constant are locally constant in this topology.

Proposition 24.1. *The topology τ_{FS} is Polish. Its Borel sets are exactly the Borel sets obtained from the canonical finite-support coefficient coding.*

Proof. Each nonsingleton component is open in a finite-dimensional Euclidean space, and hence Polish. There are countably many components. A countable topological disjoint union of Polish spaces is Polish: choose bounded complete metrics on the components and put distance 2 between distinct components. Taking the union of countable dense sets gives separability.

Every coefficient map is continuous for (24.1). Conversely, an exact support condition consists of nonzero tests at finitely many coordinates and zero tests at countably many others, so is Borel in the coefficient product. On each such Borel piece the coefficient coordinates give exactly the usual finite-dimensional Borel structure. Since there are countably many pieces, the two Borel structures agree. $\square$

This is not the order topology. The order topology on a discretely ordered uncountable arithmetic domain is discrete and therefore not separable. Nor is τ_{FS} the ambient coefficient subspace topology, where convergence need not make the exact support eventually constant. The distinction is responsible for the cancellation discontinuities below.

24.2 A general support-normalization lemma

Lemma 24.2. *Let $U \subseteq \mathbb{R}^d$ be nonempty and open, and let $(p_j)_{j \in J}$ be a finite family of real polynomials. Define a map whose output is the finite list of nonzero coordinates $p_j(u)$, with the target topologized as a disjoint union by exact nonzero support. Let $E = \{j : p_j \not\equiv 0\}$. Its discontinuity set is exactly*

$$\mathcal{Z} = \bigcup_{j \in E} \{u \in U : p_j(u) = 0\}.$$

This set is closed and nowhere dense in U . The map is continuous on countably many closed subsets covering U .

Proof. Outside $\mathcal{Z}$, all nonidentically-zero coordinates stay nonzero on a sufficiently small neighborhood, while the others are always zero. The output support is therefore constant and the coordinate formulas are continuous.

At $u \in \mathcal{Z}$, choose $j \in E$ with $p_j(u) = 0$. A nonzero real polynomial cannot vanish on a nonempty open set. Thus arbitrarily near u there are points where this coordinate is nonzero. Their output support differs from the support at u , so their images lie outside the clopen target component containing the image of u . Continuity fails. This also proves that the finite union of zero sets is nowhere dense; closedness is immediate.

For $Z \subseteq E$ and integer $k \geq 1$, use the closed set

$$F_{Z,k} = \{u \in U : p_j(u) = 0 \ (j \in Z), \quad |p_j(u)| \geq 1/k \ (j \in E \setminus Z)\}. \quad (24.2)$$

These sets cover U : at any point choose the actual zero set and then a large enough k for the finitely many remaining coordinates. On each piece, the output support is constant and the map is continuous. For a zero-dimensional component consisting of one point, the same conclusion holds with empty discontinuity set. $\square$

24.3 The exact arithmetic cancellation loci

Fix an input component $U_{S,n} \times U_{T,m}$. Extend the variable coefficient families by $a_q = 0$ for $q \notin S$ and $b_q = 0$ for $q \notin T$. For addition, the output coefficient at $r > 0$ is

$$A_r = a_r + b_r, \quad (24.3)$$

and its constant is $n + m$. For multiplication, the output constant is nm and the coefficient at $r > 0$ is

$$M_r = nb_r + ma_r + \sum_{p+q=r} a_p b_q. \quad (24.4)$$

Only finitely many exponents need to be considered: $S \cup T$ for addition and $S \cup T \cup (S + T)$ for multiplication.

For multiplication define

$$E_{S,T,n,m} = \{r \in S \cup T \cup (S + T) : M_r \not\equiv 0\},$$

where identity means polynomial identity in the independent input coefficient variables after fixing the integers n, m . This removes absent terms and zero-factor components correctly.

Theorem 24.3 (Exact cancellation theorem). *On the Polish presentation (24.1), addition and multiplication are Borel. Their discontinuity sets are given componentwise by*

$$D_+ \cap (U_{S,n} \times U_{T,m}) = \bigcup_{r \in S \cap T} \{a_r + b_r = 0\}, \quad (24.5)$$

$$D_\times \cap (U_{S,n} \times U_{T,m}) = \bigcup_{r \in E_{S,T,n,m}} \{M_r = 0\}. \quad (24.6)$$

Globally, both sets are closed and nowhere dense. Both operations are continuous on countably many closed pieces. In particular, both are continuous at every point of the common dense open set $\mathcal{A}_{\text{fin}, \geq 0}^2 \setminus (D_+ \cup D_\times)$.

Proof. On an input component, the actual normal form is obtained from (24.3) or (24.4) by deleting the zero coefficients. The integer output constant is fixed. Apply Lemma 24.2. For

addition, a coefficient occurring in only one input support is constrained to be nonzero, so the possible zero sets are precisely those in (24.5). For multiplication the expected polynomial support is exactly $E_{S,T,n,m}$.

Every input component is clopen, and these components partition the domain. A set whose intersection with each component is closed is globally closed; a set whose intersection with each nonempty open component is nowhere dense has empty interior and, when closed, is nowhere dense globally. This gives the assertions about D_+ and $D_\times$.

The pieces (24.2) are closed relative to a clopen component, hence closed in the full domain. Taking them over countably many components, zero patterns, and integers k gives countable closed covers for both operations. To see Borelness directly, the inverse image of an open set is a countable union of sets open relative to these closed pieces. Such sets are Borel (indeed F_σ in a metric space). $\square$

Example 24.4 (Addition: cancellation below the leading term). For real δ sufficiently near 0, set

$$x = X^2 + X, \quad y_\delta = X^2 + (-1 + \delta)X.$$

All these elements are positive and remain in fixed input components. But

$$x + y_\delta = 2X^2 + \delta X.$$

At $\delta = 0$ the output support loses exponent 1, so the outputs do not converge to $2X^2$ in τ_{FS} . Leading coefficients of positive inputs cannot cancel each other; the lower coefficient is what fails here.

Example 24.5 (Multiplication: a vanished middle coefficient). Let $x = X + 1$ and $y_\delta = (1 + \delta)X - 1$. Near $\delta = 0$ both inputs are positive, their supports and integer constants are fixed, and

$$xy_\delta = (1 + \delta)X^2 + \delta X - 1.$$

The middle term disappears at $\delta = 0$. Thus $(X + 1, X - 1) \in D_\times$. Multiplication by 0, on the other hand, is constant and continuous; this explains the exclusion of identically zero coefficient polynomials.

Remark 24.6 (A measurability qualification). Countable closed-piece continuity is the statement proved here. We do not identify it without further hypotheses with the Baire-one class of maps into this disconnected target. Nor does the dense-open conclusion say that an algebraic cancellation can be ignored by exact computation: deleting a single zero coefficient changes the canonical component.

25 Glazer's obstruction survives every compatible Polish recoding

We use precisely the following published result.

Theorem 25.1 (Glazer, Corollary 1 of [1]). *There is no model of Robinson arithmetic plus !Open on an uncountable Polish space in which addition is continuous and multiplication is Borel.*

Theorem 25.2 (No removal of the addition obstruction). *No Polish topology on the set $\mathcal{A}_{\text{fin}, \geq 0}$ with the same Borel sets as τ_{FS} makes addition continuous. More generally, there is no Polish presentation Borel-isomorphic to the canonical one in which transported addition is continuous.*

Proof. By Theorem 23.2, the algebraic structure is an uncountable model of the required arithmetic. By Theorem 24.3, multiplication is Borel in the canonical Borel structure. Changing the Polish topology without changing its Borel sets leaves multiplication Borel on the product; finite products of standard Borel spaces have the same product Borel structure. Continuous addition would contradict Theorem 25.1. Transporting the operations through a Borel isomorphism changes none of these hypotheses. $\square$

Corollary 25.3. *The distinction between global continuity and continuity off a closed nowhere dense set is genuine for this explicit model: the latter is attained for both operations, while the former is impossible for addition under any Borel-compatible Polish presentation.*

This is a sharp distinction between an empty discontinuity set and an explicit closed nowhere dense one. It is not a proof that our particular zero loci have minimal cardinality, minimal dimension, or a minimal descriptive rank among all recodings.

Glazer's Theorem 2 also excludes continuous multiplication for uncountable Borel models of the stronger theory $\mathbf{Q} + I\Delta_0 + \text{Exp}$ [1]. We have proved only open induction for $\mathcal{A}_{\text{fin}, \geq 0}$. The multiplication obstruction from that stronger result is therefore *not* transferred to every compatible topology on $\mathcal{A}_{\text{fin}, \geq 0}$. Multiplication fails to be continuous in our displayed topology by Theorem 24.3; whether a different compatible Polish topology can make multiplication continuous remains a separate question here.

26 Where the two parts meet [merge]

The real model of Part I is one stratum of the cone $\mathcal{A}_{\text{fin}, \geq 0}$ of section 24. Read ω as X and let

$$\iota : M_{\mathbb{R}} \longrightarrow \mathcal{A}_{\text{fin}}, \quad \iota(r, n) = rX + n,$$

which is 05's map $\iota_{\mathbb{R}}$ of theorem 9.1 in 03's notation. Put

$$C_{\mathbb{R}} = \{a \in \mathcal{A}_{\text{fin}, \geq 0} : \text{supp } a \subseteq \{1\}\} = \bigcup_{n \in \mathbb{Z}} U_{\{1\}, n} \cup \bigcup_{n \geq 0} U_{\emptyset, n}.$$

Proposition 26.1 (The real model as a stratum [merge]). (i) ι is an injective ordered additive map onto $C_{\mathbb{R}}$, fixing 0 and 1. The set $C_{\mathbb{R}}$ is clopen in $(\mathcal{A}_{\text{fin}, \geq 0}, \tau_{\text{FS}})$ and closed under addition, but not under multiplication.

- (ii) On $\{(r, n) \in M_{\mathbb{R}} : r > 0\}$, ι is a homeomorphism onto $\bigcup_n U_{\{1\}, n}$, for the topology of $M_{\mathbb{R}}$.
- (iii) At the standard numerals the two topologies differ: every ordinary integer $n \geq 0$ is isolated in τ_{FS} , but $(0, n)$ is not isolated in $M_{\mathbb{R}}$. Thus τ_{FS} induces on $C_{\mathbb{R}}$ a topology strictly finer than the one transported from $M_{\mathbb{R}}$.
- (iv) Addition is continuous on $(C_{\mathbb{R}}, \tau_{\text{FS}})$. Consequently $(C_{\mathbb{R}}, \tau_{\text{FS}})$ is another uncountable, locally compact Polish model of $\text{Th}(\mathbb{N}; 0, 1, +, <)$ with continuous addition; it is not perfect, its isolated points being exactly the standard numerals.

Proof. (i) An element of $\mathcal{A}_{\text{fin}}$ with support in $\{1\}$ is $n + aX$; by (24.1) it lies in $\mathcal{A}_{\text{fin}, \geq 0}$ when $a > 0$, or $a = 0$ and $n \geq 0$. These are exactly the values of ι . Addition is coordinatewise and the order is decided by the leading coefficient, as in (3.1). The set $C_{\mathbb{R}}$ is a union of components of the topological sum (24.1), hence open, and its complement is the union of the other components, hence also open. Finally $X \cdot X = X^2 \notin C_{\mathbb{R}}$.

(ii) Both $U_{\{1\},n}$ and the part $r > 0$ of the slice of integer coordinate n in $M_{\mathbb{R}}$ are the open ray $(0, \infty)$ with its Euclidean topology, and ι is the identity on the coefficient.

(iii) $U_{\emptyset,n}$ is a singleton component of (24.1). In $M_{\mathbb{R}}$, $(1/j, n) \rightarrow (0, n)$ by (5.1).

(iv) On a component $U_{S,n} \times U_{T,m}$ with $S, T \subseteq \{1\}$, (24.5) is a union over $r \in S \cap T \subseteq \{1\}$ of the sets $\{a_1 + b_1 = 0\}$, which are empty because both leading coefficients are positive. So $C_{\mathbb{R}}^2$ does not meet D_+ , and addition maps $C_{\mathbb{R}}^2$ into $C_{\mathbb{R}}$. A clopen subspace of a Polish space is Polish, the components are points and open rays, and the algebraic structure is that of $M_{\mathbb{R}}$, so [corollary 4.11](#) applies. $\square$

The boundary. Glazer’s Corollary 1 concerns a model of $Q+I\text{Open}$ with Borel multiplication. The ring cone $\mathcal{A}_{\text{fin}, \geq 0}$ is such a model ([theorems 23.2](#) and [24.3](#)), so no Borel-compatible Polish topology on $\mathcal{A}_{\text{fin}, \geq 0}$ makes its addition continuous ([theorem 25.2](#)). The stratum $C_{\mathbb{R}}$ is not such a model: as an additive monoid it is $M_{\mathbb{R}}$, which carries no unital semiring multiplication at all ([corollary 10.5](#)), and the multiplication of $\mathcal{A}_{\text{fin}, \geq 0}$ leaves it. On $C_{\mathbb{R}}$ addition is continuous for τ_{FS} itself; the discontinuities of addition on $\mathcal{A}_{\text{fin}, \geq 0}$ lie on the cancellation loci (24.5), which need two inputs whose supports share an exponent below their leading ones, and so lie outside $C_{\mathbb{R}}^2$. Thus [Part I](#) and [theorem 25.2](#) delimit, inside one ring, where Corollary 1 applies: the additive stratum admits continuous addition for both topologies, while the ring, with its Borel multiplication and open induction, admits it for none.

The Baire model. The image $\iota_B(M_B)$ of [theorem 9.1](#) lies in 03’s rational Hahn ring $\mathcal{H}_{\mathbb{Q}}$ of (17.1): its supports are contained in $\{1, \frac{1}{2}, \frac{1}{3}, \dots\} \cup \{0\}$ and have reverse order type at most $\omega + 1$. Its elements with infinitely many nonzero coefficients lie neither in $\mathcal{A}_{\text{fin}}$ nor in the Puiseux field $\mathcal{P}$, whose denominators are bounded. Each coefficient of $\iota_B(a, n)$ is a_j , n or 0, a continuous function on M_B , so the composite of 05’s homeomorphism $h : \mathbb{N}^{\mathbb{N}} \rightarrow M_B$ ([theorem 6.1](#)) with ι_B is a coefficient-observable standard Borel parametrization of $\iota_B(M_B)$ in the sense of [definition 17.1](#). This is consistent with [theorem 21.2](#), whose bound can be taken to be $\omega + 2$ here, while the full field $\mathbb{R}((t^{\mathbb{Q}}))$ lies on the non-Borel side of [theorem 20.2](#). The Baire model is one of the restricted support families that [section 27](#) calls for.

A correction to the placement record [write]. The placement commit 3d2177df4 and the intake dossier said that 03’s topology τ_{FS} restricts on the stratum to the topology of 04 and 05. Parts (ii) and (iii) of [proposition 26.1](#) give the exact statement: the two topologies agree on the elements with positive real coefficient and differ at the standard numerals, which τ_{FS} isolates. Both make addition continuous on the stratum, so the conclusion drawn there stands.

27 Consequences for exact notation systems and formalization

27.1 A practical separation of three design goals

The results separate three tasks that can otherwise be conflated. *Arithmetic* asks whether operations can be performed on valid inputs. *Validation* asks whether an arbitrary code is a valid well-based normal form. *Topological presentation* asks whether the operations respect a complete separable topology on the names or values. A positive answer to one is not a positive answer to the others.

In the finite-support model, every code carries an explicit finite list, and arithmetic is polynomial coefficient arithmetic followed by exact zero tests. The canonical domain is Polish, the operations are Borel, and their failure of continuity is exactly the support-changing cancellation calculated above. No general theorem of computability over arbitrary real coefficients is claimed: a real coefficient supplied by an unrestricted approximation oracle need not have decidable equality with zero.

For full Hahn fields on scattered exponent groups, support validation is a Borel predicate. This is not the same as decidability by a finite algorithm. The Π_3^0 examples already involve quantification over infinitely many rows. For dense rational exponents the obstruction is stronger: no standard Borel system of fully observable names covers every valid series.

A useful exact system must therefore limit its admissible support families, relax coefficient observability, or accept a nonstandard measurable domain. The Puiseux field demonstrates that a severe-looking support restriction may still permit real closedness. Theorem 21.2 says every observable Borel system necessarily has some uniform countable support bound, even when that bound is not built into the notation explicitly.

27.2 No paradoxes are needed

All main results are in ZFC and concern sets. For countable Γ , the ambient coefficient space $\mathbb{R}^\Gamma$ is a set of cardinality continuum. Its valid-code subset is a set whether or not it is Borel. The rational Hahn ring $\mathcal{H}_\mathbb{Q}$ and its finite subring $\mathcal{A}_{\text{fin}}$ are also sets. The construction never forms a set of all surreal numbers or quantifies over all class maps as a set.

The optional identification with Conway normal forms can be read as a set-indexed definable embedding into the proper class **No**, or carried out inside a sufficiently large set-sized normal-form universe. Standard Replacement handles the image of a fixed set of codes. No global choice, large cardinal, or class-sized Polish topology is used.

27.3 A formalization plan with an explicit dependency boundary

A Lean development can be organized into independent interfaces rather than starting from the entire proper-class surreal hierarchy.

- Layer 1. Finite algebra and normalization.** Use finitely supported rational-exponent coefficient maps with an integer constant. Establish convolution, leading-exponent sign, discreteness, and the coefficient identities (24.3)–(24.4). The exact cancellation examples are small executable tests at this layer.
- Layer 2. Series and arithmetic.** Use a set-sized Hahn or Puiseux field, prove the principal-part floor with its negative-infinitesimal case, and derive quantifier-free induction via the one-variable sign-cell argument. Newton–Puiseux is a named external dependency until its real-closedness interface is formally supplied.
- Layer 3. Topological support components.** Formalize the countable disjoint union, the polynomial normalization lemma, and the closed-piece covers. None of this requires descriptive ranks or proper-class surreals.
- Layer 4. Descriptive-set-theoretic structure.** Formalize the fixed Kleene–Brouwer embedding, Hausdorff-sum support criterion, analytic boundedness, and observable-field equivalence. The classical Hausdorff characterization and Glazer theorem must be proved or explicitly listed as hypotheses; importing them informally is not a checked proof.

This is a proposed architecture, not a claim that these layers have been implemented or that all required library APIs currently exist. The provided Python artifact checks finite rational instances, not Lean terms or infinitary assertions.

28 Further research questions and concrete next targets

The following questions are proposed here. Unless explicitly attributed, “open” means unresolved in this report, not certified as absent from the entire literature.

Question 28.1 (Continuous multiplication on the finite model). Does $\mathcal{A}_{\text{fin}, \geq 0}$ admit a Polish topology with the canonical Borel sets in which multiplication is continuous?

Theorem 25.2 settles the analogous addition question negatively. Glazer’s stronger multiplication theorem cannot be invoked without establishing its extra arithmetic hypotheses. A positive solution would isolate a concrete asymmetry between the two operations; a negative solution should identify an obstruction valid already for this particular open-induction model.

Question 28.2 (How small can the discontinuity set be?). Among all Borel-compatible Polish presentations of $\mathcal{A}_{\text{fin}, \geq 0}$, what restrictions hold on the discontinuity set of addition? Can it be countable? Can it be contained in a prescribed small-dimensional or otherwise canonical set?

The present model gives a nonempty closed nowhere dense set. Glazer excludes only its complete disappearance, not every possible reduction in complexity. This is a natural refinement of the topological theorem that can first be studied on explicit finite-support components.

Question 28.3 (Exact complexity beyond finite lexicographic rank). For a fixed countable scattered order L , compute the least Borel level of $\text{WO}(L)$ directly from an economical Hausdorff decomposition. Which Wadge degrees occur among these support predicates?

The qualitative dichotomy is settled by Theorem 18.4, while Theorem 19.1 shows that simply increasing finite order rank need not increase Borel level. A satisfactory answer should distinguish ordinal support capacity from descriptive complexity rather than treating them as interchangeable ranks.

Question 28.4 (Borel, algebraically rich subfields of the rational Hahn field). Classify Borel subfields $F \subseteq \mathbb{R}((t^{\mathbb{Q}}))$ containing all rational monomials and stable under specified truncations or algebraic-root operations. Which countable support bounds can such fields attain?

The Puiseux field is one real closed example with supports of type at most ω . The full Hahn field is excluded. An interesting target is a well-described hierarchy of truncation-closed real closed Borel subfields between them, together with provably sharp support bounds at each stage. No claim is made that every ordinal cutoff is preserved by inversion.

Question 28.5 (How much coefficient observability is necessary?). If only a selected family of coefficient combinations or truncations is required to be Borel, which versions of Theorem 20.2 survive?

The current proof uses all countably many coefficient maps to obtain one Borel map into the full product. This pinpoints a possible escape route and suggests investigating minimal observation families that still recover the binary support cylinder in Theorem 22.1.

Question 28.6 (Certified support languages). Construct useful exact languages whose valid supports carry explicit well-foundedness certificates, and determine their closure under arithmetic, root extraction, and named truncations.

The repository already contains effective restricted omnific calculi [21]. The present results suggest comparing them by their Borel parameter spaces and support bounds as well as by decidable equality. A certificate verifier for a fixed language is not a total validator for all raw rational-exponent supports; Theorem 22.1 forbids conflating those objectives.

Question 28.7 (A minimal arithmetic hypothesis for the presentation barrier). Which concrete consequences of open induction suffice to rule out continuous addition in the finite principal-part model or related integer parts?

A proof specialized to finite Puiseux principal parts might replace the full open-induction interface with explicitly stated root-approximation or division properties. This would clarify the boundary between algebraic normal-form obstructions and arithmetic induction.

Question 28.8 (Formalized transfer from support theory to arithmetic). Can the observable Hahn-field criterion and the finite-support cancellation theorem be formalized as reusable results, with the classical Hausdorff and Glazer inputs exposed as exact dependency interfaces?

The suggested first milestone is the support-sum criterion (18.1) together with the normalization Lemma 24.2. Both are modular and elementary enough to be reviewed independently before the larger descriptive-set-theoretic machinery is introduced.

[merge] Question 28.1 is the multiplicative counterpart, on $\mathcal{A}_{\text{fin}, \geq 0}$, of question 14.4. The question on the size of the discontinuity set of addition concerns all of $\mathcal{A}_{\text{fin}, \geq 0}$; on the clopen stratum $C_{\mathbb{R}}$ of proposition 26.1 that set is empty.

29 Conclusions of Part II

The intersection with Glazer's research produces a precise hierarchy rather than a blanket impossibility theorem. Full Hahn support is Borel exactly on scattered countable exponent orders. Dense alphabets carry a coanalytic-complete validation problem, and no standard Borel family with observable coefficients can exhaust supports of unbounded countable type. This obstruction persists inside one fixed interval of omnific integers.

At the same time, a real closed Puiseux subfield and its finite principal-part integer part have explicit Borel presentations. The latter even admits a Polish topology where both arithmetic operations are continuous off closed nowhere dense sets, with exact polynomial equations describing every failure. Glazer's theorem makes the remaining addition discontinuity unavoidable under all compatible Polish recordings. The positive construction and the negative result constrain one another: Borel arithmetic is attainable, globally continuous addition is not, and arbitrary well-based support introduces a separate obstruction before topological continuity can even be asked in the canonical standard-Borel setting.

Part III

Continuous Presburger arithmetic on Polish spaces

[write] This part merges two manuscripts of batch 86 that carry the same title, *Continuous Presburger Arithmetic on Polish Spaces*, but are independent texts: source 07 (subtitle *An affirmative construction for Glazer’s question, Baire-space models, and omnific realizations*, dated October 4, 2026, pinned to 0f95145cb) and source 06 (subtitle *Lexicographic models, an isolated-zero obstruction, and a boundary between additive and semiring arithmetic*, dated October 3, 2026, pinned to fa2f3e419). They share at most 0.74% of their word 8-grams, the longest shared runs being Glazer’s quoted question and bibliography entries, and their theorem sets differ.

Base and reading conventions. Source 07 is the base: its realization theorem ([theorem 34.1](#)) needs only a divisible Polish abelian group with a Borel positive cone and covers the coordinate families of source 06. Unmarked text in this part is source 07’s. Text from source 06 is marked [06]: in the titles of its sections, subsections and statements, and in the sentence that introduces an inserted passage. Text written for this report is marked [write] or [merge] as in [section 1.2](#). Section and subsection headings of source 06 were moved one level down where its sections are printed inside source 07’s.

What is printed again, and what is new. Both sources prove Part I’s main results once more; these statements are printed with their proofs, because the proofs differ in detail, and each says in its title which statement of Part I it repeats:

- the real model ([theorem 32.1](#) of 07 and [theorem 32.2](#) of 06 are [theorem 3.1\(1\)](#));
- the Baire model and its effective addition ([theorems 35.6](#) and [35.7](#) of 06 are [theorems 6.1](#), [6.2](#) and [6.4](#));
- Δ_2^0 definability ([theorem 37.7](#) of 06 and part (iii) of [theorem 34.1](#) extend [theorem 7.1](#));
- the order obstruction ([proposition 37.1](#) is [theorem 7.4](#)), compactness ([proposition 37.3](#) is [corollary 12.2](#)), and the semiring obstruction ([proposition 37.5](#) and [theorem 37.10](#) are [theorem 10.4](#), the latter without a prescribed unit);
- the omnific realizations ([theorem 40.1](#) extends [theorem 9.1](#) from ω to every countably infinite ordinal);
- the classical elimination theorem ([sections G](#) and [I](#), compare [theorem 4.9](#) and [corollary 4.11](#)).

The results new to the collection are 07’s Borel-cone theorem ([theorem 33.1](#)) and cone closure rank, the general realization theorem and the Borelness of definable sets in every continuous presentation ([theorem 34.1](#) and [proposition 34.3](#)), the ordinal family on Baire space ([theorems 35.1](#) and [35.4](#); also 06’s [theorem 35.9](#)), the count of exactly $2^{\aleph_0}$ models on Baire space (07’s [theorem 35.5](#), and 06’s [theorem 35.8](#) with topology, order, 0 and 1 fixed), the oscillation ranks ([theorem 36.1](#) and [corollary 36.2](#)), the discontinuity of min and max ([corollary 37.2](#)), a second topology with continuous predecessor ([proposition 37.4](#)), and source 06’s Polish semiring cone of $\mathbb{Z} + t\mathbb{R}[t]$ with continuous addition and multiplication that

fails open induction ([theorems 38.1](#) and [38.2](#)) and its isolated-zero theorem ([theorem 39.1](#)). [Section 42](#) locates these results with respect to Parts I and II.

A consequence for Part I’s criterion [merge]. Let D be a divisible ordered abelian group with a Polish group topology. By [theorem 5.2](#), M_D is Polish if and only if $D_{>0}$ is G_δ ; by [theorem 33.1](#), a Borel $D_{>0}$ is already Δ_2^0 . Hence

$$M_D \text{ is Polish} \iff D_{>0} \text{ is Borel} \iff D_{>0} \text{ is } \Delta_2^0.$$

The proof of [theorem 34.1](#) also shows that division by a standard integer is then automatically continuous (open mapping), a hypothesis that source 04’s [theorem 5.4](#) assumed.

Labels and notation. Labels of this part are `pma:cp:<label>` for both sources, except the six labels of source 06 that coincide with labels of source 07 (`eq:realmodel`, `sec:baire`, `sec:real`, `thm:baire`, `thm:continuum`, `thm:real`), which take the sub-prefix `pma:cp:s06:`. Source 06’s family M_s ($s \in 2^{\mathbb{N}}$) is renamed M'_s , with G'_s , H'_s , because source 07’s M_s is a different family; [section 1.3](#) lists this and the other conventions of Parts III–V. In particular t is a positive infinite element in this part (source 06’s $\mathbb{Z} + t\mathbb{R}[t]$), whereas in [Part V](#) t is a positive infinitesimal.

30 The question and the research intersection

30.1 An exact question with a concrete answer

Glazer’s *A Topological Tennenbaum Theorem* ends with the question:

Does some uncountable Polish space support a model of Presburger arithmetic with continuous addition?

This is Question 2 of the November 2023 version [1]. Here “Polish” means separable and completely metrizable. Presburger arithmetic is the complete first-order theory of addition on the natural numbers, with multiplication of two variables absent from the language.

The answer supplied by this report is affirmative. Its simplest witness is

$$M_{\mathbb{R}} = ((0, \infty) \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}), \quad (30.1)$$

with coordinatewise addition, $0 = (0, 0)$, $1 = (0, 1)$, and lexicographic arithmetic order that compares the real coordinate first. The topology is the subspace topology from $\mathbb{R} \times \mathbb{Z}_{\text{discrete}}$. The distinction between this topology and the arithmetic order is essential.

There is an even stronger realization on $\mathbb{N}^{\mathbb{N}}$, the usual Baire space. It is obtained by replacing $\mathbb{R}$ with a countable product of discrete copies of $\mathbb{Q}$, then taking the same lexicographic integer extension. No forcing, nonprincipal ultrafilter, or nonconstructive choice of an arithmetic operation is needed for either coordinate model.

Glazer also proves that certain stronger Borel arithmetics on uncountable Polish spaces cannot have either addition or multiplication continuous. His Corollary 1 already excludes continuous addition and Borel multiplication for $Q + I\text{Open}$ [1]. Our result concerns the additive theory. [Section 37](#) explains the precise boundary, including a stronger algebraic obstruction for our examples.

30.2 Why this is a substantive overlap with ProveIt

The repository snapshot examined is commit

`0f95145cb65d5dc279e2053e8707a71bb490bb59`.

Its timestamp is October 3, 2026, 18:50:05 in the America/Los_Angeles time zone. Two parts of that snapshot are especially relevant:

1. `Logic/PresburgerArithmetic` contains an executable Cooper quantifier-elimination development for formulas over ordinary integers. The main semantic correctness theorem is implemented in Lean; Rocq independently checks its normalized one-variable core [32].
2. The surreal research report on discrete initial subgroups and omnific normalization studies split ordered groups of the form $D \overrightarrow{\times} \mathbb{Z}$, including their relation to omnific integers [33].

The first supplies an existing computational and formalization framework; the second supplies the relevant algebraic shape. The new work here is to give that shape explicit Polish topologies, prove its model-theoretic and descriptive properties, and connect it to the exact published question.

The main positive theorems do not depend on accepting an unrefereed repository theorem. We prove the required ordered-group facts directly and use the classical characterization of Presburger models. The surreal connection is also given by an explicit normal-form map.

30.3 What is proved, and what is a novelty claim

The existence assertion is a theorem with a short proof. The assertion that it is the first published resolution would be a historical claim requiring different evidence. We make only the former claim. A targeted literature search found the question in Glazer's paper and found no later resolution; that is not an exhaustive priority search.

The same distinction applies to the general cone theorem, the family of Baire-space models, and the rank computations. They are proved here and are useful extensions of the construction. Their precise place in the literature remains suitable for specialist review. In particular, the underlying lexicographic construction of a $\mathbb{Z}$ -group is classical and is not presented as an invention.

30.4 The question, the overlap, and the contribution [06]

30.4.1 A precise research target [06]

Glazer's work on topological models of arithmetic gives a particularly direct intersection with ProveIt. His 2023 paper ends with the question [1, Question 2]:

Does some uncountable Polish space support a model of Presburger arithmetic with continuous addition?

Here *Presburger arithmetic* means the complete first-order theory of addition on the natural numbers, rather than full Peano arithmetic. We answer this formulation affirmatively, with

explicit models and proofs. The use of a topology different from the arithmetic order topology is essential.

The context is a sharp obstruction for stronger arithmetic. Glazer proves that neither arithmetic operation can be continuous in an uncountable Polish Borel model of $\mathbb{Q} + \text{I}\Delta_0 + \text{Exp}$; his additive obstruction already holds for $\mathbb{Q} + \text{I}\text{Open}$ when multiplication is Borel [1, Theorem 2 and Corollary 1]. The constructions below show why the induction and language hypotheses matter.

30.4.2 What is already in ProveIt [06]

We inspected the repository at commit

fa2f3e419e592a1e90c065b79d559865ec45d436.

The following connections are concrete, not merely thematic.

1. `Logic/PresburgerArithmetic` contains a Lean development of Cooper elimination and an independent Coq development of the normalized one-variable step [34]. Its current semantics use ordinary integers. The abstract $\mathbb{Z}$ -group semantics needed for the nonstandard models below are an additional formalization task.
2. The report *Discrete Initial Groups and Omnific Normalization* studies split groups $D \xrightarrow{\sim} \mathbb{Z}$ and their connection to omnific integers [35]. The algebraic form of our examples thus already has a close repository counterpart.
3. The report *Omnific Integers and Omnific–Diophantine Geometry* explicitly studies $\mathbb{Z} + t\mathbb{R}[t]$ and its realization $\mathbb{Z} + \omega\mathbb{R}[\omega]$ [36]. We equip its nonnegative cone with a useful Polish topology and prove continuity of both operations.

The lexicographic $\mathbb{Z}$ -group construction and its model theory are classical [4]; rational polynomial Presburger examples also appear in established lecture notes [24]. The contribution here is the explicit topological realization, its consequences for the stated question, and the accompanying obstruction and abundance results. We use the repository reports as documented points of contact, rather than as unverified premises for our main proofs.

30.4.3 A map of the results [06]

Model	Topology	Continuous operations	Arithmetic boundary
$(\mathbb{R} \xrightarrow{\sim} \mathbb{Z})_{\geq 0}$	Perfect, locally compact Polish	Addition; fixed standard division	Presburger; no unital semiring expansion
$(\mathbb{Q}^{\mathbb{N}} \xrightarrow{\sim} \mathbb{Z})_{\geq 0}$	Baire space	Addition, with an effective stream presentation	Presburger; no unital semiring expansion
$(\mathbb{Z} + t\mathbb{R}[t])_{\geq 0}$	Locally compact Polish; isolated standard points	Addition and multiplication	PA^- and Presburger; fails IOpen

The general obstruction in Section 39 says that the last row cannot be made perfect while retaining Polishness and continuity of both operations. The abundance theorem in Section 35.6 gives the exact possible cardinality of the collection of isomorphism types in the Baire-space setting: $2^{\aleph_0}$.

Scope of the literature check. The cited version of Glazer’s paper is arXiv:2311.13699v1. Searches of the paper, its available version record, related primary papers, and the selected repository materials did not locate a later answer or this topological construction. This is a bounded bibliographic finding, not a proof of universal priority. The mathematical assertion that our model answers the displayed question is independent of that priority issue.

31 Conventions and classical inputs

31.1 Language, order, and induction

We use $\mathbb{N} = \{0, 1, 2, \dots\}$. A model of $\text{Th}(\mathbb{N}; 0, 1, +, <)$ is a model of the *complete first-order theory* of $(\mathbb{N}; 0, 1, +, <)$, including every instance of first-order induction with parameters. In this theory,

$$x \leq y \iff \exists z (x + z = y), \quad x < y \iff \exists z (x + z + 1 = y). \quad (31.1)$$

Thus naming $<$ is a definitional convenience. A topological presentation in this report requires $+$ to be jointly continuous. It does not require the definable relation $<$ to be open or closed.

If G is an ordered abelian group, write $G_{\geq 0}$ and $G_{>0}$ for its nonnegative and positive cones. A distinguished $e \in G$ is a *least positive element* if $0 < e \leq g$ whenever $g > 0$. A $\mathbb{Z}$ -group is an ordered abelian group with a least positive element e such that $G/\mathbb{Z}e$ is divisible.

We use the following classical facts about Presburger arithmetic; a convenient precise reference is Jeřábek’s preliminary section [4].

1. A group is a model of $\text{Th}(\mathbb{Z}; 0, 1, +, -, <)$ exactly when it is a $\mathbb{Z}$ -group. Its nonnegative cone is then a model of $\text{Th}(\mathbb{N}; 0, 1, +, <)$.
2. In the language enlarged by fixed-modulus congruence predicates, formulas admit quantifier elimination: they reduce to finite Boolean combinations of integer-linear comparisons and congruences.

These inputs address *all* formulas, not merely the displayed division axioms. That is why our construction satisfies full first-order induction. We do not confuse it with second-order induction over every external subset.

31.2 Topological and descriptive notation

An F_σ set is a countable union of closed sets; a G_δ set is a countable intersection of open sets. A Δ_2^0 set is both. In metrizable spaces, open and closed sets are Δ_2^0 , finite Boolean combinations of Δ_2^0 sets are Δ_2^0 , and continuous inverse images preserve this class.

We use three standard descriptive-set-theoretic inputs: a subspace of a Polish space is Polish when it is G_δ ; a Borel injective map between standard Borel spaces has Borel image and inverse onto its image; and a nonempty zero-dimensional Polish space in which every compact subset has empty interior is homeomorphic to $\mathbb{N}^\mathbb{N}$. References and proofs of these standard tools can be found in [9, 8].

All metamathematics is carried out in ordinary ZFC. Every model constructed below is a set. Countable ordinals are actual ordinals of this metatheory, not nonstandard ordinal codes inside an arithmetic model.

31.3 Algebraic and topological preliminaries [06]

31.3.1 First-order arithmetic and induction [06]

Let

$$\text{Pres} = \text{Th}(\mathbb{N}, 0, 1, +, <).$$

Adding $0, 1, <$ to the addition-only language is harmless here because they are definable. A model of Pres satisfies every first-order induction instance in this additive language, including instances with parameters. It need not satisfy induction for formulas involving a subsequently added multiplication symbol.

We use PA^- in its standard algebraic sense: the theory of the nonnegative cones of discretely ordered commutative rings, with their inherited $0, 1, +, \cdot, <$. This convention agrees with that used in the topological-arithmetic literature [2]. We will describe all concrete models directly, so the results do not depend on choosing a particular printed list of PA^- axioms.

A discretely ordered abelian group is a linearly ordered abelian group with a least positive element, denoted by 1. The subgroup generated by 1 is a convex copy of $\mathbb{Z}$: the only elements between consecutive standard integers are the endpoints. A $\mathbb{Z}$ -group is such a group for which $G/\mathbb{Z}$ is divisible.

Lemma 31.1 (Classical Presburger characterization [06]; = [theorem 4.9](#) and [corollary 4.11](#)). *If G is a $\mathbb{Z}$ -group, then*

$$(G, 0, 1, +, -, <) \equiv (\mathbb{Z}, 0, 1, +, -, <), \quad G_{\geq 0} \models \text{Pres}.$$

Every formula in the group language is equivalent to a Boolean combination of integer-linear inequalities and congruences modulo positive standard integers.

This characterization is recalled in [4, Section 2]. Appendix I supplies a proof of the elimination step and of the transfer to the nonnegative cone. In particular, our use of this lemma includes full first-order additive induction; it does not merely check a few semiring-like equations.

Lemma 31.2 (Split construction [06]; the setup of [theorem 3.1](#)). *Let D be a divisible ordered abelian group. Give*

$$G(D) = D \times \mathbb{Z}$$

coordinatewise addition and lexicographic order with the D coordinate dominant. Then $G(D)$ is a $\mathbb{Z}$ -group with $1 = (0, 1)$, and

$$M(D) = G(D)_{\geq 0} = (D_{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}) \quad (31.2)$$

models Pres.

Proof. If $(d, z) > 0$, either $d > 0$, in which case $(d, z) > (0, 1)$, or $d = 0$ and $z \geq 1$. Thus $(0, 1)$ is least positive. The subgroup $\{0\} \times \mathbb{Z}$ is convex and the quotient is D , which is divisible. Apply Lemma 31.1.

For additional explicitness, if $n \geq 1$ is a standard integer, write $z = nq + r$ in $\mathbb{Z}$, with $0 \leq r < n$. Then

$$(d, z) = n(d/n, q) + (0, r). \quad (31.3)$$

The quotient is unique. If $(d, z) \geq 0$, then $(d/n, q) \geq 0$ as well: either $d > 0$, or $d = 0$ and $z \geq 0$, so $q \geq 0$. $\square$

31.3.2 The topologies under discussion [06]

A space is *Polish* if it is separable and admits a compatible complete metric. Its displayed metric need not itself be complete. We use the product topology for each product of topological groups, and the subspace topology on a cone unless a different topology is explicitly specified.

We write Δ_2^0 for the sets that are simultaneously F_σ and G_δ . This class is closed under finite Boolean operations and continuous inverse images. Every open or closed subset of a metrizable space belongs to this class. A G_δ subspace of a Polish space is Polish. For reference, if $X = \bigcap_n U_n$ in a complete metric space X_0 , the map

$$x \mapsto (x, (d(x, X_0 \setminus U_n)^{-1})_{n \in \mathbb{N}})$$

embeds X as a closed graph in a product of complete spaces, with the empty-complement coordinates omitted. This gives the usual complete metrizability argument.

32 The real-coordinate model

Theorem 32.1 (An affirmative construction; = [theorem 3.1\(1\)](#)). *The structure $M_{\mathbb{R}}$ of (30.1), with the operations and order described in Section 30, is an uncountable model of $\text{Th}(\mathbb{N}; 0, 1, +, <)$. Its topology is Polish, locally compact, and has no isolated points. Addition is jointly continuous. Therefore Question 2 of [1], as stated there, has an affirmative answer.*

Proof. Consider the abelian group

$$G_{\mathbb{R}} = \mathbb{R} \times \mathbb{Z}, \quad (r, n) + (s, m) = (r + s, n + m).$$

Give it the order

$$(r, n) < (s, m) \iff r < s \text{ or } (r = s \text{ and } n < m). \quad (32.1)$$

This is a translation-invariant total order. Its least positive element is $e = (0, 1)$: a positive pair either has positive real coordinate and exceeds every $(0, n)$, or has real coordinate zero and a positive integer coordinate.

The subgroup $\mathbb{Z}e$ is $\{0\} \times \mathbb{Z}$, and $G_{\mathbb{R}}/\mathbb{Z}e \cong (\mathbb{R}, +)$ is divisible. Hence $G_{\mathbb{R}}$ is a $\mathbb{Z}$ -group, and its nonnegative cone is a model of $\text{Th}(\mathbb{N}; 0, 1, +, <)$. Computing that cone gives exactly (30.1).

Topologically, $\mathbb{R} \times \mathbb{Z}_{\text{discrete}}$ is Polish, and

$$M_{\mathbb{R}} = ([0, \infty) \times \mathbb{Z}) \setminus (\{0\} \times \mathbb{Z}_{<0}). \quad (32.2)$$

The first set is closed, and the removed set is closed. Thus $M_{\mathbb{R}}$ is open in a closed Polish subspace, so is Polish. Equivalently, it is the countable topological disjoint sum of the slices

$$M_{\mathbb{R}} \cap (\mathbb{R} \times \{n\}) \cong \begin{cases} [0, \infty), & n \geq 0, \\ (0, \infty), & n < 0. \end{cases}$$

Each slice is locally compact and has no isolated points. Those properties therefore hold for the disjoint sum. The map $r \mapsto (r, 0)$ embeds $[0, \infty)$ into the model, proving uncountability.

Ambient coordinatewise addition is continuous. The sum of two nonnegative pairs is again nonnegative, so its restriction is a continuous map $M_{\mathbb{R}}^2 \rightarrow M_{\mathbb{R}}$. $\square$

[write] [Theorem 32.1](#) and its proof are those of [theorem 3.1\(1\)](#); source 07 adds the picture in [figure 1](#) and a complete metric that uses $\log r$ on the negative slices ([equation \(32.3\)](#); compare (5.2)). Source 06's account follows in [section 32.3](#).

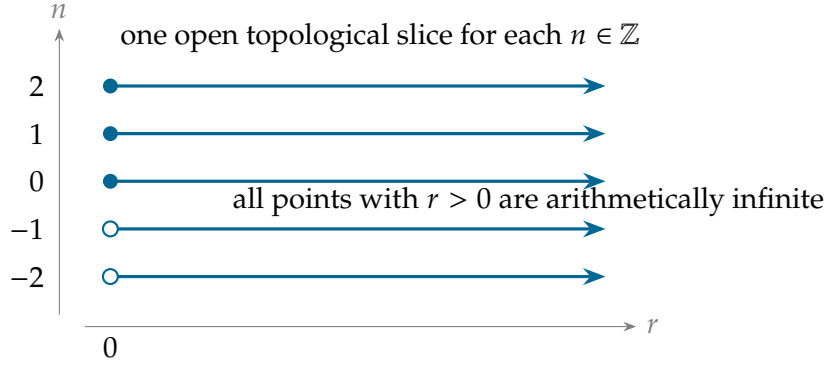


Figure 1: The space $M_{\mathbb{R}}$. Filled endpoints are present; hollow endpoints are absent. Different horizontal slices are topologically separated. Arithmetic order compares the horizontal coordinate before the slice label.

32.1 A complete metric, rather than an incomplete inherited metric

Being a Polish subspace does not mean that the restriction of the ambient Euclidean product metric is complete. For example, $(1/j, -1)$ approaches a missing endpoint in the ambient space. A compatible complete metric is easy to exhibit.

For the slice with integer coordinate n , put

$$h_n(r) = \begin{cases} r, & n \geq 0, \\ \log r, & n < 0. \end{cases}$$

Define

$$d((r, n), (s, m)) = \begin{cases} 1, & n \neq m, \\ \min\{1, |h_n(r) - h_n(s)|\}, & n = m. \end{cases} \quad (32.3)$$

The triangle inequality holds within each slice and also across slices. A Cauchy sequence is eventually in a single slice. In a nonnegative slice it is complete because $[0, \infty)$ is complete; in a negative slice the logarithm identifies it with a complete copy of $\mathbb{R}$. Thus d is complete and gives precisely the topology used in Theorem 32.1.

32.2 Division and the meaning of nonstandard induction

For a standard integer $d \geq 1$, divide the ordinary integer n as $n = dq + r$, where $0 \leq r < d$. Then

$$(a, n) = d(a/d, q) + (0, r). \quad (32.4)$$

When $(a, n) \in M_{\mathbb{R}}$, its quotient $(a/d, q)$ is also in $M_{\mathbb{R}}$. This remains true for negative n : such an input has $a > 0$, so its positive real coordinate determines positivity of the quotient.

For example,

$$(5/2, -7) = 3(5/6, -3) + (0, 2).$$

The integer coordinate is permitted to be negative in a positive element. Forbidding it would destroy predecessor and division properties and would give the wrong model.

Every (a, n) with $a > 0$ exceeds every standard numeral $(0, k)$. Nevertheless, each induction instance in the first-order additive language holds, by the $\mathbb{Z}$ -group characterization. The external set of standard numerals is proper and successor-closed, but is not definable in

this model, even with parameters: a definition would violate the corresponding induction instance. This is the usual distinction between first-order and full second-order induction, manifested in very elementary coordinates.

32.3 A perfect locally compact model [06]

Theorem 32.2 (An affirmative answer to Glazer’s question [06]; = [theorem 3.1\(1\)](#)). *With the usual topology on $\mathbb{R}$ and the discrete topology on $\mathbb{Z}$, the space*

$$M_{\mathbb{R}} = ((0, \infty) \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}) \quad (32.5)$$

is uncountable, perfect, locally compact, and Polish. With coordinatewise addition, $0 = (0, 0)$, $1 = (0, 1)$, and lexicographic order, it is a model of Pres with jointly continuous addition.

Proof. The algebra follows from Lemma 31.2 with $D = \mathbb{R}$. For each integer z , the slice with second coordinate z is clopen and is homeomorphic to

$$I_z = \begin{cases} [0, \infty), & z \geq 0, \\ (0, \infty), & z < 0. \end{cases}$$

Thus $M_{\mathbb{R}}$ is a countable topological disjoint union of locally compact Polish rays. It is second countable and has no isolated points.

One can verify complete metrizability without relying on the disjoint sum assertion. Put $\phi_z(r) = r$ for $z \geq 0$ and $\phi_z(r) = \log r$ for $z < 0$, and define

$$\rho((r, z), (s, w)) = \begin{cases} 2, & z \neq w, \\ \min\{1, |\phi_z(r) - \phi_z(s)|\}, & z = w. \end{cases}$$

This is a compatible complete metric: a Cauchy sequence eventually stays in one slice, where it corresponds to a Cauchy sequence in $[0, \infty)$ or $\mathbb{R}$. Rational points in all slices give a countable dense set.

Finally, $\mathbb{R} \times \mathbb{Z}$ is a topological group under coordinatewise addition, and its nonnegative cone is closed under addition. Restricting the ambient operation gives a continuous map $M_{\mathbb{R}} \times M_{\mathbb{R}} \rightarrow M_{\mathbb{R}}$. $\square$

Example 32.3 (Arithmetic order and topological proximity [06]). Every (r, z) with $r > 0$ is larger, arithmetically, than every standard natural number $(0, n)$. Nevertheless

$$(1/k, 0) \longrightarrow (0, 0).$$

There is no contradiction: the topology is the product topology, not the order topology. In fact, the order topology of any discretely ordered group or its nonnegative cone is discrete, since consecutive integers relative to any point isolate it.

The explicit quotient and remainder maps in (31.3) are continuous for every fixed standard n . In this example they are

$$Q_n(r, z) = (r/n, \lfloor z/n \rfloor), \quad R_n(r, z) = (0, z - n\lfloor z/n \rfloor).$$

The residue classes form a finite clopen partition. These are fixed standard-divisor operations; the assertion does not introduce division by a variable nonstandard element into Presburger arithmetic.

33 Borel orders on Polish abelian groups

The real construction can be extended whenever the positive cone of a divisible ordered Polish group is a Polish subspace. The following theorem shows that Borel definability of the order is already enough. We include the full proof because it is the descriptive engine for the general result.

Theorem 33.1 (Low complexity of an invariant Borel order). *Let A be a Polish abelian group with a translation-invariant total order. If $P = A_{>0}$ is Borel, then all four sign cones are Δ_2^0 . Both order relations $<$ and $\leq$ are Δ_2^0 subsets of A^2 . Moreover, there is a canonically defined countable decreasing chain of closed convex subgroups that decomposes P into countably many locally closed pieces.*

Proof. For a closed subgroup $H \leq A$, write

$$P_H = P \cap H, \quad C(H) = \text{cl}_H(P_H \cup \{0\}), \quad D(H) = C(H) \cap (-C(H)).$$

The set $C(H)$ is a closed additive monoid. Indeed, approximate two of its points by nonnegative points of H and use continuity of addition. Consequently $D(H)$ is a closed subgroup. Totality of the order gives

$$H = C(H) \cup (-C(H)). \quad (33.1)$$

We first show that $D(H) \subsetneq H$ whenever $H \neq \{0\}$. The strict cone P_H is nonmeagre in H . In the nondiscrete case this follows from the Baire category theorem and

$$H = P_H \dot{\cup} (-P_H) \dot{\cup} \{0\},$$

since $\{0\}$ is nowhere dense and negation is a homeomorphism. In the discrete case P_H is nonempty and hence nonmeagre.

As a Borel set, P_H has the Baire property. There is therefore a nonempty open set $V \subseteq H$ on which it is comeagre. For every $t \in V + V$, the set $V \cap (t - V)$ is nonempty and open. Both P_H and $t - P_H$ are comeagre there, so they intersect. Hence

$$V + V \subseteq P_H + P_H \subseteq P_H. \quad (33.2)$$

This is the sumset form of the category argument often associated with Piccard and Pettis; the displayed proof fixes the version needed here.

If $C(H) = H$, then $-P_H \cup \{0\}$ is dense in H . It would meet the nonempty open set $V + V \subseteq P_H$, which is impossible. Thus $C(H) \neq H$, and a fortiori $D(H) \neq H$.

Now form the transfinite recursion

$$H_0 = A, \quad H_{\xi+1} = D(H_\xi), \quad H_\lambda = \bigcap_{\xi < \lambda} H_\xi \quad (\lambda \text{ a limit}). \quad (33.3)$$

Every term is a closed subgroup, with a strict successor decrease whenever the current subgroup is nonzero. This process reaches $\{0\}$ at a countable ordinal η .

To prove the last claim, suppose instead that there were a strict decrease at every $\xi < \omega_1$. Fix a countable base $\mathcal{B}$ of A . For each ξ , choose the first basic set B_ξ meeting H_ξ and disjoint from $H_{\xi+1}$. Such a basic set exists because $H_{\xi+1}$ is closed and properly contained in H_ξ . If $\zeta > \xi$, then B_ξ misses H_ζ , whereas B_ζ meets it. The B_ξ are therefore distinct, contradicting countability.

For $\xi < \eta$, set $C_\xi = C(H_\xi)$ and

$$L_\xi = H_\xi \setminus (-C_\xi).$$

Every member of L_ξ is positive, since a nonpositive member would already lie in $-P_{H_\xi} \cup \{0\} \subseteq -C_\xi$. Equation (33.1) gives the disjoint decomposition

$$H_\xi \setminus H_{\xi+1} = L_\xi \dot{\cup} (-L_\xi).$$

A nonzero element leaves the decreasing chain at a successor stage: its first absent stage cannot be a limit, by the definition of intersection. It follows that

$$P = \bigcup_{\xi < \eta} L_\xi, \quad -P = \bigcup_{\xi < \eta} (-L_\xi). \quad (33.4)$$

Each layer is relatively open in the closed set H_ξ , and hence is F_σ in the ambient metrizable space. Both unions in (33.4) are countable. Thus P and $-P$ are F_σ , and their complements, obtained by adjoining $\{0\}$ to the opposite cone, are also F_σ . This proves all four Δ_2^0 assertions. Applying the continuous difference map $(x, y) \mapsto y - x$ proves the assertions for order relations.

Finally, every H_ξ is order-convex. The limit step is immediate. For a successor step, suppose H is convex, $h \in D(H)$, and $0 \leq x \leq h$. Then $x \in H$ and $x \in C(H)$. Also

$$-x = (-h) + (h - x) \in C(H) + C(H) \subseteq C(H).$$

Hence $x \in D(H)$, proving convexity. □

[write] The sumset step (33.2) is the Pettis argument that source 08 uses in [lemmas 47.3](#) and [53.1](#) and source 09 states as [lemma 61.1](#). For locally compact second-countable groups, part (1) of [theorem 51.2](#) in Part IV is the special case of [theorem 33.1](#), with a different proof through finite-dimensional cuts.

Remark 33.2 (The hypotheses that do the work). The topology must make A a Polish group. A standard Borel group without an established compatible Polish group topology is not enough for this proof. Borelness is used on every closed subgroup; merely knowing that the cone has the Baire property in the ambient space does not automatically give that hereditary regularity.

There is classical related work on measurable orders on locally compact abelian groups and their convex subgroup structure; see Asmar and Montgomery-Smith [22]. We use a direct category proof on Polish groups, including groups that are not locally compact. No historical priority claim for the precise formulation of Theorem 33.1 is needed for the positive construction.

33.1 A countable ordinal invariant of the order and topology

Definition 33.3. The *cone closure rank* $\rho(A, <)$ is the least η for which the recursion (33.3) reaches $H_\eta = \{0\}$. It is an invariant of the ordered topological group, not just of its abstract group.

Proposition 33.4. For every countable ordinal γ , the product group $\mathbb{R}^\gamma$ with its product topology and lexicographic order has cone closure rank γ . For infinite countable γ , these are different orders on topological groups all isomorphic to $\mathbb{R}^\mathbb{N}$.

Proof. For $\beta \leq \gamma$ put

$$K_\beta = \{x \in \mathbb{R}^\gamma : x_\xi = 0 \text{ for all } \xi < \beta\}.$$

These are closed subgroups. The positive cone is Borel, since it is the countable union, over $\beta < \gamma$, of the sets with zero prefix below β and positive β -coordinate. For $\beta < \gamma$, inside K_β , the closure of the nonnegative cone is exactly

$$\{x \in K_\beta : x_\beta \geq 0\}.$$

The inclusion from left to right follows by coordinate continuity. For the reverse inclusion, points with $x_\beta = 0$ are limits of $x + j^{-1}e_\beta$, which are positive. Intersecting with the negative closure gives $K_{\beta+1}$. At limit stages the intersection is the corresponding K_λ . Thus $H_\beta = K_\beta$ throughout, with the first zero subgroup at γ . Reindexing the countably infinite coordinate set gives the asserted topological group isomorphism. $\square$

Even very low Borel complexity therefore permits arbitrarily long countable closure recursions. The Δ_2^0 conclusion is sharp: the lexicographic positive cone of $\mathbb{R}^2$ is neither open nor closed.

34 A general Polish realization theorem

Theorem 34.1 (Divisible-group realization). *Let A be a divisible Polish abelian group carrying a translation-invariant total order with Borel positive cone. Define*

$$G_A = A \overrightarrow{\times} \mathbb{Z}, \quad M_A = (A_{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}). \quad (34.1)$$

Use coordinatewise addition, $0 = (0, 0)$, $1 = (0, 1)$, and the product subspace topology. Then:

- (i) M_A is a Polish model of $\text{Th}(\mathbb{N}; 0, 1, +, <)$, and addition is jointly continuous.
- (ii) Every fixed-standard-divisor quotient and remainder map is continuous. Partial subtraction on its arithmetic domain is continuous.
- (iii) Every relation definable with parameters in a finite Cartesian power of M_A is Δ_2^0 in the product topology.
- (iv) If A is uncountable, so is M_A .

Proof. The order on G_A is lexicographic with A the more significant coordinate. Its least positive element is $(0, 1)$, and its quotient by $\{0\} \times \mathbb{Z}$ is the divisible group A . Thus it is a $\mathbb{Z}$ -group and M_A is its nonnegative cone.

Theorem 33.1 says that $A_{>0}$ and $A_{\geq 0}$ are Δ_2^0 . In the clopen slice $A \times \{k\}$, membership in M_A means $a > 0$ if $k < 0$ and $a \geq 0$ if $k \geq 0$. Both M_A and its complement are therefore countable unions of F_σ sets. In particular M_A is G_δ in $A \times \mathbb{Z}$, so is Polish. Coordinatewise addition restricts continuously to it.

An ordered group is torsion-free. Consequently multiplication by a standard integer $d \geq 1$ is a continuous bijective homomorphism on the divisible Polish group A . Its inverse, division by d , is continuous. Here is a short justification of openness. Given a neighborhood U of zero, choose open V with $V - V \subseteq U$. Countably many translates of V cover A , so countably many translates of dV do as well. The analytic set dV has the Baire property and is nonmeagre. The difference form of the category argument then gives a neighborhood of zero inside $dV - dV \subseteq dU$. Thus multiplication by d is open.

Writing $k = dq + r$, $0 \leq r < d$, the desired maps are

$$Q_d(a, k) = (a/d, q), \quad R_d(a, k) = (0, r). \quad (34.2)$$

They are restrictions of continuous coordinate maps and take their values in M_A . They give the unique decomposition $x = dQ_d(x) + R_d(x)$. Partial subtraction is the restriction of the ambient continuous map $(x, y) \mapsto x - y$ to the subspace where $y \leq x$.

For (iii), consider affine integer-linear expressions on G_A . Their A -coordinate is a continuous affine map into A , and their integer coordinate is a continuous map into the discrete group $\mathbb{Z}$. The nonnegative cone of G_A is Δ_2^0 , by the same slice argument used for M_A . Consequently each linear comparison is Δ_2^0 ; equalities are closed. Since A is divisible, congruence modulo d depends only on the integer coordinate modulo d , so is clopen. Classical Presburger quantifier elimination reduces every parameter-definable relation to a finite Boolean combination of these atoms. Restricting to M_A^k preserves their complexity.

Finally, if A is uncountable, totality and negation show that $A_{>0}$ is uncountable. It embeds into M_A via $a \mapsto (a, 0)$. $\square$

[write] Part (ii) of [theorem 34.1](#) includes, for $A = \mathbb{R}$ and $A = \mathbb{Q}^{\mathbb{N}}$, the continuity statements of [proposition 8.1](#) and of source 04's [theorem 5.4](#); part (iii) extends [theorems 7.1](#) and [7.2](#) to every divisible Polish group with a Borel cone. Source 06 proves part (iii) for its own families in [theorem 37.7](#).

Remark 34.2. Only fixed standard divisors occur in (34.2). A function taking a *variable* divisor as input belongs to a stronger language. The theorem also distinguishes continuous partial subtraction from total truncated subtraction, which will be impossible in an uncountable Polish presentation.

34.1 Definable Borelness for arbitrary continuous presentations

The special Δ_2^0 estimate uses the ambient Polish group. A weaker conclusion holds for every Polish Presburger presentation with continuous addition, whether or not it comes from our construction.

Proposition 34.3. *Let $M \models \text{Th}(\mathbb{N}; 0, 1, +, <)$ have a Polish topology making addition continuous. Every relation definable with parameters is Borel. Every definable partial function between finite Cartesian powers has Borel domain and is Borel measurable on its domain.*

Proof. Equation (31.1) makes $<$ analytic: it is the projection of the closed equality relation $x + z + 1 = y$. Its complement $y \leq x$ is analytic for the same reason. Hence $<$ is Borel by the analytic–coanalytic characterization of Borel sets.

For a fixed $d \geq 1$ and $0 \leq r < d$, the map $x \mapsto dx + r$ is continuous and injective. Its image is Borel by the Borel injective-image theorem. These d residue classes partition M . Congruence relations are therefore Borel, as are integer-linear comparisons after negative coefficients have been moved to the other side. Quantifier elimination now gives every definable relation Borel.

The graph of a definable partial function is a Borel set. Projection from that graph to its input coordinates is Borel and injective. Its image, the domain, is Borel, and its inverse is Borel. Composing that inverse with projection to output coordinates gives the desired function. $\square$

35 A family of models on Baire space

35.1 The countable-ordinal coefficient groups

Fix a countably infinite ordinal α . Let

$$A_\alpha = \mathbb{Q}^\alpha, \quad G_\alpha = A_\alpha \overrightarrow{\times} \mathbb{Z}, \quad M_\alpha = (G_\alpha)_{\geq 0}. \quad (35.1)$$

Here $\mathbb{Q}^\alpha$ is the full Cartesian product, not the finite-support direct sum. Give each copy of $\mathbb{Q}$ the *discrete* topology. On A_α , the least coordinate at which two vectors differ determines their lexicographic comparison. Such a coordinate exists because α is well ordered. Addition and division by a nonzero integer are coordinatewise.

The product topology uses finite coordinate restrictions. In particular, when $\alpha > \omega$, it does not treat an entire transfinite prefix as one open condition. Keeping these two uses of coordinates separate is the main point in the topology proof.

Theorem 35.1 (Continuous arithmetic on Baire space; for $\alpha = \omega$, [theorem 6.1](#)). *For every countably infinite ordinal α , M_α is a continuum-sized model of $\text{Th}(\mathbb{N}; 0, 1, +, <)$ homeomorphic to $\mathbb{N}^\mathbb{N}$. Addition, every quotient and remainder by a fixed positive standard integer, and partial subtraction are continuous. Every parameter-definable relation in every finite arity is Δ_2^0 .*

Proof. The group A_α is divisible, ordered, and Polish: it is a countable product of countable discrete groups. Its topology is zero-dimensional. For each $\beta < \alpha$ define

$$C_\beta^+ = \{a : a_\gamma = 0 \ (\gamma < \beta), \ a_\beta > 0\}, \quad C_\beta^- = \{a : a_\gamma = 0 \ (\gamma < \beta), \ a_\beta < 0\}.$$

Each set is closed. The prefix-zero condition is an intersection of closed coordinate conditions, and the sign condition on a discrete rational coordinate is clopen. Therefore

$$(A_\alpha)_{>0} = \bigcup_{\beta < \alpha} C_\beta^+, \quad (A_\alpha)_{<0} = \bigcup_{\beta < \alpha} C_\beta^-$$

are F_σ , with F_σ complements after adjoining $\{0\}$. [Theorem 34.1](#) now gives Polishness and all the arithmetic and definability assertions. Cardinality is $\mathfrak{c}$ because the entire fiber $a_0 = 1$ lies in the positive cone and has countably infinitely many free rational coordinates.

It remains to identify the underlying Polish space. We prove that every nonempty open subset of M_α contains a closed noncompact cylinder. Let U be nonempty and choose $x = (a, n) \in U$. For some finite $F \subseteq \alpha$, the ambient cylinder

$$B = \{(b, n) : b_\gamma = a_\gamma \text{ for } \gamma \in F\}$$

satisfies $x \in B \cap M_\alpha \subseteq U$. Let j be the least natural number not in F . All indices $i < j$ are in F .

If the finite prefix $(a_0, \dots, a_{j-1})$ has a nonzero entry, its first nonzero entry is positive, since $a \geq 0$. Positivity is already forced; in this case additionally set $b_j = 0$. If that prefix is zero, additionally set $b_j = 1$. In both cases the resulting cylinder C fixes only finitely many coordinates, is nonempty, and has strictly positive A_α coordinate. Thus

$$C \subseteq B \cap M_\alpha \subseteq U.$$

The set C is clopen in the ambient product and closed in M_α . Infinitely many rational coordinates remain free. Projection onto any one of them maps C continuously onto the infinite discrete space $\mathbb{Q}$, so C is not compact.

If a compact subset $K \subseteq M_\alpha$ had nonempty interior U , the cylinder just constructed would be a closed subset of K and hence compact, a contradiction. Thus every compact subset has empty interior. The space is nonempty, zero-dimensional, and Polish. The Baire-space characterization gives $M_\alpha \cong \mathbb{N}^\mathbb{N}$. $\square$

[write] For $\alpha = \omega$, M_ω is Part I's Baire model M_B , and [theorem 35.1](#) is [theorem 6.1](#), where an explicit and computable homeomorphism is given ([theorems 6.2](#) and [6.4](#); also source 06's [theorems 35.6](#) and [35.7](#) below). For $\alpha > \omega$ the theorem is new to the collection, and the homeomorphism above comes from the topological characterization of Baire space.

Remark 35.2 (Why the transfinite cases require the full argument). For $\alpha = \omega$, both strict sign cones of A_α are open, since a finite prefix detects each nonzero vector's sign. When $\alpha > \omega$, a vector first nonzero at ω is not an interior point of its sign cone. The closed-piece proof works in both cases. Likewise, the cylinder argument above covers neighborhoods of such transfinite-leading vectors and of zero, not just the dense set of vectors with finite leading coordinate.

Choose a homeomorphism $h : \mathbb{N}^\mathbb{N} \rightarrow M_\alpha$ and transport the arithmetic:

$$x \oplus y = h^{-1}(h(x) + h(y)), \quad 0_{\mathbb{N}^\mathbb{N}} = h^{-1}(0), \quad 1_{\mathbb{N}^\mathbb{N}} = h^{-1}(1).$$

The operation $\oplus$ is jointly continuous. The coefficient model is explicit; the homeomorphism here is supplied by a topological characterization. We do not claim a computable homeomorphism for an arbitrary presentation of a countable ordinal.

35.2 Residue classes as copies of the whole space

Corollary 35.3 (extends [proposition 8.1](#)). *For each standard $d \geq 1$, the map*

$$x \mapsto (Q_d(x), r_d(x))$$

is a homeomorphism $M_A \rightarrow M_A \times \{0, \dots, d-1\}$, where $R_d(x) = r_d(x)1$. Each residue class is clopen and homeomorphic to M_A .

Proof. Continuity of the forward map follows from [Theorem 34.1](#). Its inverse is $(q, r) \mapsto dq + r1$, continuous because the second factor is finite discrete and addition is continuous. Division with remainder proves that the maps are inverse. $\square$

[write] For $M_A = M_\mathbb{R}$ and $M_A = M_B$ this is [proposition 8.1](#).

This is an arithmetic explanation of a topological self-similarity. It also applies to $M_\mathbb{R}$, whose countably many ray components can be rearranged into any positive finite number of copies.

35.3 An uncountable ordinal spectrum of models

For positive elements g, h of an ordered abelian group, define

$$g \asymp h \iff (\exists k, \ell \in \mathbb{N}_{>0}) (g \leq kh \text{ and } h \leq \ell g).$$

These are the Archimedean equivalence classes. We list the classes in *decreasing magnitude*: a class occurs earlier when its members exceed every standard multiple of a member of the later class.

Theorem 35.4 (An ordinal spectrum on one topological space). *The Archimedean classes of G_α , listed in decreasing magnitude, have order type $\alpha + 1$. Therefore the models M_α , for countably infinite α , are pairwise nonisomorphic, even as additive monoids. There are at least $\aleph_1$ pairwise nonisomorphic continuum-sized Presburger structures continuously presented on $\mathbb{N}^\mathbb{N}$. No continuum hypothesis is used.*

Proof. For a positive (a, n) with $a \neq 0$, let $\lambda(a, n)$ be the least $\beta < \alpha$ with $a_\beta \neq 0$. For $a = 0$, let $\lambda(0, n) = \alpha$. If two positive elements have the same leading index $\beta < \alpha$, their leading coefficients are positive rationals. Each can be made larger than the other by multiplication by a sufficiently large standard integer; lower coordinates do not affect that comparison. Thus they are Archimedean equivalent. The same holds in the integer class.

If $\lambda(g) < \lambda(h)$, then $g > kh$ for every standard positive integer k , since the leading coordinate of g is positive while that coordinate of kh is zero. Every index $\beta \leq \alpha$ occurs. This proves the asserted class order.

An isomorphism of additive positive monoids extends uniquely to an isomorphism of their group completions: represent a group element as a difference of two positives and use cancellation to check independence of the representation. The extension preserves positivity and hence order. It preserves the Archimedean class order. Thus an isomorphism between M_α and M_δ would imply $\alpha + 1 = \delta + 1$, so $\alpha = \delta$.

The set of countably infinite ordinals has cardinality $\aleph_1$. Theorem 35.1 places each corresponding model on Baire space. $\square$

The invariant uses addition. It does not by itself distinguish the pure underlying linear orders; no classification of those pure orders is claimed.

[write] (Batch 89.) Source 16 sharpens this: M_α elementarily embeds into M_β exactly when $\alpha \leq \beta$ (corollary 129.3), and it classifies the embeddings and the Polish elementary submodels of the family (Part X).

35.4 The optimal number of models is the continuum

The ordinal family gives an uncountable hierarchy with a transparent invariant. A different parameter reaches the largest possible number of isomorphism types on a fixed Baire space, without any hypothesis about the value of the continuum.

Theorem 35.5 (Sharp continuum multiplicity). *There are exactly $\mathfrak{c} = 2^{\aleph_0}$ isomorphism types of Presburger models continuously presentable on $\mathbb{N}^\mathbb{N}$. The lower bound can be realized with all parameter-definable relations Δ_2^0 and with continuous quotient and remainder by every fixed standard positive integer.*

Proof. For an irrational $s \in (1, 2)$ let $B_s = \mathbb{Q} + \mathbb{Q}s$, ordered as a subgroup of $\mathbb{R}$ but given the discrete topology. This is a countable divisible ordered abelian group. Put

$$A_s = B_s \overrightarrow{\times} \mathbb{Q}^\omega, \quad G_s = A_s \overrightarrow{\times} \mathbb{Z}, \quad M_s = (G_s)_{\geq 0}.$$

The product of the discrete coordinate topologies is Polish. The leading coordinate test makes the sign cones Borel. Theorem 34.1 gives all the arithmetic and definability properties. The finite-cylinder proof of Theorem 35.1 applies with B_s as the first coordinate, so M_s is homeomorphic to Baire space.

The subgroup

$$H_s = \{0\} \times \mathbb{Q}^\omega \times \mathbb{Z}$$

is the unique largest proper convex subgroup of G_s . Indeed it is proper and convex. If a convex subgroup contains an element g with nonzero B_s -coordinate, standard multiples of $|g|$ dominate every element of G_s , because $B_s \subseteq \mathbb{R}$ is Archimedean. Convexity forces such a subgroup to be the whole group. Therefore an isomorphism $M_s \cong M_t$, extended to group completions, induces an ordered additive isomorphism $B_s \cong B_t$.

Every increasing additive isomorphism $f : B_s \rightarrow B_t$ is multiplication by a positive real scalar λ . To see this, let $\lambda = f(1) > 0$. Additivity and divisibility give $f(q) = q\lambda$ for every rational q . For any $x \in B_s$, comparison with every rational shows that $f(x)/\lambda$ and x have the same rational cut in $\mathbb{R}$, hence are equal. Surjectivity now supplies rational a, b, c, d with

$$1 = \lambda(c + ds), \quad t = \lambda(a + bs), \quad t = \frac{a + bs}{c + ds}.$$

The denominator is nonzero, and $ad - bc \neq 0$, since otherwise t would be rational. Thus for each fixed s only countably many parameters t can yield an isomorphic model. The continuum-sized set of irrational parameters therefore yields continuum many isomorphism types in ZFC.

For the upper bound, fix $X = \mathbb{N}^{\mathbb{N}}$ and a countable dense set $D \subseteq X^2$. Since X is Hausdorff, a continuous map $X^2 \rightarrow X$ is determined by its values on D . There are at most $|X|^{\aleph_0} = \mathfrak{c}$ possible such maps. Choosing finitely many constants does not increase this bound, and all Presburger relations are determined by the addition and constants. Consequently there are at most $\mathfrak{c}$ continuously presented structures and at most $\mathfrak{c}$ isomorphism types. $\square$

[write] Source 06 proves the same count with a different family and adds that the lower bound is attained with the topology, the arithmetic order, 0 and 1 fixed ([theorem 35.8](#)); source 07's statement adds the Δ_2^0 and continuous-division clauses, and [corollary 36.2](#) below adds every oscillation rank. The two refinements are independent, so both statements are printed.

35.5 An explicit model on Baire space [06]

Let $H = \mathbb{Q}^{\mathbb{N}}$, giving every rational coordinate the *discrete* topology. Order H lexicographically: for a nonzero sequence h , its sign is the sign of its first nonzero coordinate. Put

$$G_\omega = H \overrightarrow{\times} \mathbb{Z}, \quad M_\omega = (G_\omega)_{\geq 0}.$$

The coordinatewise group H is divisible. It follows at once that $M_\omega \models \text{Pres}$ and that addition is continuous in the product subspace topology. The remaining issue is to identify that space.

Theorem 35.6 (Baire-space realization [06]; = [theorem 6.1](#)). *The space M_ω is homeomorphic to $\mathbb{N}^{\mathbb{N}}$. Consequently Baire space supports a model of Presburger arithmetic with jointly continuous addition.*

Proof. We give homeomorphisms rather than merely a cardinality argument. The strict cone $H_{>0}$ is the disjoint union of cylinders

$$C_{j,q} = \{h : h(0) = \dots = h(j-1) = 0, h(j) = q\}, \quad j \in \mathbb{N}, q \in \mathbb{Q}_{>0}.$$

Each cylinder is clopen and is homeomorphic to $\mathbb{Q}^{\mathbb{N}}$ by deleting its fixed prefix. A coordinatewise enumeration of $\mathbb{Q}$ identifies $\mathbb{Q}^{\mathbb{N}}$ with $\mathbb{N}^{\mathbb{N}}$. A countable disjoint union of Baire spaces is again Baire space, by using the first coordinate to record the component. Hence $H_{>0} \cong \mathbb{N}^{\mathbb{N}}$.

The closed cone $H_{\geq 0}$ has an equally direct description. Consider the tree of finite rational words whose first nonzero entry, if present, is positive. An all-zero word can be extended by

any nonnegative rational; a word already having positive first nonzero entry can be extended by any rational. Every node therefore has countably infinitely many successors. Choose a bijection from $\mathbb{N}$ to each successor set. The resulting rooted-tree isomorphism with $\mathbb{N}^{<\mathbb{N}}$ induces a homeomorphism of branch spaces $H_{\geq 0} \cong \mathbb{N}^{\mathbb{N}}$.

The fixed- z slice of M_ω is clopen, and equals $H_{>0}$ when $z < 0$ and $H_{\geq 0}$ when $z \geq 0$. Their countable disjoint union is therefore Baire space. Transport the arithmetic structure through any of these explicit homeomorphisms. $\square$

35.5.1 A computable stream presentation [06]

Theorem 35.7 (Effective addition [06]; = [theorems 6.2](#) and [6.4](#)). *The homeomorphism in Theorem 35.6 can be chosen computable in both directions for exact rational-coordinate streams. Under this homeomorphism, addition on $\mathbb{N}^{\mathbb{N}}$ is Type-2 computable. The constants 0, 1 and the quotient and remainder maps for each fixed standard divisor are computable as well.*

Proof. Fix computable bijections, with computable inverses, between $\mathbb{N}$ and each of $\mathbb{Q}$, $\mathbb{Q}_{>0}$, $\mathbb{Q}_{\geq 0}$, and $\mathbb{Z}$. Canonical reduced fractions provide such enumerations. Choose the nonnegative enumeration to send 0 to 0. Also fix a computable pairing bijection $\mathbb{N} \leftrightarrow \mathbb{N}^2$.

For $H_{\geq 0}$, decode successive input coordinates using the nonnegative rational enumeration until the first positive output; after that, use the enumeration of all rationals. The inverse uses exactly the same finite-prefix state. Thus each coordinate in either direction requires only finitely much input.

For $H_{>0}$, decode the first input coordinate as (j, k) , emit j zeros and then the k th positive rational, and use the remaining input to produce an arbitrary rational tail. The inverse searches for the first nonzero coordinate. This search terminates on its promised domain $H_{>0}$.

Finally, the first coordinate of a Baire input selects $z \in \mathbb{Z}$; the remaining stream selects a point of $H_{>0}$ or $H_{\geq 0}$ according to the sign of z . This defines a computable homeomorphism

$$F : \mathbb{N}^{\mathbb{N}} \longrightarrow M_\omega.$$

Coordinatewise rational addition and integer addition are computable. Therefore

$$a \oplus b = F^{-1}(F(a) + F(b))$$

is Type-2 computable. If the output integer coordinate is negative, the output rational sequence is necessarily strictly positive: a sum of nonnegative ordered-group elements has zero rational part only when both rational parts vanish, in which case both input integers are nonnegative. Thus the search used by F^{-1} always terminates when it is invoked on that branch. Fixed standard division is coordinatewise rational division together with ordinary integer division, so the same argument applies to it and to remainders. $\square$

Here “computable” concerns functions on infinite streams: every requested output coordinate is produced after finitely much input has been inspected. It does not assert a computable enumeration of this uncountable model or decidability of equality between arbitrary streams. It therefore has no conflict with the classical countable-presentation Tennenbaum theorem.

35.6 How many continuous models fit on Baire space? [06]

The existence result has substantial room for variation. The next theorem counts abstract isomorphism types, not merely different coordinate descriptions of one model.

Theorem 35.8 (Sharp abundance [06]; the count of [theorem 35.5](#)). *There are exactly $2^{\aleph_0}$ isomorphism types of Presburger models on Baire space with jointly continuous addition. The lower bound can be realized while keeping the underlying topology, arithmetic order, 0, and 1 fixed and varying only addition.*

Proof. Let $A_0 = \mathbb{Q}$ and $A_1 = \mathbb{Q} + \mathbb{Q}\sqrt{2}$, as additive ordered subgroups of $\mathbb{R}$, and give both groups discrete topology. For $s \in 2^{\mathbb{N}}$ put

$$H'_s = \prod_{n \in \mathbb{N}} A_{s(n)}, \quad G'_s = H'_s \overrightarrow{\times} \mathbb{Z}, \quad M'_s = (G'_s)_{\geq 0},$$

using the first differing coordinate for the order on H'_s . Both coefficient groups are countable and divisible. Each has countably infinitely many positive and nonnegative elements. The proof of [Theorem 35.6](#), with its coordinate-dependent enumerations, therefore applies unchanged. Hence each M'_s is a Baire-space model with continuous addition.

For positive elements of an ordered abelian group, say $x \asymp y$ if $x \leq my$ and $y \leq nx$ for some positive standard integers m, n . This is Archimedean equivalence. In G'_s , two positive elements are equivalent exactly when they have the same first nonzero coordinate, or when both are ordinary integers. Indeed, equal-rank real leading coefficients dominate one another after sufficiently large standard integer multiplication; an earlier nonzero coordinate dominates every standard multiple of a later one. The classes, ordered from largest magnitude to smallest, consequently have order type $\omega + 1$.

For each n , define the convex subgroup

$$U_n = \{(h, z) : h(0) = \cdots = h(n-1) = 0\}.$$

It is determined intrinsically by the Archimedean classes of rank at least n , and

$$U_n/U_{n+1} \cong A_{s(n)}. \tag{35.2}$$

Any additive-monoid isomorphism $f : M'_s \rightarrow M'_t$ extends to an ordered-group isomorphism of their difference groups by

$$\widetilde{f}(a - b) = f(a) - f(b).$$

This is well-defined since $a - b = c - d$ means $a + d = c + b$; every signed group element is a difference of nonnegative ones. The order is preserved because $a \leq b$ in a cone is equivalent to $\exists c (a + c = b)$.

The ordered set $\omega + 1$ has no nontrivial order automorphism, so $\widetilde{f}$ must preserve each rank and send U_n to its counterpart. The quotients in (35.2) then recover $s(n)$: A_0 has rational vector-space dimension 1, whereas A_1 has dimension 2. An additive isomorphism between divisible torsion-free groups is automatically $\mathbb{Q}$ -linear. Thus $M'_s \cong M'_t$ implies $s = t$. This gives $2^{\aleph_0}$ distinct isomorphism types.

For the fixed-order refinement, A_0 and A_1 are countable dense linear orders without endpoints. A back-and-forth construction gives an order isomorphism $\theta : A_0 \rightarrow A_1$ fixing 0: alternately place the next unused element of each enumerated order into the appropriate nonempty interval of the other. Since both topologies are discrete, θ is a homeomorphism. Apply either θ or the identity in coordinate n according to $s(n)$, and leave the integer

coordinate unchanged. This gives an order-preserving homeomorphism $M_\omega \rightarrow M'_s$ fixing 0, 1. Pulling back all the additions, and then using one homeomorphism $M_\omega \cong \mathbb{N}^\mathbb{N}$, proves the refinement.

Finally, a continuous function $\mathbb{N}^{\mathbb{N}^2} \rightarrow \mathbb{N}^\mathbb{N}$ is determined by its values on a fixed countable dense subset. There are at most $(2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0}$ such value assignments. Choices of the two constants do not enlarge this bound, and in Presburger arithmetic the order is determined by addition. Thus no more than continuum many isomorphism types are possible. $\square$

The continuous family in this theorem is not asserted to be an uncountable family of absolutely computable operations. There are only countably many algorithms. The effective assertion proved above applies to the particular rational ω -indexed model; computable variations can be handled separately, and noncomputable sequences s may serve as oracles.

35.6.1 Countably transfinite Archimedean scales [06]

There is a different useful variation, relevant to the repository's interest in ordered groups and ordinal scales.

Theorem 35.9 (Prescribed well-ordered scales [06]; = [theorems 35.1](#) and [35.4](#)). *For every countable ordinal $\alpha \geq \omega$, the positive cone*

$$M_\alpha = (\mathbb{Q}^\alpha \overrightarrow{\times} \mathbb{Z})_{\geq 0},$$

with discrete rational coordinates and the product subspace topology, is a Baire-space Presburger model with continuous addition. Its Archimedean classes, in decreasing magnitude, have order type $\alpha + 1$. In particular distinct such ordinals give nonisomorphic additive models.

Proof. For $\beta < \alpha$, let

$$C_\beta^\pm = \{h : (\forall \gamma < \beta) h(\gamma) = 0, \pm h(\beta) > 0\}.$$

Each set is closed, including at limit indices, since the preceding zero conditions are closed and the sign condition is clopen in a discrete coordinate. Both

$$M_\alpha = \bigcup_{\beta < \alpha} (C_\beta^+ \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N})$$

and its complement in $\mathbb{Q}^\alpha \times \mathbb{Z}$ are F_σ . Therefore M_α is G_δ in a Polish space, hence Polish and zero-dimensional. Its algebra and addition follow from Lemma [31.2](#).

Every nonempty open set in M_α contains a closed copy of $\mathbb{Q}^\mathbb{N}$. To see this, choose a point and a neighborhood fixing its integer coordinate and finitely many rational coordinates. If the chosen point first becomes positive at a finite coordinate j , fix all its coordinates through j . If it vanishes on every finite coordinate, choose a finite j beyond all the finitely prescribed finite indices, put zeros below j , and put 1 at j . Keep the other prescribed coordinates. In either case, the resulting finite cylinder is clopen in the ambient product, lies inside the positive cone and the chosen neighborhood, and has countably infinitely many free coordinates. Hence no compact subset of M_α has nonempty interior.

For completeness, a nonempty zero-dimensional Polish space X with this last property is homeomorphic to Baire space. Use a complete compatible metric. Every nonempty clopen set is complete and noncompact, and hence is not totally bounded. It therefore has a countable clopen cover of arbitrarily small diameter with no finite subcover. Removing earlier members from each member disjointifies it into countably infinitely many nonempty

clopen sets. Repeat within each piece, with diameters tending to zero. Each branch of the resulting $\mathbb{N}$ -branching tree determines exactly one point by completeness, and each point determines a branch. The pieces are clopen, so the resulting bijection in both directions is continuous. Apply this construction to M_α .

Finally, the same leading-coordinate argument as in Theorem 35.8 identifies the Archimedean class of a noninteger element with its least nonzero index $\beta < \alpha$; the integer class is last. Their ordered quotient is exactly $\alpha + 1$. An additive isomorphism preserves this quotient. Equality of the successor order types implies equality of α and β by removing their final elements. $\square$

The countable ordinals in this theorem describe *classes of magnitude*, not an increasing or decreasing ω_1 -sequence inside one arithmetic order. There is therefore no conflict with the uncountable-monotone-sequence obstruction in [2].

36 Unbounded oscillation at Borel level two

The Borel-level conclusion may suggest that all definable sets in these models have similarly simple topological behavior. A finer invariant shows that they do not. One fixed Presburger formula already has arbitrarily high countable boundary oscillation ranks across the models.

36.1 The derivative and its exact convention

Let X be a space and $E \subseteq X$. For a closed $F \subseteq X$, put

$$D_E(F) = \text{cl}_F(E \cap F) \cap \text{cl}_F(F \setminus E).$$

Starting with $F_0 = X$, set

$$F_{\xi+1} = D_E(F_\xi), \quad F_\lambda = \bigcap_{\xi < \lambda} F_\xi \quad (\lambda > 0 \text{ a limit}). \quad (36.1)$$

The *boundary oscillation rank* used here is the least ξ with $F_\xi = \emptyset$, when such a stage exists; otherwise we assign rank ∞ . For the characteristic function $\mathbf{1}_E$, this is the oscillation derivative at any threshold $0 < \varepsilon \leq 1$. The definition fixes the indexing convention; no identification with all possible conventions for difference-hierarchy ranks is intended.

In A_α , let u be the vector with $u_0 = 1$ and $u_\gamma = 0$ for $\gamma > 0$. Set

$$p = (u, 0), \quad E_\alpha = \{x \in M_\alpha : x < p\}.$$

This is the fixed formula $\varphi(x; y) \equiv x < y$, unary in x , with the indicated parameter p . Its chosen positive zeroth coordinate ensures that all boundary perturbations below stay inside the model.

Theorem 36.1 (Exact boundary oscillation ranks). *Write the countably infinite ordinal α uniquely as*

$$\alpha = \omega \cdot \beta + n, \quad 0 < \beta < \omega_1, \quad n < \omega.$$

For $0 < \mu \leq \alpha$, write

$$C_\mu = \{(a, z) \in M_\alpha : a_\gamma = u_\gamma \text{ for all } \gamma < \mu\}.$$

The recursion (36.1) for E_α satisfies

$$F_\xi = C_{\omega \cdot \xi} \quad (1 \leq \xi \leq \beta), \quad F_{\beta+1} = \emptyset. \quad (36.2)$$

Its rank is exactly $\beta + 1$. Consequently this one formula realizes every countable successor rank at least 2, and its ranks are unbounded below ω_1 .

Proof. Suppose first that $x = (a, z)$ differs from p at a finite first rational coordinate $k < \omega$. Fixing coordinates $0, \dots, k$ decides its comparison with p on a neighborhood of x . Such a point is not in F_1 .

Conversely, suppose that $a_k = u_k$ for every $k < \omega$. Every basic neighborhood of x fixes only finitely many coordinates. Choose an unfixed positive natural index k , change its value from 0 to -1 or to 1 , and leave everything else unchanged. All earlier coordinates still agree with u . The two resulting points are respectively below and above p , and remain positive because the zeroth coordinate stays 1. Thus $F_1 = C_\omega$.

For the successor step assume $1 \leq \xi < \beta$ and $F_\xi = C_\mu$, where $\mu = \omega \cdot \xi$. At least ω coordinates remain before α . Inside C_μ , a first residual difference at $\mu + k$, $k < \omega$, is decided by fixing the finite residual prefix $\mu, \dots, \mu + k$. These points leave the derivative. If a point agrees with u throughout the block $[\mu, \mu + \omega)$, every basic relative neighborhood leaves some $\mu + k$ unfixed. Perturbing this coordinate in the two directions gives points on both sides of p , still in C_μ and in M_α . Therefore

$$F_{\xi+1} = C_{\mu+\omega} = C_{\omega \cdot (\xi+1)}.$$

At a nonzero limit $\lambda \leq \beta$, intersecting the prefix conditions and using continuity of ordinal multiplication in its right argument gives

$$F_\lambda = \bigcap_{1 \leq \xi < \lambda} C_{\omega \cdot \xi} = C_{\omega \cdot \lambda}.$$

At stage β , only n rational coordinates and the integer coordinate are free. The resulting fiber is homeomorphic to the discrete space $\mathbb{Q}^n \times \mathbb{Z}$. Every subset of that fiber is relatively clopen, so the next derivative is empty. Each earlier displayed stage contains $\{u\} \times \mathbb{Z}$, proving that no earlier stage is empty.

Taking $\alpha = \omega \cdot \beta$ gives all stated successor ranks. □

Coordinate order α	Boundary oscillation rank of E_α
ω	2
$\omega \cdot 2$	3
ω^2	$\omega + 1$
$\omega \cdot (\omega + 1)$	$\omega + 2$
ω^ω	$\omega^\omega + 1$

The first derivative removes points where a finite prefix decides the comparison. Each later derivative removes one further block of ω coordinates. This is a topological calculation, although the defining formula itself never changes.

36.2 Explicit continuous approximations

There is also a direct approximation of $\mathbf{1}_{E_\alpha}$ by continuous functions. Choose an enumeration $\gamma_0, \gamma_1, \dots$ of α and put $S_N = \{\gamma_0, \dots, \gamma_N\}$. Sort this finite set in its ordinal order. Define $c_N(a, z)$ to be 1 exactly when comparison of the coordinates in S_N , followed by the integer coordinate, places (a, z) below $(u, 0)$.

Each $c_N : M_\alpha \rightarrow \{0, 1\}$ is continuous, because it depends on finitely many discrete coordinates. If $a \neq u$, its true first differing coordinate eventually belongs to S_N . From that point on, every $c_N(a, z)$ is the correct answer; coordinates preceding the true first difference are equal and cannot change it. If $a = u$, every $c_N(a, z)$ already equals $\mathbf{1}_{\{z < 0\}}$. Hence

$$c_N(x) \longrightarrow \mathbf{1}_{E_\alpha}(x)$$

with eventual equality at each point.

Thus these characteristic functions are explicitly Baire class 1, while their boundary oscillation ranks have no countable uniform bound across the family. Low Borel level and finite formula length do not give a uniform bound on this finer topological invariant.

Corollary 36.2 (Optimal multiplicity at every specified rank). *For every countable $\beta > 0$, there are $\mathfrak{c}$ pairwise nonisomorphic continuous Presburger models on Baire space with a parameter p for which the cut $x < p$ has boundary oscillation rank $\beta + 1$. All their definable relations are Δ_2^0 .*

Proof. In Theorem 35.5, replace the tail $\mathbb{Q}^\omega$ by $\mathbb{Q}^{\omega \cdot \beta}$ and choose $p = (1, 0, 0)$. The model remains Polish and homeomorphic to Baire space by the same closed-cylinder proof. An isomorphism still induces an isomorphism of the greatest Archimedean quotients B_s , so the continuum lower bound is unchanged.

For the cut at p , comparison at the discrete B_s -coordinate is locally decided unless that coordinate equals 1. On that fiber, the derivative removes successive blocks of ω rational coordinates, exactly as in Theorem 36.1. The first derivative fixes the B_s -coordinate to 1 and the first ω tail coordinates to zero; the ξ th derivative fixes the first $\omega \cdot \xi$ tail coordinates. The β th derivative is the discrete integer fiber and the next is empty. The rank is $\beta + 1$. $\square$

37 What continuity permits and forbids

37.1 The arithmetic order is not the Polish topology

In any Presburger model, the intrinsic order topology is discrete. For $a > 0$, the arithmetic interval $(a - 1, a + 1)$ is the singleton $\{a\}$; at zero the initial interval $[0, 1)$ is $\{0\}$. An uncountable discrete space is not second countable and hence not Polish. Accordingly, a positive answer to Glazer's question must distinguish continuity of addition from compatibility with the intrinsic order topology.

In $M_{\mathbb{R}}$,

$$x_j = (1/j, 0) \longrightarrow (0, 0), \quad x_j > (0, k) \quad \text{for every standard } k.$$

Topological convergence to zero is entirely compatible with being larger than every standard numeral. In M_α , let v_j be the coordinate unit vector at the natural index j . Then the analogous sequence $(v_j, 0)$ converges to zero and exceeds every standard numeral.

These sequences show explicitly that strict order is neither open nor closed. The ordered pair $(0, 1)$ belongs to $<$ but is approached by $(x_j, 1)$, which do not. The pairs $(1, x_j)$ belong to $<$ and converge to $(1, 0)$, which does not. The following general obstruction explains why this behavior is unavoidable.

Proposition 37.1 (Closed arithmetic order forces discreteness; = theorem 7.4). *Let M be a Hausdorff model of $\text{Th}(\mathbb{N}; 0, 1, +, <)$ with continuous successor. If $\leq$ is closed in M^2 , then the topology of M is discrete. If M is uncountable Polish with continuous addition, neither $<$ nor $\leq$ is open or closed.*

Proof. If $\leq$ is closed, then $<$ is open because its complement is the reverse non-strict order. Sections of $<$ are open. The singleton intervals just described are therefore open, so the topology is discrete.

If $<$ is open, the same singleton argument applies. If $<$ is closed, then $x \leq y \iff x < y + 1$ and continuity of successor make $\leq$ closed. If $\leq$ is open, then $x < y \iff x + 1 \leq y$ makes $<$ open. Each of these four possibilities forces discreteness. An uncountable second-countable space cannot be discrete. $\square$

[write] [Proposition 37.1](#) is [theorem 7.4](#) (source 05), and its first sentence also follows from source 04's [theorem 7.5](#), which needs no continuity of successor. The monus clause of [corollary 37.2](#) is source 04's [corollary 7.6](#); the min and max clauses are new.

Corollary 37.2 (Three individually forbidden operations). *On an uncountable Polish Presburger model with continuous addition, none of the following operations can be continuous:*

$$\min(x, y), \quad \max(x, y), \quad x \dot{-} y = \max\{x - y, 0\}.$$

Proof. In a Hausdorff space equalizers of continuous maps are closed. Each of the three operations, if continuous, would make $\leq$ closed through one of the equivalences

$$x \leq y \iff \min(x, y) = x \iff \max(x, y) = y \iff x \dot{-} y = 0.$$

Apply [Proposition 37.1](#). $\square$

This does not contradict continuity of partial subtraction: its domain is the subspace of pairs with $y \leq x$, and it is not extended across that domain's boundary by assigning the value zero.

37.2 Compactness remains impossible

The following obstruction is established in the earlier work of Enayat, Hamkins, and Wcisło [2]. We include the short jointly continuous case to locate our locally compact example relative to compact spaces.

Proposition 37.3 (= [corollary 12.2](#)). *No infinite compact Hausdorff space supports a Presburger model with jointly continuous addition. In particular, Cantor space does not.*

Proof. The set

$$C = \{(x, y, z) \in M^3 : x + z = y\}$$

is closed and compact. Its projection onto (x, y) is $\leq$, so this relation is compact and closed in the Hausdorff space M^2 . [Proposition 37.1](#) makes M discrete, contradicting compactness and infinitude. $\square$

[write] This is [corollary 12.2](#) of Part I, there with the compact-cancellative-monoid proof.

The model $M_{\mathbb{R}}$ is locally compact but not compact. Its separated ray components and its incomplete ambient-metric ends present no conflict with this obstruction.

37.3 A second topology makes predecessor continuous

In the topology of Theorem 32.1, total truncated predecessor

$$\text{pred}(x) = \begin{cases} 0, & x = 0, \\ x - 1, & x > 0 \end{cases}$$

is discontinuous at zero: $x_j = (1/j, 0) \rightarrow 0$, whereas $\text{pred}(x_j) = (1/j, -1)$ cannot converge to zero. That particular discontinuity is not forced by additive arithmetic.

Proposition 37.4. *The same algebraic model $M_{\mathbb{R}}$ admits a second Polish topology making addition, successor, and truncated predecessor continuous. Its isolated points are exactly the standard numerals.*

Proof. Use the topological disjoint sum

$$M_{\mathbb{R}}^{\sharp} = \mathbb{N}_{\text{discrete}} \sqcup ((0, \infty) \times \mathbb{Z}_{\text{discrete}}),$$

with the same underlying identification $(0, n) \leftrightarrow n$. It is a countable sum of Polish spaces. Standard points are isolated, and nonstandard points lie in open real rays, so have no isolated points.

The product splits into four clopen pieces according to whether each input is standard. On two standard inputs addition is a map on a discrete space. On a mixed piece it is, on each standard slice, translation of the integer coordinate. On two nonstandard inputs it is ordinary coordinate addition on positive real coordinates. Each restriction is continuous, and a sum involving a nonstandard input remains nonstandard. Hence addition is continuous on the whole product.

Successor and truncated predecessor act on the standard discrete part as the familiar maps, and on the nonstandard part as translations by ± 1 of the integer coordinate. They are continuous on the clopen pieces and therefore globally. $\square$

[merge] Under the identification $(a, n) \mapsto aX + n$ of section 26, $M_{\mathbb{R}}^{\sharp}$ is exactly the clopen stratum $(C_{\mathbb{R}}, \tau_{\text{FS}})$ of 03's cone in proposition 26.1, whose parts (iii) and (iv) found the same topology with continuous addition; proposition 37.4 adds that successor and truncated predecessor are continuous. It is also the subspace of polynomials of degree at most one in source 06's cone (proposition 42.1). Part I does not ask whether the discontinuity of predecessor in proposition 8.2 is forced; this proposition shows that it depends on the topology.

The two presentations are not homeomorphic: one has no isolated points and the other has countably many. Binary truncated subtraction still cannot be continuous in the second presentation, by Corollary 37.2.

37.4 The explicit models have no ordered-semiring expansion

Our examples are subject to an algebraic obstruction stronger than a mere failure of Borel regularity of multiplication.

Proposition 37.5 (Cofinal multiples obstruct multiplication; compare theorem 10.4). *Suppose $M \models \text{Th}(\mathbb{N}; 0, 1, +, <)$ has an element c such that*

- (i) $c > n1$ for every standard $n \in \mathbb{N}$;
- (ii) the standard multiples $\{nc : n \in \mathbb{N}\}$ are cofinal in M .

Then there is no multiplication, with the given addition and identity 1, making M the nonnegative cone of a discretely ordered ring. More generally, no semiring multiplication can satisfy distributivity, the given identity, and monotonicity under this arithmetic order.

Proof. Suppose such a multiplication exists. By cofinality choose a standard k with $c^2 < kc$; increasing a weak cofinal bound by one gives a strict bound. Since $c > (k + 1)1$, additive order gives $c = (k + 1)1 + u$ for some $u > 0$. Distributivity and nonnegativity of products give

$$c^2 = (k + 1)c + uc \geq (k + 1)c > kc,$$

a contradiction. $\square$

Corollary 37.6. *Neither $M_{\mathbb{R}}$ nor any M_{α} has a Peano-arithmetic expansion with the given addition, regardless of the regularity of multiplication.*

Proof. For $M_{\mathbb{R}}$ take $c = (1, 0)$. For M_{α} take $c = (u, 0)$ with $u_0 = 1$ and all other coordinates zero. In each case c exceeds every standard numeral. Given any element, choose a standard integer larger than its leading real or rational coordinate; that multiple of c strictly exceeds the element. Proposition 37.5 applies. $\square$

This proof shows exactly why the additive construction cannot be promoted to a model of full arithmetic by selecting a sufficiently irregular product. It also leaves room for studying other additive models, without the cofinal-multiple property, against Glazer's more general topological obstructions.

[write] Proposition 37.5 is a weaker form of theorem 10.4, which needs neither monotonicity nor commutativity; corollary 37.6 applies it to the whole ordinal family. Source 06's lemma 37.9 and theorem 37.10 remove the remaining hypothesis that the multiplicative unit be the named 1.

37.5 Definable regularity and an algebraic obstruction [06]

37.5.1 All additive definable relations are low in the Borel hierarchy [06]

Theorem 37.7 (Definable relations [06]; extends theorem 7.1). *In $M_{\mathbb{R}}$, in every M_{α} above, and in every M'_s of Theorem 35.8, every first-order Presburger-definable relation with parameters is Δ_2^0 in the corresponding finite product topology. Fixed-standard-modulus residue classes are clopen.*

Proof. In the ambient group $\mathbb{R} \overrightarrow{\times} \mathbb{Z}$, the nonnegative cone and its complement are each a union of an open real-coordinate condition and a closed zero-coordinate condition. Thus the cone is Δ_2^0 . For the countable rational and mixed-coefficient products, the decompositions by first nonzero index used above give the same conclusion. Equality is closed in the ambient Hausdorff topological group.

Every integer-linear affine map from a finite power of the ambient group to that group is continuous. Consequently its comparison with zero defines a Δ_2^0 set. Congruence modulo n depends only on the discrete final integer coordinate, because the other factor is divisible. Such congruences are clopen, including for affine linear combinations and parameters.

Apply the quantifier-elimination conclusion of Lemma 31.1. A formula on the nonnegative cone is translated to the group by restricting its quantified variables to that cone. Elimination produces a finite Boolean combination of the described atomic conditions. Restrict it back to the cone and use closure of Δ_2^0 under finite Boolean operations. $\square$

Remark 37.8 (Graphs versus functions [06]). Low Borel complexity of a definable graph does not imply continuity of its function. In the real example the truncated predecessor

$$p(x) = \begin{cases} 0, & x = 0, \\ x - 1, & x > 0 \end{cases}$$

is definable but discontinuous at zero: $(1/k, 0) \rightarrow 0$, whereas $p(1/k, 0) = (1/k, -1)$ stays in a different integer slice from $p(0)$. In the Baire model, let e_k be the rational sequence with 1 in coordinate k and zeros elsewhere. Then $(e_k, 0) \rightarrow 0$ gives the identical obstruction.

The order relation itself is neither open nor closed in these two basic models. Write $x_k = (e_k, 0)$ in the Baire model and $x_k = (1/k, 0)$ in the real model. In both cases $x_k > 1$ and $x_k \rightarrow 0$. Thus the pairs $(x_k, 1)$ lie outside the relation $<$ but converge to $(0, 1)$ inside it. The pairs $(1, x_k)$ lie inside $<$ but converge to $(1, 0)$ outside it.

The standard cut $\{0\} \times \mathbb{N}$ is closed in these models and is not definable, even with parameters, in the Presburger language. If it were defined by $\varphi(x, a)$, it would contain zero and be closed under successor, while omitting nonstandard elements. The induction instance for φ would then fail. This supplies a concrete distinction between topological visibility and first-order definability.

37.5.2 Why the lexicographic models cannot acquire multiplication [06]

The failure of multiplication for our first examples is stronger than the absence of a continuous or Borel multiplication.

Lemma 37.9 (The unit is forced [06]). *Suppose a nontrivial Presburger additive monoid M , with least positive element e , is given any unital semiring multiplication, with the same addition and zero. Then its multiplicative identity is e .*

Proof. The natural order $x \leq y \iff \exists z (x + z = y)$ is preserved by multiplication on either side, by distributivity. Write u for the multiplicative identity. It is nonzero, so $e \leq u$ and $0 \leq e^2 \leq eu = e$. By discreteness, e^2 is either 0 or e . If $e^2 = 0$, use division by two in the Presburger monoid to write $u = 2q + re$, where $r = 0$ or 1. Multiplication by e gives $e = ue = 2(qe)$, impossible for the least positive element. Therefore $e^2 = e$. If $u > e$, write $u = e + v$ with $v \geq e$. Then $e = ue = e^2 + ve \geq 2e$, again impossible. Thus $u = e$. $\square$

Theorem 37.10 (No semiring expansion [06]; compare [theorem 10.4](#)). *None of $M_{\mathbb{R}}$, the nontrivial M_{α} , or the M'_s admits a unital semiring multiplication with its prescribed additive monoid. This remains true without imposing any topology or prescribing the multiplicative identity in advance.*

Proof. In each example choose a with leading coordinate 1 in the first divisible coordinate and zero elsewhere. Then $a > n \cdot 1$ for every standard n . On the other hand, for every $b \in M$ there is a standard N with $b < Na$, since an ordinary integer can be chosen larger than the real leading coefficient of b . Thus standard multiples of a are cofinal in M .

Any proposed multiplication has identity 1 by Lemma 37.9. Since $a \geq n \cdot 1$, distributivity and the natural order imply $a^2 \geq na$ for every standard n . Taking $b = a^2$ in cofinality gives $a^2 < Na$ for some standard N , a contradiction. $\square$

The existing repository report contains a more general obstruction to PA expansions of nonstandard split $\mathbb{Z}$ -groups, using residue types [35]. The argument here has a different and stronger conclusion for the displayed groups: their largest Archimedean magnitude class

already rules out every unital semiring structure. The next example escapes this obstruction by having polynomial degrees of arbitrarily large magnitude.

38 A Polish semiring with full additive induction [06]

38.1 The ring and its order [06]

Let

$$A = \mathbb{Z} + t\mathbb{R}[t] = \left\{ m + \sum_{i=1}^d a_i t^i : m \in \mathbb{Z}, a_i \in \mathbb{R}, d < \omega \right\}.$$

The indeterminate t is ordered as positive infinite: a nonzero polynomial is positive exactly when its leading coefficient is positive. This is an ordered subring of $\mathbb{R}[t]$ under its leading-term order. It is discrete because a positive constant in A is a positive integer, while every positive nonconstant polynomial exceeds every integer. Put $P = A_{\geq 0}$.

Algebraically, $A/\mathbb{Z} \cong t\mathbb{R}[t]$ is a divisible ordered abelian group. Thus P is a model of Pres as an additive structure, and its inherited polynomial multiplication makes it a model of PA^- . This underlying polynomial construction has classical precedents and occurs explicitly in the repository [24, 36]. Our task is to choose and verify a topology.

38.2 The topology by degree and constant coefficient [06]

For $d \geq 1$ and $m \in \mathbb{Z}$, let

$$P_{d,m} = \{m + a_1 t + \cdots + a_d t^d : a_d > 0\}.$$

Give $P_{d,m}$ the coefficient topology $\mathbb{R}^{d-1} \times (0, \infty)$, and make each $n \in \mathbb{N}$ a singleton component. Define P to be the topological disjoint union

$$P = \mathbb{N}_{\text{discrete}} \sqcup \bigsqcup_{d \geq 1, m \in \mathbb{Z}} P_{d,m}. \quad (38.1)$$

In particular, convergence within a nonconstant stratum eventually has fixed degree and constant coefficient.

Theorem 38.1 (A continuous weak arithmetic semiring [06]). *The space in (38.1) is uncountable, locally compact, and Polish. Both polynomial addition and polynomial multiplication are jointly continuous on it. Its additive reduct satisfies full Presburger arithmetic, and its semiring structure satisfies PA^- .*

Proof. Each $P_{d,m}$ is homeomorphic to $\mathbb{R}^d$ by applying log to the positive leading coefficient. The countably many components in (38.1) are locally compact Polish spaces. A complete metric is obtained by bounding complete component metrics by 1 and putting distance 2 between components. Countable dense sets in the components give separability. The algebraic assertions were verified above.

It remains to prove joint continuity rather than only separate continuity. Products of components form an open cover of P^2 . On $P_{d,m} \times P_{e,n}$, positivity of both leading coefficients prevents cancellation in addition, so

$$\deg(p + q) = \max(d, e), \quad (p + q)(0) = m + n.$$

Thus this whole product maps into the fixed component $P_{\max(d,e),m+n}$, and the coefficient map there is linear and continuous.

On the same product,

$$\deg(pq) = d + e, \quad (pq)(0) = mn,$$

and the leading coefficient is a positive product. It maps into the fixed component $P_{d+e,mn}$, with continuous polynomial coefficient maps. If one input is a positive standard integer, its component is a singleton and the corresponding operations are coefficient translation or scalar multiplication. If an input is zero, multiplication is constant on its open singleton component. Pairs of standard integers are handled by singleton products. This proves continuity on every member of an open cover of P^2 , hence joint continuity globally. $\square$

This theorem concerns the nonnegative cone. The analogous topology on all signed polynomials would not make the entire ring a topological ring: cancellation can reduce degree. Even partial subtraction on the cone fails continuity. For instance

$$((1 + 1/k)t, t) \longrightarrow (t, t), \quad (1 + 1/k)t - t = (1/k)t \not\rightarrow 0,$$

because zero is isolated. This is consistent with the continuity just proved for addition and multiplication of nonnegative elements.

38.3 An exact failed induction instance [06]

Theorem 38.2 (Failure of open induction [06]). *In the polynomial model P , the quantifier-free formula*

$$\varphi(x; t) \equiv x^2 < t$$

defines exactly the standard natural numbers. Consequently P fails IOpen , despite satisfying every first-order additive induction instance.

Proof. If $x = n \in \mathbb{N}$, then $n^2 < t$. If x is a positive nonconstant polynomial of degree $d \geq 1$, then x^2 has degree $2d \geq 2$ and positive leading coefficient, so $x^2 > t$. Therefore $\varphi(P; t) = \mathbb{N}$. It contains zero and is closed under successor, but omits t . This falsifies the open-induction instance with parameter t . $\square$

The full omnific integer cone and the polynomial subring are different arithmetic structures. This counterexample concerns the subring $\mathbb{Z} + t\mathbb{R}[t]$ only; it does not transfer to the full omnific integer cone simply because the subring embeds there.

Proposition 38.3 (Additive definability in the polynomial topology [06]). *Every relation definable with parameters in the Presburger reduct of P is Δ_2^0 . Quotient and remainder by each fixed standard divisor are continuous.*

Proof. On a fixed product of coefficient components, an affine integer-linear expression has bounded degree and continuous coefficient functions, with fixed integer constant coefficient. Its positive and negative sign conditions are finite unions of sets where all higher coefficients vanish and a specified coefficient has the required sign; the residual constant case is also included. These sets and their complements are F_σ on each component. There are countably many clopen product components, so the conditions are Δ_2^0 globally. Congruences depend only on the integer constant term and are clopen. Apply Presburger elimination.

For standard $n \geq 1$, division sends the nonconstant coefficients a_i to a_i/n and the constant term m to $\lfloor m/n \rfloor$; the positive leading coefficient preserves degree. Remainder is the standard integer $m \bmod n$. These maps are continuous on each clopen component, hence on all of P . $\square$

39 Zero must be isolated when both operations are continuous [06]

The preceding polynomial example has isolated points, whereas our additive examples are perfect. There is an exact structural reason for this contrast. The essential subset-sum and probability method below comes from Glazer's proof of his first theorem [1]. An audit of its assumptions removes the arithmetic induction hypothesis. We also avoid any need to prove injectivity of the subset-sum limit map.

Theorem 39.1 (Isolated zero [06]). *Let A be a discretely ordered commutative unital ring, and put $M = A_{\geq 0}$. If M carries a Polish topology under which its restricted addition and multiplication are jointly continuous, then 0 is isolated.*

Corollary 39.2 ([06]). *No perfect Polish space supports a topological model of PA^- with continuous addition and multiplication. In particular Baire space supports the additive models constructed above, but cannot support such a weak-arithmetic semiring structure.*

39.1 Borel order from continuous addition [06]

We first record a point that prevents any hidden regularity assumption. For $a, b \in M$,

$$\begin{aligned} a < b &\iff \exists c \in M (b = a + c + 1), \\ a \geq b &\iff \exists c \in M (a = b + c). \end{aligned}$$

The equations define closed subsets of M^3 , because addition is continuous and M is Hausdorff. Their projections are analytic. They are complementary relations, so the analytic/coanalytic separation theorem gives that the order is Borel [8]. This argument uses no order-topology compatibility.

39.2 A uniformly convergent subset-sum map [06]

Suppose, toward a contradiction, that 0 is not isolated. Since $[0, 5] = \{0, 1, \dots, 5\}$ and finite sets are closed, one can choose $x_k > 5$ with $x_k \rightarrow 0$. Fix a complete compatible metric d on M . Set $y_0 = 1$. After choosing $y_0, \dots, y_{n-1}$, choose $k(n)$ so large that, with

$$y_n = x_{k(n)} y_{n-1},$$

we have

$$d(s, s + y_n) < 2^{-n} \quad \text{for every subset sum } s \text{ of } y_0, \dots, y_{n-1}. \quad (39.1)$$

There are only finitely many such sums. For each, continuity implies $s + x_k y_{n-1} \rightarrow s$, so a common choice exists. At the same time, ordered-ring positivity gives

$$y_n > 5y_{n-1} > 0. \quad (39.2)$$

For $r \in 2^{\mathbb{N}}$, define

$$\sigma_N(r) = \sum_{i < N} r(i) y_i.$$

Each σ_N is continuous, since it depends on finitely many bits. For $N \geq 1$ the distance between its next two partial sums is at most 2^{-N} , uniformly in r . Thus σ_N converges uniformly to a continuous function $r \mapsto z_r \in M$. Completeness is used at precisely this point.

If r, s differ only at finitely many positions, continuous addition implies the following *algebraic* identity in A :

$$z_s - z_r = \sum_i (s(i) - r(i))y_i. \quad (39.3)$$

For justification, add the finite removed terms to the appropriate sides of the finite partial-sum identity, then pass to the limit using addition only. Rearranging the resulting equality is an algebraic operation in A . No topology on A and no continuity of subtraction are being assumed.

Let $\bar{r}$ denote coordinatewise bit complementation and set

$$\Delta_r = z_r - z_{\bar{r}}$$

as notation for an element of the algebraic ring. Equation (39.3) yields

$$\Delta_s - \Delta_r = 2 \sum_i (s(i) - r(i))y_i \quad (39.4)$$

whenever r, s differ finitely.

39.3 Bounded complementary differences have probability zero [06]

Give $2^{\mathbb{N}}$ fair independent product measure μ . Put $S_j = \sum_{i < j} y_i$. External finite induction, using (39.2), gives

$$4S_j < y_j \quad (j \in \mathbb{N}). \quad (39.5)$$

Indeed it holds for $j = 0$, and $4S_{j+1} = 4S_j + 4y_j < 5y_j < y_{j+1}$.

For fixed n consider

$$E_n = \{r : |\Delta_r| \leq y_n\}.$$

This is a Borel set. To see this without treating Δ as a map into a topological ring, write its definition as the conjunction

$$z_r \leq z_{\bar{r}} + y_n, \quad z_{\bar{r}} \leq z_r + y_n.$$

All terms are continuous M -valued maps and the order is Borel.

Freeze all coordinates except $n+1, \dots, n+k$. If two assignments r, s on these coordinates differ last at $j > n$, then (39.4) and (39.5) imply

$$|\Delta_s - \Delta_r| \geq 2(y_j - S_j) > 2y_n.$$

For the last inequality, $S_j \geq y_n$ and $y_j > 4S_j$. Two values lying in the ordered interval $[-y_n, y_n]$ cannot be this far apart. Thus at most one of the 2^k assignments in each fiber lies in E_n . Integrating over the frozen coordinates gives

$$\mu(E_n) \leq 2^{-k}$$

for every k , and hence $\mu(E_n) = 0$. Therefore

$$C = \bigcap_{n \in \mathbb{N}} \{r : |\Delta_r| > y_n\} \quad (39.6)$$

is Borel and has measure one. In particular, complementary subset sums are unequal almost surely. No injectivity of $r \mapsto z_r$ is needed.

39.4 The sign is a tail event [06]

If r, s differ only at positions at most j , write $b = \Delta_s - \Delta_r$. Equations (39.4) and (39.5) give

$$|b| \leq 2S_{j+1} < 3y_j.$$

For an arbitrary n , take $m = \max(n, j) + 1$. Then

$$y_n + |b| < 4y_{m-1} < y_m.$$

If $r \in C$, it follows that $|\Delta_s| \geq |\Delta_r| - |b| > y_n$ and that Δ_s has the same sign as Δ_r . Since n was arbitrary, C and the sign on C are invariant under every finite change of coordinates.

Consequently the Borel event

$$B = \{r \in C : z_{\bar{r}} < z_r\}$$

belongs to the tail sigma algebra. Such an event has probability zero or one. One quick proof is that it is independent of the first N coordinates for every N ; a monotone-class argument makes it independent of the full sigma algebra, and in particular of itself. Thus $\mu(B) = \mu(B)^2$.

But $r \mapsto \bar{r}$ preserves measure and C , and reverses the sign of Δ_r . Since C has measure one, it follows that $\mu(B) = 1/2$, a contradiction. This proves Theorem 39.1. $\square$

Remark 39.3 (Exactly what the proof uses [06]). The proof uses the ordered-ring laws, discreteness, separability, complete metrizable, continuity, and ordinary external finite induction. It does not use induction *inside* M , standard division in M , internal exponentiation, saturation, or an integer-part theorem. It concludes that zero is isolated. It does not by itself imply that every standard numeral is isolated, since predecessor has not been assumed continuous.

40 Surreal and omnific realizations

40.1 The real model as an omnific subgroup

The most direct connection to the repository is the map

$$J_{\mathbb{R}} : G_{\mathbb{R}} \longrightarrow \mathbf{No}, \quad J_{\mathbb{R}}(r, n) = r\omega + n. \quad (40.1)$$

Here multiplication and addition on the right are surreal field operations. Conway normal forms compare the coefficient of the dominant monomial ω before the constant coefficient. Thus $J_{\mathbb{R}}$ is an ordered additive isomorphism onto $\mathbb{R}\omega + \mathbb{Z}$.

An omnific integer has a Conway normal form with no negative exponents and with integer constant coefficient. This description places $\mathbb{R}\omega + \mathbb{Z}$ inside $\mathbf{Oz}$. The standard normal-form facts used here and below are recalled in [23]; a general treatment of initial ordered subgroups appears in [5].

Transport the topology of $M_{\mathbb{R}}$ through $J_{\mathbb{R}}$. The nonnegative part of $\mathbb{R}\omega + \mathbb{Z}$ becomes an uncountable Polish additive model with continuous addition. This is a topology on a *set-sized substructure*, described by its coefficients. It is not the intrinsic surreal order topology.

For example, in this transported topology,

$$\frac{\omega}{j} \longrightarrow 0,$$

although ω/j exceeds every real number for every positive standard j . A coefficient limit is not a claim about order convergence of surreal numbers.

40.2 Normal forms for the entire ordinal family

Theorem 40.1 (Explicit omnific embedding; extends [theorem 9.1](#)). *Fix a countably infinite ordinal α . Regard each ordinal $\beta + 1$ as a surreal number and put*

$$\varepsilon_\beta = \frac{1}{\beta + 1} \quad (\beta < \alpha).$$

Then

$$J_\alpha(a, n) = \sum_{\beta < \alpha} a_\beta \omega^{\varepsilon_\beta} + n \quad (40.2)$$

is a well-defined ordered additive embedding of G_α into $\mathbf{Oz}$, sending $(0, 1)$ to 1. Its nonnegative image, with transported coefficient topology, has all the properties of [Theorem 35.1](#) and [Theorem 36.1](#).

Proof. The positive surreal exponents ε_β are strictly decreasing: ordinal inclusion agrees with surreal order, and reciprocation reverses strict order on positive elements of a field. Every subset of $\{\varepsilon_\beta : \beta < \alpha\}$ has a largest element, corresponding to its least index. Thus every support in (40.2) is a set-sized reverse well order, as required for a Conway normal form. The sum denotes this formal normal form, not a limit in an unspecified topology.

Uniqueness of coefficients proves injectivity. Addition of normal forms with a common set of monomials is coefficientwise, proving additivity. For unequal vectors, the least differing β supplies the largest monomial in their difference, so determines its sign. If all rational coordinates agree, the integer constant does. This proves order preservation and reflection.

All displayed exponents are positive, and the constant coefficient is an integer. Hence every value is omnific. The remaining assertions follow by transporting the topology and structure along this isomorphism onto its image. $\square$

[write] For $\alpha = \omega$, J_ω is Part I's map ι_B of [theorem 9.1](#), and $J_{\mathbb{R}}$ is $\iota_{\mathbb{R}}$. Source 06's realizations ([section 40.4](#)) use the exponents $1/(i + 1)$ for $\alpha = \omega$ and $\omega^{-\beta}$ for larger α ; both choices are strictly decreasing sequences of positive surreal exponents. Whether these images are initial in the sense of `isg:cf:main:presburger` is not checked here; the groups are split, so that theorem applies to them abstractly ([section 1.4](#)).

Remark 40.2 (Ordinals, reciprocals, and the omega-map). In (40.2), $1/(\beta + 1)$ is a reciprocal in the surreal field, not ordinal division. Likewise ω^x denotes Conway's monomial map. It should not be silently replaced by a different surreal exponential convention. For $\alpha = \omega$ the exponents are simply $1, 1/2, 1/3, \dots$

Corollary 40.3 (Omnific realization of the continuum family). *For $B_s = \mathbb{Q} + \mathbb{Q}s$ as in [Theorem 35.5](#) and any countably infinite ordinal α , the map*

$$J_{s,\alpha}(b, a, n) = b\omega^2 + \sum_{\beta < \alpha} a_\beta \omega^{1/(\beta+1)} + n$$

is an ordered additive embedding of $B_s \xrightarrow{\times} \mathbb{Q}^\alpha \xrightarrow{\times} \mathbb{Z}$ into $\mathbf{Oz}$. Consequently the continuum family, including the models in [Corollary 36.2](#), also has explicit omnific realizations with its topology transported from the coefficient space.

Proof. All tail exponents are positive and at most 1, so the extra exponent 2 lies strictly above them. The same normal-form argument applies, with the real coefficient $b \in B_s$ determining the leading component and the integer n the constant term. $\square$

40.3 Set size and formal foundations

For a fixed α , the domain $\mathbb{Q}^\alpha \times \mathbb{Z}$ is a set, and the exponents in (40.2) form a set. In the sign-sequence presentation of surreal numbers, the normal-form operation assigns a unique surreal set-code to each input. Replacement therefore makes the whole image a set. Its birthdays have an ordinal supremum; after increasing that bound, the entire image lies in a single set-sized initial segment of the surreal class. The same argument applies to the real model and to each fixed parameter s in the continuum family.

Accordingly, ordinary ZFC suffices to quantify over the model, all its subsets, its topology, and all its finite Cartesian powers. No “set of all surreals” is introduced. The proper class **No** is used as a definable ambient class of set-codes, not as an element of itself or as a domain of unrestricted set comprehension.

The images are additive ordered subgroups. We do not claim that the particular normal-form images are initial under surreal simplicity, closed under surreal multiplication, or closed under arbitrary surreal-analytic operations. Already $(\omega)^2$ is outside the rank-two group $\mathbb{R}\omega + \mathbb{Z}$. Proposition 37.5 also forbids manufacturing a different compatible semiring product internally in these examples.

40.4 Surreal realizations and a formalization route [06]

40.4.1 The same algebra at explicit surreal scales [06]

The connection with the repository’s surreal work is visible directly in Conway normal forms. Using the usual normal-form arithmetic, the map

$$(r, n) \mapsto r\omega + n \quad (40.3)$$

identifies $\mathbb{R} \overrightarrow{\times} \mathbb{Z}$ with an ordered additive subgroup of the omnific integers. The rational-sequence model has the realization

$$((q_i)_{i \in \mathbb{N}}, n) \mapsto n + \sum_{i \in \mathbb{N}} q_i \omega^{1/(i+1)}. \quad (40.4)$$

The exponents are strictly decreasing positive real numbers; every subset of this sequence is reverse well ordered. The expression is therefore a valid set-supported normal form. Its leading coefficient determines the sign, and addition is coefficientwise, exactly as in our lexicographic group. For arbitrary countable α , one can similarly use the strictly decreasing positive surreal exponents $\omega^{-\beta}$, $\beta < \alpha$.

Finally the substitution $t \mapsto \omega$ identifies $\mathbb{Z} + t\mathbb{R}[t]$ with the ring

$$\left\{ n + \sum_{i=1}^d a_i \omega^i : n \in \mathbb{Z}, a_i \in \mathbb{R}, d < \omega \right\},$$

which is explicitly studied in the repository [36]. These identifications use standard surreal normal-form facts, also documented in the cited repository reports [35, 36]; none is needed for the proofs of our Polish models.

Two distinctions matter. First, the subgroup in (40.4) is being used additively; no closure under surreal multiplication is asserted. Second, all Polish topologies here must be *transported* through the displayed identifications. They are not claimed to be inherited from a topology on the proper class **No**. Our constructions are set-sized throughout and need no class of all subsets, global enumeration, or paradox-prone quantification over all surreal numbers.

40.4.2 What the existing formal development certifies [06]

The repository's Presburger development supplies a promising implementation starting point. In the inspected snapshot, its `Syntax.lean` interprets formulas under valuations `List Int`. The central results in `Decision.lean`, including `holds_iff_quantifierEliminate`, therefore concern standard integer semantics. They do not already assert that an arbitrary $\mathbb{Z}$ -group satisfies every eliminated formula.

A faithful formalization can separate four layers.

1. **Algebraic semantics.** Define a discretely ordered abelian group with its least positive element and the standard division/remainder schema. Interpret affine terms and formulas in an arbitrary such group. Generalize the Cooper residue and interval lemmas so they no longer mention `Int` in their semantic valuation type.
2. **Concrete models.** Build the ordered groups $\mathbb{R} \overrightarrow{\times} \mathbb{Z}$, $\mathbb{Q}^{\mathbb{N}} \overrightarrow{\times} \mathbb{Z}$, and the additive group of $\mathbb{Z} + t\mathbb{R}[t]$. Prove the explicit division identity and transfer the generic semantics to their nonnegative cones.
3. **Topology.** Supply the product or disjoint-sum topologies, prove the specified operations continuous, and certify Polishness by the explicit homeomorphisms or complete metrics. The polynomial model requires a topology on the positive cone, not an incompatible claim about a topological signed ring.
4. **Boundary results.** Formalize the cofinal-scale obstruction and the single failed induction instance. The isolated-zero theorem can be a separate later target, adding descriptive-set-theoretic and probability dependencies.

An adjacent interface requires care. The inspected file `PAHF/PASyntax.lean` contains a shallow `PA.Model` interface with induction for every Lean predicate $\alpha \rightarrow \text{Prop}$. Such full external induction excludes the nonstandard objects of interest. The appropriate route uses raw `PreModel` operations, syntactic formulas and satisfaction, and first-order induction instances [37]. For the polynomial construction, one must also keep the additive language distinct from the language with multiplication.

These are proposed formalization steps, not claims that a formalization was run in this project. The deliverable is a self-contained mathematical article; its full proofs make the remaining kernel-checking task precise and reviewable.

41 Formalization scope and a concrete development plan

41.1 What the existing repository actually proves

The repository's executable development has a specific semantic domain. In the inspected source,

```
Formula.holds : Formula -> List Int -> Prop
```

interprets quantifiers over Lean's ordinary integer type. In particular, the existential case is an existential over `Int`. The theorem `holds_iff_quantifierEliminate` proves the computed elimination correct for these integer valuations [32].

This is relevant infrastructure, but does not formally establish that an arbitrary ordered group with a divisible quotient is a Presburger model. It also does not verify a topology

on $\mathbb{R} \times \mathbb{Z}$ or $\mathbb{Q}^\alpha$. The source was inspected during this investigation; no Lean or Rocq build was run. The new theorems in this report are ordinary mathematical proofs, not claims of a successful proof-assistant execution.

[write] The claims about the formal project hold at the repository’s current state: `Syntax.lean` defines `holds : Formula -> List Int -> Prop` with an existential over `Int` (lines 192–197), and `holds_iff_quantifierEliminate` is in `Decision.lean` (line 18); the project’s Lean files are unchanged since both pins. [Section 13.5](#) names the declarations that Part I’s porting requirement concerns. **Nothing in this part is formalized.**

41.2 A small first formalization target

The best first target is Theorem [32.1](#), with the algebra and topology proved in separate modules whose assumptions meet in one final statement. A suitable sequence is:

1. Define the coordinate type $\mathbb{R} \times \mathbb{Z}$, its lexicographic order, and the nonnegative subtype $M_{\mathbb{R}}$. Prove the least-positive property and the explicit division identity ([32.4](#)).
2. Prove, or port, the semantic theorem that a $\mathbb{Z}$ -group satisfies the complete Presburger theory. Generalize formula interpretation from `List Int` to valuations in a structure. Avoid assuming the new model satisfies the theory as an unproved axiom.
3. Put the product topology on the coordinates, independently of the lexicographic order instance. Prove closure under addition, restricted continuity, and Polishness using ([32.2](#)) or the explicit metric ([32.3](#)).
4. Combine the semantic and topological theorems. State cardinality through an explicit injection from $\mathbb{R}_{\geq 0}$, so uncountability is not hidden in a typeclass declaration.

The central implementation risk is accidentally selecting the intrinsic order topology when defining a lexicographically ordered type. That topology is discrete. The product topology used here must be installed explicitly and its compatibility claims stated only for the operations that have actually been proved continuous.

41.3 Extending the verified boundary

Once the real-coordinate construction is complete, the $\alpha = \omega$ Baire model is the next manageable target. Its strict coefficient cones are open, avoiding the transfinite Borel decomposition. A coordinate enumeration, cylinder basis, and the noncompact-cylinder lemma can be implemented without any proper-class formalization.

The general cone theorem requires more library infrastructure: Baire category for closed subgroups, Borel sets’ Baire property, countable-base arguments, and transfinite recursion on closed subgroups. The oscillation theorem needs ordinal arithmetic and a general relative-boundary recursion. Those are substantive formalization tasks, not consequences of running the existing integer decision procedure.

The surreal embedding can be isolated still further. One may first work with a set-sized Hahn additive group supported on the fixed exponent set, prove its coordinate order and addition facts, and then use a separately verified normal-form map into a universe-bounded surreal type. This avoids making the core topological result depend on a full surreal field development.

Layer	Mathematical obligation	Present status
Ordinary integer formulas	Executable Cooper elimination and semantic correctness	Existing repository sources inspected
Generic additive models	$\mathbb{Z}$ -group characterization and full theory	Classical theorem used explicitly
Real Polish model	Cone, metric, continuity, uncountability	Proved in this report
Baire ordinal family	G_δ cone, cylinder argument, nonisomorphism	Proved in this report
Oscillation ranks	Relative derivatives and ordinal calculation	Proved in this report
Omnific embedding	Normal-form existence and coefficient transport	Classical normal forms plus proof here
New Lean/Rocq certificates	Kernel-checked versions of the report's new claims	Not supplied

42 Where Part III meets Parts I and II [merge]

Source 06's ring $A = \mathbb{Z} + t\mathbb{R}[t]$, with t positive infinite, becomes under $t \mapsto X$ a subring of 03's finite principal-part ring $\mathcal{A}_{\text{fin}} = \mathbb{Z} \oplus \bigoplus_{q \in \mathbb{Q}_{>0}} \mathbb{R}X^q$ of [Part II](#): the elements with integer exponents. Both rings are ordered by the sign of the leading coefficient, so $P = A_{\geq 0} = A \cap \mathcal{A}_{\text{fin}, \geq 0}$. Write τ_P for source 06's topology ([38.1](#)) on P .

Proposition 42.1 (The stratum of degree one, and two topologies on P [merge]). *The following hold.*

(i) *The set of elements of degree at most one,*

$$P_{\leq 1} = \{x \in P : \deg x \leq 1\} = \mathbb{N} \cup \bigcup_{m \in \mathbb{Z}} P_{1,m},$$

is clopen in (P, τ_P) . The map $m + at \mapsto (a, m)$ is an isomorphism of ordered additive monoids and a homeomorphism from $P_{\leq 1}$ onto source 07's $M_{\mathbb{R}}^\#$ ([proposition 37.4](#)), and $t \mapsto X$ maps $P_{\leq 1}$ homeomorphically onto the stratum $(C_{\mathbb{R}}, \tau_{\text{FS}})$ of [proposition 26.1](#).

(ii) *On P , the topology τ_P is strictly coarser than the topology induced by 03's τ_{FS} .*

(iii) *Addition and multiplication are continuous on (P, τ_P) ([theorem 38.1](#)), but addition is not continuous on P with the topology induced by τ_{FS} .*

Proof. (i) $P_{\leq 1}$ is a union of components of ([38.1](#)). Each $P_{1,m} = \{m + at : a > 0\}$ carries the topology of $(0, \infty)$ in the coefficient a , and the points of $\mathbb{N}$ are isolated. These are exactly the components of $M_{\mathbb{R}}^\#$, namely the rays $(0, \infty) \times \{m\}$ and the isolated standard numerals, and of $C_{\mathbb{R}}$, namely $U_{\{1\},m}$ and $U_{\emptyset,n}$, $n \geq 0$. The algebra is the lexicographic algebra of $\mathbb{R} \overrightarrow{\times} \mathbb{Z}$ in all three cases.

(ii) A component $U_{S,n}$ of τ_{FS} that lies in P has $S \subseteq \{1, \dots, d\}$ with $\max S = d$, so it lies in $P_{d,n}$, and its inclusion there, which inserts zero coefficients, is continuous. Hence every τ_P -open set meets each such component in an open set, and τ_P is coarser. It is strictly coarser: $t^2 + t/j \rightarrow t^2$ in τ_P , inside the component $P_{2,0}$, while $t^2 + t/j$ and t^2 lie in the different components $U_{\{1,2\},0}$ and $U_{\{2\},0}$ of τ_{FS} .

(iii) Put $x_j = t^2 + (1 + 1/j)t$ and $y = t^2 - t$; both lie in P , and $x_j \rightarrow x = t^2 + t$ for τ_{FS} . Then $x_j + y = 2t^2 + t/j$ lies in $U_{\{1,2\},0}$ while $x + y = 2t^2$ lies in $U_{\{2\},0}$, so $x_j + y$ does not converge to $x + y$. The pair (x, y) lies on the cancellation locus ([24.5](#)) of [theorem 24.3](#). $\square$

Glazer’s Corollary 1 inside one ring. Section 26 placed the real model as a stratum of $\mathcal{A}_{\text{fin}, \geq 0}$, which satisfies $Q + \text{IOpen}$ with Borel multiplication and on which no Borel-compatible Polish topology makes addition continuous (theorem 25.2). The integer-exponent subcone P sits between the two: it carries a Polish topology with both operations continuous, and Corollary 1 does not apply to it because P fails open induction at the single formula $x^2 < t$ (theorem 38.2). In $\mathcal{A}_{\text{fin}}$ the same formula is satisfied by $x = \frac{1}{2}X^{1/2}$, which is not a standard integer. Source 06’s isolated-zero theorem (theorem 39.1) explains why τ_P isolates 0, whereas the additive models of Part I are perfect.

Cone topologies versus group topologies. The same ring is source 08’s $A_* = \mathbb{Z} + X\mathbb{R}[X]$ (example 54.1) and, under $t \mapsto \omega$, the ring $\mathbb{Z} + \omega\mathbb{R}[\omega]$ of corollary 55.1 and `odg:frac:prop:` intersection of the omnific Diophantine report. By source 08’s theorem 44.1, no Hausdorff locally compact second-countable additive-group topology on A has a Borel order, so τ_P , a locally compact Polish topology on the cone, is induced by no such group topology. Source 06 already notes that the signed analogue of τ_P is not a ring topology. This is the situation of source 08’s question 57.2 and source 09’s question 73.11, which ask when a cone topology extends to the group of differences; the Hilbert’s-tenth programme’s triage of the arrival (Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/recent_incoming_substrate_triage_20261003.md) makes the same caveat.

The same failed instance elsewhere. The formula $x^2 < X$ defining the standard cut in $\mathbb{Z} + X\mathbb{Q}[X]$ is used by batch 86’s manuscript 10, written into the report Algebra/Surreal1Numbers/docs/surreal/discrete-initial-subgroups-and-omnific-normalization.[write] [Batch-86 reciprocal notes, 3 October 2026.] There it is source 03, Sections 19–22. It is motivated by the same paper of Glazer but is not topological, and its obstruction precedes any topology: the additive group of the difference ring of a nonstandard model of $\text{I}\Delta_0$ has no nonzero homomorphism to $\mathbb{Z}$ (`isg:sc:thm:nochar`), so no such model has a unit-preserving additive map into **Oz** (`isg:sc:cor:omnific`), and by that report’s Presburger criterion the ordered group completion of its additive monoid is not initially realizable (`isg:sc:cor:fragment`). It answers neither of Glazer’s questions. For **Oz** itself the collection records existential failures of induction (`odg:thm:induction`, `odg:rem:ringinduction`); as source 06 says, the failure for P does not transfer to **Oz**.

43 Further research questions and topics

The following program separates an explicitly published question from questions suggested by the constructions. Unless stated otherwise, these are proposed directions for investigation; we do not assert that every formulation is historically new or unresolved in all existing literature.

[06] Source 06 lists twelve numbered projects rather than question environments. Each one that coincides with a question of source 07 is printed after it, with source 06’s number:

43.1 Published reverse-mathematical question

Question 43.1 (Glazer’s Question 1). Can the stronger topological Tennenbaum theorem in [1] be proved over ATR_0 ?

This report does not answer that question. The affirmative Presburger construction shows that a proposed route from a weaker metatheory to a negative Presburger conclusion cannot yield the latter conclusion as stated. It does not determine the metatheoretic strength of the theorem about arithmetic with multiplication.

[write] Source 06's project with the same subject:

6. **Exact strength of the isolated-zero principle.** Formalize the proof of Theorem 39.1 over weak subsystems of second-order arithmetic using codes for Polish spaces and continuous functions. Locate the strength of analytic/coanalytic separation and of the coded measure argument. This is related to, but does not resolve, Glazer's published Question 1 asking whether his full Theorem 2 is provable in ATR_0 [1].

[write] (Batch 87.) Source 14 claims an affirmative answer, under a stated coding convention (Part VIII, corollary 111.2). The claim is unrefereed and has not been independently reviewed; see section 1.1.

43.2 Classification and topology

Question 43.2 (Which spaces and which additive models?). Characterize the uncountable Polish spaces that support continuous Presburger addition, and characterize which continuum-sized Presburger models admit such a presentation.

The real-coordinate model, Baire-space family, and isolated-standard variant give several positive topological types. Compact spaces are excluded. A useful first refinement is to classify locally compact presentations in terms of their components and the position of the standard numerals. A separate algebraic problem asks how much of the existence theory is captured by split $\mathbb{Z}$ -groups.

[write] Source 06's project with the same subject:

4. **Which Polish spaces support additive arithmetic?** Classify the possible underlying spaces. We have a locally compact perfect example and a nowhere locally compact example. Compact Hausdorff spaces are excluded by [2]. A useful next step is to analyze connected spaces and spaces with prescribed sets of isolated points, while specifying whether a group-completion topology is also required.

[write] (Batch 87.) For locally compact cones inherited from a Borel ordered Polish group, source 15 determines the space: it is countable discrete or the countable sum of lines and closed rays (theorem 121.1); on that one space there are exactly $2^{\aleph_0}$ models (theorem 123.1). The general question stays open.

Question 43.3 (Non-split integer extensions). Which ordered extensions $0 \rightarrow \mathbb{Z} \rightarrow G \rightarrow D \rightarrow 0$ with divisible D , but without a group-theoretic splitting, have positive cones with Polish topologies making addition continuous?

The present construction starts with a splitting. General $\mathbb{Z}$ -groups need not possess it. The residue and extension invariants appearing in the repository's omnific research provide a natural algebraic language for formulating obstructions, but do not automatically decide the topological question.

[write] Source 06's project with the same subject:

2. Which additive groups admit Polish presentations? Find necessary and sufficient conditions for a continuum-sized $\mathbb{Z}$ -group to have a nonnegative cone with a Polish topology and continuous addition. Our split products are a substantial class, but a general $\mathbb{Z}$ -group need not split. In particular, investigate how a nonsplit extension of a divisible ordered group by $\mathbb{Z}$ affects the problem.

[write] (Batch 87.) Existence, for extensions coming from a countable nonsplit core: source 13's [theorem 94.1](#) gives, for every countable $\mathbb{Z}$ -group K , Polish cones of $D \overrightarrow{\times} K$ with continuous addition ($D = \mathbb{R}$ or $\mathbb{Q}^{\mathbb{N}}$), nonsplit when K is, with $2^{\aleph_0}$ nonisomorphic nonsplit examples ([corollary 94.2](#)); sources 14 and 15 give instances. Which extensions admit such topologies is not classified.

Question 43.4 (Connected examples). Does a connected uncountable Polish space support a continuous Presburger presentation?

The examples here have clopen residue classes and hence are disconnected. In an arbitrary presentation, residue classes are Borel by [Proposition 34.3](#); that alone does not make them clopen. The gap between these two regularities is a concrete place to investigate.

[write] (Batch 87.) Negative for cones inheriting a Polish group topology (note after [question 14.3](#)); open otherwise.

43.3 Definability and descriptive invariants

Question 43.5 (Ranks for a fixed model). For one fixed continuous presentation, which boundary oscillation ranks occur among parameter-definable sets? How do they depend on arity and on the chosen parameter?

[Theorem 36.1](#) varies the model and uses a parameter with positive zeroth coordinate. A full parameter spectrum requires analyzing cuts based at vectors with a transfinite first nonzero coordinate, where the boundary of the model itself affects the derivative. It should also distinguish the rank of a relation from the rank of a unary section.

Question 43.6 (Uniform definability bounds). Must every Polish Presburger presentation with continuous addition have all definable relations at some uniformly bounded Borel level? Is the Δ_2^0 bound forced, or special to the ambient-group construction?

[Proposition 34.3](#) proves Borelness in complete generality. [Theorem 34.1](#) proves the stronger level-two conclusion under an additional structural hypothesis. Closing that gap would turn the examples into a general regularity theorem.

[write] For the cones of locally compact signed groups with Borel order, source 08's [theorem 52.1](#) gives the uniform Δ_2^0 bound; for an arbitrary Polish presentation the question stays open.

Question 43.7 (Interpreting the cone closure rank). Which structural features of an ordered Polish group are detected by $\rho(A, <)$? How does the rank behave under closed convex subgroups, quotients, products, and order-preserving topological embeddings?

[Proposition 33.4](#) realizes every countable rank on real coordinate groups. A useful next goal is an intrinsic characterization of the first few ranks and precise inequalities for those operations. The rank records the interaction of order with topology; it need not agree with the algebraic Archimedean rank.

43.4 Arithmetic expansions and formal verification

Question 43.8 (Additional continuous functions). Which natural expansions of Presburger arithmetic still admit uncountable Polish presentations with all named functions continuous?

Fixed-divisor quotient and remainder are positive examples. Total truncated subtraction, minimum, and maximum are impossible individually. Predecessor depends on the presentation. Expansions by selected unary functions, or restricted coding functions, offer an intermediate program before multiplication is considered.

[write] Source 06's project with the same subject:

9. **Effective comparison and definable functions.** For the explicit computable Baire model, calibrate the effective complexity of its order, equality, minimum, and truncated predecessor. The continuous addition algorithm does not decide an infinite stream's equality to zero. A useful target is an exact classification of the discontinuity or computational strength required by each definable operation.

Question 43.9 (Removing the cofinal-multiple obstruction). Construct and analyze continuous Polish Presburger models in which no nonstandard element has cofinal standard multiples. Which of their additive structures admit an ordered-ring expansion at all?

The explicit family in this report cannot answer the latter question positively. Divisible coefficient groups with no greatest Archimedean class are natural candidates for a next stage. Any resulting multiplication must then be tested against the appropriate arithmetic axioms and Glazer's regularity obstruction.

[write] Source 06's project with the same subject:

3. **Presburger reducts of genuine PA models.** Can the additive reduct of a nonstandard model of PA have an uncountable Polish continuous-addition presentation if its multiplication is allowed to be non-Borel? Glazer's published obstruction rules out a Borel multiplication under stronger arithmetic hypotheses. Our lexicographic examples do not answer this question, because their additive structure already excludes a semiring multiplication.

Question 43.10 (Weak foundations and certified models). What is the weakest useful coded metatheory for the real and Baire constructions? Can the generic Presburger semantic theorem and the Polish topology be joined in a short audited Lean development?

The explicit coordinate formulas suggest a substantially lighter proof than the general transfinite cone theorem. A careful analysis must still specify codes for Borel sets, complete metrics, countable bases, and satisfaction of first-order formulas. Section 41 gives an implementation order with a concrete first target.

[write] Source 06's project with the same subject:

1. **Formal generic Presburger semantics.** Complete the four-layer route in Section 40.4, beginning with generic $\mathbb{Z}$ -group soundness for the repository's Cooper eliminator. The first deliverable should be a theorem about all models of the division/remainder axioms, followed by the real cone as its first nonstandard instance. This would convert the main answer into a directly auditable formal result.

Question 43.11 (Simplicity and coefficient topologies). Which initially realized omnific additive groups admit coefficient topologies of this kind, and which initiality operations preserve Polishness and continuity of addition?

Theorem 40.1 supplies an explicit ordered embedding without asserting initiality of the displayed image. Combining a proved initial realization theorem with a transported Polish topology is straightforward as an abstract existence argument; preserving a natural coefficient description during that transport is a sharper and more informative goal.

[write] Source 06's project with the same subject:

12. **Surreal topology without class-level assumptions.** Study the transported topologies on set-sized omnific subgroups and subrings such as (40.3) and (40.4). Ask which algebraic embeddings are continuous for these explicit topologies. Keep the support set, coefficient topology, and ambient set theory specified; the proper class of all surreal numbers is not itself an ordinary Polish space.

43.5 Further projects of source 06 [06]

The remaining projects of source 06 have no counterpart among source 07's questions.

5. **How many isolated points are necessary for semirings?** Theorem 39.1 forces zero to be isolated, and the polynomial example isolates every standard numeral. Determine whether a Polish topological PA^- model can have zero as its only isolated point, or classify which patterns of isolated numerals are possible. Requiring continuous predecessor is an additional hypothesis and should be treated separately.
7. **Completeness and separability thresholds.** Find sharp replacements for Polishness in the isolated-zero theorem. Completeness supplies limits of all subset sums, and separability supplies descriptive regularity of the order. One can test these roles separately using incomplete metrizable models and complete nonseparable presentations.
8. **Classification beyond cardinality.** Theorem 35.8 gives the exact number of isomorphism types but no full classification. Determine the descriptive complexity of isomorphism for suitably coded Polish Presburger presentations. Convex quotients and residue extensions provide natural invariants. Compare with the countable-model Scott analysis in [25], without assuming that countable and Polish presentations have identical classification behavior.
10. **Coefficient fields and other support families.** Replace $\mathbb{R}$ in the polynomial construction by an ordered Polish topological field whose positive cone is Polish. The same degree-stratum proof applies whenever coefficient operations are continuous. More ambitiously, determine which infinite support families allow a Polish positive-cone topology with both operations continuous, and identify where leading-term cancellation or completeness obstructs such a topology.
11. **A bridge from additive induction to open induction.** The formula $x^2 < t$ isolates one exact failure in the polynomial model. Seek a useful classification of additional multiplication principles that remain compatible with a Polish continuous semiring presentation. The class must stop before the hypotheses of Glazer's additive obstruction, but the interval between additive induction and IOpen contains many distinct theories.

[merge] Relation to Part I's questions. Source 07's question on spaces and models and its connected-carrier question, with source 06's project 4, are question 14.3 extended to all carriers; source 07's question on weak foundations and source 06's project 1 continue questions 14.1 and 14.2; the cofinal-multiple question and project 3 continue question 14.4. Source 06's

[theorem 38.1](#) answers the first sentence of source 05’s part of that question affirmatively: its cone is an explicit uncountable Polish continuous-addition Presburger model whose additive structure carries a unital semiring multiplication, continuous for the same topology (the note after [question 14.4](#) records this). It fails open induction ([theorem 38.2](#)), so the clause of that question about $Q + I\text{Open}$, and source 04’s growth-order question, stay open. Project 6 and source 07’s comment on Glazer’s Question 1 bear on the speculation discussed in [section 1.1](#).

Part IV

Local compactness forces arithmetic rigidity

[write] This part is source 08 of batch 86, *Local Compactness Forces Arithmetic Rigidity: Borel ordered groups, Presburger models, and multiplication without regularity assumptions* (dated October 3, 2026; “prepared with ChatGPT”), pinned to repository commit 883e0b3b2. All unmarked text is source 08’s; its labels carry the prefix `pma:lc:`. Source 08 cites two earlier manuscripts as “user-library” files: [40] is source 05, the base of [Part I](#), and [41] is source 03, [Part II](#). Its finite checks are shipped as `code/08-local-compactness-verify.py` with the record `data/08-local-compactness-verification_results.json` ([section K](#)). Its central theorem is complementary to source 09’s [theorem 62.1](#) in [Part V](#): both make an uncountable discretely ordered ring impossible under a topology, 08 assuming local compactness and no regularity of multiplication, 09 assuming Borel scalar maps and no local compactness. Neither result bears directly on Glazer’s Question 2, which concerns models without multiplication ([section 1.1](#)).

[write] (Batch 87.) [Part VI](#) removes local compactness: [theorem 44.1](#) and source 09’s [theorem 62.1](#) are both special cases of [corollary 78.5](#), which assumes only a Polish (indeed a Hausdorff Baire ccc) additive-group topology and a cone with the Baire property ([section 90](#)).

44 The intersection and the contribution

The point of contact is the topology of arithmetic presentations. In *A Topological Tennenbaum Theorem*, Elliot Glazer rules out continuous addition with Borel multiplication for uncountable Polish models of $Q + I\text{Open}$; his Question 2 asks for an uncountable Polish Presburger model with continuous addition [[1](#), Corollary 1 and Question 2]. ProveIt’s relevant interfaces are its Cooper-elimination development and its ordered surreal and omnific constructions [[38](#), [39](#)]. These suggest separating three issues: addition, the regularity of order, and the possibility of installing multiplication.

The earlier companion manuscript *Polish Presburger Arithmetic inside the Omnific Integers* already constructs the model

$$M_{\mathbb{R}} = (\mathbb{R}_{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}) \quad (44.1)$$

with continuous addition, and gives example-specific multiplication obstructions [[40](#)]. Repeating that construction would not be an advance. The question pursued here is structural:

What does local compactness force on every Borel ordered additive-group presentation, and can multiplication evade the obstruction by being completely nonmeasurable?

Our answer is negative for discretely ordered rings. The obstruction occurs before induction, definability of multiplication, or a choice of particular normal forms enters the argument.

[write] Equation (44.1) is Part I's real model (theorem 3.1); source 07's theorem 32.1 and source 06's theorem 32.2 prove it again.

44.1 The central theorem

Theorem 44.1 (Arithmetic rigidity). *Let A be a nonzero commutative unital ring with a total order such that addition preserves order, products of positive elements are positive, and 1 is the least positive element. Suppose that A has a Hausdorff locally compact second-countable topology making $(A, +, -, 0)$ a topological group, and that $A_{>0}$ is Borel. Then the topology is discrete and A is countable. No regularity assumption is imposed on the multiplication map.*

The proof is given in section 48. Its two essential parts are an algebraic growth argument and a classical topological structure argument. The former produces more cosets than the latter permits. In particular, it does not attempt to pass a discontinuous multiplication through a limit.

44.2 Other results proved here

The structural conclusions have independent content even when no ring multiplication exists. For signed Presburger models, we obtain an exact classification of the *pointed topological additive groups*, the local-compactness threshold for their positive cones, and a uniform Δ_2^0 bound on all first-order definable relations. We do not claim that every order itself is the standard product lexicographic order.

There is also a second, distinct obstruction. On these same additive groups, an arbitrary integral-domain multiplication, not required to respect the specified order, has only countably many scalars whose multiplication maps are Borel. For the additive type $\mathbb{R}^d \times \mathbb{Z}$, these scalars are exactly the ordinary integers. This theorem is not vacuous: an explicit algebraic transport construction supplies such ring structures.

44.3 What is classical, and what remains uncertified

Measurable orders on locally compact abelian groups were studied by Hewitt and Koshi and by Asmar and Montgomery-Smith [26, 22]. The compact-group obstruction, finite-dimensional separation arguments, and the structure theorem for locally compact abelian groups are classical inputs, not discoveries claimed here. We supply the needed proofs except for the principal structure theorem itself. Presburger quantifier elimination is likewise classical [28]; section 49 records the version needed for nonstandard groups.

The contribution proposed for independent review is the amplification and arithmetic-rigidity package, together with its classification, scalar, and omnific consequences. A bounded literature search did not establish priority for these formulations. In particular, lack of a matching search result is not evidence of a first proof. The existing companion constructions are cited as prior work, not relabelled as new.

45 Conventions, topology, and foundations

All groups in this paper are abelian and written additively. An ordered group has a translation-invariant *total* order. The associated strict positive cone is $P = G_{>0}$, and $G_{\geq 0} = P \cup \{0\}$. For a topological group, P is Borel if and only if the binary order relation is Borel: one direction takes a section at zero, and the other uses the continuous difference map $(x, y) \mapsto y - x$.

The phrase *discretely ordered* refers to the *algebraic order*: there is a least positive element, denoted 1. It does not say that the presentation topology is discrete. In a ring, $\underline{n} = n1$ denotes the ordinary integer numeral. Throughout, n, m, N used as moduli, finite indices, or bounds are external ordinary integers, not arbitrary elements of a nonstandard model. We write $|x| = \max(x, -x)$.

A Hausdorff locally compact second-countable topological group is Polish. Here, a Polish space is a separable completely metrizable space. The notation Δ_2^0 means both F_σ and G_δ in the specified topology (usually Polish); Δ_1^0 means clopen. No order topology is silently substituted for the given topology.

Definition 45.1 (Order unit). If V is an ordered additive subgroup, an *order unit* in V is an element $u > 0$ in V such that for every $v \in V$ there exists an ordinary integer $N \geq 1$ with $|v| \leq Nu$.

This is an additive notion. It neither says that u is a ring unit nor that V is closed under multiplication. The distinction is indispensable in the main proof.

All theorems are statements about sets in ZFC. When surreals are mentioned, we work with explicitly set-sized subgroups or subrings of **No** or **Oz**. We never equip the proper class of all surreals with a Polish topology, form a set of all its subsets, or apply Haar measure to a proper class. The algebraic splitting arguments use the ordinary ZFC fact that divisible abelian groups are injective. No assertion about the minimum choice or reverse-mathematical strength of the complete proof is made.

46 An algebraic amplification theorem

The key obstruction can be proved without topology. The strongest useful hypothesis is weaker than divisibility.

Lemma 46.1 (Finite elements are ordinary integers). *In a discretely ordered ring A , every element of $[0, \underline{n}]$, for an ordinary $n \in \mathbb{N}$, is one of $\underline{0}, \dots, \underline{n}$. Consequently, for an additive subgroup $V \subseteq A$ the following are equivalent:*

1. $V \cap \mathbb{Z}1 = \{0\}$;
2. for every $v \in V \setminus \{0\}$ and every ordinary $n \geq 0$, $|v| > \underline{n}$.

Every divisible additive subgroup of A satisfies these conditions.

Proof. The interval assertion follows by external induction on n . If $0 < x \leq \underline{n+1}$, then $x \geq 1$ and $0 \leq x - 1 \leq \underline{n}$; the induction hypothesis applies. This proves the equivalence because $|v| \in V$.

For the last assertion, take $v > 0$ in a divisible subgroup and an ordinary $n \geq 0$. Write $v = (n+1)w$ with $w \in V$. Ordered-group torsion-freeness and positivity imply $w > 0$, hence $w \geq 1$. Therefore $v \geq \underline{n+1} > \underline{n}$. Apply the same argument to $-v$ for negative elements. $\square$

Theorem 46.2 (Coset and polynomial amplification). *Let A be a discretely ordered ring, and let $V \subseteq A$ be a nonzero additive subgroup with $V \cap \mathbb{Z}1 = \{0\}$. Suppose $u \in V$ is an order unit. Then:*

1. $uV \cap V = \{0\}$, and the additive map

$$V \longrightarrow A/V, \quad v \longmapsto uv + V \quad (46.1)$$

is injective. In particular, $|A/V| \geq |V|$.

2. The finite-sum map

$$\Psi : \bigoplus_{j \geq 0} V \longrightarrow A, \quad (v_j)_{j \geq 0} \longmapsto \sum_{j \geq 0} u^j v_j \quad (46.2)$$

is an injective additive homomorphism. If the source is ordered by the sign of its highest-index nonzero coefficient, then Ψ is an order embedding.

3. The induced additive map $\bigoplus_{j \geq 1} V \rightarrow A/V$ is injective.

Proof. Suppose $uv \in V$ for $v \neq 0$. Since $u > 0$, the order-unit condition provides an ordinary N with

$$u|v| = |uv| \leq Nu.$$

Cancellation of the positive factor u gives $|v| \leq \underline{N}$, contrary to [lemma 46.1](#). Thus $uv \in V$ implies $v = 0$, which proves the first assertion and injectivity of (46.1).

For the second assertion, take a nonzero finite tuple and let k be its largest nonzero index. If $k = 0$, the result is immediate. For each $j < k$ choose an ordinary $N_j \geq 1$ with $|v_j| \leq N_j u$. Put $N = \sum_{j < k} N_j$. By [lemma 46.1](#), $u > 1$ and $|v_k| > \underline{N}$. The lower-degree terms satisfy

$$\left| \sum_{j < k} u^j v_j \right| \leq \sum_{j < k} u^j |v_j| \leq \sum_{j < k} N_j u^{j+1} \leq Nu^k < u^k |v_k|. \quad (46.3)$$

Therefore the sum in (46.2) is nonzero and has the sign of v_k . Additivity is distributivity. Applying the sign statement to the difference of two tuples proves the claimed order embedding.

Finally, if a finite sum $\sum_{j \geq 1} u^j v_j$ belongs to V , subtract that element as the zeroth coefficient. Injectivity of Ψ makes all coefficients zero, proving the third assertion. $\square$

Remark 46.3 (What the polynomial language does not mean). The domain of Ψ is an ordered additive direct sum. In general V is not a coefficient ring, and the theorem does not define a ring $V[t]$ or an evaluation homomorphism from such a ring. The quotient A/V is also only an additive quotient: V need not be an ideal or convex.

Corollary 46.4 (Divisible-subgroup obstruction). *If a discretely ordered ring contains an uncountable divisible additive subgroup V with an order unit, then A/V is uncountable. More precisely, A/V contains an additive copy of a countable direct sum of copies of V .*

The strength of this result is its insensitivity to the regularity of multiplication. The map (46.1) may be arbitrarily nonmeasurable; its existence as an injection is already enough for a contradiction.

47 Borel orders on locally compact additive groups

This section assembles the classical topological ingredients in the form needed later. The measurable-order literature [26, 22] contains broader structure results. Borelness is useful here because it survives restriction to every closed subgroup.

47.1 A compact subgroup cannot carry the restricted order

Lemma 47.1 (Compact-order obstruction). *A second-countable topological abelian group with a Borel invariant total order has no nontrivial compact subgroup.*

Proof. Let K be a compact subgroup. Ordered groups are torsion-free, so a nontrivial K cannot be finite. Let μ be normalized Haar measure on an infinite K , and set $P_K = K \cap G_{>0}$. Singletons have measure zero: otherwise arbitrarily many disjoint translates of one singleton would have total measure greater than one. Since inversion preserves Haar measure, the partition

$$K = P_K \dot{\cup} (-P_K) \dot{\cup} \{0\}$$

gives $p := \mu(P_K) = 1/2$.

For every $t \in K$, the two upper rays P_K and $P_K + t$ are nested. Their measures are equal by translation invariance. Thus

$$\mu(P_K \cap (P_K + t)) = p \quad (t \in K).$$

On the other hand, the relevant indicator on $K \times K$ is Borel; second countability permits the usual product-measure Fubini theorem. Consequently

$$\begin{aligned} \int_K \mu(P_K \cap (P_K + t)) d\mu(t) &= \int_K \int_K 1_{P_K}(x) 1_{P_K}(x - t) d\mu(x) d\mu(t) \\ &= \int_K 1_{P_K}(x) p d\mu(x) = p^2. \end{aligned}$$

Hence $p = p^2$, contradicting $p = 1/2$. □

47.2 The connected component is an open real vector group

Theorem 47.2 (Topological additive decomposition). *Let G be a Hausdorff locally compact second-countable abelian group with a Borel invariant total order. There are a finite $d \geq 0$ and a countable discrete torsion-free abelian group D such that*

$$G \cong_{\text{top.grp}} \mathbb{R}^d \times D. \quad (47.1)$$

The identity component $C = G^\circ$ is open and is the $\mathbb{R}^d$ factor.

Proof. The principal structure theorem for locally compact abelian groups supplies an open subgroup $H \cong \mathbb{R}^d \times K$, where K is compact; see [27, p. 4] and the classical references recorded there. The image of K is a compact subgroup of G , so lemma 47.1 forces $K = 0$. Thus $H \cong \mathbb{R}^d$ is open and connected. It is also closed, as every open subgroup is closed. Therefore $H = G^\circ = C$.

As an abstract abelian group, C is divisible and hence injective. There is a homomorphism $r : G \rightarrow C$ extending the identity of C . The restriction of r to the open neighborhood C of

zero is continuous, so r is continuous at zero and hence everywhere. Let $D = \ker r$. Each coset of C meets D in exactly one point; in particular D is discrete. The maps $(c, e) \mapsto c + e$ and $g \mapsto (r(g), g - r(g))$ are continuous inverses. Second countability makes the family of disjoint open cosets countable. Finally D is torsion-free because it is a subgroup of the ordered group G . $\square$

For completeness, the injectivity fact just used can be obtained by the usual maximal-extension argument. Given a homomorphism from a subgroup B of an abelian group into a divisible group, adjoin one new element x . If a least positive n has $nx \in B$, choose an n th root of its image; otherwise assign any image to x . Each choice extends the homomorphism to $B + \mathbb{Z}x$. Zorn's lemma then gives an extension to the whole group. Its use here is purely algebraic, with continuity recovered from the openness of C .

47.3 A Borel order on the real component is a linear flag

Lemma 47.3 (Finite-dimensional Borel-order lemma). *Every Borel invariant total order on $(\mathbb{R}^d, +)$ is, after an invertible real-linear change of coordinates, the standard lexicographic order. For $d > 0$ it has a positive order unit.*

Proof. Proceed by induction on d . The assertion is trivial for $d = 0$. Let P be the strict positive cone when $d > 0$. Ordered divisibility implies that P is stable under positive rational scaling and rational convex combinations. The sets $P, -P, \{0\}$ cover $\mathbb{R}^d$, so the Baire category theorem implies that P is not meager. Since P is Borel, it has the Baire property and is comeager in some nonempty open set U .

If $z \in U + U$, then $U \cap (z - U)$ is nonempty and open. Both P and $z - P$ are comeager in this intersection, so it contains $x \in P$ with $z - x \in P$. Hence $z \in P + P \subseteq P$. This proves that P has nonempty interior.

Let $F = \overline{P}$. Rational convexity and passage to limits show that F is a closed convex real cone. It is proper: the nonempty open set $-\text{int}(P)$ is disjoint from P , and therefore from its closure. Finite-dimensional separation yields a nonzero real-linear functional ℓ with $\ell(F) \subseteq [0, \infty)$. Totality then gives

$$\{x : \ell(x) > 0\} \subseteq P \subseteq \{x : \ell(x) \geq 0\}. \quad (47.2)$$

Indeed, an element with positive ℓ cannot be zero or negative. The restriction of the order to $\ker \ell$ is again Borel. Apply the induction hypothesis there, and extend its coordinate functionals to $\mathbb{R}^d$, with ℓ as first coordinate. Equation (47.2) and the induction hypothesis give the required lexicographic rule.

In those coordinates, $u = (1, 0, \dots, 0)$ is an order unit: for any $v = (v_1, \dots, v_d)$ choose an ordinary integer $N > |v_1|$. Then $-Nu < v < Nu$. $\square$

Remark 47.4. We did not assume that the given order respects multiplication of vectors by arbitrary positive real numbers. That compatibility is a conclusion of Borelness and finite dimensionality. Applying an ordered-vector-space classification without first establishing this point would leave a gap.

48 Proof and scope of arithmetic rigidity

Proof of theorem 44.1. Apply theorem 47.2 to the additive group of A . Its identity component C is an open real vector group $\mathbb{R}^d$, and A/C is countable. If $d > 0$, then C is uncountable and

divisible. By [lemma 47.3](#), the restricted order has an order unit $u \in C$. The amplification theorem therefore gives an injection

$$C \hookrightarrow A/C, \quad c \mapsto uc + C.$$

This contradicts countability of A/C . Hence $d = 0$. The zero subgroup is then open, so the topology is discrete. A second-countable discrete space is countable. $\square$

[write] Source 09 reaches the same conclusion, discreteness and countability, without local compactness but with Borel scalar maps and a cone with the Baire property ([theorem 62.1](#)); the two hypotheses are incomparable. Source 06's [theorem 39.1](#) is a third result of the kind, on the nonnegative cone: if addition and multiplication are both continuous for a Polish topology on the cone, then 0 is isolated.

[write] (Batch 87.) The two hypotheses are no longer incomparable in effect: both conclusions follow from [corollary 78.5](#) of [Part VI](#), which needs neither local compactness nor Borel scalars.

Corollary 48.1 (No compatible ordered multiplication). *Let G be an uncountable discretely ordered abelian group with a Borel order and a locally compact Polish group topology. There is no commutative unital ring multiplication on its underlying set, with unit the named least positive element, which respects the specified order. This remains true when multiplication is allowed to be non-Borel.*

Corollary 48.2 (Positive-cone semirings). *Let G satisfy the hypotheses of [corollary 48.1](#), and let $M = G_{\geq 0}$. There is no commutative unital semiring multiplication on M with the given addition and unit, such that the product of two nonzero elements is nonzero.*

Proof. The cancellative additive monoid M has Grothendieck group G . A commutative semiring multiplication extends by distributivity to this group: $(a-b)(c-d) = ac + bd - ad - bc$. The assumed absence of zero divisors on positive elements makes products of positive elements positive in G . It would therefore yield the ring forbidden by [corollary 48.1](#). $\square$

The group-topology hypothesis is important. A topology on a cancellative positive monoid need not automatically give its Grothendieck group the same desirable separation and local-compactness properties. Thus our result is not a theorem about every possible Polish presentation of a positive arithmetic model. Similarly, the theorem does not replace the general Polish-space result of [\[1\]](#); it is a different refinement with a stronger topological assumption and much weaker multiplicative assumptions.

The proof uses no standard-residue axiom, no square roots, no induction scheme in the ring, and no compatibility between multiplication and limits. The only induction in [lemma 46.1](#) is ordinary external induction on a finite integer. These distinctions make the theorem suitable for applications to weak arithmetic and integer parts.

49 Signed Presburger arithmetic and a uniform elimination lemma

Definition 49.1 ($\mathbb{Z}$ -group). A $\mathbb{Z}$ -group is an ordered abelian group with least positive element 1 such that for each ordinary $m \geq 1$ and each $x \in G$ there are unique $q \in G$ and $r \in \{0, \dots, m-1\}$ satisfying

$$x = mq + r1. \tag{49.1}$$

Equivalently, $\mathbb{Z}1$ is a convex subgroup and $G/\mathbb{Z}1$ is divisible. For the implication from quotient divisibility, write $x = mq + k1$, perform ordinary Euclidean division on k , and absorb its quotient into q . Uniqueness follows because $|r - s| < m$ cannot be a nonzero multiple of m in a discretely ordered group. The converse follows immediately from (49.1). Convexity of $\mathbb{Z}1$ follows from the finite-interval argument of lemma 46.1, which only needs the group order.

We use the language

$$\mathcal{L}_{\text{Pres}} = \{+, -, 0, 1, <, \equiv_m: m = 2, 3, \dots\}, \quad x \equiv_m y \iff x - y \in mG.$$

The equivalence between $\mathbb{Z}$ -groups and models of signed Presburger arithmetic, and quantifier elimination in this language, are standard [28]. We give the finite-witness argument to identify precisely what transfers from ordinary integers to nonstandard groups.

Lemma 49.2 (Coset-invariant finite witnesses). *Fix an ordinary $m \geq 1$. Let $P(y)$ be a predicate invariant under adding every element of mG . Let a_i, b_j be finite lists of elements of a $\mathbb{Z}$ -group G . Then*

$$\exists y \left(\bigwedge_i a_i < y \wedge \bigwedge_j y < b_j \wedge P(y) \right)$$

is decided by a finite test as follows. If lower bounds occur, let $a = \max_i a_i$ and test $y = a + s1$ for $1 \leq s \leq m$. If only upper bounds occur, let $b = \min_j b_j$ and test $y = b - s1$ for $1 \leq s \leq m$. If no bounds occur, test $y = r1$ for $0 \leq r < m$.

Proof. Only completeness of the candidate list needs proof. Suppose first that $y > a$ is a witness. Euclidean division of $y - a$ gives $y - a = mq + r1$, $0 \leq r < m$. If $r > 0$, then $q \geq 0$; otherwise $q \leq -1$ would imply $y - a \leq -1$. Set $s = r$ in this case. If $r = 0$, positivity implies $q \geq 1$, and we set $s = m$. In either case

$$a < a + s1 \leq y, \quad y - (a + s1) \in mG.$$

Thus $P(a + s1)$ holds and every upper bound remains satisfied. With only upper bounds, apply the reflected argument to $b - y > 0$. With no bounds, use the remainder of y modulo mG . The reverse implications are immediate because every accepted candidate is itself a witness. $\square$

Theorem 49.3 (Classical Presburger elimination, in the needed form). *The axioms of $\mathbb{Z}$ -groups admit quantifier elimination in $\mathcal{L}_{\text{Pres}}$ and form the complete theory of $(\mathbb{Z}, +, -, 0, 1, <, \equiv_m)_{m \geq 2}$. Their nonnegative cones are models of the full first-order theory $\text{Th}(\mathbb{N}; 0, 1, +, <)$, including all its induction instances.*

Proof. After disjunctive normal form, it suffices to eliminate one existential variable from a conjunction. Negated inequalities and equalities become finite combinations of strict inequalities using discreteness. Negated congruences become finite disjunctions of the other ordinary residue classes. Atoms not involving the eliminated variable are carried along.

Normalize the nonzero integer coefficients of the eliminated variable x by taking an ordinary common multiple $L > 0$ of their absolute values and replacing Lx by y , with the additional condition $y \in LG$. For example, with $a \neq 0$ and $k = L/|a|$, an inequality $ax < t$ becomes $\text{sgn}(a)y < kt$, and

$$ax + t \in nG \iff \text{sgn}(a)y + kt \in nkG.$$

The equivalence uses torsion-freeness: $kz \in nkG$ if and only if $z \in nG$. Equalities with coefficient 1 or -1 permit substitution. What remains is a finite collection of upper and lower bounds on y and a Boolean combination $P(y)$ of congruences. This predicate is invariant under mG , where m is a common multiple of the finitely many moduli, including L .

Apply [lemma 49.2](#). Choosing a maximum or minimum among a finite list can itself be expressed by a finite disjunction of comparisons. Thus the output is quantifier-free. Repeating eliminates all quantifiers. A quantifier-free sentence concerns only ordinary integer numerals, their order, and their standard residues, so it has the same truth value in every $\mathbb{Z}$ -group as in $\mathbb{Z}$. This proves completeness.

Finally, the nonnegative cone is definable, and any formula in its language translates to a group formula by restricting its quantifiers to $x \geq 0$. The cone therefore satisfies exactly the first-order sentences true in $\mathbb{N}$. In particular the translated induction instances hold; they are consequences, not additional assumptions of the construction. $\square$

[write] [Lemma 49.2](#) is Part I's coset-invariant finite search, [lemma 4.7](#) (source 05) and source 04's [theorem 4.8](#), and [theorem 49.3](#) is the classical [theorem 4.9](#); source 08 proves them again in its own notation.

Remark 49.4 (The transfer condition is coset invariance). For nonstandard G , invariance under the single ordinary shift $m1$ does not by itself imply invariance under mG . The latter is what the proof needs and what congruence formulas supply. This distinction already appears in the earlier companion manuscript [40]. ProveIt's integer-specific periodicity lemmas must not be generalized merely by replacing the carrier $\mathbb{Z}$ with an arbitrary ordered group.

50 An exact pointed topological-group classification

Theorem 50.1 (Locally compact signed Presburger groups). *Let $(G, +, <, 0, 1)$ be a $\mathbb{Z}$ -group equipped with a locally compact Polish group topology and a Borel order. There exist a finite $d \geq 0$ and a countable $\mathbb{Z}$ -group $(E, +, <_E, 0, 1_E)$ such that*

$$(G, +, 1) \cong_{\text{top.grp},1} (\mathbb{R}^d \times E, +, (0, 1_E)), \quad (50.1)$$

where E is discrete. On the identity component the given order is real-linearly conjugate to the lexicographic order.

Conversely, every such pair (d, E) admits a Borel $\mathbb{Z}$ -group order on the displayed pointed topological group. The theorem classifies the pointed topological additive groups, not all possible full orders on them.

Proof. By [theorem 47.2](#), write $G = C \oplus D$ topologically, with $C \cong \mathbb{R}^d$ and D countable discrete. A divisible subgroup cannot contain a nonzero ordinary integer by [lemma 46.1](#). Hence $C \cap \mathbb{Z}1 = 0$.

Write the named element as $1 = (v, e)$ in this splitting. The element $e \in D$ has infinite order: otherwise a nonzero ordinary multiple of 1 would belong to C . Define a homomorphism on $\mathbb{Z}e$ by $ne \mapsto nv$ and extend it to $f : D \rightarrow C$ using divisibility of C . Since D is discrete, f is continuous. The shear

$$(c, x) \mapsto (c - f(x), x)$$

is a topological group automorphism taking 1 to $(0, e)$. Transport the order along it and let $E = \{0\} \times D$ with the restricted order and named unit $1_E = e$.

The element 1_E is least positive in E , because it is least positive in the whole group. If $(0, x) = m(c, y) + r(0, e)$ is standard division in G , then $mc = 0$ forces $c = 0$. Thus the quotient lies in E , proving its standard-residue property. The component order is handled by [lemma 47.3](#).

Conversely, put the usual lexicographic order on $\mathbb{R}^d$, and compare pairs first in this vector coordinate and then in E . The element $(0, 1_E)$ is least positive; vector divisibility together with division in E gives (49.1). The order is Borel because the vector comparison is a finite union of coordinate conditions and E is countable discrete. This constructs every asserted topological type. $\square$

Remark 50.2 (Why the order caveat matters). The component G° need not be convex in the full ordered group. For example, take the ordered additive group with coordinates $(r, q, s, n) \in \mathbb{R} \times \mathbb{Q} \times \mathbb{R} \times \mathbb{Z}$, compared in that order, and give $\mathbb{R}^2 \times (\mathbb{Q} \times \mathbb{Z})$ the product topology with the second factor discrete. This is a $\mathbb{Z}$ -group: the first three coordinates are divisible and n supplies the remainder. Its connected component has $q = n = 0$, but a positive element in the q coordinate lies between positive elements at the two real scales. Thus countable discrete scales can be interleaved with real scales.

There is a precise invariant distinguishing this example from a block lexicographic product with all real coordinates first. Define

$$B_C = \{g \in G : |g| < c \text{ for every } c \in C_{>0}\}.$$

In a block product $\mathbb{R}^d \times E$ with the vector coordinate compared first, $B_C = \{0\} \times E$, so $G = C + B_C$. In the displayed interleaved example, comparison with arbitrarily small positive vectors in each real coordinate shows that B_C consists exactly of the integer coordinate. Hence $C + B_C$ misses every element with $q \neq 0$. Both C and B_C are preserved by an ordered topological-group isomorphism, so the example has no such block-order presentation. This does not affect the pointed additive-group decomposition proved above.

51 Positive cones: exact regularity and the dimension threshold

A useful feature of finite-dimensional lexicographic order is that even its arbitrary cuts have low descriptive complexity.

Lemma 51.1 (Cuts in finite lexicographic powers). *Every upper set in $\mathbb{R}^d$ with the lexicographic order is Δ_2^0 in the usual Euclidean topology. No definability hypothesis on the upper set is needed.*

Proof. Induct on d . For $d = 0$ there are only the empty and full sets. For $d > 0$, let U be a nonempty proper upper set. Consider the first coordinates of its elements. They form a nonempty upper subset of $\mathbb{R}$ with a finite lower endpoint a . The endpoint is finite because an excluded first coordinate gives a lower bound and an included one gives an upper bound for the infimum. If all first coordinates occurred, the upper-set condition would make U the whole space, contrary to assumption. For first coordinate $> a$ every point belongs to U , and for first coordinate $< a$ none does. Thus

$$U = \{x : x_1 > a\} \cup (\{a\} \times U'), \quad (51.1)$$

where U' is an upper set in lexicographic $\mathbb{R}^{d-1}$, possibly empty or full. By induction U' is both F_σ and G_δ . Equation (51.1) expresses U as an F_σ set. The corresponding formula for its complement does the same, so U is also G_δ . Empty and full upper sets are immediate. $\square$

Theorem 51.2 (Cone regularity and local compactness). *Let G satisfy [theorem 47.2](#), and write $G^\circ \cong \mathbb{R}^d$. Then:*

1. $G_{\geq 0}$ and the order relation are Δ_2^0 ;
2. $M = G_{\geq 0}$ with its subspace topology is Polish, and addition $M \times M \rightarrow M$ is continuous;
3. M is locally compact if and only if $d \leq 1$.

Proof. The cosets of $C = G^\circ$ are a countable family of clopen subsets of G . Translate each to C and use the linear coordinates of [lemma 47.3](#). In each coset the intersection with M is an upper set for the induced lexicographic order. By [lemma 51.1](#), each such intersection and its relative complement are F_σ in the coset, and hence in G . Taking countably many unions shows that both M and $G \setminus M$ are F_σ . The order relation follows by the continuous difference map. This proves the first assertion.

A G_δ subspace of a Polish space is Polish [8]. The addition on M is the restriction of the continuous group operation, with its image in the subspace M , giving the second assertion.

If $d = 0$, M is discrete. If $d = 1$, each nonempty coset slice is a real line or an open or closed half-line, by the proof of [lemma 51.1](#). Each is locally compact. As the slices are open in M , their union is locally compact.

Suppose now $d \geq 2$. The slice at the identity component is the nonnegative lexicographic cone in $\mathbb{R}^d$. If M had a compact neighborhood K of zero, then for some $\epsilon > 0$ its intersection with C would contain all points of this cone in the Euclidean ball of radius ϵ . In particular, for sufficiently large n ,

$$x_n = (\epsilon/n, -\epsilon/2, 0, \dots, 0) \in K.$$

But in G this sequence converges to

$$x = (0, -\epsilon/2, 0, \dots, 0) < 0.$$

The inclusion $M \hookrightarrow G$ is continuous, so K is compact and therefore closed in the Hausdorff space G . This would put x in $K \subset M$, a contradiction. $\square$

[write] Part (1) is the locally compact case of source 07's [theorem 33.1](#), proved here through [lemma 51.1](#); parts (2) and (3) are new. [Proposition 51.4](#) below is an instance of [theorem 7.4](#) and [proposition 37.1](#).

Corollary 51.3 (The locally compact positive Presburger case). *In the class of positive cones of locally compact Borel signed Presburger-group presentations, an uncountable cone is locally compact exactly when the connected component of its signed group is one-dimensional. Every cone in this class is Polish and has continuous addition.*

This is a classification within the stated presentation class. It does not assume, or prove, that every topological Presburger monoid arises from a locally compact topological group in this manner.

51.1 The order bound is genuinely rank two

Proposition 51.4 (Order is neither open nor closed). *For an uncountable model in [theorem 50.1](#), the strict order on $M = G_{\geq 0}$ is Δ_2^0 but neither open nor closed in M^2 . The same is true of the weak order.*

Proof. Here $d > 0$. Choose $c_n > 0$ in C with $c_n \rightarrow 0$, for example the points $(1/n, 0, \dots, 0)$ in the flag coordinates. The component is divisible, so $c_n > 1$ for every n by [lemma 46.1](#). The pairs $(0, c_n)$ belong to the strict order and converge to $(0, 0)$, which does not; the relation is not closed. The point $(0, 1)$ belongs to the strict order, whereas $(c_n, 1)$ does not and converges to it; the relation is not open. Its complement after exchanging coordinates is the weak order, giving both failures there as well. The upper Δ_2^0 bound follows from [theorem 51.2](#) and restriction to M^2 . $\square$

There is no contradiction between continuous addition and this exact rank-two behavior. Arithmetic comparison is a Borel relation in a topology that is intentionally different from the discrete order topology.

52 Definable relations and continuous standard division

Theorem 52.1 (Uniform definable regularity). *In every model of [theorem 50.1](#), all Presburger-definable subsets of G^k , with parameters and arbitrary finite k , are Δ_2^0 . The same holds for definable subsets of the positive cone M^k in its relative topology. For each ordinary $m \geq 1$, the quotient and remainder maps in [\(49.1\)](#) are continuous, both on G and on M .*

Proof. Use the pointed splitting $G \cong \mathbb{R}^d \times E$. Since the real factor is divisible,

$$mG = \mathbb{R}^d \times mE.$$

This is clopen, because E is discrete. Every congruence condition is therefore the inverse image of a clopen set under a continuous affine map. Every comparison is Δ_2^0 by [theorem 51.2](#). Equality is closed and hence Δ_2^0 in a metrizable space. Affine integer-linear terms are continuous. Quantifier elimination in [theorem 49.3](#) now expresses each definable set as a finite Boolean combination of Δ_2^0 sets, which is again Δ_2^0 .

For positive-cone formulas, restrict quantifiers to $x \geq 0$, apply group quantifier elimination, and then restrict the resulting set to M^k . This preserves the stated relative Borel class.

The remainder map r_m has the finitely many clopen fibers $r1 + mG$, $0 \leq r < m$, so it is continuous into the discrete finite set of ordinary residues. Multiplication by the ordinary integer m is a homeomorphism from G onto mG : in the product splitting it is $(v, e) \mapsto (mv, me)$, whose inverse is continuous on its image. Consequently

$$q_m(x) = (m \cdot)^{-1}(x - r_m(x)1) \tag{52.1}$$

is continuous. If $x \geq 0$, the quotient $q_m(x)$ is nonnegative; otherwise $q_m(x) \leq -1$ would imply $x \leq -m + (m - 1) = -1$. Restricting these maps to M proves the final claim. $\square$

[write] This is the statement of [theorem 7.1](#) and [theorem 34.1\(iii\)](#) for a class that also admits the interleaved, non-block orders of the remark after [theorem 50.1](#). It answers source 07's question on uniform definability bounds ([section 43](#)) within that class.

52.1 A model in every finite real dimension

Take $E = \mathbb{Z}$ and order $\mathbb{R}^d \times \mathbb{Z}$ lexicographically, with the integer coordinate last. Write

$$G_d = \mathbb{R}^d \overrightarrow{\times} \mathbb{Z}, \quad M_d = (G_d)_{\geq 0}. \tag{52.2}$$

For $x = (v, n)$,

$$r_m(v, n) = n \bmod m, \quad q_m(v, n) = (v/m, \lfloor n/m \rfloor). \quad (52.3)$$

Thus standard division is explicit and continuous. Every M_d is a Polish model of full Presburger arithmetic with continuous addition. For $d = 1$, (52.2) is exactly (44.1); for $d \geq 2$ the cone is not locally compact although the signed group is.

The case $d = 1$ therefore provides the same affirmative construction for the question cited in section 44, already given in [40]. The advance here is the classification and dimensional boundary, not a second priority claim for that example.

[write] M_1 is Part I's $M_{\mathbb{R}}$; for $d \geq 2$ the models M_d are new to the collection.

52.2 Why ordinary scalar operations are not ring multiplication

All maps $x \mapsto nx$ for ordinary $n \in \mathbb{Z}$ are continuous group homomorphisms. This remains compatible with the complete absence of an order-compatible ring multiplication in uncountable models. Presburger arithmetic only provides these externally indexed scalar maps, not a binary multiplication of arbitrary elements.

Likewise, low Borel complexity of every definable relation does not make all definable functions continuous. For instance, truncated subtraction on the positive cone can fail to be continuous. A claim of continuity requires an argument such as (52.1), not merely quantifier elimination.

53 Even unordered multiplication has few Borel scalars

The main theorem forbids multiplication preserving the order, without any measurability assumption. We now remove order compatibility instead and ask how much measurable multiplication can survive. In this section the specified Borel $\mathbb{Z}$ -group order and the multiplication may be unrelated; their named additive unit 1 is the same.

53.1 Automatic continuity and component rigidity

Lemma 53.1 (Classical automatic continuity). *A Borel homomorphism between Polish groups is continuous.*

Proof. We give the abelian case needed here. Let $h : G \rightarrow H$ be Borel and additive. For a neighborhood W of zero in H , choose an open neighborhood V with $V - V \subseteq W$. Separability gives countably many translates of V covering H . Their Borel preimages cover the Baire space G , so one preimage B is nonmeager.

A nonmeager set with the Baire property has $B - B$ as a neighborhood of zero. Indeed, it is comeager in some nonempty open set U ; for small t , the set $U \cap (t + U)$ is nonempty open, and the two translated comeager sets meet, putting t in $B - B$. It follows that $h^{-1}(W)$ contains a neighborhood of zero, because $h(B - B) \subseteq V - V$. Thus h is continuous at zero, and hence everywhere. This is the usual Baire-category automatic-continuity argument; see also [8] for the Polish-group setting. $\square$

[write] Lemma 53.1 is the first half of source 09's lemma 61.2.

Lemma 53.2 (Injective maps cannot move a component into itself from outside). *Let $G \cong \mathbb{R}^d \times E$ with E discrete, and let $C = G^\circ$. Every continuous injective additive endomorphism $h : G \rightarrow G$ satisfies*

$$h(C) = C, \quad h^{-1}(C) = C. \quad (53.1)$$

Proof. Continuity and connectedness give $h(C) \subseteq C$. The restriction to C is a continuous additive map between finite-dimensional real vector groups, hence real-linear: first use rational scalars, then continuity and rational approximation for real scalars. Injectivity makes this linear endomorphism bijective. If $h(g) \in C$, choose $c \in C$ with $h(c) = h(g)$. Injectivity gives $g = c$, proving the preimage statement. $\square$

53.2 The Borel-scalar subring

Definition 53.3. For a ring multiplication on a topological additive group, define

$$\mathcal{B}(A) = \{a \in A : L_a : A \rightarrow A, x \mapsto ax, \text{ is Borel}\}.$$

We call this the *Borel-scalar subring* once its subring property has been proved.

Theorem 53.4 (Countably many Borel scalar multipliers). *Let $(G, +, <, 0, 1)$ be a locally compact Borel Polish $\mathbb{Z}$ -group, and let A be any commutative unital integral-domain multiplication on its underlying additive group, with this same 1. Multiplication need not respect the order. Then $\mathcal{B}(A)$ is a countable closed discrete subring, and the component quotient map is injective on it. In particular:*

1. *no nonzero element of $C = G^\circ$ has a Borel scalar map;*
2. *if the pointed topological group is $\mathbb{R}^d \times \mathbb{Z}$, then $\mathcal{B}(A) = \mathbb{Z}1$;*
3. *if G is uncountable, multiplication cannot be separately Borel in every scalar, and therefore cannot be jointly Borel.*

Proof. We have $C \cap \mathbb{Z}1 = 0$ and $1 \notin C$ by the additive discreteness and divisibility argument used in [theorem 50.1](#). If $a \in C \setminus \{0\}$ and L_a were Borel, then it would be continuous by [lemma 53.1](#). It is injective because A is a domain. But $L_a(1) = a \in C$, contradicting [lemma 53.2](#). This proves $\mathcal{B}(A) \cap C = \{0\}$.

The maps L_0, L_1 are continuous. If $a, b \in \mathcal{B}(A)$, then $L_{a-b} = L_a - L_b$ is Borel, and $L_{ab} = L_a \circ L_b$ is Borel. Thus $\mathcal{B}(A)$ is a subring, and it contains $\mathbb{Z}1$. If two of its elements have the same coset modulo C , their difference belongs to $\mathcal{B}(A) \cap C$ and is zero. Its projection into the countable discrete quotient G/C is therefore injective.

Each open component contains at most one element of $\mathcal{B}(A)$. Within that component a singleton is closed, and its complement is open. Taking the union of these complements over all components proves that $\mathcal{B}(A)$ is closed in G ; the same component neighborhoods make it discrete. It is countable because the quotient is countable.

If the pointed quotient is $\mathbb{Z}$ generated by the image of 1, every $a \in \mathcal{B}(A)$ has the same image as $n1$ for some ordinary n . Injectivity of the projection gives $a = n1$. Finally, an uncountable G has $d > 0$ and nonzero elements in C , whose scalar maps are not Borel. Sections of a jointly Borel multiplication would all be Borel, so joint Borelness is impossible as well. $\square$

Remark 53.5 (Two independent obstructions). [Theorem 44.1](#) uses order-compatible multiplication, but no regularity of its scalar maps. [Theorem 53.4](#) does not require order compatibility, but bounds which scalar maps can be Borel. Neither statement is obtained merely by rephrasing the other.

54 Examples isolating the hypotheses

54.1 The countable conclusion is optimal

Every countable discretely ordered ring has a discrete Polish additive group topology with Borel order and continuous multiplication. Ordinary $\mathbb{Z}$ is the smallest example, but there are non-Archimedean ones, such as $\mathbb{Z}[X]$ ordered by the sign of the highest nonzero coefficient. Thus the conclusion is discreteness and countability, not that the ordered ring must be isomorphic to $\mathbb{Z}$.

54.2 Second countability cannot simply be deleted

Example 54.1 (An uncountable locally compact ordered ring presentation). Let

$$A_* = \mathbb{Z} + X\mathbb{R}[X] \quad (54.1)$$

with the leading-coefficient order, so X is positive and greater than all ordinary integers. This is a discretely ordered domain. Give its additive group the product decomposition

$$A_* = (\mathbb{R}X) \oplus D_*, \quad D_* := \mathbb{Z} \oplus X^2\mathbb{R}[X], \quad (54.2)$$

where $\mathbb{R}X$ has its usual real-line topology and D_* is discrete. Then the additive group is Hausdorff and locally compact, the order is Borel, and the ring is uncountable. The topology is not second-countable or separable.

Proof. The leading-coefficient rule defines an ordered domain, and every positive nonconstant polynomial exceeds all ordinary integers. Positive constants are positive integers, so 1 is least positive. The product in (54.2) is locally compact. Its discrete factor has cardinality $\mathfrak{c}$, making the family of open components uncountable.

For Borelness, a component with nonzero degree-at-least-two part has a constant sign, independent of the X coefficient and integer constant. The union of all such positive components is clopen. On components with no degree-at-least-two part, positivity is the ordinary condition $r > 0$, with the boundary $r = 0$ governed by the integer constant. These conditions and their complements are countable unions of closed sets. Thus the positive cone is in fact Δ_2^0 in this metrizable, nonseparable topology. $\square$

Here $V = \mathbb{R}X$ has the order unit X , and

$$|A_*/V| = |V| = \mathfrak{c}.$$

So the cardinal lower bound of [theorem 46.2](#) is attained. Multiplication by X takes the connected line $\mathbb{R}X$ injectively into the discrete $X^2\mathbb{R}$ coordinate. It is not continuous. It is not even Borel: the inverse images of open sets obtained from arbitrary subsets of that discrete coordinate give arbitrary subsets of the real line, including non-Borel ones. No topological-ring claim is made for this example.

54.3 Borelness of the order is also essential

Example 54.2 (A locally compact Polish additive presentation with wild order). Fix $d \geq 1$. As rational vector spaces, both $\mathbb{R}^d$ and $X\mathbb{R}[X]$ have dimension $\mathfrak{c}$. Choose a $\mathbb{Q}$ -linear isomorphism $\theta : \mathbb{R}^d \rightarrow X\mathbb{R}[X]$, and define

$$\Theta : \mathbb{R}^d \times \mathbb{Z} \longrightarrow A_*, \quad \Theta(v, n) = n + \theta(v).$$

Transport the ring structure and the leading-coefficient order of A_* along Θ , while retaining the usual product topology on $\mathbb{R}^d \times \mathbb{Z}$. This gives an uncountable discretely ordered domain with a locally compact Polish additive-group topology. Its order is necessarily non-Borel by [theorem 44.1](#).

The dimension assertion uses ordinary ZFC basis theory: an infinite dimensional vector space over a countable field has cardinality equal to its basis cardinality, and a countable direct sum of copies of $\mathbb{R}$ still has rational dimension $\mathfrak{c}$. This construction is not effective and is not proposed as a usable notation system.

Independently give the same carrier the *different*, standard Borel lexicographic $\mathbb{Z}$ -group order. The transported multiplication is not compatible with that order, but [theorem 53.4](#) applies to it and gives the exact identity

$$\mathcal{B}(A) = \mathbb{Z}1.$$

Thus the Borel-scalar theorem has concrete algebraic realizations rather than merely a conditional conclusion about nonexistent ring structures.

54.4 Other boundaries

Discreteness of the arithmetic order is indispensable: the ordinary real field is uncountable, locally compact, and has continuous operations and Borel order, but has no least positive element. Totality also matters: our compact-order and finite-dimensional arguments do not apply to a partial order without additional hypotheses.

We do not present a separable non-locally-compact counterexample with all the other hypotheses of [theorem 44.1](#). The general Polish-group problem is left as a research question, rather than being called a settled sharpness result.

[write] (Batch 87.) [Part VI](#) settles the general Polish-group problem: there is no such counterexample ([corollary 78.5](#)). Its examples on the same ring show which further hypotheses cannot be dropped ([section 84](#)).

55 Consequences for surreal and omnific presentations

55.1 A set-sized no-go theorem for actual omnific rings

ProveIt's surreal-number guide describes its construction of the ordered surreal field, its omnific integer ring, the least-positive-element property, and the constant-term map [\[39\]](#). Only elementary finite normal forms and the discrete order are needed below; none of the repository's stronger unrefereed research claims is an input.

Corollary 55.1 (Omnific additive topologies). *Let $A \subseteq \mathbf{Oz}$ be an uncountable set-sized unital subring, with its inherited ordered-ring structure. There is no Hausdorff locally compact second-countable additive-group topology on A for which the inherited order is Borel. This holds without requiring multiplication to have any topological or measurable regularity.*

Proof. Since $1 \in A$ and 1 is least positive in $\mathbf{Oz}$, it is least positive in A . Apply [theorem 44.1](#). $\square$

[merge] Source 06 equips the nonnegative cone of the same ring $\mathbb{Z} + t\mathbb{R}[t]$ with a locally compact Polish topology in which addition and multiplication are continuous ([theorem 38.1](#)).

There is no conflict with [corollary 55.1](#): that topology lives on the cone and is induced by no locally compact second-countable group topology on the ring ([section 42](#)).

This is not merely the observation that a Polish space has cardinality at most continuum. The concrete ring

$$\mathbb{Z} + \omega\mathbb{R}[\omega] = \left\{ n + \sum_{j=1}^k r_j \omega^j : n \in \mathbb{Z}, r_j \in \mathbb{R}, k < \omega \right\} \quad (55.1)$$

has cardinality exactly $\mathfrak{c}$, so its cardinality alone permits a Polish coding. Finite surreal normal forms show that it is an ordered copy of A_* in ([54.1](#)). The theorem obstructs the combination of Borel order, locally compact additive-group topology, and the actual ordered multiplication even at this set-sized level.

55.2 Finite additive fragments have excellent topology but no ring expansion

Corollary 55.2 (Finite surreal scales). *For every ordinary $d \geq 1$, let*

$$H_d = \{n + r_1\omega + \cdots + r_d\omega^d : n \in \mathbb{Z}, r_j \in \mathbb{R}\} \subseteq \mathbf{Oz}. \quad (55.2)$$

Its inherited ordered additive group, with coefficient topology $\mathbb{R}^d \times \mathbb{Z}$, is a locally compact Borel $\mathbb{Z}$ -group. Its nonnegative cone is Polish with continuous addition and continuous standard quotient and remainder. That cone is locally compact exactly for $d = 1$. All its Presburger-definable relations are Δ_2^0 .

No alternative commutative unital ring multiplication on H_d can respect this addition, order, and named unit. In particular, failure of closure under the actual surreal multiplication cannot be repaired by inventing a different order-compatible multiplication on the same carrier.

Proof. In descending coefficient order $(r_d, \dots, r_1, n)$, addition and order identify H_d with G_d of ([52.2](#)). Standard division is ([52.3](#)). Apply [theorems 51.2](#) and [52.1](#) and [corollary 48.1](#). Alternatively, the last assertion follows already from algebraic amplification with $V = \sum_{j=1}^d \mathbb{R}\omega^j$ and order unit $u = \omega^d$: $H_d/V \cong \mathbb{Z}$ is countable but V is not. $\square$

55.3 Implications for exact symbolic systems

These results distinguish three legitimate design choices. One may retain finite surreal additive scales and continuous addition, accepting that there is no compatible full multiplication. One may enlarge to a genuine set-sized ordered ring, abandoning local compactness or separability of an additive-group topology. Or one may transport the algebra to a convenient Polish additive space and accept a non-Borel inherited order. The examples in [section 54](#) make the last two alternatives mathematically explicit.

The earlier support-barrier manuscript concerns a different obstruction: which exact Hahn or omnific normal-form families admit Borel coefficient observations [[41](#)]. Our obstruction requires neither unrestricted supports nor an analysis of support validity. It already appears for the finite polynomial ring ([55.1](#)).

56 A formalization plan with explicit trust boundaries

The proof should be formalized in layers. Its algebraic core is much smaller than its locally compact group input, so formalization need not begin with Haar measure or the full theory

of real Lie groups.

Layer A: finite-order arithmetic. Formalize that $[0, n1]$ contains exactly the ordinary numerals in a discretely ordered group. Deduce that a subgroup disjoint from $\mathbb{Z}1$ has no nonzero element bounded by a standard integer, and that a divisible subgroup has this property. The induction index is a natural number in the metatheory of the formal development.

Layer B: amplification. For a subgroup V , an element $u \in V$, and the explicit order-unit predicate, prove $uv \in V \Rightarrow v = 0$. Package (46.1) as an injective additive map into the quotient additive group. Then formalize (46.3) for finitely supported coefficient families and derive the ordered direct-sum embedding. No topological assumption, infinite series, or proper class occurs in this layer.

Layer C: the topological interface. State a reusable interface consisting of an open connected subgroup C , a topological additive-group isomorphism $C \cong \mathbb{R}^d$, and a countable quotient. Prove arithmetic rigidity from this interface and the Borel finite-dimensional flag lemma. Separately discharge the interface using the compact-order obstruction and the principal LCA structure theorem. A conditional theorem with this interface is useful, but must not be advertised as an unconditional formalization of the whole locally compact result.

Layer D: Presburger transfer. Prove lemma 49.2 over an abstract $\mathbb{Z}$ -group with a predicate invariant on cosets of mG . Prove the group theory of standard division before connecting to syntax. ProveIt already supplies syntax, coefficient normalization, and an executable integer decision procedure, according to its inspected guide [38]; the semantic transfer must be proved rather than inferred from that guide.

Layer E: topology of presentations. Formalize lexicographic cuts by finite dimension, prove their F_σ and G_δ descriptions, and derive the cone dimension threshold and continuity of the standard quotient and remainder maps. The scalar theorem then reuses an automatic-continuity theorem and finite-dimensional linear algebra.

No Lean files are included as if they had been checked. The companion Python script validates finite exact-arithmetic examples, not proofs in Lean or in any other kernel. In particular it does not test Haar measure, nonmeager sets, arbitrary Borel orders, cardinal injections, or the principal LCA structure theorem.

57 Further research questions and concrete next targets

Question 57.1 (Remove local compactness). Can an uncountable discretely ordered ring have a Polish additive-group topology with Borel order when multiplication has no regularity at all?

The proof here uses local compactness to obtain a finite-dimensional open component with an order unit. A non-locally-compact Polish group can have an infinite-dimensional component or be totally disconnected without being discrete. An extension would need a substitute for that finite-dimensional argument, perhaps an uncountable subgroup to which

[theorem 46.2](#) applies and a suitably small quotient. Neither a proof nor a counterexample is claimed here.

[write] Source 09's [theorem 62.1](#) answers this question negatively when every scalar map $x \mapsto ax$ is Borel and the positive cone has the Baire property; without any regularity of multiplication it stays open.

[write] (Batch 87.) Answered negatively, with no regularity of multiplication, by [Part VI](#): sources 12 and 10 prove that a discretely ordered ring with a Hausdorff Baire additive-group topology satisfying the countable chain condition, whose positive cone has the Baire property, is countable and discrete ([corollary 78.4](#)); in particular every Polish additive-group topology with Borel order is discrete on a countable ring ([corollary 78.5](#)). The preceding note's "it stays open" is superseded. The proofs are unrefereed and not formalized.

Question 57.2 (Topological group completion of positive models). Which locally compact Polish cancellative Presburger monoids admit a Hausdorff locally compact Polish topology on their Grothendieck groups that induces the original topology on the positive cone?

A positive answer under usable hypotheses would convert [corollary 51.3](#) into a broader classification of positive arithmetic presentations. The relevant issue is topological embedding of the monoid into its group completion, not the elementary algebraic existence of that completion. Conditions on translations, quotient maps, and neighborhoods of zero are natural starting points.

[write] Source 09 asks the same question as [question 73.11](#). Source 06's cone of $\mathbb{Z} + t\mathbb{R}[t]$ is a data point ([section 42](#)): a locally compact Polish cone with continuous addition, whose group of differences has no locally compact second-countable group topology with Borel order, by [theorem 44.1](#).

[write] (Batch 87.) Sources 12 and 10 ask it again for Baire ccc completions with a Baire-property cone ([Part VI](#), questions on group completion), and source 15 for locally compact Presburger monoids ([Part IX](#)).

Question 57.3 (Classify the orders, not only the pointed groups). For a fixed countable $\mathbb{Z}$ -group E , classify Borel $\mathbb{Z}$ -group orders on $\mathbb{R}^d \times E$ up to continuous additive order isomorphism fixing 1.

The real component carries a finite flag, but discrete coordinates can occur between real scales, as the example following [theorem 50.1](#) shows. The cut description in [lemma 51.1](#) and the continuous separating homomorphisms of the classical measurable-order literature suggest finite flag data together with countable extension data. Determining which data are complete invariants is a concrete next classification problem.

Question 57.4 (Weaken Borel order carefully). Which hereditary measurability assumptions on an invariant order suffice for arithmetic rigidity? In particular, what replaces Borelness if restrictions to every compact subgroup and every closed real subspace are required to be Haar measurable or to have the Baire property?

An ambient Haar-measurable set can have uncontrolled behavior on a null closed subgroup. The present proof intentionally uses Borelness to avoid that problem. A weakening should state restriction hypotheses explicitly rather than treating ambient measurability as automatically hereditary.

[write] (Batch 87.) Answered by [Part VI](#) without any hereditary condition: the Baire property of the cone for the given topology suffices ([lemma 77.2](#) and [corollary 78.4](#)), and on a locally compact group measurability of the cone for completed Haar measure suffices ([theorem 83.3](#)). Both sources of Part VI ask for still weaker regularity.

Question 57.5 (Cardinal and topological size spectra). For nonseparable locally compact additive presentations of discretely ordered rings, what cardinalities of the component quotient can occur relative to the identity component and the weight of the topology?

Example 54.1 attains $|A/C| = |C| = \mathfrak{c}$ with a real component. The direct-sum strengthening in theorem 46.2 records more than one injection. It may impose additional ordered-group or module constraints on possible quotients even when it yields no stronger bare infinite-cardinal inequality.

[write] (Batch 87.) For Hausdorff Baire presentations with a Baire-property cone, source 12 proves $c(R) = d(R) = |R|$ (theorem 78.3) and source 10 adds extent and Lindelöf degree for uncountable rings (theorem 79.13); example 54.1 attains them. The finer spectra asked for stay open.

Question 57.6 (Realize Borel-scalar subrings). For fixed $\mathbb{R}^d \times E$, which countable subrings can arise as $\mathcal{B}(A)$ for integral-domain multiplications on the additive group?

The projection of $\mathcal{B}(A)$ into E is injective, and it contains the ordinary integers. When $E = \mathbb{Z}$ the answer is exactly $\mathbb{Z}$. For larger E , continuous scalar maps act real-linearly on the component and induce additive maps on E ; compatibility of these two actions and their extension data may give a useful algebraic realization criterion.

Question 57.7 (Effective and reverse-mathematical strength). What is the exact effective content of extracting a finite real flag from a code for a Borel order, and of proving the resulting rigidity statement over weak base theories?

The current ZFC proof uses category, separation, Haar measure, and an LCA structure theorem. It is not an analysis of their strength in second-order arithmetic. Glazer's separate Question 1 about the strength of his theorem [1] remains outside the results proved here; an argument for the locally compact fragment should not be substituted for an answer to that question.

[write] Section 1.1 discusses Glazer's Question 1 and the speculation that follows his Question 2.

[write] (Batch 87.) Source 14 (Part VIII) claims an ATR_0 proof of Glazer's Theorem 2, not independently reviewed. It is not an analysis of this question, which concerns the locally compact rigidity theorem; sources 12 and 10 ask the corresponding question for Part VI.

Question 57.8 (Certified amplification and notation design). Can the algebraic amplification theorem be kernel-checked first, and then used as an automatic obstruction to proposed finite-scale omnific or non-Archimedean ring interfaces?

A small certificate could consist of a divisible or integer-disjoint additive subgroup, an order unit, and an upper bound on the quotient's size. This would separate a universally reusable algebraic theorem from application-specific normal-form calculations. The finite-scale examples in corollary 55.2 are a natural first target, followed by the Borel-scalar theorem and the full topological interface.

58 Conclusions of Part IV

Locally compact arithmetic separates cleanly into an additive possibility and a multiplicative obstruction. Uncountable Polish Presburger models with continuous addition are available, and their locally compact signed group presentations have a precise finite-dimensional

structure. Their order relations have exact rank-two regularity, and their standard quotient and remainder operations remain continuous.

But a discretely ordered ring cannot use that topology in the uncountable second-countable case. The obstruction does not ask multiplication to be Borel: it uses multiplication only to create an impossible injection from a real component into a countable component quotient. When order compatibility is dropped instead, only countably many scalar maps can remain Borel. This gives a concrete structural connection between Glazer’s arithmetic-presentation questions and ProveIt’s finite surreal, omnific, and Presburger developments, while keeping the unresolved Polish-group and formal-verification questions explicit.

[write] (Batch 87.) The Polish-group question is answered by [Part VI](#).

Part V

Polish topologies at the Hahn–Puisseux–Levi-Civita boundary

[write] This part is source 09 of batch 86, *Polishability at the Hahn–Puisseux–Levi-Civita Boundary: Automatic order bounds, local Polish hulls, and countable omnific integer parts* (dated October 3, 2026; “prepared with AI assistance”), pinned to repository commit 4af956fe0. All unmarked text is source 09’s; its labels carry the prefix `pma:ser:.` It continues [Part II](#): source 03 asked when Hahn-series codes form a *standard Borel* space with observable coefficients, and source 09 asks when the additive group of a series field can be made *Polish* with measurable arithmetic. Its “preceding support manuscript” [44] is source 03 (Part II), and its Presburger companion [43] is source 05 (Part I). Its finite checks are shipped as `code/09-hahn-polishability-verify.py` with the record `data/09-hahn-polishability-verification.json`.

Convention [write]. In this part t is a positive infinitesimal, exponents of t that are negative give infinite elements, and the surreal realization is $t^q \mapsto \omega^{-q}$ ([equation \(71.1\)](#)). This is the reverse of [Part III](#), where t is positive infinite, and agrees with source 03’s increasing- t convention; 03’s growth variable is $X = t^{-1}$. Source 09’s macros for its fields are renamed only in the source: the Puisseux field $\mathcal{P}_k$ is printed exactly as before and is not 03’s cone $\mathcal{A}_{\text{fin}, \geq 0}$ ([section 1.3](#)).

Reading routes. Sections 61–63 contain the main obstruction and classification. Sections 64–67 explain the Puisseux/Levi-Civita contrast. Sections 68–71 connect the analysis to real closed fields, arithmetic, and ProveIt. The final sections separate proved conclusions from further research.

59 The research intersection and the questions answered

Elliot Glazer’s *A Topological Tennenbaum Theorem* [1] studies Borel arithmetic on uncountable Polish spaces. Its Theorem 2 excludes continuity of either arithmetic operation for Borel models of $\mathbb{Q} + I\Delta_0 + \text{Exp}$, and its Corollary 1 excludes continuous addition with Borel multiplication for uncountable Polish models of $\mathbb{Q} + \text{IOpen}$. This provides a specific, documented intersection with ProveIt’s work on ordered algebra, exact Hahn normal forms, omnific integers, and formal arithmetic. No inference about Glazer’s present employment or

endorsement is needed.

The ProveIt source collection distinguishes formalized constructions from unrefereed research reports. Its surreal documentation describes the normal-form realization of the surreal field and the integer-constant, nonnegative-growth normal forms of the omnific integers [42]. We use those as motivation and as a possible implementation interface, not as substitutes for proofs of the topological assertions below.

Two earlier manuscripts prepared for this research program already address Polish Presburger positive cones and descriptive support barriers [43, 44]. In particular, the preceding support manuscript discusses the coanalytic obstruction for unrestricted rational Hahn supports. Repeating that obstruction would not answer the next question: *even when exact series codes form a standard Borel space, can their additive group be made Polish without losing measurable arithmetic?*

The answer is sensitive to which series are admitted. Fix a countable ordered field k , and use t as a positive infinitesimal. Three natural fields are

$$\mathcal{P}_k = \bigcup_{n \geq 1} k((t^{1/n!})), \quad (59.1)$$

$$\mathcal{L}_k = \left\{ \sum_{q \in \mathbb{Q}} a_q t^q : \text{supp}(a) \cap (-\infty, B] \text{ is finite for every } B \in \mathbb{Q} \right\}, \quad (59.2)$$

$$\mathcal{H}_k = k((t^{\mathbb{Q}})). \quad (59.3)$$

They are, respectively, the Puiseux field, the left-finite Levi-Civita field, and the full rational Hahn field. Their inclusions are strict. The middle field, not the largest or the descriptively simplest one, admits the desired Polish arithmetic.

59.1 The central conclusions

The term *admissible* is made precise in Definition 60.1. It requires a Polish additive group, a Baire-measurable positive cone, and Borel multiplication by each fixed scalar. It does not initially require jointly continuous multiplication.

Main results. For a nonzero ordered abelian group Γ , the full Hahn field $k((t^\Gamma))$ has an admissible Polish topology if and only if k is countable and $\Gamma \cong \mathbb{Z}$ as an ordered group. For countable ordered k , $\mathcal{P}_k$ has no admissible Polish topology, while $\mathcal{L}_k$ has exactly one: its complete valuation topology. Every discretely ordered ring with an admissible Polish additive group is countable. Every integer part of an admissible Polish ordered field is countable and closed discrete.

There is also a sharp local converse. Every countable family of rational Hahn series lies in a complete separable subfield, and these Polish subfields have no maximal member. Exactly $2^{\aleph_0}$ of them are needed to cover the full Hahn field (Theorem 69.6). Over $\mathbb{R}_{\text{alg}}$ they can be chosen real closed and truncation-closed, with canonical countable omnific integer parts. Thus the Levi-Civita field is not the largest possible positive construction: the global obstruction concerns simultaneous freedom of prefixes, not any one prescribed countable support pattern.

The main advance over a support-code obstruction is quantification over *all* admissible Polish topologies on the abstract field. The Hahn classification is not defeated by abandoning coefficientwise coding. The Puiseux obstruction is stronger still in a different direction:

the code domain is already F_σ , but no compatible Polish group topology can support the required arithmetic.

For $k = \mathbb{R}_{\text{alg}}$, the real algebraic numbers, both $\mathcal{P}_k$ and $\mathcal{L}_k$ are real closed and share the same explicitly described countable integer part. Its nonnegative cone satisfies open induction. The integer-part algebra therefore does not distinguish these two sharply different topological situations.

59.2 What is and is not claimed as new

The article gives complete arguments for its central classifications and regularity conclusions. The category tools are classical. The construction and completeness of Levi-Civita fields belong to established non-Archimedean analysis; see, for example, Shamseddine–Berz [31]. Generalized series and Newton–Puisseux theory are classical, with a useful account in Poonen [30]. The coanalytic support obstruction and the negative-infinitesimal floor correction are already present in the preceding project materials.

The contribution developed here is their combination into a presentation-invariant classification, the Puisseux versus Levi-Civita Polishability separation, the local Polish-hull and exact field-cover theorems, and the resulting integer-part and floor-regularity statements. These are proposed research contributions relative to the inspected materials. They are not accompanied by a certified global priority claim. We answer the precisely formulated questions in this article; we do not claim to have newly resolved either of Glazer’s printed open questions.

60 Conventions: four different notions of presentation

All fields, groups, supports, and topological spaces in the main arguments are sets in ZFC. Every ordered field has its specified field-compatible linear order. A *discretely ordered ring* is a commutative ordered unital ring whose least positive element is 1. An *integer part* of an ordered field F is a discretely ordered unital subring $I \subseteq F$ such that each $x \in F$ has an $a \in I$ satisfying $a \leq x < a + 1$.

A Polish space is separable and completely metrizable. A set has the *Baire property* when it differs from an open set by a meager set. Every Borel set has this property. A Polish additive group has jointly continuous addition and continuous negation.

Definition 60.1 (Admissible Polish arithmetic). A topology τ on an ordered field or discretely ordered ring F is *admissible* when:

- (i) $(F, +, \tau)$ is a Polish topological group;
- (ii) $F_{>0}$ has the Baire property in τ ;
- (iii) for each $a \in F$, the map $L_a : F \rightarrow F$, $L_a(x) = ax$, is Borel.

An admissible Polish presentation through another carrier means a transported topology satisfying these conditions. It is therefore equivalent to an admissible topology on the original algebraic carrier.

The following distinctions prevent several false conclusions. A standard Borel presentation supplies measurable codes and operations, but need not supply continuous addition. A *Polishable Borel group* is one whose given Borel structure comes from a Polish group topology.

An admissible ordered-field topology may initially be allowed to change the Borel structure; our absolute nonexistence results exclude that possibility too. Finally, a Polish topology on a *positive cone* with continuous addition is only a topological monoid. It is not a Polish topology on the group of differences.

In particular, the ring theorem below cannot be applied by silently adjoining negative elements to a Polish model of Presburger arithmetic. Extending the algebra is easy; extending the topology with all the required properties is a separate problem.

For a Hahn series $f = \sum a_\gamma t^\gamma$, the support is well ordered in Γ , and

$$v(f) = \min \text{supp}(f) \quad (f \neq 0), \quad v(0) = +\infty.$$

The order is determined by the coefficient at the least support exponent. Positive exponents represent small quantities. In particular, if $\alpha < \beta$ and $c \in k_{>0}$, then $ct^\alpha > t^\beta$. We distinguish this formal order from the ordinary absolute value of a coefficient when k happens to be a subfield of $\mathbb{R}$.

61 Category forces local order bounds

The mechanism is a short application of classical category theory. We include the needed proofs to make the hypothesis boundary visible. Pettis's theorem, automatic continuity, and the open mapping theorem are classical; Chen [29] reviews their relationship and develops localic counterparts.

Lemma 61.1 (Pettis, additive form). *Let G be a Polish group and let $A \subseteq G$ have the Baire property and be nonmeager. Then $A - A$ contains a neighborhood of 0.*

Proof. There is a nonempty open set U on which A is comeager. Choose a nonempty open V and a neighborhood W of 0 so small that $V \subseteq U$ and $V - g \subseteq U$ for every $g \in W$. For each such g , both $A \cap V$ and $(g + A) \cap V$ are comeager in V . Their intersection is nonempty because a nonempty open subspace of a Polish space is Baire. An element x of the intersection satisfies $x \in A$ and $x - g \in A$, so $g \in A - A$. $\square$

Lemma 61.2 (Automatic continuity and open mapping). *A Borel homomorphism between Polish groups is continuous. A continuous surjective homomorphism between Polish groups is open.*

Proof. For the first assertion, let $\phi : G \rightarrow H$ be Borel and let U be a neighborhood of 0 in H . Choose an open neighborhood V with $V - V \subseteq U$. Separability gives a countable cover of H by translates of V . Their inverse images are Borel and cover G , so one inverse image A is nonmeager. Then $A - A \subseteq \phi^{-1}(U)$, and Lemma 61.1 gives continuity at 0.

For the second assertion, let U be a neighborhood of 0 in G , and choose an open neighborhood V with $V - V \subseteq U$. Countably many translates of V cover G , hence their images cover H . The set $\phi(V)$ is analytic, and therefore has the Baire property. At least one of its translates is nonmeager. Pettis gives a neighborhood of 0 inside $\phi(V) - \phi(V) \subseteq \phi(U)$. $\square$

The descriptive-set-theoretic facts about analytic sets used here are standard [8]. In particular, continuous images of Polish spaces are analytic, analytic sets have the Baire property, and an analytic set with analytic complement is Borel.

Lemma 61.3 (A Baire order is locally bounded). *Let G be a nontrivial linearly ordered abelian group with a Polish group topology. If $P = \{x : x > 0\}$ has the Baire property, some neighborhood of 0 is contained in a bounded order interval $[-a, a]$, with $a > 0$.*

Proof. For a discrete group, take the neighborhood $\{0\}$ and any $a > 0$. Suppose the topology is not discrete. Then $\{0\}$ is nowhere dense. Inversion is a homeomorphism and takes P to $-P$. If P were meager, then $G = P \cup (-P) \cup \{0\}$ would be meager in itself, contrary to the Baire category theorem. Thus P is nonmeager.

Choose a nonempty open set U on which P is comeager, and choose $p \in P \cap U$. Let V be a symmetric open neighborhood of 0 with $W = p + V \subseteq U$. The reflection $x \mapsto 2p - x$ maps W onto itself. Consequently, both $P \cap W$ and $(2p - P) \cap W$ are comeager in W . The set

$$A = W \cap P \cap (2p - P)$$

has the Baire property and is nonmeager. Every $x \in A$ satisfies $0 < x < 2p$. Therefore $A - A \subseteq (-2p, 2p)$. Pettis gives a neighborhood of 0 inside this interval. Take $a = 2p$. $\square$

Theorem 61.4 (Automatic order-topology refinement). *Every admissible Polish topology on an ordered field refines its order topology. More generally, the same conclusion holds for a Polish additive group topology with Baire positive cone and continuous fixed scalar maps.*

Proof. In the admissible case every L_a is continuous by Lemma 61.2. Choose a neighborhood U of 0 with $U \subseteq [-a, a]$ as in Lemma 61.3. For any $\varepsilon > 0$, put

$$c = \frac{a+1}{\varepsilon} > 0.$$

Then $L_c^{-1}(U)$ is a neighborhood of 0, and its elements satisfy

$$|x| \leq \frac{a}{c} = \varepsilon \frac{a}{a+1} < \varepsilon.$$

Thus every interval $(-\varepsilon, \varepsilon)$ contains a neighborhood of 0. Translating this statement at every point of any order interval proves that the interval is open in the given topology. $\square$

Corollary 61.5 (Uniqueness when the order topology is Polish). *If the order topology of an ordered field is a Polish group topology, then it is the only possible admissible Polish topology on that field.*

Proof. The identity from an admissible topology to the order topology is a continuous bijective group homomorphism between Polish groups. It is open by Lemma 61.2. $\square$

Corollary 61.6 (Order-free hypothesis for real closed fields). *Let F be real closed. If its additive group is Polish and its multiplication is jointly Borel, then its unique field order is Borel. Consequently, Theorem 61.4 applies without separately assuming measurable order.*

Proof. The positive cone is the image of $F \setminus \{0\}$ under the Borel map $x \mapsto x^2$, so is analytic. Its complement is the union of $\{0\}$ with the image of $F \setminus \{0\}$ under $x \mapsto -x^2$, hence is analytic too. The analytic/coanalytic criterion gives Borelness. The scalar maps are sections of multiplication and therefore Borel. $\square$

Remark 61.7. The last corollary uses *joint* Borel multiplication to obtain a Borel square map. Merely postulating Borelness of each fixed scalar map does not by itself justify substituting the variable into both multiplication arguments.

62 Discretely ordered rings and integer parts

Theorem 62.1 (Automatic discreteness). *An admissible Polish topology on a discretely ordered ring is discrete. In particular, the ring is countable.*

Proof. By Lemma 61.3, choose a neighborhood U of 0 contained in $[-a, a]$, where $a > 0$. Multiplication by $b = a + 1$ is a continuous additive homomorphism. Thus $W = L_b^{-1}(U)$ is a neighborhood of 0. If $x \neq 0$, discrete order gives $|x| \geq 1$, whence $|bx| \geq b > a$. Such an x cannot belong to W . Therefore $W = \{0\}$. A discrete separable space is countable. $\square$

[write] Theorem 62.1 and source 08's theorem 44.1 have the same conclusion under incomparable hypotheses: 09 needs Borel scalar maps and a cone with the Baire property but no local compactness, 08 needs local compactness but no regularity of multiplication. Together they answer source 08's question 57.1 under the Borel-scalar hypothesis only. The non-example below is the group of Part I's real model.

[write] (Batch 87.) Part VI answers question 57.1 without the Borel-scalar hypothesis; that hypothesis of theorem 62.1 is redundant, and both theorems are cases of corollary 78.5 (section 90). Source 12's lemma 77.2 is lemma 61.3 for Hausdorff Baire groups.

Corollary 62.2 (A non-Polishable Borel group criterion). *Let R be an uncountable standard Borel discretely ordered ring whose order, addition, negation, and multiplication are Borel. Its additive Borel group is not Polishable.*

Proof. A compatible Polish group topology would preserve the stipulated Borel order and scalar maps and would contradict Theorem 62.1. $\square$

This result needs no induction scheme. That does not make it a theorem with weaker hypotheses than Glazer's arithmetic result: here a Polish group of differences is assumed from the outset. Glazer's theorem concerns the arithmetic domain itself. These are different boundaries.

Theorem 62.3 (All integer parts are countable). *Let F have an admissible Polish topology. Every discretely ordered unital subring $I \subseteq F$ is countable and closed discrete. In particular this holds for every integer part, without any measurability assumption on I .*

Proof. By Theorem 61.4, the interval $(-1, 1)$ is open in F . Its intersection with I is $\{0\}$, so every translate of a sufficiently small order interval meets I in at most one point. For example, two distinct points of I cannot lie in an interval of radius $1/3$, because their nonzero difference has absolute value at least 1.

Each point of I therefore has an open neighborhood containing no other point of I . A countable base supplies distinct base elements for distinct points, proving countability. To see closedness, suppose $x \in \bar{I}$. The interval $(x - 1/3, x + 1/3)$ contains at most one point a of I , and must contain one by closure. If $x \neq a$, a smaller neighborhood of x avoiding a contradicts $x \in \bar{I}$. Hence $x = a \in I$. $\square$

Example 62.4 (An important non-example). The lexicographically ordered group $\mathbb{R} \times \mathbb{Z}$, with its ordinary product group topology, has a Borel discrete order: $(0, 1)$ is its least positive element. It is uncountable and Polish. This does not contradict Theorem 62.1, since no ordered-ring structure with the required scalar maps has been supplied. The scalar $a + 1$ in the proof is a genuine ring-theoretic resource.

63 A complete classification for full Hahn fields

The existence and algebraic operations of full Hahn fields are classical [30]. We use the entire well-ordered-support field, not a smaller convergence-restricted subfield.

Lemma 63.1 (A bounded increasing sequence in a noncyclic group). *Let $\Gamma \neq 0$ be an ordered abelian group. Either Γ is order-isomorphic to $\mathbb{Z}$, or there are*

$$0 < \gamma_0 < \gamma_1 < \cdots < \delta$$

with $\delta \in \Gamma_{>0}$.

Proof. Suppose first that Γ has no least positive element. Choose $0 < \varepsilon_0 < \delta$ and then recursively $0 < \varepsilon_{n+1} < \varepsilon_n$. Put $\gamma_n = \delta - \varepsilon_n$.

Suppose instead that ε is the least positive element. If every positive element is bounded by a positive integer multiple of ε , then for every $x > 0$ choose the least n with $x \leq n\varepsilon$. The difference $x - (n-1)\varepsilon$ is positive and at most ε , hence equals ε . Thus $x = n\varepsilon$ and $\Gamma = \mathbb{Z}\varepsilon$. Otherwise some δ exceeds every $n\varepsilon$, and $\gamma_n = (n+1)\varepsilon$ has the required properties. $\square$

Lemma 63.2 (Disjoint intervals in a Hahn field). *Let $H = k((t^\Gamma))$ with $\Gamma \neq 0$. If k is uncountable, its order topology has an uncountable family of pairwise disjoint nonempty open intervals. If Γ is not order-isomorphic to $\mathbb{Z}$, it has such a family of cardinality $2^{\aleph_0}$, even with centers having only 0–1 coefficients.*

Proof. For uncountable k , fix $\delta > 0$. The intervals

$$(c - t^\delta, c + t^\delta), \quad c \in k,$$

are disjoint: every positive nonzero constant dominates $2t^\delta$.

For noncyclic Γ , use Lemma 63.1. For $b \in 2^{\mathbb{N}}$ define

$$f_b = \sum_{n \geq 0} b(n)t^{\gamma_n}.$$

This is a legal Hahn series: each support is a subset of the increasing sequence (γ_n) and hence is well ordered. If $b \neq c$, let m be the first index where they differ. Then $v(f_b - f_c) = \gamma_m < \delta$, so $|f_b - f_c| > 2t^\delta$. The intervals $(f_b - t^\delta, f_b + t^\delta)$ are pairwise disjoint. $\square$

Theorem 63.3 (Full Hahn classification). *For an ordered field k and nonzero ordered abelian group Γ ,*

$$k((t^\Gamma)) \text{ admits an admissible Polish topology} \iff |k| = \aleph_0 \text{ and } \Gamma \cong \mathbb{Z} \text{ as ordered groups.}$$

In the positive case the topology is unique and is the Laurent valuation topology; all field operations, including inversion away from 0, are continuous.

Proof. An admissible topology refines the order topology by Theorem 61.4. A second-countable space has only countably many pairwise disjoint nonempty open sets, since one can choose a distinct basic open set inside each. Lemma 63.2 therefore forces k countable and $\Gamma \cong \mathbb{Z}$.

Conversely identify Γ with $\mathbb{Z}$ and use the metric

$$d(f, g) = \min\{1, 2^{-v(f-g)}\}, \quad 2^{-\infty} = 0.$$

A Cauchy sequence eventually stabilizes all coefficients of exponent at most N , for every integer N . Stabilization for $N = 0$ also gives a common finite negative principal part for its tail. The stabilized coefficients therefore define a Laurent series, and the sequence converges to it. Finite Laurent polynomials form a countable dense set because k is countable.

The valuation identities $v(fg) = v(f) + v(g)$ and $v(f + g) \geq \min\{v(f), v(g)\}$ give continuity of addition and multiplication. For $f \neq 0$, sufficiently small changes do not change $v(f)$, and

$$v(f^{-1} - g^{-1}) = v(f - g) - v(f) - v(g)$$

gives continuity of inversion. The order and valuation topologies agree: a valuation ball sufficiently far above the leading exponent of a positive ε lies inside $(-\varepsilon, \varepsilon)$, while a sufficiently small monomial interval lies inside any prescribed valuation ball. Thus the order is open and the topology is admissible. Uniqueness follows from Corollary 61.5. $\square$

[write] Source 03's [theorem 20.2](#) is the Borel counterpart: $\mathbb{R}((t^\Gamma))$ has a coefficient-observable standard Borel parametrization if and only if Γ is scattered. The two results are consistent: for $\Gamma = \mathbb{Z}^d$, $d \geq 2$, the canonical support domain is Borel (03's [theorem 19.1](#)) but no admissible Polish topology exists (the corollary below). For real coefficients, [lemma 63.2](#) excludes an admissible Polish topology on every field containing $\mathbb{R}$ and a nonzero infinitesimal, among them 03's real Puiseux field $\mathcal{P}$, which nevertheless has Borel codes ([section 23](#)).

Corollary 63.4 (The obstruction is not a coding artifact). *For any countable ordered field k , the full field $k((t^\mathbb{Q}))$ has no admissible Polish topology, regardless of how its elements are encoded. For finite lexicographic powers $\Gamma = \mathbb{Z}^d$, $d \geq 2$, the same conclusion holds even when the canonical support domain is Borel.*

Proof. Both ordered groups are noncyclic. In the second assertion the relevance of Borel support coding is only comparative; the topological conclusion uses Theorem 63.3 alone. $\square$

The excluded case $\Gamma = 0$ is deliberate. It reduces to the coefficient field itself; for example $\mathbb{R}$ with its usual topology is admissible. The nontrivial infinitesimal scale is exactly what separates distinct constants in Lemma 63.2.

For countable k and countable noncyclic Γ , the order density and cellularity of the full Hahn field are both exactly $2^{\aleph_0}$: the lemma gives the lower bound, and $|k^\Gamma| = 2^{\aleph_0}$ gives the upper bound.

64 The Puiseux obstruction: a Borel field that is not Polishable

A second category obstruction does not use an order at all.

Theorem 64.1 (Countable proper-subfield cover). *Let F be an uncountable field whose additive group has a Polish topology. Assume every fixed scalar map is Borel. Then F cannot be the union of countably many Borel proper subfields.*

Proof. Suppose $F = \bigcup_n K_n$, with each K_n a Borel proper subfield. By the Baire category theorem some K_n is nonmeager. As it is an additive subgroup, Pettis shows that $K_n = K_n - K_n$ is open. Choose $a \in F \setminus K_n$. The maps L_a and $L_{a^{-1}}$ are Borel homomorphisms, hence continuous; thus L_a is a homeomorphism and aK_n is open. But

$$K_n \cap aK_n = \{0\}.$$

Indeed, a nonzero $x = ay$ with $x, y \in K_n$ would imply $a = x/y \in K_n$. It follows that $\{0\}$ is open. The Polish group would be discrete and countable, a contradiction. $\square$

The Borel hypothesis on the covering subfields matters. A category argument cannot simply be applied to arbitrary nonmeasurable subfields. In the Puiseux application the subfields will actually be closed under *every* admissible topology, so no measurability assumption is being hidden.

Let k now be a countable ordered field, and set

$$K_n = k((t^{1/n!})), \quad \mathcal{P}_k = \bigcup_{n \geq 1} K_n.$$

The factorial convention makes the chain increasing. Each K_n is a proper subfield of $\mathcal{P}_k$, because rational monomials of denominators not dividing $n!$ occur at later stages.

Lemma 64.2 (Closed stages). *Each K_n is closed in the order topology of $\mathcal{P}_k$, and hence closed under every admissible Polish topology on $\mathcal{P}_k$.*

Proof. If $f \notin K_n$, some nonzero coefficient f_q has $q \notin (1/n!)\mathbb{Z}$. Every g agreeing with f at all exponents at most q still lies outside K_n . The set

$$\{g : v(g - f) > q\}$$

is an order-open neighborhood of f . To check order-openness, a monomial interval of exponent greater than q about any of its points lies inside it. Thus the complement of K_n is open. Theorem 61.4 transfers this closedness to any admissible topology. $\square$

Theorem 64.3 (Absolute Puiseux non-Polishability). *The Puiseux field $\mathcal{P}_k$ over a countable ordered field has no admissible Polish topology. In its canonical standard Borel presentation, even without mentioning the order, no compatible Polish additive-group topology can make all fixed scalar maps Borel.*

Proof. The field is uncountable, since $k((t))$ already contains all binary power series. For an arbitrary admissible topology, Lemma 64.2 makes the proper subfields K_n closed, and Theorem 64.1 applies. For the canonical Borel structure, K_n is defined by the vanishing of all coefficients outside $(1/n!)\mathbb{Z}$, hence is Borel. The same theorem applies directly, without an order hypothesis. $\square$

The ordinary valuation topology on $\mathcal{P}_k$ is separable and metrizable, but is not complete. More strongly, it is meager in itself: each closed stage K_n has empty interior. Given $f \in K_n$ and a valuation ball about f , add a monomial t^q of sufficiently large rational exponent with $q \notin (1/n!)\mathbb{Z}$. The resulting point is in the ball but outside K_n . Thus merely exhibiting an incomplete metric is not the reason for Theorem 64.3; the theorem rules out every admissible replacement.

65 The Levi-Civita completion: a positive and unique presentation

The classical Levi-Civita field is often defined with real coefficients. Here the coefficient field is deliberately *countable*. This changes separability decisively. Our proofs work for every countable ordered k ; the algebraic and completeness arguments also work without an order on k .

Definition 65.1 (Left-finite support). A set $S \subseteq \mathbb{Q}$ is *left-finite* if $S \cap (-\infty, B]$ is finite for every rational B . The field candidate $\mathcal{L}_k$ consists of all functions $f : \mathbb{Q} \rightarrow k$ with left-finite support, written $\sum f_q t^q$.

A nonempty left-finite set has a least element. Indeed, if $b \in S$, the finite nonempty set $S \cap (-\infty, b]$ has a least element, which is least in S . A left-finite support is either finite or an increasing sequence tending to $+\infty$. In particular, it is a legal Hahn support.

Proposition 65.2 (Field operations). *The usual coefficientwise addition and convolution multiplication turn $\mathcal{L}_k$ into a subfield of $k((t^{\mathbb{Q}}))$.*

Proof. A finite union of left-finite supports is left-finite, so addition is defined and preserves the support condition. Let S, T be nonempty left-finite sets with minima s_0, t_0 . If $s + t \leq B$, then $s \leq B - t_0$ and $t \leq B - s_0$. There are only finitely many such pairs. Thus convolution coefficients are finite sums, and the product has left-finite support. The coefficient at the least exponent of a nonzero product is the product of the two leading coefficients, so the product is nonzero.

Write a nonzero series in the form

$$f = at^q(1 + h), \quad a \in k^\times, \quad q \in \mathbb{Q}, \quad h = 0 \text{ or } v(h) = \delta > 0.$$

For $h \neq 0$, define

$$g = \sum_{j \geq 0} (-h)^j.$$

At exponents at most B , only indices j with $j\delta \leq B$ can contribute, apart from the constant term $j = 0$. There are finitely many such indices, and each corresponding power has finite support below B . Therefore g is left-finite and all coefficient manipulations are finite below any prescribed cutoff. The geometric identity gives $(1 + h)g = 1$. Hence $f^{-1} = a^{-1}t^{-q}g \in \mathcal{L}_k$. The case $h = 0$ is immediate. $\square$

Theorem 65.3 (A complete separable field topology). *For countable ordered k , the metric*

$$d(f, g) = \min\{1, 2^{-v(f-g)}\}, \quad 2^{-\infty} = 0, \quad (65.1)$$

makes $\mathcal{L}_k$ an uncountable Polish topological field. Its topology equals the order topology and is the unique admissible Polish topology on $\mathcal{L}_k$.

Proof. The valuation inequality gives the ultrametric inequality for d . Let (f_j) be Cauchy. For each integer $N \geq 0$ there is an index j_N such that all f_j , $j \geq j_N$, have exactly the same coefficients at every exponent at most N . Choose the j_N increasing. These stabilized finite truncations are compatible, and their union defines a coefficient function f . For every bound B choose an integer $N \geq \max\{0, B\}$. The support of f below B is contained in the finite stabilized support below N . Thus $f \in \mathcal{L}_k$. Stabilization also shows $f_j \rightarrow f$.

Finite rational-exponent polynomials over k form a countable set. Truncating any left-finite series above successively larger integers proves density. The binary series $\sum_{n \geq 0} b(n)t^n$ show uncountability.

Addition, multiplication, and inversion are continuous by the valuation identities used in Theorem 63.3. For completeness, continuity of multiplication at (f, g) follows from

$$(f + h)(g + u) - fg = fu + gh + hu$$

and the lower bound on the valuation of a sum. If $v(h) > v(f)$ and $f \neq 0$, then $v(f+h) = v(f)$, and the inverse-difference identity proves continuity of inversion there.

For $\varepsilon > 0$, every series of valuation strictly greater than $v(\varepsilon)$ has absolute ordered value less than ε . Conversely, an interval of radius t^{B+1} lies inside the ball $\{x : v(x) > B\}$. Translation gives equality of the two topologies. The positive cone is open, so the topology is admissible. Uniqueness follows from Corollary 61.5. $\square$

Proposition 65.4 (Completion and strict inclusions). *The valued-field completion of $\mathcal{P}_k$, and also of the rational-function field $k(t^{\mathbb{Q}})$, is $\mathcal{L}_k$. Moreover,*

$$k(t^{\mathbb{Q}}) \subseteq \mathcal{P}_k \subsetneq \mathcal{L}_k \subsetneq \mathcal{H}_k.$$

Proof. Each Puiseux support lies on a discrete rational lattice and is bounded below, so is left-finite. Both $\mathcal{P}_k$ and $k(t^{\mathbb{Q}})$ contain every finite rational-exponent polynomial. These polynomials are dense in the complete field $\mathcal{L}_k$, proving the completion assertion.

For strictness of the first displayed strict inclusion, use

$$x = \sum_{n \geq 2} t^{n+1/n}. \quad (65.2)$$

Its exponents are increasing and unbounded, so $x \in \mathcal{L}_k$. Their reduced denominators are exactly n , since $\gcd(n^2 + 1, n) = 1$. They cannot all divide one fixed factorial, so $x \notin \mathcal{P}_k$. For the other strict inclusion use

$$y = \sum_{n \geq 2} t^{1-1/n}. \quad (65.3)$$

The support is increasing of order type ω , hence well ordered, but is infinite below 1. Thus $y \in \mathcal{H}_k \setminus \mathcal{L}_k$. $\square$

The partial sums of (65.2) are valuation-Cauchy. The partial sums of (65.3) are not: their successive differences have valuations bounded above by 1. A Hahn-summable family need not be a convergent sequence of partial sums in the valuation topology. This is the precise support distinction behind the completion boundary.

Proposition 65.5 (The topology is not locally compact). *No point of $\mathcal{L}_k$ has a compact neighborhood.*

Proof. Every neighborhood of 0 contains a valuation ball $V_B = \{f : v(f) > B\}$. Fix $q > B$ and infinitely many distinct coefficients $c_j \in k$. Then $c_j t^q \in V_B$, and the mutual distances are the same positive number $\min\{1, 2^{-q}\}$. Thus no set containing V_B is totally bounded, and it cannot be compact. Translation deals with other points. $\square$

Remark 65.6 (Why real coefficients do not work). The usual real-coefficient Levi-Civita field is complete in its valuation topology but is not separable in that topology. More strongly, it has no admissible Polish topology at all: the constant-indexed disjoint intervals from Lemma 63.2 lie inside this smaller field as well. The positive construction is therefore not a claim that the conventional real-coefficient field is Polish.

66 A prefix criterion for separability

The Hahn and Levi-Civita results can be organized without referring to a particular coding of supports. Let F be any subfield of $k((t^{\mathbb{Q}}))$ that contains k and all rational monomials. For $B \in \mathbb{Q}$, let

$$\pi_B(f) = (f_q)_{q \leq B}, \quad V_B = \{f \in F : v(f) > B\}.$$

The prefix $\pi_B(f)$ may be infinite. It need not itself represent an element of F .

Theorem 66.1 (Countable prefixes characterize order separability). *The following conditions are equivalent:*

- (i) *the valuation topology of F is separable;*
- (ii) *the order topology of F is second countable;*
- (iii) *for every integer $N \geq 0$, the set $\pi_N(F) = \{\pi_N(f) : f \in F\}$ is countable.*

Every admissible Polish topology on F forces these conditions.

Proof. The order and valuation topologies agree by the monomial comparison in Theorem 65.3. The prefix classes at cutoff N are exactly the open cosets of V_N . A dense countable set meets every such coset, proving (i) $\Rightarrow$ (iii).

Conversely, choose one representative of each prefix class for every integer $N \geq 0$. The union D of these countably many countable sets is dense: every valuation neighborhood contains a coset at an integer cutoff and hence one of the chosen representatives. The cosets themselves give a countable base. This proves (iii) $\Rightarrow$ (i),(ii), and (ii) $\Rightarrow$ (i) is immediate.

Finally, in an admissible topology each V_N is open by Theorem 61.4. Its pairwise disjoint cosets are therefore countable in number. This gives (iii) even when the candidate Polish topology is strictly finer than the valuation topology. $\square$

For $\mathcal{L}_k$, each prefix is a finite rational-exponent polynomial, so there are only countably many. The same holds for $\mathcal{P}_k$. For $\mathcal{H}_k$, a binary family supported below a fixed bound gives continuum many different prefixes. Thus the prefix criterion captures the Hahn obstruction but cannot by itself capture the Puiseux obstruction: separability and complete metrizability remain distinct questions.

A useful sufficient condition follows. If F has countably many prefixes at each cutoff and is complete for the valuation metric, then its order topology is Polish and admissible; it is then the unique admissible topology. Whether all admissible intermediate-field presentations must arise this way is left as a sharply stated further question in Section 73.

67 Exact descriptive complexity of the three code domains

In this section k is a countable ordered field with the discrete topology, and all domains are regarded as subsets of the Polish product $k^{\mathbb{Q}}$. The phrase “complete” refers to continuous reductions, with zero-dimensional Polish source spaces. This is a complexity statement about coefficient codes; it is not a statement that their subspace topology is the desired group topology.

Theorem 67.1 (Puiseux codes). *The coefficient-code domain $\mathcal{P}_k \subseteq k^{\mathbb{Q}}$ is Σ_2^0 -complete.*

Proof. For positive integers n and nonnegative integers M , the set of all coefficient functions supported on

$$(1/n!)\mathbb{Z} \cap [-M, \infty)$$

is closed in $k^{\mathbb{Q}}$. Every such support defines a Puiseux series, and every Puiseux series occurs for some n, M . Hence $\mathcal{P}_k$ is F_σ .

Let $\text{FIN} \subseteq 2^{\mathbb{N}}$ be the set of binary sequences with finite support, a standard Σ_2^0 -complete set. Define a continuous map by placing the bit $b(m)$ at exponent $1 - 1/(m + 2)$ and placing zero at every other exponent. Finite support gives a polynomial. Infinite support gives infinitely many nonzero exponents in the bounded interval $[1/2, 1)$, which cannot lie in one discrete rational lattice. The map reduces FIN to $\mathcal{P}_k$. $\square$

Theorem 67.2 (Levi-Civita codes). *The coefficient-code domain $\mathcal{L}_k \subseteq k^{\mathbb{Q}}$ is Π_3^0 -complete.*

Proof. At a fixed integer cutoff $N \geq 0$, having finite support below N is F_σ : take the union over finite $S \subseteq \mathbb{Q} \cap (-\infty, N]$ of the closed condition that all coefficients below N outside S vanish. Intersect over N to obtain a Π_3^0 definition.

The set

$$\text{FIN}^{\mathbb{N}} = \{b \in 2^{\mathbb{N} \times \mathbb{N}} : \text{each row } b(n, \cdot) \text{ has finite support}\}$$

is Π_3^0 -complete. Place the coefficient $b(n, m)$ at

$$q(n, m) = n + 1 - \frac{1}{m + 2}, \quad (67.1)$$

and zero elsewhere. These rational exponents are pairwise distinct. If one row is infinite, its support is infinite below $n + 1$, so the resulting code is not left-finite. If every row is finite, only finitely many rows can contribute below any prescribed bound, and each contributes finitely many terms. The map is continuous and gives the required reduction. $\square$

[write] These are the Borel levels of source 03's [theorem 19.1](#) ($\text{WO}(\mathbb{Z})$ is Σ_2^0 -complete, $\text{WO}(\mathbb{Z}_{\text{lex}}^d)$ is Π_3^0 -complete for $d \geq 2$), attained here by different sets.

For completeness, the usual hardness inputs can be obtained directly. Given an F_σ set $A = \bigcup_n C_n$ in Baire space, take the C_n closed and increasing. At stage s , let $m_s(x)$ be the least $n \leq s$ whose closed-set tree is compatible with the length- s prefix of x , or $s + 1$ if there is none. The values m_s are nondecreasing. Output a 1 exactly when the value increases. This is a continuous map into $2^{\mathbb{N}}$; it has finitely many 1s exactly when $x \in A$. Coordinatewise application to a countable intersection of F_σ sets proves the completeness of $\text{FIN}^{\mathbb{N}}$.

Theorem 67.3 (Full rational Hahn codes; classical support obstruction). *The coefficient-code domain $\mathcal{H}_k \subseteq k^{\mathbb{Q}}$ is Π_1^1 -complete. Consequently, it has no standard Borel parametrization with Borel coefficient observations and exactly one code for each series.*

Proof. A support fails to be well ordered exactly when there is an infinite strictly decreasing sequence of rational exponents with nonzero coefficients. Existence of that sequence is an analytic condition. Thus $\mathcal{H}_k$ is coanalytic.

For hardness use the standard Π_1^1 -complete set WF of well-founded trees on $\mathbb{N}$ [8]. Give $\mathbb{N}^{<\mathbb{N}}$ the Kleene–Brouwer order: proper extensions precede their prefixes, and incompatible sequences are compared at their first differing coordinate. This is a countable linear order, so it embeds into $\mathbb{Q} \cap (0, 1)$. Fix such an embedding e . For a tree T , put coefficient 1 at $e(s)$ exactly when $s \in T$, and put zero elsewhere. The map is continuous. An infinite branch gives a descending chain in the Kleene–Brouwer order. Conversely, well-founded induction on a

well-founded tree shows that the order below each node is a countable ordered sum of the child orders, followed by the node itself; it is therefore a well-order. Hence $T \in WF$ exactly when the resulting support is well ordered.

If a standard Borel space parametrized all Hahn series uniquely and every coefficient were Borel, its coefficient map into $k^{\mathbb{Q}}$ would be a Borel injection with range exactly $\mathcal{H}_k$. The Lusin–Souslin theorem would make that range Borel, contradicting completeness. $\square$

[write] [Theorem 67.3](#) is a second proof of the non-scattered half of source 03’s [theorems 18.4](#) and [20.2](#) (via [lemma 18.2](#)) and of the collection’s lightface results `onot:neg:lem:kbtree` and `onot:neg:thm:pionesone`; source 09 credits “the preceding project report” and claims no priority for it.

The last theorem is included to complete the comparison, not to reclaim the project’s earlier coanalytic support result [\[44\]](#). It is also logically weaker than the absolute topological obstruction for a different reason: it requires coefficient-observable codes, whereas [Theorem 63.3](#) does not.

Proposition 67.4 (The Polish Levi-Civita topology preserves the Borel codes). *The valuation topology on $\mathcal{L}_k$ has the same Borel sets as the subspace Borel structure inherited from $k^{\mathbb{Q}}$.*

Proof. Each coefficient map is continuous for the valuation topology: fixing the entire prefix up to its exponent fixes that coefficient. Thus coefficient Borel sets are valuation-Borel. In the other direction, take the countable base consisting of balls centered at finite rational-exponent polynomials with nonnegative integer cutoffs. A basic ball is defined by countably many coefficient equalities. It is therefore Borel in the coefficient structure. Every valuation-open set is a countable union of these basic balls, proving the converse inclusion. $\square$

Both $\mathcal{P}_k$ and $\mathcal{L}_k$ carry Borel field operations in their coefficient structures. For $\mathcal{L}_k$ this follows from [Proposition 67.4](#) and continuous valuation arithmetic. On $\mathcal{P}_k$ the same Borel equality holds for its countable valuation base; its field operations are valuation-continuous. This does not imply that $\mathcal{P}_k$ is completely metrizable as a group.

Field	Exact coefficient-code complexity	Admissible Polish arithmetic
$\mathcal{P}_k$	Σ_2^0 -complete	None; proper-subfield cover obstruction
$\mathcal{L}_k$	Π_3^0 -complete	Exists and is unique; valuation completion
$\mathcal{H}_k$	Π_1^1 -complete	None; uncountably many separated prefixes

Table 2: Countable ordered coefficients and rational exponents throughout. A low Borel rank does not predict Polishability.

68 Real closedness and completion without a defect argument

To connect these fields to integer-part arithmetic, we specialize to a real closed coefficient field. A useful proof avoids importing a full theory of henselian defect. The sole substantial algebraic input is the classical Newton–Puiseux theorem in characteristic zero: for an algebraically closed field K of characteristic zero, $\bigcup_n K((t^{1/n!}))$ is algebraically closed. Poonen gives this as Corollary 7 [\[30\]](#).

Lemma 68.1 (Completion preserves algebraic closedness). *The metric completion of a non-Archimedean valued algebraically closed field is algebraically closed.*

Proof. Write $|\cdot|$ for the multiplicative real-valued absolute value; it is not the truncated metric. Let K be the field and $\widehat{K}$ its completion. Take a monic polynomial

$$P(X) = X^d + \sum_{i < d} a_i X^i \in \widehat{K}[X], \quad d \geq 1.$$

Choose monic polynomials $P_n \in K[X]$ of degree d whose coefficients converge to those of P , and positive real numbers $\varepsilon_n \rightarrow 0$ such that every coefficient of $P_{n+1} - P_n$ has absolute value at most ε_n . There is a common real bound $M \geq 1$ for 1 and for the absolute values of all coefficients of all P_n . Every root z of any P_n satisfies $|z| \leq M$: otherwise the monic term would strictly dominate every lower-degree term.

Choose a root z_0 of P_0 . Having chosen z_n , factor $P_{n+1}(X) = \prod_{j=1}^d (X - w_j)$ over K . Since $P_n(z_n) = 0$,

$$\prod_{j=1}^d |z_n - w_j| = |P_{n+1}(z_n)| \leq \varepsilon_n M^{d-1}.$$

Some root w_j therefore satisfies $|z_n - w_j| \leq (\varepsilon_n M^{d-1})^{1/d}$. Choose it as z_{n+1} . Consecutive distances tend to zero, and the ultrametric inequality then makes (z_n) Cauchy: the distance between any two late terms is at most the largest intervening consecutive distance. Let $z \in \widehat{K}$ be its limit. Continuity of polynomial evaluation and coefficient convergence give $P(z) = 0$. Thus every nonconstant polynomial has a root, which is equivalent to algebraic closedness. $\square$

Theorem 68.2 (Real closed Puiseux and Levi-Civita fields). *If k is a real closed ordered field, then $\mathcal{P}_k$ and $\mathcal{L}_k$ are real closed. In particular, this applies to the countable field $k = \mathbb{R}_{\text{alg}}$.*

Proof. Let $K = k(i)$, which is algebraically closed because k is real closed. Splitting coefficients into their real and imaginary parts gives

$$\mathcal{P}_K = \mathcal{P}_k(i), \quad \mathcal{L}_K = \mathcal{L}_k(i).$$

For the first identity, finitely many denominator bounds can be replaced by one common bound; for the second, a finite union of left-finite supports is left-finite. By Newton–Puiseux, $\mathcal{P}_K$ is algebraically closed. By Proposition 65.4 and Lemma 68.1, so is $\mathcal{L}_K$. The fields $\mathcal{P}_k$ and $\mathcal{L}_k$ are ordered and adjoining i makes them algebraically closed; the standard characterization of real closed fields therefore proves the assertion. $\square$

Corollary 68.3 (A ring-language separation). *For $k = \mathbb{R}_{\text{alg}}$, the field $\mathcal{P}_k$ has no Polish additive-group topology with jointly Borel multiplication, even without a separately specified measurable order. The field $\mathcal{L}_k$ does have such a topology, and it is unique.*

Proof. Real closedness makes the order Borel by Corollary 61.6. Apply Theorems 64.3 and 65.3. $\square$

Real closedness here is a classical algebraic property proved using a classical input; it is not the proposed novelty. Its role is to remove the order-measurability hypothesis from the final field-language contrast.

69 Local Polish hulls, unbounded support ranks, and an exact cover number

The failure of Polishability for the full Hahn field is global, not an obstruction to accommodating any prescribed countable collection of its elements. This section proves that much stronger local positive statement. In particular, the Levi-Civita field is a minimal completion, not a maximal Polish subfield of the Hahn field.

Lemma 69.1 (Metric completeness of the rational Hahn field). *For every field k , $k((t^{\mathbb{Q}}))$ is complete for the metric $d(f, g) = \min\{1, 2^{-v(f-g)}\}$.*

Proof. For a Cauchy sequence, every prefix up to an integer $N \geq 0$ eventually stabilizes, just as in Theorem 65.3. This time a stabilized prefix need not have finite support, but it has well-ordered support because it is a prefix of a Hahn series. The compatible stabilized prefixes define a coefficient function f . Its support is well ordered: given any nonempty subset S of that support, choose $s \in S$ and an integer $N \geq \max\{0, s\}$. The nonempty set $S \cap (-\infty, N]$ has a least element since it lies in the stabilized well-ordered prefix. That element is also least in S , as all remaining points exceed N . Thus f is a Hahn series, and stabilization shows that it is the limit of the original sequence. $\square$

Theorem 69.2 (Countable-parameter Polish hulls). *Let k be countable and ordered, $H = k((t^{\mathbb{Q}}))$, and $A \subseteq H$ countable. Then*

$$F_A = \overline{k(t^{\mathbb{Q}})(A)}^v \subseteq H \quad (69.1)$$

is a complete separable ordered subfield containing $\mathcal{L}_k \cup A$. Its valuation topology is its unique admissible Polish topology. It is the smallest valuation-closed subfield of H containing $k(t^{\mathbb{Q}}) \cup A$. Every valuation-complete separable intermediate field $\mathcal{L}_k \subseteq F \subseteq H$ arises in this way.

Proof. The algebraic field $k(t^{\mathbb{Q}})(A)$ is countable. Its closure is separable and is complete by Lemma 69.1. Closure preserves addition and multiplication by continuity, and preserves inversion at every nonzero limit by the inverse-difference identity. Thus F_A is a subfield. The closure of $k(t^{\mathbb{Q}})$ in H is exactly $\mathcal{L}_k$: it is dense in $\mathcal{L}_k$, and $\mathcal{L}_k$ is complete and hence closed in the metric space H . Therefore $\mathcal{L}_k \subseteq F_A$. The valuation and order topologies agree, and Corollary 61.5 gives uniqueness. Minimality follows directly from taking closure. Finally, for a complete separable intermediate field F , choose a countable valuation-dense subset $A \subseteq F$. Completeness makes F closed in H , and (69.1) then gives $F_A = F$. $\square$

Taking $A = \{y\}$ for the bounded infinite support in (65.3) immediately gives

$$\mathcal{L}_k \subsetneq F_{\{y\}} \subsetneq H.$$

The second strict inclusion follows from Theorem 63.3. Thus bounded infinite supports do not individually prevent Polish field arithmetic. What is impossible is allowing all independently varying bounded prefixes at once.

Theorem 69.3 (A support bound for every separable family). *Let $X \subseteq H = k((t^{\mathbb{Q}}))$ be valuation-separable, and let $D \subseteq X$ be countable and dense. Then*

$$\text{ot}(\text{supp } f) \leq \beta_D := \sup_{d \in D} \text{ot}(\text{supp } d) < \omega_1 \quad (f \in X). \quad (69.2)$$

Here the empty support has order type 0. In particular, every admissibly Polish intermediate field has a uniform countable support-order bound. There is no uniform countable bound valid for all Polish hulls F_A .

Proof. For $f \in X$ and every integer $N \geq 0$, density supplies a $d_N \in D$ with $v(f - d_N) > N$. Their prefixes through N agree. Hence

$$\text{ot}(\text{supp } f \cap (-\infty, N]) \leq \text{ot}(\text{supp } d_N) \leq \beta_D.$$

These prefixes are increasing initial segments whose union is the support of f , so their order types have supremum $\text{ot}(\text{supp } f)$. This proves (69.2). The supremum defining β_D is countable because D and all rational supports are countable.

For an admissible topology, a countable dense set remains valuation-dense by Theorem 61.4, so the preceding argument applies. For the last assertion, let α be any countable ordinal. Embed its linear order into $\mathbb{Q} \cap (0, 1)$ and let h_α have coefficient 1 on that support and 0 elsewhere. It belongs to the Polish hull $F_{\{h_\alpha\}}$ and has support order type α . $\square$

[write] Source 03's theorem 21.2 proves a uniform countable support bound for every Borel coefficient-observable family by analytic boundedness; theorem 69.3 is the elementary counterpart for separable families, as source 09 says below.

This bound uses separability and prefix stabilization, not analytic boundedness or a Borel-domain hypothesis. It is a specialized elementary counterpart of the support bounds in the preceding Borel-support manuscript. It also makes the failure of maximality explicit.

Corollary 69.4 (No maximal admissibly Polish intermediate field). *Every admissibly Polish field F with $k(t^\mathbb{Q}) \subseteq F \subseteq H$ is properly contained in another such field. The larger field can have an element whose support order type exceeds all support order types in F .*

Proof. Choose a countable dense set D in the admissible topology of F and let β_D be as in Theorem 69.3. Choose a Hahn series h with support order type $\beta_D + 1$. The hull $F_{D \cup \{h\}}$ contains the valuation closure of D , and hence contains F . It also contains $h \notin F$. Theorem 69.2 proves that it is admissibly Polish. $\square$

69.1 Real-closed, truncation-closed hulls

With $k = \mathbb{R}_{\text{alg}}$, a hull can be required to preserve both real closedness and all rational truncations. Full Hahn real closedness follows from the classical algebraic-closedness theorem for Hahn fields with algebraically closed coefficients and divisible exponent group [30, Corollary 4], applied after adjoining i .

Theorem 69.5 (A strengthened hull with a countable truncation core). *Let $H = \mathbb{R}_{\text{alg}}((t^\mathbb{Q}))$ and let $A \subseteq H$ be countable. There exist a countable real closed truncation-closed subfield $B \subseteq H$ and a field $C = \overline{B}^v$ such that*

$$k(t^\mathbb{Q}) \cup A \subseteq B \subseteq C,$$

C is real closed and admissibly Polish, and every rational prefix of every $f \in C$ belongs to B :

$$T_{\leq q}(C) \subseteq B \quad (q \in \mathbb{Q}). \quad (69.3)$$

In particular, C is truncation-closed and contains $\mathcal{L}_k \cup A$.

Proof. Start with the countable field $B_0 = k(t^\mathbb{Q})(A)$. Given B_n , adjoin $T_{\leq q}b$ for all $b \in B_n$ and all $q \in \mathbb{Q}$, then take its real closure inside the real closed field H to obtain B_{n+1} . Each step is countable: there are countably many truncations and countably many elements algebraic over a countable field. The union $B = \bigcup_n B_n$ is a countable field, is truncation-closed, and is real closed. For the last assertion, any finite list of polynomial coefficients and any positive

element occur at one stage, and the requisite odd-degree roots and positive square roots occur at the next stage.

The field $C = \overline{B}^v$ is complete and separable, hence admissibly Polish by the same argument as Theorem 69.2. For $f \in C$ and $q \in \mathbb{Q}$, choose $b \in B$ with $v(f - b) > q$. Then $T_{\leq q} f = T_{\leq q} b \in B$, proving (69.3). To prove real closedness, regard $B(i)$ inside $H(i)$. The coefficientwise valuation satisfies

$$v(a + ib) = \min\{v(a), v(b)\} \quad (a, b \in H).$$

Consequently its completion is $C(i)$. Since B is real closed, $B(i)$ is algebraically closed; Lemma 68.1 makes $C(i)$ algebraically closed. As C is ordered, it is real closed. $\square$

69.2 An exact global invariant

Let $\text{pcov}_k(H)$ be the least cardinality of a family of admissibly Polish intermediate subfields $k(t^{\mathbb{Q}}) \subseteq F \subseteq H$ whose union is H . The topology on a covering field need not initially be the inherited valuation topology.

Theorem 69.6 (The Polish field-cover number). *For every countable ordered coefficient field k ,*

$$\text{pcov}_k(k((t^{\mathbb{Q}}))) = 2^{\aleph_0}.$$

For $k = \mathbb{R}_{\text{alg}}$, the same minimum holds even if all covering fields are required to be real closed, truncation-closed, and Polish in their valuation topology.

Proof. For the upper bound, $|H| = 2^{\aleph_0}$, and the fields $F_{\{h\}}$ from Theorem 69.2, one for each $h \in H$, cover H . For real algebraic coefficients use the stronger hulls from Theorem 69.5 instead.

For the lower bound, use the continuum-sized binary family

$$h_b = \sum_{n \geq 0} b(n)t^{1-1/(n+2)}, \quad b \in 2^{\mathbb{N}}.$$

Distinct centers differ in absolute ordered value by more than $2t$. In any one admissibly Polish intermediate field, the intervals of radius t about the centers it contains are nonempty, open, and pairwise disjoint. It therefore contains only countably many members of this binary family. A union of $\kappa < 2^{\aleph_0}$ such fields contains at most $\kappa \cdot \aleph_0 < 2^{\aleph_0}$ members of the family, and cannot cover H . This cardinal calculation is in ZFC and uses no continuum hypothesis. $\square$

The local and global statements now fit together exactly. Every countable collection of Hahn data is contained in one well-behaved Polish field; no such field is maximal; nevertheless continuum many are necessary to cover all Hahn data. These conclusions are not consequences of the mere cardinality $|H| = 2^{\aleph_0}$: the lower bound uses the fixed-scale separation forced by arithmetic and order.

70 A countable omnific integer part and its exact floor regularity

Throughout this section $k = \mathbb{R}_{\text{alg}} \subseteq \mathbb{R}$. Define

$$\mathcal{I}_k = \mathbb{Z} \oplus \bigoplus_{q \in \mathbb{Q}_{<0}} kt^q. \quad (70.1)$$

Thus an element has finitely many negative-exponent terms and an ordinary integer constant term. The direct sum is literal finite support; replacing it by arbitrary Hahn support would change the ring.

Theorem 70.1 (A common integer part). *The ring $\mathcal{I}_k$ is countable and discretely ordered. It is an integer part of both $\mathcal{P}_k$ and $\mathcal{L}_k$. If*

$$f = p + c + u,$$

where p is the negative principal part, $c = f_0 \in k$, and u has only positive exponents, then

$$\lfloor f \rfloor_{\mathcal{I}_k} = p + \lfloor c \rfloor - \begin{cases} 1, & c \in \mathbb{Z} \text{ and } u < 0, \\ 0, & \text{otherwise.} \end{cases} \quad (70.2)$$

Proof. Finite rational supports with coefficients in a countable field form a countable set. Addition and multiplication preserve finite negative support and integer constant terms, so $\mathcal{I}_k$ is a countable unital subring. If a nonzero element has a negative leading exponent, its absolute ordered value exceeds every positive ordinary integer. Otherwise it is a nonzero ordinary integer. Hence 1 is its least positive element.

For $f \in \mathcal{L}_k$, left-finiteness makes p finite, and every nonzero u is smaller in absolute ordered value than every positive constant of k . If $c \notin \mathbb{Z}$, both $c - \lfloor c \rfloor$ and $\lfloor c \rfloor + 1 - c$ are positive constants; the infinitesimal u cannot change either inequality. If $c \in \mathbb{Z}$ and $u \geq 0$, the floor is $p + c$. If $c \in \mathbb{Z}$ and $u < 0$, the floor is $p + c - 1$. In every case the stated element a satisfies $a \leq f < a + 1$. Uniqueness follows because distinct elements of $\mathcal{I}_k$ differ by at least 1 in absolute ordered value. Since $\mathcal{P}_k \subseteq \mathcal{L}_k$ and contains $\mathcal{I}_k$, the same result holds in $\mathcal{P}_k$. $\square$

[write] With $t^q \mapsto \omega^{-q}$, formula (70.2) is the collection's floor rule `odg : thm : floor`, proved in Lean for every surreal, and source 03's lemma 23.1. For $k = \mathbb{R}_{\text{alg}}$, the ring $\mathcal{I}_k$ is source 03's finite principal-part ring $\mathcal{A}_{\text{fin}}$ with coefficients restricted to $\mathbb{R}_{\text{alg}}$ (with $X = t^{-1}$), a countable subring of it; it is also the Shepherdson ring used by batch 86's manuscript 10 in Algebra/SurrealNumbers/docs/surreal/discrete-initial-subgroups-and-omnific-normalization.

Formula (70.2), particularly its correction for a negative infinitesimal at an integer constant, is a specialization of the omnific normal-form floor rule already documented in ProveIt [42]. It is not claimed as a new floor formula. The following topological consequences are the point of its use here.

Theorem 70.2 (Exact discontinuity set and Baire class). *Give $\mathcal{L}_k$ its unique admissible Polish topology and $\mathcal{I}_k$ the discrete topology. The floor map $\lfloor \cdot \rfloor_{\mathcal{I}_k} : \mathcal{L}_k \rightarrow \mathcal{I}_k$ is Baire class one. Its set of discontinuities is exactly $\mathcal{I}_k$, a countable closed nowhere-dense subset of $\mathcal{L}_k$.*

Proof. Its fibers are the half-open intervals $[a, a + 1)$ for $a \in \mathcal{I}_k$. If $f \notin \mathcal{I}_k$ and $a = \lfloor f \rfloor_{\mathcal{I}_k}$, then $a < f < a + 1$, so a sufficiently small order neighborhood of f lies in the same fiber. The floor is continuous there. At $a \in \mathcal{I}_k$, the sequence $a - t^n$ converges to a , while

$$\lfloor a - t^n \rfloor_{\mathcal{I}_k} = a - 1 \neq a = \lfloor a \rfloor_{\mathcal{I}_k}.$$

Thus the discontinuity set is exactly $\mathcal{I}_k$.

For $n \geq 0$ let $T_n f = \sum_{q \leq n} f_q t^q$, a finite polynomial. The map T_n into its discrete, countable range is locally constant: $v(f - g) > n$ implies $T_n f = T_n g$. Consequently

$$F_n(f) = \lfloor T_n f \rfloor_{\mathcal{I}_k}$$

is a continuous map $\mathcal{L}_k \rightarrow \mathcal{I}_k$. These maps converge pointwise to the floor, and indeed are eventually equal to it at each input. The negative principal part and constant are already fixed for all $n \geq 0$. When the constant is an integer and the positive tail is nonzero, the sign needed for the correction is determined by the first positive exponent, which is included once n is large enough. In all other cases no such correction can change. This proves the Baire class one assertion.

Closed discreteness follows from Theorem 62.3. Its interior is empty because every neighborhood of $a \in \mathcal{I}_k$ contains $a + t^n \notin \mathcal{I}_k$ for large n . Hence it is nowhere dense. $\square$

Example 70.3 (Boundary tests). The formulas include

$$\lfloor -t \rfloor_{\mathcal{I}_k} = -1, \quad \lfloor 7 - t^{1/3} \rfloor_{\mathcal{I}_k} = 6, \quad \lfloor 7 + t^{1/3} \rfloor_{\mathcal{I}_k} = 7,$$

and

$$\lfloor 3t^{-2} - 5t^{-1/2} + \frac{3}{2} - t^{10} \rfloor_{\mathcal{I}_k} = 3t^{-2} - 5t^{-1/2} + 1.$$

A finite-precision procedure that ignores a small negative tail at an integer constant is wrong exactly at the kind of boundary detected by the theorem. The continuous approximants above permit eventual correction; they do not provide an input-uniform stage at which correction is finished.

70.1 Floors in the larger Polish hulls

The strengthened hulls of Theorem 69.5 support the same regularity without restricting their negative principal parts to finite support.

Theorem 70.4 (Canonical integer parts in real closed Polish hulls). *Let $B \subseteq C \subseteq \mathbb{R}_{\text{alg}}((t^{\mathbb{Q}}))$ be as in Theorem 69.5. Then*

$$J_C = \{f \in C : \text{supp } f \subseteq \mathbb{Q}_{\leq 0}, f_0 \in \mathbb{Z}\}$$

is a countable closed discrete integer part of C , and $J_C \subseteq B$. Its floor is Baire class one as a map into discrete J_C , with discontinuity set exactly J_C . Under the surreal normal-form embedding it is a subring of the omnific integers.

Proof. For $f \in C$, the principal part $p = T_{\leq 0}f - f_0$ belongs to B by (69.3). Thus the formula

$$\lfloor f \rfloor_{J_C} = p + \lfloor f_0 \rfloor - \mathbf{1}_{\{f_0 \in \mathbb{Z}, f - T_{\leq 0}f < 0\}}$$

defines an element of C with no positive exponents and integer constant. The same leading-term comparisons as in Theorem 70.1 prove the integer-part inequalities and discrete order. Ring closure follows from convolution: negative exponents cannot contribute to the constant term. Every $j \in J_C$ equals $T_{\leq 0}j$, so lies in the countable set B . Closed discreteness also follows from Theorem 62.3.

For each nonnegative integer n , the map $T_{\leq n} : C \rightarrow B$, with B given the discrete topology, is locally constant. The maps $f \mapsto \lfloor T_{\leq n}f \rfloor_{J_C}$ are therefore continuous and are eventually equal to the floor at each input. Only the first nonzero positive term is needed to decide the possible correction. The interval fibers and the sequence $a - t^n \rightarrow a$ give the exact discontinuity set as before. Finally, replacing t^q by ω^{-q} turns the nonpositive support and integer constant conditions into the omnific normal-form conditions. $\square$

Remark 70.5 (The target topology matters). The Baire class one assertion is about a *discrete* integer-part target, not just its inclusion in the field. It is stronger than merely producing continuous real-valued approximants to the ordinary floor on $\mathbb{R}$. Indeed, the ordinary floor $\mathbb{R} \rightarrow \mathbb{Z}_{\text{discrete}}$ is not Baire class one: every continuous map from connected $\mathbb{R}$ to discrete $\mathbb{Z}$ is constant, and a pointwise convergent sequence of constant maps has a constant limit. The locally constant prefix maps are a genuine non-Archimedean resource.

70.2 The common integer part satisfies open induction

The classical relationship between integer parts of real closed fields and open induction is often called the integer-part characterization of open induction. For our specific ring the needed implication has a short proof.

Proposition 70.6 (Open induction). *The nonnegative cone $\mathcal{I}_{k,\geq 0}$ satisfies $Q + \text{IOpen}$ in the language of ordered arithmetic, with parameters allowed in the induction scheme.*

Proof. The elementary axioms of discretely ordered semiring arithmetic hold in the non-negative cone of any discretely ordered unital ring. Every positive $a \in \mathcal{I}_k$ has predecessor $a - 1 \geq 0$.

A quantifier-free formula in one variable with parameters from $\mathcal{I}_k$ is a Boolean combination of polynomial equations and inequalities. In the real closed field $\mathcal{L}_k$, the real roots of its finitely many nonzero polynomials partition the line into finitely many intervals and points on which the formula has constant truth value. Every nonempty trace of one of these pieces on $\mathcal{I}_{k,\geq 0}$ has a least element: the lower endpoint, when included and already in $\mathcal{I}_k$, or the first integer-part element above that endpoint, with 0 serving as a lower bound when needed. The floor property supplies this first element. Therefore every nonempty quantifier-free definable subset of $\mathcal{I}_{k,\geq 0}$ has a least element.

If a quantifier-free formula held at 0 and were preserved by successor but failed somewhere, its counterexamples would have a least element a . Then $a > 0$ and the formula holds at $a - 1$, contradicting preservation by successor. This is open induction. $\square$

[write] This is the route of `dsn:cor:openinduction` and of source 03's [theorem 23.2](#), which proves open induction for $\mathcal{A}_{\text{fin}}$; the Shepherdson model is classical [6].

Corollary 70.7 (Arbitrary countable support complexity in countable arithmetic). *For every countable ordinal α , there is a real closed Polish truncation-closed field $C \subseteq \mathbb{R}_{\text{alg}}((t^{\mathbb{Q}}))$ whose countable canonical integer part J_C contains an element with support order type α . The nonnegative cone of J_C satisfies $Q + \text{IOpen}$.*

Proof. Embed α into $\mathbb{Q} \cap (-2, -1)$ and take the Hahn series with coefficient 1 on that support. Put it in a strengthened hull using Theorem 69.5. Its support is negative and its constant is 0, so it belongs to J_C . The proof of Proposition 70.6 applies verbatim with this integer part and its real closed ambient field. $\square$

There is no conflict with Glazer's result: the arithmetic model $\mathcal{I}_{k,\geq 0}$ is countable. Its uncountable real closed ambient field $\mathcal{L}_k$ is not a discretely ordered arithmetic model. Moreover, $\mathcal{P}_k$ and $\mathcal{L}_k$ share this same integer part even though only the latter supports admissible Polish field arithmetic.

70.3 Why the finite integer part does not extend to the full Hahn field

The element

$$z = t^{-2}y = \sum_{n \geq 2} t^{-1-1/n} \in \mathcal{H}_k,$$

where y is from (65.3), has infinitely many negative terms. For every $a \in \mathcal{I}_k$, at least one negative coefficient of $z - a$ remains nonzero; its leading exponent is negative, so $|z - a| > 1$. Hence $\mathcal{I}_k$ is not an integer part of $\mathcal{H}_k$.

Replacing finite negative support in (70.1) by arbitrary well-ordered negative support gives an integer part of $\mathcal{H}_k$, with the same coefficientwise floor rule. This larger ring is uncountable: binary choices along the support of z already produce continuum many elements. Theorem 62.3 supplies another obstruction to admissible Polish arithmetic on this full Hahn field. Here that argument agrees with, but is less general than, Theorem 63.3.

71 Surreal realization and a formalization interface

71.1 A set-sized embedding, not a topology on a proper class

With $k = \mathbb{R}_{\text{alg}} \subseteq \mathbb{R}$, the normal-form assignment

$$\iota \left(\sum_{q \in \mathbb{Q}} a_q t^q \right) = \sum_{q \in \mathbb{Q}} a_q \omega^{-q} \quad (71.1)$$

realizes the three fields as set-sized subfields of the surreal numbers. Well-ordered support in the q -variable becomes reverse-well-ordered support in the Conway growth exponents $-q$. The normal-form rules preserve addition, multiplication, and order; t corresponds to ω^{-1} . The image of $\mathcal{I}_k$ has only positive growth exponents and an integer constant term, so lies in $\mathbf{Oz}$.

This is a formal normal-form embedding. It does not assert that an arbitrary infinite series can be evaluated at an infinite real or surreal argument by ordinary analytic convergence. In particular, the direction of the exponent change in (71.1) is essential. The full proper class **No** is never asserted to be a Polish space. The series carriers have cardinality continuum and can be constructed as subsets of the set $k^{\mathbb{Q}}$. Only afterwards are they identified with surreal normal forms.

The project documentation is an interface to these structures, not a proof that the new topological statements have already been formalized. The reported kernel checks in ProveIt concern the repository's declared constructions and theorems. No new kernel run was performed for this article.

71.2 A modular Lean development plan

A formalization need not begin with surreal arithmetic. Most of the proof is independent of it. A suitable division into modules is as follows.

Category and ordered groups. Formalize the Baire-order local-bound lemma over a linearly ordered abelian group, separately from the field specialization. Its interface should return an order bound $a > 0$ and a neighborhood inclusion, not an implicit assertion that the ambient

topology is the order topology. Import a proved Pettis lemma and Borel-homomorphism automatic continuity at clearly stated instances.

Ordered rings and fields. Prove automatic discreteness using one scalar $a + 1$, and prove the field refinement theorem using $(a + 1)/\varepsilon$. These are different lemmas: quotient scalars are not available in a general ordered ring. Then prove closed discreteness and countability of any integer-part subring without assuming its carrier is measurable.

Series and topology. Construct left-finite coefficient functions as a subtype of $\mathbb{Q} \rightarrow k$. Give finite-below-cutoff witnesses explicitly. Prove convolution finiteness, the inverse construction, valuation identities, and metric completeness. The countable dense subspace consists of finitely supported functions. A separate theorem identifies the metric and order topologies.

Obstruction certificates. Formalize the binary family from Lemma 63.2; the only infinite support needed for the negative result has order type at most ω . For Puiseux, package the general countable proper-subfield-cover theorem, then prove each lattice-support stage is closed. This separation prevents a proof of incompleteness of one metric from being mistaken for a proof of absolute non-Polishability.

Integer parts and measurable floors. First prove (70.2) in the left-finite series subtype. Then identify this ring with the corresponding set-sized subring of the repository's omnific integers. The exact discontinuity theorem and the pointwise eventually constant sequence (F_n) can be formalized independently of the full surreal field.

Descriptive complexity. The explicit maps in (67.1) and the FIN reduction are the finite combinatorial portion. The well-founded-tree reduction and Lusin–Souslin step belong in a separate descriptive-set-theory layer. They are not required for the core Hahn classification or the positive Levi-Civita construction.

This architecture leaves no proper-class quantification in the main proof. Universe parameters are needed only when identifying the set-sized series carrier with a universe-relative surreal implementation. No additional large cardinal assumption is used.

72 Consequences for exact symbolic arithmetic

An exact coefficient stream and a valuation-Cauchy name provide different information. A coefficient stream reports one coefficient at a time. A valuation-Cauchy name eventually fixes *all* coefficients below a specified cutoff. In a left-finite representation, certifying a completed prefix therefore includes a support bound, not just a list of currently observed nonzero terms.

Theorem 65.3 supports continuous arithmetic in the second representation. Proposition 67.4 says that the measurable information is the same, but it does not turn these representations into the same effective oracle. No computability claim follows merely from Borel equivalence.

For an exact rational-coefficient prototype one can represent finite truncations by dictionaries $q \mapsto a_q$ with $q, a_q \in \mathbb{Q}$. When multiplying to cutoff B , lower bounds on the two supports determine which input prefixes suffice: terms of the first factor above $B - v(g)$, and terms of the second above $B - v(f)$, cannot affect the result. For inversion of $at^q(1 + h)$, a positive

lower bound $\delta \leq v(h)$ bounds the number of geometric powers needed below any fixed cutoff. These are mathematically certified finite reductions, not numerical approximations to real coefficients.

Floor is different. Its Baire class one approximants may change once the first nonzero infinitesimal term is revealed at an integer boundary. No continuous exact floor algorithm on the whole field can exist, because the mathematical function is discontinuous there. The theorem pinpoints the necessary limitation instead of attributing it to a poor choice of code.

The accompanying Python program uses rational exponents and coefficients to check finite algebraic identities, truncated geometric inverses, the floor inequalities, and finite instances of the support witnesses. The deterministic run passed 20,116 finite assertions; its output records the counts by test family. These tests catch implementation mistakes in the supplied examples; they cannot establish Baire category, quantification over all Polish topologies, infinite support completeness, or real closedness. The conventional proofs are the basis of those assertions.

73 Further research questions and concrete targets

The following questions are proposed next steps. They are not asserted to be previously published open problems, and no novelty claim is inferred from failure to locate an answer in the inspected sources.

Question 73.1 (Intermediate-field rigidity). Let k be countable and ordered, and let $\mathcal{P}_k \subseteq F \subseteq k((t^{\mathbb{Q}}))$. If F has an admissible Polish topology, must its valuation topology be complete, and must the admissible topology equal the valuation topology?

The prefix criterion gives countably many prefixes, but it does not prove completeness of a coarser topology. A proof must close precisely that gap; one cannot appeal to the open mapping theorem until both topologies are known to be Polish.

Question 73.2 (Classification of Polish hulls). Within a fixed standard Borel family of coefficient-observable parameters, what is the descriptive complexity of equality and inclusion between the countable-parameter hulls F_A ? Can one construct useful canonical dense cores or invariants for their ordered-field isomorphism types?

The existence of proper Polish extensions of $\mathcal{L}_k$ and the absence of maximal ones are settled by Theorem 69.2 and Corollary 69.4. The remaining issue is classification, not existence. The ambient parameter space must be specified: the full Hahn coefficient domain is not standard Borel.

Question 73.3 (Minimal scalar hypotheses). How few Borel scalar maps suffice to force the order topology in the rational-exponent setting? Can a specified countable set of monomial multiplications replace all scalar maps under additional algebraic or truncation-closure assumptions?

The current proof uses a scalar depending on the local order bound and the target interval. An answer should identify which part of that freedom is essential, rather than dropping scalar regularity altogether.

Question 73.4 (Order-free non-real-closed fields). For countable ordered k that is not real closed, can the Hahn and Puiseux nonexistence statements be derived from a Polish additive group and jointly Borel multiplication alone, without requiring the displayed order to have the Baire property?

Corollary 61.6 settles the real closed case. In general a field may have several orders, and the image-of-squares argument does not identify one positive cone.

Question 73.5 (Other support completions). Extend the left-finite construction and its classification to countable ordered exponent groups of higher rank. Which support restrictions give separable complete field topologies, and which admit a countable cofinal valuation scale but still fail by a proper-subfield cover?

The inverse proof in Proposition 65.2 explicitly uses $n\delta \rightarrow +\infty$ in an Archimedean exponent group. It cannot be copied unchanged into a lexicographic higher-rank group.

Question 73.6 (Descriptive ranks beyond the trichotomy). Which Borel ranks can occur for coefficient-code domains of admissibly Polish subfields of $k((t^{\mathbb{Q}}))$? Is there a useful algebraic characterization of when a Borel support family yields a Polishable additive group?

Table 2 shows that a smaller rank is not sufficient. A productive refinement should combine support closure, completeness of prefixes, and the proper-subfield obstruction.

Question 73.7 (The least support bound of a one-parameter hull). For $y = \sum_{n \geq 2} t^{1-1/n}$, determine the least countable ordinal bounding all support order types in $F_{\{y\}}$. How does that least bound change when real closure and closure under all rational truncations are imposed before metric completion?

Theorem 69.3 proves existence of a bound but does not compute it. A solution would require quantitative control of support order types under convolution, inversion, algebraic extension, and stabilized-prefix limits. It would turn the qualitative local-hull theorem into an ordinal complexity calculation.

Question 73.8 (Other integer parts and floor discontinuities). Classify integer parts of $\mathcal{L}_k$ and the larger real closed Polish hulls up to field automorphisms. Which floors admit canonical or effective locally constant approximants for presentations not built from truncations?

Countability and closed discreteness hold for every integer part by Theorem 62.3. The discontinuity set of its floor is also its integer part in the order topology. The strengthened hulls retain explicit truncation approximants. The remaining constructive issue is canonical or effective approximation for other presentations, together with a classification of the integer parts.

Question 73.9 (A uniform effective presentation). For computably presented k , construct a certified Type-2 representation of $\mathcal{L}_k$ with computable arithmetic, explicit precision transfer for products and inverses, and a finite-mind-change floor procedure for $k = \mathbb{R}_{\text{alg}}$.

The representation must include effective support-completion information. Bare enumerations of nonzero coefficients do not suffice to know that a prefix is complete. Any claimed effective embedding into Baire space should state which oracle representation is being used.

Question 73.10 (Formal proof milestones). Can the local order-bound theorem, automatic discreteness, and full Hahn classification be formalized as a small reusable library independent of the full surreal development? What is the minimal imported descriptive-set-theory interface for the exact code-complexity theorems?

These are concrete implementation tasks. The first milestone is the category-to-order bridge; the coanalytic lower bound should not block the rest of the project.

Question 73.11 (Positive cones versus groups of differences). Characterize Borel discretely ordered semirings whose positive cones admit Polish continuous addition but whose groups of differences are not Polishable with measurable multiplication. Which hypotheses force a topology to extend from the cone to its group completion?

This returns directly to the boundary emphasized by Glazer’s arithmetic questions and the earlier Presburger constructions. Theorem 62.1 provides an obstruction on the group side, not a classification of monoid presentations.

[write] This is source 08’s [question 57.2](#); source 06’s cone of $\mathbb{Z} + t\mathbb{R}[t]$ is a data point ([section 42](#)).

Question 73.12 (Weak choice and category). Which portions of the classification survive in weaker choice settings? Separate the dependent choices used to build the bounded increasing exponent sequence, the Baire theorem, and the selections of representatives of countably many prefix classes.

The present paper works in ZFC and makes no choice-free assertion. A foundational analysis would be a distinct project rather than an automatic consequence of the set-sized constructions.

74 Conclusions of Part V

The main dividing line is not whether exact infinite series can be assigned Borel codes. It is whether their arithmetic, order, and completeness fit within a separable Polish group. A Baire-measurable order and Borel scalar arithmetic force local order bounds and hence the order topology. Full Hahn support then admits too many mutually separated prefixes, except in the countable-coefficient Laurent case. Puiseux support avoids that obstruction but fails a different category test because it is a countable union of proper measurable subfields. Left-finite completion avoids both obstructions and supplies a unique Polish topological field. Countable-parameter hulls go further: arbitrary countable Hahn data fit into a Polish subfield, even with real closedness and truncation closure over real algebraic coefficients. There is no maximal Polish intermediate field, while the exact cover number of the full field is the continuum.

The integer-part consequences connect this topological classification to arithmetic and surreal normal forms, including arbitrary prescribed countable support types inside suitable countable integer parts. A Polish ordered field cannot hide an uncountable discrete integer part, even a nonmeasurable one. The real algebraic-coefficient Levi-Civita field nevertheless supports a countable omnific integer part with open induction, and its floor has exactly controlled discontinuities. These results supply a precise next layer beyond Borel support analysis while leaving topology rigidity on intermediate fields and the broader positive-cone classification open.

Part VI

Baire order forces arithmetic rigidity

[write] This part merges two manuscripts of batch 87 that prove the same principal theorem by different proofs. Source 12 is *Baire-Category Rigidity for Discretely Ordered Rings: Beyond local compactness, without measurable multiplication* (dated October 3, 2026; “research manuscript

prepared with ChatGPT for Vladimir Reshetnikov”), pinned to repository commit 0f084afa9. Source 10 is *Baire Order Forces Arithmetic Rigidity: Discretely ordered rings with Polish addition and completely unrestricted multiplication* (same date and wording of authorship), pinned to 6fef5383b. They share under 0.8% of their word 8-grams; neither is an edition of the other. Both continue [Part IV](#): they answer its [question 57.1](#) negatively, without any regularity of multiplication.

The principal theorem. A discretely ordered ring whose additive group carries a Hausdorff Baire topology satisfying the countable chain condition, and whose strict positive cone has the Baire property, is countable, and the topology is discrete. In particular this holds for every Polish additive-group topology with Borel order. Source 12 proves it as [corollary 78.4](#) by packing $|R|$ disjoint open translates ([lemma 78.1](#)); source 10 proves it (its Theorem 5.1, [corollary 78.4](#)) by the ordered quotient embedding $A/\text{Fin}(A) \hookrightarrow A/H_a$ of [theorem 79.7](#). Both use an order-bounded neighbourhood of 0 ([lemma 77.2](#)), which is source 09’s [lemma 61.3](#) of [Part V](#), extended from Polish to Hausdorff Baire groups.

Base and reading conventions. Source 12 is the base, because on the shared statements it has the weaker hypotheses: its cardinal theorem ([theorem 78.3](#)) needs neither the chain condition nor uncountability; its semiring theorem ([theorem 81.2](#)) does not assume commutativity; its non-Polishability theorem ([theorem 82.2](#)) assumes no regularity of multiplication; and it has a second, Haar-measure route ([theorem 83.3](#)). Unmarked text in this part is source 12’s. Text from source 10 is marked [10]: in the titles of its sections, subsections and statements, and in the sentence that introduces an inserted passage. Source 10’s sections are printed as subsections of source 12’s sections on the same subject, one level down; its section on continuous scalar maps is a section of its own ([section 85](#)). Text written for this report is marked [write] or [merge] as in [section 1.2](#).

What is printed once. The principal theorem ([corollary 78.4](#); source 10’s proof is printed as a second proof in [section 79.2](#)); the bounded-neighbourhood lemma (source 10’s proof, which avoids Pettis’s lemma, is a second proof); the open principal convex subgroup ([proposition 79.1](#), source 10’s Proposition 4.3); the absence of an ordered-ring expansion ([corollary 81.1](#), source 10’s Corollary 5.3 for Polish topologies); the three topologies on $\mathbb{Z} + t\mathbb{R}[t]$ ([section 84](#); source 10’s Examples 9.1, 9.3 and 9.4, whose additional remarks are printed as marked passages); and the completion without a discrete order ([proposition 86.2](#), source 10’s Proposition 9.2, with its proof as a second proof). Where source 10 states a sharper or differently scoped version, both are printed: the exact Polish criterion ([corollary 78.6](#)), the unit-image spectrum ([theorem 80.1](#) with an explicit bound, source 10’s [theorem 80.4](#) with three further clauses), signed Presburger models ([corollary 81.3](#) with a Borel cone, source 10’s [corollary 81.4](#) with a Baire-property cone) and the omnific corollary ([corollary 86.3](#); source 10’s [corollary 86.4](#) adds extent and Lindelöf degree).

What is new to the collection. The theorems of this part are new to the collection except where a note names an earlier statement. They make several results of Parts IV and V corollaries, among them [theorems 44.1](#) and [62.1](#); [section 90](#) lists them with the deductions. Neither source claims historical priority.

Labels and notation. Labels of this part are `pma:br:<label>` for both sources, except the seven labels of source 10 that coincide with labels of source 12 (`cor:omnific`, `cor:polish`,

sec:category, sec:conventions, sec:embeddings, sec:questions, thm:main), which take the sub-prefix pma:br:s10:. Where source 10 states a result that source 12 also states, source 10's label is a second label of source 12's statement: pma:br:lem:localbound is lemma 77.2, pma:br:prop:openhull is proposition 79.1, pma:br:s10:thm:main is corollary 78.4, pma:br:cor:noexp is corollary 81.1, pma:br:prop:completion is proposition 86.2, and source 10's examples ex:incomplete, ex:nonccc, ex:nonbp are labels of source 12's sections 84.1 to 84.3. Source 10's references to its displays eq:localbound and eq:polynomialring, which are not printed, point to source 12's (77.1) and (84.1). One label was added to an unlabelled statement of source 12 (pma:br:cor:signed-presburger). Source 10's $\text{Fin}(A)$, the ordinary-integer-bounded part, is printed with roman type and is not Part II's set Fin of finite sets; its unit-image spectrum $\mathcal{S}(G, e)$ is printed $\mathcal{U}(G)$, source 12's name for the same set. Section 1.3 lists these and the other conventions of Parts VI–IX; in particular source 12's t and source 10's X are the same positive *infinite* variable of $\mathbb{Z} + t\mathbb{R}[t]$, as in Part III.

75 The research intersection and the question answered

The point of contact is the topology of arithmetic structures: how much arithmetic can coexist with a well-behaved additive topology? Glazer's *A Topological Tennenbaum Theorem* rules out uncountable Polish models of $\mathcal{Q} + \text{IOpen}$ with continuous addition and Borel multiplication [1, Corollary 1]. His Question 2 asks about continuous addition for uncountable Polish Presburger models. ProveIt contains complementary interfaces: integer Presburger elimination and ordered surreal and omnific constructions [51, 52]. This suggests examining what happens when an additive presentation is required to support a full, order-compatible multiplication.

An earlier project report established a local-compactness obstruction [49]. Its first research question in Section 14 asks whether an uncountable discretely ordered ring can carry a Polish additive-group topology with Borel order when multiplication has no regularity assumption. The answer proved here is *no*. More strongly, Borelness can be weakened to the Baire property of a single cone, and Polishness can be weakened to Baire plus the countable chain condition.

The principal result

Baire-category rigidity. If a discretely ordered ring R has a Hausdorff Baire additive-group topology and a positive cone with the Baire property, then

$$c(R) = d(R) = |R|.$$

If this topology satisfies the countable chain condition, then R is countable and the topology is discrete.

There is no regularity hypothesis on multiplication.

Here $c(R)$ is cellularity and $d(R)$ is density. The cardinal equality is not merely an upper-bound obstruction: the proof explicitly constructs $|R|$ disjoint open translates. It identifies a mechanism that persists far beyond the finite-dimensional locally compact setting of the previous report.

75.1 What advances beyond the earlier work

The local-compactness argument used structural information about locally compact abelian groups. The proof below needs no decomposition into real components, no finite-dimensional flag, and no regularity of scalar multipliers. Its central interface is just an order-bounded neighborhood. This makes the argument both more general and shorter to formalize. The main consequences developed here are as follows.

1. A universal packing theorem gives exact density and cellularity, with both Baire-category and Haar-measure routes to its hypothesis.
2. The images of a least positive element under *all* ordered additive self-embeddings are order-bounded in every uncountable Baire ccc presentation; no regularity of the embeddings is required.
3. An uncountable such ordered group admits no compatible unital ordered-ring multiplication, and its nonnegative cone admits no unital semiring multiplication with the prescribed addition and unit.
4. Polynomial examples explain why Baire category, ccc, and regularity of order cannot simply be deleted. A natural completion destroys discrete order even before topology is considered.

These statements are established within the manuscript. We do not claim that each auxiliary lemma is historically new. Pettis's theorem and its measure counterpart are classical; the contribution here is their use with discrete arithmetic separation, the resulting sharp conclusions, and the resolution of the specified project question.

75.2 The scope is not Glazer's entire setting

A positive arithmetic structure has an additive *monoid*. Our central hypothesis is a topology on its signed additive *group*, with continuous inversion as well as continuous addition. Algebraically taking differences does not, by itself, construct a Polish group topology. Therefore the theorem below is not a replacement for Glazer's theorem under weaker hypotheses on the same objects. The carrier and topology requirements are different.

Glazer's Question 1 concerns provability in ATR_0 [1, Question 1]; we do not settle that question. Nor do we re-present the earlier affirmative Presburger construction [50] as a new answer. The exact open question answered here is the one in the preceding local-compactness report.

[write] Source 12's "earlier project report" [49] is source 08, [Part IV](#); its "first research question in Section 14" is [question 57.1](#), and the question on weakening Borel order is the fourth question of that section ([question 57.4](#)). The "earlier affirmative Presburger construction" [50] is source 05, [Part I](#). Source 12's pin precedes the placement of Parts III–V, so it did not see source 09 ([Part V](#)).

75.3 The intersection and the result as source 10 states it [10]

The topic is the boundary between topological presentations of arithmetic and the algebra of non-Archimedean integer systems. It connects a specific part of Elliot Glazer's published research with two concrete parts of ProveIt, rather than relying on a speculative account of his interests. In *A Topological Tennenbaum Theorem*, Glazer proves, in particular, that

an uncountable Polish model of $\mathbb{Q} + \text{IOpen}$ cannot have continuous addition and Borel multiplication [1, Corollary 1]. ProveIt develops Cooper elimination for Presburger arithmetic and an ordered surreal/omnific algebra [54, 55]. These are natural interfaces at which to ask whether an additive presentation can be expanded by multiplication.

Two earlier project manuscripts supply the immediate starting point. *Local Compactness Forces Arithmetic Rigidity* proves a locally compact, second-countable version without regularity assumptions on multiplication, and explicitly asks whether local compactness can be removed [56]. *Polishability at the Hahn–Puisseux–Levi–Civita Boundary* isolates the category argument that makes a Baire order locally bounded [57]. The present article combines that local bound with a different quotient construction. The quotient has a countable kernel even when the open subgroup is neither divisible nor finite dimensional.

Main conclusion. Let A be a nonzero discretely ordered ring. If $(A, +)$ is a Polish topological group and $\{x \in A : x > 0\}$ has the Baire property, then A is countable and its topology is discrete. Multiplication is an arbitrary algebraic operation compatible with the ordered-ring axioms; it is not required to be measurable.

Thus the answer to the general Polish-group question posed in [56] is negative: there is no uncountable example. We actually prove the stronger Baire–ccc statement. Here *ccc* means that every family of pairwise disjoint nonempty open sets is countable.

75.3.1 What changes in the proof

The earlier locally compact argument finds an open real vector component and uses multiplication to move an uncountable divisible subgroup into its quotient. That approach makes finite-dimensional structure appear essential. Instead, for a positive element a , consider

$$H_a = \{x \in A : |x| \leq na \text{ for some ordinary } n \in \mathbb{N}\}.$$

This is a convex additive subgroup. Its elements need not be bounded by a *single* multiple of a ; the ordinary integer n may depend on x . Nevertheless,

$$ax \in H_a \iff |x| \leq n1 \text{ for some ordinary } n.$$

For a discretely ordered ring, the right side says exactly that $x \in \mathbb{Z}1$. Therefore multiplication by a , followed by the additive quotient map, has the countable kernel $\mathbb{Z}1$ on *all* of A . When H_a is open, ccc makes the target countable. No continuity of this map is needed: an algebraic injection is enough.

75.3.2 Scope and contribution ledger

Item	Status in this article
Category local boundedness	Classical category mechanism, already used in [57]; reproved here for Hausdorff Baire groups.
Principal-hull quotient embedding	Main algebraic observation, proved for arbitrary ordered rings in section 79.1 .
Baire-ccc arithmetic rigidity	Main extension beyond [56]; proved without any multiplication regularity in section 79.2 .
Full-cardinality discrete witnesses	Quantitative strengthening for nonseparable presentations; proved in section 79.3 .
Presburger examples	Standard divisible-plus-integer constructions, including examples already used in project reports; not claimed as new.
Universal no-expansion theorem	A consequence for <i>every</i> uncountable Polish signed Presburger group with a Baire positive cone.
Historical priority	Not established by the literature check.
Machine verification	No new Lean/Rocq formalization is claimed; exact-arithmetic tests are supplementary only.

The signed-group hypothesis matters. A Polish topology on a positive arithmetic monoid does not automatically extend to a Polish topology on its group of differences. We return to this distinction in [section 81.4](#). We also do not claim to settle Glazer’s reverse- mathematical Question 1, which concerns an ATR_0 proof of his Theorem 2 [1].

[write] Source 10’s [56] is source 08, [Part IV](#), and [57] is source 09, [Part V](#); the category argument it credits to [57] is [lemma 61.3](#). The “general Polish-group question posed in” [56] is [question 57.1](#).

76 Conventions and elementary ordered algebra

All structures and topological spaces in the theorems are sets, and the ambient foundation is ZFC. We use the usual standard natural numbers when writing an external countable union or a sequence indexed by $\mathbb{N}$.

Definition 76.1. A *discretely ordered ring* is a nonzero unital ring R with a total order such that addition preserves order, the product of two strictly positive elements is strictly positive, and 1 is the least strictly positive element. Unless stated otherwise, rings are commutative.

This is the standard discrete-order convention used in weak arithmetic [47]. It is not a statement that a supplied topology is discrete. In fact, distinguishing those two meanings of discreteness is central to the problem.

The ring order makes the additive group ordered and torsion-free. For $x \in R$ put $|x| = \max(x, -x)$. Directly from the order axioms,

$$x \neq y \implies |x - y| \geq 1, \quad |xy| = |x||y|. \quad (76.1)$$

There are no zero divisors: the absolute value of a product of nonzero elements is positive.

Lemma 76.2 (Ordinary integers form a convex subring). *If R is discretely ordered and $0 \leq x \leq n$ for an ordinary natural number n , then x is one of $0, 1, \dots, n$. Consequently, either $R = \mathbb{Z}$ as an ordered unital ring, or there is $c \in R$ with $c > n$ for every ordinary $n \in \mathbb{N}$.*

Proof. The first assertion follows by external induction on n . If $x > 0$, then $x \geq 1$, so $0 \leq x - 1 \leq n - 1$. For the second assertion choose an element outside the ordinary integer subring and take its absolute value. If it were bounded by any ordinary natural number, the first assertion would put it in that subring. Thus its absolute value is larger than all ordinary natural numbers. $\square$

This uses no induction axiom *inside* the ring. It is an ordinary metatheoretic proof about a finite integer interval.

Definition 76.3. An *additive-group topology* on R is a topology making $(R, +, 0, -)$ a topological group. We always require it to be Hausdorff. We make no compatibility assumption between this topology and the order topology. Write $P = \{x \in R : x > 0\}$.

A set has the *Baire property* if its symmetric difference with an open set is meager. A space is *Baire* if no nonempty open set is meager in the space. These are distinct notions: a positive cone may have the Baire property in a space that is itself meager.

A space satisfies the *countable chain condition*, abbreviated ccc, if every family of pairwise disjoint nonempty open sets is countable. Its cellularity $c(X)$ is the supremum of the cardinalities of such families. Its density $d(X)$ is the least cardinality of a dense subset. For any space,

$$c(X) \leq d(X) \leq |X|. \quad (76.2)$$

Indeed, each member of a disjoint open family meets a dense set, and the chosen intersection points are different. Separability implies ccc. A Polish space is separable and completely metrizable, and is Baire.

For Borelness, regularity of the binary order and regularity of its cone are equivalent in an additive topological group: the cone is a section of the order relation, and

$$x < y \iff y - x \in P.$$

Our category theorem needs only the Baire property of P itself. It does not quantify over the Borel complexity of every definable relation.

76.1 Conventions of source 10 [10]

Throughout the main development, an *ordered ring* is a nonzero commutative unital ring A with a translation-invariant total order such that $x, y > 0$ implies $xy > 0$. This is the strict ordered-domain convention. It implies cancellation of positive factors in inequalities. We do not use the weaker convention allowing positive zero divisors. The noncommutative extension is recorded in [remark 78.9](#).

Write $\underline{n} = n1_A$ for an ordinary integer represented in A . An ordered ring is *discretely ordered* when 1_A is its least positive element. This is an algebraic property, not a statement about the topology. The standard examples and the connection to weak arithmetic are discussed in [47]. All uses of induction on an ordinary integer in this paper are external finite mathematical arguments; no first-order induction axiom inside A is assumed.

The topology τ is a topology on the additive group: addition and negation are continuous. Unless stated separately, multiplication is not a topological operation. The topology need not agree with the order topology. We write

$$P = \{x \in A : x > 0\}, \quad |x| = \max(x, -x).$$

The set P has the Baire property if it differs from an open set by a meager set. This condition on the positive cone is exactly the regularity used. In particular, a Borel positive cone suffices. We do not silently replace it by a stronger hereditary condition on every closed subspace, nor do we need such a condition.

A space is *Baire* when each countable intersection of open dense sets is dense. In a Baire space, no nonempty open set is meager. Open subspaces of a Baire space are Baire. A Polish space is separable and completely metrizable, so it is Baire and ccc. For the latter fact, a countable dense set meets each nonempty member of a disjoint open family at a different point.

All structures in the theorems are sets, and all arguments take place in ZFC. Quotients such as A/H are quotients of *additive groups*, unless explicitly specified otherwise. No ideal property is presumed. When **No** or **Oz** appears, we apply the result only to specified set-sized subrings or to a fixed universe-bounded formal model. There is no Polish topology on a proper class in the sense used here.

77 From category to an order-bounded neighborhood

The following elementary form of the Pettis lemma is the only category input beyond the definition of a Baire space. We include the proof to make the later dependency boundary explicit [45].

Lemma 77.1 (Pettis difference lemma). *Let G be a Baire topological abelian group. If $A \subseteq G$ has the Baire property and is nonmeager, then $A - A$ contains an open neighborhood of 0.*

Proof. There is a nonempty open set U on which A is comeager. For any $h \in U - U$, the intersection $U \cap (h + U)$ is nonempty and open. Both A and $h + A$ are comeager in that intersection. Since a nonempty open subspace of a Baire space is Baire, they meet there. Thus $h = x - y$ for some $x, y \in A$. We have proved

$$U - U \subseteq A - A.$$

The set $U - U$ is open and contains 0. □

Pettis alone is not enough: differences of an arbitrary positive set need not be order-bounded. A reflection supplies the missing bounded set.

Lemma 77.2 (Bounded-neighborhood lemma). *Let G be a nontrivial linearly ordered abelian group with a Hausdorff Baire group topology. If its strict positive cone P has the Baire property, then there are $a > 0$ and an open neighborhood W of 0 such that*

$$0 \in W \subseteq (-a, a). \tag{77.1}$$

Proof. If the topology is discrete, take $W = \{0\}$ and any $a > 0$. Suppose it is nondiscrete. A Hausdorff topological group with an isolated point is discrete, so $\{0\}$ is closed with empty interior and hence nowhere dense. Because

$$G = P \dot{\cup} \{0\} \dot{\cup} (-P)$$

and inversion is a homeomorphism, P cannot be meager: otherwise G would be meager in itself, contrary to the Baire assumption.

Choose a nonempty open U in which P is comeager, and choose $u \in U \cap P$. Reflection $x \mapsto 2u - x$ is a homeomorphism. Therefore $2u - P$ is comeager in $2u - U$. The open set

$$V = U \cap (2u - U)$$

contains u . On V , both P and $2u - P$ are comeager. Hence

$$A = P \cap (2u - P) = (0, 2u)$$

is nonmeager and has the Baire property. By [lemma 77.1](#), there is an open neighborhood W of 0 with $W \subseteq A - A$. If $x, y \in A$, then $-2u < x - y < 2u$. Thus $W \subseteq (-2u, 2u)$, and $a = 2u$ works. $\square$

[write] [Lemma 77.2](#) is source 09's [lemma 61.3](#), stated there for Polish groups and proved by the same reflection-and-Pettis argument; [lemma 77.1](#) is the Baire-group form of [lemma 61.1](#). Source 12 states both for Hausdorff Baire groups and does not cite source 09, which it had not seen. Source 10 states the lemma as its Lemma 4.2 (pma:br:lem:localbound is a second label of [lemma 77.2](#)), credits the Polish version to source 09, and proves it without Pettis's lemma, as follows.

The category mechanism is elementary and is related to the usual Pettis sumset argument; [29] discusses Pettis's theorem and its connections with automatic continuity. We supply the full argument because its hypotheses are central to the result. The Polish version of the local-bound lemma was already isolated in [57].

Lemma 77.3 (Comeager semigroups have interior [10]). *Let G be a Baire topological group, written additively and assumed abelian here. Suppose a set P is comeager in a nonempty open set O and satisfies $P + P \subseteq P$. Then $O + O \subseteq P$.*

Proof. Take $z \in O + O$. The set $W = O \cap (z - O)$ is nonempty and open. The sets P and $z - P$ are both comeager in W : the second assertion uses that reflection followed by translation is a homeomorphism. Because W is Baire, these two comeager subsets intersect. Choose x in their intersection. Then $x \in P$ and $z - x \in P$, hence $z = x + (z - x) \in P + P \subseteq P$. $\square$

Second proof of lemma 77.2 [10]. Source 10 writes c and U for the bound a and the neighbourhood W .

For a discrete group use $U = \{0\}$ and any $c > 0$. Otherwise every singleton is closed with empty interior and hence nowhere dense. Inversion takes the positive cone P to $-P$. If P were meager, the partition $G = P \cup (-P) \cup \{0\}$ would make G meager in itself, a contradiction. Thus P is nonmeager. Since it has the Baire property, it is comeager in some nonempty open set O .

Choose $p \in O \cap P$. By [lemma 77.3](#), $O + O \subseteq P$. The element $2p$ lies in this open set, so there is a symmetric neighborhood U of 0 with $2p + U \subseteq O + O$. For $u \in U$, both u and $-u$ belong to U . Therefore $2p + u > 0$ and $2p - u > 0$. These inequalities say $-2p < u < 2p$. Set $c = 2p$. $\square$

Remark 77.4. The lemma applies to ordered groups, not just rings. The order may have many Archimedean levels, the group may be infinite-dimensional or non-locally-compact, and its positive cone need not be open, closed, or Borel. Inversion and translation are used; multiplication is absent.

The proof needs one locally comeager part of the positive cone, then a symmetric overlap. This is why there is no need to classify Borel orders or to restrict them to real subspaces or compact subgroups.

78 Algebraic packing and maximal cellularity

Lemma 78.1 (Arithmetic packing). *Let R be a discretely ordered ring with an additive-group topology. Suppose that an open neighborhood W of 0 satisfies $W \subseteq (-a, a)$ for some $a > 0$. Set $b = 2a + 1$. Then*

$$\{br + W : r \in R\} \quad (78.1)$$

is a family of $|R|$ pairwise disjoint nonempty open sets.

Proof. Every translate is open and nonempty. Suppose $r \neq s$. By discrete order, $|r - s| \geq 1$, and therefore

$$|b(r - s)| = b|r - s| \geq b = 2a + 1 > 2a. \quad (78.2)$$

On the other hand, $W - W \subseteq (-2a, 2a)$. If $br + w_1 = bs + w_2$ for some $w_1, w_2 \in W$, then $b(r - s) = w_2 - w_1 \in W - W$, contradicting (78.2). Thus the translates are pairwise disjoint. Since they are nonempty, distinct indices give distinct sets. $\square$

Remark 78.2 (Why no scalar continuity is hidden). It is not asserted that bW is open, that $r \mapsto br$ is continuous, or that its graph is measurable. The open sets are *translates of one fixed open set W* . The scalars merely choose their centers.

Theorem 78.3 (Maximal cellularity and density). *Let R be a discretely ordered ring with a Hausdorff Baire additive-group topology. If $P = \{r : r > 0\}$ has the Baire property, then there is a family of $|R|$ pairwise disjoint nonempty open subsets of R . In particular,*

$$\boxed{c(R) = d(R) = |R|.} \quad (78.3)$$

Proof. Use lemma 77.2 to obtain $W \subseteq (-a, a)$, and apply lemma 78.1. This proves $|R| \leq c(R)$. Combine this with (76.2). $\square$

Corollary 78.4 (Ccc rigidity). *Under the hypotheses of theorem 78.3, if the topology satisfies ccc, then R is countable and the topology is discrete.*

Proof. The explicit family in lemma 78.1 and ccc imply countability. A countable Hausdorff nondiscrete topological group has every singleton nowhere dense, so it is meager in itself. This contradicts the Baire assumption. Hence the group is discrete. $\square$

Corollary 78.5 (Polish rigidity; answer to the earlier question). *Every discretely ordered ring with a Polish additive-group topology and a positive cone having the Baire property is countable and discrete. In particular, an uncountable discretely ordered ring cannot have a Polish additive-group topology with Borel order, regardless of the regularity of multiplication.*

Proof. A Polish space is Baire and separable; separability implies ccc. Borel sets have the Baire property. Apply corollary 78.4. $\square$

[write] Corollary 78.4 is source 10's main theorem, its Theorem 5.1 (pma:br:s10:thm:main); source 10's proof, through the principal-hull quotient embedding, is printed in section 79.2. With corollary 78.5 it answers question 57.1 of Part IV negatively. Source 10 states the Polish case as an exact criterion:

Corollary 78.6 (Exact Polish presentation criterion [10]). *A discretely ordered ring admits a Polish additive-group topology with a Baire-property positive cone if and only if its underlying set is countable. Every such topology is discrete.*

Proof. Polish spaces are Baire, Hausdorff, and ccc, so necessity and uniqueness of the possible topology follow from [corollary 78.4](#). Conversely, a countable ring with the discrete topology is a Polish topological ring, and all subsets, in particular its positive cone, are Borel. $\square$

Here “uniqueness” refers to the topology for a fixed underlying ring: it must be the discrete topology. The theorem does not classify countable discretely ordered rings up to isomorphism. There are many nonstandard ones, including $\mathbb{Z}[X]$ ordered with X positive infinite.

Remark 78.7 (Countable does not mean standard). A countable nonstandard discretely ordered ring can carry the discrete Polish topology. The corollary does not assert that the ring is $\mathbb{Z}$; it asserts that its cardinality is countable and its given Baire ccc group topology is discrete.

Corollary 78.8 (A separable category dichotomy). *Let a discretely ordered ring have a separable Hausdorff additive-group topology and a positive cone with the Baire property. Then either it is countable discrete, or it is meager in itself.*

Proof. If the topology is Baire, use [corollary 78.4](#). Otherwise some nonempty open U is meager in the group. Let D be countable and dense. For each x , the nonempty open set $x - U$ meets D , so $x \in D + U$. Thus $G = \bigcup_{d \in D} (d + U)$ is a countable union of meager sets. $\square$

The same packing proof works for noncommutative discretely ordered unital rings, provided the order axioms make products of positives positive: left multiplication by $b > 0$ is order-preserving and satisfies $|bx| = b|x|$. Commutativity of multiplication is not essential to [theorem 78.3](#).

Remark 78.9 (Noncommutative extension [10]). The main proof also applies to an associative unital ring with a total translation-invariant order and strictly positive products, without commutativity. Use left multiplication $x \mapsto ax$. For $a > 0$, $|ax| = a|x|$, positive left multiplication reflects inequalities, and ordinary integer multiples of 1 commute with a . These are all the multiplicative facts used in the kernel computation. The quotient maps remain homomorphisms of the abelian additive groups. We keep the commutative convention elsewhere to match the arithmetic and omnific applications.

79 Principal convex subgroups and a second obstruction

There is a useful purely algebraic form of the argument. It reveals why an open convex subgroup cannot rescue an uncountable ring presentation.

For an ordered abelian group G and $a > 0$, define the principal convex subgroup

$$H_a = \{x \in G : \text{for some ordinary } n \geq 1, |x| \leq na\}. \quad (79.1)$$

It is a subgroup by the triangle inequality, and it is convex by construction. The indexing quantifier is external and ranges over the ordinary natural numbers.

Proposition 79.1 (An open principal convex subgroup). *Every nontrivial ordered Hausdorff Baire topological abelian group whose positive cone has the Baire property has an open subgroup H_a of the form (79.1). If the topology is ccc, G/H_a is countable.*

Proof. Choose $W \subseteq (-a, a)$ using [lemma 77.2](#). Then $W \subseteq H_a$. A subgroup containing a neighborhood of the identity is open, because it is the union of its translates of that neighborhood. Its cosets are disjoint nonempty open sets, so ccc makes the coset space countable. $\square$

[write] Source 10 states [proposition 79.1](#) for ordered rings, with $a \geq 1$ so that $\text{Fin}(A) \subseteq H_a$ (its Proposition 4.3, `pma:br:prop:openhull`), and adds that the open subgroup H_a is also closed, its complement being a union of open cosets. Its proof is the one above. Source 10 continues [10]:

It is important that H_a , rather than the interval $[-a, a]$, is the subgroup used in the proof. The interval need not be a subgroup. Conversely, H_a need not be bounded by one fixed element of A . Only membership in some ordinary multiple of $[-a, a]$ is needed.

Theorem 79.2 (Every principal convex quotient is as large as the ring). *Let $R \neq \mathbb{Z}$ be a discretely ordered ring. For every $a > 0$,*

$$[R : H_a] = |R|. \quad (79.2)$$

More explicitly, choose $c > n$ for every ordinary natural number n , and put $b = (a + 1)c$. The map

$$R \longrightarrow R/H_a, \quad r \longmapsto br + H_a \quad (79.3)$$

is injective.

Proof. Such a c exists by [lemma 76.2](#). For every ordinary $n \geq 1$,

$$b = (a + 1)c > (a + 1)n > an.$$

If $r \neq s$, then $|r - s| \geq 1$, so $|b(r - s)| \geq b > na$ for every ordinary n . Therefore $b(r - s) \notin H_a$, proving injectivity in (79.3). The quotient has at most $|R|$ elements, so its cardinality is exactly $|R|$. $\square$

No ideal is being asserted here. H_a is an additive convex subgroup; R/H_a is used only as an additive quotient. Confusing a convex subgroup with a ring ideal would obscure the purpose of the construction.

Combining [proposition 79.1](#) and [theorem 79.2](#) gives another proof of the countability part of [corollary 78.4](#) for $R \neq \mathbb{Z}$. This route uses an infinite element to make a quotient injection. By contrast, the packing proof uses only the finite expression $2a + 1$, works uniformly for $R = \mathbb{Z}$, and directly attains the full cellularity bound. It is the preferred proof of the main theorem.

Example 79.3 (Polynomial interpretation). In $\mathbb{Z} + t\mathbb{Q}[t]$, ordered by the leading coefficient with t infinite, H_{t^d} consists exactly of polynomials of degree at most d . Multiplication by t^{d+1} sends every nonzero difference outside that subgroup. Thus a principal convex quotient retains at least as much cardinality as the entire ring, despite discarding a complete finite range of growth degrees.

For real rather than rational coefficients, the same description holds: each individual leading coefficient is bounded by an ordinary integer. This observation will also be useful in the sharpness constructions.

79.1 The principal-hull quotient embedding [10]

79.1.1 The ordinary-integer-bounded part

Definition 79.4 ([10]). For an ordered ring A , define

$$F = \text{Fin}(A) = \{x \in A : \exists n \in \mathbb{N} (|x| \leq \underline{n})\}. \quad (79.4)$$

For $a > 0$, define its principal convex additive hull by

$$H_a = \{x \in A : \exists n \in \mathbb{N} (|x| \leq na)\}. \quad (79.5)$$

The quantifiers over $\mathbb{N}$ in these definitions are external ordinary quantifiers, not quantifiers over a nonstandard arithmetic model.

Lemma 79.5 ([10]). *The set F is a convex additive subgroup and a unital subring of A . For every $a > 0$, H_a is a convex additive subgroup, the smallest such subgroup containing a . If $a \geq 1$, then $F \subseteq H_a$.*

Proof. The triangle inequality gives closure under addition, and absolute value gives closure under negation. Bounds by ordinary integers multiply, so $|x| \leq \underline{n}$ and $|y| \leq \underline{m}$ imply $|xy| \leq \underline{nm}$. Thus F is a subring containing 1. If $|z| \leq |x|$ and x belongs to either hull, the same bound puts z in that hull; this proves convexity. Any convex additive subgroup containing a contains all na and all intervals $[-na, na]$, hence contains H_a . Finally, $\underline{n} \leq na$ when $a \geq 1$. $\square$

Lemma 79.6 (Finite elements in a discretely ordered ring [10]). *If A is discretely ordered, then $\text{Fin}(A) = \mathbb{Z}1_A$.*

Proof. For each ordinary $n \geq 0$, we prove externally that $[0, \underline{n}] = \{0, \dots, \underline{n}\}$. At $n = 0$ this is immediate. If $0 < x \leq \underline{n} + 1$, discreteness gives $x \geq 1$, and consequently $0 \leq x - 1 \leq \underline{n}$. Apply the induction hypothesis. Negating deals with negative finite elements. Conversely, every ordinary integer is bounded by its ordinary absolute value. $\square$

The loss of finiteness in F is precisely what permits uncountable ordered fields such as $\mathbb{R}$ to escape the arithmetic obstruction. The next theorem separates this algebraic issue from topology entirely.

79.1.2 A quotient embedding with exactly known kernel

Theorem 79.7 (Principal-hull amplification [10]). *Let A be an ordered ring and $a > 0$. The additive homomorphism*

$$T_a : A \longrightarrow A/H_a, \quad T_a(x) = ax + H_a \quad (79.6)$$

has kernel exactly $\text{Fin}(A)$. It induces an ordered additive embedding

$$\Phi_a : A/\text{Fin}(A) \hookrightarrow A/H_a, \quad x + \text{Fin}(A) \longmapsto ax + H_a. \quad (79.7)$$

In particular,

$$|A/\text{Fin}(A)| \leq |A/H_a|. \quad (79.8)$$

Proof. Distributivity makes T_a additive. Since $a > 0$, absolute values and positive cancellation give the equivalences

$$\begin{aligned} T_a(x) = 0 &\iff ax \in H_a \\ &\iff \exists n \in \mathbb{N} (|ax| \leq na) \\ &\iff \exists n \in \mathbb{N} (a|x| \leq a\underline{n}) \\ &\iff \exists n \in \mathbb{N} (|x| \leq \underline{n}) \\ &\iff x \in \text{Fin}(A). \end{aligned}$$

The induced additive map is therefore well defined and injective.

Both quotient orders are defined because their subgroups are convex. A positive nonzero class $x + \text{Fin}(A)$ has a representative satisfying $x > \underline{n}$ for every ordinary n . Hence $ax > na$ for every n . Every member of H_a is at most some na , so $ax + H_a$ is positive and nonzero. Applying the same reasoning to $-x$ proves that signs, and hence all comparisons of distinct classes, are preserved. $\square$

Remark 79.8 (Why this is not a continuity argument [10]). The map T_a is used only as a homomorphism of abstract additive groups. It may be discontinuous or nonmeasurable. The quotient A/H_a need not be a ring, and multiplication by a need not descend to any endomorphism of A/H_a . What descends is exactly the map from $A/\text{Fin}(A)$ specified in (79.7). No other descent is asserted.

Corollary 79.9 ([10]). *If an ordered ring has an additive order unit $a > 0$, so that $H_a = A$, then $A = \text{Fin}(A)$. A discretely ordered ring with an additive order unit is therefore just its standard integer subring.*

Proof. The target in (79.7) is zero, so the source is zero. Apply lemma 79.6 in the discrete case. $\square$

Remark 79.10 (The change from the earlier amplification theorem [10]). The order-unit theorem in [56] uses a subgroup avoiding nonzero ordinary integers in order to obtain an injective map *before* quotienting its domain. Here there is no reason to avoid those integers. They are precisely the kernel, and their countability is harmless. Taking the principal convex hull makes an order unit available without any local compactness, vector structure, or divisibility assumption.

[write] Lemma 79.6 is the first assertion of lemma 76.2 and of source 08's lemma 46.1. Theorem 79.7 generalizes part (1) of source 08's theorem 46.2: it needs neither an integer-free subgroup nor an order unit, and it holds in every ordered ring.

79.2 Arithmetic rigidity without local compactness [10]

Source 10's main theorem is corollary 78.4; its proof is a second proof of that corollary.

Second proof of corollary 78.4 [10]. By proposition 79.1, some principal convex subgroup H_a is open. Its cosets are pairwise disjoint nonempty open sets, so ccc implies that A/H_a is countable. By theorem 79.7 and lemma 79.6, there is an injection

$$A/\mathbb{Z}1_A \hookrightarrow A/H_a.$$

Thus $A/\mathbb{Z}1_A$ is countable. Each coset of $\mathbb{Z}1_A$ is countable, so A is countable.

A countable Hausdorff Baire group must be discrete. Otherwise it has no isolated point, since any isolated point can be translated to 0 and then to every other point. Every singleton would consequently be nowhere dense, and their countable union would make A meager in itself. This contradicts the Baire hypothesis. $\square$

79.3 The cardinal theorem for general ordered rings [10]

The argument has a useful form before discreteness of the order is imposed. It measures the topological size needed to accommodate the quotient by ordinary-integer-bounded elements.

For an infinite space X , let $\text{cell}(X)$ be the supremum of the cardinalities of disjoint nonempty open families, $\text{dens}(X)$ the least cardinality of a dense subset, and $\text{ext}(X)$ the supremum of the cardinalities of closed discrete subsets. We use the usual infinite-cardinal convention for the Lindelöf degree $L(X)$: every open cover has a subcover of size at most $L(X)$.

Theorem 79.11 (Relative cardinal obstruction [10]). *Let A be an ordered ring with a Hausdorff additive-group topology. Suppose some neighborhood of 0 is order bounded. Then there is an open convex additive subgroup H such that*

$$A/\text{Fin}(A) \hookrightarrow A/H \quad (79.9)$$

is an ordered additive embedding. In particular,

$$|A/\text{Fin}(A)| \leq |A/H| \leq \text{cell}(A) \leq \text{dens}(A). \quad (79.10)$$

Moreover, A has a closed discrete subset of cardinality $|A/H|$. The order-bounded-neighborhood hypothesis follows from a Baire group topology and a Baire-property positive cone.

Proof. Choose $a \geq 1$ so that the neighborhood lies in $[-a, a]$, and put $H = H_a$. The proof of [proposition 79.1](#) shows that H is open without needing Baire once the neighborhood bound is given. Apply [theorem 79.7](#). The cosets form a disjoint nonempty open family, proving the middle bound. A dense set must meet every member of any such family, proving the last bound.

Choose one representative from each coset and call the resulting set T . Each point of T is isolated in T by its coset. To see that T is closed, take $x \notin T$. Its coset contains exactly one selected point t . Since the group is Hausdorff, the coset with t removed is an open neighborhood of x disjoint from T . Thus T is closed and discrete, and $|T| = |A/H|$. The last sentence is [lemma 77.2](#). $\square$

Corollary 79.12 (Cardinality of the finite part [10]). *For a Hausdorff Baire ccc additive presentation of an ordered ring with a Baire-property positive cone,*

$$|A| = |\text{Fin}(A)| = \max(\aleph_0, |[0, 1]_A|). \quad (79.11)$$

Consequently a countable unit interval forces a countable, discrete additive group.

Proof. By [theorem 79.11](#), $A/\text{Fin}(A)$ is countable. The group $\text{Fin}(A)$ contains $\mathbb{Z}1$ and is infinite. A countable union of its cosets has the same cardinality as $\text{Fin}(A)$, giving the first equality.

Every ordinarily bounded element lies in an ordinary integer translate of $[0, 1]$. Indeed, inside a fixed finite interval $[-n, n]$, one can choose the largest ordinary integer k not exceeding that element, using only a finite search. Hence

$$\text{Fin}(A) = \bigcup_{k \in \mathbb{Z}} (k + [0, 1]_A).$$

This proves the upper bound for the second equality; the reverse bound follows from the inclusions of $[0, 1]$ and $\mathbb{Z}1$ in $\text{Fin}(A)$. A countable Hausdorff Baire group is discrete as in [corollary 78.4](#). $\square$

This relative formulation does not exclude the usual real field: its finite part is all of $\mathbb{R}$, and its unit interval already has its full cardinality. Nor does it exclude every non-Archimedean ordered field. Its conclusion is an exact restriction on how much cardinality can occur beyond the finite part in a ccc presentation.

Theorem 79.13 (Maximal topological size [10]). *Let A be an uncountable discretely ordered ring equipped with a Hausdorff Baire additive-group topology and a Baire-property positive cone. Then some open convex additive subgroup H has*

$$|A/H| = |A|. \quad (79.12)$$

There is a closed discrete subset of A of size $|A|$, and

$$\text{cell}(A) = \text{dens}(A) = \text{ext}(A) = L(A) = |A|. \quad (79.13)$$

Proof. Every coset of $\mathbb{Z}1$ is countable. Since A is uncountable, cardinal arithmetic gives $|A/\mathbb{Z}1| = |A|$. By [theorem 79.11](#), an open H satisfies $|A| \leq |A/H| \leq |A|$, proving (79.12) and the existence of the closed discrete transversal.

The cosets witness $\text{cell}(A) \geq |A|$, and no disjoint family can have more members than there are points. A dense set must meet every coset, so $\text{dens}(A) = |A|$. The closed discrete transversal gives the equality for extent. Finally, the coset cover has no proper subcover and has $|A|$ members, while any open cover of a space of size $|A|$ has a subcover of size at most $|A|$, by choosing one covering set per point. This proves the equality for $L(A)$. $\square$

Thus a nonseparable counterexample cannot be merely a little larger than a Polish space in its supply of disjoint open sets. It must have as many disjoint open cosets, and as many points in one closed discrete set, as it has elements altogether. The explicit locally compact example in [section 84.2](#) attains this conclusion.

[write] [Theorem 79.13](#) and source 12's [theorem 78.3](#) overlap: source 12 proves $c(R) = d(R) = |R|$ for every discretely ordered ring with a Hausdorff Baire topology and a Baire-property cone, with no uncountability assumption; source 10 adds, for uncountable rings, an open convex subgroup of full index, a closed discrete set of full size, and the equalities for extent and Lindelöf degree. Source 10's cell and dens are source 12's c and d .

79.4 A complementary bounded-subgroup amplification [10]

The principal-hull theorem is the shortest route to the main result. A different amplification statement retains information about multiple independent additive copies and is useful for algebraic certificates. Here a subgroup is bounded by one element, which is stronger than having an order unit.

Theorem 79.14 (Bounded-subgroup amplification [10]). *Let A be a discretely ordered ring, and let $H \leq (A, +)$ satisfy $|h| \leq c$ for every $h \in H$, where $c \geq 0$. If $b > c$, then $bH \cap H = \{0\}$ and the additive map*

$$H \longrightarrow A/H, \quad h \longmapsto bh + H$$

is injective. With $a = c + 2$, the finite-sum map

$$\Psi : \bigoplus_{j \geq 0} H \longrightarrow A, \quad (h_j) \longmapsto \sum_{j \geq 0} a^j h_j \quad (79.14)$$

is an ordered additive embedding when the source is ordered by the sign of its highest-index nonzero coordinate. Its restriction to the positive-index summands remains injective after quotienting by H .

Proof. If $h \neq 0$, then $|h| \geq 1$, so $|bh| = b|h| \geq b > c$. Thus $bh \notin H$. This proves the first assertion and the coset injection.

For the second, take a nonzero finite tuple and let k be the largest index with $h_k \neq 0$. If $k = 0$, the conclusion is immediate. For $k > 0$, the magnitude of its lower terms is at most

$$B = c \sum_{j < k} a^j.$$

The geometric-sum identity and $a - 1 = c + 1$ give

$$(a - 1)B = c(a^k - 1) < (a - 1)a^k.$$

Indeed, subtracting the middle expression from the right gives $a^k + c > 0$. Cancelling $a - 1 > 0$ yields $B < a^k$. Since $|h_k| \geq 1$, the leading term has magnitude $a^k|h_k| \geq a^k > B$. The sum is therefore nonzero and has the sign of h_k . Distributivity gives additivity; applying the sign conclusion to differences gives the order embedding.

If a positive-index sum belongs to H , subtract that element as the zeroth coefficient in (79.14). Injectivity of Ψ forces every coefficient to vanish. $\square$

No divisibility, convexity, ideal property, or integer-disjointness is assumed of H . The source is an additive direct sum, not a polynomial ring with coefficient ring H . Its ordered embedding gives more structure than a cardinal inequality, although countable direct sums do not increase an infinite cardinal.

79.4.1 Two initial approaches become corollaries

A non-Archimedean topological group has a basis at the identity of open subgroups; this terminology is standard in Polish group theory [53]. If the additive group in [corollary 78.4](#) is non-Archimedean, a bounded neighborhood contains an open bounded subgroup H . Then [theorem 79.14](#) injects H into the countable coset space A/H , yielding countability. This is a valid proof in that case, but the principal-hull argument makes the non-Archimedean assumption unnecessary.

Likewise, an open real topological vector subgroup can be treated by order-unit amplification. If $U \subseteq [-u, u]$ is a neighborhood in that subgroup, then $v/n \rightarrow 0$ implies $|v| \leq nu$ for some ordinary n , making u an order unit. Divisibility makes each nonzero member larger in magnitude than every ordinary integer. Multiplication by u then injects the subgroup into its additive quotient, as in [56]. This works in infinite as well as finite dimension. Again, [corollary 78.4](#) is stronger and needs neither vector structure nor an open subgroup of a prescribed kind.

These two arguments help explain the discovery path, but they should not be mistaken for the endpoint. Arbitrary Polish additive groups, including mixed or otherwise unclassified ones, are covered by the principal-hull proof.

[write] [Theorem 79.14](#) is a variant of parts (2) and (3) of source 08's [theorem 46.2](#), with a bound by one element in place of an integer-free subgroup with an order unit.

80 Arbitrary self-embeddings have bounded unit images

The topological obstruction has an interpretation that does not mention multiplication. It restricts purely algebraic ordered self-embeddings.

Let G be an ordered abelian group with a least positive element e . Write $\text{Emb}_{+,<}(G)$ for the set of injective additive maps $f : G \rightarrow G$ that preserve strict order. The maps are not required to preserve e , nor to be continuous or measurable. Define the unit-image spectrum

$$\mathcal{U}(G) = \{f(e) : f \in \text{Emb}_{+,<}(G)\}.$$

This terminology refers only to a distinguished least positive element; G need not be a ring.

Theorem 80.1 (Bounded unit-image spectrum). *Suppose G is uncountable and has a Hausdorff Baire ccc group topology, an invariant total order, and a positive cone with the Baire property. If e is its least positive element, then $\mathcal{U}(G)$ is bounded above in G . In fact, if $W \subseteq (-a, a)$ is the neighborhood from lemma 77.2, then*

$$f(e) \leq 2a \quad \text{for every } f \in \text{Emb}_{+,<}(G). \quad (80.1)$$

Proof. Suppose $f(e) > 2a$. For distinct $x, y \in G$, discreteness gives $|x - y| \geq e$. Additivity and order preservation imply

$$|f(x) - f(y)| = f(|x - y|) \geq f(e) > 2a.$$

Consequently the sets $f(x) + W$, indexed by all $x \in G$, are pairwise disjoint by the same difference argument as in lemma 78.1. This contradicts ccc because G is uncountable. $\square$

Corollary 80.2 (Cofinal unit movement is impossible). *An uncountable discretely ordered abelian group with a family of ordered additive self-embeddings whose values at the least positive element are cofinal cannot have a Hausdorff Baire ccc group topology with a Baire-property positive cone.*

Proof. A cofinal subset cannot be bounded above in a nontrivial ordered group, since adding a positive element always produces a larger element. Apply theorem 80.1. $\square$

For a discretely ordered ring, the maps $x \mapsto bx$ for $b > 0$ are ordered additive embeddings, and their values at 1 range over the entire positive cone. The main ccc obstruction is therefore a special case of this cofinal-unit-movement obstruction. The formulation is useful for other algebraic structures equipped with an expansive family of endomorphisms, even before a ring structure is supplied.

80.1 A spectrum can be infinite and still bounded

Consider

$$G_{\mathbb{R}} = \mathbb{R} \overrightarrow{\times} \mathbb{Z}, \quad e = (0, 1), \quad (80.2)$$

with the product topology $\mathbb{R}_{\text{usual}} \times \mathbb{Z}_{\text{discrete}}$ and lexicographic order in which the real coordinate is more significant. This is an uncountable Polish ordered group with least positive element e . Its cone is

$$(\mathbb{R}_{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{Z}_{>0}),$$

a Δ_2^0 set.

For each ordinary $k \geq 1$, the map $(r, n) \mapsto (r, kn)$ is an ordered additive embedding. Thus $\mathcal{U}(G_{\mathbb{R}})$ is infinite, although all these images are less than $(1, 0)$. In fact these are exactly the possible images of e . To see this, an additive map from the divisible group $\mathbb{R}$ to $\mathbb{Z}$ is zero. Therefore an ordered additive embedding restricts on $\mathbb{R} \times \{0\}$ to $r \mapsto (\alpha(r), 0)$, where α is an increasing additive injection. It follows that $\alpha(r) = \lambda r$ for some $\lambda > 0$: first prove the identity on rationals, then squeeze an arbitrary real between rationals. Write $f(e) = (b, k)$.

Positivity excludes $b < 0$. If $b > 0$, choose $r > 0$ with $\lambda r < b$; then $(r, 0) > e$ but $f(r, 0) < f(e)$, a contradiction. Thus $b = 0$ and $k \geq 1$. Hence

$$\mathcal{U}(G_{\mathbb{R}}) = \{(0, k) : k \in \mathbb{Z}_{>0}\}. \quad (80.3)$$

This illustrates why [theorem 80.1](#) gives an *order bound*, not a finite bound on the number of embeddings or possible unit images.

80.2 All four clauses of the bounded spectrum [10]

Source 10 proves [theorem 80.1](#) as clause (3) of the following theorem; clauses (1), (2) and (4) are additional. Its own section reads [10]:

The proof uses less than a ring multiplication. This gives an independent result about ordered additive groups and explains which part of multiplication is incompatible with the topology.

Definition 80.3 ([10]). For a discretely ordered abelian group $(G, +, <, e)$ with least positive e , define the *unit-image spectrum*

$$\mathcal{U}(G) = \{f(e) : f : G \longrightarrow G \text{ is an additive order embedding}\}.$$

No continuity or measurability is required of f . For $a > 0$, define $H_a = \{x : |x| \leq na \text{ for some } n \in \mathbb{N}\}$.

Theorem 80.4 (Bounded unit-image spectrum [10]). *Let G be an uncountable discretely ordered abelian group with a Hausdorff Baire ccc group topology and a Baire-property positive cone. Then:*

1. *If $f : G \rightarrow G$ is an additive order embedding and $a = f(e)$, there is an ordered additive embedding $G/\mathbb{Z}e \hookrightarrow G/H_a$.*
2. *For every $a \in \mathcal{U}(G)$, the subgroup H_a is not open.*
3. *There is $b > 0$ such that $0 < a < b$ for every $a \in \mathcal{U}(G)$.*
4. *Each subgroup H_a with $a \in \mathcal{U}(G)$ is meager.*

Proof. For the first assertion use $x \mapsto f(x) + H_a$. Additivity and order reflection imply

$$|f(x)| \leq na \iff f(|x|) \leq f(ne) \iff |x| \leq ne.$$

The ordinarily bounded elements of a discretely ordered abelian group are exactly $\mathbb{Z}e$, by the finite-interval proof of [lemma 79.6](#), which uses no multiplication. The kernel is therefore $\mathbb{Z}e$. The sign proof for the quotient is the same as in [theorem 79.7](#).

If H_a were open, ccc would make G/H_a countable. The injection would make $G/\mathbb{Z}e$, and then G , countable, contrary to hypothesis. For the third assertion, the local-bound lemma provides $b > 0$ with H_b open. If $a \geq b$, then $H_b \subseteq H_a$, so H_a is open. Thus no such a can lie in the spectrum.

Finally, each interval $[-na, na]$ has the Baire property, since it is formed from translates of the positive cone, their complements, and singletons. Therefore H_a , their countable union, has the Baire property. A subgroup with the Baire property that is nonmeager in a Baire group is open: if it is comeager in an open set O , the usual intersection argument shows that $O - O$ is contained in the subgroup. This is an open neighborhood of zero. Since the second assertion excludes openness, H_a is meager. $\square$

The final assertion concerns *each* hull separately. It does not say that the spectrum itself, or the union of all these hulls, is meager. Such a union may have uncountably many terms.

Every positive scalar in an ordered ring gives an additive order embedding by multiplication. Thus its unit-image spectrum contains all positive elements. The bounded-spectrum theorem prohibits this in an uncountable Baire ccc discrete ordered group, independently of any compatibility among the individual embeddings. Associativity and the existence of one binary multiplication are not needed for this reduct obstruction.

This is also a useful self-embedding result for a set-sized integer-like ordered group. A Polish presentation may allow many discontinuous additive order embeddings, but they cannot send the least positive element to arbitrarily large scales. It is not a theorem about field self-embeddings of **No**: those fix 1, and **No** is not discretely ordered in the first place.

[write] [Corollary 80.2](#) contains source 15's [theorem 119.2](#) of [Part IX](#): a positive endomorphism T with $T(e) > 0$ is an ordered additive embedding, so endomorphisms with $T_a(e) = a$ for every $a > 0$ give cofinal unit images, and Part VI needs only a Baire-property cone and no regularity of the maps (note in [section 119](#)). The exact spectrum (80.3) extends Part I's [theorem 11.1](#) (unital endomorphisms of $M_{\mathbb{R}}$) to all ordered self-embeddings of $G_{\mathbb{R}} = \mathbb{R} \overrightarrow{\times} \mathbb{Z}$, not necessarily unital.

81 Ring expansions, positive semirings, and Presburger arithmetic

Corollary 81.1 (No ordered-ring expansion). *Let G be an uncountable ordered abelian group with least positive element e , a Hausdorff Baire ccc group topology, and a positive cone having the Baire property. No multiplication on the set G makes $(G, +, \cdot, 0, e, <)$ a discretely ordered unital ring.*

Proof. Such a multiplication would satisfy the hypotheses of [corollary 78.4](#), contradicting uncountability. Equivalently, its positive left multipliers would contradict [corollary 80.2](#). $\square$

The multiplication is ruled out algebraically relative to the prescribed addition, order, and unit. The statement is stronger than saying that a particular candidate multiplication is not Borel.

[10] Source 10 states this corollary for Polish topologies (its [Corollary 5.3](#), `pma:br:cor:noexp`) and adds:

This excludes the existence of the multiplication itself, not merely its continuity or Borel measurability. It is therefore stronger than a regularity obstruction on the specified additive presentation. It does not, however, remove the topological-group hypothesis or the Baire condition on order.

Theorem 81.2 (No positive-semiring expansion). *Under the hypotheses of [corollary 81.1](#), let $M = G_{\geq 0}$. There is no associative unital semiring multiplication on M with its given addition, zero, and unit e .*

Proof. Suppose there were such a multiplication. Every $g \in G$ is a difference of two elements of M , and addition on M is cancellative. Extend the multiplication to differences by

$$(a - b)(c - d) = (ac + bd) - (ad + bc), \quad a, b, c, d \in M. \quad (81.1)$$

We verify that it is independent of representatives. If $a - b = a' - b'$, then $a + b' = a' + b$ in M . Multiplying this identity on the right by $c \in M$ and using distributivity gives $ac + b'c = a'c + bc$, so $ac - bc = a'c - b'c$ in G . The same holds with d ; subtracting gives

independence of the first difference. Left distributivity proves independence of the second. Expanding finite products proves associativity and both distributive laws on G ; e remains the unit. Thus this is a unital ring extension.

Products of nonnegative elements are nonnegative because they belong to M . For strictly positive $x, y \in M$, the least-positive-element property gives $x = e + u$ and $y = e + v$ with $u, v \in M$. Therefore

$$xy = e + u + v + uv \geq e > 0.$$

The extended ring has the original discrete order. This contradicts [corollary 81.1](#). $\square$

This proof does not assume commutative semiring multiplication: the formula and positivity argument still work, and the main packing theorem also has the required noncommutative version. Additive cancellation is available because M is a cone in a group.

81.1 Application to signed Presburger models

A signed Presburger model is a model of the complete ordered additive theory of the integers. In particular, it is an ordered abelian group with a least positive element 1 and the standard-integer division-with-remainder laws. We use only the ordered-group and least-positive-element properties in the obstruction; quantifier elimination and induction are unnecessary.

Corollary 81.3. *An uncountable signed Presburger model with a Polish additive-group topology and a Borel positive cone has no compatible unital ordered-ring expansion. Its nonnegative cone has no unital semiring expansion with its prescribed addition and unit.*

Proof. Apply [corollary 81.1](#) and [theorem 81.2](#). $\square$

The group $G_{\mathbb{R}}$ in (80.2) is a concrete example of the additive behavior that survives. For a fixed ordinary $m > 0$, divide the real coordinate by m and use the ordinary residue of the integer coordinate. This provides the familiar split signed Presburger example; its positive counterpart was studied in the earlier report [50]. The new general theorem explains the obstruction to multiplication independently of its particular coordinate system.

81.2 The group-completion boundary

For an abstract cancellative commutative monoid, the algebraic Grothendieck group always exists. Given a topology on the monoid, several additional questions arise: whether the difference quotient is Hausdorff, whether addition and inversion are continuous, whether the original cone embeds topologically, and whether the resulting group is Baire and ccc. None is supplied by the purely algebraic construction (81.1).

A useful conditional consequence is therefore precise. If a positive arithmetic presentation admits an order-compatible embedding as the nonnegative cone of a Hausdorff Baire ccc topological group with a Baire-property positive cone, and its carrier is uncountable, then no unital semiring multiplication can extend its addition. But without that group-topology hypothesis, this manuscript makes no such general claim. The classical literature on topological semigroup embeddings [48] is relevant to that remaining boundary; no unverified embedding criterion from that literature is being used here.

81.3 Presburger arithmetic: presentations but no expansions [10]

81.3.1 Signed models and the universal obstruction

A $\mathbb{Z}$ -group is a discretely ordered abelian group with named least positive element 1 in which every element has a unique standard remainder modulo each positive ordinary integer m :

$$\forall g \exists!(q, r) \in G \times \{0, 1, \dots, m-1\} : \quad g = mq + r1, \quad (81.2)$$

The usual interpretation is as a model of signed Presburger arithmetic. Cooper elimination reduces formulas to inequalities and congruences; the relevant finite-residue mechanism is the one formalized in ProveIt [54]. For clarity, the transfer argument needed for this abstract characterization is included in [section P](#).

Corollary 81.4 (All uncountable Polish signed Presburger groups [10]). *Let G be an uncountable $\mathbb{Z}$ -group with a Polish additive-group topology and a Baire-property positive cone. It has no expansion to a discretely ordered ring with the same addition, order, and named 1. The proposed multiplication may be arbitrary and need not be definable.*

Proof. Apply [corollary 81.1](#). The standard division axioms of a $\mathbb{Z}$ -group are not needed for this obstruction. $\square$

The last observation is useful: the obstruction belongs to the weaker ordered-group reduct. Presburger quantifier elimination explains why many such reducts exist and are well behaved, but the contradiction requires no quantifier elimination or induction.

81.3.2 Divisible-plus-integer examples

Let D be a divisible linearly ordered abelian group. Put

$$G_D = D \overrightarrow{\times} \mathbb{Z}, \quad 1_G = (0, 1), \quad (81.3)$$

where the first coordinate is compared first. The element $(0, 1)$ is least positive. If $g = (d, n)$ and $m \geq 1$ is ordinary, write $n = mq + r$ with $0 \leq r < m$. Then

$$(d, n) = m(d/m, q) + r(0, 1). \quad (81.4)$$

Uniqueness follows from torsion-freeness and the ordinary remainder uniqueness in $\mathbb{Z}$. Thus G_D is a $\mathbb{Z}$ -group.

Three useful choices are

$$D = \mathbb{R}, \quad D = \mathbb{Q}^{\mathbb{N}}, \quad D = \ell^2(\mathbb{N}, \mathbb{R}).$$

Use the usual order on $\mathbb{R}$, and on either sequence space compare at the first nonzero coordinate. Use the ordinary real topology in the first case, the product of discrete rational topologies in the second, and the norm topology in the third. These are divisible ordered groups; the order is defined additively and need not be a lattice order in a functional-analytic sense.

Each G_D is an uncountable Polish additive group with a Borel order. For the sequence cases its strict positive cone is

$$P_D = \bigcup_{j \geq 0} \{d : d_0 = \dots = d_{j-1} = 0, d_j > 0\}. \quad (81.5)$$

In $\mathbb{Q}^{\mathbb{N}}$ each displayed condition is clopen. In ℓ^2 , each term is the intersection of a closed set with an open set and is F_σ , since the space is metrizable. Thus P_D is F_σ . Its complement is $-P_D \cup \{0\}$, also F_σ , so P_D is Δ_2^0 . The corresponding cone in G_D is

$$(P_D \times \mathbb{Z}) \cup (\{0\} \times \mathbb{Z}_{>0}),$$

again Δ_2^0 . The same conclusion is immediate for $D = \mathbb{R}$.

These constructions, including the real-coordinate model, belong to the existing Presburger project context and are not claimed as new here [56]. What [corollary 81.4](#) adds is a theorem covering *all* uncountable Polish signed models with the stated order regularity, not only these examples or the locally compact ones.

81.3.3 Positive cones still give Polish arithmetic monoids

Since the nonnegative cone $G_D^{\geq 0}$ is G_δ in a Polish space, it is Polish with its subspace topology. Addition restricts continuously, and this cone is a model of nonnegative Presburger arithmetic. For $D = \mathbb{R}$ it is explicitly

$$M_{\mathbb{R}} = (\mathbb{R}_{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}). \quad (81.6)$$

Thus additive Polish arithmetic is not ruled out. What fails is its ordered-ring expansion on the corresponding signed group.

For these explicit cones, even an ordered semiring multiplication compatible with the given addition and order is impossible when its ring of differences uses the displayed signed ordered group. The extension of such a multiplication to differences is algebraic; the compatible Polish signed topology is already provided by G_D . This is more specific than asserting the same conclusion for every Polish monoid, a step not justified by the present proof.

[write] [Corollary 81.4](#) assumes a Baire-property cone where source 12's [corollary 81.3](#) assumes a Borel one; the semiring clause of the latter is [theorem 81.2](#). The models G_D with $D = \mathbb{R}$ and $D = \mathbb{Q}^{\mathbb{N}}$ are Part I's G_D ([section 3](#)); their cones are $M_{\mathbb{R}}$ and $M_{\mathbb{B}}$. Source 10 credits these constructions to source 08 [56]; they are source 05's, [theorem 3.1](#). The case $D = \ell^2(\mathbb{N}, \mathbb{R})$ is new here.

81.4 Positive monoids, group completion, and the exact boundary [10]

The group structure is used twice in the category proof: inversion exchanges P and $-P$, and reflection $x \mapsto z - x$ is a homeomorphism. Neither operation is available on an arbitrary positive monoid. These are substantive uses, not notational conveniences.

For a cancellative commutative monoid M , its algebraic Grothendieck group can be formed from pairs (x, y) , with

$$(x, y) \sim (x', y') \iff x + y' = x' + y,$$

interpreted as formal differences. A semiring multiplication extends algebraically to this group by

$$(x - y)(u - v) = (xu + yv) - (xv + yu).$$

However, none of these formulas proves that the quotient has a compatible Hausdorff Polish group topology, or that the original positive cone embeds with its original topology. Quotient topologies can fail the required separation or completeness properties.

Proposition 81.5 (Conditional completion obstruction [10]). *Let M be an uncountable cancellative ordered semiring whose signed ring of differences A is discretely ordered. Suppose A admits a Polish additive-group topology with a Baire-property positive cone. Then such a semiring cannot exist. In particular this applies whenever a specified topological group completion supplies those properties.*

Proof. The natural map from M to A is injective by cancellation, so A is uncountable. The ring then contradicts [corollary 78.6](#). $\square$

For the explicit models $G_D^{\geq 0}$ in [section 81.3](#), the signed topology is already available. For an arbitrary positive model, it is not. Therefore the present proof does not by itself remove the Borel multiplication assumption from Glazer’s positive-model theorem. The relationship between those results is a comparison of different hypothesis sets, not an unconditional strengthening in every setting.

Glazer’s Question 2 asks for an uncountable Polish Presburger model with continuous addition [1]. The additive examples recalled here belong to the already developed affirmative constructions in the project; they are not presented as a new resolution in this manuscript. Our new conclusion is their universal signed ordered-ring non-expandability under Baire comparison, and the general rigidity mechanism behind it.

82 Borel structures, Polishability, and meager embeddings

Definition 82.1. For this section, a *Polish group presentation of a fixed Borel structure* is a Polish additive-group topology whose Borel sets are exactly the specified Borel sets. This is a requirement, not an automatic consequence of having a standard Borel group.

Theorem 82.2 (No compatible Polish group presentation). *Let an uncountable discretely ordered ring have a specified standard Borel structure in which its positive cone is Borel. There is no Polish additive-group topology generating that Borel structure. No Borelness of multiplication is assumed.*

Proof. In such a topology the cone would remain Borel, so [corollary 78.5](#) would force the carrier to be countable. $\square$

The result distinguishes two failures. A displayed metric may merely fail to be complete; [theorem 82.2](#) rules out *every* compatible Polish group topology preserving the prescribed Borel sets. It does not rule out a Polish topology after the measurable structure and the order’s regularity have been changed.

Theorem 82.3 (Borel ordered-ring subgroups are meager when uncountable). *Let K be a Polish abelian group. Suppose $A \subseteq K$ is an uncountable Borel subgroup, equipped algebraically with a discretely ordered ring structure whose addition is the inherited group addition. If its positive cone is Borel as a subset of K , then A is meager in K .*

Proof. If A were nonmeager, [lemma 77.1](#) would show that $A - A = A$ contains a neighborhood of 0. Thus A is open. An open subgroup is also closed, since its complement is a union of open cosets. Consequently A , with the inherited topology, is a Polish group. Its positive cone is Borel, so [corollary 78.5](#) makes A countable, contradicting the hypothesis. $\square$

Corollary 82.4 (Closed subrings and integer parts). *In the setting of [theorem 82.3](#), a closed subgroup admitting a discretely ordered ring structure with Borel cone must be countable and discrete in its subspace topology. More generally, an uncountable Borel integer part of an ordered field with a Polish additive topology and Borel order must be meager in the ambient additive group.*

Proof. A closed subgroup of a Polish group is Polish, so the first statement is [corollary 78.5](#). An integer part is a discretely ordered subring; its positive cone is the intersection of its carrier with the field's positive cone. Thus the second statement follows from [theorem 82.3](#). $\square$

An integer part here means a discretely ordered subring I such that for every field element x there is $i \in I$ with $i \leq x < i + 1$. The approximation clause is not needed for this corollary; only discrete order and subring structure are used. This is another instance where weaker algebraic hypotheses suffice than full arithmetic.

83 A parallel Haar-measure theorem

The bounded-neighborhood mechanism can also be obtained from measure rather than category. This gives a separate improvement of the order regularity hypothesis in the locally compact setting. We use a regular Haar measure and its completion; no assumption about restrictions to null subgroups is imposed.

Lemma 83.1 (Positive-measure differences). *Let G be a Hausdorff locally compact abelian group with regular Haar measure μ . If a compact set K has $\mu(K) > 0$, then $K - K$ contains a neighborhood of 0.*

Proof. Choose an open $U \supseteq K$ of finite measure with $\mu(U) < 2\mu(K)$. Compactness and continuity of addition give a neighborhood W of 0 such that $K + W \subseteq U$. For $h \in W$, the sets K and $K + h$ both lie in U , so

$$\mu(K \cap (K + h)) \geq 2\mu(K) - \mu(U) > 0.$$

Thus they meet, and $h \in K - K$. $\square$

This is the compact-set form of the classical Steinhaus difference-set principle [\[46\]](#); the displayed argument establishes the locally compact version actually needed here.

Lemma 83.2 (Haar-measurable order gives a bounded neighborhood). *Let G be a nontrivial ordered abelian group with a Hausdorff locally compact group topology. If its positive cone is measurable for the completion of a regular Haar measure, then 0 has an order-bounded open neighborhood.*

Proof. The discrete case is immediate. Otherwise $\mu(\{0\}) = 0$: an atom at 0, together with translation invariance and a compact neighborhood of finite measure, would force that neighborhood to be finite, and the Hausdorff group would then be discrete.

Choose a compact set of positive finite measure. Its partition by P , $-P$, and $\{0\}$ shows that one of the first two pieces has positive measure. Invert the negative piece if necessary. Inner regularity for completed Haar-measurable sets provides a compact $B \subseteq P$ with $0 < \mu(B) < \infty$.

All integrations that follow may be performed in a sigma-compact open subgroup containing B . Such a subgroup exists: take a compact identity neighborhood, adjoin B and $-B$, and take the subgroup generated by their union. It is a countable union of compact sets and contains an open identity neighborhood. Haar measure restricted to it is sigma-finite. Tonelli's theorem and translation invariance give

$$\begin{aligned} \int_G \mu(B \cap (a - B)) d\mu(a) &= \int_G \int_G \mathbf{1}_B(x) \mathbf{1}_B(a - x) d\mu(x) d\mu(a) \\ &= \mu(B)^2 > 0, \end{aligned} \tag{83.1}$$

where the integrands vanish outside that subgroup and its relevant compact sumsets. Hence there is an a such that $K = B \cap (a - B)$ has positive measure. It is compact. For any $x \in K$, both x and $a - x$ are positive, so $0 < x < a$ and in particular $a > 0$. Thus $K \subseteq (0, a)$ and $K - K \subseteq (-a, a)$. Apply [lemma 83.1](#). $\square$

Theorem 83.3 (Haar-measurable arithmetic rigidity). *Let a discretely ordered ring have a Hausdorff locally compact additive-group topology. If its positive cone is measurable for the completion of a regular Haar measure, then*

$$c(R) = d(R) = |R|.$$

If the topology is ccc, it is countable and discrete. In particular, this conclusion holds for second-countable locally compact additive-group topologies with Haar-measurable order.

Proof. Combine [lemmas 78.1](#) and [83.2](#) with [\(76.2\)](#). Under ccc the ring is countable. Locally compact Hausdorff spaces are Baire, so the same singleton argument as in [corollary 78.4](#) makes it discrete. $\square$

The earlier report asked how to weaken Borel order without assuming that ambient measurability automatically passes to every closed subgroup [[49](#), Section 14]. This proof avoids that issue: it works directly with a positive-measure compact subset of the ambient cone. The sole restriction used for Tonelli is to an *open* sigma-compact subgroup containing that compact subset. No conclusion about measurability on a null closed subgroup is required.

[write] [Theorem 83.3](#) answers source 08's [question 57.4](#) for ambient Haar measurability, and [lemma 77.2](#) answers it for the Baire property: in both routes the regularity of the cone for the given topology suffices, and no condition on restrictions to compact subgroups or closed real subspaces is needed. [Theorem 83.3](#) also strengthens [theorem 44.1](#) from Borel to Haar-measurable order, and removes second countability from its cardinal part.

84 Sharpness: three topologies on one polynomial ring

The same algebraic example isolates the role of the principal hypotheses. Let

$$A = \mathbb{Z} + t\mathbb{R}[t] = \left\{ z + \sum_{j=1}^N r_j t^j : z \in \mathbb{Z}, r_j \in \mathbb{R}, N \in \mathbb{N} \right\}. \quad (84.1)$$

Order a nonzero polynomial by the sign of its leading coefficient, so that t is larger than every ordinary real constant wherever that comparison is made in $\mathbb{R}[t]$. This is an ordered ring: leading coefficients multiply, and comparison is invariant under translation. If a positive element has degree zero, it is a positive integer; if it has positive degree, it exceeds every ordinary integer. Hence 1 is its least positive element. The cardinality of A is $\mathfrak{c}$.

[write] Source 10 prints the same three topologies on this ring (written $\mathbb{Z} + X\mathbb{R}[X]$) as its Examples 9.1, 9.3 and 9.4; their labels are second labels of the three subsections below. The ring is source 08's A_* ([example 54.1](#)) and source 06's $\mathbb{Z} + t\mathbb{R}[t]$ ([theorem 38.1](#)).

84.1 A separable metrizable ring that is meager in itself

Identify a polynomial with its coefficient sequence and give A the subspace topology from

$$\mathbb{Z}_{\text{discrete}} \times \mathbb{R}^{\mathbb{N}_{>0}}.$$

The higher-coordinate sequence has finite support. The ambient space is metrizable and second countable, so A is separable and metrizable. Addition and inversion are continuous. Multiplication is jointly continuous as well, since its n th coefficient is the finite polynomial expression

$$(fg)_n = \sum_{i=0}^n f_i g_{n-i}. \quad (84.2)$$

Continuity into the product follows coordinate by coordinate; the product lies again in the finite-support subring.

For $N \geq 0$, let A_N be the set of polynomials of degree at most N . It is closed: all higher coefficients vanish. It has empty interior, because a basic open set constrains only finitely many coordinates, and a nonzero term can be inserted at a still higher degree. Thus

$$A = \bigcup_{N \geq 0} A_N \quad (84.3)$$

is meager in itself.

Nevertheless the cone has low Borel complexity. For $d \geq 1$, the condition

$$f_j = 0 \text{ for all } j > d, \quad f_d > 0$$

is the intersection of a closed set and an open set. In a metrizable space every open set is F_σ , so this condition is F_σ . The positive degree-zero polynomials form a countable union of closed singletons. Taking the countable union over d proves that P is F_σ . Its complement is $(-P) \cup \{0\}$, also F_σ . Therefore P is Δ_2^0 .

This example has continuous multiplication, not merely arbitrary multiplication. It shows that the Baire hypothesis cannot be replaced by metrizability or separability. By [theorem 82.2](#), there is also no Polish additive-group topology generating the same Borel structure: the defect is not repaired by choosing another compatible complete group metric.

[10] Source 10 adds:

This example shows why a separable metrizable topology with Borel order is not enough, even when multiplication happens to be continuous. The failure occurs in the Baire hypothesis, not in the algebra. For an even smaller example, $\mathbb{Z}$ with its p -adic subspace topology is a countable, nondiscrete, non-Archimedean topological ring; every subset is Borel, but it is again meager in itself.

84.2 A locally compact Baire ring presentation without ccc

Decompose the additive group as

$$A = \mathbb{R}t \oplus D, \quad D = \mathbb{Z} \oplus \bigoplus_{j \geq 2} \mathbb{R}t^j, \quad (84.4)$$

and give it the product topology with $\mathbb{R}t$ usual and D discrete. This is a Hausdorff locally compact metrizable topological group and is Baire. It is not separable: the distinct sets $\mathbb{R}t + d$, $d \in D$, are disjoint open subsets.

Its cone is again Δ_2^0 . When the high-degree part of d is nonzero, its leading coefficient determines the sign of $rt + d$ independently of r . Such a coset is either wholly positive or wholly negative. When $d = z \in \mathbb{Z}$, positivity is determined by $r > 0$, or by $r = 0$ and $z > 0$. Thus the positive cone is the union of an open set and the closed set of positive integer points

at $r = 0$. The negative cone has the analogous form. Metrizable gives the claimed Δ_2^0 complexity.

Choosing rational r in each open coset gives a dense set of size $\mathfrak{c}$. The disjoint cosets give the matching lower bound. Hence

$$c(A) = d(A) = |A| = \mathfrak{c},$$

exactly as [theorem 78.3](#) predicts. The ring multiplication is not continuous in this topology. Indeed, $t/n \rightarrow 0$ whereas multiplication by t sends that sequence to t^2/n , which does not converge to 0 in the discrete D coordinate. This is permitted by the theorem, which assumes nothing about multiplication.

[write] This is source 08's [example 54.1](#). [10] Source 10 adds the extent and the Lindelöf degree: by [theorem 79.13](#), or directly by choosing one point on each line, $\text{cell}(A) = \text{dens}(A) = \text{ext}(A) = L(A) = |A| = \mathfrak{c}$. It notes:

This is not asserted to be a topological ring. The continuity of addition is all the main theorem assumes, and multiplication in this example fails continuity for many fixed scalars; see [section 85](#).

84.3 A Polish additive topology with a non-Baire positive cone

Let $V = t\mathbb{R}[t]$, regarded only as a rational vector space. Its dimension is $\mathfrak{c}$: it contains a copy of the rational vector space $\mathbb{R}$, and its cardinality is $\mathfrak{c}$. The vector space $\mathbb{Q}^{\mathbb{N}}$ also has dimension $\mathfrak{c}$. To justify the latter assertion, its cardinality is $\mathfrak{c}$; over a countable field, an infinite-dimensional vector space has cardinality the maximum of its dimension and $\aleph_0$. A countable-dimensional one would be countable. These facts force the dimension to be $\mathfrak{c}$.

Using bases, choose a rational-linear isomorphism

$$\psi : V \longrightarrow \mathbb{Q}^{\mathbb{N}}.$$

Transport the Polish group topology of $\mathbb{Z}_{\text{discrete}} \times \mathbb{Q}_{\text{discrete}}^{\mathbb{N}}$ to the additive group $A = \mathbb{Z} \oplus V$ through $z + v \mapsto (z, \psi(v))$. The ring and its original leading-coefficient order remain algebraically unchanged. The new topology is Polish and makes addition and inversion continuous. By [corollary 78.5](#), its positive cone does *not* have the Baire property.

This is an existence construction using choice of vector-space bases, not a computable coding. It shows that regularity of order cannot be omitted from the theorem. No claim is made that multiplication is Borel in the transported presentation.

[write] This is a variant of source 08's [example 54.2](#), which transports $\mathbb{R}^d \times \mathbb{Z}$ and calls the order necessarily non-Borel; here the transported topology is that of $\mathbb{Q}^{\mathbb{N}} \times \mathbb{Z}$ and the conclusion is that the cone lacks the Baire property. [10] Source 10's version of the example ends:

The isomorphism is chosen using vector-space bases. It is not an algorithm, and the construction gives no Borel multiplication or regular comparison procedure. Its point is exact: additive Polishability alone does not prohibit uncountable discretely ordered rings. Order regularity is an indispensable part of the theorem.

84.4 Other indispensable distinctions

The ordinary real field is an uncountable Polish ordered ring with continuous operations and open positive cone. It does not contradict our theorem because it has no least positive element. Thus discrete order is essential.

[10] Source 10 adds:

The usual real field is an uncountable Polish topological ring with open positive cone. It escapes [corollary 78.4](#) because it has elements strictly between 0 and 1. In the relative theorem, its finite part is already all of the ring. This is the correct boundary, not a failure of Baire category or of the chain condition.

The group $G_{\mathbb{R}}$ from (80.2) is uncountable, Polish, locally compact, discretely ordered as a group, and has a Δ_2^0 positive cone. It lacks any compatible unital ordered-ring expansion by [corollary 81.1](#). Thus continuous addition and regular discrete order alone are not contradictory. It is the combination with an unrestricted positive multiplier family that creates the packing.

85 Continuous scalar maps beyond the separable setting [10]

Once ccc is imposed, the main theorem already gives discreteness. The following observation is instead meaningful for nondiscrete, necessarily non-ccc Baire presentations such as [section 84.2](#).

Proposition 85.1 (A bounded subring of continuous scalars [10]). *Let A be a discretely ordered ring with a nondiscrete Hausdorff Baire additive-group topology and a Baire-property positive cone. Define*

$$\text{Sc}_{\text{cont}}(A) = \{a \in A : L_a : A \rightarrow A, L_a(x) = ax, \text{ is continuous}\}.$$

Then $\text{Sc}_{\text{cont}}(A)$ is a unital subring containing $\mathbb{Z}1$, and for some $c > 0$ it is contained in $[-c, c]$.

Proof. The identities $L_{a+b} = L_a + L_b$, $L_{-a} = -L_a$, and $L_{ab} = L_a \circ L_b$ prove the subring assertion. Integer scalar maps are continuous because the additive operations are continuous. Choose a neighborhood $U \subseteq [-c, c]$ by local boundedness. If $|a| > c$, then for every nonzero x , discreteness gives $|ax| = |a||x| \geq |a| > c$. Hence $L_a^{-1}(U) = \{0\}$. Continuity of L_a would make $\{0\}$ a neighborhood and the group discrete. Therefore such a cannot belong to $\text{Sc}_{\text{cont}}(A)$. $\square$

Corollary 85.2 ([10]). *A discretely ordered ring with a Hausdorff Baire additive topology, a Baire-property positive cone, and continuous multiplication by every fixed scalar is discrete. No separability or ccc assumption is needed.*

Proof. If the topology were nondiscrete, [proposition 85.1](#) would bound all of A by one c , contradicting the existence of $c + 1$. $\square$

In [section 84.2](#), the continuous-scalar subring is exactly $\mathbb{Z}$. Every integer works. If a has positive degree k and leading coefficient $\alpha \neq 0$, the restriction of L_a to the open line $\mathbb{R}X$ has degree- $(k + 1)$ coefficient $r \mapsto \alpha r$ in the discrete factor D . This is not continuous, so L_a is not continuous. Thus the bounded-scalar conclusion is nonvacuous and can be exact.

We deliberately do not identify Borel scalar maps with continuous ones in this nonseparable discussion. Standard automatic-continuity statements for Borel homomorphisms between Polish groups do not license that substitution for an arbitrary nonseparable target. In the Polish case of the main theorem, every map is already continuous because the topology has been proved discrete.

[write] [Corollary 85.2](#) generalizes source 09's [theorem 62.1](#) in another direction: continuous scalar maps replace Borel ones, and no separability is assumed. Compare source 08's [theorem 53.4](#), which counts the Borel scalars of an integral-domain multiplication unrelated to the order.

86 Completion, units, and omnific applications

86.1 Completion may destroy discrete order algebraically

The additive completion of the coefficient-topology example in [section 84.1](#) is

$$\widehat{A} = \mathbb{Z} + t\mathbb{R}[[t]]. \quad (86.1)$$

The higher coefficients may now have infinite support. The topology is the full product topology, and (84.2) defines a jointly continuous multiplication. Every coefficient sequence is the limit of its finite truncations, so A is dense. Completeness of the product additive group gives the asserted completion.

One might try to extend the leading-degree discrete order to this Polish ring. The obstruction is stronger than failure of Borelness: *there is no discrete ring order on this completion at all.*

Lemma 86.1 (Units in a discretely ordered ring). *Every unit of a discretely ordered ring is 1 or -1 .*

Proof. If $uv = 1$ and $u > 0$, then $v > 0$, and discreteness gives $u, v \geq 1$. If $u > 1$, then $u \geq 2$, and hence $uv \geq 2$, a contradiction. Thus $u = 1$. If $u < 0$, apply the positive case to $-u$ and $-v$. $\square$

Proposition 86.2 (No discrete order on the coefficient completion). *The ring $\widehat{A} = \mathbb{Z} + t\mathbb{R}[[t]]$ admits no unital discrete ring order.*

Proof. The element $1 - t$ is a unit with inverse

$$1 + t + t^2 + \cdots.$$

This identity is a formal coefficient identity, not an analytic convergence assertion. It also follows from the exact finite identity

$$(1 - t) \sum_{j=0}^N t^j = 1 - t^{N+1} \quad (86.2)$$

by passing to the coefficient limit. But $1 - t$ is neither 1 nor -1 , contrary to [lemma 86.1](#). $\square$

Second proof [10]. Truncations show that the polynomial ring is dense in the product coefficient group of all such formal series. This product is completely metrizable, and the coefficientwise ring operations extend by the finite convolution formulas.

In any discretely ordered ring, the only units are 1 and -1 . Indeed, if $uv = 1$, then $|u|, |v| \geq 1$ and $1 = |u||v| \geq |u| \geq 1$, so $|u| = 1$. In $\widehat{A}$, however, the element $1 - X$ is a unit with inverse $\sum_{n \geq 0} X^n$. It is neither 1 nor -1 as a formal series. This contradicts the necessary unit property. $\square$

The example separates three issues: the finite-polynomial ring has a discrete order and a useful incomplete topology; its completion has excellent topological ring operations; and the completion loses the algebraic possibility of discrete order. It would be incorrect to attribute this last failure only to descriptive-set-theoretic complexity.

86.2 Set-sized omnific subrings

ProveIt's surreal development describes the omnific ring as the normal-form elements with nonnegative exponents and integer constant term; it records in particular the least-positive-element property [52]. The present theorem uses only the resulting ordered-ring laws. It does not use the deeper claims in the repository's unrefereed reports.

Corollary 86.3 (Omnific presentation obstruction). *Let B be any uncountable set-sized unital subring of the omnific integers, with its inherited order. It admits no Polish additive-group topology in which the inherited positive cone has the Baire property. More generally, any Hausdorff Baire additive-group topology with such a cone satisfies $c(B) = d(B) = |B|$.*

Proof. An ordered unital subring of a discretely ordered ring is discretely ordered: its positive elements are positive in the larger ring and therefore at least 1. Apply [theorem 78.3](#) and [corollary 78.5](#). $\square$

[10] Source 10 proves the same corollary with the cardinal equalities of [theorem 79.13](#):

The inspected ProveIt surreal README describes the construction of the omnific ring, its least positive element 1, its constant-term homomorphism to $\mathbb{Z}$, and the fact that an omnific integer bounded by an ordinary real is an ordinary integer [55]. The first two properties already make its set-sized unital ordered subrings eligible for the present theorem. We do not need the full normal-form theory, constant-term retraction, or real closedness of the surreal field for the obstruction.

Corollary 86.4 (Set-sized omnific presentations [10]). *Let A be a set-sized unital ordered subring of $\mathbf{Oz}$. If its additive group is given a Polish group topology and the inherited positive cone has the Baire property, then A is countable and the topology is discrete. If instead the topology is Hausdorff and Baire and A is uncountable, a Baire-property positive cone forces a full-cardinality closed discrete subset and the equalities of [theorem 79.13](#).*

Proof. The inherited order is discrete, with least positive element 1. Apply [corollary 78.6](#) and [theorem 79.13](#). $\square$

A frequent coding pattern is a closed additive subgroup of a countable product of countable discrete groups. Such a group is Polish: use a complete product metric and restrict it to the closed subgroup. Its finite-coordinate agreement subgroups are open and form a basis. Thus an uncountable exact-stream representation of a set-sized omnific subring in this format cannot simultaneously give continuous additive group operations and a Baire comparison predicate.

This does not rule out countable notation systems, incomplete spaces of finite expressions, non-Baire comparisons, non-Polish presentation spaces, or purely algebraic formalizations. A kernel proof of field or ring laws does not produce a Polish presentation or a measurable order algorithm. Likewise, being homeomorphic to Baire space as a bare space does not by itself give a non-Archimedean group topology.

Finite polynomial expressions

$$z + r_1\omega + r_2\omega^2 + \cdots + r_N\omega^N$$

provide the direct connection with [\(84.1\)](#). Finite normal-form arithmetic identifies their ring and order with $\mathbb{Z} + t\mathbb{R}[t]$ through $t \mapsto \omega$. The sharpness examples are thus also examples

on an explicit set-sized omnific subring; one does not need the full proper class of surreal numbers to see the obstruction.

An additive fragment such as $\mathbb{R}\omega + \mathbb{Z}$ behaves differently. It is closed under addition but not under multiplication, since ω^2 lies outside it. Giving that fragment the topology of $G_{\mathbb{R}}$ is compatible with a regular discrete additive order. The theorem explains why expanding it to a ring while retaining that additive presentation is impossible.

86.3 Foundations and universe boundaries

No theorem above treats the proper class **Oz** as a Polish space. A Polish space is set-sized. To discuss the entire surreal or omnific class in a class theory, one must separately specify which collections of classes and which class maps are allowed; none of that is needed for [corollary 86.3](#).

In a type-theoretic implementation, fix a universe containing the carrier of the ring or additive group and a universe large enough for its topology and cardinal invariants. The packing map is a set-indexed family within those universes. If a universe-relative construction of surreals is used, first identify the desired set-sized unital subring and its order. A birthday cutoff or a finite list of permitted scales must not be assumed to be a subring without proving closure.

The finite-polynomial example can instead be defined from scratch, without any surreal ontology: its coefficient sequences are finitely supported functions on the ordinary naturals, with an integer constant term. This is a particularly clean target for a foundationally modest formalization of the examples.

86.3.1 Universe discipline [10]

A proper class such as the full **No** or **Oz** is not an underlying set for the theorems here. In a set-theoretic treatment, first choose a set-sized subring closed under the needed operations. In a dependent- type-theoretic treatment, first specify a universe level and a type carrying the actual ordered ring, rather than a type of arbitrary syntax with an unproved interpretation map. The theorem then concerns that type's underlying set in a metatheoretic model.

The quotient $A/\mathbb{Z}1$ is a set-sized additive quotient. No “set of all class functions”, class-valued probability space, or topology on all proper-class subsets is formed. These restrictions are part of the statement, not a later repair to its proof.

87 Formalization architecture and proof audit

The repository snapshot used for the interface review is

`0f084afa9be466593ffa6aff37749e78c0adc733`.

The inspected Presburger README describes an integer-specific Cooper eliminator with syntax, normalization, elimination, and a decision procedure. The inspected surreal README records the ordered omnific construction and distinguishes proved-library interfaces from research reports [51, 52]. We have not built the repository or independently verified those advertised audit results.

The central theorem can be formalized independently of Cooper elimination, nonstandard-model syntax, or the full surreal construction. Its proof should be split into interfaces that expose exactly what is used.

87.1 The algebraic interface

The first component is a discretely ordered ring and its absolute-value inequalities. The core certificate is the statement

$$r \neq s, a > 0 \implies 2a < |(2a + 1)(r - s)|. \quad (87.1)$$

Next, a set-level lemma says that if $W \subseteq (-a, a)$, the translates $(2a + 1)r + W$ are disjoint for distinct indices. Neither lemma needs Baire category, cardinal arithmetic, topological continuity of scalar multiplication, or properties of a particular ring.

For Lean, a robust design would expose the least-positive-element property explicitly as a hypothesis on the ordered ring, rather than assuming that a convenient name for a discrete-order typeclass matches the required semantics. The proof only needs the implication $0 < x \implies 1 \leq x$. A Rocq development can package precisely the same fields in a record. This paragraph is a design specification, not a claim that any particular library API has been tested.

87.2 The topological and category interface

The second component concerns an ordered additive group with a Hausdorff group topology. It should prove, in order:

1. a nondiscrete Hausdorff topological group has nowhere-dense singletons;
2. a nonmeager Baire-property set has differences containing a neighborhood of zero;
3. reflection of a locally comeager part of the positive cone produces a nonmeager interval $(0, 2u)$;
4. a Baire-property positive cone yields an order-bounded neighborhood.

The last statement should be written for groups rather than rings, so that the self-embedding theorem can reuse it without introducing unneeded multiplication.

The ccc consequence need not formalize general cardinal invariants first. It can directly turn the family of translates into an injection from the carrier into a countable family. The exact cardinal equalities can be added later. This keeps the first kernel-checking target small.

87.3 Separate modules and theorems

A reasonable dependency order is:

Module	Mathematical boundary
DiscreteOrderGap	(76.1) and (87.1)
DisjointTranslates	The packing lemma, with an arbitrary bounded set
BaireOrderedGroup	Pettis and the bounded-neighborhood lemma
ArithmeticRigidity	Baire ccc and Polish conclusions
UnitImageSpectrum	All ordered additive self-embeddings, no regularity
PositiveExpansion	Difference-ring construction and semiring obstruction
PolynomialExamples	Finite-support ring and the completion unit witness
OmnificApplication	An adapter using proved subring and discrete-order laws

The measure theorem belongs in a later module, because it adds Haar measure, compact inner regularity, and Tonelli. It is logically independent of the category route to the principal theorem.

87.4 Trust boundaries of this package

The package contains mathematical proofs and an executable exact arithmetic script. It does not contain a new Lean or Rocq proof, and no such toolchain was run during preparation. The tests check finite polynomial identities in $\mathbb{Z} + t\mathbb{Q}[t]$. They do not certify Baire category, all Borel sets, uncountable cardinalities, arbitrary semiring completions, or a theorem for every ordered ring.

The most important audit questions are therefore mathematical. Does one mistakenly use continuity of $r \mapsto br$? No: only translates of W are used. Does the reflection require multiplication? No: $2u - x$ is formed in the additive group. Is an open convex subgroup assumed to be an ideal? No. Is a positive monoid silently given a Polish group completion? No. Is the proper class of all omnific integers treated as a set? No. The manuscript's statements are formulated to keep each of these distinctions visible.

87.5 A small algebraic core before topology [10]

A sensible implementation order is the following.

Layer A: finite intervals and hulls. Prove finite-interval exhaustion from least positivity of 1; define $\text{Fin}(A)$ and H_a using ordinary natural numbers; prove their additive subgroup and convexity properties. Keep the hull's external ordinary index distinct from an element of a nonstandard model.

Layer B: the exact kernel. Build the additive map $x \mapsto ax + H_a$ and prove that its kernel is $\text{Fin}(A)$. Obtain the quotient injection by a general additive-group quotient theorem. This layer should depend only on ordered algebra, not topology, measurable spaces, or surreal normal forms. The bounded-direct-sum theorem can be implemented independently using finite supports and a largest nonzero index.

Layer C: the category interface. Prove [lemmas 77.2](#) and [77.3](#). The only topological input needed by the algebraic layer is one bounded neighborhood. Explicitly expose that interface before connecting it to the Baire-property positive cone. This makes the theorem reusable with other sources of local boundedness.

Layer D: quotient cardinality and discreteness. Prove that an open subgroup has at most countably many cosets in a ccc group. Combine the injection with countability of $\mathbb{Z}1$. Finish with the elementary countable-Hausdorff-Baire-group lemma. A separate cardinal module can produce the closed discrete transversal and the maximal-cardinality results using choice.

Layer E: syntax and applications. Connect the theorem to the actual omnific ordered ring and to abstract $\mathbb{Z}$ -groups. ProveIt’s Presburger development exposes affine syntax, coefficient normalization, a normalized Cooper step, and a full Lean sentence decision procedure, while its Rocq development independently checks the decisive one-variable step [54]. Transfer to abstract $\mathbb{Z}$ -group semantics is a separate theorem; an integer implementation alone is not evidence that the transfer has been formalized.

No new source file is labeled “Lean verified” in this package. The repository was inspected at the snapshot recorded in the source audit; its builds were not rerun. The mathematical arguments above do not rely on the repository’s unrelated research claims.

88 Further research questions and concrete next targets

The following questions are proposed research directions arising from the proved results. Except for the earlier project questions explicitly identified in [sections 75](#) and [83](#), we do not claim that a complete literature search has established their open status.

[10] Source 10 introduces its twelve questions as follows. Each one that asks the same as a question of source 12 is printed after it; the other four follow in [section 88.1](#).

The general Polish signed-group question is resolved by [corollary 78.4](#); it should no longer appear on the open list. The following questions identify extensions not settled by this paper. Unless attributed explicitly, they are proposed research targets, not assertions that a problem’s literature-wide open status has been certified.

[write] The question that both sources answer is [question 57.1](#) of [Part IV](#); their answers are [corollaries 78.4](#) and [78.5](#). Source 08’s [question 57.4](#) is answered as stated (note after [theorem 83.3](#)); the questions below on weaker regularity ask for more.

Question 88.1 (Positive monoids beyond Polishable signed groups). Can an uncountable Polish continuous-addition positive arithmetic model admit an algebraically compatible unital semiring multiplication when no compatible Baire ccc signed-group presentation with regular order exists?

The new obstruction shows exactly where a positive example would have to escape: failure of an appropriate group topology or failure of order regularity after passage to differences. Requiring stronger arithmetic, such as $Q + \text{IOpen}$, also invokes Glazer’s obstruction if multiplication is Borel. The non-Borel multiplication case and the weaker algebraic case should be kept separate.

[10] Source 10 asks:

Question 88.2 (Positive arithmetic without a signed topology [10]). Can an uncountable Polish positive arithmetic monoid with continuous addition and suitably regular order carry an ordered semiring multiplication when no regularity of that multiplication is assumed?

The task is to recover a replacement for the group reflection argument, or to construct a genuine example escaping it. A result must state its arithmetic axioms and must not silently

assume a Polish group completion. Glazer's positive-model work is the closest published starting point, but its Borel multiplication hypothesis changes the available tools.

Question 88.3 (A usable group-completion criterion). Which conditions on a Polish cancellative ordered monoid ensure that its algebraic difference group has a Hausdorff Baire ccc topology with a Baire-property cone and the required compatibility with the original presentation?

The first useful outcome would be a sufficient criterion stated in terms of translations and the difference quotient map. A proof must check the quotient topology rather than appealing only to algebraic cancellation. A converse would identify the precise gap between the signed theorem here and the positive-monoid setting.

[10] Source 10 asks:

Question 88.4 (A usable group-completion criterion [10]). Which Polish cancellative Presburger monoids embed topologically as the positive cone of a Hausdorff Polish topological group?

A promising formulation would use translation embeddings, local neighborhood compatibility, and a complete uniform structure on formal differences. Both the topology on the original monoid and the regularity of the completed order must be tracked. Such a criterion would turn [proposition 81.5](#) into a practical obstruction.

[write] Both are source 08's [question 57.2](#), asked also by source 09 as [question 73.11](#), for Baire ccc completions with a Baire-property cone.

Question 88.5 (The topology of regular ordered additive groups). For a fixed Polish abelian group, which discrete invariant orders have a Baire-property positive cone, and which open principal convex subgroups can they produce?

[Proposition 79.1](#) provides an open principal convex subgroup with countable quotient. This is necessary, not a classification. One can ask whether recursive data from open convex subgroups and their ordered quotients give complete invariants in natural classes such as non-Archimedean Polish groups.

Question 88.6 (Exact unit-image spectra). Which subsets of the positive cone occur as $\mathcal{U}(G)$ for uncountable Baire ccc discretely ordered groups with regular cone?

The spectrum is bounded by [theorem 80.1](#); [\(80.3\)](#) gives one exact infinite example. Possible next invariants include its cofinality within the convex subgroup it generates, closure under ordinary multiplication of indices when available, and the effects of restricting to continuous or elementary self-embeddings. These restrictions need not produce the same spectrum.

[10] Source 10 asks:

Question 88.7 (The intrinsic unit-image spectrum [10]). For an uncountable Baire ccc discretely ordered group with regular order, how can $\mathcal{U}(G)$ be classified algebraically and descriptively?

The spectrum is order bounded and each of its principal hulls is meager. Determining when the spectrum itself is meager, Borel, analytic, or definable requires additional information. One should first analyze explicit divisible-plus-integer examples, then groups without a global additive order unit.

Question 88.8 (Cardinal and component spectra). For a fixed uncountable discretely ordered ring, which Hausdorff Baire regular-order group topologies realize the forced equality $c(R) = d(R) = |R|$, and what weights and connected-component structures can occur?

The equality itself is settled here. The remaining realization problem is more refined: prescribed local compactness, completeness, metrizability, or topology on selected coefficient subgroups. [Section 84.2](#) shows that local compactness can coexist with exact maximal cellularity on a continuum-sized ring.

[10] Source 10 asks:

Question 88.9 (Optimal nonseparable presentations [10]). Which discretely ordered rings admit nondiscrete Hausdorff Baire additive topologies with Borel order, and which full-cardinality open quotient structures can occur?

The polynomial example realizes maximal cellularity with an open real line. The cardinal lower bound is exact, but it does not classify possible local groups or orders. A classification for rings $\mathbb{Z} + I$ with a specified divisible additive ideal I would make the boundary concrete.

[write] This refines source 08's [question 57.5](#): the equality $c = d = |R|$ is now proved.

Question 88.10 (Regularity weaker than Baire or Haar measurability). What assumptions on an ordered positive cone are sufficient to force an order-bounded neighborhood without requiring the full Baire property or completed Haar measurability?

The proof isolates a concrete intermediate property: some order-bounded set has a difference set containing a neighborhood of zero. A productive weakening should yield that property directly. In non-locally-compact groups, any proposed measure substitute needs a carefully specified notion of null set and invariance; there is no general locally finite Haar measure to import unchanged.

[10] Source 10 asks:

Question 88.11 (Alternative notions of regular order [10]). Can the category hypothesis be replaced by a useful measure-theoretic or definability hypothesis that still forces a bounded neighborhood?

A positive result needs an explicit substitute for the nonmeager-cone argument. Ambient measurability alone should not be assumed to survive restriction to arbitrary null or meager closed subgroups. The algebraic kernel theorem is already available once a local bound is proved.

[write] Source 08's [question 57.4](#) asked for this with hereditary conditions; Part VI answers it with the Baire property or completed Haar measurability of the cone itself.

Question 88.12 (Definability of integer parts in Polish fields). What descriptive complexities can occur for uncountable integer parts inside ordered fields with Polish additive topology and Borel order?

A Borel integer part is meager by [corollary 82.4](#), and its inherited Borel ordered-group structure is not compatibly Polishable when uncountable. These obstructions leave room for finer questions about Borel ranks, analytic presentations, and the complexity of floor-selection maps. The field, its topology, and its order should be fixed before these complexities are compared.

Question 88.13 (Which completions preserve discrete order?). For natural additive topologies on discretely ordered subrings, which completions avoid new units and retain any discrete ring order?

The unit witness in [proposition 86.2](#) gives an algebraic necessary condition independent of Borel complexity. It is not sufficient: avoiding nontrivial units need not produce an order, and producing an order need not make it regular. A useful theory would separate these three successive obstructions.

[10] Source 10 asks:

Question 88.14 (Completion with controlled units [10]). Which dense discretely ordered subrings admit useful completions without introducing new nontrivial units or other elementary obstructions to discrete order?

The coefficient completion in [proposition 86.2](#) fails before measurability is considered. A systematic completion audit could separate algebraic failures from the category obstruction and could inform the design of exact symbolic versus stream-based omnific interfaces.

Question 88.15 (Reverse mathematics and effective witnesses). What is the exact proof-theoretic strength of the bounded-neighborhood lemma when the order cone is supplied by a Borel code or by an explicit Baire-property witness?

Supplying an open set and a coded meager exceptional set may change the strength of the problem compared with extracting such witnesses from arbitrary Borel codes. The packing stage is elementary once the bound is supplied. An analysis of this signed fragment must not be confused with a solution to Glazer's published ATR_0 question.

[10] Source 10 asks:

Question 88.16 (Reverse-mathematical strength [10]). What weak subsystem suffices for a coded version of [corollary 78.4](#), including Baire-property witnesses and the countability of the relevant coset space?

The main proof uses neither Haar measure nor a principal structure theorem for locally compact groups. Its critical formal issues are category on coded spaces, selecting countably many coset witnesses, and passing from countable quotient and kernel to countable domain. This is a new target; no consequence for Glazer's distinct ATR_0 question is claimed.

[write] Compare source 08's [question 57.7](#). Source 14 ([Part VIII](#)) claims an ATR_0 proof of Glazer's Theorem 2, a different theorem; neither source of Part VI draws on it.

Question 88.17 (Choice and pathological Polish presentations). How much choice is required to construct the Polish additive presentation in [section 84.3](#), and what regularity principles rule out analogous examples over weaker foundations?

The displayed construction uses rational vector-space bases. It provides no effective presentation and no assertion about minimal choice strength. Studying it over ZF with specified choice or regularity axioms would connect the arithmetic obstruction with the foundational side of this research intersection.

Question 88.18 (A small kernel-checked theorem). Can the gap inequality, bounded-neighborhood lemma, and ccc consequence be formalized as a compact independent development, then instantiated at a set-sized omnific subring?

The initial milestone should be the packing lemma with its full hypothesis audit. The next is the reflection argument, not the much larger machinery of arithmetic quantifier elimination. The module boundaries in [section 87](#) make this a concrete implementation project rather than a request to formalize an unspecified entire theory.

[10] Source 10 asks:

Question 88.19 (Certified impossibility interfaces [10]). Can the exact-kernel theorem become a reusable kernel-checked obstruction certificate for proposed arithmetic presentations?

A certificate could expose a positive scale, an open principal hull, a countable kernel, and a bound on the quotient size. This is much smaller than a formalization of the entire surrounding theory and would allow ProvelT to reject incompatible combinations of algebraic and topological interface claims.

[write] Compare source 08's last question ([question 57.8](#)).

88.1 Further questions of source 10 [10]

Question 88.20 (Beyond cardinality: quotient order types [10]). What additional restrictions does the ordered embedding $A/\text{Fin}(A) \hookrightarrow A/H_a$ impose on cofinalities, convex-subgroup chains, and ordered ranks?

The main theorem uses only cardinality. The embedding preserves order, so it may yield sharper obstructions for proposed arithmetic reducts whose quotients have small rank even when both cardinalities are large. A useful first target is a uniform rank calculation for iterated polynomial and lexicographic extensions.

Question 88.21 (The bounded continuous-scalar subring [10]). Which subrings can occur as $\text{Sc}_{\text{cont}}(A)$ in a nondiscrete Baire presentation with regular order?

It must contain the ordinary integers and be bounded in the ambient order. The example here realizes exactly $\mathbb{Z}$. Constructing examples with prescribed larger scalar subrings, or finding new algebraic restrictions on them, would sharpen the link with topological algebra. Borel and continuous scalars must be kept separate outside the separable setting.

Question 88.22 (Countable unit intervals in general ordered rings [10]). How restrictive is the relative criterion $|A| = \max(\aleph_0, |[0, 1]_A|)$ when combined with model-theoretic conditions on A ?

The theorem isolates the finite part as the possible location of uncountability. Investigating ordered fields, integer-part extensions, and definable subrings with controlled bounded intervals may yield applications beyond discretely ordered arithmetic.

Question 88.23 (Definability of the transported counterexample [10]). What is the best possible descriptive complexity of an ordered-ring comparison relation after transporting its addition to $\mathbb{Q}^{\mathbb{N}} \times \mathbb{Z}$?

The cone cannot have the Baire property, so it cannot be Borel or analytic in that Polish presentation. The basis-selected construction does not provide any finite projective definition. The problem should be treated with explicit set-theoretic assumptions rather than assuming that a nonconstructive transport is an effective code.

89 Conclusions of Part VI

The decisive topological resource is an order-bounded neighborhood, not local compactness. Category regularity of an invariant positive cone provides such a neighborhood in every

Hausdorff Baire group. Discrete ring arithmetic then amplifies the unit gap into a maximal family of disjoint open translates:

$$\text{regular cone} \implies W \subseteq (-a, a) \implies \{(2a + 1)r + W : r \in R\} \text{ disjoint.}$$

This resolves the earlier local-compactness-removal question and gives the sharp equality $c(R) = d(R) = |R|$. A separate measure argument obtains the same conclusion from ambient Haar measurability in locally compact groups.

The consequences are both restrictive and diagnostic. They prohibit arbitrary ordered-ring and positive-semiring expansions of a broad class of Polish additive presentations, constrain all ordered additive self-embeddings, and explain why natural coefficient completions of omnific polynomial fragments must fail somewhere. At the same time, the examples show that uncountable continuous addition, nonseparable Baire ring presentations, and pathological Polish ordered algebra can all exist outside the precise hypothesis boundary.

Source 10's conclusion [10]. The decisive improvement is to retain the ordinary-integer kernel instead of trying to eliminate it before forming a quotient. A Baire order provides an open principal convex additive hull. Multiplication by its defining positive scale sends the quotient by finite elements injectively into its open coset space. In discretely ordered arithmetic, finite elements are exactly the ordinary integers. A countable open coset space therefore leaves no room for an uncountable ring.

This resolves the project's remove-local-compactness question for signed additive groups, strengthens it from Polish to Baire-ccc topologies and from Borel order to the Baire property, and gives an exact cardinal obstruction outside the separable setting. It also identifies a purely ordered-group limitation on additive self-embeddings. The remaining boundary is not local compactness or infinite dimension; it is the passage from positive monoids to signed topological groups, along with the precise regularity of order.

90 Where Part VI meets Parts I–V [merge]

Part VI was written against Parts I and IV (source 12) and Parts I, IV and V (source 10). Its principal theorem has hypotheses that several earlier results of this report satisfy, so those results become corollaries. Each deduction below is one line; none uses more than is stated.

Proposition 90.1 (Earlier results that follow from Part VI [merge]).

1. *Theorem 44.1* (source 08): a Hausdorff locally compact second-countable group is Polish and a Borel cone has the Baire property, so [corollary 78.5](#) applies. So does [corollary 55.1](#), by [corollary 86.3](#).
2. [Corollaries 48.1 and 48.2](#) (source 08) are the locally compact cases of [corollary 81.1](#) and [theorem 81.2](#); the latter needs neither commutativity nor the absence of zero divisors.
3. [Theorem 62.1](#) (source 09): an admissible topology is a Polish additive-group topology with a Baire-property cone, so [corollary 78.5](#) gives the conclusion; the Borel-scalar part of admissibility is not needed. [Corollary 62.2](#) is likewise a case of [theorem 82.2](#), which needs no Borelness of multiplication.
4. Part I's [corollary 10.5](#), for the cones $M_{\mathbb{R}}$ and $M_{\mathbb{B}}$, is a case of [theorem 81.2](#): their groups G_D are uncountable Polish groups with Δ_2^0 cones. Part I's [theorem 10.4](#) itself is purely algebraic and is not a case.

Proof. Each item is the stated specialization; the hypotheses compared are those of the cited statements. $\square$

Not every earlier result becomes a corollary. Source 09's [theorem 62.3](#) concludes that every discretely ordered subring of a field with an admissible topology is countable and closed discrete, with no measurability assumption on the subring; source 12's [corollary 82.4](#) concludes only meagerness, for Borel integer parts, but without Borel scalars. The two are incomparable. Source 08's [theorem 53.4](#) concerns multiplications unrelated to the order, to which Part VI does not apply. The positive-cone results of Part III ([theorems 38.1](#) and [39.1](#)) concern topologies on the cone, not on its group of differences; both sources of Part VI state that their theorems do not apply to a positive monoid without a compatible signed group topology ([sections 81.2](#) and [81.4](#)).

Questions. Part VI answers [question 57.1](#) negatively and [question 57.4](#) as stated; notes after both questions say so. It bears on [question 57.5](#) and asks [question 57.2](#) again. It does not bear on Glazer's questions: both sources state that they answer neither.

Part VII

Polish Presburger arithmetic beyond the split construction

[write] This part is source 13 of batch 87, *Polish Presburger Arithmetic Beyond the Split Construction: Nonsplit models, exact continuity, and optimal computation from streams* (dated October 4, 2026; "research report prepared with ChatGPT"), pinned to repository commit eab47e330, one commit before Parts I and II were placed. All unmarked text is source 13's; its labels carry the prefix pma:nsp:, and nine labels were added to unlabelled statements so that notes can cite them. It continues [Part I](#) and [Part III](#): it starts from the split models of Part I, read as the three batch-86 manuscripts in docs/incoming, and answers four of Part I's questions, wholly or in part: [question 14.8](#) (for parameter-free formulas, exactly), [question 14.11](#) (its existence half), [question 14.13](#) and [question 14.16](#) (for joint uniform continuity, negatively); for countable cores it also answers the existence part of Part III's [question 43.3](#), which it did not see. Notes after the relevant results say which.

Printed once here. The countable-core transfer ([theorem 94.1](#)) places an arbitrary countable $\mathbb{Z}$ -group K , with the discrete topology, below $\mathbb{R}$ or $\mathbb{Q}^{\mathbb{N}}$. With $D = \mathbb{R}$ and a nonsplit core it is also source 14's [proposition 113.2](#) and source 15's [theorem 122.2](#); those are printed in [Parts VIII](#) and [IX](#) as flagged instances (note after the theorem).

What is recalled and what is new. Source 13's Section 2 ([section 92](#)) recalls Part I. Its nonsplit core K is a member of the factorial family of the discrete-initial-subgroups-and-omnific-normalization report, its algebraic splitting criterion is part of that report's `isg:cf:main:presburger`, and its omnific criterion is that theorem with `isg:cf:cor:unit`; source 13 says so in each case. New to the collection are the localization presentation of K and its fixed division, the countable-core transfer, automatic continuity of residues and the

topological splitting criterion, the continuity compiler and exact discontinuity sets, the target-valued Baire classification, the rank and exact revision theorems, and the closed-translate metric obstruction. Source 13 claims no priority.

Notation. Source 13's M_ω , D_ω and $\mathbb{Q}^\omega$ are Part I's M_B , D_B and $\mathbb{Q}^\mathbb{N}$; its $G(D)$, $M(D)$ are Part I's G_D , M_D ; its lexicographic sums $\oplus_{\text{lex}}$ are printed $\overline{\times}$, and its $\backslash \text{dotminus}$ as $\dot{-}$. In [section 95](#) the letter D is the divisible kernel $\ker r$, not a coefficient group, and r the canonical residue map; $\mu(f)$ in [section 100](#) is a revision number. [Section 1.3](#) lists these.

Provenance. Source 13's finite diagnostics are shipped as `code/13-beyond-split-verify.py` with the record `data/13-beyond-split-verification_results.json`, and its source record as `data/13-beyond-split-provenance.json` ([sections R](#) and [S](#)).

91 The research intersection and the contribution

The relevant intersection is topological and effective model theory of weak arithmetic. Glazer's paper [\[1\]](#) asks whether an uncountable Polish space can support a model of Presburger arithmetic with continuous addition. It also proves strong obstructions once multiplication and suitable induction are available. The distinction between continuous basic operations and the regularity of definable operations is therefore mathematically consequential.

Provelt provides two direct connections. Its Presburger project formalizes an executable Cooper elimination procedure over the ordinary integers, with an independent Rocq proof of the finite-residue core [\[59, 60, 61\]](#). Its surreal project develops the ordered field of surreal numbers and the omnific integers [\[62\]](#). These connections are established by mathematical sources and actual repository material; the inaccessible LinkedIn profile supplied in the request is not used to infer a current appointment or professional affiliation.

91.1 What was already in the repository

At the inspected revision, the incoming directory contained three related research packages. Two independently develop Polish Presburger models [\[63, 64\]](#); the third concerns Borel presentations and support barriers in Hahn and omnific arithmetic [\[65\]](#). The repository's own triage records a README-level inventory rather than a validation of those proofs [\[66\]](#). For this article the three LaTeX manuscripts were extracted and read directly.

The first two packages already contain the positive cones

$$M_D = (D_{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}), \quad D = \mathbb{R} \text{ or } \mathbb{Q}^\omega,$$

continuous addition and fixed division, low Borel complexity of all definable relations, examples of discontinuous definable operations, and embedding results. The first package even classifies all self-embeddings of its rational-stream model. Repeating these claims would not constitute a further result for the present request.

Instead, the present work addresses their questions about nonsplit presentations, exact discontinuity profiles, optimal finite revisions of truth values, and complete metrics with uniform addition. The connection to omnific arithmetic uses the repository's existing classification: splitting over the standard integers is exactly the obstruction to a unital additive ordered realization in **Oz** [\[67\]](#). The new models provide concrete Polish applications of that obstruction.

Topic	Starting point in the inspected reports	Result developed here
Polish Presburger existence	Split real and Baire constructions	Explicit nonsplit Polish models; automatic continuity of the canonical retraction
Definable regularity	Δ_2^0 relations; selected operations	Exact finite continuity compiler for every definable total function
Computation from streams	A bound by the number of atomic tests	A bound by linear rank; optimal arity bound and an exact finite criterion
Complete metrics	No invariant complete metric	No complete metric with uniformly equicontinuous translations at zero

[write] The three “incoming” packages are batch 86’s manuscripts 04 ([63]), 05 ([64]) and 03 ([65]); one commit after source 13’s pin they were placed in this report, and they are now [Part I](#) (sources 05 and 04) and [Part II](#) (source 03). The “repository’s own triage” [66] is a routing note of the Hilbert’s-tenth programme ([section E](#)). The “existing omnific splitting criterion” [67] is `isg:cf:main:presburger` of the `discrete-initial-subgroups-and-omnific-normalization` report. The four questions taken up are Part I’s [question 14.11](#), [question 14.13](#), [question 14.8](#) and [question 14.16](#); “no invariant complete metric” in the table is [proposition 8.4](#). Source 13 did not see Parts III–VI.

91.2 Scope of the claims

An “open question” from an earlier repository manuscript means a question that that manuscript explicitly left unresolved. It does not assert that the question had independently become a recognized open problem throughout the literature. The present results settle the specified cases with proofs. They do not resolve Glazer’s separate question about the reverse-mathematical strength of his multiplication theorem, and they do not classify arbitrary Polish topological monoids.

Several ingredients are classical: the theory of $\mathbb{Z}$ -groups, Presburger quantifier elimination, elementary facts about Polish spaces, and the ordered-group universality of the surreal numbers. These are cited where used. Their applications and the additional arguments are supplied in full. In particular, a computational diagnostic is never substituted for an infinitary proof.

92 Arithmetic models and their two topologies

We work in ordinary classical mathematics. A Polish space is separable and admits a compatible complete metric. Unless stated otherwise, all groups and parameter tuples are sets. The appearances of **No** and **Oz** will be confined to embeddings with set-sized domain and image.

92.1 Presburger groups and positive cones

Write

$$\text{Pr}_{\mathbb{N}} = \text{Th}(\mathbb{N}; 0, 1, +), \quad \text{Pr}_{\mathbb{Z}} = \text{Th}(\mathbb{Z}; 0, 1, +, -, <).$$

Order is definable in the monoid language by $x \leq y \iff \exists z (x + z = y)$. A $\mathbb{Z}$ -group is a linearly ordered abelian group G with least positive element 1_G such that $G/\mathbb{Z}1_G$ is divisible.

The classical characterization says that these are exactly the models of $\text{Pr}_{\mathbb{Z}}$; their nonnegative cones are exactly the associated models of $\text{Pr}_{\mathbb{N}}$ [4, Section 2]. This includes the full first-order induction scheme with parameters, not only the elementary equations for addition.

For every standard integer $m \geq 1$ and every $g \in G$, there are unique $q \in G$ and $r \in \{0, \dots, m-1\}$ such that

$$g = mq + r1_G. \quad (92.1)$$

To see existence, divisibility of the quotient gives $g = mq_0 + k1_G$; ordinary integer division of k gives the asserted remainder. For uniqueness, if $m(q - q') = (r' - r)1_G$, convexity of the standard integers forces $q - q'$ to be standard and the usual remainder argument applies. If $g \geq 0$, then $q \geq 0$: a negative quotient would be at most -1_G and would make $mq + r1_G < 0$.

After congruence predicates are named, quantifier elimination reduces each formula, uniformly over $\mathbb{Z}$ -groups, to a finite Boolean combination of affine inequalities, equalities, and congruences [58, 4]. Transferring this theorem to a positive cone means relativizing each quantified variable to $x \geq 0$ before elimination. No multiplication of two variables is introduced.

The word “uniformly” matters. A subset of a nonstandard group that is invariant under adding a standard integer m need not be invariant under adding every member of mG . Presburger congruence predicates have the latter, full-coset invariance. The earlier packages correctly isolate that condition when adapting Cooper’s finite-boundary proof [63, 64]. The formal source inspected here evaluates its syntax specifically in $\mathbb{Z}$; a model-uniform formalization remains a separate task.

92.2 The real and stream coefficient groups

Let D be either $\mathbb{R}$ with its usual topology and order, or

$$D_\omega = \mathbb{Q}^{\mathbb{N}}$$

with each rational factor discrete. The latter carries the first-difference lexicographic order: a nonzero sequence is positive exactly when its first nonzero coefficient is positive. Both groups are divisible Polish topological groups. Their strict positive and negative cones are open; for D_ω they are countable unions of finite-prefix cylinders.

Let $G_D = D \overrightarrow{\times} \mathbb{Z}$, with the D coordinate dominant and the integer coordinate discrete. Its positive cone is M_D as above, and $1 = (0, 1)$. The integer quotient in (92.1) is explicit:

$$q_m(u, n) = (u/m, \lfloor n/m \rfloor), \quad r_m(u, n) = n \bmod m. \quad (92.2)$$

These maps, as well as addition, are continuous in the specified topology. We denote the real model by $M_{\mathbb{R}}$ and the stream model by M_ω .

For clarity, the arithmetic order is not the topology being imposed. Every arithmetic element has a successor, while zero is topologically nonisolated. For example,

$$(1/j, 0) \longrightarrow 0 \quad \text{in } M_{\mathbb{R}}, \quad (e_j, 0) \longrightarrow 0 \quad \text{in } M_\omega.$$

Here e_j is the stream with a single coefficient 1 at position j . Each displayed nonzero element is greater in arithmetic order than every standard integer. Convergence of these elements is one reason that arithmetic order and continuous subtraction cannot be silently assumed.

92.3 Polishness, Baire space, and definable relations

The cone M_D is open in the closed Polish subspace $D_{\geq 0} \times \mathbb{Z}$: its complement there is $\{0\} \times \mathbb{Z}_{<0}$, a closed set. Open subspaces of Polish spaces are Polish. In the real model the integer slices are copies of $[0, \infty)$ or $(0, \infty)$, so the space is perfect and locally compact.

The stream model is zero-dimensional and nowhere locally compact. Every neighbourhood contains a closed cylinder with one positive coordinate fixed and infinitely many free rational coordinates. Such a cylinder is homeomorphic to $\mathbb{Q}^{\mathbb{N}}$ and cannot be compact, since a free rational coordinate has an infinite discrete image. The standard characterization of Baire space therefore gives $M_\omega \cong \mathbb{N}^{\mathbb{N}}$ [8]. The earlier packages provide explicit coding homeomorphisms as well [63, 64].

Every parameter-definable relation on M_D is Δ_2^0 in this topology. Indeed, a strict affine inequality is the union of an open leading-sign condition and a closed leading-zero condition with a discrete integer test. Its complement has the same form. Equalities are closed and congruences are clopen, since $mG_D = D \times m\mathbb{Z}$. Quantifier elimination then gives the claim. This background bound concerns sets and graphs; it does not by itself prove a target-valued Baire-class bound for functions. That distinction will become explicit below.

[write] This section recalls Part I, as source 13 says: M_D and its Polishness are [theorems 3.1](#) and [5.2](#), the homeomorphism $M_\omega \cong \mathbb{N}^{\mathbb{N}}$ is [theorem 6.1](#), and the Δ_2^0 bound is [theorems 7.1](#) and [7.2](#). Source 13's $D_\omega = \mathbb{Q}^{\mathbb{N}}$ and M_ω are Part I's D_B and M_B (the same group, order and topology); its $\mathbb{Q}^\omega$ is $\mathbb{Q}^{\mathbb{N}}$.

93 A nonsplit countable arithmetic core

The split constructions in the incoming reports do not exhaust the Polish possibilities. Nonsplit countable $\mathbb{Z}$ -groups are classical, and the repository already contains an explicit profinite-parameter family [67, section “An explicit factorial family of nonsplit models”]. The following localization presentation is a particularly elementary member of that family. Its elements and its fixed-division algorithms use ordinary rational arithmetic only. The new topological step below is to transfer arbitrary countable arithmetic cores to perfect Polish universes while preserving their nonsplitting obstruction.

Put

$$A = \mathbb{Z}_{(2)} = \{a/b : a \in \mathbb{Z}, b \in \mathbb{N}_{>0} \text{ odd}\}, \quad B = \mathbb{Z}[1/2] = \{c/2^\ell : c \in \mathbb{Z}, \ell \in \mathbb{N}\},$$

and let $K = A \times B$ with coordinatewise addition. Its distinguished unit is $e = (1, 1)$. Define

$$\pi : K \longrightarrow \mathbb{Q}, \quad \pi(u, v) = v - u.$$

Order K by declaring

$$(u, v) >_K 0 \iff v - u > 0 \text{ or } (u = v \text{ and } u > 0). \quad (93.1)$$

The second case involves an ordinary integer, since $A \cap B = \mathbb{Z}$.

Proposition 93.1 (Explicit nonsplit $\mathbb{Z}$ -group; a member of the family of `isg:cf:prop:family`). *The ordered group $(K, +, <_K, e)$ is a model of the ordered-group version of Presburger arithmetic. The sequence*

$$0 \longrightarrow \mathbb{Z} \xrightarrow{n \mapsto ne} K \xrightarrow{\pi} \mathbb{Q} \longrightarrow 0$$

is exact and does not split. In fact,

$$\text{Hom}(K, \mathbb{Z}) = 0, \quad \text{Div}(K) = 0,$$

where $\text{Div}(K)$ is the largest divisible subgroup.

Proof. The equality $A \cap B = \mathbb{Z}$ follows by writing a rational in lowest terms: its denominator must be both odd and a power of two. Every rational belongs to $A + B$. Indeed, write its denominator as $2^\ell b$ with b odd, and choose integers s, t with $s2^\ell + tb = 1$; then

$$\frac{a}{2^\ell b} = \frac{as}{b} + \frac{at}{2^\ell}.$$

Replacing the first summand by its negative shows that π is onto. Its kernel is exactly $\mathbb{Z}e$.

Formula (93.1) is the lexicographic extension of the usual orders on $\mathbb{Z}e$ and $K/\mathbb{Z}e = \mathbb{Q}$. It therefore defines a translation-invariant linear order. The subgroup $\mathbb{Z}e$ is convex; e is the least positive element; and the quotient is divisible. The standard characterization of $\mathbb{Z}$ -groups now applies: a discretely ordered abelian group is a Presburger group precisely when its quotient by the subgroup generated by its least positive element is divisible [4, §2].

For completeness, the two assertions about homomorphisms and divisibility are elementary. The group A is divisible by 3^j for every j . Consequently, the image of any $a \in A$ under a homomorphism $A \rightarrow \mathbb{Z}$ would be divisible by every 3^j , hence would be zero. The same argument with powers of two applies to B . Thus every homomorphism $K = A \times B \rightarrow \mathbb{Z}$ is zero. A splitting over $\mathbb{Z}e$ would give a retraction taking e to 1, which is impossible.

There is no nonzero infinitely 2-divisible element of A , and no nonzero infinitely 3-divisible element of B . For example, if $a/b \neq 0$ is in lowest terms with b odd, $(a/b)/2^j$ fails to belong to A for all sufficiently large j . Hence a divisible subgroup of either factor is zero. Projection of a divisible subgroup of K to either factor is divisible, so the divisible subgroup itself is zero. $\square$

93.1 Exact relation to the repository's profinite family

Let $\varepsilon_2 \in \widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$ have 2-adic coordinate 1 and every odd-prime coordinate 0. The canonical residue homomorphism of K is

$$r_K(u, v)_2 = u, \quad r_K(u, v)_p = v \quad (p \text{ odd}),$$

where each rational is interpreted in the indicated p -adic integer ring. In particular, $r_K(-1, 0) = -\varepsilon_2$.

This identifies K explicitly with the repository family member G_{ε_2} . To verify the identification without any extension classification, let S_n be the ordinary residue of ε_2 modulo $n!$, and put

$$H = \bigcup_{n \geq 1} (\mathbb{Z}(1/n!, S_n/n!) + \mathbb{Z}(0, 1)) \subseteq \mathbb{Q} \overrightarrow{\times} \mathbb{Q}.$$

The map $T(u, v) = (v - u, v)$ is a unital ordered group embedding. Writing $n! = 2^h m$ with m odd, the congruences $S_n \equiv 1 \pmod{2^h}$ and $S_n \equiv 0 \pmod{m}$ show that every displayed generator of H belongs to $T[K]$. Conversely, write $T(u, v) = (k/n!, l/n!)$ for sufficiently large n . The condition $u \in A$ gives $2^h \mid l - k$, and $v \in B$ gives $m \mid l$. Hence $n! \mid l - kS_n$, proving

$$T(u, v) = k(1/n!, S_n/n!) + \frac{l - kS_n}{n!}(0, 1) \in H.$$

Thus $T[K] = H = G_{\varepsilon_2}$. The localization presentation is a transparent specialization of the existing family, rather than an independent claim of algebraic priority.

[write] As source 13 says, K is a member of the factorial family of the discrete-initial-subgroups-and-omnific-normalization report: $K \cong G_{\varepsilon_2}$ in the notation of `isg:cf:prop:family`, and its nonsplitting is the case $\zeta = \varepsilon_2 \notin \mathbb{Z}$ of `isg:cf:thm:familytest`. The localization coordinates and the fixed-division algorithm below are source 13's.

93.2 An exact fixed-division algorithm

Let $n \geq 1$, and write $n = 2^k m$ with m odd. For an input $x = (u, v) \in K$, choose presentations

$$u = \frac{a}{b} \quad (b \text{ odd}), \quad v = \frac{c}{2^\ell}.$$

There is a unique integer $r \in \{0, \dots, n-1\}$ satisfying

$$\begin{aligned} r &\equiv ab^{-1} \pmod{2^k}, \\ r &\equiv c(2^\ell)^{-1} \pmod{m}. \end{aligned} \tag{93.2}$$

Congruences modulo 1 impose no condition. Both indicated inverses exist, and the Chinese remainder theorem gives r . Define

$$q_n^K(u, v) = \left(\frac{u-r}{n}, \frac{v-r}{n} \right), \quad r_n^K(u, v) = r. \tag{93.3}$$

Then $q_n^K(u, v) \in K$ and

$$(u, v) = nq_n^K(u, v) + re.$$

The first congruence in (93.2) removes exactly the forbidden power-of-two factor from the denominator of $(u-r)/n$; the second removes exactly the forbidden odd factor from the denominator of $(v-r)/n$. Conversely, membership of both quotient coordinates in A and B forces these two congruences. This also proves uniqueness of q and r directly.

If $x \geq_K 0$, then $q_n^K(x) \geq_K 0$. If $\pi(x) > 0$, use $\pi(q_n^K(x)) = \pi(x)/n > 0$. If $\pi(x) = 0$, then $x = te$ with an integer $t \geq 0$, and the formula reduces to ordinary Euclidean division of t . Thus this is also the fixed-division algorithm for the nonnegative cone.

94 Transferring every countable core to a perfect Polish universe

The preceding core can be replaced by any countable Presburger group. Let $(K, +, <, e)$ be such a group, equipped with its discrete topology, and let D be either

$D = \mathbb{R}$ with its usual topology, or $D = \mathbb{Q}^{\mathbb{N}}$ with its product of discrete topologies.

In the second case, order D lexicographically at the first differing coordinate. Let

$$G = D \overrightarrow{\times} K, \quad 1_G = (0, e), \quad P = G_{\geq 0},$$

and give G the product topology. Its nonnegative cone is

$$P = (D_{>0} \times K) \cup (\{0\} \times K_{\geq 0}). \tag{94.1}$$

Theorem 94.1 (Countable-core transfer). *The ordered group G is a $\mathbb{Z}$ -group, and P is an uncountable Presburger model with continuous addition. The space P is perfect and Polish. It is locally compact when $D = \mathbb{R}$, and is homeomorphic to Baire space when $D = \mathbb{Q}^{\mathbb{N}}$.*

All fixed standard quotient and remainder functions on P are continuous. Every Presburger-definable relation on any finite power of P is Borel; indeed, it belongs to Δ_2^0 in these two presentations. The map $k \mapsto (0, k)$ embeds K as a closed discrete elementary substructure of G .

There is a unit-preserving homomorphism $G \rightarrow \mathbb{Z}$ if and only if there is one $K \rightarrow \mathbb{Z}$. In particular, taking the core of Proposition 93.1 gives nonsplit models in both topological types, and their largest divisible subgroup is exactly $D \times \{0\}$.

Proof. The element $(0, e)$ is least positive. The quotient by its integer multiples is $D \oplus (K/\mathbb{Z}e)$, which is divisible. This proves the arithmetic claim by the characterization of $\mathbb{Z}$ -groups.

Both choices of D are Polish topological groups. For both, $D_{>0}$ is open and $D_{\geq 0}$ is closed. For rational streams, a nonzero sequence's sign is fixed by its first nonzero coordinate, which proves the open-cone assertion. The slice of P at a fixed $k \in K$ is $D_{\geq 0}$ when $k \geq 0$, and $D_{>0}$ when $k < 0$. It is therefore Polish, and P is a countable topological disjoint union of Polish spaces, hence Polish. Equivalently, P is a G_δ subset of $D \times K$. It need not be closed there: if $k < 0$, positive $d_j \rightarrow 0$ give $(d_j, k) \in P$ converging in G to $(0, k) \notin P$. No completeness claim is made for the restriction of the ambient product metric.

Addition is the restriction of the continuous group addition on G . Every point of either cone $D_{\geq 0}$ or $D_{>0}$ is nonisolated. This is immediate for $\mathbb{R}$; for rational streams, one can change a sufficiently late coordinate while preserving every prescribed initial coordinate and, when necessary, keep the first nonzero coordinate positive. Hence P is perfect and uncountable.

For $D = \mathbb{R}$, the slices are $[0, \infty)$ and $(0, \infty)$, so P is locally compact. For rational streams, the slices, and hence P , are zero-dimensional. No nonempty open subset of P is compact: inside every nonempty basic neighborhood one can find a closed cylinder in which a later rational coordinate is unrestricted; its projection onto the infinite discrete space $\mathbb{Q}$ is onto, contradicting compactness. For clarity at the all-zero stream, first choose a sufficiently late coordinate to be 1, then leave a still later coordinate unrestricted. Thus P is nowhere locally compact. The classical characterization of Baire space as the nonempty zero-dimensional Polish space with no nonempty compact open subsets yields $P \cong \mathbb{N}^{\mathbb{N}}$.

Write $k = nq_n^K(k) + r_n^K(k)e$, with $0 \leq r_n^K(k) < n$. On G set

$$q_n^G(d, k) = (d/n, q_n^K(k)), \quad r_n^G(d, k) = r_n^K(k).$$

These are continuous, since division by n is continuous on D and every map from the discrete factor K is continuous. For a nonnegative input, the quotient remains nonnegative, by the sign of d/n when $d > 0$ and by the ordinary Presburger division axiom in K when $d = 0$. Restriction gives the desired maps on P .

The strict order on G is

$$(d, k) < (d', k') \iff d < d' \text{ or } (d = d' \text{ and } k < k').$$

The first condition is open; the equality condition is closed; and the condition on the discrete coordinates is clopen. Hence the order is Δ_2^0 . Linear equality is closed, congruence modulo each standard n is clopen, and integer-affine maps are continuous. Presburger quantifier elimination expresses every definable relation as a finite Boolean combination of these atomic relations [4, §2]. Finite Boolean combinations of Δ_2^0 sets remain Δ_2^0 . Restricting to the cone and its finite powers preserves the conclusion. This argument concerns definable sets, and by itself does not assert a target-valued Baire class for every definable function.

The copy $\{0\} \times K$ is closed and discrete. It is elementary because it is itself a unit-containing $\mathbb{Z}$ -subgroup, or directly by the same quantifier elimination. Finally, every homomorphism $D \rightarrow \mathbb{Z}$ is zero because D is divisible. Thus any unit-preserving homomorphism $G \rightarrow \mathbb{Z}$ restricts to one on K , and any such map on K extends by zero on D . For the explicit reduced core, projecting a divisible subgroup of G to K gives zero, while $D \times \{0\}$ is divisible. This proves the last assertion. $\square$

The transfer is algebraically nonsplit over the distinguished standard $\mathbb{Z}$, although it is deliberately a product with a discrete *countable core*. These two statements are compatible. Confusing them would erase exactly the distinction resolved by the construction.

[write] **One construction in three sources.** With $D = \mathbb{R}$ and a countable nonsplit core, [theorem 94.1](#) is the construction of source 14's [proposition 113.2 \(Part VIII\)](#), whose core K_* is the member G_{ζ_*} of the same factorial family, and of source 15's [theorem 122.2 \(Part IX\)](#), whose core is the difference group of an arbitrary countable Presburger model. The construction is printed here, once, in its general form; Parts VIII and IX print those two statements with their own proofs, flagged as instances, because they add the omnific obstruction and the boundary identification respectively. With [theorem 95.4](#) below, the theorem answers the existence half of Part I's [question 14.11](#) and, for countable cores, Part III's [question 43.3](#): every ordered extension $0 \rightarrow \mathbb{Z} \rightarrow G \rightarrow D \oplus (K/\mathbb{Z}) \rightarrow 0$ obtained from a countable nonsplit $\mathbb{Z}$ -group K has a Polish positive cone with continuous addition. Neither question asks only for existence, and neither classification is attempted.

94.1 Continuum many topological examples

Corollary 94.2 (Many nonsplit Polish arithmetic structures). *There are $2^{\aleph_0}$ pairwise nonisomorphic Presburger structures on locally compact perfect Polish spaces having continuous addition and every standard fixed-division operation, with no retraction onto the standard integers. There are likewise $2^{\aleph_0}$ such structures whose underlying spaces are homeomorphic to Baire space.*

Proof. Use the countable cores G_{ζ} from the repository's factorial family, with $\zeta \in \widehat{\mathbb{Z}} \setminus \mathbb{Z}$, and apply [Theorem 94.1](#). The family has $2^{\aleph_0}$ extension classes modulo endpoint-preserving isomorphism: the parameters ζ, η define the same such class exactly when $\zeta - \eta \in \mathbb{Z}$. Every nonsplit core is reduced, meaning that its largest divisible subgroup is zero [67, section "An explicit factorial family of nonsplit models"].

For either fixed choice of D , the maximal divisible subgroup of $D \overrightarrow{\times} G_{\zeta}$ is $D \times \{0\}$. Thus a unital group isomorphism between two transferred groups would induce a unital *abstract group* isomorphism between their reduced quotients G_{ζ} and G_{η} . Such an isomorphism induces multiplication by some nonzero rational on the quotient $\mathbb{Q}$ of the cores. There are only countably many possibilities. For each such quotient map, at most one endpoint-marked extension class is possible: comparing two isomorphisms with the same quotient map gives an endpoint-preserving isomorphism between their targets. Consequently, each abstract unital group isomorphism type accounts for at most countably many marked extension classes, leaving $2^{\aleph_0}$ different transferred types. An isomorphism of positive cones would extend to their Grothendieck groups, so the same conclusion holds for the Presburger monoids.

Only an abstract quotient by the maximal divisible subgroup is used in this argument. That subgroup need not be convex, and no ordered quotient by it is asserted. $\square$

[write] By source 15's [theorem 121.1 \(Part IX\)](#), the locally compact models of this corollary all

live on the one space $X_{\mathbb{C}}$; source 15's [theorem 123.1](#) gives $2^{\aleph_0}$ split models on that space with standard boundary. The two continuum families are disjoint: one is nonsplit, the other split.

95 Canonical residues and automatic topological splitting

The algebraic criterion recalled below is already present in Provelt [67]; the residue kernel is standard [4]. The topological consequences are developed separately.

In this section G is a $\mathbb{Z}$ -group: a discretely ordered abelian group with named least positive element 1_G such that $G/\mathbb{Z}1_G$ is divisible. Standard division with remainder gives, for every integer $m \geq 1$, a homomorphism

$$r_m : G \longrightarrow \mathbb{Z}/m\mathbb{Z}, \quad r_m(1_G) = 1,$$

whose kernel is mG . The maps are compatible, giving the canonical residue homomorphism

$$r : G \longrightarrow \widehat{\mathbb{Z}} = \varprojlim_m \mathbb{Z}/m\mathbb{Z}.$$

Let

$$D = \ker r = \bigcap_{m \geq 1} mG.$$

Lemma 95.1 (The divisible kernel). *The group D is the largest divisible subgroup of G , and $D \cap \mathbb{Z}1_G = \{0\}$.*

Proof. The group G is torsion-free. If $x \in D$ and $m \geq 1$, there is a unique $y \in G$ with $my = x$. For every $n \geq 1$, choose $z \in G$ with $mnz = x$. Cancellation gives $y = nz$, so $y \in \bigcap_n nG = D$. Hence D is divisible. Conversely, every divisible subgroup lies in each mG , and therefore in D . If $k1_G \in D$ for $k \in \mathbb{Z}$, then m divides k for every standard positive m , which forces $k = 0$. $\square$

Theorem 95.2 (Algebraic splitting criterion; part of `isg:cf:main:presburger`). *The following conditions are equivalent:*

1. *There is a group homomorphism $\rho : G \longrightarrow \mathbb{Z}$ with $\rho(1_G) = 1$.*
2. *$G = D \oplus \mathbb{Z}1_G$ as abelian groups.*
3. *$r(G)$ is precisely the standard diagonal copy of $\mathbb{Z}$ in $\widehat{\mathbb{Z}}$.*

When these conditions hold, ρ is unique, its kernel is D , and the group decomposition is lexicographic for the induced order on D .

Proof. Assume (1) and put $K = \ker \rho$. The map $k \mapsto k + \mathbb{Z}1_G$ is an isomorphism $K \cong G/\mathbb{Z}1_G$: subtracting $\rho(g)1_G$ from g proves surjectivity, and the intersection is trivial because $\rho(1_G) = 1$. The quotient is divisible, so K is divisible and $K \subseteq D$. Conversely, every homomorphism from a divisible group to $\mathbb{Z}$ is zero, so $D \subseteq K$. Thus $K = D$, proving (2).

Assume (2). Every $d \in D$ has zero residue modulo every m , so $r(d + k1_G) = r(k1_G)$. Since all standard integer elements occur, this proves (3).

Assume (3). For each $g \in G$, there is a unique standard $k \in \mathbb{Z}$ with $r(g) = r(k1_G)$. Define $\rho(g) = k$. Injectivity of the diagonal map $\mathbb{Z} \rightarrow \widehat{\mathbb{Z}}$ and the homomorphism property of r show that ρ is a homomorphism with $\rho(1_G) = 1$. This proves (1).

Every such retraction vanishes on D and fixes the integer summand, so it is unique. Finally, $\mathbb{Z}1_G$ is convex in G . Any nonzero element of D lies outside this convex subgroup, and its sign is unchanged by adding a standard integer. Thus the decomposition has the order of $D \times \mathbb{Z}$. $\square$

[write] [Theorem 95.2](#) is part of `isg:cf:main:presburger`, as the opening sentence of this section says; [lemma 95.3](#) and [theorem 95.4](#) are new.

95.1 Automatic continuity in Polish topological groups

The next theorem assumes a *group* topology: addition and inversion on the entire signed group G are continuous. It is not a theorem about an arbitrary topology on the positive monoid. No Borelness or compatibility assumption for the arithmetic order is needed in this theorem.

Lemma 95.3 (Finite residues are automatically continuous). *If a $\mathbb{Z}$ -group G carries a Polish group topology, every mG is open and closed, every r_m is continuous, r is continuous, and D is a closed Polish subgroup.*

Proof. The map $g \mapsto mg$ is continuous, so its image mG is analytic and has the Baire property. There are exactly m cosets, represented by $0, 1_G, \dots, (m-1)1_G$. The Baire category theorem implies that mG is nonmeagre.

For completeness, a subgroup A of a Polish group which has the Baire property and is nonmeagre is open. There is a nonempty open set V on which A is comeagre. For all sufficiently small u , the open overlap $V \cap (u + V)$ is nonempty. On that overlap, A and $u + A$ are both comeagre, hence intersect. Thus $u \in A - A = A$, proving that A contains a neighbourhood of 0. Apply this fact to $A = mG$.

An open subgroup is also closed, since its other cosets are open. The fibres of r_m are these open cosets, proving continuity. The inverse-limit map r is continuous because all its coordinates are continuous. Its kernel D is closed; a closed subgroup of a Polish group is Polish. $\square$

Theorem 95.4 (Topological splitting criterion). *Let G be a $\mathbb{Z}$ -group carrying a Polish group topology. The conditions of [Theorem 95.2](#) are equivalent to the existence of a continuous unit-preserving retraction $\rho : G \rightarrow \mathbb{Z}$, where $\mathbb{Z}$ is discrete. When they hold, the canonical map*

$$D \times \mathbb{Z} \longrightarrow G, \quad (d, k) \longmapsto d + k1_G$$

is a topological group isomorphism and an isomorphism of the corresponding lexicographically ordered groups.

Proof. Only automatic continuity and the topological assertion require proof. Under algebraic splitting, D is closed by [Lemma 95.3](#), and its cosets $D + k1_G$, $k \in \mathbb{Z}$, cover G . If D had empty interior, every such closed coset would be nowhere dense, contradicting the Baire category theorem. Thus D is open. Its cosets are the fibres of ρ , so ρ is continuous into discrete $\mathbb{Z}$. The displayed map is continuous, and its inverse is

$$g \longmapsto (g - \rho(g)1_G, \rho(g)),$$

which is continuous. The order statement is already proved in [Theorem 95.2](#). $\square$

In particular, every positive cone with the topology inherited from a Polish topological $\mathbb{Z}$ -group has continuous fixed residues. Since each residue class occurs, such a cone is disconnected (already reduction modulo 2 gives a nontrivial clopen partition). This does not settle the connected-universe question for arbitrary Polish monoids with continuous addition: one would first need an appropriate Polish group completion.

[write] (Batch 87.) This settles Part I's [question 14.3](#) and Part III's [question 43.4](#) negatively for cones whose topology is inherited from a Polish topological $\mathbb{Z}$ -group: such a cone is disconnected. Source 15's [theorem 121.1](#) gives the locally compact case again, with the exact carrier. For a topology on the cone alone both questions stay open, as source 13 says.

95.2 Compatibility with the nonsplit example

The nonsplit construction with $L = \mathbb{Z}_{(2)} \times \mathbb{Z}[1/2]$ and diagonal unit $e = (1, 1)$ is consistent with all of these conclusions. In $G = \mathbb{Q}^\omega \overrightarrow{\times} L$, with L discrete, the maximal divisible subgroup is $D = \mathbb{Q}^\omega \times \{0\}$. It is clopen. Its quotient is the countable discrete group L , rather than $\mathbb{Z}$. The finite residue maps are continuous, but their coherent values cannot be represented by a single standard integer for every element. The failure of an integer-coordinate retraction is algebraic, not a failure of continuity of individual residues.

96 The existing omnific criterion and its new Polish applications

The equivalence in this section is recalled from [67, Presburger initiality criterion and unit-preserving omnific embeddings], which proves a stronger result about initial images. It is not claimed as a new theorem here. The normal-form proof below is included to make the application independent of that stronger initiality machinery.

The following equivalence is included as background and an independently checkable explanation of the obstruction, not as a new classification. The repository manuscript [67], theorem labelled `isg:cf:main:presburger`, already proves a stronger criterion with *initial* realizations in **No** and **Oz**. Here we need only ordered additive embeddability, and the sufficiency proof uses a common monomial shift instead of initial-realization theory.

An omnific integer has a Conway normal form whose exponents are nonnegative and whose coefficient at exponent zero is an ordinary integer [5, Section 6]. Denote its constant coefficient by

$$c_0 : \mathbf{Oz} \longrightarrow \mathbb{Z}.$$

Normal-form addition is coefficientwise, so c_0 is additive and $c_0(1) = 1$. In fact it is a unital ring homomorphism: a sum of two nonnegative exponents is zero only when both are zero. Only its additive property is needed below. No continuity or measurability of c_0 is assumed.

Lemma 96.1 (Moving a set-sized group into purely infinite surreals). *Every set-sized divisible ordered abelian group D admits an ordered additive embedding into the group of purely infinite omnific integers.*

Proof. Use the standard embedding theorem that every divisible ordered abelian group has an isomorphic copy as an ordered additive subgroup of **No**; the stronger initial-subgroup result is recalled in [5, Introduction]. Let $i : D \rightarrow \mathbf{No}$ be such an embedding. Since D is a set and every surreal has a set-sized Conway support, the union

$$S = \bigcup_{d \in D} \text{supp}(i(d))$$

is a set of surreal exponents. Choose a surreal β satisfying

$$\beta > 0, \quad \beta > -s \quad (s \in S).$$

For example, the surreal cut $\{\{0\} \cup \{-s : s \in S\} \mid \emptyset\}$ has this property. Multiplication by the positive monomial ω^β shifts every exponent s to $\beta + s > 0$. Thus

$$j(d) = \omega^\beta i(d)$$

has only strictly positive exponents, is an omnific integer, and has zero constant coefficient. Multiplication by a fixed positive surreal preserves addition and strict order, so j is the desired embedding. $\square$

Theorem 96.2 (Omnific criterion, recalled; `isg:cf:main:presburger`, `isg:cf:cor:unit`). *Let G be a set-sized $\mathbb{Z}$ -group with least positive element 1. The following conditions are equivalent.*

1. *There is a unital additive homomorphism $G \rightarrow \mathbf{Oz}$, with no assumption of injectivity or order preservation.*
2. *There is a unital ordered additive embedding $G \hookrightarrow \mathbf{Oz}$.*
3. *There is an additive retraction $\rho : G \rightarrow \mathbb{Z}$ with $\rho(1) = 1$.*
4. *G is isomorphic, as an ordered group with distinguished unit, to $D \overrightarrow{\times} \mathbb{Z}$ for some divisible ordered abelian group D .*
5. *The canonical residue homomorphism $r : G \rightarrow \widehat{\mathbb{Z}}$ has image exactly the standard diagonal copy of $\mathbb{Z}$.*

When these conditions hold, the retraction is unique and $\ker \rho$ is the largest divisible subgroup of G . The same conditions characterize the existence of a unital additive homomorphism from $G_{\geq 0}$ to $\mathbf{Oz}$, or of a unital ordered additive embedding from $G_{\geq 0}$ to $\mathbf{Oz}_{\geq 0}$.

Proof. Condition (ii) implies (i). If f witnesses (i), then $\rho = c_0 \circ f$ witnesses (iii).

Suppose (iii). Algebraically,

$$G = \ker \rho \oplus \mathbb{Z}1, \quad g = (g - \rho(g)1) + \rho(g)1.$$

The kernel $D = \ker \rho$ is divisible. One can see this from $D \cong G/\mathbb{Z}1$, or directly as follows. If $d = nq + s1$ with $0 \leq s < n$, then $0 = n\rho(q) + s$, so $s = 0$ and $\rho(q) = 0$. Thus $d/n \in D$.

Every nonzero $d \in D$ is larger in absolute value than every standard integer. Otherwise, discreteness and a finite induction would place d in $\mathbb{Z}1$, contradicting $D \cap \mathbb{Z}1 = 0$. Consequently the sign of $d + n1$ for $d \neq 0$ is the sign of d . This proves that the displayed algebraic decomposition has the lexicographic order required in (iv).

Suppose (iv). Choose $j : D \rightarrow \mathbf{Oz}$ as in Lemma 96.1. Then

$$f(d, n) = j(d) + n$$

is additive and preserves the unit. Its nonzero purely infinite part dominates every integer, so it is an ordered embedding. This proves (ii).

For the residue formulation, recall that for each standard $n > 0$ there is a unique homomorphism $r_n : G \rightarrow \mathbb{Z}/n\mathbb{Z}$ taking 1 to 1; the compatible system forms $r : G \rightarrow \widehat{\mathbb{Z}}$ [4, §2]. If (iii) holds, uniqueness gives $r_n(g) = \rho(g) \bmod n$, so $r(G)$ is exactly diagonal $\mathbb{Z}$. Conversely,

under (v), define $\rho(g)$ to be the unique integer whose diagonal residue is $r(g)$. This is the required additive retraction.

Two retractions differ by a homomorphism vanishing on $\mathbb{Z}1$. Their difference therefore factors through the divisible group $G/\mathbb{Z}1$, and every homomorphism from a divisible group to $\mathbb{Z}$ is zero. This proves uniqueness. A divisible subgroup of G is killed by ρ ; since the kernel itself is divisible, it is the largest one.

Finally, G is the Grothendieck group of its nonnegative cone. An additive map from the cone to the additive group of $\mathbf{Oz}$ extends uniquely by $\tilde{f}(a - b) = f(a) - f(b)$; this is well defined because $a - b = a' - b'$ implies $a + b' = a' + b$. Thus a unital cone homomorphism implies (i). Conversely, the ordered embedding constructed above restricts to an ordered embedding of the cones. $\square$

[write] The equivalence of (i), (ii) and (iii) is `isg:cf:cor:unit`, and $(iii) \Leftrightarrow (iv) \Leftrightarrow (v)$ is contained in `isg:cf:main:presburger`; source 13 recalls them and supplies a direct normal-form proof.

96.1 A two-element certificate excluding even a homomorphism

For the explicit core, the failure is visible without any abstract decomposition. In its nonnegative cone put

$$x = (-1, 0), \quad y = (0, 1) = x + e.$$

Both are positive because $\pi(x) = \pi(y) = 1$. The element x is divisible in the positive cone by 3^j for every j , with quotient $(-3^{-j}, 0)$; the element y is divisible there by 2^j for every j , with quotient $(0, 2^{-j})$.

Suppose a unital additive map $f : K_{\geq 0} \rightarrow \mathbf{Oz}$ existed. The integer $c_0(f(x))$ would be divisible by every 3^j , so it would be zero. Similarly, $c_0(f(y)) = 0$. But $y = x + e$ would give

$$0 = c_0(f(y)) = c_0(f(x)) + c_0(f(e)) = 0 + 1,$$

a contradiction. These same two points, placed in the fiber $D = 0$, prove the obstruction for both perfect Polish models of Theorem 94.1. In particular, neither a different topology nor a loss of injectivity can repair the unital additive embedding problem.

96.2 The precise set/class boundary

The theorem does not quantify over a set of all surreal numbers. It can be read in a conservative class framework such as NBG with the usual surreal construction, while G , D , $i[D]$, and S in the proof are sets. In a set-theoretic coding, surreal numbers are set-sized sign sequences and the embedding has a set-sized graph. Replacement bounds the birthdays of its image and the supports occurring in the proof by a suitable ordinal. The cut defining β has set-sized left and right sides, so it is a legitimate surreal cut. After construction, all elements of the final embedded group also belong to a sufficiently large set-sized birthday stage. No assertion that the whole proper class carries a Polish topology is involved.

The set-size hypothesis is used decisively when taking the union of all supports and choosing a common monomial shift. This proof does not provide the same conclusion for arbitrary proper-class groups. It also proves an additive and order-theoretic criterion only. The embedding produced by multiplying a group copy by one monomial is not asserted to preserve a multiplication operation or to have an initial image in the simplicity tree.

96.3 Relation to the incoming research questions

The countable-core transfer answers the existence part of the incoming “beyond split topological models” question. It supplies explicit uncountable Polish examples with continuous fixed divisions for which no integer-coordinate retraction exists at all. The omnific criterion identifies exactly why the incoming additive realizations inside **Oz** cannot cover these examples: preserving the standard unit forces the entire coherent residue system to come from one ordinary integer. Thus the obstruction is algebraic and persists before imposing any regularity on a proposed embedding.

The countable nonsplit algebra and the general omnific criterion have explicit prior counterparts in the repository. The contribution proposed here is their transfer to perfect Polish models, together with elementary CRT coordinates and a short divisibility certificate. The searches recorded for this project do not establish historical priority in the entire literature.

[write] The “incoming ‘beyond split topological models’ question” is Part I’s [question 14.11](#); the existence half is answered here, the characterization asked for there is not.

97 Finite continuous branches for every definable function

Finite affine descriptions of Presburger functions are classical; see [28]. We prove the precise candidate form needed for the topological arguments, directly from quantifier elimination.

Let D be a divisible ordered abelian group, and put $G(D) = D \overrightarrow{\times} \mathbb{Z}$, with positive cone $M(D)$. Write q_m for Euclidean quotient by a positive standard integer m :

$$q_m(d, n) = \left(\frac{d}{m}, \left\lfloor \frac{n}{m} \right\rfloor \right).$$

For the two topological models $D = \mathbb{R}$ and $D = \mathbb{Q}^\omega$, every $q_m : G(D) \rightarrow G(D)$ is continuous. Parameters are held fixed unless they are explicitly made variables.

Theorem 97.1 (Finite continuous branches). *Every Presburger-definable total function $f : M(D)^k \rightarrow M(D)^r$, with a finite parameter tuple a , admits a finite definable partition $C_1, \dots, C_s$ of its domain and candidate maps $F_j : G(D)^k \rightarrow G(D)^r$ such that $f = F_j$ on C_j . Each coordinate of each candidate has the form*

$$q_m \left(c + \sum_i b_i x_i + \sum_t c_t a_t \right), \quad m \in \mathbb{Z}_{>0}, \quad b_i, c_t \in \mathbb{Z}, \quad c \in \mathbb{Z}1. \quad (\text{B1})$$

The candidates and partition are obtainable by a finite syntactic procedure from a formula for the graph. The same assertion holds with arbitrary ambient inputs $G(D)^k$ provided the output is still in $M(D)^r$. In the two topological models every candidate is continuous into the ambient group. A candidate need not be $M(D)$ -valued outside its own cell.

Proof. First take a single output variable y . Relativize the original formula to the ordered group and eliminate quantifiers. Conjoin $y \geq 0$ and put the resulting formula into disjunctive normal form. Replace an equality by two inequalities, a disequality by a disjunction of strict inequalities, and strict inequalities by weak inequalities with a unit shift. Each disjunct is a conjunction of linear inequalities, congruence conditions, and negated congruence conditions.

For a particular disjunct, take L to be a common positive multiple of the absolute values of all its nonzero coefficients of y , with $L = 1$ if needed. Substitute $u = Ly$ and add $L \mid u$.

Multiplying an inequality by a positive standard integer makes its coefficient of u equal to 1 or -1 . A congruence with nonzero coefficient b is transformed using

$$d \mid (by + t) \iff d \frac{L}{|b|} \mid \left(\operatorname{sgn}(b)u + \frac{L}{|b|}t \right). \quad (\text{B2})$$

This equivalence follows by torsion-freeness from $d \mid w \iff cd \mid cw$ for positive standard c . Terms independent of u are left unchanged.

Let m be a common multiple of L and all transformed moduli. Split into the finitely many standard residues $u \equiv r \pmod{m}$. Within one such residue, all congruence literals reduce to conditions on the inputs and parameters. The remaining constraints on u have the form

$$u \geq b_1, \dots, u \geq b_p, \quad u \leq c_1, \dots, u \leq c_q, \quad u \equiv r \pmod{m}, \quad (\text{B3})$$

where all b_i, c_i are affine group terms. The lower-bound list is nonempty because $y \geq 0$ supplied $u \geq 0$.

Fix an input for which this disjunct has a solution. Let $B = \max_i b_i$, and let $s \in \{0, \dots, m-1\}$ be the unique standard integer such that $B + s \equiv r \pmod{m}$. Then $u_0 = B + s$ is the least element of this residue class above B . To check this in an arbitrary nonstandard group, if $u \geq B$ has the same residue, write $u = B + s + mv$. If $v < 0$, discrete order gives $v \leq -1$, hence $u \leq B + s - m < B$, a contradiction. Thus $u \geq u_0$. Because some solution satisfies the upper bounds, so does u_0 . It therefore satisfies the entire disjunct. Since that disjunct lies inside the graph of a function, its solution is unique, and the value of y is exactly $q_L(B + s)$. The expression B is one of the finitely many terms b_i , and s belongs to a finite standard set. Enumerating all such possibilities over all disjuncts gives the candidates (B1).

For vector outputs, apply this argument to each coordinate graph, obtained by existentially quantifying the other coordinates, and take finite products of candidate lists. To make a partition, assign an input to the first candidate which satisfies the original graph. This is a finite definable partition: quotient functions are themselves Presburger definable, so their use here can be eliminated from formulas. Continuity of the candidates follows from continuity of affine group maps and of q_m . $\square$

Audit of the nonstandard step. The proof used the least element of a fixed congruence class above a given bound, explicitly justified from division and discrete order. It did not use an iteration of successor a nonstandard number of times, or infer invariance modulo mG from invariance under one standard translation. This is the distinction highlighted in the archived manuscripts' finite-residue proofs.

98 An exact compiler for continuity loci

Let D denote either $\mathbb{R}$ with its usual topology and order, or $\mathbb{Q}^\omega$ with its product-of-discrete topology and lexicographic order. Put

$$G = D \overrightarrow{\times} \mathbb{Z}, \quad M = G_{\geq 0} = (D_{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}).$$

Coordinates of an input tuple are written $x = (u, n) \in M^k$, where $u \in D^k$ and $n \in \mathbb{Z}^k$. All definability below means Presburger definability, with a fixed finite tuple of parameters, unless an expanded coordinate language is explicitly specified. Coordinate projections themselves are not presumed to be Presburger definable.

We use the finite-branch normal form: a definable total map $f : M^k \longrightarrow M^r$ admits a finite definable partition $C_1, \dots, C_s$ and maps $F_1, \dots, F_s : G^k \longrightarrow G^r$ such that $f|_{C_j} = F_j|_{C_j}$ and each F_j is continuous. Its D -coordinates are rational-affine expressions in u ; its integer coordinates are integer parts of rational-affine expressions in n , with any needed fixed residue partition absorbed into C_j . The extensions F_j need not have values in M away from C_j . This causes no problem for the continuity criterion, which compares limits in the ambient Hausdorff group G^r .

Apply quantifier elimination to the cells. List their affine group forms as

$$L_i(u, n) = (\ell_i(u), z_i(n)), \quad \ell_i(u) = a_i \cdot u + b_i, \quad i = 1, \dots, t.$$

Here $a_i \in \mathbb{Q}^k$, $b_i \in D$ is a rational-linear combination of parameter coordinates, and after clearing denominators z_i is an integer-affine expression. Atomic formulas compare L_i with zero or test an integer congruence. When $\ell_i(u) \neq 0$, the sign of L_i is the sign of $\ell_i(u)$. When $\ell_i(u) = 0$, it is the sign of $z_i(n)$. An equality $L_i = 0$ requires both coordinates to vanish. A congruence depends only on integer coordinates.

98.1 Rational directions describe all local sign possibilities

Fix $x = (u, n) \in M^k$. The integer tuple n is constant on a neighbourhood of x . Let

$$I(x) = \{i : \ell_i(u) = 0\}, \quad B(x) = \{q : u_q = 0\}.$$

For $i \notin I(x)$, the sign of ℓ_i is locally constant. For $i \in I(x)$, a perturbation $\delta \in D^k$ changes its value to $a_i \cdot \delta$. The only local domain restrictions are $\delta_q \geq 0$ for $q \in B(x)$: an input with zero D -coordinate has nonnegative integer coordinate, and may move to zero or positive D -coordinate, but not to negative D -coordinate.

Lemma 98.1 (Rational directions suffice). *A finite pattern of signs of the forms $a_i \cdot \delta$, together with the restrictions $\delta_q \geq 0$ for $q \in B(x)$, is realizable in D^k if and only if it is realizable by a rational vector $v \in \mathbb{Q}^k$.*

Proof. For $D = \mathbb{R}$, the conditions are a finite homogeneous system of rational linear equalities, strict inequalities, and weak inequalities. A nonempty rational polyhedral relatively open region contains a rational point. More explicitly, solve the equality subsystem over $\mathbb{Q}$, restrict to its rational basis, and approximate a real solution closely enough to preserve the finitely many strict signs and the face determined by any vanishing weak inequalities.

For $D = \mathbb{Q}^\omega$, suppose δ realizes the pattern. For each nonzero form, include the coordinate at which its first nonzero entry occurs, and do likewise for each nonzero δ_q appearing in a domain restriction. Choose N beyond all these finitely many coordinates and put

$$v_q(\varepsilon) = \sum_{j=0}^N \delta_q(j) \varepsilon^j.$$

Every required nonzero linear form in these polynomials has the sign of its first nonzero coefficient when $\varepsilon > 0$ is sufficiently small. Every required zero form is identically zero, since all its sequence coordinates were zero. A sufficiently small positive rational ε therefore gives the required rational vector. Conversely, a rational vector is realized in $\mathbb{Q}^\omega$ by putting it in a single coordinate. $\square$

For the topological use, rational directions are not just witnesses to abstract consistency. In the real model, $\delta = \varepsilon v$ tends to zero as $\varepsilon \downarrow 0$. In the stream model, put $\delta_q^{(N)} = v_q e_N$, where e_N has a single 1 in coordinate N . Then $\delta^{(N)} \rightarrow 0$. These perturbations preserve all inactive nonzero signs eventually and realize the chosen active signs exactly. If all required signs are zero, the constant zero perturbation is allowed; this correctly accounts for a cell which contains x itself.

Theorem 98.2 (Exact local accessibility). *The following are equivalent for a cell C_j :*

1. $x \in \overline{C_j}$, with closure relative to M^k .
2. There is $v \in \mathbb{Q}^k$ satisfying $v_q \geq 0$ for $q \in B(x)$ such that the Boolean formula for C_j is true after: freezing all integer coordinates at n ; retaining the signs of the inactive $\ell_i(u)$; and replacing each active sign by $\text{sgn}(a_i \cdot v)$.

Proof. A rational witness gives arbitrarily close points in the cell by the perturbations just constructed. Conversely choose a point of C_j sufficiently close that the finitely many inactive signs and all integer coordinates agree with those of x . Its displacement realizes the required homogeneous sign pattern. Apply the rational-direction lemma. $\square$

This theorem is an algorithm: enumerate the finitely many assignments of signs $-1, 0, 1$ to the active forms, evaluate the Boolean cell formula, and decide feasibility of the associated rational linear system. Strict inequalities introduce no difficulty; a homogeneous feasible system with finitely many strict positive requirements can be rescaled so that all those positive quantities are at least 1, and ordinary rational linear feasibility then applies. If there are no strict requirements, solve the weak system directly. Rational Fourier–Motzkin elimination is one possible implementation.

98.2 The continuity criterion and its finite output

Theorem 98.3 (The branch-limit criterion). *For $x \in M^k$,*

$$f \text{ is continuous at } x \iff \forall j (x \in \overline{C_j} \implies F_j(x) = f(x)). \quad (\text{C})$$

Proof. Suppose f is continuous and $x \in \overline{C_j}$. Since the domain is metrizable there is a sequence $x_m \in C_j$ tending to x . In G^r , $f(x_m) = F_j(x_m) \rightarrow F_j(x)$ by continuity of F_j , and $f(x_m) \rightarrow f(x)$ by continuity of f . Uniqueness of limits yields the equality. The value $F_j(x)$ is allowed initially to lie outside M^r ; the conclusion rules that possibility out for accessible branches.

Conversely, assume every accessible branch agrees at x . For each inaccessible branch choose a neighbourhood of x avoiding its cell. There are only finitely many such branches. On every remaining branch, continuity of its ambient extension at x gives the usual neighbourhood estimate around $f(x)$. Intersect the finitely many neighbourhoods. This proves continuity into G^r , hence into its subspace M^r . $\square$

In particular, the discontinuity set is exactly

$$\bigcup_{i,j=1}^s \{x \in C_i : x \in \overline{C_j}, F_i(x) \neq F_j(x)\}. \quad (\text{D})$$

Using rational accessibility to compile the closure conditions produces a finite formula for this set. Its atoms are signs and equalities of rational-affine D -coordinate expressions, and

Presburger conditions on integer coordinates. It is crucial to describe this as an *external coordinate formula*. It need not be a formula in the original Presburger structure.

Corollary 98.4 (Exact form of discontinuity sets). *The discontinuity set of every definable $f : M^k \rightarrow M^r$ is a Δ_2^0 set. More strongly, it is a finite union of locally closed sets and is contained in a finite union of closed nowhere dense affine D -coordinate hyperplanes. Thus it is nowhere dense; its closure is nowhere dense as well. It need not itself be closed.*

Proof. In both coefficient models, each strict sign set of a continuous D -affine form is open and each zero set is closed. Integer conditions are clopen because the integer-coordinate space is discrete. Finite Boolean combinations therefore form finite unions of intersections of an open and a closed set. Such locally closed sets and their finite unions are $F_\sigma \cap G_\delta$ in a metrizable space.

If x lies on none of the nonconstant hyperplanes $\ell_i(u) = 0$ associated with the cell partition, all cell truth values are locally constant. Forms constant in the variable u also have locally constant truth values: they either have fixed nonzero sign, or are decided by the locally constant integer tuple. Hence f locally agrees with one continuous candidate and is continuous. Each nonconstant hyperplane is closed and nowhere dense relative to M^k : vary an input coordinate with nonzero coefficient by an arbitrarily small positive perturbation, retaining membership in M . In the stream model such a perturbation may be placed arbitrarily far out in the sequence. A finite union remains closed nowhere dense. $\square$

98.3 Two exact examples and necessary qualifications

Define

$$f(x, y) = \begin{cases} x, & y = 0, \\ 0, & y \neq 0. \end{cases}$$

This map is Presburger definable, and in both models its exact discontinuity set is

$$\text{Disc}(f) = \{(x, 0) : x \neq 0\}.$$

At a point of this set, let the second input approach zero through $(\varepsilon, 0)$ in the real model or through $(e_N, 0)$ in the stream model. The outputs become zero, whereas $f(x, 0) = x \neq 0$. If $y \neq 0$, the function is locally zero since the complement of the closed singleton $\{0\}$ is open. At $(0, 0)$ every output is either zero or the first input, so continuity holds. This discontinuity set is not closed.

For truncated subtraction

$$x \dot{-} y = \max(x - y, 0),$$

write $x = (u, n)$ and $y = (v, m)$. The compiler gives

$$\text{Disc}(\dot{-}) = \{((u, n), (v, m)) : u = v, n \neq m\}. \quad (\text{M})$$

If $u - v \neq 0$, its sign is stable and one of the two continuous branches is selected nearby. If $u = v$ and $n = m$, both ambient branches agree at zero. If $u = v$ and $n \neq m$, arbitrarily small admissible perturbations of the D -coordinates realize both signs of their difference; the branch limits are the distinct points $(0, n - m)$ and zero in G . This proves the formula, including boundary inputs with $u = v = 0$.

The set in (M) is not Presburger definable, even with parameters. If it were, its section at $y = 0$, with the point $x = 0$ added, would define exactly the standard copy of $\mathbb{N}$ in M . That set

contains zero and is closed under successor, so the induction scheme with those parameters would force it to be all of M , a contradiction. Thus an exact continuity compiler must be allowed to output the expanded coordinate description, rather than claiming definability in the input language.

Strict signs cannot be removed from the output language. The archived research question proposed descriptions involving equalities of divisible-coordinate affine expressions and residue conditions. That fragment is too small in both models. Choose a parameter $a = (\alpha, c)$ with $\alpha > 0$, and define

$$g_a(x, y) = \begin{cases} 1, & x > a \text{ and } y = 0, \\ 0, & \text{otherwise.} \end{cases}$$

Writing $x = (u, n)$, the exact discontinuity set is

$$\text{Disc}(g_a) = \{(x, 0) : u \geq \alpha\}.$$

For $u > \alpha$ the output at $y = 0$ is one and nearby nonzero y gives zero. For $u < \alpha$ a neighbourhood forces $x < a$, so the map is zero. At $u = \alpha$, approaching from larger divisible coordinate gives one with $y = 0$, while approaching through nonzero y gives zero, regardless of the integer comparison at the point. At $y \neq 0$, the map is locally zero. These observations prove the formula in both models.

In either model restrict to $x = (t\alpha, 0)$ with rational $t \geq 0$, and to $y = 0$. Each affine equality in this remaining scalar variable holds at at most one t or at every t , because D is torsion-free and $\alpha \neq 0$. A finite Boolean combination therefore defines a finite or cofinite subset of $\mathbb{Q}_{\geq 0}$. All residue conditions are constant on this slice. But the resulting threshold $t \geq 1$ is neither finite nor cofinite. Thus the equality-and-residue-only proposal has a negative answer; the additional sign predicates in the compiler are necessary.

[write] (Batch 87.) This answers Part I's [question 14.13](#): the discontinuity set of every definable total function is computed from the graph ([theorem 98.3](#), [theorem 98.5](#)), and the fragment proposed there, equalities of divisible-coordinate affine expressions and residue conditions, is too small, by the example g_a . The formula (M) is Part I's [\(8.2\)](#) ([theorem 8.3](#)). [Corollary 98.4](#) generalizes that theorem to every definable function. The output is a formula in an expanded two-sorted coordinate language, not a Presburger definition, as source 13 stresses (and the Hilbert's-tenth programme's review of the arrival notes, [section E](#)).

98.4 Deciding global continuity after coordinate elimination

The output formula (D) lives in a two-sorted language. The first sort is the divisible ordered abelian group D , with rational scalar maps; the second is Presburger integer arithmetic. Each atomic relation belongs entirely to one of the two sorts. The domain condition is the finite Boolean combination

$$\bigwedge_{q=1}^k (u_q > 0 \vee (u_q = 0 \wedge n_q \geq 0)).$$

The apparent interaction between the sorts occurs only through Boolean combinations. Consequently, after distributing a finite disjunction, existential quantification of all input coordinates separates into ordinary divisible-ordered-group feasibility and ordinary Presburger integer feasibility.

Theorem 98.5 (Effective global continuity criterion). *Given a Presburger definition of a total function, with a finite parameter tuple, one can effectively construct a finite Boolean combination Ψ of:*

- signs and equalities of rational-linear expressions in the D -coordinates of the parameters; and
- Presburger conditions on the integer coordinates of the parameters,

such that the function is globally continuous exactly when Ψ holds. The same procedure compiles continuity at a named point by treating that point's coordinates as additional parameters.

Proof. Apply (D), add the domain condition, and existentially quantify the inputs. Perform quantifier elimination for divisible ordered abelian groups and for Presburger arithmetic, then negate the result. All preceding transformations are finite syntactic procedures. The rational accessibility feasibility tests depend only on the rational coefficients of the active forms and the selected zero-coordinate face; they therefore may be performed at compilation time for every possible finite sign pattern. $\square$

This yields a genuine ordinary decision algorithm when the needed parameter sign/equality tests are decidable, in particular for rational real parameters and for exactly represented finitely supported rational sequences. With arbitrary infinite input or parameter streams, it yields an oracle reduction, not an ordinary decision procedure. A zero-sequence test is already discontinuous, so that distinction cannot be removed. For the stream model, each finitely many affine sequence forms can be scanned until its first nonzero coordinate is found, treating the form as zero in the meantime; the Boolean answer has a computable eventually stable approximation. A separate sharper rank argument may improve the mind-change bound; it is not needed for the compiler theorem.

99 Exact target-valued Baire classification

Here “Baire class one” means a pointwise limit of continuous maps into the stated target space, with that target's actual topology. It is not being used as a synonym for having F_σ inverse images of open sets. That distinction is decisive for the real-coordinate model.

99.1 The stream model: explicit locally constant approximations

Let $D_\omega = \mathbb{Q}^\omega$ with discrete rational coordinates and lexicographic order. Let $T_N : G(D_\omega) \rightarrow G(D_\omega)$ retain the first N rational coordinates, set all later rational coordinates to zero, and retain the integer coordinate. Its image is the subgroup $G_N \cong \mathbb{Q}^N \vec{\times} \mathbb{Z}$. This is an additive projection and commutes with every q_m . It need not preserve the positive cone: a positive element whose first nonzero rational coordinate occurs after N , and whose integer coordinate is negative, is sent to a negative element. The following construction explicitly accounts for that issue.

Theorem 99.1 (Locally constant approximation with eventual exact prefixes). *Every parameter-definable total function $f : M(D_\omega)^k \rightarrow M(D_\omega)^r$ is a pointwise limit of locally constant functions f_N with the same domain and target. Given a graph formula and streams for its parameters, these functions can be computed uniformly: f_N requires only the first N rational coordinates and the integer coordinates of the inputs and parameters. For each fixed input x , there is N_x such that*

$$f_N(x) = T_N f(x) \quad (N \geq N_x). \quad (\text{B4})$$

Proof. Let $R(x, a, y)$ be an ambient group formula for the graph, explicitly including $y \in M^r$. Replace it by the following totalized graph: if there is a unique $y \in M^r$ satisfying $R(x, a, y)$,

return that y ; otherwise return the zero tuple. This defines a function $F : G^{k+p} \rightarrow M^r$ for every input and parameter tuple in every $\mathbb{Z}$ -group. It agrees with f at the original parameter tuple and at inputs in the original domain. The definition is a first-order formula using existence and uniqueness; no choice operation is needed.

Evaluate this same totalized formula in the finite-coordinate model G_N on $(T_N x, T_N a)$, and include its output in $M(D_\omega)^r$. Call the resulting output $f_N(x)$. It always lies in the target, even when a truncated input or parameter lies outside the positive cone. Every f_N depends on finite coordinate data and is therefore locally constant. It is effective because arithmetic and order in $\mathbb{Q}^N \vec{\times} \mathbb{Z}$ are decidable and the finite-branch theorem provides a finite list of possible outputs.

Apply the finite-branch theorem and quantifier elimination to the totalized graph, with the parameters treated as variables. In every atomic cell condition, a nonzero affine rational-coordinate stream has a first nonzero coordinate. After that coordinate is retained, its sign is fixed forever. A zero stream remains zero in every truncation. Integer equalities and congruences are unchanged. Consequently every atomic truth value, and hence every cell truth value, eventually agrees with its value at (x, a) . There are only finitely many atoms. For all sufficiently large N , the selected candidate is therefore the same candidate F_j as at (x, a) . Since affine group terms and fixed quotients commute with T_N ,

$$f_N(x) = F_j(T_N x, T_N a) = T_N F_j(x, a) = T_N f(x).$$

This proves (B4). Coordinate truncations converge in the product topology, proving the asserted pointwise convergence. $\square$

The theorem gives an actual target-valued Baire-one approximation, rather than only a classification of the graph or its coordinate functions. The integer output coordinate stabilizes, and each desired finite output prefix is eventually exact. There is no claimed effective bound on the stabilization stage. A stream which vanishes for a long initial segment may reveal its first nonzero coordinate arbitrarily late.

A dimension bound on branch revisions. Suppose there are k input streams and p parameter streams. Compile the totalized function into a common finite list of quantifier-free cell conditions, and let ρ be the rational rank of the coefficient vectors of all affine D -coordinate forms occurring in their comparisons. Then the selected candidate index in the approximation above changes at most $\rho \leq k + p$ times. The entire output integer tuple likewise changes at most ρ times.

Indeed, the finite-rank truth theorem, Theorem 100.1, applies simultaneously to all the cell atoms. A new stage at which an as-yet-zero leading form becomes nonzero contributes an independent row functional on the ρ -dimensional span of comparison forms, so there are at most ρ such stages. Between them every atom and hence the selected branch is unchanged. For a fixed candidate, its output integer tuple depends only on the input and parameter integer tuples, which truncation leaves unchanged. This proves both bounds. It does not bound the number of distinct full stream outputs: even a single fixed branch can acquire further nonzero output coordinates at infinitely many stages.

99.2 The real model: a complete criterion and a sharp obstruction

Write $M_{\mathbb{R}} = M(\mathbb{R})$. For $n = (n_1, \dots, n_k) \in \mathbb{Z}^k$, the input slice of fixed integer coordinates is

$$X_n = \prod_{i=1}^k I_{n_i} \times \{n\}, \quad I_t = \begin{cases} [0, \infty), & t \geq 0, \\ (0, \infty), & t < 0. \end{cases} \quad (\text{B5})$$

Each X_n is nonempty, connected, and clopen in $M_{\mathbb{R}}^k$. For a vector-valued map, let $\pi_{\mathbb{Z}}f$ denote the entire tuple of output integer coordinates.

Theorem 99.2 (Exact real-model Baire classification). *For a Presburger-definable total function $f : M_{\mathbb{R}}^k \rightarrow M_{\mathbb{R}}^r$, the following are equivalent:*

1. f is Baire class one with target $M_{\mathbb{R}}^r$.
2. f belongs to some countable Baire class with that target.
3. For every $n \in \mathbb{Z}^k$, the map $\pi_{\mathbb{Z}}f$ is constant on X_n .

Thus every definable map admitted anywhere in the countable target-valued Baire hierarchy already belongs to class one. Other definable maps are outside that hierarchy altogether.

Proof. Necessity. For every continuous $g : X_n \rightarrow M_{\mathbb{R}}^r$, its integer-coordinate map is continuous into the discrete space $\mathbb{Z}^r$, so connectedness forces it to be constant. This property is preserved by pointwise limits. Indeed, suppose g_m have constant integer tuples c_m on X_n and $g_m \rightarrow g$ pointwise. At one fixed point of X_n , convergence in the discrete coordinate forces c_m eventually to equal one tuple c . The same sequence c_m occurs at every point, so $\pi_{\mathbb{Z}}g$ is identically c throughout X_n . Transfinite induction over the usual countable Baire hierarchy proves the claim for every countable Baire class, establishing (2) $\Rightarrow$ (3).

Sufficiency. In both coefficient models every definable cell is Δ_2^0 . For completeness, a group equality is closed; a strict or weak order condition is a finite union of an open strict sign condition on the D -coordinate and a closed zero condition with a clopen integer test. Both it and its complement are F_{σ} . Congruence predicates are clopen. Quantifier elimination and finite Boolean closure prove the assertion.

On a metric space the real-valued characteristic function of a Δ_2^0 set C is Baire one. To see this directly, choose increasing closed exhaustions $A_m \uparrow C$ and $B_m \uparrow (X \setminus C)$. When both are nonempty, the continuous functions

$$u_m(x) = \frac{d(x, B_m)}{d(x, A_m) + d(x, B_m)}$$

converge to $\mathbf{1}_C$. The denominator is positive because the two closed sets are disjoint. If A_m is empty use $u_m = 0$; if only B_m is empty use $u_m = 1$. These conventions preserve eventual convergence.

Apply this construction to the finite partition in the finite-branch theorem. If H_j is the real-coordinate part of the continuous candidate F_j , then the real-coordinate part h of f is

$$h = \sum_{j=1}^s \mathbf{1}_{C_j} H_j.$$

Replacing each characteristic function by its continuous approximation gives continuous maps h_m into $\mathbb{R}^r$ converging pointwise to h .

Under condition (3), write the output integer tuple as $w(n)$; it is a continuous function of the input because the slices X_n are clopen. For $m \geq 1$ and output coordinate j , replace $h_{m,j}$ by

$$\tilde{h}_{m,j}(x) = \begin{cases} \max\{h_{m,j}(x), 0\}, & w_j(n) \geq 0, \\ \max\{h_{m,j}(x), 1/m\}, & w_j(n) < 0. \end{cases}$$

These are continuous, land in the required real-coordinate half-lines, and converge to h_j . Thus $x \mapsto (\tilde{h}_m(x), w(n))$ are continuous maps into the actual target $M_{\mathbb{R}}^r$ converging pointwise to f . This proves (3) $\Rightarrow$ (1); (1) $\Rightarrow$ (2) is immediate. $\square$

A parameter-free example. The definable function

$$\text{sgn}_M(x) = \begin{cases} 0, & x = 0, \\ 1, & x > 0 \end{cases}$$

has a Δ_2^0 graph and Δ_2^0 inverse images of open sets. On $M_{\mathbb{R}}$ it belongs to no countable target-valued Baire class: its integer output is zero at the endpoint of X_0 and one elsewhere on that connected half-line. On $M(D_\omega)$ it is Baire class one by the explicit approximation theorem. This isolates the precise reason that graph complexity alone cannot settle the target-valued question.

An effective addition to the classification. The criterion (3) can itself be compiled into the external two-sorted coordinate language used by the continuity theorem: assert that there are no two inputs with the same integer tuple but different integer output tuples. Substitute the finite branch candidates, split the finite cell cases, and eliminate the real-coordinate quantifiers by divisible-ordered-group elimination and the integer quantifiers by Presburger elimination. The result is a finite Boolean combination of rational-linear sign tests on the real parts of the parameters and Presburger conditions on their integer parts. Consequently membership in the target-valued Baire hierarchy is decidable for parameter-free definitions, and for parameter presentations with decidable required real sign/equality tests. Arbitrary real streams do not supply those tests as uniformly decidable operations.

99.3 Topological extension to a countable discrete core

Corollary 99.3 (Regularity over every countable discrete core). *Let K be any countable $\mathbb{Z}$ -group with the discrete topology, let $D = \mathbb{R}$ or $D = \mathbb{Q}^\omega$ with the respective topologies used above, and put*

$$G = D \overrightarrow{\times} K, \quad P = G_{\geq 0}.$$

For every parameter-definable total map $f : P^k \rightarrow P^r$, the following topological conclusions continue to hold:

1. *There are finitely many continuous ambient quotient-affine candidates. The rational-direction accessibility criterion and branch-limit continuity criterion remain valid with the core coordinates frozen. Consequently $\text{Disc}(f)$ is a finite union of locally closed sets contained in finitely many closed nowhere dense affine D -coordinate hyperplanes.*
2. *For $D = \mathbb{Q}^\omega$, there are locally constant maps $f_N : P^k \rightarrow P^r$ such that, for each x , eventually $f_N(x) = T_N f(x)$; here T_N truncates only the rational stream and leaves the K -coordinate unchanged. In particular f is target-valued Baire class one.*

3. For $D = \mathbb{R}$, membership in any countable target-valued Baire class is equivalent to class one, and holds exactly when the output K -tuple is constant on every slice with fixed input K -tuple.

Proof. The group G is a $\mathbb{Z}$ -group, and its standard quotient maps are

$$q_m(d, b) = (d/m, q_m^K(b)).$$

They are continuous because K is discrete. The finite-branch proof uses only Presburger elimination, discrete order, and standard division, so it applies unchanged. On a neighbourhood of an input, all core coordinates are fixed. Every core affine equality, order test, and congruence therefore has a fixed truth value. The only varying signs are the same rational-affine D -coordinate forms as before. At a zero D -coordinate the domain again allows exactly nonnegative directions. The accessibility proof, branch-limit proof, and nowhere-dense hyperplane argument are consequently unchanged. Core conditions are clopen, so the finite locally closed description follows as well.

In the stream case, totalize the graph on the ambient group and evaluate it on truncated inputs and parameters in $\mathbb{Q}^N \vec{\times} K$. This is again a $\mathbb{Z}$ -group, so the totalized output exists uniquely and lies in its positive cone. The resulting map is locally constant, because it depends only on a finite rational prefix and discrete core data. All core tests remain fixed while the leading-coordinate tests eventually stabilize. Quotient-affine candidates commute with truncation, giving $f_N(x) = T_N f(x)$ eventually, exactly as before.

In the real case the fixed-core slices are products of $[0, \infty)$ and $(0, \infty)$, according to the signs of their core coordinates. They are connected and clopen. Projection to the discrete target K^r therefore gives the same transfinite constancy obstruction. For sufficiency, approximate the leading real coordinates and clamp them to $[0, \infty)$ or to positive values according to the constant output core coordinate. The earlier construction then produces continuous maps into P^r converging pointwise to f . $\square$

This extension is purely topological. No computable presentation of K is assumed. Countability and discreteness alone do not supply an algorithm for evaluating core arithmetic or make the displayed locally constant approximants uniformly computable. Effective versions must separately specify a presentation of K and the available operations and decision procedures.

100 The exact number of changes of mind

The earlier Provelt report [64] proved a finite-change truth theorem and bounded the number of changes by the number of equality and order atoms after quantifier elimination. Its question entitled “Optimal finite-mind-change complexity” asked for the smallest possible bound for an individual formula. We first give a bound by linear dimension, which is sharp as a function of the number of input streams. We then give a finite, effective calculation of the optimum for each parameter-free formula.

100.1 Presentation and counting convention

Let

$$D = \mathbb{Q}^N, \quad G = D \vec{\times} \mathbb{Z}, \quad M = G_{\geq 0},$$

where a nonzero rational stream is positive when its first nonzero coefficient is positive, and the D coordinate precedes the integer coordinate in the order on G . Give G the product

topology in which each rational coordinate and the integer coordinate are discrete. An element of M has the form $x = (h, n)$, with

$$h \geq_{\text{lex}} 0, \quad h = 0 \implies n \geq 0.$$

Thus $n < 0$ guarantees that h has a positive first nonzero coefficient. The earlier report provides a computable homeomorphism between this presentation and a total coding by Baire space. A valid coordinate code can also be converted to that Baire code effectively: when $n < 0$, search for its guaranteed positive leading coefficient before producing the corresponding negative-slice tag.

A *truth approximation* for a relation $f : M^d \rightarrow \{0, 1\}$ is a sequence of continuous maps $A_s : M^d \rightarrow \{0, 1\}$ satisfying

$$\lim_{s \rightarrow \infty} A_s(\bar{x}) = f(\bar{x}) \quad \text{for every } \bar{x} \in M^d.$$

Its number of changes on $\bar{x}$ is

$$\#\{s \in \mathbb{N} : A_{s+1}(\bar{x}) \neq A_s(\bar{x})\}.$$

The initial output is not counted as a change. A computable approximation means that the stage maps are uniformly Type-2 computable. Each output must use only finite input information, but its computation may inspect an input-dependent finite number of coordinates. In particular, the definition does not require an output before the integer coordinates have been read.

Write $\mu(f)$ for the least uniform change bound among continuous approximations, and $\mu_{\text{eff}}(f)$ for the corresponding minimum among computable approximations. The constructions below show that these minima exist for the definable relations under consideration.

100.2 The dimension bound

Translate a Presburger formula on M to the ordered group G by relativizing its quantified variables to the positive cone. Effective quantifier elimination for $\mathbb{Z}$ -groups then produces a Boolean combination of atoms

$$t_j(\bar{x}) > 0, \quad t_j(\bar{x}) = 0, \quad m_\ell \mid u_\ell(\bar{x}),$$

where each affine term has integer coefficients. For an order or equality atom write

$$t_j(\bar{x}) = \sum_{i=1}^d c_{ji}x_i + b_j \mathbf{1}, \quad c_j = (c_{j1}, \dots, c_{jd}) \in \mathbb{Z}^d.$$

Repeated occurrences do not need to be duplicated in the following rank. Define

$$U = \text{span}_{\mathbb{Q}}\{c_j : j \text{ is an order or equality atom}\}, \quad r = \dim_{\mathbb{Q}} U.$$

Only these atoms enter r ; congruence coefficients do not.

Theorem 100.1 (Linear dimension bounds truth changes; sharpens [theorem 7.3](#)). *For every Presburger formula with d external stream variables, effective quantifier elimination and rational Gaussian elimination compute a uniform truth approximation with at most*

$$r \leq d$$

changes on every input. Here an external stream variable can be a free variable or a parameter supplied as an oracle. In particular, a parameter-free formula in k free variables requires at most k changes, independently of its length and its quantifier alternation. A formula in k variables with p supplied stream parameters admits the uniform bound $k + p$.

Proof. Write $x_i = (h_i, n_i)$. First read all integer coordinates. The integer coordinate of t_j is the exactly known integer

$$N_j = \sum_i c_{ji} n_i + b_j,$$

and its stream coordinate is $\sum_i c_{ji} h_i$. Congruence atoms are decided immediately from the corresponding integer coordinate, since

$$mG = D \times m\mathbb{Z}.$$

Let

$$v_a = (h_1(a), \dots, h_d(a)) \in \mathbb{Q}^d \quad (a \in \mathbb{N}).$$

At stage s , use the sign of the first nonzero value among $c_j \cdot v_0, \dots, c_j \cdot v_{s-1}$ if there is one. Otherwise use the sign of N_j . This gives an approximation to every order atom. For equality use the answer $N_j = 0$ until a nonzero stream coefficient is seen, at which point the answer is permanently false. Evaluate the fixed Boolean combination at each stage to obtain A_s .

There are finitely many atoms. Every nonzero stream has a first nonzero coefficient, so every atom eventually becomes permanently correct. Consequently A_s converges to the desired truth value.

To count changes, restrict the row functional $c \mapsto c \cdot v_a$ to U , and denote the resulting element of U^* by ℓ_a . Put

$$W_s = \text{span}_{\mathbb{Q}}\{\ell_0, \dots, \ell_{s-1}\} \subseteq U^*.$$

An order or equality atom has no witnessed nonzero stream coefficient exactly when its coefficient vector is annihilated by all members of W_s . If $\ell_s \in W_s$, every such coefficient vector is also annihilated by ℓ_s . Hence no unresolved atom can become resolved at that stage. Resolved atoms never change again, and congruence atoms are already fixed. Thus

$$A_{s+1} \neq A_s \implies \dim W_{s+1} > \dim W_s.$$

The dimension starts at zero and never exceeds r , which proves the bound. The construction is uniform and reads only finite prefixes at each stage. All its preprocessing is effective. $\square$

The proof also gives the input-dependent bound $\dim \text{span}\{\ell_a : a \in \mathbb{N}\}$, which can be strictly smaller than r . It does not provide a computable time at which the last change has occurred. A new independent row can first appear arbitrarily far along the input streams.

Many atoms, only two possible changes. For example, the formula

$$((x + y = 0) \text{ XOR } (x = y)) \wedge \bigwedge_{m=1}^{32} (mx < (m+1)y \vee x = y)$$

has many order atoms, but all their coefficient vectors lie in $\mathbb{Q}^2$. On inputs with integer coordinates zero, take row $v_2 = (1, 1)$, row $v_7 = (1, 0)$, and all other rows zero. The displayed approximation has successive relevant answers 0, 1, 0: the first row resolves $x + y$, while the later row resolves $x - y$. The many inequalities remain true after the first row. Both rank increases can therefore matter to the final Boolean answer.

100.3 Sharpness of the bound by arity

Theorem 100.2 (Exactly d changes can be necessary). *For every $d \geq 1$, the parameter-free Presburger relation*

$$P_d(x_1, \dots, x_d) \iff \text{an odd number of the statements } x_i \neq 0 \text{ hold}$$

satisfies

$$\mu(P_d) = \mu_{\text{eff}}(P_d) = d.$$

Thus the dimension bound is sharp even when no order atoms or quantifiers are used. For any proposed computable approximation with fewer than d changes, a violating input can be chosen computable and with every stream of finite support. For arbitrary continuous stage maps the same lower-bound argument gives an existential violating input.

Proof. The upper bound follows from the preceding theorem. The parity relation is a finite Boolean combination of the equality atoms $x_i = 0$; it is therefore a Presburger formula, with no parity operation added to the language.

For the lower bound restrict to the closed slice on which all integer coordinates are zero. Initially let every stream be zero. Its truth value is zero, so convergence gives a stage t_0 at which the proposed approximation outputs zero there. By continuity, the value at that stage is fixed on some finite-prefix neighborhood of this tuple.

Insert a single positive rational coefficient, sufficiently far out in the first stream to remain in this neighborhood. The tuple still belongs to M^d , its first coordinate is now nonzero, and its truth value has toggled to one. Convergence gives a later stage $t_1 > t_0$ outputting one. Choose a finite-prefix neighborhood that fixes the outputs at both chosen stages. Insert a sufficiently delayed positive coefficient in the second, previously zero, stream. Its truth value toggles again.

Continue in this way through the d streams. At each step preserve all previously selected output values, and choose the next correct output at a strictly later stage. The final tuple has finite support in each stream and satisfies

$$A_{t_j}(\bar{x}) = j \bmod 2 \quad (0 \leq j \leq d).$$

There must therefore be at least one change between every pair of successive selected stages, giving at least d changes altogether.

If the stage maps are computable, the construction is effective relative to their index. At each step the current finite-support input is computable, so search for a stage with its known correct truth value. The computation terminates by convergence. Record the finitely many input coordinates queried by all the selected computations and place the next positive coefficient beyond them. This preserves their outputs and gives a computable final input. Mere continuity supplies the requisite neighborhoods without an effective procedure for finding them, which is enough for the noneffective lower bound. $\square$

100.4 An exact finite calculation for every parameter-free formula

The rank bound is often not the individual optimum. For example, $x_1 = 0 \wedge \dots \wedge x_d = 0$ has coefficient rank d , but a single change suffices: begin with the answer determined by the integer coordinates and change permanently to false when a nonzero stream coefficient appears. The following finite construction computes the exact optimum instead of just a rank bound.

Fix a parameter-free formula $\varphi(\bar{x})$ and an effectively obtained quantifier-free group formula Ψ as above. Include standard numerals in the affine constants b_j . Put

$$C = \{c_j \neq 0 : c_j \text{ occurs in an order or equality atom}\} \cup \{e_1, \dots, e_d\},$$

where e_i is the i th coordinate vector. Retain just one copy of each vector. Including the coordinate vectors is essential: they record the positivity requirements defining M^d .

Integer regimes. The truth of Ψ at fixed stream signs depends on the integer tuple $\bar{n} \in \mathbb{Z}^d$ only through finitely many data:

1. the bits $\eta_i = [n_i < 0]$;
2. the signs $\alpha_j = \text{sgn}(c_j \cdot \bar{n} + b_j)$ for its order and equality atoms;
3. the truth values β_ℓ of its congruence atoms evaluated on $\bar{n}$.

A choice of all these data is an *integer regime* R . There are finitely many choices. Whether a regime is nonempty is an ordinary Presburger decision problem over $\mathbb{Z}$: ask whether there exists an integer tuple having the stipulated signs and congruences. Discard every infeasible regime.

The finite sign-pattern poset. For a feasible regime R , let $\mathcal{P}_R$ be the set of vectors

$$\sigma(v) = (\text{sgn}(c \cdot v))_{c \in C} \in \{-1, 0, 1\}^C$$

realized by rational vectors $v \in \mathbb{Q}^d$ satisfying

$$v_i \geq 0 \quad \text{for all } i, \quad v_i > 0 \quad \text{whenever } \eta_i = 1.$$

Order these patterns by

$$\sigma \leq \tau \iff \forall c \in C (\sigma_c \neq 0 \implies \tau_c = \sigma_c).$$

In other words, passing upward can replace zeros by signs but cannot change an already nonzero sign. This is a finite partially ordered set. It is computable: enumerate the finitely many proposed sign vectors and decide the corresponding finite systems of rational linear equalities and strict inequalities. Rational linear elimination suffices. Since the systems are homogeneous, one can alternatively replace their strict positive inequalities by inequalities at least one whenever there is a strict inequality; a common positive rescaling proves equivalence.

Label each $\sigma \in \mathcal{P}_R$ by a bit $\lambda_R(\sigma)$, obtained by evaluating Ψ as follows. For an order atom, use the nonzero stream sign σ_{c_j} when available and otherwise its integer sign α_j . An equality atom holds exactly when both those signs are zero. A congruence atom has the fixed value β_ℓ . For a zero coefficient vector set $\sigma_0 = 0$ by convention.

Let

$$a(R) = \max_{\sigma_0 < \dots < \sigma_q \text{ in } \mathcal{P}_R} \#\{j \in \{1, \dots, q\} : \lambda_R(\sigma_j) \neq \lambda_R(\sigma_{j-1})\}, \quad N_\varphi = \max_{R \text{ feasible}} a(R).$$

Singleton chains are allowed and contribute zero. The finite maximum is effectively computable, for example by dynamic programming on the finite partially ordered sets.

Lemma 100.3 (Finite sign compression). *For any finite $C \subseteq \mathbb{Q}^d$, every sign pattern*

$$(\text{sgn}_{\text{lex}}(\sum_i c_i h_i))_{c \in C}$$

of a tuple of rational streams is realized by a single rational vector. If C contains the coordinate vectors, the rational vector has exactly the same zero and positive coordinate pattern as the streams.

Proof. Choose a finite prefix long enough to witness the first nonzero coefficient of every nonzero linear combination indexed by C . Let its rows be $v_0, \dots, v_{L-1}$. For a sufficiently small positive rational ε , set

$$v = v_0 + \varepsilon v_1 + \dots + \varepsilon^{L-1} v_{L-1}.$$

For each $c \in C$ whose stream is nonzero, the first nonzero coefficient of the polynomial $c \cdot v$ determines its sign for all sufficiently small positive ε . Finitely many such requirements can be satisfied simultaneously. A combination that is identically zero annihilates every row and remains zero. Applying this also to the coordinate vectors proves the final assertion. If all combinations are zero, the zero vector suffices. $\square$

Theorem 100.4 (Exact computable change complexity). *For every parameter-free Presburger formula φ on this presentation,*

$$\mu(\mathbf{1}_\varphi) = \mu_{\text{eff}}(\mathbf{1}_\varphi) = N_\varphi \leq r \leq d.$$

Thus the minimum possible number of changes of mind is computable from the formula. It is the maximum number of truth-value alternations along a chain of feasible sign patterns, maximized over its finitely many feasible integer regimes.

Proof. First, the sign-compression lemma shows that $\mathcal{P}_R$ describes exactly the finite stream-sign patterns occurring on the integer slice of any tuple $\bar{n}$ in regime R . Coordinate signs enforce precisely the necessary conditions: every stream is lexicographically nonnegative, and a stream is nonzero when its integer coordinate is negative. Conversely, a feasible rational vector gives a valid tuple by placing that vector in the first row and making all later rows zero.

For the upper bound, read the integer coordinates and hence their regime. If $n_i < 0$, scan until the positive leading coefficient of h_i has been witnessed. Validity of the input guarantees termination. Choose L beyond all these witnessed indices; if there are no negative integer coordinates, use $L = 0$. This finite preliminary scan is part of the computation of the first output, so it does not itself count as a change.

At stage s truncate all streams after their first $L + s$ coefficients, and let σ_s be their pattern on C . Each truncated tuple is still in the same integer slice of M^d , so by sign compression $\sigma_s \in \mathcal{P}_R$. Revealing a new row cannot change a previously witnessed nonzero leading sign. Therefore

$$\sigma_0 \leq \sigma_1 \leq \sigma_2 \leq \dots$$

Output $A_s = \lambda_R(\sigma_s)$. Every relevant linear combination eventually has its correct sign, and only finitely many combinations are involved, so these outputs converge to truth. After repeated patterns are removed, their path is a chain in $\mathcal{P}_R$. Consequently it has at most $a(R) \leq N_\varphi$ changes. All the computations are effective and inspect finite input information.

For the lower bound, choose a feasible regime R and a chain attaining N_φ . Remove consecutive portions with constant label to obtain

$$\sigma_0 < \sigma_1 < \dots < \sigma_{N_\varphi}, \quad \lambda_R(\sigma_{j+1}) \neq \lambda_R(\sigma_j).$$

Fix an integer tuple $\bar{n}$ in that regime and a one-row rational tuple realizing σ_0 . For any continuous truth approximation, choose a stage t_0 outputting its correct label.

Suppose a finite-support tuple with pattern σ_j has been chosen and the outputs at the earlier selected stages have been fixed. Take a finite-prefix neighborhood preserving all those outputs. At a coordinate beyond both that prefix and the current finite support, append a rational row v realizing σ_{j+1} . Every previously nonzero linear combination retains its old leading coefficient and hence its sign. Every combination previously zero acquires the sign of $c \cdot v$. By the defining extension order, the resulting tuple has exactly pattern σ_{j+1} . The coordinate vectors in C ensure that the tuple remains in M^d with the same integer regime.

Its truth value is the opposite label, so convergence yields a strictly later stage $t_{j+1} > t_j$ with that value. Continue until the end of the chain. The final tuple preserves alternating outputs at $t_0 < \dots < t_{N_\varphi}$ and therefore forces at least N_φ changes. This lower bound holds even for arbitrary continuous stage maps. With computable stage maps, integer and rational witnesses can be found effectively, and the finite-prefix neighborhoods can be recovered from finite oracle-query traces. The same argument then constructs a computable finite-support witness.

Finally, along any chain of realizable sign patterns, let Z_j be the span of the order/equality coefficient vectors still assigned zero. These subspaces form a descending chain. Whenever one of these forms becomes nonzero, the descent is strict: a rational vector realizing the new pattern annihilates all still-zero forms but does not annihilate the newly nonzero one. A truth-value change requires at least one order/equality form to become nonzero, because the integer regime and all congruence values are fixed. Thus every truth change strictly decreases $\dim Z_j$, which starts at most r . This proves $N_\varphi \leq r \leq d$ and completes the argument. $\square$

100.5 A sharp obstruction when parameters are fixed

Theorem 100.5 (Uniform parameter optimization is impossible). *For a fixed parameter $p \in M$, let $f_p(x) = \mathbf{1}_{x < p}$. Then*

$$\mu(f_p) = \begin{cases} 0, & p = 0, \\ 1, & p > 0. \end{cases}$$

The same values hold for the minimum among approximations computable relative to a supplied name of p . There is no Type-2 computable procedure that obtains this minimum uniformly from the name of p . Even when p is promised computable, there is no algorithm that obtains the minimum from an index computing its stream.

Proof. The relation is empty when $p = 0$, so its characteristic function is constant. For every p , the single inequality $x - p < 0$ has coefficient rank one in the two supplied streams and therefore has a uniform one-change approximation relative to p .

Suppose $p > 0$. The element $x = p - 1$ belongs to M : if the stream of p is positive, subtracting one does not change that fact, and if its stream is zero, its integer coordinate is at least one. Let e_N be the stream whose only nonzero coefficient is a one at coordinate N , and put $\delta_N = (e_N, 0)$. In the product topology, $\delta_N \rightarrow 0$, whereas in the arithmetic order $\delta_N > 1$ for every N . Thus

$$x + \delta_N \longrightarrow x, \quad f_p(x) = 1, \quad f_p(x + \delta_N) = 0.$$

Consequently f_p is not continuous. A zero-change approximation is constant in its stage index and equals its first continuous stage map, so none can approximate this function. Its optimum is exactly one.

The map from p to this optimum is discontinuous at zero, as witnessed by $\delta_N \rightarrow 0$. It therefore cannot be Type-2 computable. For the stronger index statement, given a machine index e , define a computable stream h_e that has coefficient one at its halting time if the machine halts and has all other coefficients zero. Computing its j th coefficient requires only a bounded simulation through step j . The element $p_e = (h_e, 0)$ is always a valid computable parameter, and

$$\mu(f_{p_e}) = 1 \iff \text{machine } e \text{ halts.}$$

A uniform algorithm for the minimum would decide the halting problem. $\square$

Scope of the resolution. This theorem completely answers the earlier report’s individual-formula optimization question for parameter-free formulas in the specified rational-stream presentation. A formula with parameters that are allowed to vary as additional input streams is covered after those parameters are made additional free variables. For one fixed arbitrary parameter tuple, the dimension bound remains valid uniformly relative to its supplied names, but the exact finite-poset calculation just proved is not asserted: fixing an infinite parameter stream can prevent the free delayed tail modifications used in the lower bound. The last theorem shows that a uniform algorithm for arbitrary fixed computable parameters is impossible, even for a single unary order formula.

The finite procedure is a decidability result about optimal change counts, not an efficient quantifier-elimination or truth algorithm in the ordinary running-time sense. It can require a large quantifier-free formula, many integer regimes, and many feasible sign patterns. Nor does a bound on changes identify a finite stopping stage for every input. These distinctions are essential to the statement.

Further questions suggested by the theorem. Classify fixed parameter tuples or restricted formulas for which a finite effective replacement for the sign-pattern calculation exists, despite the general obstruction just proved. Study the complexity of computing N_φ from a formula already in quantifier-free form, and compare the finite poset with effective finite difference hierarchies. Seek syntactic fragments for which the integer regimes and sign-pattern chains can be analyzed without full elimination. Finally, formalize the finite-sign compression lemma, rank argument, and delayed-tail lower bound as independently checkable steps in ProveIt.

[write] (Batch 87.) This answers Part I’s [question 14.8](#) exactly for parameter-free formulas ([theorem 100.4](#)) and negatively for uniform optimization at a fixed parameter ([theorem 100.5](#)). The rank bound of [theorem 100.1](#) sharpens the atom-count bound of [theorem 7.3](#). A revision count bounds changes of the output, not running time, the number of stream coordinates read, or arithmetic operations, and no stage after which the answer is final is computed; the Hilbert’s-tenth programme’s review of the arrival makes the same distinction ([section E](#)). Part I’s [question 14.15](#) on sharp oracle bounds is related and stays open.

101 A completeness obstruction beyond invariant metrics

Let M be a commutative cancellative monoid with identity 0. Assume that its algebraic order

$$x \leq y \iff (\exists z \in M) x + z = y$$

is a linear order. In particular it is antisymmetric. Define truncated subtraction by

$$x \dot{-} y = \begin{cases} z, & y + z = x, \quad y \leq x, \\ 0, & x < y. \end{cases}$$

Cancellation makes z unique. Let M carry a metrizable topology in which addition is continuous.

Definition 101.1. For a compatible metric d on M , say that translations are *uniformly equicontinuous at the identity* if, for every $\varepsilon > 0$, there is a neighbourhood U of 0 such that

$$d(b + u, b) < \varepsilon \quad (b \in M, u \in U). \quad (\text{UE0})$$

The same neighbourhood U must work for all b . Joint uniform continuity of addition for the product metric $\max\{d(x, x'), d(y, y')\}$ implies (UE0) by comparing (b, u) with $(b, 0)$.

Theorem 101.2 (Closed translates forced by complete uniformity). *Suppose $x \dot{-} y$ is continuous at (a, a) for every $a \in M$. If a compatible complete metric d satisfies (UE0), every translate $c + M$ is closed in M .*

Proof. Fix $c \in M$ and a convergent sequence $x_n = c + b_n \rightarrow x$. We prove that (b_n) is d -Cauchy. Given $\varepsilon > 0$, select U as in (UE0). Continuity of truncated subtraction at (x, x) , where its value is 0 , gives a neighbourhood V of x such that

$$s, t \in V \implies s \dot{-} t \in U.$$

For all sufficiently large n, m , both x_n, x_m belong to V . If $b_m \leq b_n$, write $b_n = b_m + z$. Cancellation implies $z = x_n \dot{-} x_m$, so $z \in U$, and

$$d(b_n, b_m) = d(b_m + z, b_m) < \varepsilon.$$

If $b_n \leq b_m$, interchange the indices and use symmetry of d . The order is total, so these cases cover every pair. Thus (b_n) is Cauchy. Completeness gives $b_n \rightarrow b \in M$. Continuity of addition gives $x_n = c + b_n \rightarrow c + b$. Uniqueness of limits yields $x = c + b \in c + M$. Metric spaces are sequential, so $c + M$ is closed. $\square$

Corollary 101.3 (No complete metric with uniformly continuous addition; strengthens [proposition 8.4](#)). *Under the hypotheses of the preceding theorem, suppose some translate $c + M$ is not closed. No compatible complete metric satisfies (UE0); in particular no compatible complete metric makes addition jointly uniformly continuous.*

The theorem uses ordinary continuity at each individual diagonal point. It does not assume a uniform modulus for truncated subtraction. Nor does it require translations to be isometries or their inverses to be uniformly continuous.

101.1 Application to the real and rational-stream cones

Put $G = H \overrightarrow{\times} \mathbb{Z}$, with least positive element $1_G = (0, 1)$, and let $M = G_{\geq 0}$ carry the subspace topology of $H \times \mathbb{Z}$. Here either $H = \mathbb{R}$ with its usual topology, or $H = \mathbb{Q}^\omega$ with discrete rational factors and first-difference lexicographic order.

Lemma 101.4. *Truncated subtraction on M is continuous at every diagonal point.*

Proof. Fix $a = (h, k)$. Near (a, a) both integer coordinates equal k , because the integer factor is discrete. For $x = (u, k)$ and $y = (v, k)$, therefore,

$$x \dot{-} y = ((u - v)^+, 0), \quad w^+ = \max\{w, 0\}.$$

For $H = \mathbb{R}$ the positive-part map is continuous. For $H = \mathbb{Q}^\omega$, it is continuous at 0: if the first N coordinates of w vanish, the first N coordinates of w^+ also vanish. Since $u - v \rightarrow 0$ as $(x, y) \rightarrow (a, a)$, the displayed formula proves the claim in both cases. $\square$

Every nonzero element of M has a predecessor, so $1_G + M = M \setminus \{0\}$. This translate is not closed. Indeed, $((1/n), 0) \rightarrow 0$ in the real example, and $(e_n, 0) \rightarrow 0$ in the rational-stream example, where e_n is the rational stream with a single 1 in coordinate n . All these points are nonzero. Consequently Corollary 101.3 applies.

Both spaces are nevertheless Polish and addition is continuous. The obstruction concerns the simultaneous demands of a compatible complete metric and uniform continuity. The incomplete restriction of an invariant metric from the ambient group has uniformly continuous addition, so completeness is essential.

[write] (Batch 87.) Corollary 101.3 answers Part I's question 14.16 negatively for joint uniform continuity of addition, and for uniform equicontinuity of translations at the identity, in $M_{\mathbb{R}}$, $M_{\mathbb{B}}$ and the countable-core models; it strengthens proposition 8.4. The question's other parts (uniform continuity of all translations in one direction, the distortion of their inverses) stay open.

102 Consequences, examples, and the formalization boundary

The results separate three different forms of regularity. An arithmetic operation can be continuous, uniformly continuous for an incomplete metric, or uniformly continuous for a complete compatible metric. The first two hold for addition in the explicit cones; the third is impossible. A definable map can have a simple graph while failing to be a pointwise limit of continuous maps into its actual target. Finally, a truth value can be eventually correct after finitely many revisions without there being an effective stage at which correctness can be announced permanently.

102.1 A compact comparison of the models

Property	$M_{\mathbb{R}}$	M_{ω}	Nonsplit countable-core models
Perfect Polish carrier	Yes, locally compact	Yes, Baire space	Either type, by choice of coefficient group
Addition and fixed division	Continuous	Continuous	Continuous
All definable relations	Δ_2^0	Δ_2^0	Δ_2^0
Retraction onto standard integers	Unique, continuous	Unique, continuous	Absent for the explicit core
Unital additive map to Oz	Exists	Exists	Impossible for the explicit core
Every definable map target-valued Baire one	No; exact slice criterion	Yes, explicit finite-prefix approximants	Stream: yes; real: exact core-slice criterion
Complete metric with uniform addition	Impossible	Impossible	Impossible for both explicit coefficient choices

The metric conclusion extends to the countable-core cones without any new arithmetic assumption. Near a diagonal point the two discrete core coordinates agree, so truncated subtraction is $((u - v)^+, 0_K)$. The diagonal continuity proof is unchanged. The successor image again omits exactly zero, which is nonisolated.

The no-homomorphism result concerns **Oz**, not all of **No**. For example, the countable core itself has the unital ordered additive embedding

$$(u, v) \mapsto (v - u)\omega + v$$

into **No**. Its finite constant coefficient need not be an integer. This example pinpoints why the omnific restriction matters.

102.2 What the continuity procedure returns

For a graph formula defining f , the procedure has a concrete finite output: a Boolean expression describing $\text{Disc}(f)$ in a language with one divisible ordered group sort and one discrete integer sort. It can also decide global continuity in the absence of arbitrary undecidable parameter comparisons. A full implementation would perform the following transformations.

1. Relativize the graph to the signed group and eliminate its quantifiers; extract a finite list of affine quotient candidates.
2. Partition by the first successful candidate, and eliminate the remaining quantifiers in its cell conditions.
3. For each possible pattern of zero and nonzero leading forms, solve the finite rational directional-feasibility problems for the accessible cells.
4. Compare every accessible candidate with the actual candidate at the point, using the exact continuity criterion.

5. If global continuity is desired, eliminate the input coordinates in the two separate sorts.

This output language is necessary, even for a simple operation. In the monus example the condition that two leading coordinates are equal detects standard finite difference, which is unavailable as a Presburger definition of the standard cut. Thus the compiler is an analysis tool for the presentation, not an additional definability theorem inside the original arithmetic.

102.3 What can be formalized directly in Provelt

The inspected Lean syntax interprets affine expressions in lists of ordinary integers, and its central theorem quantifies over `List Int` [60, 61]. It therefore cannot be cited as an existing proof that every nonstandard $\mathbb{Z}$ -group validates the same elimination algorithm. That semantic generalization has to be implemented and proved.

A useful implementation order is as follows. First define an abstract discretely ordered abelian group with a named least positive unit and standard division with remainder. Prove the full-coset finite-boundary lemma and the model-uniform elimination theorem. These algebraic steps support the finite-candidate theorem and the arithmetic correctness of all the explicit models.

Next formalize rational sign feasibility, finite sign compression, and the rank argument. These are finite algebraic statements plus an elementary eventual-stabilization argument. The exact minimum theorem then separates into a finite poset calculation and the delayed-tail lower-bound construction. The existing integer eliminator can decide the feasibility of each integer regime once its relation to the abstract semantics has been established.

The topological component can be developed independently: the positive-cone construction, the finite-branch continuity lemma, the rational-direction realization, and the closed-translate metric theorem. The last of these is particularly small and has no dependence on surreal numbers, descriptive set-theoretic classification, or Presburger quantifier elimination.

Finally, the omnific application needs the additive constant map and an ordered embedding of a set-sized divisible group into **No**. The explicit two-element contradiction only needs the former. It is therefore a shorter initial formalization target than the general surreal embedding criterion.

103 Further research questions and concrete next steps

The questions below arise after the proved results. They are not presented as already solved, and their scopes are narrower than a blanket request for a new theory of all Polish arithmetic.

Question 103.1 (Fixed-parameter revision complexity). For a fixed parameter tuple, classify the exact degree of difficulty of computing the optimal revision number. The uniform decision problem is already obstructed by the threshold example above. Which representations with certified zero tests, rational dependence data, or finite support restore an effective finite classification?

The next attainable case is a tuple with explicitly supplied finite supports. Its known initial coordinates can be incorporated into the sign-pattern calculation, after which later perturbations see zero parameter tails. The proof must account for the transition between these fixed initial coordinates and the freely variable tail.

Question 103.2 (Complexity of the exact finite algorithms). How hard is it to compute the optimal revision number or the exact continuity condition from a quantifier-free graph? Can the sign-pattern search be compressed using the intersection lattice of the rational hyperplanes, rather than enumerating all ternary patterns?

The theorems prove decidability, not a good running-time bound. It is useful to separate the cost of Presburger elimination from the combinatorial cost of feasible sign patterns and alternating chains. A small verified implementation of the latter would already be a substantive addition to the repository.

Question 103.3 (Effective computation for countable cores). For $D \vec{\times} K$, where K is a countable $\mathbb{Z}$ -group equipped with the discrete topology, classify the oracle information needed for an effective continuity compiler and for finite-prefix evaluation of definable maps. How does the answer depend on the chosen presentation of K and the names of its parameters?

The purely topological continuity and Baire conclusions extend by the same finite-branch proofs, as established above. Computation must distinguish knowing a core element as a discrete topological point from having algorithms for its arithmetic. A countable carrier does not by itself provide decidable operations or order. A useful first case is a computable presentation with a decidable atomic diagram and named standard quotient functions.

Question 103.4 (Connected Polish monoid presentations). Can a connected uncountable Polish space support Presburger addition when no signed Polish group presentation is supplied?

The automatic continuity of standard residues settles the negative answer for cones inheriting a Polish group topology. It does not settle the question for arbitrary monoid topologies. The essential missing step is a suitable group-completion theorem, or an example showing why such a completion cannot be demanded.

[write] This is Part I's [question 14.3](#) and Part III's [question 43.4](#), restricted to monoid topologies with no signed Polish group presentation; see the note after [theorem 95.4](#).

Question 103.5 (Rigidity with continuous addition). Which rigid Presburger models admit a Polish topology with continuous addition and Borel order? How does the answer change when one demands a Polish topology on the entire signed group?

The split models have nontrivial positive dilations of their divisible part. The nonsplit construction here retains such a divisible coefficient part and is not a rigid example. Jeřábek's structural theory [4] gives algebraic constraints, but these do not automatically produce or exclude a prescribed Polish topology.

[write] Part I's [question 14.12](#) asks the first sentence.

Question 103.6 (Open divisible subgroups and continuous core decompositions). In which Polish Presburger groups with Borel order is the maximal divisible subgroup $\ker r$ open? When does the algebraic decomposition admit a continuous splitting with a countable discrete reduced core?

The transfer examples have an open divisible factor and a countable reduced core. The automatic-continuity theorem proves a different, narrower statement: if a retraction onto the standard integers exists algebraically, it is continuous. It does not classify arbitrary countable cores or make every algebraic decomposition topological.

Question 103.7 (Sufficiency in the metric problem). For a cancellative linearly ordered Polish monoid with diagonal continuity of monus, is closedness of all translates part of a useful sufficient criterion for a compatible complete metric with uniformly continuous addition? What additional hypotheses are necessary?

The closed-translate theorem supplies a necessary condition only. One should first study concrete subclasses, such as cones with continuous ambient subtraction or locally compact monoids, and keep the uniform structure as part of the data.

Question 103.8 (Embeddings between different countable cores). Describe continuous elementary embeddings between $D \vec{\times} K$ and $D' \vec{\times} K'$, especially when the cores are reduced and nonsplit. Which information in the coherent residue image is preserved by a topological or ordered embedding?

The automatic continuity theorem shows that all finite residue maps are topologically visible on a Polish signed group. The distinction between their separate continuity and a single integer retraction is therefore a natural invariant to retain in such a classification.

Question 103.9 (Minimal foundational and formalization requirements). What is the weakest convenient subsystem in which the explicit stream model, the rank bound, and the finite-pattern optimization can be proved for coded inputs? Can the mathematical and computational parts be isolated into small reusable proof-assistant modules?

This is a question about the positive construction and its classification algorithms. It is separate from Glazer’s published reverse-mathematical question about the strength of the topological Tennenbaum theorem. A proof of one would not be a proof of the other.

[write] Compare Part I’s [question 14.2](#). Source 14 ([Part VIII](#)) claims an answer to Glazer’s published question; it is a different theorem from the positive construction, as source 13 says.

Part VIII

An ATR_0 bound for Glazer’s Question 1 (claimed)

[write] This part is source 14 of batch 87, *Diophantine Definability and Topological Arithmetic: An ATR_0 proof of Glazer’s theorem, cofinal unary discontinuities, and nonsplit Presburger models* (dated October 2026, its records October 4; “prepared with ChatGPT”), pinned to repository commit 6fef5383b. All unmarked text is source 14’s; its labels carry the prefix `pma:atr:`.

What this part claims, and its status

Source 14 claims that Glazer’s Theorem 2 (no model of $\mathbf{Q} + \mathbf{I}\Delta_0 + \mathbf{Exp}$ on an uncountable Polish space has continuous addition or continuous multiplication when both are Borel) is provable in $\mathbf{ATR}_0$, under the coded-model convention of its Section 2 ([section 105.2](#)): an affirmative answer to Glazer’s Question 1, which this report printed as open ([question 43.1](#)). It proves more: discontinuous translations and discontinuous multiplications by internal powers of two occur cofinally ([theorem 104.1](#)).

This answer is printed **as claimed**. The manuscript is AI-assisted and unrefereed. Parts of its $\mathbf{ATR}_0$ bookkeeping rest on inputs it declares rather than proves (uniform perfect-set and order homogenization, coded Baire approximation, analytic separation, and the evaluation of countable Borel iterations; note in [section 105.4](#)), and its author did not see the Gaifman–Dimitracopoulos chapter whose internal MRDP theorem it relies on, taking the statement from Chen, Müller and Yokoyama. Neither the intake nor the repository’s Hilbert’s-tenth programme, which reviewed the archive’s provenance and read the MRDP input and the automatic-Borel argument, has reviewed the whole proof ([section E](#)). It is an upper bound only. Nothing in it is formalized.

The construction it shares with Parts VII and IX. Source 14’s Section 10 ([section 113](#)) builds the locally compact Polish Presburger model $(\mathbb{R} \overrightarrow{\times} K_*)_{\geq 0}$ on a countable nonsplit $\mathbb{Z}$ -group K_* . That group is the member G_{ζ_*} , $\zeta_* = \sum_k k!$, of the factorial family of the discrete-initial-subgroups-and-omnific-normalization report, and the omnific obstruction it proves is that report’s `isg:cf:cor:unit`; source 14 cites neither (notes after [lemma 113.1](#) and [proposition 113.3](#)). The construction is source 13’s countable-core transfer ([theorem 94.1](#)), printed in general in [Part VII](#); [proposition 113.2](#) is printed here with its own proof as a flagged instance.

Notation. Source 14’s countable group D of its Section 10 is printed K_* , and the generic countable $\mathbb{Z}$ -group of the commentary on its question on nonsplit Presburger topology is printed K , because D is the divisible coefficient group in Parts I and VII; its map $\Psi = \omega\Phi$ is printed Φ_ω , because Ψ is Part I’s Baire homeomorphism; its Cantor space 2^ω is printed $2^\mathbb{N}$. In Sections 7 and 8 of source 14 ([sections 110](#) and [111](#)) the letter D is a Borel set of Cantor space, and $e(x) = 2^x$ is the internal exponential, not a unit; t_n , q_n , X_n are its root scales, unrelated to the variable t of Parts III and V. The equation tags (3.1)–(10.6) are source 14’s own: (3.x) are in [section 106](#), (5.x) in [section 108](#), (6.x) in [section 109](#), (7.x) in [section 110](#), (8.x) in [section 111](#) and (10.x) in [section 113](#).

Provenance. Source 14’s finite checks are shipped as `code/14-atr0-tennenbaum-verify_finite_lemmas.py`, which prints its record (`data/14-atr0-tennenbaum-finite_check_results.txt`); its proof audit is `14-atr0-tennenbaum-PROOF_AUDIT.txt`. Its source audit is not shipped, because it contains a third party’s personal profile address; its verified content is summarized in [section E](#).

104 The question, the intersection, and the contributions

Elliot Glazer’s research in set theory and formal systems, together with his work on foundations of proof assistants, makes the interface between arithmetic models, definability,

and formalization a direct match for Provolt; an institutional author biography confirms these interests [77]. The particularly useful source is his *A Topological Tennenbaum Theorem* [1]. The relevant part of Provolt is its work on Diophantine exponentiation, arithmetic coding, and Polish models of omnific arithmetic [73, 75].

In ordinary mathematics, Glazer proves that a model of $\mathbf{Q} + \mathbf{I}\Delta_0 + \mathbf{Exp}$ on an uncountable Polish space, with both arithmetic operations Borel, cannot have either operation continuous. His Question 1 asks whether this conclusion is provable in $\mathbf{ATR}_0$. The unresolved issue identified there concerns the descriptive complexity of exponentiation and the choices used in the multiplicative argument. We give a proof using internal MRDP, Borel regularization, and category.

Here and below, *uncountable* is external to the arithmetic object structure; its precise coded meaning is specified in Section 105.2. Arithmetic comparisons refer to the model's order. A Polish space is a separable space admitting a complete compatible metric. Neither that topology nor its metric is assumed to agree with the arithmetic order topology.

Theorem 104.1 (Main theorem: cofinal unary discontinuity). *Work in $\mathbf{ATR}_0$, with the coded-presentation convention of Section 105. Let $\mathcal{M}$ have an uncountable Polish carrier M and Borel addition and multiplication.*

- (i) *If $\mathcal{M} \models \mathbf{Q} + \mathbf{IOpen}$, then for every $b \in M$ there is an $a \geq b$ such that*

$$\tau_a : M \longrightarrow M, \quad \tau_a(x) = a + x$$

is discontinuous.

- (ii) *If $\mathcal{M} \models \mathbf{Q} + \mathbf{I}\Delta_0 + \mathbf{Exp}$, then for every $b \in M$ there is an $a \geq b$ such that*

$$\lambda_a : M \longrightarrow M, \quad \lambda_a(x) = 2^a x$$

is discontinuous.

In particular neither addition nor multiplication is separately continuous, and neither is jointly continuous, under the hypotheses of (ii).

The two witnesses a need not be the same. *Cofinal* here means unbounded in the arithmetic order; it makes no assertion about topological density, cardinality, or category of the set of parameters.

Part (i) uses a weak theory already covered by Glazer's additive obstruction. The substantive new connection for Question 1 is the passage from a *provably correct internal* Diophantine definition to a Borel exponentiation map. Cofinality and the unary formulation are additional consequences of the proof supplied here.

A second contribution records a concrete distinction that is useful for the repository's surreal program.

Theorem 104.2 (A nonsplit Presburger boundary; an instance of theorem 94.1). *There is an explicitly described uncountable, locally compact, perfect Polish model of Presburger arithmetic with jointly continuous addition whose standard-integer subgroup is not a direct summand of its Grothendieck group. The model has an ordered additive embedding into the surreal numbers using expressions*

$$x\omega^2 + q\omega + p \quad (x \in \mathbb{R}, p, q \in \mathbb{Q}),$$

but has no additive map to the canonical omnific integers that sends its unit to 1.

The abstract existence of nonsplit $\mathbb{Z}$ -groups is classical [4]; we supply an elementary explicit example, its Polish realization, and the omnific obstruction. We do not claim the first discovery of nonsplit Presburger groups.

104.1 What was already present in the repository

The audit used the snapshot

```
6fef5383b126be343cccc9baf47081ef37ab5afe.
```

The inspected Polish-arithmetic article already gives affirmative constructions for Glazer’s Presburger Question 2, including real and Baire-space carriers, detailed definability results, and omnific realizations. It explicitly does not answer Question 1. The three original incoming Glazer-related bundles were also inspected; none contains the internal-MRDP argument developed here [75, 76].

The existing Lean module on Diophantine exponentiation proves the graph theorem over standard $\mathbb{N}$, with interfaces including `naturalPowGraph_diophantine` and `power_polynomial_exists`. This is relevant mathematical infrastructure, but it is not itself an internal proof over $\text{I}\Delta_0 + \text{Exp}$. Confusing those two statements would leave a real gap in the present argument. Section 114 explains the distinction.

104.2 Contribution and dependency ledger

Statement	Role	Status here
Internal MRDP over $\text{I}\Delta_0 + \text{Exp}$	Correctness in nonstandard models	Classical input [68, 69]
Borel exponentiation	Removes the projective complexity obstacle	Proved from that input
Common Cantor restriction	Regularizes countably many Borel maps together	Proof and coding supplied
Cofinal unary obstructions	Strengthens the continuity conclusion	Full proofs supplied
ATR_0 upper bound	Answers the question in the cited 2023 paper	Proof with explicit coding
Nonsplit Polish model and omnific obstruction	Complementary Presburger application	Explicit construction and proof
An optimal lower bound or Lean verification	Further research	Not claimed

[write] “The inspected Polish-arithmetic article” [75] is this report as it stood at source 14’s pin 6fef5383b: then the delivered text of batch-86 source 05, now the base of Part I. Parts II–V were added later the same day, and none of them answers Question 1 (section 1.1). The “three original incoming Glazer-related bundles” are batch-86 manuscripts 03, 04 and 05, now Parts I and II; [76] is a routing note of the Hilbert’s-tenth programme. The Lean declarations named here exist at the current state in `Logic/PeanoArithmetic/ListCoding/Lean/PAListCoding/ExponentiationDiophantine.lean` and are proved there for the standard natural numbers; as source 14 says, they are not the internal statement it needs.

105 Framework and the established inputs

105.1 Object arithmetic and external mathematics

The object language contains $0, 1, +, \cdot, <$, with the usual discretely ordered semiring conventions. The order may equivalently be recovered from addition:

$$x < y \iff \exists z (x + z + 1 = y).$$

We use IOpen for induction for quantifier-free formulas and $\text{I}\Delta_0$ for induction for bounded formulas. Write

$$T_{\text{exp}} = \text{Q} + \text{I}\Delta_0 + \text{Exp}.$$

The exponentiation axiom is the ordinary first-order assertion of total exponentiation through finite-sequence coding, not an assumption that an external exponential operation has already been supplied as a Borel map. The standard equivalent formulations of elementary arithmetic may be used.

The external theory ATR_0 is second-order arithmetic with arithmetical transfinite recursion. Its natural role is that Borel codes may require transfinite recursion to evaluate. The object theory and the external theory do different jobs: T_{exp} controls the algebra in M , while ATR_0 controls codes for the Polish space and its Borel data.

105.2 Coded-model convention

A coded Polish space is specified by a countable dense metric presentation and its completion; a point is represented by a suitable fast Cauchy name. Equality of names means equality of the represented points. Borel maps are specified by standard Borel codes, including uniform codes for inverse images of a countable basis. All functions and relations must respect equality of point names.

For the reverse-mathematical statement, a coded space or coded Borel subset A is *uncountable* when no coded sequence of point names covers it. In symbols, this is

$$\neg \exists (x_n)_{n \in \mathbb{N}} \forall x (x \in A \longrightarrow \exists n \ x = x_n),$$

where membership is interpreted using the supplied Borel code. The word external does not introduce an ambient cardinality predicate unavailable in second-order arithmetic.

The notation $\mathcal{M} \models T_{\text{exp}}$ is ordinary semantic notation in the mathematical exposition. For the reverse-mathematical claim it is interpreted *instance by instance*: every particular first-order formula needed by the proof is translated directly into second-order arithmetic, with its variables ranging over point names. We do not postulate a truth predicate for all first-order formulas on an uncountable carrier.

Lemma 105.1 (Fixed-proof interpretation). *Let π be a fixed finite first-order derivation of a sentence ψ from a theory T , and let Γ_π be the finite collection of nonlogical axiom instances actually used. For a coded real-domain structure, the direct translations of those instances imply the direct translation of ψ . No uniform satisfaction predicate is required.*

Proof. Translate each of the finitely many formulas in π . Object quantifiers become quantifiers over valid point names, and atomic formulas use the specified equality and operation relations.

Logical axioms remain valid under this interpretation; modus ponens and generalization remain valid as well. Thus the translated finite derivation proves

$$\bigwedge_{\theta \in \Gamma_\pi} \theta^M \longrightarrow \psi^M.$$

This is a fixed syntactic translation. It neither constructs a set of truths nor proves uniform soundness for arbitrary proof codes. The elementary logic involved is available already in the base second-order framework, once the presentation laws are assumed. $\square$

Consequently the main argument can be stated with a sufficient fixed finite arithmetic package, giving an actual second-order sentence. Using the familiar model-theoretic notation is harmless provided this convention is kept explicit.

105.3 The exact internal MRDP input

The theorem of Gaifman and Dimitracopoulos needed here is the following proof-theoretic form of MRDP [68]. Its exact statement is explicitly recorded in the recent research paper of Chen, Müller, and Yokoyama: see its Section 1.4 and Definition 5.2 [69].

Theorem 105.2 (Gaifman–Dimitracopoulos; established input). *For each fixed bounded arithmetic formula $\theta(\bar{x})$, there are fixed polynomial terms $P(\bar{x}, \bar{u})$ and $Q(\bar{x}, \bar{u})$, with natural-number coefficients, such that*

$$T_{\text{exp}} \vdash \theta(\bar{x}) \iff \exists \bar{u} \, P(\bar{x}, \bar{u}) = Q(\bar{x}, \bar{u}).$$

The same conclusion holds for each fixed Σ_1 formula by including its initial existential variables in $\bar{u}$.

Only one standard instance, for the graph of 2^x , is essential to the multiplicative theorem. No uniform formalization of the whole MRDP theorem inside the external theory is needed. The theorem’s *internal correctness* is essential: a polynomial that merely agrees with exponentiation on standard integers would be insufficient.

The original 1982 contribution is a chapter in monograph no. 30 of *L’Enseignement Mathématique*, not an article in journal volume 30. Its bibliography is confirmed by the author’s publication list [78]. An accessible scan of the original chapter was not obtained in this audit; the precise input above is corroborated by [69], rather than inferred from a casual summary.

[write] [Theorem 105.2](#) is a declared input. Source 14’s author did not see the original chapter [68] and took the statement from [69]; the intake did not check either source.

105.4 Coded descriptive-set-theoretic inputs

We use the following standard results in their coded forms, available in ATR_0 ; the relevant implementation of the perfect-set and order steps is also described in Glazer’s Optimization section [1, 70, 71].

1. Disjoint analytic sets admit a Borel separator. In particular, an analytic set with analytic complement has a Borel code.
2. Borel sets have the property of Baire, with an open approximant and a coded countable cover of the error by closed nowhere-dense sets.

3. An uncountable Borel subset of a Polish space admits a continuous injective Cantor map. For a Borel linear order, this may be refined to an increasing continuous Cantor map.
4. The preceding perfect-set/order construction can be carried out for a coded sequence of uncountable Borel sets without adding a countable-choice principle for arbitrary relations.

For the order refinement, pull the arithmetic order back along a Cantor injection, color pairs by agreement with lexicographic order, and use perfect-set homogeneity for a Borel two-coloring of pairs. Reverse the Cantor order if necessary. The uniform version asserts the existence of one real coding the sequence of embeddings; it does not assert an effective or Borel selection operator on all input codes. Apply the pseudohierarchy construction relative to the join of the input codes, then one tagged Baire approximation to the pullback pair-colorings, a common perfect tree avoiding all errors, and the coded open-color homogeneous-tree construction in each section. This is the common coding and perfect-tree construction used in Glazer's optimization.

These are declared inputs, not hidden consequences of elementary metric completeness. The new uses of them, particularly the tagged continuity construction, are proved below.

[write] These four coded inputs, together with [lemma 108.2](#) and the evaluation of countable Borel iterations along a pseudohierarchy in [section 112](#), are where the ATR_0 bookkeeping of source 14 rests on statements it declares rather than proves. Whether each holds in ATR_0 in the uniform form used is part of what an independent review would have to check.

106 Internal Diophantine definitions become Borel maps

106.1 The functional projection principle

Lemma 106.1 (Total analytic functions). *Let X, Y be Polish spaces. Suppose $R \subseteq X \times Y$ is analytic and for every $x \in X$ there is exactly one y with $R(x, y)$. Then the function $f(x) = y$ has a Borel graph and is a Borel map. From the analytic data and the asserted totality and uniqueness, the needed codes can be obtained in ATR_0 .*

Proof. The complement of the graph is analytic because

$$\neg R(x, y) \iff \exists z (R(x, z) \wedge z \neq y).$$

Here totality supplies an output, and uniqueness identifies all other pairs as nongraph pairs. Analytic separation therefore gives a Borel graph.

To obtain a map code explicitly, let $(U_j)_{j \in \mathbb{N}}$ be a countable basis of Y . Form the tagged sets

$$\begin{aligned} A &= \{(j, x) : \exists y (R(x, y) \wedge y \in U_j)\}, \\ B &= \{(j, x) : \exists y (R(x, y) \wedge y \notin U_j)\}. \end{aligned}$$

They are analytic, disjoint, and exhaust $\mathbb{N} \times X$. One separator is therefore exactly A . Its sections give all the codes for $f^{-1}(U_j)$ at once. This avoids a countable sequence of independently chosen separators. $\square$

Theorem 106.2 (Automatic Borel definability). *Suppose $\mathcal{M} \models T_{\text{exp}}$ has a Polish carrier and Borel arithmetic operations. If a fixed Σ_1 formula $\varphi(\bar{x}, y)$ provably defines a total single-valued function*

in T_{exp} , then its interpretation in $\mathcal{M}$ is a Borel map. Also, any relation whose positive and negative definitions are provably Σ_1 over T_{exp} is Borel.

Proof. Use Theorem 105.2 to write

$$\varphi(\bar{x}, y) \longleftrightarrow \exists \bar{u} P(\bar{x}, y, \bar{u}) = Q(\bar{x}, y, \bar{u})$$

inside T_{exp} . Polynomial evaluation in $\mathcal{M}$ is Borel: the polynomial is a fixed finite expression built from Borel operations. Equality of two values is a closed relation on the carrier, so the equality locus is Borel. Projection over the finite tuple $\bar{u}$ makes the displayed relation analytic. Lemma 106.1 applies.

For a relation with positive and negative Σ_1 definitions, apply internal MRDP to both definitions and then analytic separation. The transfer of each fixed arithmetic derivation is justified by Lemma 105.1. $\square$

Corollary 106.3. *Every fixed bounded arithmetic relation is Borel in a Borel presentation of a model of T_{exp} .*

Proof. A bounded formula and its negation are both bounded. Apply the relational part of Theorem 106.2. $\square$

This corollary is not obtained by pretending that an internally bounded quantifier ranges over an externally finite set. In an uncountable nonstandard model, an interval $[0, a]$ can be uncountable. The Diophantine translation is doing substantive work.

106.2 Exponentiation is an instance of the principle

For clarity, one may use the standard remainder-based sequence coding. Let

$$\beta(u, v, i) = u \bmod (1 + (i + 1)v).$$

The assertion $\beta(u, v, i) = r$ can be expanded using a bounded quotient witness and a polynomial equality, so it is bounded arithmetic syntax. The graph formula

$$\begin{aligned} E(x, y) \iff \exists u \exists v (\beta(u, v, 0) = 1 \wedge \beta(u, v, x) = y \\ \wedge \forall i < x \beta(u, v, i + 1) = 2\beta(u, v, i)) \end{aligned} \quad (3.1)$$

is Σ_1 after expanding the remainder notation. All intermediate remainders and quotient witnesses can be bounded by terms in the code parameters.

The usual arithmetic properties, proved internally in T_{exp} , are

$$e(0) = 1, \quad e(1) = 2, \quad e(a + b) = e(a)e(b), \quad a < b \implies e(a) < e(b), \quad (3.2)$$

where $e(x)$ is the unique y satisfying $E(x, y)$. Only fixed proofs of these facts and of totality and uniqueness are used.

Corollary 106.4 (Borel internal exponentiation). *In every Borel Polish presentation of a model of T_{exp} , the internally defined function $e(x) = 2^x$ is Borel. This conclusion, including a Borel-map code, follows in ATR_0 .*

Proof. Apply Theorem 106.2 to the fixed formula E . $\square$

The auxiliary Diophantine witnesses need not be unique. Only the output y must be unique. Nor do we require that those witnesses depend continuously or Borel-measurably on x . The argument concerns projection and the resulting function, not a selection of polynomial solutions.

107 A common Cantor set of continuity

The next lemma explains precisely how the Borel code for e replaces higher-complexity choices.

Lemma 107.1 (Simultaneous Cantor regularization). *Let Y be a Polish space and let $(H_n)_{n \in \mathbb{N}}$ be a uniformly coded sequence of Borel maps $2^{\mathbb{N}} \rightarrow Y$. There is a continuous increasing injection $F : 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$, for the lexicographic order, such that every $H_n \circ F$ is continuous. The construction is available in ATR_0 .*

Proof. Let (U_j) be a countable basis of Y . Apply the Baire-property construction to the single Borel set

$$A = \{(n, j, r) : H_n(r) \in U_j\} \subseteq \mathbb{N} \times \mathbb{N} \times 2^{\mathbb{N}}.$$

It gives open approximants $V_{n,j}$ and, uniformly, a countable family of closed nowhere-dense sets covering every symmetric difference

$$H_n^{-1}(U_j) \Delta V_{n,j}.$$

Let G be the complementary dense G_δ subset of $2^{\mathbb{N}}$. On G , every basis preimage for every H_n is relatively open. Thus every $H_n|_G$ is continuous.

Write $G = \bigcap_\ell W_\ell$, with each W_ℓ dense open. Construct nonempty clopen cylinders V_s , $s \in 2^{<\omega}$, so that the two successors lie inside $V_s \cap W_{|s|}$, are disjoint, occur in left-to-right lexicographic order, and have lengths tending to infinity with $|s|$. Each branch has a unique intersection point. The branch map F is continuous, injective, increasing, and has range in G . Therefore all $H_n \circ F$ are continuous. $\square$

The use of a single tagged set matters in weak metatheory. It produces one coded family of approximations and errors; it does not appeal to unrestricted countable choice. The conclusion also does not say that the original H_n are continuous on all of Cantor space.

108 Arithmetic scales above any prescribed parameter

We next replace informal fractional powers by exact floor operations. This also makes the cofinal strengthening transparent.

108.1 Borel arithmetic order, division, and square roots

Lemma 108.1. *In a Borel presentation of a model of $\mathbf{Q} + \mathbf{IOpen}$, the arithmetic order is Borel. Floor division by a positive element, its remainder, and the function $x \mapsto \lfloor \sqrt{x} \rfloor$ are Borel. Truncated subtraction is also Borel.*

Proof. Both $x < y$ and its complement have existential definitions:

$$x < y \iff \exists z (x + z + 1 = y), \quad x \geq y \iff \exists z (y + z = x).$$

They are analytic, so separation gives a Borel order.

Open induction supplies Euclidean quotient and remainder and the largest q with $q^2 \leq x$. For example, the quantifier-free inequality $q^2 > x$ holds at $q = x + 1$; the least-transition argument supplied by open induction gives a least such q . The division argument uses $dq > x$, with $d > 0$, in the same way.

The characterizing graphs

$$x = dq + r, \quad r < d, \quad \text{and} \quad q^2 \leq x < (q+1)^2$$

are Borel and define unique total outputs on their stated domains. For coding purposes, extend quotient and remainder by zero at divisor zero, so their domains are the Polish space M^2 . The tagged inverse-open argument of Lemma 106.1 supplies map codes. Truncated subtraction has the Borel graph

$$(y \leq x \wedge y + z = x) \vee (x < y \wedge z = 0),$$

which again gives a unique total output. No continuity of subtraction is asserted. $\square$

Lemma 108.2 (Uniform increasing Cantor maps; established input). *For a Borel linear order on an uncountable Polish space, every uncountable Borel subset contains an increasing continuous Cantor image. Given a coded sequence of such subsets, the images can be supplied by a single real coding a sequence of embeddings in ATR_0 .*

This is the perfect-set/order-homogenization input described in Section 105. Its uniform form is the one used in Glazer's weak-metatheory optimization [1]. No uncountable choice of points is involved.

108.2 The exact scale construction

Fix any threshold $b \in M$. An increasing Cantor image in M supplies a c_0 for which $[0, c_0]$ is uncountable: take its upper endpoint. Enlarge it to

$$c = \max\{c_0, (b+2)^4\}.$$

Define, for external $n \in \mathbb{N}$,

$$t_0 = c, \quad t_{n+1} = \lfloor \sqrt{t_n} \rfloor, \quad q_n = \lfloor c/t_n \rfloor. \quad (5.1)$$

Every interval $[0, t_n]$ is uncountable. To verify this without induction on an assertion of uncountability, fix an arbitrary external n . At each level $i < n$, quotient and remainder upon division by $t_{i+1} + 1$ inject $[0, t_i]$ into $[0, t_{i+1}]^2$, since

$$t_i < (t_{i+1} + 1)^2.$$

Iteration along the finite binary tree gives an injection $[0, c] \hookrightarrow [0, t_n]^{2^n}$. A coded enumeration of $[0, t_n]$ would enumerate its finite Cartesian power. Reconstruct the tree upward with $x = (t_{i+1} + 1)q + r$, retaining results in $[0, c]$ and using zero otherwise. Every element of $[0, c]$ would occur, contradicting its uncountability. Correctness of these finite calculations uses arithmetic induction relative to their evaluation traces, not induction on a projective uncountability predicate. In particular every t_n exceeds every standard integer, all the divisions in (5.1) are legitimate, and $q_n \geq 1$.

Set

$$X_n = \{q_n i : i < t_{n+2}\}. \quad (5.2)$$

Each X_n is uncountable because multiplication by $q_n > 0$ is injective. It is Borel by the direct description

$$X_n = \{x < q_n t_{n+2} : \text{rem}(x, q_n) = 0\}.$$

Also $X_n \subset [0, c)$, because $q_n t_n \leq c$ and $t_{n+2} < t_n$.

The scale separation has a short exact proof:

$$q_{n+1} \geq q_n t_{n+1} \geq q_n t_{n+2}^2 > 8q_n t_{n+2}. \quad (5.3)$$

The first inequality follows from $c \geq q_n t_n \geq q_n t_{n+1}^2$, and the last uses $t_{n+2} > 8$.

We use the shifted sets X_{n+1} . By Lemma 108.2, choose uniformly increasing continuous injections

$$f_n : 2^{\mathbb{N}} \longrightarrow X_{n+1}.$$

Write $\epsilon_k = \chi_{\{k\}}$, so $\epsilon_k \rightarrow 0^\omega$. For a suitable increasing Cantor restriction F , or simply $F = \text{id}$, put

$$x_n = f_n(F(0^\omega)), \quad h_{k,n} = f_n(F(\epsilon_k)) - x_n. \quad (5.4)$$

The subtraction denotes the unique nonnegative arithmetic difference, not an assumed continuous operation.

Lemma 108.3 (Separated perturbations). *The parameters in (5.4) satisfy*

$$b < h_{k,n}, \quad q_{n+1} \leq h_{k,n} < q_{n+1} t_{n+3}, \quad x_n < c. \quad (5.5)$$

Every choice $h_n = h_{k_n,n}$ satisfies

$$h_{n+1} > 8h_n. \quad (5.6)$$

If $F = \text{id}$, then $x_n + h_{k,n} \rightarrow x_n$. If e is Borel and F is obtained by Lemma 107.1 for $H_n = e \circ f_n$, then

$$e(x_n + h_{k,n}) \longrightarrow e(x_n). \quad (5.7)$$

Proof. The differences are positive multiples of q_{n+1} , because both endpoints lie in X_{n+1} . They are smaller than $q_{n+1} t_{n+3}$. Moreover

$$q_1 \geq t_1 \geq (b+2)^2 > b,$$

and (5.3) makes the subsequent q_n increase. The scale inequality gives (5.6), independently of the choices of k_n . The convergence assertions follow from the specified continuity and from $\epsilon_k \rightarrow 0^\omega$. $\square$

Although t_n decreases and remains internally infinite for every standard n , no paradox is involved. The recursion is externally countable, and internal nonstandard intervals need not behave like standard finite intervals.

109 A quantitative fusion lemma

The following metric estimate will serve both operations. Suppose finite vertices $V_s \in M$, $s \in 2^{<\omega}$, are constructed level by level, with

$$V_{s0} = V_s.$$

Assume vertices at each level are pairwise distinct. Put

$$\Delta_0 = 1, \quad \Delta_n = \min(\{1\} \cup \{d(V_s, V_t) : s, t \in 2^n, s \neq t\}) \quad (n > 0).$$

Zero extensions imply $\Delta_{n+1} \leq \Delta_n$.

Lemma 109.1 (Fusion estimate). *If*

$$d(V_s, V_{s1}) < 2^{-n-4} \Delta_n \quad (s \in 2^n), \quad (6.1)$$

then

$$z_r = \lim_N V_{r|N}$$

exists for every $r \in 2^{\mathbb{N}}$, uniformly in r , and the map $r \mapsto z_r$ is a continuous injection. More precisely,

$$d(V_{r|N}, z_r) \leq 2^{-N-3} \Delta_N, \quad (6.2)$$

and if $r|N \neq s|N$, then

$$d(z_r, z_s) \geq (1 - 2^{-N-2}) \Delta_N > 0. \quad (6.3)$$

Proof. For $i \geq N$, $\Delta_i \leq \Delta_N$. The sum of the tail edge bounds is at most

$$\Delta_N \sum_{i=N}^{\infty} 2^{-i-4} = 2^{-N-3} \Delta_N.$$

Completeness gives the limit and (6.2). Since $\Delta_N \leq 1$, the bound also gives a common Cauchy rate. The finite-level maps $r \mapsto V_{r|N}$ are locally constant, so uniform convergence makes the limit map continuous.

If the prefixes at level N differ, their vertices are at least Δ_N apart. Subtracting the two tail bounds gives (6.3). Thus distinct branches have distinct limits. $\square$

At an implementation level, the search for (6.1) need not compare arbitrary real numbers exactly. Choose rational lower bounds for the finitely many positive separation distances and dovetail a search for certified strict inequalities. The convergence used below guarantees termination at each stage. This is a recursive construction with supplied codes, not an arbitrary sequence of choices.

110 Cofinal discontinuity of additive translations

We first give a full additive proof, including the category lemma that will be reused.

Lemma 110.1 (Category zero–one law for finite changes). *If $D \subseteq 2^{\mathbb{N}}$ has the property of Baire and membership in D is invariant under changing finitely many coordinates, then D is meager or comeager.*

Proof. If D is nonmeager, it is comeager in some nonempty basic cylinder. Finite bit flips take this cylinder to every cylinder of the same length. Invariance makes D comeager in every one of them. There are finitely many such cylinders and they cover Cantor space, so D is comeager. $\square$

Theorem 110.2. *Under the hypotheses of Theorem 104.1(i), discontinuous translations occur cofinally in the arithmetic order.*

Proof. Fix $b \in M$, and suppose for contradiction that every translation τ_a with $a \geq b$ is continuous. Use Section 108 with $F = \text{id}$. For every $m \geq 2c$, the translating constant $m - x_n$ is larger than c , hence larger than b . Therefore

$$m + h_{k,n} = (m - x_n) + (x_n + h_{k,n}) \longrightarrow m. \quad (7.1)$$

Only a fixed translation is used in this passage to the limit.

Choose $h_n = h_{k_n, n}$ recursively and set

$$V_s = 2c + \sum_{i < |s|, s(i)=1} h_i.$$

At stage n , (7.1) allows k_n to make every one of the finitely many new edges satisfy (6.1). The growth $h_{n+1} > 8h_n$ makes all subset sums at a fixed level distinct. For example, induction gives the integer inequality

$$7 \sum_{i < n} h_i < h_n;$$

the largest differing summand dominates all earlier ones. Lemma 109.1 gives a continuous injection $r \mapsto z_r$.

Identify subsets of $\mathbb{N}$ with their characteristic functions. For $j \notin A$, the exact finite-sum identities and continuity of the fixed translation by $h_j > b$ give

$$z_{A \cup \{j\}} = z_A + h_j. \quad (7.2)$$

This identity, not continuity of order or subtraction, controls the limits.

Let r^* denote coordinatewise complementation and define

$$E_n = \{r : z_{r^*} < z_r \leq z_{r^*} + h_n\}, \quad D = \{r : \forall n (z_{r^*} + h_n < z_r)\}. \quad (7.3)$$

These are Borel sets.

Let $r(j) = 0$ and let s be obtained by flipping the j th bit. Equation (7.2) says

$$z_s = z_r + h_j, \quad z_{r^*} = z_{s^*} + h_j. \quad (7.4)$$

If $r, s \in E_n$ and $j > n$, then

$$z_{s^*} + h_j < z_r, \quad z_r + h_j \leq z_{s^*} + h_n.$$

Adding and cancelling yields $2h_j < h_n$, which is impossible. Thus E_n is disjoint from its image under any sufficiently far bit flip. If E_n were nonmeager, it would be comeager in some cylinder. A flip beyond that cylinder's length would give two disjoint comeager subsets of the same cylinder. Hence each E_n is meager.

The set D is invariant under a single bit flip. In the notation (7.4), $r \in D$ immediately implies $s \in D$. Conversely, suppose $s \in D$ and fix n . Choose $k = \max(n, j) + 1$; growth gives

$$h_k > h_n + 2h_j.$$

Then

$$z_{s^*} + h_k < z_s = z_r + h_j$$

implies, after comparison and cancellation,

$$z_{r^*} + h_n = z_{s^*} + h_j + h_n < z_r.$$

Thus $r \in D$. Iterating proves finite-change invariance.

The sets D and $D^* = \{r^* : r \in D\}$ are disjoint. Their union is comeager, because $z_r \neq z_{r^*}$ by injectivity, and

$$2^{\mathbb{N}} \setminus (D \cup D^*) \subseteq \bigcup_n (E_n \cup E_n^*). \quad (7.5)$$

Complementation is a homeomorphism. Lemma 110.1 says D is meager or comeager. In the first case both D and D^* are meager, contradicting their comeager union. In the second case they are disjoint comeager sets, another contradiction. This proves the theorem. $\square$

The proof never takes an arithmetic inequality through a metric limit. In particular it does not assert $z_r \geq 2c$ because every approximating vertex has that property.

111 Cofinal discontinuity of power-of-two multipliers

We now prove the part that resolves the weak-metatheory question.

Theorem 111.1 (A Borel exponential obstruction). *Suppose $\mathcal{M} \models \mathbf{Q} + \mathbf{IOpen}$ has an uncountable Polish carrier and Borel arithmetic operations. Suppose in addition there is a Borel function $e : M \rightarrow M$ satisfying (3.2). Then, for every $b \in M$, some fixed multiplication map $x \mapsto e(a)x$, with $a \geq b$, is discontinuous.*

Proof. Fix b , and assume that multiplication by $e(a)$ is continuous for every $a \geq b$. Construct c, t_n, q_n, X_n, f_n as in Section 108. Apply Lemma 107.1 to all maps $e \circ f_n$, and define $x_n, h_{k,n}$ using the resulting common F . We have (5.7).

For each $m \geq 2c$, the parameter $m - x_n > c > b$. Using only continuity of multiplication by the fixed element $e(m - x_n)$, we obtain

$$\begin{aligned} e(m + h_{k,n}) &= e(m - x_n)e(x_n + h_{k,n}) \\ &\longrightarrow e(m - x_n)e(x_n) = e(m). \end{aligned} \quad (8.1)$$

Choose $h_n = h_{k_n,n}$ recursively, write $y_n = e(h_n)$, and put

$$V_s = e(2c) \prod_{i < |s|, s(i)=1} y_i. \quad (8.2)$$

Each current vertex is $e(m_s)$, with $m_s = 2c + \sum_{i < |s|, s(i)=1} h_i \geq 2c$. Thus (8.1) allows all new edges at stage n to satisfy (6.1).

The arithmetic identities for e give

$$y_0 \geq 2, \quad y_{n+1} > y_n^5. \quad (8.3)$$

Let $Q_n = \prod_{i < n} y_i$. Induction gives $Q_n^4 < y_n$: it holds at $n = 0$, and

$$Q_{n+1}^4 = Q_n^4 y_n^4 < y_n^5 < y_{n+1}.$$

Consequently the largest differing index in two finite products dominates all preceding factors. The vertices at every level are distinct. Lemma 109.1 again produces a continuous injection $r \mapsto z_r$.

For $j \notin A$, continuity of multiplication by $y_j = e(h_j)$, whose exponent is $h_j > b$, gives

$$z_{A \cup \{j\}} = y_j z_A. \quad (8.4)$$

Every z_A is strictly positive. If $z_A = 0$, flip any bit. Equation (8.4), and cancellation of the nonzero factor y_j when the bit is removed, give a second input whose limit is zero. This contradicts injectivity. The carrier is the nonnegative arithmetic domain, so nonzero means positive. No order-closedness assumption is used.

Define direct comparison events

$$E_n = \{r : z_{r^*} < z_r \leq y_n z_{r^*}\}, \quad D = \{r : \forall n (y_n z_{r^*} < z_r)\}. \quad (8.5)$$

They are Borel. With $r(j) = 0$ and s its one-bit flip, (8.4) gives

$$z_s = y_j z_r, \quad z_{r^*} = y_j z_{s^*}. \quad (8.6)$$

If $r, s \in E_n$ and $j > n$, then

$$y_j z_{s^*} < z_r, \quad y_j z_r \leq y_n z_{s^*}.$$

Multiplying the first inequality by y_j , combining, and cancelling $z_{s^*} > 0$, yields $y_j^2 < y_n$. This contradicts (8.3). The same cylinder argument used in Section 110 therefore makes each E_n meager.

The set D is invariant under one-bit flips. If $r \in D$, (8.6) immediately implies $s \in D$. For the converse, suppose $s \in D$, fix n , and put $k = \max(n, j) + 1$. Then

$$y_k > y_n y_j^2.$$

We have

$$y_j z_r = z_s > y_k z_{s^*} > y_n y_j^2 z_{s^*}.$$

Cancelling $y_j > 0$ proves $z_r > y_n z_{r^*}$, as required.

The category contradiction now finishes the proof: D, D^* are disjoint, have a comeager union by

$$2^{\mathbb{N}} \setminus (D \cup D^*) \subseteq \bigcup_n (E_n \cup E_n^*),$$

and are interchanged by a homeomorphism. Finite-change invariance gives the meager-or-comeager dichotomy, which is incompatible with these two facts. $\square$

Corollary 111.2 (The answer to Glazer’s Question 1, as claimed by source 14). *With the coding convention of Section 105, Glazer’s topological Tennenbaum theorem for $\mathbb{Q} + \text{ID}_0 + \text{Exp}$ is provable in ATR_0 . In fact Theorem 104.1 holds.*

Proof. Corollary 106.4 supplies the Borel function e and its internally proved algebraic properties. Apply Theorem 111.1 and Theorem 110.2. Joint continuity implies continuity of every unary section, so the original continuity conclusion follows as well. $\square$

Status of this answer [write]

Corollary 111.2 is a **claimed** affirmative answer to Glazer’s Question 1, which this report printed as open (question 43.1; section 1.1). It is an unrefereed, AI-assisted manuscript’s claim, an upper bound under the coded-model convention of section 105.2, and it has **not been independently reviewed**: not by the intake, and not by the Hilbert’s-tenth programme’s scoped review of the same archive, which read the internal-MRDP input and the automatic-Borel argument and states that it does not certify the claimed resolution (section E). Its ATR_0 bookkeeping rests partly on declared inputs (note in section 105.4), and its author did not see the Gaifman–Dimitracopoulos chapter it relies on. No reversal is claimed: ACA_0 is not shown insufficient.

[write] Glazer names two places where his proof of Theorem 2 uses more than ATR_0 [1, p. 8]: the comprehension used to define a partial map g from preimages under $\exp \circ f_n$ in his Claim 2, and $\Sigma_2^1\text{-AC}_0$ to choose the points $s_{k,n}$. Source 14 replaces the first by the Borel exponentiation of corollary 106.4 and the common continuity restriction of lemma 107.1, and the second by the finite searches of the fusion estimate (lemma 109.1). Its additive part, theorem 110.2, strengthens Glazer’s Corollary 1, which Glazer already proves in ATR_0 [1, p. 7], from measure to category and to cofinal unary maps.

112 Why the proof stays in ATR_0

The new bridge is short, but its metatheoretic interpretation needs care. The following table states the roles of the ingredients.

Operation	Data or justification	Choice or truth issue
Internal exponentiation formula	One fixed MRDP proof instance	Finite proof translation only
Polynomial evaluation	Finite composition of Borel operations	No solution selector
Exponentiation map code	One tagged analytic separator	No separate separators chosen
Arithmetic scales	Borel floor-root and quotient maps	Coded recursive evaluation
Increasing Cantor maps	Uniform perfect/order construction	Use the coded uniform form
Common continuity restriction	One tagged Baire approximation	One exceptional cover
Metric fusion	Finite searches with a common tail bound	No arbitrary real choices
Tail contradiction	Baire property and finite flips	No measure theory

112.1 Countable iteration of Borel functions

The sequence $t_{n+1} = \lfloor \sqrt{t_n} \rfloor$ should not be justified by an unexamined appeal to dependent choice. Evaluation of the finitely many Borel maps involved can be organized along a common countable well-order α . Successive evaluations can then be performed along $\alpha \cdot \omega$, with each block using the point name produced by the preceding one. This is arithmetical transfinite recursion, precisely the available principle. The rational search used in fusion is handled within the same coded evaluation framework.

112.2 Uniform meager covers

The passage from the individual meagerness of E_n to the meagerness of their union also has a uniform implementation. Apply the Baire-property construction to

$$\{(n, r) : r \in E_n\} \subseteq \mathbb{N} \times 2^{\mathbb{N}}.$$

Because each section is meager, each open approximant section must be empty: a nonempty open subset of Cantor space cannot be meager. The uniformly supplied error cover therefore covers every section. Its sections, together with their complement-images, supply the one countable cover needed in (7.5) and its multiplicative analogue.

112.3 An upper bound, not a reversal

An affirmative answer to the question about provability in ATR_0 is an upper bound on the metatheory required. Nothing here proves that the theorem is equivalent to ATR_0 over RCA_0 , or that ACA_0 cannot prove a suitably restricted version. Those are separate questions, sensitive to how uncountability, the Borel data, and the arithmetic hypotheses are coded.

Similarly, the proof does not construct a global satisfaction predicate for T_{exp} , a set of all Borel functions, or a collection of all models on an uncountable carrier. The cofinality

assertions themselves can be written as second-order formulas about point names and the supplied Borel codes.

112.4 Three tempting shortcuts that are insufficient

First, a Diophantine representation proved only on standard $\mathbb{N}$ does not automatically identify the internally defined function in an arbitrary nonstandard model. Theorem 105.2 is the remedy.

Second, a Borel graph is not the chosen endpoint of the argument. The tagged basis-preimage construction explicitly gives the Borel-map information needed for regularization.

Third, transferring the topology along $e : M \rightarrow e[M]$ does not immediately reduce multiplication to addition. The Borel subspace $e[M]$ need not be Polish in the inherited topology, and a Polish refinement can destroy the continuity one hoped to preserve. The direct metric fusion works in the original complete carrier.

113 A nonsplit Polish Presburger model

This supplementary section is ordinary set-theoretic mathematics. The ATR_0 upper-bound claim concerns the coded topological-arithmetic theorem above; no formalization of the surreal statements in ATR_0 is asserted.

The preceding obstruction needs more arithmetic than addition alone. To make that boundary concrete, we now construct a Presburger model with continuous addition whose algebra differs from the split examples already investigated in Provel.

113.1 A countable nonsplit ordered group

For $n \geq 1$, define

$$s_n = \sum_{k=1}^{n-1} k!, \quad e_0 = (1, 0), \quad a_n = \left(\frac{s_n}{n!}, \frac{1}{n!} \right) \in \mathbb{Q}^2. \quad (10.1)$$

Let K_* be the subgroup generated by $e_0, a_1, a_2, \dots$. Order $\mathbb{Q}^2$ by the second coordinate first and then by the first coordinate:

$$(p, q) > 0 \iff q > 0 \text{ or } (q = 0 \text{ and } p > 0).$$

Lemma 113.1 ($K_* = G_{\zeta_*}$ of `isg:cf:prop:family`). *The group K_* is countable, e_0 is its least positive element, and projection onto the second coordinate induces*

$$K_*/\mathbb{Z}e_0 \cong \mathbb{Q}.$$

The exact sequence $0 \rightarrow \mathbb{Z}e_0 \rightarrow K_ \rightarrow \mathbb{Q} \rightarrow 0$ does not split.*

Proof. Countability is immediate. If $i \leq N$, then

$$a_i = \frac{N!}{i!} a_N - \frac{s_N - s_i}{i!} e_0, \quad (10.2)$$

and both coefficients are integers. Hence every finite linear combination of generators has the form $me_0 + ka_N$ for suitable integers m, k and N . Its second coordinate is zero exactly

when $k = 0$. Thus the kernel of the projection is $\mathbb{Z}e_0$. The image contains $1/n!$ for every n , so it is all of $\mathbb{Q}$. An element with positive second coordinate is larger than e_0 ; an element with zero second coordinate is an integer multiple of e_0 . This proves the least-positive claim.

The generators obey

$$(n+1)a_{n+1} = a_n + e_0. \quad (10.3)$$

If a homomorphism $\rho : K_* \rightarrow \mathbb{Z}$ satisfied $\rho(e_0) = 1$, write $b_n = \rho(a_n)$. Iterating (10.3) gives

$$n!b_n = b_1 + s_n. \quad (10.4)$$

Both s_n and $n! - s_n$ tend to infinity. For the second assertion, $s_n \leq 2(n-1)!$ for $n \geq 3$, so $n! - s_n \geq (n-2)(n-1)!$. For sufficiently large n , (10.4) would therefore put the integer multiple $n!b_n$ strictly between 0 and $n!$. This is impossible. A splitting of the displayed exact sequence would give such a retraction, proving nonsplitting. $\square$

[write] K_* is the group G_{ζ_*} , $\zeta_* = \sum_k k!$, of the discrete-initial-subgroups-and-omnific-normalization report (isg:cf:eq:explicitparameter), with the two coordinates exchanged: source 14's $a_n = (s_n/n!, 1/n!)$ and $e_0 = (1, 0)$ are that report's $x_n = (1/n!, S_n/n!)$ and $e = (0, 1)$, with $S_n = s_n$, and relation (10.3) of lemma 113.1 is its transition relation with every digit equal to 1. This lemma is therefore isg:cf:prop:family with isg:cf:thm:familytest for ζ_* , proved with the same estimate $s_n \leq 2(n-1)!$ (isg:cf:lem:noninteger). Source 14 does not cite that report; its statement that it does not claim the first discovery of nonsplit Presburger groups stands.

This supplies all algebra needed here without invoking the classification of extensions of $\mathbb{Q}$ by $\mathbb{Z}$.

113.2 The Polish carrier

Let

$$G = \mathbb{R} \overrightarrow{\times} K_*$$

as an ordered group, with the real coordinate having priority. Give it the product group topology $\mathbb{R} \times K_{*,\text{discrete}}$. Its distinguished unit is $(0, e_0)$. Let $M = G_{\geq 0}$, with the inherited topology and addition.

Proposition 113.2 (an instance of theorem 94.1). *The structure M is an uncountable, locally compact, perfect Polish model of Presburger arithmetic with jointly continuous addition. Its group completion is G , and the standard-integer subgroup of G is not a direct summand.*

Proof. The group K_* is countable discrete, hence Polish. The slices of M are

$$M \cap (\mathbb{R} \times \{d\}) = \begin{cases} [0, \infty) \times \{d\}, & d \geq 0, \\ (0, \infty) \times \{d\}, & d < 0. \end{cases} \quad (10.5)$$

These slices are clopen in M . They are locally compact Polish spaces without isolated points. A countable topological disjoint union of Polish spaces is Polish: choose bounded complete metrics on the slices and a fixed positive distance between distinct slices. This also proves local compactness and perfectness directly. The carrier is uncountable because any one slice is uncountable. Addition is the restriction of continuous group addition on $\mathbb{R} \times K_{*,\text{discrete}}$.

The least positive element of G is $(0, e_0)$, and

$$G/\mathbb{Z}(0, e_0) \cong \mathbb{R} \oplus \mathbb{Q}$$

is divisible. The standard characterization of Presburger groups as discretely ordered abelian groups with divisible quotient by the subgroup generated by 1 therefore applies [4, Section 2]. Its nonnegative cone is a model of Presburger arithmetic.

Every element of a linearly ordered group is a difference of nonnegative elements, so G is the Grothendieck group of M . A retraction $G \rightarrow \mathbb{Z}$ fixing its unit would restrict on $\{0\} \times K_*$ to the forbidden retraction in Lemma 113.1. Thus the extension remains nonsplit. $\square$

[write] Proposition 113.2 is the case $D = \mathbb{R}$, $K = K_*$ of source 13's theorem 94.1, printed there in general; source 14's proof is its own. Source 15 uses the same construction in theorem 122.2.

In fact every first-order definable relation, with parameters, is Δ_2^0 in this presentation. On each clopen discrete fiber of a finite power of M , a linear inequality is a real open or closed half-space, with its equality boundary decided by the order of K_* . Both the inequality and its complement are F_σ after taking the countable union of fibers. Congruence classes are clopen unions of fibers: $nG = \mathbb{R} \times nK_*$, so divisibility of the real coordinate leaves the real part unrestricted. Presburger quantifier elimination [4], and closure of Δ_2^0 under finite Boolean operations, prove the assertion. Thus this is a Borel arithmetic presentation at the additive level.

113.3 A surreal realization and an omnific obstruction

For $(x, (p, q)) \in G$, set

$$\Phi(x, (p, q)) = x\omega^2 + q\omega + p. \quad (10.6)$$

These are ordinary finite Conway normal forms.

Proposition 113.3 (second clause: `isg:cf:cor:unit`). *The map $\Phi : G \rightarrow \mathbf{No}$ is additive, injective, preserves order, and sends the distinguished unit to 1. There is nevertheless no additive map $M \rightarrow \mathbf{Oz}$ that sends the distinguished unit to 1.*

Proof. Addition of these finite normal forms is coefficientwise. The sign of their difference is determined by its first nonzero coefficient in the order $\omega^2, \omega, 1$. This is exactly the order on G , so Φ is an ordered additive embedding and has the stated unit value.

For the second assertion, recall the defining normal-form property of the canonical omnific integers: their constant coefficient is an integer, and their normal forms have no negative exponents [72]. Taking that coefficient gives an additive map

$$\text{ct} : \mathbf{Oz} \longrightarrow \mathbb{Z}, \quad \text{ct}(1) = 1.$$

Suppose $f : M \rightarrow \mathbf{Oz}$ were additive with $f(1) = 1$. All $(0, a_n)$ lie in M , since their second coordinate inside K_* is positive. Thus $b_n = \text{ct}(f(0, a_n)) \in \mathbb{Z}$ would satisfy

$$(n+1)b_{n+1} = b_n + 1$$

by (10.3), and hence the impossible relation (10.4). No extension of f to the group completion is needed for this contradiction. $\square$

Preservation of the unit is essential to the obstruction. The map

$$\Phi_\omega(x, (p, q)) = \omega\Phi(x, (p, q)) = x\omega^3 + q\omega^2 + p\omega$$

is an ordered additive embedding into $\mathbf{Oz}$, but sends the distinguished unit to ω . Its constant term is zero.

[write] The second assertion of [proposition 113.3](#) is the case $G = K_*$ of `isg:cf:cor:unit` (a $\mathbb{Z}$ -group admits an additive map to **Oz** sending its least positive element to 1 only if it splits), and the embedding Φ_ω is the one that report records after that corollary, $(a, b) \mapsto a\omega^2 + b\omega$ on G_ζ . Source 13 gives a two-element certificate of the same obstruction for its core ([Part VII](#)).

The distinction is algebraic. The topology under discussion is the explicitly given Polish topology on G and M . The image of Φ is discrete in the surreal order topology, since its ordered-group gaps are at least 1. Consequently Φ is not a homeomorphic realization of the perfect Polish carrier in that order topology.

The use of surreals creates no collection of all surreal numbers as a set. Formula (10.6) describes a set-indexed family of finite normal forms; it can be handled within a sufficiently large set-sized stage, or as a class-valued map in a conservative class framework. The nonexistence argument is stronger and more general: it applies to any target additive group that has a unit-preserving additive map to $\mathbb{Z}$.

114 Consequences and a concrete formalization route

114.1 What the result says about representations

In a Borel arithmetic presentation, internal exponentiation can be discontinuous, but it cannot have arbitrary projective complexity: its provable Diophantine definition supplies Borel regularity. This amount of regularity is enough to find a common perfect set on which the required countably many exponential maps are continuous. Completeness then turns small metric perturbations into a Cantor family of limits. Arithmetic growth and finite-bit symmetries make that family impossible if the relevant translations remain continuous.

The result distinguishes three notions that a representation theorem must keep separate:

1. correctness on standard natural numbers;
2. correctness in every model of a specified weak arithmetic;
3. regularity of the interpreted operations in an external topological presentation.

The passage from the second to the third is the main bridge of this article. The first alone does not imply either.

The cofinal strengthening says more than a failure somewhere in a two-variable function. No final arithmetic segment of translation parameters can consist entirely of continuous maps; the same holds for exponents of continuous powers-of-two multipliers. In particular the failure cannot be confined to a bounded set of arithmetic constants.

114.2 Existing interfaces and missing obligations

The relevant Lean file is

```
Logic/PeanoArithmetic/ListCoding/Lean/PAListCoding/
ExponentiationDiophantine.lean.
```

It imports Mathlib's Diophantine development and proves the standard-natural graph statement using `Dioph.pow_dioph` [73]. The corresponding Coq file provides explicit formula and single-equation interfaces [74].

Layer	Existing or established material	Additional obligation
Standard graph	$\mathbb{N}$ -valued exponentiation is Diophantine	Keep the standard-domain scope explicit
Internal graph	Gaifman–Dimitracopoulos theorem	Extract one weak-arithmetic proof for $E(x, y)$
Presentation semantics	Borel arithmetic operations on a Polish space	Interpret the finite object proof on point names
Borel decoding	Theorem 106.2	Formalize tagged analytic separation
Perfect restriction	Lemma 107.1	Formalize uniform Baire approximations and fusion
Continuity obstruction	Sections 110–111	Formalize limits, one-bit identities, and category

A practical development should proceed in this order.

1. **Fix the arithmetic sentence.** Choose one exact coded-sequence definition E and one pair of polynomial terms. Record the finite weak-arithmetic proof obligations connecting them. Do not silently substitute a theorem about $\mathbb{N}$.
2. **Separate interpretation from topology.** Prove the fixed-proof interpretation lemma for the chosen syntax and point-equality relation. This module should have no claims about continuity.
3. **Prove the abstract Borel decoding lemma.** Its inputs are analytic relation codes, totality, and uniqueness. The result should include uniform inverse-basis codes.
4. **Formalize the metric and category core.** The fusion estimate and one-bit contradiction can be verified in ordinary mathematical foundations before reverse-mathematical code bookkeeping is added.
5. **Assemble the exact upper bound.** Audit every use of countable sequences. Tagged constructions should replace implicit choices, and the final statement should name the coding convention.

No Lean or Coq file in this package is presented as a completed formalization of these obligations. The included program checks finite arithmetic estimates and the explicit nonsplit presentation; it does not check the infinite-dimensional semantic theorem.

[write] The two repository files named here are unchanged at the current state. Placement of this report beside them gives Part VIII no formal status; none of its statements is formalized.

115 Further research questions and topics

115.1 Reverse-mathematical strength

Question 115.1. For a fixed, precise coding of Borel arithmetic presentations, is the topological Tennenbaum statement equivalent to ATR_0 over RCA_0 , or does a weaker subsystem suffice?

The present proof provides an upper bound. A useful first step is to separate the contribution of analytic separation from that of the perfect-set/order lemma. Restrictions on the Borel ranks of the supplied operations may allow a sharper statement. A proposed reversal must specify the meaning of uncountability and the available presentation codes; otherwise different formulations can conceal different principles.

115.2 Weaker object arithmetic

Question 115.2. Can the power-of-two multiplier obstruction be established under a strictly weaker arithmetic hypothesis than $\text{I}\Delta_0 + \text{Exp}$, without assuming a total Borel exponential homomorphism as extra data?

Theorem 111.1 isolates exactly the usable external substitute. The remaining task is to obtain enough internally correct exponential behavior in weaker arithmetic. This should not be confused with a claimed proof of MRDP in $\text{I}\Delta_0$; that is a separate longstanding problem [69]. Partial exponentiation cuts are a natural intermediate setting.

115.3 How complicated are the discontinuous parameters?

There is an immediate general upper bound. If $F : P \times X \rightarrow Y$ is Borel between Polish spaces, discontinuity of the section F_p is equivalent to

$$\exists x \exists \varepsilon \in \mathbb{Q}_{>0} \exists (x_k)_{k \in \mathbb{N}} \forall k \left[d_X(x_k, x) < 2^{-k} \wedge d_Y(F(p, x_k), F(p, x)) \geq \varepsilon \right].$$

The displayed witness relation is Borel, so the discontinuous-parameter set is analytic and the continuous-parameter set is coanalytic. The countable sequence of witnesses can be selected in ATR_0 , which already proves $\Sigma_1^1\text{-AC}_0$ [70, Theorem V.8.3]. Apply this to the jointly Borel families $F(a, x) = a + x$ and $F(a, x) = e(a)x$.

Question 115.3. For the arithmetic families above, is the coanalytic upper bound sharp, or must the parameter sets

$$\{a : \tau_a \text{ is continuous}\}, \quad \{a : \lambda_a \text{ is continuous}\}$$

be Borel under some natural additional hypotheses? Can their complements be forced to be nonmeager, or must one settle for arithmetic cofinality?

The theorem proves cofinal discontinuity, which does not settle a topological size question. The outstanding problem is sharper classification for arithmetic families, beyond the general analytic/coanalytic bound just proved.

115.4 Borel-rank bounds for internally computable functions

Question 115.4. Given a fixed internal proof that a Σ_1 graph is total and single-valued, what bounds can one give on the Borel complexity of its interpretation in terms of the Borel codes for the primitive arithmetic operations?

The analytic-separation argument gives existence of codes, not a small fixed finite rank. A quantitative version could turn a qualitative impossibility theorem into an analysis of the representation cost of arithmetic decoding. One should distinguish graph ranks from ranks of inverse images of open sets.

115.5 The computational content of Borel decoding

Question 115.5. What is the uniform computational strength of the operation that takes coded Borel arithmetic and an internal Diophantine correctness proof to a Borel exponentiation code?

A precise formulation can compare this operation with analytic separation or transfinite Borel evaluation. The fixed-proof theorem shows that uniform arithmetic truth is not required. Determining whether separation is genuinely necessary for the full code-construction problem would clarify the source of noncomputability.

115.6 Nonsplit Presburger topology

Question 115.6. How can continuous unit-preserving homomorphisms and elementary embeddings between the countable-slice Presburger models $(\mathbb{R} \overrightarrow{\times} K)_{\geq 0}$ be classified? Which additional specified topological or definability conditions constrain the extension class or force splitting?

The slice argument already applies to every countable $\mathbb{Z}$ -group K with its discrete topology. In particular, an extension $0 \rightarrow \mathbb{Z} \rightarrow K \rightarrow V \rightarrow 0$ with V countable, divisible, and ordered carries the lexicographic order with convex kernel $\mathbb{Z}$, and yields the same locally compact Polish positive cone and definability bounds. General existence is therefore settled by this construction. Classification of maps, extension classes under extra constraints, and other Polish carriers remains a separate task.

[write] The general existence statement is source 13's [theorem 94.1](#). With it, this answers the existence part of Part III's [question 43.3](#) for countable cores.

115.7 Which arithmetic fragments fit inside the omnific integers?

Question 115.7. For set-sized Presburger groups admitting a retraction to $\mathbb{Z}$, which additional conditions allow unit-preserving omnific embeddings with specified definability, topology, or initiality in the surreal simplicity order?

Proposition [113.3](#) supplies a necessary obstruction: the constant-term map forces such a retraction. The question concerns these extra requirements, rather than bare ordered additive existence. The explicit formula (10.6) also makes clear that an algebraic surreal realization need not carry the desired topology. Any prescribed topology and notion of initiality must be stated before an embedding classification is attempted.

115.8 A proof-assistant milestone with a finite target

Question 115.8. Can one extract and verify one explicit $\text{I}\Delta_0 + \text{Exp}$ -proof of the Diophantine correctness of $E(x, y)$, together with its finite semantic interpretation?

This is more focused than immediately formalizing every part of MRDP or all reverse mathematics of Borel sets. It would close the exact gap between the repository's standard-natural exponentiation theorem and the principal new application in this article. The tagged Borel lemma and the metric/category core can then be developed independently.

116 Verification, scope, and research status

The mathematical review concentrated on the points where the argument could otherwise fail:

1. internal, rather than merely standard, Diophantine correctness;

2. Borel-map codes in addition to graph codes;
3. uniformity of the perfect restriction and meager covers;
4. positivity of multiplicative limits without order continuity;
5. exact one-bit identities and inclusive comparison endpoints;
6. control of all continuity parameters above an arbitrary threshold;
7. the distinction between a fixed finite proof translation and a global satisfaction construction.

Separate in-session proof audits checked the arithmetic/descriptive bridge and the metric/category argument. This is a mathematical review record, not a substitute for independent external scrutiny.

The accompanying Python program uses exact integers and rational numbers to check finite instances of the root-scale inequalities, subset-sum separation, product-growth estimates in logarithmic coordinates, factorial identities, and the finite obstruction to a proposed retraction. These computations test concrete algebraic details; they neither simulate a nonstandard model nor prove the infinite topological theorem.

The claim about the question’s status is deliberately precise. The argument answers Question 1 *as stated in the inspected 2023 paper*. The current repository manuscripts inspected for this task leave that question open. Searches did not locate a later arXiv version or a separate indexed solution. This is not an exhaustive bibliographic priority determination.

The final package includes the complete LaTeX source, its compiled PDF, build instructions, the finite-check program and its results, and concise source and proof-audit records.

Part IX

Locally compact cones and the limits of multiplication

[write] This part is source 15 of batch 87, *Polish Arithmetic: Multiplication Obstructions and Local Compactness*, subtitled *Structural extensions of the Presburger constructions at the intersection of Elliot Glazer’s work and ProveIt* (dated October 2026, its source snapshot and proof audit October 4; “research manuscript prepared for Vladimir Reshetnikov with AI assistance”), pinned to repository commit `0f084afa9`, before Parts III–V were placed. All unmarked text is source 15’s; its labels carry the prefix `pma:lcc:.` It continues [Part IV](#): it classifies the locally compact cones of Borel ordered Polish groups without assuming a locally compact ambient group.

What is new, and what is a special case of Part VI. New to the collection are the classification of locally compact inherited cones with the exact carrier X_{lc} ([theorems 120.5](#) and [121.1](#)), the compact-open isolated-zero theorem without a group topology ([theorem 121.4](#)), the boundary theorems ([theorems 122.1](#) and [122.2](#)), the exact count of $2^{\aleph_0}$ models on X_{lc} with standard boundary ([theorem 123.1](#)) and its omnific realization ([theorem 124.3](#)). Source 15’s Section 3, the “induction-free obstruction to Borel multiplication” ([section 119](#)), and its [theorems 119.6](#) and [124.1](#) are special cases of [Part VI](#), which source 15 did not see; they

are printed, with their different proofs, and flagged in their titles. Its background results repeat Part I and Part III, also flagged. The construction of [theorem 122.2](#) is source 13's countable-core transfer ([Part VII](#)).

Notation. Source 15's lexicographic products are printed $\overrightarrow{\times}$. In [theorem 122.1](#) its $H = G^0$ and C are source 08's C and B_C (letters exchanged; tempting false reading: C is the identity component, as in [Part IV](#)). Its M_L, G_L, D_L, V_L , with an italic countable linear order L , are not Part I's Laurent model $M_L = M_L$, and its Φ_L is not source 14's Φ . Its $\mathcal{A}(\cdot)$, the ordered set of Archimedean classes, has the meaning of the same symbol in [proposition 10.7](#). [Section 1.3](#) lists these.

Provenance. Source 15 ships no executable checks, by design ([section 125](#)); its build script is `code/15-polish-arithmetic-build.sh` and its proof audit `15-polish-arithmetic-proof_audit.txt`. Its source manifest is not shipped; see [section E](#).

117 The research intersection and the contribution

The intersection chosen here is the topological model theory of weak arithmetic, together with ordered additive groups and set-sized fragments of Conway's omnific integers. It is supported by Glazer's own mathematical paper, rather than inferred from a professional profile. In *A Topological Tennenbaum Theorem*, Glazer proves strong nonexistence results for uncountable Polish models of arithmetic and asks whether an uncountable Polish space can support Presburger arithmetic with continuous addition [1]. The ProveIt repository contains both an integer Cooper elimination development and a research manuscript on Polish Presburger models inside the omnific integers [83, 84, 85].

The repository manuscript already supplies an affirmative construction for Glazer's Question 2 as it is stated in the cited 2023 paper. In particular, the nonnegative cone of $\mathbb{R} \overrightarrow{\times} \mathbb{Z}$ works when the real coordinate has its usual topology and the integer coordinate is discrete. That construction, its Baire-space analogue, and the basic affirmative answer are background for this article. Repeating them would not constitute further progress. The present questions are what their locally compact generalisations look like, whether their additive structures can support multiplication, and how much arithmetic variation survives a fixed underlying topology.

117.1 Five principal results

The following statements give a guide to the proofs. A Borel order below means that the order relation is Borel for the specified topology. It does not mean that the topology is the order topology.

1. **Local compactness classification.** For a Borel ordered abelian Polish group G , local compactness of $G_{\geq 0}$ at zero forces G to be countable discrete or topologically $\mathbb{R} \times A$ with A countable discrete and torsion-free. These conditions are also sufficient. If G has a least positive element and is uncountable, its cone is homeomorphic to

$$X_{lc} = \left(\bigsqcup_{j \in \mathbb{N}} \mathbb{R} \right) \sqcup \left(\bigsqcup_{j \in \mathbb{N}} [0, \infty) \right). \quad (117.1)$$

2. **An absolute multiplication obstruction in this case.** If H is an uncountable Archimedean additive subgroup of a discretely ordered unital ring R , multiplication by any fixed $h > 0$ in H induces an injection $H \hookrightarrow R/H$. The locally compact groups above have an Archimedean real subgroup of countable index. Consequently their positive cones admit no unital semiring multiplication with the same addition and prescribed unit, whether measurable or not.
3. **A Borel multiplication obstruction without local compactness.** In a Polish abelian group with Borel discrete order and least positive element e , positive Borel endomorphisms cannot send e to every positive element unless the group is discrete. In particular, a nondiscrete such cone admits no semiring multiplication for which each fixed right-multiplier map is Borel. This is a group-theoretic extension of Glazer's probabilistic mechanism, with a stronger topological hypothesis and substantially weaker arithmetic assumptions.
4. **Optimal arithmetic diversity on one locally compact space.** There are exactly $2^{\aleph_0}$ isomorphism classes of full Presburger models with continuous addition on the space X_{lc} . An explicit family encodes subsets of $\mathbb{N}$ in its orders of Archimedean classes and admits finite-normal-form realizations in the omnific integers.
5. **Arithmetic on the topological boundary.** For an uncountable locally compact inherited Presburger cone, the endpoints of its closed-ray components form a countable discrete elementary initial submodel. Every countable Presburger model can occur as this boundary. Even when the boundary is fixed to the standard natural numbers, the continuum lower bound in the preceding item persists.

These results settle structural follow-up questions inside a precisely specified class of presentations. They do not settle the existence of all possible connected or zero-dimensional locally compact Polish Presburger presentations. A topology on a positive monoid need not extend to a Polish group topology on its difference group. That distinction is essential and will be demonstrated by a counterexample.

117.2 Existing results and the status of the extensions

The probabilistic construction of rapidly convergent, rapidly growing subseries, the use of complementation, and the final zero-one-law contradiction come from Glazer [1, Theorem 1]. Automatic continuity of Borel homomorphisms is classical [45]. Measurable orders on locally compact abelian groups have an extensive earlier theory, including the work discussed by Asmar and Montgomery-Smith [22]; Borel ordered-group classification also appears in Christensen, Kanovei, and Reeken [79]. Neither the no-compact-subgroup lemma nor the real-line order lemma is offered as a new discovery.

The contribution developed here is the exact condition that the *cone itself* be locally compact, its application to arithmetic carriers, the cardinal-index obstruction, the boundary interpretation and its realization converse, the optimal family on the classified carrier, and a unified statement of the multiplier barrier at the level of positive endomorphisms. The proofs are supplied in full. A targeted source search did not establish publication priority for these exact combined statements. They should be treated as proved claims in an unrefereed research manuscript, with the standard dependencies explicitly identified, rather than as a certified historical first or an announced breakthrough.

Topic	Earlier material	Present treatment
Continuous Presburger models	Real and Baire coefficient constructions in the repository	Used as established background
Topological Tennenbaum mechanism	Glazer’s zero-one-law proof	Positive-endomorphism formulation and Borel group consequences
Local compactness	A real-coordinate example; follow-up classification questions	Exact classification of inherited cones
Multiplication	Special algebraic obstructions	General Borel obstruction; absolute obstruction for the classified cones
Number of models	Earlier families on Baire space	Sharp continuum family on one locally compact carrier
Manifold boundary	Endpoints visible in the real example	Elementary initial submodel; every countable model realized

117.3 The repository snapshot

All local source claims refer to the inspected ProveIt checkout at commit

`0f084afa9be466593ffa6aff37749e78c0adc733`.

The earlier Polish article has Git blob `88f0244d0f1155cf44cd7d4915a3a5dfa3e50d03` at this snapshot. Its actual source was read; descriptive README filenames and page counts were not taken as authoritative for the current combined manuscript. The inspection is a mathematical and source-level audit. No new Lean or Rocq build is asserted, and no repository changes are part of this deliverable. The accompanying source manifest records the files used.

[write] “The earlier Polish article” [84] is this report as it stood at source 15’s pin `0f084afa9`: then the delivered text of batch-86 source 05, now the base of [Part I](#). Parts III–V were placed after that pin, so source 15 did not see them; its “earlier families on Baire space” are Part I’s, and Part III has since added continuum families on Baire space ([theorem 35.5](#)). Its “follow-up classification questions” are Part I’s [questions 14.3](#) and [14.5](#). The source manifest is not shipped, because it contains a third party’s personal profile address; its verified content is summarized in [section E](#).

118 Arithmetic and topology: the precise framework

118.1 Discrete order is independent of the chosen topology

Definition 118.1. A *discretely ordered abelian group with distinguished unit* is an ordered abelian group $(G, +, 0, <)$ with a least positive element 1. Its nonnegative cone is the cancellative commutative monoid $M = G_{\geq 0}$.

For every standard n , the interval $[0, n1]$ consists exactly of $0, 1, \dots, n1$. The assertion follows by external induction, using the absence of an element strictly between successive translates of 0 and 1. It does not bound nonstandard elements by standard integers. The order topology of such a group or cone is discrete, since each point has immediate neighbours, with the usual endpoint modification at zero. Thus an uncountable Polish presentation must use a different topology.

A *Polish group* is a topological group whose topology is separable and completely metrizable. An arbitrary compatible complete metric need not be translation invariant. A Borel order can be discontinuous: $\mathbb{R} \overrightarrow{\times} \mathbb{Z}$ with product topology is the basic example. For a topological group, the positive cone is Borel if and only if the entire order relation is Borel, since $x \leq y$ exactly when $y - x$ is in that cone.

We use the phrase *inherited group presentation* to mean exactly that $M = G_{\geq 0}$ has the subspace topology of a specified Polish abelian group G with Borel order. The phrase carries a genuine hypothesis. It is not a theorem that every continuous-addition Polish monoid has such a presentation.

118.2 The coefficient construction and full induction

Let D be a divisible ordered abelian group. Its unique rational division makes it a $\mathbb{Q}$ -vector space. Set

$$G_D = D \overrightarrow{\times} \mathbb{Z}, \quad M_D = (D_{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}), \quad 1 = (0, 1). \quad (118.1)$$

The D coordinate is compared first. In particular, a negative integer coordinate is compatible with being positive whenever the D coordinate is positive.

Proposition 118.2 (The standard algebraic certificate; = [corollary 4.11](#)). *For every divisible ordered abelian group D , M_D is a model of the entire first-order theory of $(\mathbb{N}, 0, 1, +, <)$.*

Proof. The group G_D has least positive element $(0, 1)$. For every standard $m \geq 2$, each (d, k) has the unique decomposition

$$(d, k) = m(d/m, \lfloor k/m \rfloor) + (0, r), \quad 0 \leq r < m, \quad (118.2)$$

where r is the ordinary remainder of k modulo m . For positive input, the quotient is nonnegative: it has positive D coordinate if $d > 0$, and an ordinary nonnegative integer coordinate if $d = 0$.

These are the axioms of a $\mathbb{Z}$ -group: a discretely ordered abelian group with exactly the standard residue classes modulo every standard positive integer. The classical quantifier-elimination theorem says that the theory of $\mathbb{Z}$ -groups in the language expanded by these congruence predicates is complete and is the theory of the ordered integers; see [82, Chapter 8] and the elimination method of [58]. Relativizing quantifiers to the nonnegative cone gives the asserted elementary equivalence. In particular every first-order Presburger induction instance holds in M_D . $\square$

This use of quantifier elimination is a classical dependency, not a claim that verifying a few division equations alone proves arbitrary induction. It also exposes the formalization boundary. ProveIt's current executable semantics quantify over Int ; the nonstandard semantic theorem must be supplied before substituting an arbitrary group G_D for that type. In particular, invariance under $x \mapsto x + m$ for standard iterations is weaker than invariance under every shift in the full subgroup mG_D .

118.3 Exactly when the inherited coefficient cone is Polish

Equip D with a Polish group topology and $\mathbb{Z}$ with the discrete topology. Write $P = D_{\geq 0}$. Each fixed-integer slice of M_D is homeomorphic to P or $P \setminus \{0\}$.

Proposition 118.3 (Positive-cone criterion; = [theorem 5.2](#)). *The space M_D is Polish in the inherited product topology if and only if P is G_δ in D . In that case addition on M_D is continuous. If the condition holds, the strict positive cone and the order relation on G_D are also Borel.*

Proof. If M_D is Polish, its clopen slice at integer coordinate zero is a Polish subspace of D equal to P . A subspace of a Polish space is Polish exactly when it is G_δ . Conversely, if P is G_δ , then $P \setminus \{0\}$ is G_δ as well. The countable topological sum of the resulting Polish slices is Polish. For example, choose bounded complete metrics on the slices and set the distance between distinct slices equal to a fixed larger constant. Addition is the restriction of the continuous ambient group operation; the cone is closed under addition. $\square$

For $D = \mathbb{R}$, the slice at $k \geq 0$ is $[0, \infty)$ and the slice at $k < 0$ is $(0, \infty)$. Consequently $M_{\mathbb{R}}$ has the space X_{lc} of [\(117.1\)](#). It is locally compact and has no isolated points. The map $r \mapsto \log r$ identifies an open ray with $\mathbb{R}$ and supplies a compatible complete metric there. This explains how a subspace defined using open rays can still be completely metrizable.

For contrast, on $D = \mathbb{R}^2$ with its Euclidean topology and lexicographic order,

$$P = \{(x, y) : x > 0\} \cup \{(0, y) : y \geq 0\}$$

is G_δ and F_σ , but is not locally compact at zero. Every relative neighbourhood of zero contains a sequence $(1/j, -\varepsilon)$ for some fixed $\varepsilon > 0$ whose ambient limit $(0, -\varepsilon)$ is missing. A compact relative neighbourhood would contain that limit, a contradiction. Thus local compactness of the coefficient group is weaker than local compactness of its cone.

[write] [Propositions 118.2](#) and [118.3](#) are Part I's [theorem 3.1](#) and [corollary 4.11](#) and [theorem 5.2](#), proved again; source 15 calls them background. The lexicographic $\mathbb{R}^2$ example is the step $d \geq 2$ of part (3) of source 08's [theorem 51.2](#).

119 An induction-free obstruction to Borel multiplication

The construction of Polish positive cones with continuous addition raises a natural strengthening question: can their addition be supplemented by a measurable multiplication? When the topology comes from a Polish additive group, the answer has a particularly strong form. No arithmetic induction axiom is needed, and even separate Borel measurability in one argument is already too much.

The probabilistic mechanism below is adapted from the proof of Glazer's first topological Tennenbaum theorem [[1](#), Theorem 1]. We give every step of the group version. The changes relevant here are that Borel multiplier maps become continuous by automatic continuity, limits are taken in the ambient additive group, and a finite-block probability estimate removes the need to construct an injective map from Cantor space. We present this as a consequence and extension of that mechanism; we do not assert independent historical priority for its ingredients.

[write] **A special case of Part VI.** The obstruction of this section is subsumed by [Part VI](#), which source 15 did not see. A positive additive endomorphism T with $T(e) > 0$ is an ordered additive embedding: $x > 0$ gives $x \geq e$ and $T(x) \geq T(e) > 0$. So the endomorphisms T_n of

[theorem 119.2](#) have cofinal unit images, and source 12's [corollary 80.2](#) already excludes an uncountable group with a Baire-property cone, a Hausdorff Baire ccc group topology and such a family, with no regularity of the T_a ; a countable Polish group is discrete. Likewise [corollary 119.3](#) is a case of [theorem 81.2](#), [corollary 119.4](#) of [corollary 78.5](#) (and source 09's [theorem 62.1](#) with right instead of left scalars), and [corollary 119.5](#) of [theorem 82.2](#); none of the Part VI versions needs Borel multiplier maps. [Theorem 119.6](#) is a case of source 10's [theorem 79.7](#) and of part (1) of source 08's [theorem 46.2](#). The section is printed because its proof of [theorem 119.2](#) is different: it adapts Glazer's zero-one-law argument rather than packing open sets, and its proof of [lemma 119.1](#) is source 08's [lemma 53.1](#) and source 09's [lemma 61.2](#) again. Its mechanism is that of source 06's [theorem 39.1](#) in Part III.

119.1 Automatic continuity of individual multiplier maps

Recall that an ordered abelian group is an abelian group equipped with a translation-invariant linear order. An endomorphism $T : G \rightarrow G$ is *positive* if $x \geq 0$ implies $T(x) \geq 0$. If $e > 0$ is the least positive element, then

$$0 \leq x \leq ne \implies x \in \{0, e, \dots, ne\} \quad (n \in \mathbb{N}).$$

This elementary assertion is proved by ordinary, external induction on the standard integer n . It is not an induction scheme imposed on the ordered group as an arithmetic structure.

Lemma 119.1 (Borel homomorphisms are continuous; = [lemma 53.1](#)). *Let G be a Polish group and H a second-countable topological group. Every Borel homomorphism $f : G \rightarrow H$ is continuous.*

Proof. We recall the category argument associated with the Piccard–Pettis theorem [45]. First, if $A \subseteq G$ has the Baire property and is nonmeagre, there is a nonempty open set O in which A is comeagre. For every g sufficiently close to the identity, $O \cap gO$ is nonempty and open. The sets A and gA are comeagre in this intersection, so they meet. Consequently AA^{-1} contains a neighborhood of the identity.

Now let V be an identity neighborhood in H , and choose an open identity neighborhood W such that $WW^{-1} \subseteq V$. Second countability implies that the subspace $f(G)$ is Lindelöf. The cover $\{f(g)W \cap f(G) : g \in G\}$ therefore has a countable subcover, with centers $f(g_j)$. If $A = f^{-1}(W)$, then

$$G = \bigcup_{j \in \mathbb{N}} g_j A.$$

The set A is Borel and cannot be meagre, by the Baire category theorem. Thus AA^{-1} contains an identity neighborhood, and

$$AA^{-1} \subseteq f^{-1}(WW^{-1}) \subseteq f^{-1}(V).$$

This proves continuity at the identity, hence everywhere. $\square$

For a ring whose addition makes it a Polish group, the map $x \mapsto xy$, for any fixed y , is an additive homomorphism. Accordingly, a Borel hypothesis on this one-variable map already gives all the multiplication continuity used below. Joint continuity of multiplication is neither assumed nor needed.

119.2 The positive-endomorphism obstruction

Theorem 119.2 (Borel multiplier obstruction; a case of [corollary 80.2](#)). *Let G be a Polish abelian topological group with a Borel, translation-invariant linear order and a least positive element e . Suppose that for every $a > 0$ there exists a positive Borel additive endomorphism $T_a : G \rightarrow G$ such that*

$$T_a(e) = a.$$

There is no required measurability or algebraic coherence in the parameter a . Then G is discrete and consequently countable.

Proof. Suppose that G is not discrete. Fix a complete metric d inducing its topology. We do not assume that d is translation invariant.

A positive null sequence. There is a sequence of nonzero points $v_j \rightarrow 0$. Put $u_j = |v_j|$, where this absolute value means $\max(v_j, -v_j)$ in the given group order. Since both v_j and $-v_j$ tend to zero, so does u_j , even though the absolute-value map need not be continuous. The finite set $\{e, 2e, \dots, 5e\}$ does not contain zero. After discarding finitely many terms, discreteness of the order therefore gives

$$u_j > 5e, \quad \text{and hence} \quad u_j \geq 6e \quad \text{for every } j.$$

Rapid topological convergence and rapid order growth. We recursively construct positive elements $y_0, y_1, \dots$, beginning with $y_0 = e$, such that

$$y_n > 5y_{n-1} \quad (n \geq 1) \tag{119.1}$$

and

$$d(s, s + y_n) < 2^{-n} \quad \text{for every} \quad s \in \left\{ \sum_{i < n} \varepsilon_i y_i : \varepsilon_i \in \{0, 1\} \right\}. \tag{119.2}$$

Suppose the elements before y_n have been chosen. Lemma [119.1](#) implies that $T_{y_{n-1}}$ is continuous, so $T_{y_{n-1}}(u_j) \rightarrow 0$. For each of the finitely many s in (119.2), continuity of addition gives

$$d(s, s + T_{y_{n-1}}(u_j)) \rightarrow 0.$$

Choose j large enough for all these inequalities and set $y_n = T_{y_{n-1}}(u_j)$. Positivity and additivity imply

$$y_n \geq T_{y_{n-1}}(6e) = 6y_{n-1} > 5y_{n-1}.$$

This completes the recursion.

All binary subseries converge. For $r \in 2^{\mathbb{N}}$, define

$$s_N(r) = \sum_{i < N} r(i) y_i.$$

If $N \geq 1$, consecutive partial sums either agree or differ by the addition of y_N . Thus, uniformly in r ,

$$d(s_N(r), s_L(r)) \leq \sum_{i=N}^{L-1} 2^{-i} \quad (L > N \geq 1).$$

Completeness gives a limit

$$z(r) = \lim_{N \rightarrow \infty} s_N(r) \in G,$$

and the convergence is uniform in r . Since each s_N is continuous on Cantor space, so is z . Write $\bar{r}(i) = 1 - r(i)$, and put

$$D(r) = z(r) - z(\bar{r}).$$

The map $D : 2^{\mathbb{N}} \rightarrow G$ is continuous, and

$$D(\bar{r}) = -D(r). \quad (119.3)$$

If r and s differ on only finitely many coordinates, continuity of the group operations gives the exact identity

$$D(r) - D(s) = 2 \sum_i (r(i) - s(i)) y_i. \quad (119.4)$$

To see this, the corresponding identity for finite partial sums is constant once the partial sums include every changed coordinate; then take limits.

Order separation on a finite block. Let $S_m = \sum_{i < m} y_i$. From (119.1), induction gives

$$4S_m < y_m \quad (m \geq 1). \quad (119.5)$$

Indeed $4y_0 < y_1$, and $4S_{m+1} = 4S_m + 4y_m < 5y_m < y_{m+1}$. If r and s differ finitely and m is their greatest coordinate of disagreement, then (119.4) implies

$$|D(r) - D(s)| \geq 2(y_m - S_m) > y_m \quad (m \geq 1). \quad (119.6)$$

Give $2^{\mathbb{N}}$ the fair independent Bernoulli product probability measure. Fix $n \geq 0$ and $k \geq 1$, and consider the block $B = \{n+1, \dots, n+k\}$. Once all coordinates outside B are fixed, at most one of the 2^k assignments on B can satisfy

$$|D(r)| \leq y_n.$$

Otherwise, two such assignments would have greatest disagreement $m \in B$, and (119.6) would give

$$|D(r) - D(s)| > y_m \geq y_{n+1} > 5y_n,$$

whereas the triangle inequality in the ordered group would give $|D(r) - D(s)| \leq 2y_n$. The events are Borel because D is continuous and the order is Borel. Integrating the finite conditional counting bound yields

$$\mathbb{P}(|D(r)| \leq y_n) \leq 2^{-k}.$$

As k is arbitrary, this probability is zero. Countable additivity now gives

$$\mathbb{P}(\forall n \ |D(r)| > y_n) = 1. \quad (119.7)$$

In particular, $\mathbb{P}(D(r) = 0) = 0$. No injectivity assertion about z was necessary.

A tail event with probability one half. Define

$$A = \{r : \forall n \ D(r) > y_n\}.$$

By (119.7), almost every r belongs either to A or to its reflected event $\{r : \forall n \ -D(r) > y_n\}$. Complementation preserves product measure and interchanges these two disjoint events by (119.3). Consequently

$$\mathbb{P}(A) = \frac{1}{2}.$$

We verify directly that A is a tail event. Suppose $r \in A$ and s differs from r only at coordinates at most m . Equations (119.4) and (119.5) give

$$|D(s) - D(r)| \leq 2S_{m+1} < y_{m+1}.$$

For any given n , choose $j > \max(n, m + 1)$. The growth condition implies $y_j > y_n + y_{m+1}$, and therefore

$$D(s) \geq D(r) - |D(s) - D(r)| > y_j - y_{m+1} > y_n.$$

Thus $s \in A$. Reversing r and s proves invariance under every finite coordinate change.

For each N , membership in this Borel set is consequently determined by the coordinates with index at least N : replace the first N coordinates by zero to obtain a Borel test on the remaining tail. Hence A belongs to the tail sigma-algebra. Kolmogorov's zero-one law says that its probability is zero or one. Indeed, a tail event is independent of every event determined by finitely many coordinates. The monotone-class theorem extends this independence to all events in the product Borel sigma-algebra, including A itself. Thus $\mathbb{P}(A) = \mathbb{P}(A)^2$, contradicting $\mathbb{P}(A) = 1/2$. This proves that G is discrete. Every discrete second-countable space is countable, completing the proof. $\square$

119.3 Semiring expansions and the difference group

Corollary 119.3 (No Borel semiring expansion of a nondiscrete Polish cone; a case of theorem 81.2). *Let G be a nondiscrete Polish abelian group with a Borel, translation-invariant linear order and least positive element 1. Put $M = G_{\geq 0}$, with its inherited Borel structure. There is no semiring multiplication*

$$\mu : M \times M \longrightarrow M$$

having the original addition and zero, having 1 as multiplicative identity, and satisfying the following weak regularity condition: for every fixed $a \in M$, the map $x \mapsto \mu(x, a)$ is Borel. In particular, there is no jointly Borel such multiplication. The positive cone M need not itself be Polish.

Proof. The maps

$$x^+ = \max(x, 0), \quad x^- = \max(-x, 0)$$

are Borel, take values in M , and satisfy $x = x^+ - x^-$. For $a > 0$, define

$$T_a(x) = \mu(x^+, a) - \mu(x^-, a).$$

This is a Borel map $G \rightarrow G$. It is precisely the additive extension of $x \mapsto \mu(x, a)$ from M to its difference group $G = M - M$. For completeness, if $b - c = b' - c'$ with $b, c, b', c' \in M$, then $b + c' = b' + c$. Distributivity gives

$$\mu(b, a) + \mu(c', a) = \mu(b', a) + \mu(c, a),$$

so the difference of the two products is independent of the chosen representation. This also proves additivity of the extension. For $x \geq 0$, $T_a(x) = \mu(x, a) \geq 0$, and $T_a(1) = a$. Theorem 119.2 now contradicts nondiscreteness. $\square$

If μ is jointly Borel, its usual extension to the whole group can be written explicitly as

$$\begin{aligned}\mu_G(x, y) &= \mu(x^+, y^+) + \mu(x^-, y^-) \\ &\quad - \mu(x^+, y^-) - \mu(x^-, y^+).\end{aligned}\tag{119.8}$$

Thus no measurable selection theorem is required for the passage to differences. The formula is Borel and biadditive. The usual well-definedness argument above, applied in both variables, proves that it is the unique ring multiplication extending μ . Associativity and the unit law extend from M by distributivity. For the obstruction itself, only the individual T_a are needed.

Corollary 119.4 (Polish discretely ordered rings; a case of [corollary 78.5](#)). *Let R be a unital ordered ring whose additive group is Polish and whose order is Borel. Suppose 1 is its least positive element and, for every fixed $a \in R$, right multiplication $x \mapsto xa$ is Borel. Then the additive topology of R is discrete and R is countable.*

No form of IOpen, bounded induction, division with remainder, roots, or exponentiation occurs in these statements. Indeed, Theorem [119.2](#) does not assume a multiplication operation at all: it assumes only the listed positive additive endomorphisms. The order's least positive element, Polish completeness, measurability of the order, and the multiplier regularity supply the obstruction.

Corollary 119.5 (Non-Polishability of the additive Borel group; a case of [theorem 82.2](#)). *Let R be an uncountable discretely ordered unital ring on a standard Borel space, with Borel addition, negation, order, and individual right-multiplier maps. Its additive Borel group is not Polishable: there is no Polish group topology on $(R, +)$ whose Borel sets are the given ones.*

Proof. In any such topology the order and multiplier maps would remain Borel. Corollary [119.4](#) would force R to be countable. $\square$

119.4 An algebraic cardinality obstruction

For an important subclass, measurability can be removed completely. A subgroup H of an ordered abelian group is called *Archimedean* if for all $h, k > 0$ in H , some standard positive integer n satisfies $k < nh$.

Theorem 119.6 (Archimedean subgroup versus quotient cardinality; a case of [theorem 79.7](#)). *Let R be a discretely ordered unital ring, and let $H \subseteq (R, +)$ be an uncountable Archimedean subgroup, with its induced order. Then there is an injective additive homomorphism*

$$H \longrightarrow (R, +)/H.$$

In particular,

$$|H| \leq |R/H|.$$

Thus an ordered abelian group having an uncountable Archimedean subgroup of countable index cannot be the additive ordered group of a discretely ordered unital ring with the prescribed least positive element as unit.

Proof. Every $a > 0$ in H is larger than every standard integer multiple of 1. Otherwise, $0 < a \leq n1$ for some standard n , and discreteness of the ring order gives $a = k1$ for a positive integer k . Archimedeaness would then imply that each $b > 0$ in H is bounded by

$ma = mk1$ for some standard integer m . Discreteness again gives $b \in \mathbb{Z}1$, and hence $H \subseteq \mathbb{Z}1$, contradicting uncountability.

Fix $h > 0$ in H . For any $a > 0$ in H , positivity of multiplication yields

$$ah > nh \quad \text{for every standard } n.$$

Because H is Archimedean, the multiples nh are cofinal in its positive part. Consequently ah is larger than every element of H , and in particular $ah \notin H$. If $a < 0$, applying this argument to $-a$ likewise gives $ah \notin H$. The additive homomorphism

$$a \mapsto ah + H$$

therefore has trivial kernel. □

For example, suppose a coefficient group is algebraically $D = \mathbb{R} \oplus A$, where A is countable, and its order restricts to an Archimedean order on the displayed copy of $\mathbb{R}$. In $G = D \overrightarrow{\times} \mathbb{Z}$, with the integer summand least significant and its positive generator as the proposed unit, take

$$H = \mathbb{R} \oplus \{0\} \oplus \{0\}.$$

Then H is uncountable and $G/H \cong A \oplus \mathbb{Z}$ is countable. There is no compatible semiring multiplication on the positive cone, even with no regularity requirement whatever. This applies to coefficient decompositions of this form supplied by a classification of *locally compact positive cones*. Local compactness of D alone is a different hypothesis and does not by itself supply this decomposition with a countable quotient.

119.5 Scope and sharpness of the hypotheses

The countable discrete case is possible. The usual ordered ring $\mathbb{Z}$, with its discrete topology, satisfies all the hypotheses and all its operations are continuous. Thus the conclusion cannot be strengthened to nonexistence of every such ring.

Complete metrizability matters. Give the usual ordered ring $\mathbb{Z}$ its 2-adic topology, whose basic neighborhoods are the congruence classes modulo 2^k . Addition, negation, and multiplication are continuous. The usual order is Borel: every subset of the countable T_1 space $\mathbb{Z}^2$ is a countable union of closed singletons. The element 1 remains the least positive element algebraically, while the topology is nondiscrete.

This example is not Polish. It is a countable metrizable space without isolated points, so every singleton is nowhere dense; the Baire category theorem rules out any compatible complete metric. The familiar 2-adic metric is explicitly incomplete: the integer partial sums

$$q_N = \sum_{j=0}^N 4^j = \frac{4^{N+1} - 1}{3}$$

are Cauchy and converge in the 2-adic completion to $-1/3$, which is not an integer. This is the same final-digits phenomenon used in the topological arithmetic constructions of Enayat, Hamkins, and Wcisło [2, Section 2].

A least positive element matters. The ordinary ordered field $\mathbb{R}$, with its usual topology, is an uncountable Polish additive group with continuous multiplication and a Borel order. It has no least positive element. Its null sequences have small order magnitude, so the recursion forcing simultaneous order growth and topological convergence cannot begin.

The ambient group topology is a real hypothesis. The corollaries concern a topology making *both* addition and negation continuous on the full difference group. A Polish topology on a positive cone alone does not automatically have this property. One should therefore not infer a blanket nonexistence theorem for all uncountable Polish models of the purely algebraic theory of nonnegative parts of discretely ordered rings. For any application, the original addition, the distinguished element 1, the Borel order, and the specific compatible Polish group topology must all be identified.

Here is an explicit example demonstrating the distinction. The polynomial construction is included as a boundary example; no novelty is claimed for it. Let

$$R_0 = \mathbb{Z} + t\mathbb{R}[t],$$

and order a nonzero polynomial by the sign of its highest-degree coefficient. This makes R_0 a discretely ordered ring in which t is larger than every integer and 1 is the least positive element. Let $M_0 = (R_0)_{\geq 0}$. Give M_0 the topological disjoint-union decomposition

$$M_0 = \mathbb{N} \sqcup \bigsqcup_{d \geq 1} ((0, \infty) \times \mathbb{R}^{d-1} \times \mathbb{Z}).$$

The factor $(0, \infty)$ records the leading coefficient of a degree- d polynomial, the middle factors its other nonconstant coefficients, and $\mathbb{Z}$ its constant coefficient. The constants have the discrete topology. This is an uncountable Polish space, being a countable topological sum of Polish spaces, and zero is isolated. It is also locally compact. Each stratum is open and locally compact: near a nonconstant polynomial, fix its integer constant coefficient and choose a bounded coefficient box with closure keeping the leading coefficient strictly positive. That closure is compact inside the same stratum. An isolated constant has its singleton as a compact neighborhood.

Addition and multiplication are jointly continuous. On a fixed pair of nonconstant degree strata, addition has degree equal to the larger input degree and multiplication has degree equal to their sum: positive leading coefficients prevent cancellation. Within these fixed strata the coefficient formulas are continuous polynomial functions. Addition of a constant and multiplication by a positive constant have the same property. The remaining cases involve the isolated zero or two isolated constants.

In fact its additive reduct is a model of *full* Presburger arithmetic. The convex subgroup generated by 1 in R_0 is the constant copy of $\mathbb{Z}$, and

$$R_0/\mathbb{Z} \cong t\mathbb{R}[t]$$

is a divisible ordered abelian group. The standard $\mathbb{Z}$ -group characterization therefore applies [4, Section 2]. More explicitly, for each standard integer $m > 0$, write the constant coefficient c of $p = c + \sum_{i=1}^d a_i t^i$ as $c = mq + r$, with $0 \leq r < m$. Then

$$p = m \left(q + \sum_{i=1}^d \frac{a_i}{m} t^i \right) + r$$

is its unique division by m with standard remainder. When $p \geq 0$, its quotient is also nonnegative. These division schemes, together with the discrete ordered-group axioms, give the complete Presburger theory on the nonnegative cone.

Thus M_0 is an uncountable locally compact Polish model of full Presburger arithmetic admitting a continuous semiring multiplication. It is a model of the algebraic theory of

nonnegative parts of discretely ordered rings, but it fails open induction in the language *with multiplication* explicitly. The quantifier-free formula with parameter t ,

$$\varphi(x) \equiv x^2 < t,$$

holds in M_0 exactly for the ordinary nonnegative integer constants. Hence $\varphi(0)$ and $\forall x (\varphi(x) \Rightarrow \varphi(x+1))$ hold, while $\varphi(t)$ does not.

The additive Borel group of R_0 , using the corresponding coefficient Borel structure, cannot carry a compatible Polish group topology. Besides Corollary 119.5, this has a direct category proof. Put

$$G_d = \{p \in R_0 : \deg p \leq d\}, \quad d \geq 0,$$

including zero in every G_d . These are Borel additive subgroups and $R_0 = \bigcup_d G_d$. In a hypothetical Polish group topology with these Borel sets, one G_d would be nonmeagre. The Piccard–Pettis argument makes that subgroup open. A second-countable group has at most countably many cosets of an open subgroup. But

$$G_{d+1}/G_d \cong (\mathbb{R}, +)$$

already supplies uncountably many such cosets, a contradiction. The obstruction therefore concerns the passage from a Polish positive monoid to a Polish difference group, not the existence of the displayed positive semiring itself.

[write] The degree-stratum cone M_0 is batch-86 source 06's cone of $\mathbb{Z} + t\mathbb{R}[t]$ in Part III, with the same topology: [theorems 38.1](#) and [38.2](#), and its group of differences is not compatibly Polishable for the reason given in [section 42](#); source 15 claims no novelty for it, and had not seen Part III. The 2-adic example is new to this report; source 10 makes the same remark for the p -adic topology ([section 84.1](#)).

120 Classification of locally compact coefficient presentations

The positive-cone construction has a surprisingly rigid locally compact case. The result below classifies the topological spaces obtained by this construction, including presentations whose coefficient group was not initially assumed locally compact. Its scope is precisely the inherited product topology; arbitrary Polish presentations of Presburger arithmetic are not covered.

Let D be a divisible abelian Polish group with a translation-invariant linear order $<$ whose positive cone is Borel. Write

$$P = D_{\geq 0}, \quad P^* = D_{> 0}, \quad M_D = (P^* \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}),$$

and give M_D the topology inherited from $D \times \mathbb{Z}$, with $\mathbb{Z}$ discrete. Arithmetic is inherited from the lexicographic ordered group $D \overrightarrow{\times} \mathbb{Z}$, and the distinguished arithmetic zero is $(0, 0)$. In this section no preliminary assumption that P is G_δ is needed: in the cases classified as locally compact, Polishness follows from the classification itself.

120.1 A measurable-order obstruction to compact subgroups

Lemma 120.1 (compare [lemma 47.1](#)). *Let G be a second-countable Hausdorff topological group with a translation-invariant linear order whose positive cone is Borel. If the order is invariant under left translations, then G has no nontrivial compact subgroup. In particular, this applies to the ordered abelian groups used here.*

Proof. An ordered group is torsion-free, so a nontrivial compact subgroup K would be infinite. Its normalized Haar measure μ is atomless: by translation invariance, one positive point mass would give the same positive mass to every point of the infinite group.

First consider the abelian case, which is all that is needed below. The two sets $K_{>0}$ and $K_{<0}$ have the same Haar measure, because inversion preserves Haar measure on a compact group. They partition $K \setminus \{0\}$, so each has measure $1/2$. Let X, Y, Z be independent Haar-distributed points of K . Ties have probability zero, and exchangeability gives

$$\Pr(X < Z \text{ and } Y < Z) = \frac{1}{3}.$$

Conditioning on $Z = z$, translation invariance makes $X - z$ and $Y - z$ independent Haar-distributed points. The same probability must therefore be $(1/2)^2 = 1/4$, a contradiction.

For a group with a merely left-invariant order, replace $X - z$ and $Y - z$ by $z^{-1}X$ and $z^{-1}Y$; inversion still interchanges the positive and negative cones and preserves normalized Haar measure on a compact group. The identical argument applies. $\square$

The argument isolates the obstruction: a translation-invariant measurable linear order is incompatible with a uniform distribution on a nontrivial compact subgroup. No continuity of the order relation is assumed.

120.2 Local compactness at zero determines the coefficient group

Lemma 120.2. *Let D be a Hausdorff ordered abelian topological group. If its nonnegative cone P is locally compact at zero in the subspace topology, then D is locally compact.*

Proof. There is a compact set $K \subseteq P$ which contains a neighborhood of zero relative to P . Choose an ambient open neighborhood W of zero such that $P \cap W \subseteq K$, and put $U = W \cap (-W)$. Every $x \in U$ belongs to $K \cup (-K)$: if $x \geq 0$, then $x \in P \cap W$, and otherwise $-x \in P \cap W$. Thus $K \cup (-K)$ is a compact neighborhood of zero in D . Translation gives local compactness everywhere. $\square$

Lemma 120.3. *Let C be the nonnegative cone of a translation-invariant linear order on the additive group $\mathbb{R}^n$. If C is locally compact at zero in the Euclidean subspace topology, then $n \leq 1$.*

Proof. Choose a compact $K \subseteq C$ and a Euclidean open neighborhood W of zero such that $C \cap W \subseteq K$. Since K is closed in $\mathbb{R}^n$ and $K \subseteq C$, the set $C \cap W = K \cap W$ is closed relative to W .

The cone C is invariant under multiplication by every positive rational number. Indeed, positive integral multiples preserve sign in an ordered abelian group, and division in $\mathbb{R}^n$ is unique. If $x_j \in C$ converges to x , choose a positive rational q small enough that $qx \in W$. For all sufficiently large j , $qx_j \in C \cap W$. Relative closedness yields $qx \in C$, hence $x \in C$. Thus C is globally closed.

If $n \geq 2$, the sets

$$C \cap S^{n-1} \quad \text{and} \quad (-C) \cap S^{n-1}$$

are nonempty disjoint closed sets covering the connected sphere S^{n-1} . This is impossible. Nonemptiness follows by taking a nonzero vector and its negative; closedness also implies that rescaling a positive vector to the sphere preserves its sign, by rational approximation. $\square$

Lemma 120.4 (= lemma 47.3 for $d = 1$, with the Baire property). *A translation-invariant linear order on the additive group $\mathbb{R}$ whose positive cone has the Baire property is either the usual order or its reverse.*

Proof. Let B be its strict positive cone. The sets B , $-B$, and $\{0\}$ partition $\mathbb{R}$, so B is nonmeager. Since B has the Baire property, it is comeager in some nonempty bounded open interval I . For every $t \in I + I$, the interval $I \cap (t - I)$ is nonempty. The sets B and $t - B$ are both comeager there, so their intersection is nonempty. Consequently

$$I + I \subseteq B + B \subseteq B.$$

The interval $I + I$ does not contain zero, and thus lies entirely in one of the two usual half-lines. Positive rational dilations of a nonempty open interval in $(0, \infty)$ cover $(0, \infty)$; the analogous statement holds on the negative half-line. The cone B is invariant under these dilations. It therefore contains one usual strict half-line, and the partition $B \sqcup (-B) = \mathbb{R} \setminus \{0\}$ prevents it from containing anything else. $\square$

The only structural theorem about locally compact groups needed next is the classical principal structure theorem: a locally compact abelian group has an open subgroup topologically isomorphic to $\mathbb{R}^n \times K$, with $n < \omega$ and K compact. We use the standard statement in Morris [80, Chapter 6, Theorem 25]; it is also explicitly recalled by Gabrielyan and Morris [81, p. 68].

Theorem 120.5 (Locally compact coefficient classification; extends [theorem 51.2\(3\)](#)). *For D and M_D as above, the following conditions are equivalent.*

1. M_D is locally compact at its arithmetic zero $(0, 0)$.
2. M_D is locally compact.
3. As a topological group, D is either a countable discrete $\mathbb{Q}$ -vector space or $\mathbb{R} \times A$, where A is a countable discrete $\mathbb{Q}$ -vector space.

In the first case M_D is a countably infinite discrete space. In the second case M_D is an uncountable locally compact Polish space, and

$$M_D \cong \left(\bigsqcup_{j \in \mathbb{N}} \mathbb{R} \right) \sqcup \left(\bigsqcup_{j \in \mathbb{N}} [0, \infty) \right) \quad (120.1)$$

as a topological space. Both families of connected components are countably infinite.

Proof. The slice with integer coordinate zero is a clopen subspace of M_D homeomorphic to P . Condition (1) therefore makes P locally compact at zero. By [Lemma 120.2](#), D is locally compact. The principal structure theorem supplies an open subgroup $H \cong \mathbb{R}^n \times K$. [Lemma 120.1](#) forces $K = \{0\}$, so $H \cong \mathbb{R}^n$ is open.

An ordered divisible abelian group is naturally a vector space over $\mathbb{Q}$. The subgroup H is a $\mathbb{Q}$ -subspace. Choose an algebraic vector-space complement A , so $D = H \oplus A$. It is discrete in its subspace topology because $H \cap A = \{0\}$ and H is open. There are only countably many H -cosets: a countable dense subset of D meets every open coset. Thus A is countable. The addition map $H \times A \rightarrow D$ is a topological isomorphism, since on each clopen slice it is translation from H onto an open coset.

The intersection $P \cap H$ is locally compact at zero and is the nonnegative cone of the restricted order on $H \cong \mathbb{R}^n$. [Lemma 120.3](#) gives $n \leq 1$. If $n = 0$, D is countable and discrete. If $n = 1$, $D \cong \mathbb{R} \times A$. This proves (1) $\Rightarrow$ (3).

For the converse, the discrete case is immediate. Suppose $D \cong \mathbb{R} \times A$ with A countable discrete. The restricted order on the open subgroup $\mathbb{R} \times \{0\}$ is Borel. By [Lemma 120.4](#), after

changing the orientation of the real coordinate it is the usual order. For each $a \in A$, each section

$$S_a = \{r \in \mathbb{R} : (r, a) > 0\}$$

is an upper set for the usual real order. It is consequently one of

$$\emptyset, \quad \mathbb{R}, \quad (b, \infty), \quad [b, \infty) \quad (b \in \mathbb{R}).$$

For $a = 0$ it is $(0, \infty)$. The corresponding nonnegative section at $a = 0$ is $[0, \infty)$; at $a \neq 0$ the strict and nonstrict sections agree.

The (a, k) -slices of M_D , for $(a, k) \in A \times \mathbb{Z}$, are clopen and have one of these interval types, with the replacement $(0, \infty) \mapsto [0, \infty)$ exactly when $a = 0$ and $k \geq 0$. Every nonempty slice is locally compact and homeomorphic either to $\mathbb{R}$ or to $[0, \infty)$. There are countably many slices. In particular, the slices $a = 0, k < 0$ give infinitely many line components, while $a = 0, k \geq 0$ give infinitely many closed-ray components. This proves (3) $\Rightarrow$ (2), the homeomorphism (120.1), and Polishness. Finally, (2) $\Rightarrow$ (1) is immediate. $\square$

120.3 Consequences and the precise boundary of the result

Corollary 120.6. *The inherited coefficient construction produces no uncountable zero-dimensional locally compact space. Every uncountable locally compact carrier it produces is homeomorphic to the carrier obtained from $D = \mathbb{R}$.*

The classification also excludes connected uncountable carriers inside this construction; the discrete integer coordinate already supplies clopen slices. It gives a stronger conclusion than an existence obstruction for one proposed space: the entire locally compact homeomorphism type is fixed.

It is essential not to infer that the open connected subgroup $\mathbb{R} \times \{0\}$ must be convex in the arithmetic order. For example, put the lexicographic order with real coordinate first on $D = \mathbb{R} \times \mathbb{Q}$ and give $\mathbb{Q}$ the discrete topology. Then $(0, 1)$ lies between $(-1, 0)$ and $(1, 0)$, although it does not lie in that connected subgroup. The coefficient cone remains locally compact, as the theorem predicts.

Likewise, the theorem does not assert that every uncountable Polish model of Presburger arithmetic with continuous addition has the carrier (120.1). A presentation whose topology is not inherited from a Polish coefficient group can evade these hypotheses. The broader inherited-group statement is proved next. Further questions are collected in Section 126.

[write] (Batch 87.) Within inherited coefficient presentations, [corollary 120.6](#) answers Part I's [question 14.5](#) negatively and the remark after it excludes connected carriers ([question 14.3](#)); [theorem 121.1](#) below extends both to every inherited Polish group cone. For presentations not inherited from a Polish ordered group both questions stay open ([question 126.1](#)). [Theorem 120.5](#) extends part (3) of source 08's [theorem 51.2](#) by assuming local compactness of the cone rather than of the coefficient group, and fixes the carrier.

121 The classification extends to arbitrary inherited group cones

The coefficient group need not be divisible, and the ambient ordered group need not split as a coefficient group followed by its standard integer part. The same arguments classify every inherited positive cone in a Borel ordered abelian Polish group. Write X_{lc} for the countable disjoint union of lines and closed rays displayed in (120.1).

Theorem 121.1 (General inherited-cone classification; extends [theorems 47.2](#) and [50.1](#)). *Let G be an abelian Polish group with a translation-invariant linear order whose nonnegative cone P is Borel. The following are equivalent:*

1. P is locally compact at zero in the topology inherited from G ;
2. P is locally compact in that topology;
3. as a topological group, G is either countable discrete, or $G \cong \mathbb{R} \times A$ for a countable discrete torsion-free abelian group A .

If these equivalent local-compactness conditions hold and, in addition, G has a least positive element e and P is uncountable, then P is homeomorphic to X_{lc} .

Proof. By Lemma [120.2](#), local compactness of P at zero gives local compactness of G . The principal structure theorem and Lemma [120.1](#) supply an open subgroup $H \cong \mathbb{R}^n$. The cone $P \cap H$ is locally compact at zero, so Lemma [120.3](#) gives $n \leq 1$.

Here is a direct splitting argument that does not assume divisibility of G . Since an ordered abelian group is torsion-free, it embeds in its rational hull $G_{\mathbb{Q}} = G \otimes_{\mathbb{Z}} \mathbb{Q}$, a $\mathbb{Q}$ -vector space. The subgroup H is divisible and is therefore already a $\mathbb{Q}$ -subspace of this hull. Choose a vector-space complement B to H in $G_{\mathbb{Q}}$ and put $A = G \cap B$. For $g \in G$, its decomposition $g = h + b$ has $h \in H \subseteq G$, so $b = g - h$ lies in A . Hence $G = H \oplus A$. The subgroup A is torsion-free, is discrete because H is open, and is countable because there are only countably many open H -cosets. Addition gives the topological isomorphism $H \times A \cong G$, as in the earlier proof. This proves (1) $\Rightarrow$ (3).

For (3) $\Rightarrow$ (2), the discrete case is immediate. In the other case Lemma [120.4](#) orients the real subgroup in its usual direction. In each clopen coset of that subgroup, the positive cone is an upper interval. Each nonempty such interval is a line or an open or closed ray, and is locally compact. Condition (2) plainly implies (1).

Suppose now that G has a least positive element e and is uncountable. Then $H \cong \mathbb{R}$. For every $h \in H$ with $h > 0$ and every standard $n \geq 0$, divisibility of H and minimality of e give

$$\frac{h}{n+1} \geq e, \quad \text{and consequently} \quad h \geq (n+1)e > ne.$$

In particular, $H \cap \mathbb{Z}e = \{0\}$, so the cosets $H + ke$, $k \in \mathbb{Z}$, are pairwise distinct. In an oriented real coordinate on H , their intersections with P are

$$P \cap (H + ke) \cong \begin{cases} [0, \infty), & k \geq 0, \\ (0, \infty), & k < 0. \end{cases}$$

The dominance inequality just proved determines the sign away from the real origin; the sign of ke determines whether that endpoint belongs. All other coset sections are again empty, lines, or open or closed rays. There are countably many such clopen sections, with infinitely many line components and infinitely many closed-ray components already supplied by the standard integer translates. Thus $P \cong X_{lc}$. $\square$

Corollary 121.2. *Suppose a locally compact Polish model of Presburger arithmetic is the nonnegative cone, with its inherited topology, of a Borel ordered abelian Polish group. If the model is uncountable, its carrier is X_{lc} . In particular, an uncountable zero-dimensional locally compact continuous-addition Presburger model cannot admit such an inherited Polish group presentation.*

The condition concerns a compatible topology on the entire group of differences, not merely the existence of that group algebraically. Every cancellative commutative monoid has an algebraic group of differences; it is precisely the compatible Polish group topology and inherited-cone condition that are additional here. No algebraic splitting of a general Presburger group over its standard $\mathbb{Z}$ -subgroup is assumed.

[write] [Theorem 121.1](#) strengthens source 08's [theorems 47.2](#) and [50.1](#): no ambient local compactness is assumed, and the homeomorphism type of the cone is determined. For Part III's [question 43.2](#) it settles the locally compact inherited case.

121.1 A compact-open obstruction without a group topology

One useful obstruction survives even when no ambient topological group is available.

Lemma 121.3 (compare [lemma 12.1](#)). *Let M be a Hausdorff commutative cancellative topological monoid whose only invertible element is zero. Its only nonempty compact subsemigroup is $\{0\}$.*

Proof. Let C be a nonempty compact subsemigroup. For each $a \in C$, the set $a + C$ is a nonempty compact subset of C . These sets have the finite intersection property: for $a_1, \dots, a_m \in C$, their intersection contains $a_1 + \dots + a_m + C$. Choose

$$p \in \bigcap_{a \in C} (a + C).$$

Since $p \in C$, the intersection condition includes $p \in p + C$. Thus $p = p + c$ for some $c \in C$, and cancellation against $p = p + 0$ gives $c = 0$. We have shown that every such C contains zero.

For any $a \in C$, the translate $a + C$ is itself a nonempty compact subsemigroup: the sum of $a + x$ and $a + y$ equals $a + (a + x + y)$, with $a + x + y \in C$. Applying the previous paragraph to $a + C$ gives $0 \in a + C$. Hence a is invertible in M and must be zero. Therefore $C = \{0\}$. $\square$

Theorem 121.4 (Compact-open neighborhoods force isolated zero). *Let M be a Hausdorff commutative cancellative topological monoid with jointly continuous addition and no nonzero invertible elements. If zero has a compact open neighborhood, then zero is isolated. Consequently, zero is isolated in every zero-dimensional locally compact Hausdorff continuous-addition Presburger presentation.*

Proof. Let K be a compact open neighborhood of zero. For each $x \in K$, joint continuity at $(x, 0)$ supplies neighborhoods U_x of x and V_x of zero such that $U_x + V_x \subseteq K$. Finitely many U_x cover K . Intersecting their corresponding V_x , and then intersecting with K , gives an open neighborhood V of zero satisfying

$$V \subseteq K, \quad K + V \subseteq K.$$

If $v \in V$, every positive integer multiple of v lies in K . The closure of this set of multiples is a nonempty compact subsemigroup: K is closed because it is compact in a Hausdorff space, and joint continuity preserves the subsemigroup condition under closure. [Lemma 121.3](#) says that this closure is $\{0\}$. Since it contains v , we have $v = 0$. Hence $V = \{0\}$ is open.

Finally, a zero-dimensional locally compact Hausdorff space has a compact open neighborhood at each point: inside an open neighborhood with compact closure choose a clopen neighborhood, which is then compact. A Presburger monoid satisfies cancellation and has no nonzero additive units, so the first assertion applies. $\square$

This last conclusion concerns the topology at zero only. It does not rule out zero-dimensional locally compact Presburger presentations with isolated zero, and it does not assert that every standard natural number is isolated: continuity of addition alone does not make every translation an open map.

[write] [Theorem 121.4](#) extends Part I's [corollary 12.2](#) (no compact carrier) and [lemma 12.1](#); compare source 06's [theorem 39.1](#), which isolates 0 when addition and multiplication are both continuous.

122 Arithmetic on the manifold boundary

The countably many endpoints of the locally compact carrier have an intrinsic arithmetic meaning. Here $\partial_{\text{man}}P$ denotes the boundary of P as a one-dimensional topological manifold with boundary, rather than its boundary inside the ambient group. On X_{lc} these are exactly the endpoints of the closed-ray components.

Theorem 122.1 (The boundary is an elementary initial submodel). *Let G be a Borel ordered abelian Polish group whose nonnegative cone P , in its inherited topology, is an uncountable locally compact model of Presburger arithmetic. Put $H = G^0$, the connected component of zero, and define*

$$C = \{g \in G : |g| < h \text{ for every } h \in H \text{ with } h > 0\}.$$

Then $H \cong \mathbb{R}$, C is a countable discrete convex subgroup containing the standard integer subgroup, and

$$\partial_{\text{man}}P = C_{\geq 0}.$$

With its inherited arithmetic, this boundary is a countable elementary initial submodel of P and has the discrete subspace topology.

Proof. [Theorem 121.1](#) gives an open subgroup isomorphic to $\mathbb{R}$. An open subgroup is also closed; since this one is connected, it is exactly $H = G^0$.

The set C is a subgroup. If $c, d \in C$ and $h \in H$ is positive, then $|c| < h/2$ and $|d| < h/2$, whence $|c + d| < h$; closure under negation is immediate. It is convex because $0 \leq x \leq |c|$ with $c \in C$ implies $x < h$ for every positive $h \in H$. No nonzero element of H lies in C , so $C \cap H = \{0\}$. The quotient G/H is countable, and the map $C \rightarrow G/H$ is injective. Thus C is countable. It is discrete since H is open and intersects it only at zero. If e is the least positive element, the dominance inequality in [Theorem 121.1](#) shows that every standard multiple of e belongs to C .

Each H -coset meets P in a line or an open or closed ray, or not at all. Suppose $c \in C_{\geq 0}$. For $h \in H$, the sign of $c + h$ is positive when $h > 0$, negative when $h < 0$, and nonnegative when $h = 0$. Therefore

$$P \cap (c + H) = c + H_{\geq 0},$$

and c is the endpoint of this closed ray. Conversely, if $x \in P$ is such an endpoint in its H -coset, then $x - h < 0$ for every positive $h \in H$. Consequently $0 \leq x < h$ for every such h , so $x \in C$. This proves the asserted identification with the intrinsic manifold boundary.

Finally, G is a Presburger ordered group, or $\mathbb{Z}$ -group, and C inherits its least positive element e . Division stays inside C . More explicitly, if

$$c = mq + re, \quad 0 \leq r < m,$$

with $c \in C$ and a standard positive integer m , then $mq = c - re \in C$. Convexity gives $q \in C$, since $|q| \leq m|q| = |mq|$. Thus C is itself a $\mathbb{Z}$ -group and its inclusion into G preserves and

reflects every congruence predicate. Quantifier elimination for Presburger arithmetic in the language with congruence predicates makes this inclusion elementary. Restricting to nonnegative cones gives $C_{\geq 0} \preceq P$. Convexity makes it an initial submodel. $\square$

The word “initial” is arithmetic: elements below a boundary element are again boundary elements. It does not mean that the endpoint set is a connected interval in the ambient topology; that endpoint set is discrete.

[write] In source 08’s notation (remark after [theorem 50.1](#)) source 15’s $H = G^0$ is C and its C is B_C : the letters are exchanged. The theorem shows that in the locally compact case B_C is a countable discrete convex elementary initial submodel.

Theorem 122.2 (Every countable Presburger model occurs as a boundary; the construction of [theorem 94.1](#)). *For every countable model N of Presburger arithmetic, there is an uncountable locally compact inherited Polish-group presentation P with continuous addition such that*

$$P \cong X_{lc} \text{ topologically,} \quad \partial_{\text{man}} P \cong N \text{ arithmetically.}$$

Proof. Let $C = N - N$ be the countable ordered difference group of N and give C the discrete topology. It is a $\mathbb{Z}$ -group. Set

$$G = \mathbb{R} \overrightarrow{\times} C$$

and give it the product topology, with the real coordinate dominant in the arithmetic order. This is an abelian Polish group with Borel order and least positive element $(0, 1_C)$. Since

$$G/\mathbb{Z}(0, 1_C) \cong \mathbb{R} \overrightarrow{\times} (C/\mathbb{Z}1_C)$$

is divisible, G is a $\mathbb{Z}$ -group and $P = G_{\geq 0}$ is a full Presburger model. Addition is continuous. The general classification gives its carrier X_{lc} . Its connected component is $\mathbb{R} \times \{0\}$, and the infinitesimal subgroup relative to that component is exactly $\{0\} \times C$. The preceding theorem identifies its manifold boundary with $\{0\} \times N$. $\square$

[write] The model in this proof is source 13’s countable-core transfer with $D = \mathbb{R}$ and $K = N - N$ ([theorem 94.1](#)); the identification of its boundary with N is source 15’s, through [theorem 122.1](#).

123 Continuum many models on one locally compact carrier

The topological rigidity in Theorem [120.5](#) does not imply rigidity of the arithmetic structure. We now exhibit the maximum possible number of pairwise nonisomorphic continuous-addition Presburger models on the single carrier

$$X_{lc} = \left(\bigsqcup_{j \in \mathbb{N}} \mathbb{R} \right) \sqcup \left(\bigsqcup_{j \in \mathbb{N}} [0, \infty) \right).$$

The ingredients are ordinary finite-support ordered groups; no novelty claim is made for the Hahn-group method itself.

123.1 A recoverable continuum family of countable orders

For a subset $S \subseteq \mathbb{N}$, put

$$k_n(S) = \begin{cases} 1, & n \notin S, \\ 2, & n \in S, \end{cases} \quad L_S = \sum_{n \in \mathbb{N}} (\mathbb{Z} + \mathbf{k}_n(S)),$$

where $+$ and $\sum$ denote ordered sums, $\mathbb{Z}$ has its usual two-sided order, and $\mathbf{k}$ is a finite chain of size k . The countable order L_S has neither a least nor a greatest element.

For elements of a linear order, define finite condensation by

$$x \sim_{\text{fin}} y \iff [\min(x, y), \max(x, y)] \text{ is finite.}$$

This is an equivalence relation with convex classes and is preserved by all order isomorphisms. In L_S its classes are, in succession,

$$\mathbb{Z}_0, \mathbf{k}_0(S), \mathbb{Z}_1, \mathbf{k}_1(S), \mathbb{Z}_2, \mathbf{k}_2(S), \dots$$

Indeed, two points in one integer block have a finite interval between them, as do two points in one finite block. An interval crossing from an integer block into the following finite block contains an infinite terminal segment of that integer block. An interval crossing from a finite block into the following integer block contains an infinite initial segment of the latter. Every interval crossing farther contains one of these infinite intervals.

Consequently the condensation quotient has order type ω . Every automorphism of ω is the identity, and an order isomorphism of L_S with L_T must match the corresponding finite classes. Their cardinalities recover $k_n(S)$ for every n , and hence recover S . Thus

$$L_S \cong L_T \iff S = T. \quad (123.1)$$

123.2 Encoding an order into Archimedean classes

For a countable linear order L , let

$$V_L = \bigoplus_{\ell \in L} \mathbb{Q}$$

be the group of finitely supported rational functions on L . For a nonzero $v \in V_L$, let $\lambda(v)$ be the least point in its finite support and declare

$$v > 0 \iff v(\lambda(v)) > 0.$$

Every nonempty finite subset of a linear order has a least element, so this definition requires no well-ordering of L . It gives the usual lexicographic ordered-group structure, with earlier indices dominating later indices. The group V_L is countable and divisible. Equip it with the discrete topology and define

$$D_L = V_L \overrightarrow{\times} \mathbb{R}, \quad G_L = D_L \overrightarrow{\times} \mathbb{Z}, \quad M_L = (G_L)_{\geq 0}.$$

The topology of D_L is the product of the discrete topology on V_L and the usual topology on $\mathbb{R}$; the topology on M_L is inherited from $D_L \times \mathbb{Z}$. The coefficient order is Borel: each nonzero positive V_L -coset is wholly positive, each negative coset is wholly negative, and the zero coset uses the usual real order.

The group G_L is a discretely ordered abelian group with least positive element $1 = (0, 0, 1)$, and its quotient by $\mathbb{Z}1$ is the divisible ordered group D_L . Equivalently, for every standard

$m \geq 2$, division by m has precisely the usual residue representatives $0, \dots, m-1$. In coordinates it is given by

$$(d, z) = m \left(\frac{d}{m}, \left\lfloor \frac{z}{m} \right\rfloor \right) + (0, z \bmod m).$$

For a nonnegative input the quotient is again nonnegative. By the standard ordered-group characterization of Presburger models, M_L is a model of the full first-order theory of $(\mathbb{N}, 0, 1, +, <)$, including its entire induction scheme. Addition is continuous because it is the restriction of the group operation. Theorem 120.5 gives $M_L \cong X_{lc}$ as a topological space.

For completeness, the invariant distinguishing these models can be checked directly. For positive elements of an ordered abelian group, write

$$x \asymp y \iff (\exists m, n \in \mathbb{N}_{>0}) (x \leq my \text{ and } y \leq nx).$$

This is Archimedean equivalence. Order its classes by dominance:

$$[x] \triangleleft [y] \iff x > ny \text{ for every } n \in \mathbb{N}_{>0}. \quad (123.2)$$

Thus a more dominant class is earlier. The condition is independent of the positive representatives and is invariant under ordered-group isomorphisms. In V_L , two positive elements are Archimedean equivalent exactly when they have the same least support index. Equality of the indices allows integer multiples of either positive leading coefficient to dominate the other; inequality of the indices makes the earlier coordinate dominate every positive integer multiple of the later one. Hence

$$\mathcal{A}(V_L) \cong L, \quad \mathcal{A}(D_L) \cong L + 1, \quad \mathcal{A}(G_L) \cong L + 2,$$

where $\mathcal{A}$ denotes the set of positive Archimedean classes with the convention (123.2). The two appended classes come respectively from the real summand and the integer summand.

An isomorphism of Presburger monoids $M_L \rightarrow M_{L'}$ extends uniquely to an isomorphism of their Grothendieck groups, by $x - y \mapsto h(x) - h(y)$. It preserves the positive cones, and therefore preserves the ordered groups and their Archimedean-class orders. For $L = L_S$, the order L_S has no greatest element. In $L_S + 2$, the greatest class and its immediate predecessor are therefore precisely the two appended classes. Removing those two intrinsically specified classes recovers L_S . There is no absorption of an end block. We conclude from (123.1) that

$$M_{L_S} \cong M_{L_T} \text{ as Presburger structures} \iff S = T. \quad (123.3)$$

Theorem 123.1 (Sharp number of models on one locally compact carrier; compare theorem 35.5). *There are exactly $2^{\aleph_0}$ isomorphism classes of full Presburger models with continuous addition whose underlying topological space is X_{lc} . The lower bound is realized by inherited coefficient presentations.*

Proof. The lower bound is (123.3). Choose a homeomorphism from each of the displayed carriers to the fixed space X_{lc} and transport its arithmetic structure.

For the upper bound, a continuous map from X_{lc}^2 to X_{lc} is determined by its restriction to a countable dense subset, because the target is Hausdorff. Since $|X_{lc}| = 2^{\aleph_0}$, there are at most

$$(2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0}$$

such maps. Choosing the constants 0 and 1 does not increase this bound. The order is recovered from addition by $x \leq y \iff (\exists z) x + z = y$, so it contributes no additional choices. Hence there are at most continuum many such structures, and in particular at most continuum many isomorphism classes. $\square$

This separates two classification problems. The locally compact carrier is unique inside the uncountable inherited coefficient construction, but its arithmetic structures achieve the largest multiplicity permitted by continuity on a separable space. Further questions are to determine the topological Presburger automorphism groups of these presentations and to determine the descriptive complexity of isomorphism on the full space of continuous Presburger operations on X_{lc} . Within the displayed S -indexed family the isomorphism relation has already been determined: it is exactly equality of the parameters.

In fact, the family above fixes even the boundary arithmetic. In $G_{L_S} = V_{L_S} \overrightarrow{\times} \mathbb{R} \overrightarrow{\times} \mathbb{Z}$, the subgroup infinitesimal relative to the open real component is precisely the final $\mathbb{Z}$ summand. Thus $\partial_{\text{man}} M_{L_S}$ is the standard model $\mathbb{N}$ for every S . The continuum lower bound therefore persists after prescribing the boundary to be standard arithmetic.

[write] The analogous counts on Baire space are Part III's [theorems 35.5](#) and [35.8](#). Source 13's [corollary 94.2](#) gives $2^{\aleph_0}$ nonsplit locally compact models; by [theorem 121.1](#) they too live on X_{lc} , whereas the family above is split.

124 Consequences for multiplication and for omnific arithmetic

124.1 The locally compact obstruction is entirely algebraic

The topological classification and the cardinality theorem combine to give a conclusion stronger than the Borel multiplier obstruction in this particular setting.

Theorem 124.1 (No multiplication on an uncountable locally compact inherited cone; a case of [theorem 81.2](#)). *Let G be an abelian Polish group with a Borel translation-invariant linear order and least positive element 1. Suppose that its positive cone $M = G_{\geq 0}$, with the inherited topology, is uncountable and locally compact. There is no unital semiring multiplication on M having its existing addition, zero, and distinguished unit 1. No continuity or measurability of the proposed multiplication is assumed. Neither Presburger induction nor multiplication commutativity is required for this conclusion.*

Proof. By [Theorem 121.1](#), G has an open subgroup $H \cong \mathbb{R}$ and G/H is countable. The order restricted to H is usual or reversed by [Lemma 120.4](#); therefore H is an uncountable Archimedean subgroup in its induced order.

A proposed multiplication on M extends uniquely by distributivity to its additive group of differences G . The construction is the algebraic version of [\(119.8\)](#), and does not use any Borel hypothesis. Products of nonnegative elements remain nonnegative. Moreover, the product of two strictly positive elements is strictly positive. Indeed, each can be written as $1 + x$ with $x \geq 0$, and

$$(1 + x)(1 + y) = 1 + x + y + xy > 0.$$

Thus the extension makes G a discretely ordered unital ring. [Theorem 119.6](#) then injects the uncountable group H into the countable quotient G/H , a contradiction. $\square$

The topology is used to locate the Archimedean subgroup of countable index. Once that subgroup has been found, the obstruction survives every change of topology. In particular, refining one of these cones' topologies to isolate its standard natural numbers cannot make its *unchanged additive structure* support a semiring multiplication. The polynomial example in [Section 119](#) has a different ordered additive group and therefore does not contradict this theorem.

124.2 Explicit finite normal forms in the omnific integers

The family from Section 123 can be placed inside the omnific integers without invoking a topology on a proper class, infinite support convergence, or a global choice of normal forms. All the normal forms needed for each element are finite. We use the standard convention that an omnific integer has real coefficients at strictly positive surreal exponents and an integer coefficient at exponent zero. The inspected Provel development realizes this as a coefficient pullback inside its surreal normal-form ring [85].

Lemma 124.2. *Every countable linear order L admits an order-reversing embedding $q : L \rightarrow \mathbb{Q} \cap (1, 2)$.*

Proof. Enumerate L . Assign exponents one at a time, maintaining the required reversed inequalities. At each finite stage, the already assigned elements above and below the next element determine an open interval between finitely many rational bounds; use 1 or 2 when one side has no assigned bound. The interval is nonempty because the finite assignment respects the order. Density of $\mathbb{Q}$ supplies a new rational strictly inside it. The union of these finite assignments is the desired injection. $\square$

For $v \in V_L$, $r \in \mathbb{R}$, and $k \in \mathbb{Z}$, define

$$\Phi_L(v, r, k) = \sum_{\ell \in \text{supp}(v)} v(\ell) \omega^{q(\ell)} + r\omega + k. \quad (124.1)$$

The convention is that zero coefficients are omitted. The finite support has a well-defined greatest exponent, and every exponent $q(\ell)$ lies strictly between 1 and 2.

Theorem 124.3 (A set-sized omnific realization; compare theorems 9.1 and 40.1). *The map Φ_L is an injective order-preserving additive homomorphism from*

$$G_L = V_L \overrightarrow{\times} \mathbb{R} \overrightarrow{\times} \mathbb{Z}$$

into the omnific integers, and sends its distinguished unit to 1. Its nonnegative image is an additive Presburger substructure of the nonnegative omnific integers. With the topology transported from M_L , this image is a locally compact Polish model on X_{lc} . For the orders L_S , these images yield continuum many nonisomorphic such arithmetic structures with standard boundary.

Proof. The expression (124.1) has no negative exponent and has integer constant term, so it is an omnific integer. Uniqueness of finite normal forms gives injectivity. Addition is coefficientwise, giving additivity and preservation of the unit.

If $v \neq 0$, let ℓ_0 be its least support index. Since q reverses the order, $q(\ell_0)$ is the greatest exponent appearing in the first sum. It also exceeds the exponent 1 of the real summand and the exponent 0 of the integer summand. The sign of the normal form is therefore the sign of $v(\ell_0)$. If $v = 0$ and $r \neq 0$, its sign is the sign of r ; if $v = r = 0$, its sign is the sign of k . This is exactly the lexicographic order on G_L .

Transport the topology along the resulting bijection between M_L and its positive image. The bijection is then a homeomorphism by definition, and the already proved Presburger axioms, continuity of addition, local compactness, and boundary statements are preserved. Each image is a set of cardinality continuum: its parameters range over the set $V_L \times \mathbb{R} \times \mathbb{Z}$, not over all surreal numbers. $\square$

These are additive substructures, with the full Presburger theory verified by their $\mathbb{Z}$ -group certificate. They are not subrings for the usual omnific multiplication. For example,

$$\omega = \Phi_L(0, 1, 0), \quad \omega \cdot \omega = \omega^2,$$

and uniqueness of normal forms shows that ω^2 is outside the image: every exponent allowed by (124.1) is less than 2. Theorem 124.1 says more than this failure of closure. No alternative multiplication on the same positive image can satisfy the unital semiring laws with its existing addition and unit.

The topology just transported is a coefficient topology. It is distinct from the native order topology of the omnific integers, which is discrete because 1 is least positive. ProveIt's separate discreteness development concerns the native fine order topology; it provides no contradiction to the nondiscrete coefficient topology used here [86]. An assertion of "a Polish topology on the omnific integers" without a set-sized carrier and an explicit topology would obscure both of these distinctions.

125 From mathematical proofs to a ProveIt formalization

The repository offers a concrete starting point for formalizing these results, but the distinction between a source-inspected existing declaration and a proposed future theorem must be maintained. The source audit establishes which declarations are present at the pinned commit. The proofs in this article are mathematical proofs; no new Lean kernel certificate accompanies them.

125.1 Existing formal interfaces

The Presburger files are under the directory

```
Logic/PresburgerArithmetic/Lean/PresburgerArithmetic/.
```

The actual semantics in `Syntax.lean` use valuations of type `List Int`; quantified variables range over `Int`. In `Decision.lean`, the following declarations occur in the namespace `PresburgerArithmetic.Formula` [83]:

- `quantifierEliminate`: the elimination procedure;
- `holds_iff_quantifierEliminate`: its semantic equivalence;
- `decideSentence_eq_true_iff`: correctness for sentences;
- `presburgerArithmetic_decidable`: the resulting decision instance.

These support the executable integer decision procedure. They do not by themselves instantiate that procedure over an arbitrary nonstandard ordered group.

The omnific files are under `Algebra/SurrealNumbers/Surreal/Foundations/`. The inspected `OmnificIntegers.lean` supplies the following chain of interfaces [85]:

1. `normalFormHahnHom` turns the normal-form map into a ring homomorphism, and `nonnegativeSupportSubring` restricts the permitted exponents.
2. `constantCoeff` extracts the real constant coefficient; `omnificSubring` pulls it back along the integer embedding. The abbreviation `OmnificInteger` is a universe-indexed type, and `omnificToSurreal` embeds it into the actual surreal construction.
3. `omnificToSurreal_injective` records injectivity; `omnificConstantCoeff` extracts the integer constant; `omnificPurelyInfiniteIdeal` is its kernel; and `omnificQuotientEquiv` identifies the quotient with $\mathbb{Z}$.

4. In `OmnificPurelyInfiniteModule.lean`, `omnificPurelyInfiniteAddEquiv` transfers the established purely infinite support module. The real and rational module instances provide coefficient divisibility.

These interfaces fit the finite normal-form construction above. Their existence is source evidence for a formalization route, not evidence that the new Polish topology or classification is already implemented.

[write] The declarations named in this subsection exist at the current state under these names. They are about the ordinary integers and the omnific coefficient construction; none of source 15's theorems is formalized.

125.2 A dependency order for implementation

Stage 1: algebraic certificates and the cardinal obstruction. Introduce a semantic interface for discretely ordered abelian groups with standard residue representatives. Prove closure of positive cones, the unique division statement, and extension of additive monoid maps to differences. Then formalize Theorem 119.6: after the bounded integer-interval lemma, its essential object is the additive map $a \mapsto ah + H$ and the proof that its kernel is trivial. This is the shortest substantial result in the article and needs no measure theory or descriptive set theory.

Stage 2: nonstandard Presburger semantics. Parameterize the interpretation of affine terms and congruences by the ordered group. The elimination proof must use the actual subgroup mG , including nonstandard shifts. For example,

$$x \equiv y \pmod{m} \iff x - y \in mG$$

is the semantic congruence relation. Proving a formula invariant under standard iterations of the translation $x \mapsto x + m$ only covers $m\mathbb{Z}1$, which can be a proper subgroup of mG . Generalized elimination or a formal model-theoretic completeness argument closes this gap. Its output should include elementary embeddings of pure $\mathbb{Z}$ -subgroups, the interface needed by the boundary theorem.

Stage 3: topology and local compactness. Build the coefficient models as explicit types with transported product topologies. Prove the G_δ criterion and the compact-neighborhood reflection lemma. The Euclidean closed-cone argument is a small independent target. The full classification then requires the principal LCA structure theorem and the measurable-order compact-subgroup obstruction. Availability of these results in the particular dependency version must be checked before estimating a formalization budget; this manuscript does not assume that they are already packaged in the required form.

Stage 4: the arithmetic boundary and continuum family. Formalize the infinitesimal subgroup C relative to the open real component, its convexity and countability, and its identification with the endpoints. Prove purity under division and then apply the semantic theorem from Stage 2. For the continuum family, finite support makes the ordered-group construction elementary; the recoverable finite-condensation pattern supplies an explicit nonisomorphism certificate. The final omnific embedding uses finite normal forms and the already inspected coefficient interfaces.

Stage 5: the Borel multiplier theorem. This stage requires Borel automatic continuity, a complete metric, Cantor product probability, and a tail zero-one law. The proof is organized so that each dependency can be isolated. The finite-block estimate only counts 2^k assignments and avoids a measurable inverse or an injective Cantor-space coding theorem. The order and the topology must remain separate structures in the formal statement: assuming an order topology would force discreteness at the outset and erase the content of the result.

125.3 What verification establishes

The source comparison fixes the existing background; the proofs check the new implications; and the counterexamples test the boundaries of the hypotheses. Separate AI-assisted mathematical reviews focused on the rational-hull splitting, endpoint identification, quotient injection, finite-block probability argument, and recovery of the continuum parameters. That review is useful error control, but it is not a substitute for an independent referee or a proof assistant.

Finite numerical checks could illustrate the geometric growth inequality, but they would not verify the uncountable cardinalities, Polish completeness, elementarity, or all induction instances. The article therefore supplies deductive arguments for those claims and does not present finite computation as a mathematical certificate.

126 Further research questions and a proposed program

The following are proposed questions arising from the results. Their status is deliberately local: they are not all asserted to be previously published open problems. Where a question comes from Glazer's paper or the earlier repository manuscript, that origin is stated. Questions already settled above are separated from questions outside the proved hypotheses.

126.1 Questions now settled within the inherited-group framework

For an uncountable Presburger positive cone inherited from a Borel ordered abelian Polish group, the locally compact carrier is exactly X_{lc} . Thus connected and zero-dimensional locally compact carriers are excluded in this framework. Every such cone forbids every unital semiring multiplication on the same addition and unit. The fixed carrier supports the maximum possible number, continuum, of isomorphism classes, even with standard boundary. Its intrinsic boundary is always a countable elementary initial submodel, and the set of possible boundary models is exactly the set of countable Presburger models. These are complete answers to the corresponding restricted classification, expansion, multiplicity, and boundary questions.

[write] (Batch 87.) For Part I's [question 14.5](#) and [question 14.3](#), and Part III's [question 43.2](#) and [question 43.4](#), these settle the inherited-group case only.

126.2 Presentations outside Polish group completions

Question 126.1 (Zero-dimensional locally compact arithmetic). Does any uncountable zero-dimensional locally compact Polish space support a full Presburger model with jointly continuous addition?

This extends a follow-up question in the repository manuscript [84]. The classification proves that any affirmative example cannot inherit its topology from a Borel ordered Polish group. Theorem 121.4 adds that its zero must be isolated. The remaining case is therefore precise: an example with isolated zero and no compatible inherited Polish group presentation. A global nonexistence proof must use information beyond the compact-open neighborhood argument.

[write] This is Part I's question 14.5, re-scoped by corollary 121.2 and theorem 121.4.

Question 126.2 (Connected Polish arithmetic). Does a connected uncountable Polish space admit a full Presburger model with jointly continuous addition? What changes if local compactness or finite covering dimension is imposed?

The inherited locally compact case is excluded above. Connectedness without a group-completion hypothesis remains outside these proofs. One possible approach is to study the successor map and the images of translations as topological embeddings; addition is cancellative, but its inverses on these images need not be continuous automatically. The degree-stratum polynomial example is disconnected and therefore does not answer this question.

[write] Compare Part I's question 14.3, Part III's question 43.4 and source 13's question 103.4.

Question 126.3 (A usable group-completion criterion). Which intrinsic conditions on a locally compact Polish Presburger monoid ensure that its topology is inherited from a Polish group topology on its algebraic group of differences?

A useful answer should be stated using translations, the partial difference map on ordered pairs, and neighborhood structure, so it can be checked without first constructing a group topology. The polynomial example shows that cancellation, joint continuity of addition, local compactness, and a full Presburger reduct do not suffice. This question would identify exactly where the classification can be applied to presentations given without coefficient coordinates.

[write] This is source 08's question 57.2 (= source 09's question 73.11); sources 12 and 10 ask it again in Part VI.

126.3 Arithmetic invariants beyond the carrier

Question 126.4 (Classification with a prescribed boundary). Fix a countable Presburger model N . Classify the locally compact inherited-group presentations with manifold boundary isomorphic to N , up to arithmetic isomorphism and up to topological arithmetic isomorphism.

Existence is settled by Theorem 122.2, and there are already continuum many classes when $N = \mathbb{N}$. The open classification problem is to find complete invariants. Natural data include the ordered quotient above the infinitesimal subgroup, the placement of the real component, and extension data for the standard integer subgroup. A homeomorphism of carriers alone cannot recover these data.

Question 126.5 (Threshold orders and automorphism groups). For a fixed countable torsion-free abelian group A , classify Borel translation-invariant linear orders on $\mathbb{R} \times A$ whose cones are discretely ordered Presburger models. Determine their groups of continuous Presburger automorphisms.

Each real-coset section is empty, full, or a ray. Its threshold, whether an endpoint is present, and the arithmetic relations between cosets satisfy compatibility conditions under

addition. Making these conditions explicit could turn the topological classification into an arithmetic parametrization. The real component need not be convex, so assuming a single lexicographic product would miss part of the problem.

Question 126.6 (Complexity of isomorphism). In a precise standard Borel coding of continuous binary operations and distinguished constants on X_{lc} , what is the descriptive complexity of the subset defining Presburger models? On that subset, what is the complexity of arithmetic isomorphism and of topological arithmetic isomorphism?

The family $S \mapsto M_{L_S}$ has its own isomorphism relation completely determined: it is equality of the parameters. Cardinality alone does not determine the complexity of isomorphism on the larger space of presentations. A classification should distinguish continuous changes of coordinates from abstract model isomorphisms.

126.4 Strengthening and formalizing the obstructions

Question 126.7 (Smaller endomorphism hypotheses). In Theorem 119.2, how far can the assumption that every positive a is $T_a(e)$ be weakened?

The proof actually uses only enough positive endomorphisms to continue a sequence that is rapidly increasing in order while topologically summable. A useful refinement would give an intrinsic condition on the positive endomorphism orbit of e that is both necessary and sufficient for this recursion. Merely replacing the orbit condition by an unexplained cofinality assumption would leave the needed control of neighborhoods unproved.

[write] (Batch 87.) Answered by Part VI: by source 12's theorem 80.1, in an uncountable group with a Hausdorff Baire ccc topology and a Baire-property cone every ordered additive self-embedding f satisfies $f(e) \leq 2a$ for the a of lemma 77.2. So a single positive endomorphism T with $T(e) > 2a$ already forces countability, and no regularity of T is needed.

Question 126.8 (Regularity of the order). Which conclusions survive when a Borel order is replaced by a weaker measurable or category-theoretic regularity assumption?

Different steps use different regularity. The real-line lemma uses the Baire property; the compact-subgroup lemma uses Haar measurability; the probabilistic obstruction needs the inverse images of order intervals under the Cantor limit map to be measurable. A satisfactory weakening should state these requirements on the relevant maps and subgroups, since an arbitrary weakening of “Borel” need not be preserved by restriction or inverse image.

[write] For the multiplier obstruction, the Baire property of the cone suffices (Part VI).

Question 126.9 (Algebraic obstructions beyond one real subgroup). What other additive invariants prevent a discretely ordered group from admitting any unital ordered-ring multiplication?

The injection $H \hookrightarrow G/H$ compares the size of an Archimedean subgroup with its quotient. One can ask for analogues using a chain of Archimedean layers, ranks of quotients, or several independent multiplier images. A useful classification would explain which failures are purely algebraic and which arise only from measurability. The polynomial ring supplies an example where an infinite hierarchy of larger growth layers permits multiplication.

Question 126.10 (Minimal metatheory and Glazer's Question 1). What is the weakest subsystem of second-order arithmetic that proves the particular obstruction theorems in this article? How does it compare with Glazer's original question about proving his Theorem 2 in ATR_0 ?

Glazer’s Question 1 is an explicitly published problem [1, Further Work, Question 1]. This article does not answer it. Its classical proofs use completeness, Baire category, product probability, and, in the classification, algebraic complements and LCA structure. The phrase “induction-free” in Section 119 refers to the *object arithmetic being modeled*, not to the strength of the metatheory used to prove the theorem. Determining the latter requires a separate reverse-mathematical analysis.

[write] Source 14 (Part VIII) claims an affirmative answer to Glazer’s Question 1, not independently reviewed; source 15’s own statement that it does not answer the question stands.

Question 126.11 (A checked formalization). Can the algebraic cardinality theorem, the generalized Presburger semantic certificate, and the boundary theorem be formalized in ProveIt as a coherent first milestone?

These three components connect the repository’s current integer elimination and omnific coefficient interfaces to the new structural results. They also divide naturally into algebra, model theory, and topology. The probability theorem can follow as a second milestone once the needed measure-theoretic interfaces are explicit.

126.5 Recommended order of work

The most concrete next project is the algebraic obstruction and its formal certificate: the statement is short, the proof has few dependencies, and the conclusion is stronger than a continuity obstruction. Classification of presentations with a prescribed boundary is the next mathematical project, because the boundary supplies an intrinsic invariant that the carrier alone misses. The broad existence questions for connected or zero-dimensional presentations should be pursued together with a group-completion criterion, with the polynomial example serving as a test case for every proposed general theorem.

Part X

Elementary embeddings and Polish submodels of lexicographic Presburger arithmetic

[write] This part is source 16, manuscript 02 of batch 89, *Elementary Embeddings and Polish Submodels of Lexicographic Presburger Arithmetic: Matrix normal forms, closed images, and two sharp ordinal thresholds* (23 pages, A4, dated October 3, 2026; “prepared for Vladimir Reshetnikov”; its delivery calls it AI-assisted, with internal AI-assisted proof reviews). It arrived in commit 7c0f2d9f9, as archive Glazer_ProveIt_Embeddings_and_Thresholds, and was placed in 23adb85f9; it is pinned to 0f084afa9, 70 commits before the placement. Its file prefix in this report is 16-presburger-embeddings-. It continues Parts I and III: it takes the family $M_\alpha = (\mathbb{Q}^\alpha \vec{\times} \mathbb{Z})_{\geq 0}$ of theorems 35.1 and 35.9 (α a countable ordinal, rational coordinates discrete) and classifies its elementary embeddings, its elementary submodels that are Polish in the inherited topology, and two ordinal thresholds, for continuity of all self-embeddings ($\alpha \leq \omega$) and for closedness of all their images ($\alpha < \omega \cdot 2$). All unmarked text in this part is source 16’s; its twelve sections are Sections 127–138 (its Section k is Section $126 + k$), and its two appendices are sections W and X.

What was already here. At its pin this report was the delivered base text of [Part I](#) (source 05), and the manuscripts of Parts III–V were archives in docs/incoming; source 16 read both, and credits the construction of M_α to them. It did not see source 04: that manuscript was placed in 3d2177df4, an ancestor of the pin, but as a member of a merge it was not staged, and its text entered the report only with the write ae7b9ab74, after the pin. So the case $\alpha = \beta = \omega$ of source 16’s embedding results proves again, without credit, source 04’s [proposition 11.3](#) (all unital additive maps are elementary embeddings $T \times \text{id}$), [theorem 11.4](#) (the increasing-positive-pivot matrices, automatically continuous) and [corollary 11.5](#) (closed images, clopen exactly when finitely many pivots are omitted), and source 05’s [theorem 11.7](#) is the simplest case of its [theorem 135.1](#). Its model construction ([proposition 128.3](#)) is [theorem 35.1](#) (source 07) and [theorem 35.9](#) (source 06), and its omnific realization ([section 137](#)) is [theorem 40.1](#). Each of these is printed with source 16’s proof and flagged in its title or in a note. New to the report are the transfinite cases: the classification of continuous embeddings and of all Polish elementary submodels for every countable α , the pivot normal form, ambient orbits and the lattice, the two thresholds, the automorphism counts and the intersection of all self-copies. Priority is not established.

Use of choice. The discontinuous and non-Borel maps of [sections 133](#) and [134](#) are built from a nonzero $\mathbb{Q}$ -linear functional on $\mathbb{Q}^\omega$ vanishing on the finite-support vectors, which source 16 obtains with the axiom of choice (a Hamel basis of a quotient). So the negative halves of clauses (5) and (6) of [theorem 127.1](#), [theorem 133.3](#), the value 2^c in [theorem 133.4](#), [proposition 133.5](#) and [theorem 134.2](#) are theorems of ZFC, not of ZF; [remark 133.2](#) [merge] shows that ZF with dependent choice does not prove them unless it is inconsistent.

Questions answered. Source 16 answers, for its family, Part I’s [question 14.9](#) (Polish elementary-submodel lattices; its “Research question 8”) and [question 14.14](#) (embedding categories beyond [theorem 11.4](#), for ordinal-indexed rational products), and the clause of [question 14.6](#) on continuous order-preserving endomorphisms, for the groups $\mathbb{Q}^\alpha$ (its “Research question 5”). Dated notes after those questions say what is answered. Its questions on connected carriers and on arbitrary Polish presentations repeat questions of Parts I, III, IV, VII and IX that Parts VII and IX answer in part; notes after them point there instead of reopening them.

Labels and notation. Every label of source 16 takes the prefix `pma:emb:`. Labels were added to its unlabelled corollary on orbit counts, two examples and its twelve questions, so that notes can cite them, and `pma:emb:rem:choice` was written for this report; four questions of [Part I](#) were labelled for the notes (`pma:pr:q:coefficientgroups`, `pma:pr:q:lattices`, `pma:pr:q:surrealcutoffs`, `pma:pr:q:embeddingcategories`). The fourth block of the table in [section 1.3](#) lists the symbols. In short: source 16’s $H_\alpha = \mathbb{Q}^\alpha$ is Part III’s A_α , kept because A is its matrix; its G_α and M_α are Part III’s; its $D = \mathbb{Q}^\omega$ and D_{fin} are printed D_B and D_B^{fin} (Part I’s Baire coefficient group); its shear S_φ is printed Sh_φ , because S is its pivot subset; its restriction symbol is printed $\upharpoonright$; its lexicographic sum is printed $\vec{\times}$; in two proofs local letters were renamed (S, K to N, Γ in the proof of [theorem 131.1](#); K to K_α in the proof of [theorem 134.1](#), where it is the terminal tail of [section 135](#)). No normalization changed.

Files. Its finite checks are shipped as `code/16-presburger-embeddings-verify_examples.py` with the recorded run data `data/16-presburger-embeddings-verification_results.json` (11,243 exact checks); its build record, source audit and proof-review record as

data/16-presburger-embeddings-BUILD_REPORT.json, 16-presburger-embeddings-SOURCE_AUDIT.md and 16-presburger-embeddings-PROOF_REVIEW.md. The Hilbert’s-tenth programme routed the archive without a correctness finding (section E). Nothing in this part is formalized.

127 The research intersection and the contribution

The useful intersection is topological model theory of arithmetic, with a second connection to ordered Hahn groups and the omnific integers. Glazer’s *A Topological Tennenbaum Theorem* [1] rules out uncountable Polish models of sufficiently strong arithmetic with the indicated regularity of their operations. Its final Question 2 asks whether an uncountable Polish space can support Presburger arithmetic with continuous addition. This is a question about addition and first-order induction in the additive language; multiplication is absent.

The current ProveIt corpus already contains affirmative constructions. The maintained manuscript [92] treats the positive cones of $\mathbb{R} \overrightarrow{\times} \mathbb{Z}$ and $\mathbb{Q}^\omega \overrightarrow{\times} \mathbb{Z}$. It classifies elementary submodels of the real-coordinate model and constructs shift self-embeddings of the rational-sequence model. Its further-research questions ask for classifications of continuous coefficient-group endomorphisms and of Polish elementary-submodel lattices. A separate incoming manuscript [93] already introduces the countable-ordinal family used here. We therefore take the construction of the models as prior work and study their structure.

This distinction matters. Repeating the positive construction would not constitute a new answer for this repository. Nor does the present article settle Glazer’s separate Question 1 about proving his theorem in ATR_0 . Instead, we solve a precise family of follow-up questions and exhibit two different failures of topological rigidity.

[write] At source 16’s pin 0f084afa9 the “maintained manuscript” [92] was source 05’s delivered text, now the base of Part I (its real-model results are theorems 11.1 and 11.2, the shift theorem 11.7); the “separate incoming manuscript” [93] was source 06, now part of Part III, whose thm:ordinals is theorem 35.9. The two questions it names are Part I’s questions 14.6 and 14.9. The self-embeddings of the rational-sequence model were already classified, by source 04, which source 16 did not see (introduction of this part).

127.1 A statement of the main answer

Fix a countable ordinal α . All products below are full products, not finite-support sums. Set

$$H_\alpha = \mathbb{Q}^\alpha, \quad G_\alpha = H_\alpha \times \mathbb{Z}, \quad \mathbf{1} = (0, 1), \quad M_\alpha = (G_\alpha)_{\geq 0}. \quad (127.1)$$

The first nonzero rational coordinate determines the sign in H_α ; in G_α , the H_α coordinate dominates the integer coordinate. Every rational coordinate and the integer coordinate have the discrete topology. The topology on M_α is the subspace topology from G_α .

Theorem 127.1 (Structural classification and thresholds). *For countable ordinals α, β the following hold.*

- (1) M_α is a Polish model of the full first-order theory of $(\mathbb{N}; 0, 1, +, <)$, with continuous addition. For $\alpha \geq \omega$ its underlying space is homeomorphic to Baire space.
- (2) Every map $M_\alpha \rightarrow M_\beta$ preserving 0, 1, and addition is an elementary embedding. Such a map exists exactly when $\alpha \leq \beta$.

- (3) *The Borel embeddings are exactly the continuous embeddings. They are described by row-finite rational matrices with strictly increasing pivot map $p : \alpha \rightarrow \beta$ and positive pivot entries. Every such embedding has closed image.*
- (4) *The elementary submodels of M_α are precisely $M_V = (V \overrightarrow{\times} \mathbb{Z})_{\geq 0}$ for $\mathbb{Q}$ -linear subspaces $V \leq H_\alpha$. The inherited topology is Polish exactly when V is closed. Closed subspaces have canonical pivot subsets $S \subseteq \alpha$ and are order-topologically isomorphic to $\mathbb{Q}^S$. The corresponding submodel is isomorphic to $M_{\text{otp}(S)}$.*
- (5) *Every elementary self-embedding of M_α is continuous if and only if $\alpha \leq \omega$.*
- (6) *Every elementary self-embedding of M_α has closed image if and only if $\alpha < \omega \cdot 2$.*

All claims are proved below. Parts (1) and the underlying arithmetic construction are included for completeness and attributed to the existing work. The ordered normal forms, their model-theoretic consequences, and the precise combination of thresholds are the focus of this article.

[write] **What this report already contained.** Clause (1) is [theorem 35.1](#) (source 07) and [theorem 35.9](#) (source 06) for infinite α ; for finite α it is elementary. For $\alpha = \beta = \omega$, the first sentence of clause (2) is source 04's [proposition 11.3](#) (proved there for arbitrary divisible coefficient groups, as [theorem 129.1](#) is here), and clauses (3) and (5) are source 04's [theorem 11.4](#) and [corollary 11.5](#). That the M_α are pairwise nonisomorphic is [theorem 35.4](#). Clause (4) has the real model's [theorem 11.2](#) as its counterpart. The rest, in particular everything about $\alpha > \omega$, is new to the report.

Coordinate rank	All self-embeddings continuous?	All self-embedding images closed?	Number of automorphisms
0	Yes	Yes	1
$0 < \alpha < \omega$	Yes; all are onto	Yes	$\aleph_0$
$\alpha = \omega$	Yes	Yes	$\mathfrak{c}$
$\alpha = \omega + n$, $0 < n < \omega$	No	Yes	$2^{\mathfrak{c}}$
$\alpha \geq \omega \cdot 2$	No	No	$2^{\mathfrak{c}}$

127.2 Relation to established mathematics

Quantifier elimination for Presburger arithmetic is classical [3]; we give the needed uniform argument in [section W](#). Ordered Hahn groups and their automorphisms have a substantial literature. Kuhlmann and Serra [87] develop the valuation and skeleton framework and matrix descriptions in relevant Hahn settings. Serra's dissertation [88], especially its discussion of strong additivity, already explains closely related automatic behavior for discrete ranks and failures beyond them. Continuity for discrete-field products and preservation of formal infinite sums are also studied by Freni [89]. The unordered closed-subspace theory belongs to classical linearly compact duality; see, for example, [90].

We do not claim these general mechanisms as new. The proofs here specialize them, make the positive-cone and elementary-embedding interfaces explicit, and extract the exact orbit, lattice, and closed-image consequences. The source audit found no statement of this complete package in the inspected ProveIt reports or cited sources. That limited audit does not establish historical priority.

[write] (Batch 89.) For the repository the audit was incomplete through no fault of its own: the case $\alpha = \beta = \omega$ of the dictionary, the matrix classification, automatic continuity and

the image topology was already proved by source 04 (sections 11.2 and 11.3), which had been placed but not yet written into this report at source 16's pin. The package is new for transfinite α and as a package.

127.3 Foundations and conventions

We work in ZFC. Every α is a set ordinal, and every product, matrix, subspace, and collection of submodels appearing in a fixed instance is a set. An elementary submodel always has the *inherited* topology unless another topology is specified. A map called Borel is Borel measurable between the indicated Polish spaces. "Clopen" means both closed and open. We use $\mathfrak{c} = 2^{\aleph_0}$.

The product topology is essential. At transfinite coordinate ranks it can differ sharply from both the lexicographic order topology and the Hahn valuation topology. The discontinuous maps below are discontinuous for the product topology specified in (127.1).

128 The arithmetic and topology of the models

128.1 Why the positive cone satisfies full Presburger arithmetic

Definition 128.1. A $\mathbb{Z}$ -group is an ordered abelian group with least positive element 1, such that for every standard positive integer m and every g there are unique q and $r \in \{0, \dots, m-1\}$ with $g = mq + r1$.

The group G_α is a $\mathbb{Z}$ -group. It is discretely ordered because any element with positive coefficient part is greater than every standard integer, whereas the zero coefficient part leaves the usual order on $\mathbb{Z}$. For division, write $n = mq + r$ in $\mathbb{Z}$. Then

$$(h, n) = m(h/m, q) + (0, r). \quad (128.1)$$

The coefficient group is uniquely divisible. The same formula works in the positive cone: if $h > 0$ then $h/m > 0$, and if $h = 0$ then ordinary nonnegative division applies.

The classical quantifier-elimination theorem says that the theory of $\mathbb{Z}$ -groups eliminates quantifiers after adding the predicates $g \equiv_m h$, for every standard $m \geq 2$. Its complete theory is that of $(\mathbb{Z}; 0, 1, +, -, <)$. Relativizing group formulas to the nonnegative cone gives the complete theory of $(\mathbb{N}; 0, 1, +, <)$. This includes every first-order induction instance in the additive language. It is much stronger than verifying finitely many associativity or division axioms.

In our split groups, congruences have a particularly simple meaning:

$$(h, n) \equiv_m (h', n') \iff n \equiv n' \pmod{m}. \quad (128.2)$$

This identity will make elementary embeddings almost entirely a problem about ordered $\mathbb{Q}$ -vector spaces.

128.2 Polishness and the role of the missing boundary

The group H_α is a countable product of countable discrete Polish spaces. Choose a bijective enumeration (i_k) of an infinite α ; a compatible complete metric is

$$d(h, h') = \sum_{k \geq 0} 2^{-k-1} \mathbf{1}_{h_{i_k} \neq h'_{i_k}}.$$

This enumeration specifies a topology, not the lexicographic priority of coordinates, which remains the ordinal order on α .

For $i < \alpha$, the condition $h_j = 0$ for every $j < i$ is closed. Since each rational coordinate is discrete,

$$H_\alpha^{>0} = \bigcup_{i < \alpha} \{h : h_j = 0 \ (j < i), \ h_i > 0\} \quad (128.3)$$

is F_σ . The negative cone is likewise F_σ , so the positive and nonnegative cones are both G_δ as well. It follows that

$$M_\alpha = (H_\alpha^{>0} \times \mathbb{Z}) \cup (\{0\} \times \mathbb{N}) \quad (128.4)$$

is G_δ in G_α : finite unions of G_δ sets are G_δ in a metric space. A G_δ subspace of a Polish space is Polish [8]. Addition is the restriction of the continuous group operation and preserves the cone.

Remark 128.2. The cone need not be closed in its ambient group. For example, when $\alpha \geq \omega$, $(e_k, -1) \in M_\alpha$ converges to $(0, -1) \notin M_\alpha$, where e_k has one nonzero coordinate at k . Thus completeness is obtained from the G_δ theorem, not by claiming that the displayed ambient metric remains complete on the cone.

Proposition 128.3 (Baire-space carrier; for $\alpha \geq \omega$, [theorems 35.1](#) and [35.9](#)). *If $\alpha \geq \omega$ is countable, M_α is homeomorphic to $\mathbb{N}^\mathbb{N}$. If $\alpha < \omega$, it is countable and discrete.*

Proof. The finite case is immediate. In the infinite case the cone is zero-dimensional and Polish. We show that no compact subset has nonempty interior. Let a basic coefficient cylinder, with integer coordinate fixed and only finitely many rational coordinates prescribed, meet the cone. It contains a smaller cylinder in which positivity is already decided at a finite natural-number coordinate. To see this, use an already prescribed positive first nonzero coordinate if it occurs before the first unprescribed natural coordinate; otherwise put a positive rational at that first unprescribed coordinate and zero at the earlier unprescribed ones. A prescribed negative first nonzero coordinate before this point would contradict the assumed nonempty intersection with the cone.

Choose a later natural coordinate that is still unprescribed. It can take every rational value while all these points remain in the smaller cylinder. The projection of a compact set to a discrete rational coordinate is finite. Therefore the cylinder cannot be contained in a compact set. In particular there are no isolated points and the space is nowhere locally compact. The standard characterization of Baire space as the nonempty zero-dimensional Polish space that is nowhere locally compact now applies [8]. $\square$

[write] The cylinder argument is the one of [theorem 35.1](#) (source 07) and [theorem 35.9](#) (source 06); for $\alpha = \omega$, [theorem 6.1](#) gives an explicit, computable homeomorphism.

129 From arithmetic homomorphisms to ordered linear embeddings

The following observation removes any ambiguity about which arithmetic maps are being classified.

Theorem 129.1 (The homomorphism dictionary; = [proposition 11.3](#) [04]). *Let A, B be divisible ordered abelian groups and let $M_A = (A \overrightarrow{\times} \mathbb{Z})_{\geq 0}$, $M_B = (B \overrightarrow{\times} \mathbb{Z})_{\geq 0}$. Every map $f : M_A \rightarrow M_B$ preserving 0, 1, and addition has the unique form*

$$f(a, n) = (T(a), n), \quad (129.1)$$

where $T : A \rightarrow B$ is a strictly order-preserving $\mathbb{Q}$ -linear embedding. Conversely every such T defines such an f , and f is elementary in the full Presburger language.

Proof. The cancellative positive cone generates its ordered group by differences. Hence f extends uniquely to a group homomorphism $F : A \times \mathbb{Z} \rightarrow B \times \mathbb{Z}$ fixing $(0, 1)$: define $F(u - v) = f(u) - f(v)$ for $u, v \geq 0$. This is well-defined because $u - v = u' - v'$ implies $u + v' = u' + v$.

The maximal divisible subgroup of the split group is

$$\bigcap_{m \geq 1} m(A \times \mathbb{Z}) = A \times \{0\}.$$

A homomorphism sends this subgroup into the analogous subgroup of the target. Consequently $F(a, n) = (T(a), n)$ for an additive map T , which is $\mathbb{Q}$ -linear because divisibility is unique in ordered groups.

If $a > 0$, the element $(a, -1)$ belongs to M_A . Its image $(T(a), -1)$ can belong to M_B only if $T(a) > 0$. Thus T strictly preserves positivity. Applying this to differences proves that T is injective and preserves and reflects order.

Conversely such a T plainly preserves the positive cone and the operations. The extended map preserves and reflects equality, order, and every congruence by (128.2). Quantifier elimination therefore makes the group embedding elementary. Restricting quantified variables to the nonnegative cone proves elementarity of f . $\square$

[write] This is source 04's [proposition 11.3](#), with the same hypotheses and conclusion. Both proofs extend f to the group of differences and use the maximal divisible subgroup $A \times \{0\}$; they obtain strict positivity of T differently (source 04 from $x = y + 1 + u$ for $x > y$, source 16 from the positive element $(a, -1)$).

Corollary 129.2 (for M_B , [corollary 11.5](#) [04]). *For the models M_α , “unital additive homomorphism” and “elementary embedding” describe the same maps. Injectivity, order preservation, and elementarity need not be imposed separately.*

129.1 The valuation spectrum

For $0 \neq h \in H_\alpha$, put

$$v(h) = \min\{i < \alpha : h_i \neq 0\}.$$

Two nonzero elements have the same value exactly when their absolute values are Archimedean equivalent, meaning that each is bounded by a standard positive integer multiple of the other. Indeed their first nonzero rational coefficients have a finite rational ratio. If $v(x) < v(y)$, then $|x| > n|y|$ for every positive integer n .

An order-preserving linear embedding $T : H_\alpha \rightarrow H_\beta$ therefore induces a strictly increasing map

$$p : \alpha \longrightarrow \beta, \quad p(i) = v(T(e_i)), \quad v(T(h)) = p(v(h)). \quad (129.2)$$

The last equality follows from Archimedean equivalence. In particular, an embedding exists only if $\alpha \leq \beta$. If that ordinal inequality holds, inserting zeros outside an initial copy of α gives a continuous ordered linear embedding. Thus elementary embeddability among this family is exactly ordinal comparison.

Corollary 129.3 (An ordinal hierarchy on a fixed topological carrier; extends [theorem 35.4](#)). For $\omega \leq \alpha, \beta < \omega_1$, all the carriers M_α are homeomorphic to Baire space, but

$$M_\alpha \text{ elementarily embeds into } M_\beta \iff \alpha \leq \beta.$$

The same equivalence holds when the embedding is required to be continuous or Borel. Two members are isomorphic exactly when their indices agree.

[write] Its last sentence is [theorem 35.4](#) (source 07; source 06's [theorem 35.9](#)), proved there by the Archimedean classes; the embeddability equivalence is new.

130 Borel maps, finite dependence, and matrix normal forms

130.1 A countable-index Baire lemma

Lemma 130.1. A subgroup with the Baire property and countable index in a Polish group is open. In particular a Borel subgroup of countable index is open, and a G_δ subgroup of a Polish group is closed.

Proof. Countably many cosets cover the group. The Baire category theorem implies that the subgroup is nonmeagre. A set with the Baire property that is nonmeagre is comeagre in some nonempty open set. The usual Baire argument then shows that its difference set contains a neighborhood of the identity: sufficiently small translates of that open set overlap, and the two comeagre subsets have a common point there. For a subgroup the difference set is the subgroup itself, so it is open.

For the final assertion, a G_δ subgroup K is dense and comeagre in its closure L , which is a Polish group. Every translate $x + K$, $x \in L$, is also comeagre in L and intersects K . Hence $x \in K - K = K$, so $K = L$. $\square$

Lemma 130.2 (Continuous functionals). A $\mathbb{Q}$ -linear functional $\ell : \mathbb{Q}^I \rightarrow \mathbb{Q}$, where I is countable and all factors are discrete, is continuous if and only if it depends on finitely many coordinates. If it is Borel, it is continuous.

Proof. For continuity, the inverse image of zero contains a basic open subgroup in which finitely many coordinates vanish. The functional therefore factors through that finite coordinate projection. Conversely every finite rational linear combination of coordinates is continuous. If ℓ is Borel, its kernel is a Borel subgroup of countable index, since $\mathbb{Q}$ is countable. Apply [lemma 130.1](#). $\square$

Theorem 130.3 (Borel equals continuous for arithmetic embeddings). For an elementary embedding $f : M_\alpha \rightarrow M_\beta$ and its coefficient map T , the following are equivalent:

$$f \text{ is Borel}; \quad T \text{ is Borel}; \quad T \text{ is continuous}; \quad f \text{ is continuous}.$$

Proof. The coefficient positive and negative cones are Borel by [\(128.3\)](#). If f is Borel, define $T(h)$ by taking the coefficient part of $f(h, 0)$ when $h \geq 0$, and the negative of the coefficient part of $f(-h, 0)$ otherwise. This is a Borel definition on two Borel pieces. Every coordinate of a Borel T is a Borel linear functional, hence continuous by [lemma 130.2](#). Product-topology continuity follows. A continuous T restricts to a continuous f , and any continuous f is Borel. $\square$

130.2 The exact matrices

Definition 130.4. A matrix $A = (a_{ji})_{j < \beta, i < \alpha}$ is row-finite if for each j only finitely many a_{ji} are nonzero. It has increasing positive pivots if there is a strictly increasing $p : \alpha \rightarrow \beta$ with

$$a_{ji} = 0 \quad (j < p(i)), \quad a_{p(i),i} > 0. \quad (130.1)$$

Theorem 130.5 (Matrix classification; for $\alpha = \beta = \omega$, [theorem 11.4](#) [04]). *The continuous elementary embeddings $M_\alpha \rightarrow M_\beta$ are exactly*

$$(h, n) \mapsto (Ah, n), \quad (Ah)_j = \sum_{i < \alpha} a_{ji} h_i, \quad (130.2)$$

where A is row-finite with increasing positive pivots. Each displayed sum is finite. The pivot map is uniquely determined by the embedding.

Proof. For a continuous T , [lemma 130.2](#) supplies each matrix row. The columns are $T(e_i)$. Their first nonzero entries obey (130.1) by (129.2), and positivity gives positive pivots.

Conversely let $h \neq 0$ have first nonzero coordinate i . All coordinates of Ah before $p(i)$ vanish: columns indexed below i are multiplied by zero, while those indexed at least i have no entry that early. At the pivot itself,

$$(Ah)_{p(i)} = a_{p(i),i} h_i.$$

Its sign is the sign of h_i . Thus A is a strictly ordered linear embedding, and row-finiteness gives continuity. Apply [theorem 129.1](#). $\square$

[write] For $\alpha = \beta = \omega$ this is source 04's [theorem 11.4](#), which is stronger there: it assumes no continuity ([theorem 133.1](#) below) and adds that the map is onto exactly when p is the identity.

130.3 Why the images are closed even at limit ranks

Lemma 130.6 (Finite ancestry). *Suppose each vertex i of a well-ordered index set has finitely many designated predecessors, all strictly below i . Then each vertex has only finitely many ancestors obtained by repeatedly taking designated predecessors.*

Proof. Form the tree of finite predecessor paths starting at the vertex. It is finitely branching. An infinite such tree would have an infinite branch by König's lemma, giving an infinite descending sequence in a well-order. Thus the tree and its set of labels are finite. $\square$

Theorem 130.7 (A continuous left inverse and a closed image; for $\alpha = \beta = \omega$, [corollary 11.5](#) [04]). *Every matrix embedding in [theorem 130.5](#) admits a continuous $\mathbb{Q}$ -linear left inverse $L : H_\beta \rightarrow H_\alpha$. Its group image is closed, and its cone image is a closed elementary submodel with inherited Polish topology.*

Proof. At row $p(i)$, columns indexed above i vanish. Define recursively

$$(Ly)_i = \frac{1}{a_{p(i),i}} \left(y_{p(i)} - \sum_{k < i} a_{p(i),k} (Ly)_k \right). \quad (130.3)$$

Each sum is finite. Although the recursion is indexed by an ordinal, [lemma 130.6](#) shows that each output coordinate ultimately depends on only finitely many input coordinates. Thus L is a continuous row-finite linear map. Transfinite induction in [\(130.3\)](#) gives $LA = \text{id}$.

Now AL is a continuous projection and

$$\text{im } A = \{y : ALy = y\} = \ker(\text{id} - AL),$$

which is closed in the Hausdorff group H_β . The cone image is $(\text{im } A \times \mathbb{Z}) \cap M_\beta$, because order is reflected. It is closed in the Polish space M_β , and L restricts to its continuous inverse parametrization. $\square$

The finite-ancestry argument is important. A naive sequence of transfinite row subtractions could alter a limit coordinate infinitely often. Row-finiteness alone does not justify that naive procedure. Equation [\(130.3\)](#) and the well-founded dependency argument supply the missing control.

[write] For $\alpha = \beta = \omega$ the recursion [\(130.3\)](#) is source 04's recovery formula [\(11.7\)](#), where no ancestry lemma is needed because every pivot row has finitely many predecessors, and the closed image is part of [corollary 11.5](#).

131 All elementary submodels and their Polish topology

Theorem 131.1 (Elementary submodels are coefficient subspaces; extends [theorem 11.2](#)). *The elementary submodels of M_α are precisely*

$$M_V = (V \overrightarrow{\times} \mathbb{Z})_{\geq 0}, \quad V \leq_{\mathbb{Q}} H_\alpha.$$

The model M_V is Polish in the inherited topology if and only if V is closed in H_α .

Proof. If V is a rational linear subspace, $V \overrightarrow{\times} \mathbb{Z}$ is a $\mathbb{Z}$ -group and its inclusion preserves and reflects congruences as well as order. Quantifier elimination makes its cone elementary.

Conversely let $N \preccurlyeq M_\alpha$ and set $\Gamma = N - N$ inside G_α . If $a, b \in N$ and $a - b \geq 0$, the equation $b + x = a$ has a unique solution in the ambient cone. Elementarity puts that solution in N . Thus $N = \Gamma \cap M_\alpha$. The subgroup Γ contains $(0, n)$ for every standard integer n . Subtracting integer coordinates yields $\Gamma = V \times \mathbb{Z}$ for an additive subgroup $V \leq H_\alpha$. The unique standard division of $(v, 0)$ has remainder zero and quotient $(v/m, 0)$. For $v \geq 0$ elementarity puts this quotient in N ; negatives follow by subtraction. Hence V is divisible and therefore a rational linear subspace.

If V is closed, M_V is closed in M_α and is Polish. Conversely, if M_V is Polish, its integer-coordinate-zero slice is clopen in M_V and is homeomorphic to $V^{\geq 0}$. Thus $V^{\geq 0}$ is a Polish subspace of the Polish space H_α , so it is G_δ . Its negative is also G_δ , and

$$V = V^{\geq 0} \cup (-V^{\geq 0})$$

is G_δ . By [lemma 130.1](#) the subgroup V is closed. $\square$

Polishness here cannot be replaced by merely admitting some abstract Polish presentation. A nonclosed V might be algebraically isomorphic to a closed space, but the inherited topology is the issue being classified.

[write] For the real model this is source 05's [theorem 11.2](#), with the same proof; there only two elementary submodels are Polish, here one for each closed subspace of H_α .

131.1 The finite-support annihilator

Write $H_\alpha^\circ = \mathbb{Q}^{(\alpha)}$ for the finite-support dual, with pairing

$$\langle c, h \rangle = \sum_{i < \alpha} c_i h_i.$$

For $V \leq H_\alpha$ define $V^\perp = \{c \in H_\alpha^\circ : \langle c, v \rangle = 0 \text{ for all } v \in V\}$.

Lemma 131.2 (Cylinder separation). *For every linear subspace $V \leq H_\alpha$,*

$$\text{cl}(V) = (V^\perp)^\perp.$$

Proof. Each finite-support functional is continuous, so its kernel contains $\text{cl}(V)$ whenever it contains V . For the reverse inclusion, if $h \notin \text{cl}(V)$ then a finite coordinate set F separates h from V : $h \upharpoonright F \notin \pi_F(V)$. Finite-dimensional linear algebra supplies a functional on $\mathbb{Q}^F$ vanishing on $\pi_F(V)$ but not on $h \upharpoonright F$. Extend it by zero outside F . $\square$

131.2 Canonical ordered normal form

Theorem 131.3 (Pivot normal form of a closed subspace). *Let $V \leq H_\alpha$ be closed. There is a unique subset $S \subseteq \alpha$, its valuation spectrum*

$$S = \{v(x) : 0 \neq x \in V\},$$

with the following description. For each $i \notin S$ there are unique finitely many coefficients $b_{is} \in \mathbb{Q}$, $s \in S$, $s < i$, such that

$$V = \left\{ x : x_i = \sum_{\substack{s \in S \\ s < i}} b_{is} x_s \text{ for every } i \notin S \right\}. \quad (131.1)$$

The coordinates $(x_s)_{s \in S}$ are arbitrary. The resulting parametrization $\mathbb{Q}^S \rightarrow V$ is an ordered topological linear isomorphism.

Proof. Let ℓ_i be the coordinate functional restricted to V . Scan the ordinal α in increasing order, selecting i precisely when ℓ_i is not in the algebraic span of the previously selected functionals. Let S be the selected set. The selected functionals are linearly independent. Every unselected ℓ_i is a finite linear combination of selected ℓ_s , $s < i$; the coefficients are unique by independence. This proves the necessary equations in (131.1).

Given arbitrary $c \in \mathbb{Q}^S$, define $x_s = c_s$ at selected coordinates and define the remaining x_i by those finite equations. We claim $x \in V$. Any finite-support functional annihilating V becomes, after substituting these equations, a finite linear combination of the independent selected functionals which vanishes on V . Every coefficient in that combination is therefore zero, so the original functional also vanishes at x . Since V is closed, lemma 131.2 gives $x \in V$.

All coordinates of the parametrization are finite linear combinations of the c_s , so it is continuous; its inverse is coordinate restriction. If c first becomes nonzero at $s \in S$, all earlier unselected coordinates are combinations of earlier, zero, selected coordinates. Thus the first nonzero coordinate of x is s and has value c_s . The parametrization preserves order. It also shows that the selected set is exactly the valuation spectrum, making S intrinsic. $\square$

This proof uses finite linear duality to handle limit coordinates. It does not select successive vectors and hope that an infinite elimination process converges at the limit.

Corollary 131.4 (Isomorphism types). *Every inherited-Polish elementary submodel of M_α is order-topologically and arithmetically isomorphic to a unique M_β , $\beta \leq \alpha$. All such β occur. For $\alpha = \omega$, a Polish elementary submodel is either a countable discrete M_k for $k < \omega$, or an elementary self-copy of M_ω .*

Proof. Use $\beta = \text{otp}(S)$ in [theorem 131.3](#) and [theorem 129.1](#). Uniqueness follows from the Archimedean rank of the coefficient group. A coordinate subspace supported on an initial segment β realizes every $\beta \leq \alpha$. $\square$

132 Orbits, quotients, and the complete submodel lattice

Theorem 132.1 (Ambient conjugacy and quotient topology). *Let V, W be closed coefficient subspaces of H_α .*

- (1) *A continuous ordered linear automorphism sends V to W if and only if their pivot subsets agree. The same criterion holds for arbitrary ambient arithmetic automorphisms of M_α sending M_V onto M_W .*
- (2) *If S is the pivot subset of V , then as topological vector spaces*

$$H_\alpha/V \cong \mathbb{Q}^{\alpha \setminus S}.$$

- (3) *M_V is clopen in M_α exactly when $\alpha \setminus S$ is finite. Otherwise it is closed and nowhere dense.*

Proof. Let E_S be the coordinate subspace supported on S . Using the coefficients in [\(131.1\)](#), define

$$(U_V x)_s = x_s \ (s \in S), \quad (U_V x)_i = x_i + \sum_{s < i, s \in S} b_{is} x_s \ (i \notin S).$$

This is row-finite lower triangular with diagonal one. Its inverse subtracts the same finite combinations, since selected coordinates are fixed. It is a continuous ordered automorphism and sends E_S onto V . If V, W have the same S , use $U_W U_V^{-1}$.

Every ordered automorphism induces an order automorphism of the ordinal rank α , and this must be the identity. Therefore it preserves valuation spectra even if no regularity is assumed. This proves the converse assertion about orbits.

Conjugating by U_V reduces the quotient to H_α/E_S , which is the product on the complementary coordinates. The subspace E_S is open precisely when only finitely many coordinates are required to vanish. If it is not open it has empty interior, as does any closed subgroup which is not open.

For the positive cone, openness of V implies openness of M_V . Conversely suppose M_V is open. At $(0, 0)$ it contains a neighborhood specified by finitely many coefficient coordinates; thus the nonnegative half of the corresponding basic open subgroup is contained in V . Negating shows that the entire subgroup lies in V , so V is open. To prove nowhere density when the missing set is infinite, conjugate to E_S . Given a point of M_{E_S} and a neighborhood prescribing finitely many coordinates F , choose $j \in (\alpha \setminus S) \setminus F$. Adding the positive coefficient vector e_j preserves the cone and the prescribed coordinates, but leaves E_S . Thus no point of M_{E_S} is interior, and the nonopen closed submodel is nowhere dense. $\square$

[write] For the image of a self-embedding of M_ω , the dichotomy of (3) is source 04's [corollary 11.5](#); here it holds for every closed elementary submodel and every countable α .

Remark 132.2. If $\alpha \setminus S$ is infinite countable, the quotient is a *full product* of copies of $\mathbb{Q}$. Its algebraic dimension over $\mathbb{Q}$ is $\mathfrak{c}$, not $\aleph_0$. Counting the missing pivots as though they were a Hamel basis would give an incorrect codimension.

Corollary 132.3. *For $\alpha = \omega$ there are exactly $\mathfrak{c}$ ambient automorphism orbits of Polish elementary submodels, and already $\mathfrak{c}$ orbits of uncountable elementary self-copies. In the latter case they are all abstractly isomorphic, but need not be conjugate inside the ambient model.*

Proof. The pivot subsets are exactly the subsets of ω ; the infinite ones give self-copies. There are $\mathfrak{c}$ of each. All automorphisms of M_ω will be shown continuous in [section 133](#), so the same count holds without a topological qualification. $\square$

Theorem 132.4 (The lattice). *The inherited-Polish elementary submodels of M_α form a complete lattice. Under $V \mapsto M_V$ its operations are*

$$\bigwedge_t M_{V_t} = M_{\bigcap_t V_t}, \quad \bigvee_t M_{V_t} = M_{\text{cl}(\sum_t V_t)}.$$

The lattice is order anti-isomorphic to the lattice of all linear subspaces of the finite-support space $\mathbb{Q}^{(\alpha)}$, by taking annihilators.

Proof. Intersections of closed linear subspaces are closed, and the smallest closed linear subspace containing a family is the closure of its algebraic sum. The correspondence with elementary submodels follows from [theorem 131.1](#).

The annihilator map reverses inclusion and is injective on closed subspaces by [lemma 131.2](#). If $W \leq \mathbb{Q}^{(\alpha)}$, then $W^\perp$ is closed. If $c \notin W$, ordinary linear algebra on $\mathbb{Q}^{(\alpha)}/W$ gives a functional which vanishes on W but not on c . Every algebraic functional on the finite-support space is evaluation against a point of the full product $\mathbb{Q}^\alpha$. Thus $(W^\perp)^\perp = W$, proving surjectivity and the claimed anti-isomorphism. $\square$

This answers the repository's elementary-submodel-lattice question for the displayed family. It does not classify the lattices of arbitrary Polish presentations of arbitrary Presburger models.

[write] The question is Part I's [question 14.9](#) (source 05's); a dated note after it records this answer.

133 The first threshold: automatic continuity fails at $\omega + 1$

Theorem 133.1 (Automatic continuity through ω ; for $\alpha = \omega$, [theorem 11.4](#) [04]). *Every elementary self-embedding of M_α is continuous when $\alpha \leq \omega$. If $\alpha < \omega$, every such embedding is an automorphism. For $\alpha = \omega$, all embeddings are exactly the increasing-positive-pivot matrices; row-finiteness follows automatically from their pivot condition.*

Proof. Finite products are discrete. A strictly increasing self-map of a finite ordinal is the identity, so the corresponding finite triangular matrix has nonzero diagonal and is invertible.

Now let $T : H_\omega \rightarrow H_\omega$ be any ordered linear embedding, without a regularity assumption. Its pivot map $p : \omega \rightarrow \omega$ is strictly increasing and hence $p(n) \geq n$. If h vanishes in its first n coordinates, then either $h = 0$ or $v(h) \geq n$. Equation (129.2) implies that $T(h)$ vanishes in its first $p(n)$ coordinates. These finite-coordinate kernels form the neighborhood basis of zero, proving continuity. The matrix description follows from [theorem 130.5](#). In any output row j only columns with $p(i) \leq j$ can contribute; there are finitely many of them. $\square$

[write] For $\alpha = \omega$ this is the automatic continuity of source 04's [theorem 11.4](#) and [corollary 11.5](#), proved there by the same filtration argument ([11.6](#)); the finite case is new to the report.

133.1 A functional invisible on every finite-support vector

Let

$$D_B = \mathbb{Q}^\omega, \quad D_B^{\text{fin}} = \mathbb{Q}^{(\omega)}.$$

The finite-support subspace is dense and proper. Choose a nonzero linear functional

$$\varphi : D_B \longrightarrow \mathbb{Q}, \quad \varphi(D_B^{\text{fin}}) = 0. \quad (133.1)$$

Such a functional is obtained by choosing a nonzero functional on D_B/D_B^{fin} and composing with the quotient map. This is an algebraic use of Choice. A continuous functional vanishing on a dense subspace would vanish everywhere, so φ is discontinuous. By [lemma 130.2](#) it is not Borel.

[write] **Choice.** The functional φ is the only nonconstructive object of this part. Everything built from it, [theorem 133.3](#), the value 2^c in [theorem 133.4](#), [proposition 133.5](#), [theorem 134.2](#), [example 136.4](#) and the negative halves of clauses (5) and (6) of [theorem 127.1](#), is proved in ZFC. The next remark shows that choice cannot be removed altogether.

Remark 133.2 (Discontinuous self-embeddings need a set without the Baire property [merge]). Work in ZF with dependent choice. If some elementary embedding $f : M_\alpha \rightarrow M_\beta$ between members of the family (α, β countable) is not continuous, then some subset of a Polish space does not have the Baire property. Consequently, in a model of ZF with dependent choice in which every set of reals has the Baire property, which exists if ZF is consistent [96], every elementary embedding between members of the family is continuous, given by a matrix as in [theorem 130.5](#) and has closed image ([theorem 130.7](#)); there the thresholds of clauses (5) and (6) of [theorem 127.1](#) do not occur.

Proof. By [theorem 129.1](#), $f = (T, \text{id})$ with T an ordered $\mathbb{Q}$ -linear embedding, and f is continuous if T is. If T is not continuous for the product topology, some coordinate functional $\ell = \pi_j \circ T : H_\alpha \rightarrow \mathbb{Q}$ is not continuous. Its kernel is a subgroup of countable index of the Polish group H_α , because $\mathbb{Q}$ is countable. If the kernel had the Baire property it would be open by [lemma 130.1](#), whose proof uses only countably many choices, and then ℓ , constant on the cosets of an open subgroup, would be continuous. Hence the kernel lacks the Baire property, and H_α is homeomorphic to Baire space or countable discrete. In the model described, every coordinate functional is therefore continuous, and [theorems 130.5](#) and [130.7](#) apply. $\square$

This bears on source 16's [question 138.3](#): a non-Borel ordered automorphism of $M_{\omega+1}$ yields a set of reals without the Baire property. Whether a weaker hypothesis than choice produces one, and whether the converse holds, is not settled here. The remark was written for this report and has not been independently reviewed.

Theorem 133.3 (A non-Borel elementary automorphism). *For every countable $\alpha > \omega$, define*

$$(\text{Sh}_\varphi x)_i = \begin{cases} x_\omega + \varphi(x \upharpoonright \omega), & i = \omega, \\ x_i, & i \neq \omega. \end{cases} \quad (133.2)$$

This is an ordered linear automorphism of H_α , with inverse $\text{Sh}_{-\varphi}$. Its induced automorphism of M_α is neither continuous nor Borel. It fixes every finite-support coefficient vector.

Proof. Linearity and the inverse formula are immediate. If the first nonzero coordinate of x is below ω , that coordinate is unchanged and still decides the sign. If all coordinates below ω vanish, the added functional is zero, so every coordinate of x is unchanged. Thus the first nonzero coordinate and its sign are preserved in every case.

The coordinate- ω functional of Sh_φ contains the noncontinuous summand $\varphi(x \upharpoonright \omega)$, so Sh_φ is discontinuous. [Theorem 130.3](#) makes the induced cone automorphism non-Borel. Finally [\(133.1\)](#) proves the finite-support assertion. $\square$

Together [theorems 133.1](#) and [133.3](#) prove the exact first threshold. The shear preserves the Hahn valuation v , and is continuous for the associated finer valuation topology. There is no contradiction: in the product topology, no neighborhood can demand that all the infinitely many coordinates below ω vanish.

133.2 The size of the discontinuous automorphism group

Theorem 133.4 (Exact cardinalities). *For countable α ,*

$$|\text{Aut}(M_\alpha)| = \begin{cases} 1, & \alpha = 0, \\ \aleph_0, & 0 < \alpha < \omega, \\ \mathfrak{c}, & \alpha = \omega, \\ 2^{\mathfrak{c}}, & \alpha > \omega. \end{cases}$$

For every infinite countable α , exactly $\mathfrak{c}$ automorphisms are continuous, equivalently Borel.

Proof. Finite-rank automorphisms are finite triangular rational matrices with positive diagonal, giving a countably infinite set in nonzero rank. At rank ω , matrices supply at most $\mathfrak{c}$ maps and arbitrary positive rational diagonal sequences supply at least $\mathfrak{c}$.

The quotient $D_{\mathbb{B}}/D_{\mathbb{B}}^{\text{fin}}$ has cardinality $\mathfrak{c}$: $D_{\mathbb{B}}$ has that cardinality and each coset is countable. Over a countable field an infinite-dimensional vector space has the cardinality of its Hamel basis, so this quotient has dimension $\mathfrak{c}$. Its linear functionals to $\mathbb{Q}$ number $\aleph_0^{\mathfrak{c}} = 2^{\mathfrak{c}}$. Distinct functionals give distinct shears. There are at most $\mathfrak{c}^{\mathfrak{c}} = 2^{\mathfrak{c}}$ maps on a set of cardinality $\mathfrak{c}$, which proves equality for $\alpha > \omega$.

Continuous maps on a separable space into a Hausdorff space of cardinality $\mathfrak{c}$ are determined by their values on a countable dense subset, so there are at most $\mathfrak{c}$ of them. Positive rational diagonal sequences again attain this bound. Use [theorem 130.3](#) for the Borel assertion. $\square$

[write] For $\alpha = \omega$, source 04's [corollary 11.5](#) identifies the automorphisms as the infinite lower triangular rational matrices with positive diagonal; the count $\mathfrak{c}$ and the counts for $\alpha > \omega$ are new. The value $2^{\mathfrak{c}}$ uses choice (note above). The dimension count for $\mathbb{Q}^\omega/\mathbb{Q}^{(\omega)}$ is the rational, full-product counterpart of the collection's `rvs:eu:thm:fullcodim` and `rvs:eu:cor:continuum` (real-vector-space-structure: the finite-support part of a real Hahn field has codimension equal to its dimension), and that report's subsection "Product topology versus algebraic dimension" contrasts, as the paragraph after [theorem 133.3](#) does here, the coefficientwise topology, in which the finite-support vectors are dense, with the valuation topology.

133.3 A clopen submodel can become dense under an algebraic automorphism

Proposition 133.5. *For $\alpha > \omega$, the shear Sh_φ sends the clopen coefficient subspace $\{x : x_\omega = 0\}$ to*

$$W_\varphi = \{y : y_\omega = \varphi(y \upharpoonright \omega)\}, \quad (133.3)$$

which is dense, proper, and non-Borel. The corresponding elementary cone submodel is also dense and non-Borel, and is not Polish in its inherited topology.

Proof. The image equation is immediate. A nonzero rational functional onto $\mathbb{Q}$ is surjective. Every fiber of φ is dense, since it contains a translate of the dense subspace D_B^{fin} . Thus any finite cylinder and any prescribed value of coordinate ω meet W_φ . It is proper because the value at ω is determined by the preceding block.

The subgroup W_φ has countable index: it is the kernel of $y \mapsto y_\omega - \varphi(y \upharpoonright \omega)$, onto $\mathbb{Q}$. If it were Borel, [lemma 130.1](#) would make it open and hence closed, contradicting its being dense and proper.

For density in the cone, any finite cylinder meeting the cone contains one whose positivity is decided in the first ω coordinates, as in [proposition 128.3](#). The prescribed functional value can still be attained there: first choose a vector in the desired fiber, then change finitely many coordinates to meet the cylinder and force a positive first free coordinate. These changes do not change φ . If the cone submodel were Borel, its integer-zero slice and the negative of that slice would make W_φ Borel. Finally apply [theorem 131.1](#). $\square$

For $\alpha = \omega + 1$, this does *not* yet give a nonclosed image of a self-embedding: the coordinate hyperplane has rank ω , not $\omega + 1$. That distinction leads to the second threshold.

134 The second threshold: nonclosed self-images start at $\omega \cdot 2$

Theorem 134.1 (Closed images below the second infinite block). *If $\alpha < \omega \cdot 2$, every elementary self-embedding of M_α has closed image, even when the embedding is non-Borel.*

Proof. For $\alpha \leq \omega$ use [theorems 130.7](#) and [133.1](#). Otherwise $\alpha = \omega + n$ with $0 < n < \omega$. Let T be the ordered coefficient embedding and p its increasing rank map. The last n positions must be fixed by p : an increasing self-map of an ordinal sends i to at least i , and there is no room to move any of the final finitely many positions to the right. Also every finite rank maps below ω , since infinitely many increasing finite-rank images cannot fit in a finite terminal block.

Let $K_\alpha = \{x : x \upharpoonright \omega = 0\} \cong \mathbb{Q}^n$. The restriction $T \upharpoonright K_\alpha$ is finite triangular with nonzero diagonal, hence maps K_α onto itself. The quotient map induced by T on $H_\alpha/K_\alpha \cong H_\omega$ is an ordered embedding B . By the first-threshold theorem, B is continuous and has closed image. Moreover

$$\text{im } T = \pi^{-1}(\text{im } B),$$

where $\pi : H_\alpha \rightarrow H_\omega$ is the coordinate projection. One inclusion is immediate; for the other, realize the desired quotient coordinate by an image under T and correct the remaining terminal coordinates using the surjectivity of $T \upharpoonright K_\alpha$. Thus $\text{im } T$ is closed. Its cone is closed in M_α . $\square$

Theorem 134.2 (A dense proper non-Borel elementary self-copy). *For every countable $\alpha \geq \omega \cdot 2$ there is an elementary self-embedding of M_α whose image is dense, proper, and non-Borel. In particular its image is not Polish in the inherited topology.*

Proof. Use the functional φ from (133.1). Define $T : H_\alpha \rightarrow H_\alpha$ by

$$\begin{aligned} (Tx)_k &= x_k & (k < \omega), \\ (Tx)_\omega &= \varphi(x \upharpoonright \omega), \\ (Tx)_{\omega+k+1} &= x_{\omega+k} & (k < \omega), \\ (Tx)_i &= x_i & (\omega \cdot 2 \leq i < \alpha). \end{aligned} \tag{134.1}$$

This is linear and injective. If the first nonzero input coordinate lies in the first block, its position and sign are unchanged. If it is $\omega + k$, the first block is zero and the new functional coordinate is zero, so the first nonzero output is at $\omega + k + 1$, with the same coefficient. If it lies at or after $\omega \cdot 2$, all earlier output coordinates vanish and that coordinate is unchanged. Therefore T is an ordered embedding.

Its image is exactly the graph space W_φ of (133.3), with all coordinates other than ω unconstrained. Indeed the shift of the second block can be reversed on every coordinate after ω , and the later blocks are fixed. Proposition 133.5 proves all the asserted topological and Borel properties. The cone map is elementary by theorem 129.1. $\square$

The two thresholds answer different questions. At $\omega + 1$, an automorphism can destroy the topological regularity of a submodel while still mapping the entire ambient structure onto itself. At $\omega \cdot 2$, there is enough space after the first limit coordinate to shift an infinite tail and make the graph itself a self-copy. Theorems 134.1 and 134.2 establish that this extra space is necessary.

135 The intersection of self-copies and clopen approximation

Every ordinal α has a unique expression

$$\alpha = \lambda + n,$$

where $n < \omega$ and λ is zero or a limit ordinal. Define its terminal coefficient tail

$$K_\alpha = \{x \in H_\alpha : x_i = 0 \text{ for } i < \lambda\} \cong \mathbb{Q}^n.$$

Theorem 135.1 (The unavoidable terminal tail; extends theorem 11.7). *The intersection of the images of all elementary self-embeddings of M_α is exactly*

$$(K_\alpha \overrightarrow{\times} \mathbb{Z})_{\geq 0}.$$

The same intersection is obtained using only continuous embeddings with clopen image. Thus the intersection is the standard copy of $\mathbb{N}$ exactly when α is zero or a nonzero limit ordinal. For $0 < \alpha < \omega$ it is the entire model.

Proof. The final n ranks are fixed by every increasing self-map of α , as in theorem 134.1. The restriction of any coefficient embedding to K_α is finite triangular and onto, so every image contains this tail.

If $s < \lambda$, deleting the coordinate s from α leaves an ordered set of order type α . Explicitly, $s + \omega \leq \lambda$; shift $s + k$ to $s + k + 1$ for $k < \omega$ and fix all coordinates outside $[s, s + \omega)$. This gives an order isomorphism from α onto $\alpha \setminus \{s\}$. Repeated deletion proves the corresponding finite-removal assertion. The coordinate hyperplane $x_s = 0$ is therefore a clopen ordered self-copy of H_α , and its cone is a clopen elementary self-copy. Intersect these hyperplanes over all $s < \lambda$. Exactly K_α remains. If $\lambda = 0$ there are no such coordinates, and the preceding surjectivity argument already gives the assertion. $\square$

[write] For $\alpha = \omega$ the terminal tail is 0 and the intersection is the standard copy, as in source 05's [theorem 11.7](#), which obtains it from the iterates of one shift; the theorem extends that to all self-embeddings and every countable α .

Theorem 135.2 (Exactly which closed submodels are intersections of clopen self-copies). *Let $V \leq H_\alpha$ be closed. Then M_V is an intersection of clopen elementary self-copies of M_α if and only if $K_\alpha \subseteq V$. When this condition holds, a descending countable sequence of such self-copies suffices.*

Proof. Necessity follows from [theorem 135.1](#). For sufficiency, enumerate the countably many coordinates and let F_m be the first m of them. Put

$$V_m = \pi_{F_m}^{-1}(\pi_{F_m}(V)).$$

These are descending clopen linear subspaces and $\bigcap_m V_m = V$ by closedness. They contain K_α . Each V_m has only finitely many missing pivots, because it contains the subgroup vanishing on F_m . None of the missing pivots is in the terminal tail, since that tail is contained in V_m . Deleting finitely many nonterminal positions leaves the order type α unchanged. By [theorem 131.3](#), each V_m is therefore a continuous ordered self-copy of H_α . Their cones have the required properties. $\square$

In particular every Polish elementary submodel of M_ω is an intersection of descending clopen self-copies. This includes finite-rank submodels and the standard copy. At $\omega + 1$, the standard copy cannot arise this way: every self-copy contains the last rational scale.

136 Concrete examples and what finite calculations can certify

Example 136.1 (A clopen shift and a nowhere-dense self-copy; = [example 11.6](#) [04]). In H_ω , the shift

$$(Tx)_0 = 0, \quad (Tx)_{j+1} = x_j$$

has pivot map $p(j) = j + 1$ and image $x_0 = 0$, which is clopen. The embedding

$$(Ux)_{2j} = x_j, \quad (Ux)_{2j+1} = 0$$

has pivot map $p(j) = 2j$ and closed nowhere-dense image. Both induce elementary self-copies of the same Baire-space arithmetic model. The difference is the finiteness or infinitude of the missing pivot set.

Example 136.2 (A non-coordinate closed submodel). Let

$$V = \{x \in \mathbb{Q}^\omega : x_{2j+1} = x_{2j} + 2x_{2j-2} \ (j \geq 0)\},$$

where $x_{-2} = 0$. Its pivot set is the even coordinates. The map

$$c \mapsto x, \quad x_{2j} = c_j, \quad x_{2j+1} = c_j + 2c_{j-1}, \quad c_{-1} = 0,$$

is a continuous ordered parametrization. The model M_V is a nowhere-dense elementary self-copy. Its quotient coefficient space is another full rational product. Replacing the right-hand side by zero gives a conjugate submodel, because the pivot set is unchanged.

Example 136.3 (A finite projection normal form). Consider the equations in five rational coordinates

$$x_1 = 2x_0, \quad x_3 = x_0 - x_2,$$

with x_0, x_2, x_4 free. The pivot subset is $\{0, 2, 4\}$. The ambient triangular automorphism

$$(x_0, x_1, x_2, x_3, x_4) \mapsto (x_0, x_1 + 2x_0, x_2, x_3 + x_0 - x_2, x_4)$$

sends the corresponding coordinate subspace to this graph. Its inverse subtracts the two indicated expressions. This is the finite-coordinate version of [theorem 132.1](#).

Example 136.4 (Two independent infinite stages). At rank $\omega + 1$, the shear $(x, z) \mapsto (x, z + \varphi(x))$ is onto, although non-Borel. At rank $\omega \cdot 2$, write the two blocks as (x, y) . The map

$$(x, (y_0, y_1, \dots)) \mapsto (x, (\varphi(x), y_0, y_1, \dots))$$

is an elementary self-embedding onto a dense graph. The extra infinite block makes room for insertion; one terminal coordinate does not.

The package includes an exact rational script checking finite instances of the triangular inverse, the pivot parametrization, order preservation, and standard quotient and remainder. Every calculation uses rational arithmetic. These checks are useful for detecting indexing or sign errors in examples.

They cannot verify the existence of φ , the transfinite normal-form theorem, Baire category arguments, or quantifier elimination on infinite structures. In fact every wild shear above agrees with the identity on all finite-support vectors. No test suite confined to such vectors can distinguish it from the identity. The distinction is established by the proof and its hypotheses.

137 Realization in the omnific integers and a formalization route

137.1 Set-sized fixed-support normal forms

The ProveIt surreal development gives an additional concrete interpretation of the coefficient models [\[95\]](#). For $i < \alpha$, regard the ordinal $i + 1$ as a positive surreal and put

$$\gamma_i = (i + 1)^{-1}.$$

Then $\gamma_i > 0$ and $\gamma_i > \gamma_j$ whenever $i < j$. The exponent set is reverse well-ordered. Conway normal forms therefore give a well-defined map

$$\iota_\alpha(h, n) = \sum_{i < \alpha} h_i \omega^{\gamma_i} + n. \quad (137.1)$$

Zero coefficients are omitted. Normal-form addition is coefficientwise, and the leading term decides the order. Thus ι_α is an additive ordered embedding. Every exponent in its nonconstant part is positive and the constant term is an ordinary integer, so its values are omnific integers in the standard normal-form description [\[91, 95\]](#).

[write] For infinite α the map ι_α is J_α of [theorem 40.1](#) (source 07), with the same exponents $1/(i + 1)$, and for $\alpha = \omega$ it is Part I's ι_B ([theorem 9.1](#)); the set-size argument of the next paragraph is source 07's [\(section 40\)](#). Source 16 adds the finite α . [\[95\]](#) is the Surreal project's README.

The coefficient topology is transported along ι_α . It is not being identified with a topology on the entire proper class **No**. Every fixed image is a set: its domain $\mathbb{Q}^\alpha \times \mathbb{Z}$ is a set, and Replacement collects its normal-form values. Taking the supremum of their birthdays

supplies a set ordinal bounding the image. No collection of all surreal subclasses or class-sized matrix algebra is required.

This realization preserves the additive ordered structure used in the paper. It does not assert ring closure of the fixed exponent support. Matrix maps and wild shears should accordingly be read as additive arithmetic maps of these fragments.

137.2 Why the limit-coordinate obstruction is meaningful here

At rank $\omega + 1$, the monomial indexed by ω lies after all the monomials with finite indices. The shear changes its coefficient by a linear functional of the entire earlier infinite coefficient block. Every finite sum of monomials is fixed, yet some infinite normal forms move. This is an explicit instance of the distinction between ordinary linearity and strong additivity in Hahn-type settings [87, 88, 89].

At rank ω , every coordinate has only finitely many predecessors. The combination of order preservation and the full product structure forces the finite dependence proved above. At rank $\omega + 1$, valuation preservation alone no longer forces product-topology continuity.

137.3 A tractable proof-assistant decomposition

The repository's existing Presburger decision procedure works over ordinary integers [94]. Its correctness should not be silently treated as a formalization of arbitrary nonstandard $\mathbb{Z}$ -groups. A suitable extension has the following modules.

1. Define $\mathbb{Z}$ -groups with standard division and prove generic Presburger quantifier elimination. The periodic predicates in the Cooper step must be invariant under complete congruence classes, not merely under a single standard successor shift.
2. Prove the positive-cone/group-completion dictionary and the maximal divisible subgroup identity. These isolate the arithmetic content of [theorems 129.1](#) and [131.1](#).
3. Formalize lexicographic products on a well-ordered index type, their valuation spectrum, and row-finite linear maps. The main algebraic certificate is the positive pivot condition.
4. Formalize finite-support dual separation and the greedy coordinate-functional construction. This avoids uncontrolled transfinite row operations.
5. Add the Polish G_δ and Baire-category lemmas, then derive the topology of the cone and closed subgroup results.
6. Treat the non-Borel constructions in a separate module whose explicit hypothesis is a nonzero functional annihilating the finite-support subspace. Prove the shear and graph theorems from that hypothesis before proving its existence using a basis theorem.

This route keeps the constructive matrix calculations separate from the Choice-dependent existence of wild maps. The present package contains ordinary proofs and executable examples, not completed Lean or Rocq implementations of these modules.

[write] Module 1 is the porting requirement of [section 13.5](#), already named by sources 05 and 04: the repository's Cooper lemmas assume invariance under one standard shift and are stated over the integers. No module of this list exists in the repository.

138 Further research questions and proposed next steps

The following questions are deliberately more precise than a request for a general classification of all Polish arithmetic. Each states what has been settled here and what is still missing. Except where the original source is named, they are proposed questions, not certifications that the literature contains no answer.

Question 138.1 (Effective normal forms). For computable row-finite presentations of closed subspaces of $\mathbb{Q}^\omega$, under which effective hypotheses can the pivot set, the reduced coefficients, and a continuous left inverse be computed? The proof uses finite-dimensional linear dependence, but an arbitrary closed-set code need not provide the needed dependence information. Determine the exact oracle or limit-computation complexity.

Question 138.2 (Computable ordinal ranks). For computable ordinals $\alpha > \omega$, give effective versions of the finite-ancestry argument from a row-finite triangular matrix code. A mathematical well-order guarantees termination of the dependency recursion; a useful certified implementation should expose that well-foundedness and quantify the cost of inverse rows.

Question 138.3 (Weak choice and the first threshold). What weak choice principle suffices to produce a nonzero $\mathbb{Q}$ -linear functional on $\mathbb{Q}^\omega / \mathbb{Q}^{(\omega)}$? Does the existence of a non-Borel ordered automorphism of $M_{\omega+1}$ characterize a familiar principle over a choiceless base? The shear theorem reduces one direction to a very specific linear-functional existence statement. This connects directly to Glazer's interest in choice principles.

[write] [Remark 133.2](#) [merge] gives one direction: a non-Borel ordered automorphism of $M_{\omega+1}$ yields a set of reals without the Baire property, so ZF with dependent choice does not prove that one exists, if ZF is consistent. The rest of the question is open.

Question 138.4 (All algebraic endomorphisms at higher ranks). For $\alpha > \omega$, classify the noncontinuous ordered embeddings by successor and limit blocks. The continuous maps are completely described here, and the two thresholds are exact, but the complete algebraic endomorphism monoid at ranks such as ω^2 remains to be analyzed in a usable normal form.

[write] This is what remains of the endomorphism clause of Part I's [question 14.6](#) for the groups $\mathbb{Q}^\alpha$: the continuous order-preserving endomorphisms are classified ([theorem 130.5](#)), the discontinuous ones for $\alpha > \omega$ are not.

Question 138.5 (Which dense elementary subspaces are self-images?). At $\omega \cdot 2$ one dense graph is realized as a self-image. For a general $\mathbb{Q}$ -subspace $V \leq H_\alpha$, characterize when V is order-isomorphic to H_α . Closedness allows the finite-support annihilator theorem; without it, the valuation spectrum alone no longer controls the space.

Question 138.6 (Definability complexity of images). Can one prescribe the descriptive complexity of a proper elementary self-image at a fixed higher rank? Every Borel embedding has a closed image, but an image of a non-Borel embedding can itself be closed. Determine which nonclosed Borel, analytic, or projective coefficient subspaces can arise as algebraic self-images.

Question 138.7 (Other coefficient groups). Replace the rational factors by other ordered Archimedean groups, or mix discrete rational and usual real factors. Which versions of the orbit classification and the two thresholds survive? The rational proof uses unique divisibility, a countable discrete codomain for every coordinate, and finite-dimensional linear duality; each has a different role.

[write] This is the part of Part I's [question 14.14](#) (source 04) that source 16 leaves open; it answers the rest for countable ordinal index sets with rational coefficients (note after that question).

Question 138.8 (Orders beyond countable ordinals). For a countable linear index order that is not well-ordered, work with a genuine Hahn product of well-ordered supports. The full product is no longer the appropriate carrier. Determine when it is a standard Borel or Polish space and how rank automorphisms alter the classification. Existing Hahn-group theory provides the natural starting point.

Question 138.9 (Arbitrary Polish Presburger presentations). Which Polish continuous-addition Presburger models admit a Polish group completion with a discrete standard integer coordinate? The split family here has both properties by construction. A general answer would identify the reach of the subspace method and the precise additional phenomena in nonsplit models.

[write] Answered in part. Among Polish group presentations, source 13's [theorem 95.4](#) shows that a continuous integer-coordinate retraction exists exactly for the algebraically split ones, where it is automatic, and its [theorem 94.1](#) gives nonsplit Polish models; this is Part I's [question 14.11](#) (note there). Whether a given Polish cone has a Polish group completion at all is source 08's [question 57.2](#), asked again in Parts V, VI and IX; it is open.

Question 138.10 (Connected carriers). Can a connected uncountable Polish space support a continuous-addition Presburger model? None of the rational-product carriers here is connected. The matrix classification does not settle this broader question from the repository.

[write] This is Part I's [question 14.3](#), Part III's [question 43.4](#) and source 13's [question 103.4](#). It is answered negatively for cones whose topology is inherited from a Polish topological $\mathbb{Z}$ -group (note after [question 14.3](#)) and is open for other monoid topologies.

Question 138.11 (Formalization as the next deliverable). Implement the rational- ω case first: the homomorphism dictionary, automatic continuity, positive-pivot matrices, closed images, and the closed-submodel correspondence. This is a finite collection of theorem interfaces and gives the largest immediate gain before introducing transfinite coordinate ranks.

[write] Compare Part I's [question 14.1](#); at $\alpha = \omega$ the listed interfaces are those of [proposition 11.3](#), [theorems 11.2](#) and [11.4](#), and [corollary 11.5](#).

Question 138.12 (Sharp surreal birthday bounds). For the fixed support in [\(137.1\)](#), replace the abstract Replacement bound by useful explicit birthday estimates uniform in α . Then formalize the transported coefficient topology inside one specified set-sized surreal cutoff.

[write] For $\alpha = \omega$ this is the first sentence of Part I's [question 14.10](#) (source 05).

A Worked examples and an audit of the hypotheses (source 05)

A.1 A negative constant term is compatible with positivity

In $M_{\mathbb{R}}$, take

$$x = (\tfrac{3}{2}, -17), \quad y = (\tfrac{1}{4}, 6).$$

Both are positive, and

$$x + y = (\tfrac{7}{4}, -11).$$

Division of x by 5 gives

$$q_5(x) = (\frac{3}{10}, -4), \quad r_5(x) = 3,$$

because $-17 = 5(-4) + 3$. The quotient is positive because its real coordinate is positive. A definition that incorrectly demanded a nonnegative integer coordinate for every element would break the predecessor and division axioms.

A.2 A nonstandard finite-candidate calculation

Let $L = (\frac{1}{2}, -5)$, $U = (\frac{2}{3}, 0)$, and require

$$y \equiv (\frac{7}{13}, 2) \pmod{3G}.$$

The real coefficient of a congruence parameter is irrelevant because $\mathbb{R}$ is divisible. Among $L, L+1, L+2$, the candidate $L+1$ has integer coordinate $-4 \equiv 2 \pmod{3}$, so it is a witness. Every candidate lies below U , since its leading coefficient is $1/2 < 2/3$. No iteration a nonstandard number of times is used: the finite search is justified by full invariance on $3G$ -cosets.

A.3 A Baire convergence that order cannot detect continuously

Let e_j be the rational unit vector at coordinate j . Then

$$(e_j, 0) \rightarrow (0, 0), \quad (e_j, 0) > n\mathbf{1} \quad \text{for every fixed standard } n.$$

The initial coefficients eventually agree with zero to every prescribed finite length. In omnific notation this sequence is

$$\omega, \quad \omega^{1/2}, \quad \omega^{1/3}, \quad \dots,$$

which converges to zero in the *specified coefficient topology*, not in the arithmetic order topology. This is a concrete illustration, not a claim about an unspecified topology on surreal numbers.

A.4 Hypothesis checklist

Potential pitfall	Resolution in the proofs
Only the division axioms are checked	Theorem 4.9 , corollary 4.11 , and proposition 4.12 establish the complete theory and every additive induction instance.
An open ray is given an incomplete metric	Section 5 provides a compatible logarithmic metric on each open ray and a complete metric on the whole sum.
The intended topology is silently replaced by the order topology	Both topologies are explicitly distinguished; the arithmetic order topology is discrete.
Standard-shift invariance is used with a nonstandard quotient	Example 4.4 exhibits the failure; lemma 4.7 uses invariance on mG -cosets instead.
The all-zero rational branch is lost	The nonnegative-cone tree in (6.2) explicitly includes it.

Potential pitfall	Resolution in the proofs
Computing an output tag secretly requires deciding equality of streams	For a negative output integer coordinate, a negative input supplies a finite bound on the first positive coefficient.
A type-theoretic “computable model” is confused with a countable one	The presentation is Type-2 on all Baire sequences; not every element is computable, and order has only a limit decision procedure.
The omnific image is assumed closed under multiplication	It is only an additive fragment. Theorem 10.4 rules out even an alternative semiring multiplication on the same carrier.
A proper class is treated as a Polish space	Only explicit set-sized images and a sufficiently large set-sized birthday cutoff are used.
Existing repository proofs are credited with new theorems	The formalization roadmap distinguishes the inspected integer theorems from the additional nonstandard and topological obligations.
Numerical testing is mistaken for a proof	The exact test scope is documented in section 13 ; no formal verification claim is made for this article.
An affirmative proof is mistaken for certified historical priority	The source question and version are identified; independent priority and refereeing remain separate from the mathematical proof.

B Reproduction and the scope of the executable checks (source 04)

The package requires a standard LaTeX installation for the article and Python 3.10 or later for the companion code. The Python program uses only the standard library and exact `Fraction` arithmetic. From the package directory, run

```
python3 verification.py --output verification_results.json
pdflatex -interaction=nonstopmode -halt-on-error polish_presburger.tex
pdflatex -interaction=nonstopmode -halt-on-error polish_presburger.tex
```

A Makefile supplies equivalent targets. The delivered run used seed 20261003 and passed 75,079 assertions. Its breakdown is:

Check category	Assertions
Rational, integer, and pairing codes	6,468
Finite group and order identities	9,955
Exact quotient and remainder	2,151
Boundary witness transfer	21,037
Finite candidate comparison	1,200
External periodicity counterexample	2,321
Positive-pivot matrix tests	18,000
Monus integer-coordinate obstruction	400
Baire-code round trips	97
Baire quotient and remainder prefixes	9,700
Baire addition prefixes	3,750
Total	75,079

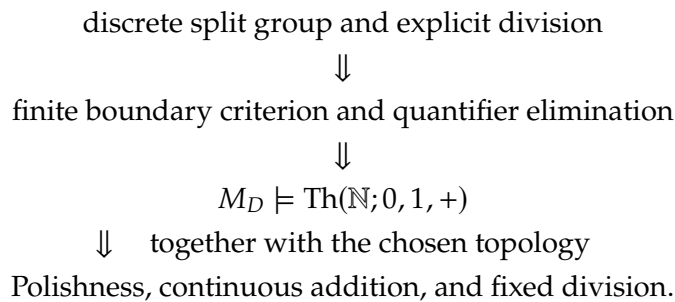
The finite group tests use rational coefficient vectors and signed integer coordinates. The boundary tests examine witnesses from a specified finite pool and compare them with the exact candidate set. They do not exhaust an infinite group. The matrix tests use finite initial pieces of matrices satisfying the positive-pivot conditions. The stream tests check finite prefixes of deterministic infinite oracles, including long zero runs. The monus tests verify exact coordinate obstructions along finite portions of the sequences used in the proof, not topological convergence by numerical approximation.

The model-theoretic, infinite-matrix, and topological statements rest on the proofs in the manuscript. Tests are useful for detecting sign errors, incorrect negative quotients, wrong coordinate offsets, and mismatched coding branches, but a successful run is not a substitute for those proofs.

[write] The commands above use source 04's delivery names. In this report its program is `code/04-polish-presburger-verification.py`, its recorded run data `data/04-polish-presburger-verification_results.json`, its Makefile `code/04-polish-presburger-Makefile` (which still names `polish_presburger.tex`, `verification.py` and `verification_results.json`), and its manuscript is merged into `this article.tex`. The program's default output is `verification_results.json` in the working directory, so it is rerun with an explicit output path outside the report (README). The intake reran it on a copy with Python 3.14.4: 75,079 assertions passed, and the record was identical up to line endings.

C A compact proof-dependency map (source 04)

The answer can be read independently of the optional extensions:



For the real example, [section 5.4](#) completes the argument at that point. For literal Baire space, add only the two chart codings in [section 6.3](#). The computation of the monus locus,

the embedding classification, the growth-order obstruction, and the surreal realization are subsequent results; none is a hidden prerequisite for the affirmative construction.

D Source audit, contribution ledger, and reproducibility (source 03)

D.1 Repository material inspected

The repository branch was resolved to the commit printed in Section 16. The following guides were inspected in selected ranges at that revision:

```
Algebra/SurrealNumbers/README.md
Algebra/SurrealNumbers/docs/README.md
Algebra/SurrealNumbers/docs/
foundations-and-computation/omnific-notations/README.md
```

The first guide supplies the context of the omnific ring, its least positive element, and the floor boundary correction. The notations guide reports the existing Kleene–Brouwer support-validity classification and carefully states the limited scope of its finite checks. The present article does not certify all statements in either guide.

The external mathematical inputs are Newton–Puiseux, the classical Hahn-field construction, Hausdorff’s characterization of countable scattered orders, standard descriptive-set-theoretic facts, and Glazer’s Corollary 1. Analytic boundedness and the needed Kleene–Brouwer reduction are proved in the text. Shepherdson’s theorem is contextualized and its needed direction is also proved directly.

D.2 Contribution ledger

Statement	Status in this report
General coanalytic support hardness	Prior classical mechanism; already recorded for omnific notations in ProveIt. Reproved to make the reductions explicit.
Scattered/support dichotomy	Proved from Hausdorff theory and the fixed tree reduction; global priority not asserted.
Observable full-Hahn criterion	Proved, including Borel field operations on the scattered side and exclusion of even repeated observable names on the nonscattered side.
Exact $\mathbb{Z}^d$ Borel levels	Proved with explicit continuous reductions and a separate sharp support-order bound.
Uniform observable support bound	Classical analytic boundedness specialized to coefficient families; a proof is supplied.
Fixed-interval binary localization	Explicit application of the support reduction, not advertised as a new foundational lower-bound method.
Principal-part integer part and floor	Shepherdson-type classical model; negative-infinitesimal correction also already in ProveIt.

Statement	Status in this report
Exact cancellation loci	Proved for the stated Polish finite-normal-form topology, including closed-piece continuity and zero-factor cases.
No compatible continuous addition	Direct application of Glazer after all model and Borel hypotheses have been verified.
Formal verification	No Lean proof delivered. Python checks only finite exact-rational identities, order tests, floor cases, and finite embedding data.

D.3 Finite verification artifact

The archive contains `verify.py`, a Python 3.10+ standard-library-only program, and the actual result `verification.json`. The recorded run used Python 3.13.5, deterministic seed 2601003, and passed **84,388 assertions**. These include 6,000 ring-law checks, 2,000 closure checks, 2,000 positive-cone checks, 999 discreteness checks, 7,500 floor-interval checks, 63 negative-infinitesimal boundary checks, 25 cancellation checks, 65,792 rational Kleene–Brouwer embedding checks on 256 nodes, and 9 coherence and branch-approximation checks.

The finite embedding is an initial segment of one fixed midpoint construction: nodes are enumerated by the weight $\sum_i (s_i + 1)$, and each new node is inserted between its previously assigned neighbors. This checks the direction of the order reversal and the rational band. It does not test the well-foundedness of arbitrary infinite trees.

To reproduce:

```
python verify.py --output verification.json
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The bibliography is internal, so no BibTeX run or network access is required. The finite checks are useful for catching sign, support, and boundary mistakes; they neither prove descriptive completeness nor replace the mathematical proofs or an independent review.

[write] In this report source 03’s program is `code/03-borel-presentations-verify.py`, its recorded run data `data/03-borel-presentations-verification.json`, and its provenance record `data/03-borel-presentations-provenance.json`; the record still describes 03’s own 23-page PDF, which is not shipped. Its manuscript is merged into `article.tex`. The intake reran the program on a copy with Python 3.14.4: 84,388 assertions passed, and the record differed from the delivered one only in the interpreter version and line endings.

E Provenance of this report

Sources of Parts I and II. Three manuscripts of batch 86 of the repository’s intake, all dated October 3, 2026, all AI-assisted (03: “prepared with ChatGPT”), arrived together in commit 31fdc6571 and were placed in commit 3d2177df4. Each inspected the repository at its own pin; the pins are 28 to 35 commits before the placement.

Source	Archive	Pin	Contributes
05	polish-presburger-glazer	a11efab09	base of Part I: sections 2 to 13 , its questions, sections A and 15
04	Polish_Presburger_Glazer_ProveIt	3ad5f878c	every passage marked [04] in Part I ; sections 2.3, 4.3, 4.5, 5.2, 5.4, 6.3, 7.3, 8.1, 10.1, 11.2, 11.3 and 13.4 , its questions, sections B and C
03	glazer-proveit-research	c5b4881fa	Part II except the notes and section 26 ; section D

Written for this report: [section 1](#) (except the quoted status statements), [section 13.5](#), the merged questions of [section 14.1](#), [section 26](#), the notes marked [write] or [merge], and these appendices.

Where the merge of Parts I and II chose.

- *Base.* Source 05, for the weaker hypotheses of its Polish criterion and Δ_2^0 theorem ([section 1.2](#)).
- *Printed once.* The main theorem (05's [theorem 3.1](#), with 04's statement as [remark 3.2](#)); the division algorithm ([proposition 4.2](#) = 04's Lemma 2.2); full Presburger arithmetic of the cone ([corollary 4.11](#) = 04's Corollary 3.5); the predecessor and truncated-subtraction loci ([proposition 8.2](#) and [theorem 8.3](#) = 04's Proposition 7.4 and Theorem 7.3); the omnific realization ([theorem 9.1](#) = 04's Proposition 10.1, same proof); the no-semiring theorem and its corollary ([theorem 10.4](#) and [corollary 10.5](#) = 04's Theorem 9.3 and Corollary 9.4, whose model-specific proof coincides with 05's); the real endomorphisms ([theorem 11.1](#) = 04's Theorem 8.2).
- *Second proofs from 04.* [Proposition 4.2](#) (uniqueness by coordinates), [lemma 4.7](#) (as [theorem 4.8](#)), [theorem 4.9](#) (as [theorem 4.10](#)), [theorem 6.1](#) (as [theorems 6.3 and 6.4](#)), [theorem 7.1](#) (as [theorem 7.2](#)), [proposition 8.2](#), [theorem 8.3](#) (with the nowhere-density clause), [theorem 10.4](#), [theorem 11.1](#).
- *Both printed.* The order obstructions [theorem 7.4](#) (05) and [theorem 7.5](#) (04), whose hypotheses are incomparable ([remark 7.7](#)); the counterexamples [example 4.4](#) (05) and [proposition 4.6](#) (04); the Polish criterion [theorem 5.2](#) (05) and the transfer theorem [theorem 5.4](#) (04), which alone gives continuous division for general D .
- *Questions.* Four questions asked by both 05 and 04 are merged ([section 14.1](#)); 05's question on canonical surreal cutoffs and topologies and 04's on coefficient topologies inside surreal arithmetic overlap only partly and are printed separately. 03's eight questions are printed as delivered. Part I has seventeen questions, Part II eight.
- *Not printed separately.* 04's title page, abstract (its non-claims are in [section 1.5](#)), reading routes, Definition 2.1 (05's setup), the opening paragraph of its Section 9 (Glazer's Corollary 1, in [section 1.1](#)), and the statements and proofs listed under "printed once"; 05's and 03's title pages and abstracts (their status statements are in [section 1.5](#)). The merged abstract replaces them.
- *Renamed symbols.* Listed in [section 1.3](#); no normalization was changed.
- *Added.* [Proposition 26.1](#) and the discussion in [section 26](#); the remark [remark 7.7](#); Glazer's speculation ([section 1.1](#)), which no source mentions; the notes citing the collection's

labels in [Part II](#).

Intake checks for Parts I and II. The intake ran the three delivered suites on copies (Python 3.14.4, standard library only): 03 passed 84,388 assertions in 11 groups, 04 passed 75,079 assertions in 13 categories, 05 passed 17,930 checks in 18 families. Their records agree with the shipped ones up to interpreter strings and line endings. The three delivered checksum manifests verified and were not shipped. It also checked by hand that $M_{\mathbb{R}}$ is a countable clopen sum of rays and a model of $\text{Th}(\mathbb{N}; 0, 1, +)$, the Polish clause of [theorem 11.2](#), [theorem 7.5](#), and the worked examples of [section A](#); no false statement was found. For this report the following references were checked against their publishers' records on October 3, 2026: Glazer's arXiv record (one version, no journal reference); Tserunyan's notes (the file at the cited address is dated November 26, 2025); Paran and Vo (*Israel J. Math.* 273 (2026), no. 2, 979–1000, online December 11, 2025); Enayat, Hamkins and Wcisło (*Fund. Math.* 256 (2022), 171–193); Jeřábek (*Math. Logic Quart.* 65 (2019), no. 1, 108–115); Haase (*ACM SIGLOG News* 5(3) (2018), 67–82); Alvir and Rossegger (published in *J. Symb. Log.* 85 (2020), 1079–1101, added here). The other classical references were not re-checked.

Correction. [Section 26](#) corrects the placement record's description of how 03's topology restricts to the real model.

Sources of Parts III–V. Four further manuscripts of batch 86 arrived together in commit fb8414869 and were placed in commit 0bd0e5527, with manuscript 10 of the same arrival, which went to Algebra/SurrealNumbers/docs/surreal/discrete-initial-subgroups-and-omnific-normalization. They are numbered by archive time. Their pins are 33 to 43 commits before the placement; the repository files they inspected are unchanged between the pins and the placement. Sources 08 and 09 are AI-assisted by their own statement ("prepared with ChatGPT", "prepared with AI assistance"); 06 and 07 describe themselves as research reports prepared for Vladimir Reshetnikov.

Source	Archive	Pin	Contributes
07	polish_presburger_research	0f95145cb	base of Part III : every unmarked passage, its questions, sections G and H
06	polish_presburger_article	fa2f3e419	every passage marked [06] in Part III , among them sections 38 and 39 ; its twelve projects; sections I and J
08	glazer_proveit_local_compactness	883e0b3b2	Part IV except the notes; section K ; staged code, data, sources and proof audit (prefix 08-local-compactness-)
09	Polish_Hahn_Puisseux_Levi_Civita_Research	4af956fe0	Part V except the notes; sections L and M ; staged code, data and sources (prefix 09-hahn-polishability-)

Written for this report in Parts III–V: the introductions of the three parts, [section 42](#) with [proposition 42.1](#), the notes marked [write] or [merge], the additions to [section 1](#) (the paragraph on Parts III–V and Glazer's speculation, the second block of the notation table, the status boxes), the note after [question 14.4](#), and the additions to these appendices.

Where the merge of [Part III](#) chose.

- *Base.* Source 07, for the weaker hypotheses of its realization theorem ([Part III](#)).
- *Printed again with a flag.* Results of Part I proved again by 06 or 07 are printed with their proofs, and each title names the statement of Part I it repeats (listed in the introduction of [Part III](#)). Source 06's [theorem 35.9](#) repeats 07's [theorems 35.1](#) and [35.4](#) with a different homeomorphism argument.
- *Both printed.* The two continuum theorems ([theorems 35.5](#) and [35.8](#)), whose constructions and refinements differ.
- *Placement.* Source 06's Sections 1 to 6 and 9 are printed as subsections of source 07's Sections 1, 2, 3, 6, 6, 8 and 9 respectively, its own subsections one level lower; its Sections 7 and 8 are sections of their own. Its twelve projects follow source 07's question with the same subject, the other five in [section 43.5](#).
- *Renamed.* Source 06's M_s, G_s, H_s ($s \in 2^{\mathbb{N}}$) are M'_s, G'_s, H'_s ; macros of all four sources are mapped to this report's ([section 1.3](#)); no normalization changed.
- *Labels added* to unlabelled statements so that notes can cite them: `pma:lc:q:remove1c`, `pma:lc:q:completion`, `pma:ser:q:cones`.
- *Not printed separately.* The four title pages and abstracts (their status statements are in [section 1.5](#); the merged abstract replaces them), source 07's table-of-contents material, and the section headings of 06 that were replaced by subsection headings. Source 07's figure, source 06's map of results and source 09's reading routes are printed.

Intake checks for Parts III–V. The suites of 08 and 09 were rerun on copies (Python 3.14.4, standard library only, about one second each): 08 passed 24,470 cases in four families, 09 passed 20,116 assertions in 29 families. Their records are JSON-equal to the shipped ones and differ only in line endings. Source 08's checksum manifest verified at placement (8 of 8) and was not shipped; 06, 07 and 09 ship none, and 06 and 07 ship no code. For this report the proofs of source 06's [theorem 39.1](#) and [theorems 38.1](#) and [38.2](#) and the oscillation table after [theorem 36.1](#) were checked by hand, with no error found, and the following records were checked on October 3, 2026: Block's arXiv record (v3 of June 16, 2026, the version cited by 06); Chen's arXiv record (one version, 2021, no journal reference). The arXiv record of Ehrlich and Kaplan lists no journal reference; the journal reference that source 07 gives was not re-checked. The other references of 06–09 were not re-checked.

Reviews elsewhere in the repository. The Hilbert's-tenth programme triaged the arrival fb8414869 for its own search in Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/recent_incoming_substrate_triage_20261003.md (commit 39e03adc5), after a README-only note on manuscripts 03–05 (review_recent_incoming_glazer_scope.md in the same directory, commit 0f3b760ae). Both are routing decisions, not proof audits; the later one notes, as [section 42](#) does, that source 06's cone topology is not automatically a locally compact signed-group topology.

Stale statements and addresses. Source 07's statement that the repository's text files never mention Glazer or Tennenbaum was true at its pin and is corrected after it ([section H](#)). Source 09 cites two repository files by unversioned addresses; the note in [section M](#) gives the pinned ones. Sources 08 and 09 cite sources 05 and 03 as “user-library” PDFs; in this report they are Parts I and II.

Sources of Parts VI–IX. Five manuscripts of batch 87 arrived in commit `fa0a0576e`, with a sixth (manuscript 02, written into `SetTheory/Cardinals/docs/reports/ordinals-and-order-types/measurable-box-games`), and were placed in commit `d151b39ca`. They are numbered by archive time in the batch and by file prefix in this report. Their pins are 44 to 51 commits before the placement. The repository files they inspected are unchanged between their pins and the placement, except that the Presburger and surreal READMEs gained the catalogue’s pointers to this report in `2b7b388ba`.

Source	Archive (batch 87 no.)	Pin	Contributes
12	Baire_Category_Arithmetic_Rigidity (03)	<code>0f084afa9</code>	base of Part VI : every unmarked passage, its questions; sections N and O ; staged code, data and audits (prefix <code>12-baire-category-</code>)
10	Glazer_ProveIt_Baire_Rigidity (01)	<code>6fef5383b</code>	every passage marked [10] in Part VI , among them sections 79.1 , 79.3 and 79.4 and section 85 ; its twelve questions; sections P and Q ; staged code, data and audit (prefix <code>10-baire-order-</code>)
13	polish_presburger_frontiers (04)	<code>eab47e330</code>	Part VII except the notes; sections R and S ; staged code and data (prefix <code>13-beyond-split-</code>)
14	glazer_atr0_research (05)	<code>6fef5383b</code>	Part VIII except the notes and the status box; sections T and U ; staged code, data and proof audit (prefix <code>14-atr0-tennenbaum-</code>)
15	polish_arithmetic_package (06)	<code>0f084afa9</code>	Part IX except the notes; section V ; staged build script and proof audit (prefix <code>15-polish-arithmetic-</code>)

Written for this report in Parts VI–IX: the introductions of the four parts, [section 90](#) with [proposition 90.1](#), the notes marked [write] or [merge] (in Parts VI–IX and at the questions and statements of Parts I and III–V that they answer or make stale), the status box of [Part VIII](#), the additions to [section 1](#) (the paragraph on Parts VI–IX and Question 1, the sources table, the third block of the notation table, the status boxes), and the additions to these appendices.

Where the merge of Part VI chose.

- *Base.* Source 12, for the weaker hypotheses of its cardinal, semiring and non-Polishability theorems and its second, Haar route ([Part VI](#)).
- *Printed once.* The principal theorem ([corollary 78.4](#) = source 10’s Theorem 5.1); the bounded-neighbourhood lemma ([lemma 77.2](#) = source 10’s Lemma 4.2); the open principal hull ([proposition 79.1](#) = source 10’s Proposition 4.3); the absence of an ordered-ring expansion ([corollary 81.1](#) = source 10’s Corollary 5.3); the three topologies on $\mathbb{Z} + t\mathbb{R}[t]$ (= source 10’s Examples 9.1, 9.3, 9.4); the completion ([proposition 86.2](#) = source 10’s Proposition 9.2). Source 10’s labels are second labels of these statements; its two displays `eq:localbound` and `eq:polynomialring` are not printed, and references to them point to source 12’s displays.
- *Second proofs from source 10.* [Lemma 77.2](#) (without Pettis’s lemma), [corollary 78.4](#) (through the principal-hull quotient embedding), [proposition 86.2](#).

- *Both printed.* The Polish corollaries ([corollary 78.5](#) and source 10's exact criterion [corollary 78.6](#)); the cardinal theorems ([theorems 78.3](#) and [79.13](#)); the spectrum theorems ([theorems 80.1](#) and [80.4](#)); the signed Presburger corollaries ([corollaries 81.3](#) and [81.4](#)); the omnific corollaries ([corollaries 86.3](#) and [86.4](#)).
- *Placement.* Source 10's Sections 1, 2, 3, 6, 7, 8, 10 and 12, and the last two subsections of its Section 13, are printed as subsections of source 12's Sections 1, 2, 5, 5, 5, 7, 6, 7, 11 and 12 respectively, their own subsections one level lower. Its Section 4 is divided between source 12's Sections 3 and 5, its Section 5 between Sections 4, 5 and 7, its Section 9 (examples) is merged into Section 10 and Proposition 9.2 into Section 11, and the first subsection of its Section 13 is printed in Section 11. Its Section 11 is a section of its own; its questions follow source 12's questions on the same subject; its conclusion follows source 12's.
- *Renamed.* Source 10's `\Fin` macro (printed `Fin`) and its $\mathcal{S}(G, e) (\rightarrow \mathcal{U}(G))$; no normalization changed.
- *Labels added* to unlabelled statements: `pma:br:cor:signed-presburger`; and `pma:br:s10:sec:category`, `pma:br:sec:examples` for source 10's sections whose text is merged; `pma:br:sec:q10` for the subsection of source 10's further questions; and `pma:br:sec:meet`, `pma:br:prop:meet` for the merge section.

Where Parts VII–IX chose. Each is one source, printed whole with notes. The construction $\mathbb{R} \overrightarrow{\times} K$ with a countable nonsplit core is printed once in general ([theorem 94.1](#)); source 14's [proposition 113.2](#) and source 15's [theorem 122.2](#) keep their own proofs and are flagged in their titles. Source 14's Section 10 group D is printed K_* , the generic D of its nonsplit-topology question K , its Ψ as Φ_ω and its 2^ω as $2^{\mathbb{N}}$; source 13's `\MB` is printed M_ω as in its delivery, and its lexicographic sums, like those of sources 12, 14 and 15, as $\overrightarrow{\times}$. Source 15's Section 3 and its Theorems 3.6 and 8.1, special cases of [Part VI](#), are printed with flags. Labels were added to unlabelled statements of sources 13 (nine) and 14 (one, `pma:atr:subsec:inputs`). The `\Needspace` commands of sources 14 and 15 (page-break hints for a package this report does not load) were dropped.

Intake checks for Parts VI–IX. The four delivered suites were rerun on copies (Python 3.14.4, standard library only): source 10 passed 29,041 checks in seven families (seed 20261003), source 12 36,989 in fifteen (seed 20261003), source 13's diagnostics reported "all_passed": true, and source 14 printed "EXACT FINITE CHECKS: PASS"; each record equals the shipped one after removing the carriage returns that Windows text mode adds. Source 15 ships no executable checks. The checksum manifests of sources 10 (7 of 7), 12 (8 of 8) and 14 (8 of 8) verified at placement and were not shipped; source 13's `provenance.json` hashes (three archives and three repository blobs) and source 15's manifest hashes (its two deliverables and seven repository files at its pin) also verified. The intake read the main chains of sources 10, 12 and 15 and of source 14's [theorem 106.2](#) to [corollary 111.2](#), without finding a gap; that reading is not a review, and in particular it did not check source 14's declared inputs or its `ATR0` coding.

Files of the deliveries not shipped. Besides the PDFs, manuscripts, delivery READMEs and checksum manifests, two provenance files are not shipped because each contains a third party's personal profile address, which no tracked file carries: source 14's `SOURCE_AUDIT.txt` and source 15's `source_manifest.json`. Both survive in the archives of the arrival commit.

Their verified content, without that address, is as follows. Source 14’s audit records that the supplied profile page could not be retrieved and that the research overlap was grounded instead in Glazer’s paper and a public author biography; that the whole of Glazer’s paper was inspected, including Theorem 2, Corollary 1, the Optimization section and both questions, and that no later arXiv version or indexed solution of Question 1 was found (not an exhaustive search); that the Gaifman–Dimitracopoulos chapter is a contribution to monograph no. 30 of *L’Enseignement Mathématique*, not obtained, and that the statement used was checked against Chen, Müller and Yokoyama (*J. Symb. Log.* 90 (2025), p. 485 and Definition 5.2); the coded descriptive set theory of Simpson’s Sections V.3–V.4 and Theorem V.8.3; the repository paths inspected at the pin 6fef5383b (the Lean and Coq exponentiation modules, this report’s `article.tex`, the Hilbert’s-tenth scope note) and the three batch-86 archives at their arrival 31fdc6571; and the claim boundaries repeated in [section F](#). Source 15’s manifest records its pin 0f084afa9 with a clean working tree; the seven repository files it inspected with their blob identifiers and SHA-256 values (all recomputed and matching at the pin); its ten external references with what was inspected for each (Christensen–Kanovei–Reeken: abstract and record only; Morris: metadata, the structure theorem being checked on p. 68 of Gabrielyan–Morris); the sizes and SHA-256 values of its source and PDF (matching); and its validation summary (32 pages, eleven questions, fourteen references, no unresolved references, AI-assisted review passes and no proof-assistant certificate).

Reviews elsewhere in the repository (Parts VI–IX). The Hilbert’s-tenth programme reviewed the six archives of fa0a0576e for its own search in commit f15962acd: `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_baire_polish_arithmetic_intake.md`, with a receipt (`review_baire_polish_arithmetic_intake.json`) and an independent audit (`review_baire_polish_arithmetic_intake_independent.md`). It authenticated every archive and member, read all READMEs and recorded spans of each manuscript, ran no code, and found no new universal polynomial or arithmetic schedule. It is not a proof review. Its findings bearing on this report are recorded where they apply: source 14’s internal-MRDP input and automatic-Borel argument were read, but the claimed ATR_0 resolution is not certified ([Part VIII](#)); source 13’s revision counts bound changes of output, not running time, coordinates read or arithmetic operations, with no known final stage, and its continuity procedure outputs an expanded coordinate language (notes in [Part VII](#)); sources 10 and 12 concern signed group topologies and grant no Polish group completion to a positive monoid; source 15’s classification and cardinality proofs were not reviewed there.

Stale statements and addresses (Parts VI–IX). Source 13 describes batch 86’s three manuscripts as packages in `docs/incoming` and cites them by addresses at its pin; they are Parts I and II here, and the archives, retired in 3d2177df4, remain at those addresses. Sources 13, 14 and 15 describe this report as it stood at their pins (Part I’s base text); sources 10 and 12 cite sources 08 and 09 as “user-library” files. Source 13 cites four repository files by unversioned addresses; the notes in its bibliography entries give the pinned ones. Notes after the statements concerned correct each of these.

Source of Part X. (Batch 89.) Batch 89 had five archives: four arrived in commit 7c0f2d9f9 and one in 8ea27d6c0, and all five were placed in 23adb85f9. Manuscript 02 is source 16 of this report; the other four went to `surreal-well-orders` (Part XV), `birthday-cutoffs-and-hereditary-sets` (Parts VI and VII) and the research-report collection’s `naming-elementary-embeddings`. Source 16’s pin 0f084afa9 is 70 commits before the

placement. Of the repository files it read, this report's `article.tex` has since become the merged report, the Presburger and surreal READMEs changed only by the catalogue's counts and pointers, and the batch-86 archives it compared were placed in `0bd0e5527`; the other files are unchanged. No two batch-89 manuscripts share more than 0.33% of their word 8-grams, and source 16 shares 1.24% with this report (Presburger phrasing).

Source	Archive (batch 89 no.)	Pin	Contributes
16	Glazer_ProveIt_Embedding s_and_Thresholds (02)	0f084afa9	Part X except its introduction, the notes and remark 133.2 ; sections W and X ; staged code, data, source audit and proof review (prefix 16-presburger-embeddings-)

Written for this report in Part X: its introduction, the notes marked [write], [remark 133.2](#) [merge] with the note after it, the labels added, the additions to [section 1](#) (the paragraph on Part X, its sources table, the fourth block of the notation table, the status box), the notes at Part I's questions on coefficient groups, lattices, surreal cutoffs and embedding categories and the pointers after [questions 14.3](#) and [14.11](#), at the end of [section 11.3](#) and after [theorem 35.4](#), the four bibliography notes, the entry [96], and these paragraphs.

Where Part X chose.

- *One source, printed whole.* Source 16's twelve sections are Sections [127–138](#) and its two appendices [sections W](#) and [X](#). Its title page, abstract and table of contents are not printed separately: its status paragraph is in [section 1.5](#), and the abstract of this report summarizes it.
- *Printed again with a flag.* Results of Parts I and III that source 16 proves again keep its proofs and name the earlier statement in their titles: [proposition 128.3](#) ([theorems 35.1](#) and [35.9](#)), [theorem 129.1](#) and [corollary 129.2](#) ([proposition 11.3](#) and [corollary 11.5](#)), [theorems 130.5](#) and [133.1](#) for $\alpha = \beta = \omega$ ([theorem 11.4](#)), [theorem 130.7](#) for $\alpha = \beta = \omega$ ([corollary 11.5](#)) and the first example of [section 136](#) ([example 11.6](#)). [Theorems 131.1](#) and [135.1](#) and [corollary 129.3](#) are flagged as extending [theorems 11.2](#), [11.7](#) and [35.4](#), and the omnific map ι_α is identified with [theorem 40.1](#) in a note. Source 16's quantifier-elimination appendix is printed as [section W](#), a fourth write-up of a classical theorem already printed three times.
- *Questions.* Source 16's twelve questions are printed in its own part, numbered with the report's shared counter; those that repeat questions of other parts carry notes pointing to them, and the questions they answer carry dated notes.
- *Renamed.* Source 16's D , D_{fin} , S_φ , its `\res` and its roman lex, and local letters in two proofs ([section 1.3](#)); its H_α is kept. No normalization changed. Its bibliography entries for Glazer, Enayat–Hamkins–Wcisło, Haase and Kechris are this report's; the others carry the prefix `s16:`.
- *Labels added* to unlabelled statements: `pma:emb:cor:orbitcount`, `pma:emb:ex:shifts`, `pma:emb:ex:stages` and the twelve `pma:emb:q:` labels; and the four Part I question labels listed in [section 1.2](#).

Intake checks for Part X. Source 16’s suite was rerun on a copy (Python 3.14.4, standard library only): 11,243 exact checks in thirteen categories pass (seed 231113699), and the record equals the shipped one after removing the carriage returns that Windows text mode adds. Source 16 delivered no checksum manifest; the hashes of its build record (data/16-presburger-embeddings-BUILD_REPORT.json) verified, 8 of 8. The write read every proof of source 16 without finding a gap; that reading is not a review.

Review elsewhere in the repository (Part X). The Hilbert’s-tenth programme routed the five batch-89 archives in commit 6aadaa4f2: Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_topology_surreal_arrivals_20261003.md, with a receipt (review_topology_surreal_arrivals_20261003.json). It authenticated every archive member, read bounded ranges of source 16 (its exclusion of multiplication and of ring closure, the dependence of the non-Borel shears on a functional annihilating the finite-support vectors, its questions on effective normal forms), ran no code and found no new Diophantine source. It is a routing decision, not a proof review, and it has no correctness finding.

Stale statements and files of the delivery (Part X). Source 16 describes this report as it stood at its pin (Part I’s base text), whose “Research questions 5 and 8” are Part I’s [questions 14.6](#) and [14.9](#), and cites the batch-86 archives in docs/incoming, retired since; its novelty statement omits source 04’s results, which it had no way to see. Notes in [Part X](#) and in its bibliography entries correct each of these. Not shipped: its PDF, its manuscript (merged) and its delivery README; it delivered no checksum manifest. Its source audit, shipped as delivered, says that a supplied LinkedIn profile could not be retrieved (no address is given) and cites a public Epoch AI article as Glazer’s author biography, the article this report cites as [\[77\]](#).

F Limitations and non-claims of the sources

Every limitation stated by a source is kept. They are grouped by source; the items written for the merge come last.

Source 05.

- 05.1 The construction answers Glazer’s Question 2 *as written*; independent bibliographic priority has not been established. The literature search verified the question in the source PDF and checked the arXiv record; it did not establish the absence of an earlier solution.
- 05.2 Classical Presburger elimination, the $\mathbb{Z}$ -group facts and elementary topological-group facts are not claimed as new; the lexicographic algebra is classical.
- 05.3 The manuscript is AI-assisted, not independently refereed and not Lean-verified; no new Lean or Rocq formalization is included; the proposed Lean modules are a plan, not existing files.
- 05.4 The executable tests are finite instances and stream prefixes; they do not quantify over arbitrary elements of $\mathbb{Q}^{\mathbb{N}}$, and finite prefixes do not decide equality of infinite streams.
- 05.5 No answer to Glazer’s Question 1 is claimed; no uncountable Polish model of Peano arithmetic; no continuous arithmetic order on these nondiscrete spaces; no terminating

decision of equality of arbitrary infinite sequences (comparison is approximated with at most one change).

- 05.6 The omnific maps are additive and order-preserving, not ring embeddings; the topologies are specified coefficient topologies, not a canonical topology on $\mathbf{Oz}$; no topology on a proper class is used; all constructions are set-sized in ZFC.
- 05.7 Neither model admits any unital semiring multiplication with the given addition and unit; nothing contradicts Glazer's obstruction for stronger arithmetic, and the no-semiring argument does not replace it.
- 05.8 The repository's integer Cooper lemmas are correct on $\mathbf{Int}$; the existence of the integer eliminator does not certify the theorem for a new carrier; the original integer theorem need not be weakened or retracted.
- 05.9 The questions are proposed directions, not certified as open in the literature; the user's repository was not modified.

Source 04.

- 04.1 The construction answers Question 2 as printed; historical priority has not been established; the literature check did not establish that the construction has never appeared elsewhere.
- 04.2 The manuscript is unrefereed; no new theorem is represented as machine-checked; Lean/Rocq certification is not supplied.
- 04.3 The classical $\mathbb{Z}$ -group characterization and quantifier elimination are not claimed as new.
- 04.4 The difference between one-step periodicity and residue invariance is a porting requirement, not a reported bug in the integer proof.
- 04.5 Baire coordinates encode the arithmetic structure and are not base digits; the stream API cannot decide totality of an arbitrary oracle, and partial, negative or noninteger entries are not supported.
- 04.6 The source audit was limited to the named files at its pin; no repository files were modified and no repository-wide build was performed; the formalization stages are proposals, not assertions about existing declarations.
- 04.7 The tests are finite, exact and partial; the model-theoretic, infinite-matrix and topological statements rest on the proofs.
- 04.8 No third-party font files or paper copies are included.

Source 03.

- 03.1 A research draft, not refereed and not a Lean proof; global priority and independent correctness review are not established.
- 03.2 The repository's coanalytic support obstruction and the negative-infinitesimal floor correction are credited, not claimed as new; the integer-part model is of Shepherdson type, not new; analytic boundedness and the Hausdorff characterization are classical.

- 03.3 The full-Hahn nonpresentation results need observable coefficients and coverage of the whole field; they do not exclude unobservable Borel presentations or Borel real closed subfields.
- 03.4 Only addition is shown impossible to make continuous; Glazer's multiplication obstruction (his Theorem 2) is not transferred, since only open induction is proved for $\mathcal{A}_{\text{fin}, \geq 0}$; continuous multiplication under another compatible topology is open.
- 03.5 The zero loci are not claimed minimal in cardinality, dimension or descriptive rank; countable closed-piece continuity is not identified with the Baire-one class.
- 03.6 The dichotomy is per fixed L , not about the class of scattered orders, nor uniform in a code of L ; the Hausdorff decomposition used need not be computable; the uniform support bound need not be computable from a code.
- 03.7 No general computability over arbitrary real coefficients is claimed.
- 03.8 All results are set-sized in ZFC; no class-sized Polish topology, global choice or large cardinal is used.
- 03.9 The Python checks are finite (they do not test well-foundedness of infinite trees, completeness, or the external theorems); the formalization plan is proposed only; no repository files were changed and the whole repository was not audited.

Source 07.

- 07.1 The construction answers Question 2 as printed; the claim that it is the first published resolution is not made. The literature search was targeted, not exhaustive; the precise place in the literature of the cone theorem, the Baire family and the rank computations is left to specialist review.
- 07.2 The lexicographic construction of a $\mathbb{Z}$ -group is classical; classical model theory and standard descriptive set theory are inputs; no historical priority for the formulation of [theorem 33.1](#) is needed or claimed, measurable orders on locally compact groups having been studied by Asmar and Montgomery-Smith.
- 07.3 No new result has been checked by Lean or Rocq; no Lean or Rocq build was run; the repository was used read-only; no new formal theorem is claimed; no publication priority or external peer review is claimed.
- 07.4 Only first-order induction is claimed, not second-order induction over external subsets.
- 07.5 No computable homeomorphism is claimed for an arbitrary presentation of a countable ordinal; no classification of the pure underlying orders is claimed; the oscillation-rank convention is fixed and not identified with every convention for difference-hierarchy ranks.
- 07.6 The omnific images are not claimed initial, closed under surreal multiplication or under surreal-analytic operations; the transported topology is not the surreal order topology; every model is set-sized and the class **No** is used only as a definable ambient class.
- 07.7 Glazer's Question 1 is not answered, and the metatheoretic strength of his theorem with multiplication is not determined.
- 07.8 The LinkedIn page supplied with the brief could not be read; the repository search covered `.tex`, `.md` and `.lean` files at the pin, not every branch or binary artifact.

- 07.9 The questions are proposed directions, not asserted to be historically new or unresolved in all of the literature.
- 07.10 No finite numerical experiment is offered as evidence; compilation and visual inspection verify the document, not the mathematics; the proof checks made during preparation are not external refereeing or proof-assistant certification.

Source 06.

- 06.1 No machine-checked certification and no unrestricted publication-priority claim is made; the literature search is a bounded bibliographic finding, not a proof of universal priority.
- 06.2 The lexicographic $\mathbb{Z}$ -group construction and its model theory are classical (Jeřábek); rational polynomial Presburger examples appear in lecture notes (Cook–Pitassi); the polynomial ring has classical precedents and occurs in the repository.
- 06.3 The computability claims are Type-2 claims about streams: no computable enumeration of the uncountable model and no decidability of equality of streams are asserted, so there is no conflict with Tennenbaum’s theorem; the continuum family is not claimed computable.
- 06.4 The ordinal scales are classes of magnitude, not ω_1 -sequences in one order; there is no conflict with the obstruction of Enayat, Hamkins and Wcisło.
- 06.5 The failure of open induction concerns $\mathbb{Z} + t\mathbb{R}[t]$ only and does not transfer to **Oz**; the analogous topology on the signed ring is not a ring topology, and partial subtraction on the cone is not continuous.
- 06.6 The isolated-zero theorem does not imply that every standard numeral is isolated; it uses no induction inside the model, no standard division, exponentiation, saturation or integer-part theorem.
- 06.7 The surreal embeddings are additive; no closure under surreal multiplication is asserted; the topologies are transported, not inherited from a topology on the class **No**.
- 06.8 The formalization steps are proposed, not run; the repository reports are points of contact, not premises; no full-repository audit or Lean rebuild was made.
- 06.9 The projects are not certified as unsolved in the literature.
- 06.10 The proof reviews made during preparation are mathematical checks, not formal certification or peer review.

Source 08.

- 08.1 Bibliographic priority has not been established; no independently certified breakthrough is claimed; the manuscript is AI-assisted, unrefereed and not Lean-checked; a missing search result is not evidence of a first proof.
- 08.2 The compact-group obstruction, finite-dimensional separation and the locally compact abelian structure theorem are classical; the principal structure theorem is imported, not reproved; Presburger elimination is classical; Hewitt and Koshi’s paper was not read in full.

- 08.3 A Hausdorff locally compact second-countable topology on the signed additive group is assumed; the result is not a theorem about every Polish presentation of a positive arithmetic model and does not replace Glazer's general theorem.
- 08.4 The real model is credited to the companion manuscript (source 05); the classification is of pointed topological additive groups, not of all orders or of all topological monoids.
- 08.5 No separable non-locally-compact counterexample is presented; the general Polish-group case and Glazer's Question 1 are not settled.
- 08.6 The transported examples are not effective; no topological-ring claim is made for the nonseparable example.
- 08.7 The finite tests cannot establish statements about all real points, arbitrary Borel sets, local compactness, Haar measure or infinite cardinalities; no Lean file is included as if checked, and a conditional formalization with an interface must not be advertised as unconditional.
- 08.8 The companion manuscripts were not independently audited; the repository guides identify interfaces only; no repository file was modified or built.
- 08.9 The proofs are in ZFC; no claim is made about the choice or reverse-mathematical strength needed; no topology, Haar measure or power set is applied to a proper class.

Source 09.

- 09.1 The proofs are conventional and unrefereed; nothing is Lean- or Rocq-verified; historical novelty is not established; the literature review was targeted and no absence-of-result argument is used as evidence.
- 09.2 Category theorems, Hahn and Puiseux constructions, Newton–Puiseux, real-closedness inputs, the Levi-Civita construction, the coanalytic support obstruction and the omnific floor rule are credited, not claimed as new.
- 09.3 No new answer to either of Glazer's printed questions is claimed, and Glazer is not a collaborator or endorser; the LinkedIn page was not used.
- 09.4 The conventional real-coefficient Levi-Civita field is not claimed to be Polish.
- 09.5 The 20,116 finite checks are not proofs of the category, completeness, descriptive or real-closedness statements.
- 09.6 The repository's formalization ledger could not be fetched; no claim-by-claim audit, local checkout or build was made; the pin is a locator.
- 09.7 The proofs are in ZFC with no choice-free assertion; no computability claim follows from Borel equivalence; no Polish topology on a proper class is used.
- 09.8 The questions are proposed next steps, not previously published open problems.

Source 12.

- 12.1 An AI-assisted, unrefereed manuscript, not an independently certified breakthrough and not a Lean- or Rocq-checked development; no such toolchain was run; historical priority is not established, and targeted searches do not prove absence from the literature.

- 12.2 The Pettis and Steinhaus difference principles are classical, and only the needed forms are proved; the bounded-neighbourhood lemma may be viewed as an application of classical regularity methods; Steinhaus is not credited with the locally compact generalization; Rothman is cited as background, and no theorem from it is assumed.
- 12.3 The theorems concern a topology on the signed additive group; they give no automatic Polish group completion of an arbitrary positive monoid and do not replace Glazer's theorem under weaker hypotheses on the same objects.
- 12.4 Glazer's ATR_0 question is not settled, and the earlier affirmative Presburger construction is not presented again as new.
- 12.5 "Countable" does not mean standard; the non-Polishability theorem does not exclude a Polish topology after the Borel structure and the regularity of the order are changed.
- 12.6 The pathological Polish presentation is a choice construction, not a computable coding, and its multiplication is not claimed to be Borel.
- 12.7 The finite checks, in $\mathbb{Z} + t\mathbb{Q}[t]$, do not certify Baire category, Borel sets, uncountable cardinalities, semiring completions or a theorem for every ordered ring.
- 12.8 No theorem treats the proper class **Oz** as a Polish space; the omnific corollary is set-sized and uses no unrefereed repository claim.
- 12.9 The repository was not built and its advertised audit results were not verified; reading its interfaces is not a check of the whole repository; the profile page supplied with the request was not available, and nothing personal is inferred from it.
- 12.10 The questions are proposed directions; except for the earlier project questions identified, their open status is not certified.

Source 10.

- 10.1 Unrefereed and AI-assisted; bibliographic priority is not established; no theorem is claimed Lean- or Rocq-checked, no new source file is labelled verified, and the repository's builds were not rerun.
- 10.2 The category mechanism is classical and its Polish version was already isolated by source 09; the divisible-plus-integer Presburger examples belong to the existing project constructions and are not claimed as new.
- 10.3 The theorem concerns signed additive-group topologies: it gives no Polish group completion of a positive arithmetic monoid and does not remove the Borel-multiplication hypothesis from Glazer's positive-model theorem; the comparison is between different hypothesis sets.
- 10.4 Glazer's Question 1 is not settled, and the affirmative Presburger examples are not presented as a new answer to Question 2.
- 10.5 "Uniqueness" in [corollary 78.6](#) is of the topology, not of the ring; countable nonstandard rings are not classified.
- 10.6 The transported example is chosen with vector-space bases, not by an algorithm, and gives no Borel multiplication or regular comparison.

- 10.7 Clause (4) of [theorem 80.4](#) concerns each hull separately, not the spectrum or the union of the hulls; the theorem is not about field self-embeddings of **No**.
- 10.8 Borel and continuous scalar maps are not identified outside the separable setting.
- 10.9 The omnific corollary is set-sized, with no Polish topology on a proper class; countable notation systems, incomplete presentation spaces, non-Baire comparisons and purely algebraic formalizations are not ruled out.
- 10.10 The exact tests do not verify category, uncountability, arbitrary orders, basis choices or any infinite theorem; the profile page supplied with the request was not used for personal details; the earlier project manuscripts were read as project documents, not refereed publications.
- 10.11 The questions are proposed targets, not certified open.

Source 13.

- 13.1 Conventional proofs with limited exact diagnostics; no first-publication priority, refereeing or new proof-assistant verification is claimed; an “open question” means one left open by a repository manuscript.
- 13.2 The affirmative existence construction is credited to the earlier manuscripts; the countable nonsplit algebra and the omnific criterion have prior repository counterparts and are not claimed as new; $\mathbb{Z}$ -groups, Presburger elimination, Polish-space facts and the ordered-group universality of **No** are classical.
- 13.3 Glazer’s reverse-mathematical question is not resolved; arbitrary Polish topological monoids are not classified; the connected-universe question for monoid topologies is not settled.
- 13.4 Discontinuity sets need not be closed or Presburger-definable; the compiler’s output is an external coordinate formula, an analysis tool and not a definability theorem; with arbitrary infinite parameters it is an oracle reduction, not a decision procedure.
- 13.5 A bound on revisions gives no stopping stage and no running-time bound; the exact finite calculation is not asserted for one fixed arbitrary parameter; decidability is not efficiency; the rank bound need not be the exact revision complexity.
- 13.6 The countable-core extension is topological; no computable presentation of the core is assumed.
- 13.7 The omnific criterion is additive and order-theoretic and set-sized; the monomial-shift embedding is not asserted to preserve multiplication or to be initial; the no-homomorphism result concerns **Oz**, not **No**.
- 13.8 The closed-translate theorem gives a necessary condition only.
- 13.9 `verify.py` is a finite diagnostic, not an implementation of quantifier elimination or a proof of the infinite theorems; the repository’s Lean semantics over `List Int` is not a proof for nonstandard $\mathbb{Z}$ -groups; the profile page supplied with the request was inaccessible and not used.
- 13.10 The questions are not presented as solved, and are narrower than a theory of all Polish arithmetic.

Source 14.

- 14.1 The answer to Question 1 is claimed for the question as stated in Glazer’s 2023 paper, under the coded-model convention of its Section 2; it is an upper bound only: no reversal, and no claim that ACA_0 fails or that the theorem is equivalent to ATR_0 .
- 14.2 Not mechanically verified and not externally refereed; the proof audits are in-session reviews; no exhaustive priority search (no later arXiv version or indexed solution was found).
- 14.3 The internal MRDP theorem is an established input; its original chapter was not obtained, the statement being corroborated by Chen, Müller and Yokoyama; analytic separation, coded Baire approximation and uniform perfect-set and order homogenization are declared inputs.
- 14.4 The standard- $\mathbb{N}$ Lean MRDP theorem does not prove the internal statement; no Lean or Coq file is a formalization of the obligations.
- 14.5 No global satisfaction predicate, set of all Borel functions or set of all models on an uncountable carrier is used.
- 14.6 “Cofinal” means unbounded in the arithmetic order, not dense, large or nonmeager.
- 14.7 Section 10 is ordinary set-theoretic mathematics, not formalized in ATR_0 ; nonsplit $\mathbb{Z}$ -groups are not a new discovery; preservation of the unit is essential (Φ_ω embeds into $\mathbf{Oz}$); Φ is not a homeomorphism onto its image in the surreal order topology.
- 14.8 The finite checks prove no infinite theorem and do not simulate a nonstandard model.
- 14.9 The questions are proposed directions.

Source 15.

- 15.1 The basic affirmative answer to Question 2, the real and Baire constructions and the special multiplication obstructions are prior project material; quantifier elimination, automatic continuity, the structure theorem for locally compact abelian groups, the compact-subgroup and real-line order lemmas and finite-support techniques are classical; the probabilistic mechanism is Glazer’s, extended, not independently invented.
- 15.2 The classification covers inherited group topologies only, not all Polish monoid topologies; the open real component need not be convex; local compactness of the coefficient group alone does not suffice.
- 15.3 The compact-open theorem concerns the topology at zero only: it does not rule out zero-dimensional locally compact presentations with isolated zero, nor make every numeral isolated.
- 15.4 The manifold boundary differs from the boundary in the ambient group; “initial” is meant arithmetically.
- 15.5 The omnific images are additive substructures, not subrings, and carry a transported coefficient topology, not the native order topology.
- 15.6 No Lean or Rocq build; AI-assisted review passes are not refereeing; publication priority is not certified; Glazer’s Question 1 is not answered; “induction-free” refers to the modeled arithmetic, not to the metatheory.

- 15.7 The degree-stratum example carries no novelty claim; no finite computation is offered as a certificate, and none is shipped.
- 15.8 The questions are not all asserted to be previously published open problems.

Source 16.

- 16.1 AI-assisted, internally reviewed by two AI agents (record shipped as 16-presburger-embeddings-PROOF_REVIEW.md), not refereed and not a literature-certified breakthrough; no new Lean or Rocq formalization; its six proposed modules are a plan, and the repository's integer Presburger procedure is not a formalization of arbitrary $\mathbb{Z}$ -groups.
- 16.2 The classification and the thresholds concern the family M_α only; the lattices of arbitrary Polish presentations of arbitrary Presburger models are not classified.
- 16.3 The family M_α and its Baire-space carrier are prior work, and the positive construction is not a new answer to Glazer's Question 2; Glazer's Question 1 is not resolved.
- 16.4 Hahn-group matrix methods and strong additivity (Kuhlmann–Serra, Serra), formal infinite sums (Freni) and linearly compact duality (Castellano–Giordano Bruno) are credited and not claimed; Presburger quantifier elimination is classical, proved in the appendix only for completeness.
- 16.5 Historical priority is not established; the source audit was bounded, not exhaustive.
- 16.6 The finite checks (11,243 assertions) cover finite matrix identities, example normal forms, signs and standard division; they do not verify infinite products, ordinal recursion, Baire category, elementarity or the existence of the non-Borel functional, and no test confined to finite-support vectors can tell a wild shear from the identity.
- 16.7 Multiplication is absent; the omnific realization is additive and ordered, with a transported coefficient topology, without ring closure and without a topology on the proper class **No**.
- 16.8 The discontinuity statements concern the product topology (the shears are continuous for the valuation topology); Polishness is that of the inherited topology, and a nonclosed subspace may be abstractly isomorphic to a closed one.
- 16.9 The non-Borel constructions use choice; the questions are proposed directions, not certified open problems.

The merge.

- M.1 The affirmative answer concerns Question 2 as printed. Glazer speculated the opposite; this report draws no conclusion about his Question 1 or about any variant of Question 2 with further hypotheses.
- M.2 [Proposition 26.1](#) and the observations of [section 26](#) are elementary consequences of the sources' theorems; they are not presented as new results about Glazer's problem.
- M.3 Citing a collection label does not certify that label: of those cited, only `odg:thm:floor` is formalized.
- M.4 Nothing in this report is formalized. The pointer from the formal project `Logic/PresburgerArithmetic` and the placement in the `Surreal` collection confer no formal status.

- M.5 The reading of Glazer’s speculation for Parts III–V in [section 1.1](#) is editorial; no source of Parts III–V draws it, and nothing in them answers Question 1.
- M.6 [Proposition 42.1](#) and the observations of [section 42](#) are elementary consequences of the sources’ theorems, not new results on Glazer’s problem.
- M.7 The note after [question 14.4](#) re-scopes one sentence of that question; the rest of it, and every other question of the report except where a note says otherwise, stays open.
- M.8 (Batch 87.) Items M.1 and M.5 were written for Parts I–V. Part VIII prints source 14’s claimed affirmative answer to Glazer’s Question 1; this report has not reviewed that answer and draws no conclusion from it beyond the reading in [section 1.1](#).
- M.9 [Proposition 90.1](#) and the observations of [section 90](#) are elementary consequences of the sources’ theorems.
- M.10 The identifications of source 14’s K_* with G_{ζ_*} and of its omnific obstruction with isg:cf:cor:unit are this report’s; source 14 does not make them (source 13 does for its core).
- M.11 Titles flagged “a case of” in Parts VIII and IX compare hypotheses only; the proofs of those sources are printed and are different.
- M.12 The notes that answer or re-scope questions of Parts I, III and IV record what the batch-87 sources prove; those answers are unrefereed and unformalized like the rest of the report. A question not named in such a note, and every part of a named question that a note does not say is answered, stays open.
- M.13 (Batch 89.) The notes of [Part X](#) that identify its results for $\alpha = \omega$ with source 04’s and source 07’s, and the statements there of what is new to the report, are this report’s; source 16 does not make them.
- M.14 [Remark 133.2](#) was written for this report and has not been independently reviewed; it rests on Shelah’s consistency theorem [96] as an input and does not decide which weak choice principle suffices for the discontinuous maps.
- M.15 The notes answering Part I’s questions on coefficient groups (one clause), lattices and embedding categories record what source 16 proves for its family; those answers are unrefereed and unformalized, and the rest of each question stays open.

G An explicit induction check after quantifier elimination (source 07)

For readers concerned that discrete order and division might conceal a missing induction argument, the standard quantifier-elimination input also yields a direct least-element proof.

Let G be any $\mathbb{Z}$ -group with positive cone M . After quantifier elimination, a unary definable set with parameters is a finite Boolean combination of linear inequalities and congruences. Division by fixed standard coefficients changes each inequality into an upper or lower bound on the variable. Taking a common standard modulus and partitioning at finitely many endpoints writes the set as a finite union of cells

$$I \cap \{x \in M : x \equiv r \pmod{d}\},$$

where I is an arithmetic interval, possibly with an infinite end. Empty cells may be discarded.

Every nonempty cell has a least element. Intersect with M and replace a strict finite lower bound by its successor; if there is no effective lower bound, use zero. This gives a closed lower endpoint $a \in M$. Choose the unique standard $t \in \{0, \dots, d-1\}$ such that $a + t \equiv r \pmod{d}$. Then $a + t$ is the first element of that residue class at or above a . If the cell is nonempty, $a + t$ also satisfies its upper bound, since it precedes any other candidate in the cell. Thus it is the cell's minimum. Taking the least of finitely many minima shows that every nonempty unary definable subset of M has a minimum.

Now let a definable $D \subseteq M$ contain zero and be closed under successor. If its complement is nonempty, choose its least element b . Then $b > 0$ and its predecessor $b - 1$ lies in D . Successor closure puts b in D , a contradiction. This proves every first-order induction instance with parameters. The only imported step is classical quantifier elimination, explicitly identified in Section 31; no topological assumption enters.

[write] This is a second route to the induction clause of [corollary 4.11](#); source 06's [section I](#) and source 08's [theorem 49.3](#) prove the elimination itself again.

H Source audit, proof dependencies, and scope (source 07)

H.1 Primary source checks

The supplied LinkedIn page could not be read directly during this investigation. The selection of topic rests on Glazer's named mathematical paper, whose full text was inspected, rather than on inferred biographical details. The exact wording and number of its Presburger question were checked on its final page.

The literature check covered the current arXiv record, the earlier Enayat–Hamkins–Wcisło paper and author page, and targeted searches combining Presburger arithmetic, Polish spaces, Glazer, and continuous addition. No later resolution of the cited question was located. This is a report of a limited search, not a certification that the problem remained open until the date of this manuscript.

General Baire presentations with low-complexity definable relations are already part of the background discussed in [2]. The additional content here is continuous addition together with explicit coordinate models, their ordinal and topological invariants, and their omnific realizations.

H.2 Repository inspection and duplication boundary

At the pinned snapshot, a search of the checked-out `.tex`, `.md`, and `.lean` files found no occurrence of “Glazer” or “Tennenbaum”. The initial-subgroup report does cite the earlier topological-arithmetic paper, but the inspected material does not give the Polish Presburger constructions used here. This statement concerns the searched text formats and snapshot; it is not a search of every historical branch or every binary artifact.

[write] *Correction of a stale statement.* The search result above was true at source 07's pin `0f95145cb`. It is no longer true. The word “Glazer” first occurs, in archive names, in the Hilbert's-tenth programme's scope note `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_recent_incoming_glazer_scope.md` (commit `0f3b760ae`), and Glazer's paper, whose title names Tennenbaum, is discussed in this report since its placement commit `3d2177df4`. Part I there gives the real and Baire models of

sections 32 and 35 independently of source 07.

The exact formalization boundary was checked in:

- `Logic/PresburgerArithmetic/Lean/PresburgerArithmetic/Syntax.lean`: the interpretation `Formula.holds` uses ordinary integer valuations.
- `Logic/PresburgerArithmetic/Lean/PresburgerArithmetic/Decision.lean`: the elimination correctness theorem quantifies over `List Int`.
- `Logic/PresburgerArithmetic/Lean/PresburgerArithmetic/Audit.lean`: the file requests dependency reports for the existing integer theorems.

The repository was used read-only. No claim is made that this report has been merged into it or that any new formal theorem has been accepted by its kernel.

H.3 Dependency structure of the new arguments

Conclusion	Required ingredients
Affirmative real model	Lexicographic coordinate algebra; classical $\mathbb{Z}$ -group theorem; an explicit Polish topology
General cone regularity	Baire category; regularity of Borel sets; countable bases; closure and transfinite recursion
General Polish realization	Cone regularity; group division; classical Presburger quantifier elimination
Baire-space models	Countable products; direct G_δ cone computation; noncompact cylinders; Baire-space characterization
Ordinal nonisomorphism	Leading-coordinate comparison; group completion; Archimedean classes
Oscillation calculation	Finite-coordinate neighborhoods; relative closures; countable ordinal arithmetic
No ordered-semiring expansion	Distributivity; positivity; a cofinal family of standard multiples
Omnific realization	Set-sized Conway normal forms; coefficientwise addition and leading-term order

The real example does not depend on the general cone theorem, the transfinite Baire family, or surreal normal forms. The direct proof of the Baire family's sign-cone complexity likewise makes it independent of the general cone theorem. These independent routes reduce the chance that a single auxiliary claim could undermine the positive answer.

H.4 Review status

The coordinate model, the Borel-cone recursion, the transfinite Baire family, and the boundary oscillation calculation were subjected to separate mathematical proof checks during preparation. Those checks specifically examined induction, completeness versus the inherited metric, transfinite leading coordinates, limit stages, and the distinction between partial and truncated subtraction. They are independent reasoning checks, not external refereeing or proof-assistant certification.

No finite numerical experiment is offered as evidence for an infinite model-theoretic theorem. The deliverable consists of the proofs, identified classical inputs, exact source

references, and a reproducible LaTeX source. Compilation and visual inspection verify the document, not the mathematics.

I A self-contained Presburger elimination argument (source 06)

This appendix proves the algebraic characterization used in the main constructions. It is included to make the induction claim fully checkable without relying only on a reference to a characterization theorem.

I.1 Division in a $\mathbb{Z}$ -group

Let G be a discretely ordered abelian group with least positive element 1 and divisible quotient $G/\mathbb{Z}$. Fix a positive standard integer n . For each $g \in G$, divisibility of the quotient gives $h \in G$ with $g - nh = k \cdot 1$ for some standard integer k . Write $k = nq + r$ in ordinary $\mathbb{Z}$, where $0 \leq r < n$. Then

$$g = n(h + q \cdot 1) + r \cdot 1.$$

This representation is unique: if two remainders r, r' in the specified finite interval differ by n times a group element, discreteness forces that element to be zero. Explicitly, a positive difference would be at least n , and a negative one at most $-n$, whereas $|r - r'| < n$.

Conversely, this division property makes $G/\mathbb{Z}$ divisible. Thus we can work with the ordered-group axioms, discreteness, and this schema of standard division algorithms.

I.2 Eliminating one quantified variable

Expand the language by predicates $D_n(u)$, for standard $n \geq 1$, with the interpretation $u \in nG$. Every group term is integer linear. To eliminate an existential quantifier, first put its quantifier-free matrix in disjunctive normal form and treat each conjunction separately. Split disequalities into the two strict inequalities and use discreteness to replace weak inequalities by strict inequalities with a shift of 1. Atoms not involving the quantified variable are retained as parameter conditions.

Suppose the variable is x . Let L be the least common multiple of the nonzero absolute coefficients of x in the conjunction, or 1 if there are none. Replace Lx by y and add $D_L(y)$. For an inequality or equality involving $ax + u$, multiply by the positive integer $L/|a|$; its coefficient of y becomes 1 or -1 . For a divisibility literal, use

$$D_n(ax + u) \iff D_{nm}(\text{sign}(a)y + mu), \quad m = L/|a|. \quad (\text{I.1})$$

This is valid because, in a torsion-free group, $v \in nG$ if and only if $mv \in nmG$. The same equivalence handles negated divisibility. Thus every occurrence of y has coefficient 1 or -1 .

If an equality specifies y , substitute its value. Otherwise the conjunction consists of finitely many lower bounds $a_i < y$, upper bounds $y < b_j$, and positive or negative congruence conditions, together with parameter-only literals.

Let N be a common multiple of all congruence moduli, including L , and split into the finitely many possibilities $y \equiv r \pmod{N}$ for $0 \leq r < N$. In each branch, a condition $D_n(\pm y + u)$ is equivalent to the parameter condition $D_n(\pm r + u)$. Only the one residue condition modulo N and the bounds now involve y .

If lower bounds are present, divide into finitely many cases choosing their greatest value a ; this is expressed by parameter comparisons. Among

$$a + 1, a + 2, \dots, a + N$$

there is exactly one member of the chosen residue class. It is the least member of that class above a . To justify minimality, if $y > a$ has the same residue and lies below that candidate, their positive difference is a multiple of N smaller than N , contradicting discreteness. Thus, when upper bounds are also present, existence is equivalent to the finite disjunction

$$\bigvee_{k=1}^N \left(D_N(a + k - r) \wedge \bigwedge_j (a + k < b_j) \right).$$

If there are no upper bounds, the candidate always works. If there are upper bounds but no lower bounds, a residue-class element exists among the N positions just below the least upper bound. If neither type of bound is present, the standard representative r itself works.

This eliminates one existential quantifier. Universal quantifiers are handled by negation, and induction on formulas gives quantifier elimination in the expanded language. The output consists of Boolean combinations of integer-linear comparisons and standard congruences.

I.3 Completeness and induction in the nonnegative cone

A closed quantifier-free formula in this language concerns only standard numerals and their congruences and comparisons. Its truth is the same in every $\mathbb{Z}$ -group and in ordinary $\mathbb{Z}$. Consequently every $\mathbb{Z}$ -group is elementarily equivalent to $\mathbb{Z}$.

The structure $G_{\geq 0}$ is interpreted in G by restricting quantifiers to $x \geq 0$. Its addition, constants, and order are the restrictions of those in G . Every sentence true in $(\mathbb{N}, 0, 1, +, <)$ therefore holds in the cone.

In particular, for every additive formula $\varphi(x, \bar{a})$ the sentence

$$\forall \bar{a} \left[(\varphi(0, \bar{a}) \wedge \forall x (\varphi(x, \bar{a}) \rightarrow \varphi(x + 1, \bar{a}))) \rightarrow \forall x \varphi(x, \bar{a}) \right]$$

holds. Parameters are covered because their universal closure is part of the sentence. Multiplication is absent from this language; the argument therefore makes no assertion about the polynomial model's formula $x^2 < t$.

[write] This is a second route to [theorem 4.9](#) and the induction clause of [corollary 4.11](#), by the finite search of [lemma 4.7](#).

J Provenance and verification record (source 06)

Mathematical dependencies. The main constructions use ordered-group algebra, elementary Presburger elimination, standard facts about Polish spaces, and explicit coordinate arguments. The isolated-zero theorem additionally uses analytic/coanalytic separation and fair product measure. Its tail zero-one step is proved in the text. The surreal realizations are explanatory embeddings and are not premises of any of the main topological theorems.

Independent scrutiny. The real and Baire models, the polynomial topology, the distinction between additive and open induction, and the isolated-zero argument were subjected to

separate proof reviews during preparation. Those reviews specifically checked complete metrizable rather than mere separability, negative integer slices, the parameter version of induction, joint continuity, cancellation, and the tail-event invariance. The final proof omits an unnecessary Cantor-injectivity claim and never assumes continuous subtraction. These reviews are mathematical checks, not formal proof-assistant certification or external peer review.

Repository audit. The repository comparison used its path inventory and selected relevant files at the pinned commit. No full-repository theorem audit or Lean rebuild was performed. Existing exploratory reports are credited for their stated algebraic content, with the main results here proved independently.

Reproducibility. This source is self-contained and uses standard $\text{\LaTeX}$ packages. It can be compiled with `latexmk -pdf polish_presburger.tex`. No external image, private data, bibliography processor, or downloaded repository is needed to reproduce the PDF. The accompanying ZIP contains this source, its compiled PDF, and a short build-and-status readme.

[write] Source 06 shipped only its $\text{\LaTeX}$ source, its PDF and a short plain-text readme. None of them is shipped here (the source is merged into this report; the archive survives in commit fb8414869); there is no program to rerun.

K Contribution ledger, source audit, and reproducibility (source 08)

K.1 Contribution ledger

Statement or ingredient	Status in this manuscript
Compact-group order obstruction; finite-dimensional separation; LCA principal structure theorem	Classical material, credited in section 47 . The first two are proved in the needed form. The LCA principal structure theorem is an explicitly imported theorem.
Presburger quantifier elimination	Classical. A complete finite-witness argument is supplied to control the nonstandard-group transfer.
The real positive-cone example $M_{\mathbb{R}}$	Previously constructed in the companion manuscript [40] ; not claimed as new.
Integer-disjoint order-unit amplification and ordered direct sums	Derived and proved here, theorem 46.2 ; global priority unverified.
Arithmetic rigidity without multiplication regularity	Derived and proved here, theorem 44.1 ; global priority unverified. No induction assumption.
Pointed group classification; cone dimension threshold	Derived here from the classical structure inputs. Not a classification of all full orders or all topological monoids.

Statement or ingredient	Status in this manuscript
Uniform Δ_2^0 definability and continuous standard division	Derived for the classified presentations. Standard elimination and topological facts are not claimed as new.
Countable Borel-scalar subring; exact answer for $E = \mathbb{Z}$	Derived and proved here, theorem 53.4 ; does not assume order-compatible multiplication.
General Polish-group version; full order classification; Lean proof	Not established here. Listed as further research.

K.2 What was actually inspected

The repository snapshot used for the interfaces is

883e0b3b22263bd9d92fdbbf10376ca4b652b862.

The inspected guides were the root `README.md`, `Logic/PresburgerArithmetic/README.md`, and `Algebra/SurrealNumbers/README.md`. The first arithmetic guide identifies the syntax, normalization, Cooper, elimination, and decision layers and their stated audit scope. The surreal guide distinguishes constructed objects, mapped formal results, and unrefereed research reports. Those guides were used to identify interfaces, not as evidence that the new theorems in this manuscript have already been formalized. No repository files were modified, and no full repository build was run.

The Glazer source was the publicly available 2023 arXiv version of [1]; the supplied LinkedIn profile was not needed as mathematical evidence. Classical cross-checks used the primary papers or author notes listed below, including the measurable-order discussion in Section 3 of [22] and the LCA structural statement on page 4 of [27]. The bibliographic entry for [26] is recorded through its explicit discussion and citation in [22]; no claim of an independent full reading of that 1983 paper is made. The prior companion manuscripts were retrieved from the user's library.

K.3 Finite verification

The included `verify.py` uses Python's standard-library `fractions.Fraction`. Its output on the supplied version is:

Test family	Cases passed
Standard division in rational samples of $\mathbb{R}^d \times \mathbb{Z}$	7,595
Coset-invariant finite-witness tests	9,000
Polynomial amplification tuples	1,875
Lexicographic order triples	6,000
Total	24,470

The finite-witness tests compare the candidate construction with an independent oracle on rational lexicographic vector coordinates and integer intervals. The polynomial tests compute in $\mathbb{Q}[X]$, with $V = \mathbb{Q}X$ and u a positive rational multiple of X ; they check finite

injectivity, the highest-coefficient sign rule, and lower-term domination. All comparisons are exact, with fixed random seeds for the sampled tests.

These checks can catch sign, indexing, and residue mistakes. They cannot establish any statement involving all real points, arbitrary Borel sets, local compactness, Haar measure, or infinite cardinalities. The mathematical proofs, not the tests, carry those conclusions.

K.4 Build instructions

The bundle contains the source and compiled PDF, this verification script, its JSON output, a source-provenance file, and a short build script. With a standard TeX Live installation including the New PX packages:

```
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
python3 verify.py --output verification_results.json
```

The PDF should then be rendered and visually checked. No external font files, third-party source PDFs, or unexecuted Lean proof claims are included.

[write] In this report the program is code/08-local-compactness-verify.py, its recorded output data/08-local-compactness-verification_results.json, the build script code/08-local-compactness-build.sh, and the source-provenance file and proof audit 08-local-compactness-sources.md and 08-local-compactness-proof_audit.md. They are byte-identical to the delivery and still use the delivery names verify.py, verification_results.json and article.tex; the build script builds a file article.tex in its own directory and must not be run in the report directory. The intake reran the program on a copy with Python 3.14.4: the four families and the total of 24,470 cases agree with the shipped record, which differs from the new output only in line endings.

L Contribution ledger and proof dependencies (source 09)

Statement or ingredient	Status and dependency boundary
Pettis, automatic continuity, open mapping	Classical tools; short proofs are included in Section 61. Their use is not claimed as new.
Local order bounds and forced order topology	Proved here from the category tools, with Baire order and scalar hypotheses explicit. Priority of this formulation has not been certified.
Full Hahn classification	Central presentation-invariant deduction, Theorem 63.3; depends only on the order bridge and the explicit separated binary family, plus classical existence of Hahn fields.
Proper-subfield cover and Puiseux obstruction	Proved here, Theorems 64.1 and 64.3; the measurability of covering subfields is essential and is verified in the application.
Levi-Civita field and completeness	Classical construction; explicit proofs are supplied. The uniqueness and comparison with non-Polishable Puiseux arithmetic use the category bridge.

Statement or ingredient	Status and dependency boundary
Exact code domains	Explicit FIN and row-finiteness reductions are given. The full Hahn coanalytic obstruction is credited to classical descriptive methods and the preceding project report, not reclaimed as new.
Real closedness	Classical Newton–Puiseux input plus a supplied proof that completion preserves algebraic closedness. Not a new algebraic theorem.
Polish hulls and exact cover number	Proved here using countable dense field cores, prefix stabilization, and the fixed-scale binary obstruction. The hull construction uses a standard completion mechanism; the local/global classification is explicitly developed in Section 69.
Countable integer parts	A consequence of the forced order topology, with no measurability assumption on the integer part.
Floor formula	Existing omnific normal-form principle, proved here in the restricted field for clarity.
Floor regularity	Explicit continuous approximants and exact discontinuity sets for Levi-Civita and strengthened Polish hulls, Theorems 70.2 and 70.4; not a kernel check.
Open induction	Classical integer-part implication, proved directly for the constructed ring; no uncountable arithmetic model is asserted.
Executable verification	Finite rational arithmetic regression tests only. They do not verify the set-theoretic or topological quantifiers.

The logical core can be read as

$$\begin{aligned} &\text{Pettis} \implies \text{local order bounds} \\ &\implies \text{order-topology refinement} \\ &\implies \left\{ \begin{array}{l} \text{full Hahn classification,} \\ \text{countability of integer parts,} \\ \text{uniqueness of complete order presentations.} \end{array} \right. \end{aligned}$$

The proper-subfield-cover theorem is a separate branch from Pettis and automatic continuity. Descriptive completeness and real closedness are independent supplementary branches; neither is needed for the main Hahn classification.

M Source audit and reproducibility (source 09)

Research scope. The research-interest overlap was based on Glazer’s mathematical paper [1], not on a speculative interpretation of a social-media profile. The article does not imply that he reviewed, collaborated on, or endorsed this manuscript. The public LinkedIn page supplied in the request was not needed to substantiate the mathematical connection.

Repository scope. The root and surreal READMEs were inspected, together with repository metadata and the directory structure of the surreal implementation. The repository head recorded during the audit was

4af956fe015e043c5e163ee5b4d2cea23e6157ea

This pin is a reproducible repository locator, not a claim to have rebuilt the project at that commit. The main proofs here are self-contained and do not depend on assuming a repository theorem beyond the explicitly credited normal-form background. The large formalization ledger could not be fetched through the available connector; no claim-by-claim audit of that ledger is represented as completed.

Earlier reports. The Library copies of the previous Polish Presburger and Borel-support manuscripts were checked at the abstract/contribution level to avoid presenting the same results as new. The present article does not use their proposed theorems as unproved premises. References [43, 44] are unpublished project manuscripts, not independently validated literature.

Literature scope. The primary mathematical sources below were accessed during preparation. The review was targeted, not exhaustive. The sources establish the classical inputs and the research connection; the priority of the combined classification and its refinements remains open to specialist review. In particular, no absence-of-search-result argument is used as evidence of novelty.

Build and checks. The archive contains this self-contained `article.tex`, the compiled PDF, `verify.py`, its machine-readable output `verification.json`, a source record, and a build README. Compile with a standard TeX Live installation using

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
python3 verify.py --output verification.json
```

The Python program uses only the standard library. It records deterministic finite test counts and fails on a violated identity. No numerical tolerance is used: all tested coefficients and exponents are exact rational numbers. The PDF was also rendered for layout inspection. Font packages are referenced by the TeX source; no font files are distributed in the archive.

[write] In this report the program is `code/09-hahn-polishability-verify.py`, its recorded output `data/09-hahn-polishability-verification.json`, the build script `code/09-hahn-polishability-build.sh`, and the source record `09-hahn-polishability-SOURCES.md`; they are byte-identical to the delivery and use the delivery names. The intake reran the program on a copy with Python 3.14.4: 20,116 assertions in 29 families with seed 20261003, equal to the shipped record up to line endings. Source 09's bibliography and source record cite the surreal README and root README by unversioned `blob/main` addresses, beside the pin above; the pinned versions are <https://github.com/VladimirReshetnikov/ProveIt/blob/4af956fe015e043c5e163ee5b4d2cea23e6157ea/Algebra/SurrealNumbers/README.md> and the root README.md at the same commit. Both files are unchanged between the pin and the placement commit `0bd0e5527`.

N Reproduction and the scope of the exact checks (source 12)

[write] Source 12’s checks are shipped as `code/12-baire-category-verify.py`, which writes `verification_results.json` beside itself; the recorded run is `data/12-baire-category-verification_results.json`. Run it on a copy (README). The build commands below name the delivered `article.tex`, whose text is merged into this report.

Run `python verify.py` with Python 3.10 or later. The script uses only the standard library and exact rational arithmetic. Its random seed is fixed at 20261003. The accompanying run completed **36,989 exact checks**, all passing.

The checks exercise leading-coefficient order, the least-positive-unit property, positivity and absolute values of products, distributivity, the separation inequality (87.1), difference-ring products, finite convolution locality, degree escape from principal convex subgroups, and the finite identity (86.2). Coefficients are rational and all supports are finite. The tested ring is $\mathbb{Z} + t\mathbb{Q}[t]$, not an approximation whose floating-point comparisons might obscure signs.

Check family	Count
Packing separation	2,399
Distinct-point unit separation	2,399
Convex-quotient degree escape	1,618
Sampled convex-quotient bounds	6,472
Finite geometric inverse identities	101
Ten further algebraic families, 2,400 each	24,000
Total	36,989

Some candidate pairs are equal, so the two separation checks have 2,399 instances rather than 2,400; equal pairs are correctly excluded by the theorem hypothesis. Similarly the quotient check excludes equal pairs. The JSON output records the full family breakdown.

These checks can detect implementation mistakes in the examples and sign errors in formulas. They cannot establish the statements quantified over all rings, all points of an uncountable space, or arbitrary Baire sets. Those statements rest on the proofs in the body of the article.

The source compiles with an ordinary LaTeX installation including `newpx`, `tcolorbox`, and `cleveref`. Run

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

A third pass may be used after substantial changes to the contents. The bibliography is embedded in the source, so no separate BibTeX step is needed.

O Provenance and contribution ledger (source 12)

The public research anchor is Glazer’s 2023 arXiv paper. The supplied LinkedIn page was not available for inspection, so no employment, affiliation, or personal-biography claim is inferred from it. The research overlap is grounded in the paper’s mathematical content.

The earlier local-compactness report was read from the user’s Library, including the exact research-question passage in its LaTeX source. It is an earlier AI-assisted project manuscript, not a peer-reviewed reference. Its role is to identify a concrete previously unanswered project question; the proof given here is independent of its main theorem. The Presburger companion is cited to avoid duplicating its construction or claiming it again as a new result.

The ProveIt interface review used the pinned repository revision stated in [section 87](#). The relevant README blob hashes are `a29a6b54c98ae5dbf5e3267a96f10dc5e1514a79` for Presburger arithmetic and `4f830b663aa28748468a9382829d0e4d90e7f8ba` for the surreal development. Reading those interfaces is not an independent check of the whole repository.

Targeted searches combined discrete ordered rings with Polish groups, Baire category, measurable order, and local compactness. They did not identify a source establishing the exact general theorem formulated here. Such a search does not prove absence from the literature or establish priority. In particular, the bounded-neighborhood lemma may be viewed as an application of classical regularity methods; its presentation here does not make those methods new.

Statement or ingredient	Status in this manuscript
Pettis and Steinhaus difference principles	Classical; needed forms proved explicitly
Bounded-neighborhood reflection	Proved here; no priority assertion
Packing and $c = d = R $	Principal established result
Removal of local compactness	Resolves the identified earlier project question
Ambient Haar-measurable version	Proved without hereditary restriction assumptions
Convex quotient and unit spectrum	Additional proved algebraic/topological consequences
Polish positive monoids in full generality	Not settled; group-completion boundary retained
Glazer’s ATR_0 question	Not settled
Lean/Rocq verification of this article	Not performed
Historical breakthrough or priority	Not certified

[write] “The earlier local-compactness report” is source 08, [Part IV](#); “the Presburger companion” is source 05, [Part I](#). The README blob hashes are those of `Logic/PresburgerArithmetic/README.md` and `Algebra/SurrealNumbers/README.md` at source 12’s pin `0f084afa9`.

P A semantic Presburger transfer argument (source 10)

This appendix justifies the abstract arithmetic language used in [section 81.3](#) without treating a decision procedure on ordinary integers as an automatic theorem about all ordered groups.

Let G be a $\mathbb{Z}$ -group. For a positive ordinary integer m , a predicate $P(x)$ will be called *coset-periodic modulo mG* if

$$x - y \in mG \implies (P(x) \leftrightarrow P(y)).$$

This is stronger than merely knowing $P(x + m) = P(x)$ and iterating the identity an ordinary finite number of times. The distinction matters in a nonstandard group: an arbitrary element of mG need not be an ordinary integer multiple of m .

Lemma P.1 (Finite interval reduction [10]). *Suppose P is coset-periodic modulo mG , and L, U are finite lists of elements of G . Then*

$$\exists x ((\forall \ell \in L \ell < x) \wedge (\forall u \in U x < u) \wedge P(x))$$

is equivalent to a disjunction of at most m candidate conditions, once the relevant finite extremum of the bounds is chosen. This is a logical reduction; an effective algorithm additionally requires effective comparison and evaluation of P .

Proof. If L is nonempty, put $b = \max L$ and test $b + 1 + r$ for ordinary $0 \leq r < m$. Given a witness x , discreteness gives $x \geq b + 1$. Divide $x - (b + 1)$ by m in G :

$$x - (b + 1) = mq + r, \quad 0 \leq r < m.$$

Necessarily $q \geq 0$: otherwise $q \leq -1$, so $mq + r \leq -m + (m - 1) < 0$. Thus $y = b + 1 + r \leq x$, and y lies above every lower bound and below every upper bound. Since $x - y \in mG$, $P(y)$ holds. The converse is immediate because every accepted candidate is a witness.

If L is empty and U is nonempty, use $b = \min U$ and test $b - 1 - r$, dividing $b - 1 - x$ for a given witness. This moves the witness upward without crossing an upper bound. If both lists are empty, test the standard residues $0, \dots, m - 1$, since every element of G is congruent to exactly one of them modulo mG . $\square$

To eliminate a quantified variable from a Presburger formula, first put the formula in disjunctive normal form. Terms are affine integer combinations of variables. Negated equalities and inequalities can be converted to finite combinations of strict bounds; discreteness permits the usual shifts by 1. Congruence conditions and their negations form a Boolean combination invariant on the cosets of a suitable mG .

For completeness, coefficients of the quantified variable can be normalized by choosing a positive ordinary common multiple k of their nonzero absolute values, substituting $y = kx$, and imposing $y \in kG$. An inequality $ax < t$ with $a > 0$ becomes $y < (k/a)t$ after multiplication by the positive integer k/a ; negative coefficients are handled by reversing the bound. A divisibility atom $d \mid ax + t$ becomes

$$dk \mid ay + kt.$$

Under $y = kx$ this equivalence follows by cancellation of k in the torsion-free group. The new congruence Boolean combination, including $k \mid y$, is invariant on cosets of mG for a common multiple of its ordinary moduli, even when the quotient elements are nonstandard. **Lemma P.1** now eliminates the existential quantifier. Recursion eliminates all quantifiers; universals are handled by negation. The finite maximum or minimum choice can itself be replaced by a finite disjunction specifying which listed bound is extremal; it does not add a new primitive operation to the language.

A resulting closed quantifier-free sentence has only ordinary integer constants. Their comparisons have their usual truth values, and an ordinary divisibility equation has a group solution exactly when it has an ordinary integer solution: any such quotient is ordinarily bounded and therefore an ordinary integer. Hence every $\mathbb{Z}$ -group has the same signed first-order arithmetic theory as $\mathbb{Z}$.

In the examples G_D , this proof also shows that every definable relation with parameters is Δ_2^0 . Integer affine maps are continuous, order comparisons are Δ_2^0 , and $mG_D = D \times m\mathbb{Z}$ is clopen. Finite Boolean combinations preserve Δ_2^0 . Standard quotient and remainder maps are continuous as well, directly from (81.4). These are explanatory consequences of the construction, not additional claims of historical novelty.

[write] [Lemma P.1](#) is source 08's [lemma 49.2](#) and the coset-invariant finite search of Part I ([lemma 4.7](#)), proved again.

Q Proof audit and reproducible checks (source 10)

Q.1 The shortest proof chain

The entire Polish obstruction has the following dependency chain:

$$\begin{aligned} \text{Baire group} + \text{Baire positive cone} &\implies \text{bounded neighborhood} \\ &\implies \text{open principal convex hull } H_a \\ &\implies |A/\mathbb{Z}1| \leq |A/H_a| \leq \aleph_0 \\ &\implies A \text{ countable} \implies A \text{ discrete.} \end{aligned}$$

The middle injection is algebraic. The first and last steps are category arguments. The countable quotient is topological. No step requires applying a continuity or measurability theorem to multiplication.

Q.2 Potential mistakes explicitly excluded

There is no claim that an order interval is a subgroup, that a principal convex subgroup is an ideal, or that an arbitrary scalar map is continuous. Ordinary finite bounds are not internal nonstandard bounds. The group topology includes continuous negation. Baire-property order is assumed for the given topology, not an unrelated transported one. Countability of a quotient is obtained from open cosets, not from an uncountable union of small fibers. The last use of Baire is essential: a countable metrizable group need not be discrete.

The quotient injection is well defined because the kernel is computed exactly, not merely bounded from above. Order preservation of the quotient uses convexity. The cardinal equalities for uncountable rings use $|A/\mathbb{Z}1| = |A|$, not an assumption that quotienting by a countable subgroup always preserves every finite or countable cardinal.

Q.3 Finite exact-arithmetic tests

The companion `verify_examples.py` uses rational coefficients and deterministic random inputs. It checks the principal-hull kernel identity in $\mathbb{Z} + X\mathbb{Q}[X]$ with $a = X^k$, leading-sign preservation for bounded-subgroup amplification with $H = \mathbb{Z} + \mathbb{Q}X$, $c = X^2$, $a = X^2 + 2$, and standard division/remainder identities in finite rational-coordinate models of G_D . It also checks the truncated geometric-inverse identity underlying [proposition 86.2](#).

The tests use exact rational arithmetic. They do not verify category, uncountability, arbitrary orders, vector-space basis choices, or any infinite theorem. They are regression checks for formulas and examples; the proofs in the article are the justification of the theorems. The generated JSON result records the seed, counts, and scope.

Q.4 Source and priority audit

The external research anchor is Glazer’s arXiv paper, with its theorem, corollary, and question numbering checked against the text. The LinkedIn URL supplied in the request was not used to infer unverified personal or current-employment details. The selected intersection is supported by the published paper itself.

The ProveIt snapshot inspected for this article is

6fef5383b126be343cccc9baf47081ef37ab5afe.

The relevant files are the Presburger and surreal READMEs listed in the bibliography. Their descriptions are attributed to the repository; this session did not independently rebuild or certify them. The earlier local-compactness and Hahn manuscripts were read as project documents, not as refereed publications.

The literature check did not establish that the main algebraic or topological deductions are new in the historical literature. Their status here is a proved contribution relative to the inspected project materials and a resolution of the explicitly posed project question. A submission should undergo independent proof review and a fuller priority search before making a stronger novelty claim.

[write] The “companion `verify_examples.py`” is shipped as `code/10-baire-order-verify_examples.py`, with the recorded run data `data/10-baire-order-verification_results.json`; source 10’s audit is `10-baire-order-SOURCE_AUDIT.md`. The “earlier local-compactness and Hahn manuscripts” are sources 08 and 09, [Parts IV](#) and [V](#).

R Source audit and contribution ledger (source 13)

The three incoming ZIPs were read at repository revision

eab47e3301e0ed2eb8eb09824b90a6e34af7b0.

Their bytes matched the archive identities recorded by the repository’s triage. The subsequent overlap check read the omnific-normalization article at

4f2d7897a114f65a30a1920fb03492d5f9d69d76.

The same omnific source blob was also verified at the earlier revision. The two retrieval revisions are recorded separately; this article does not imply that the changing repository was frozen under one revision throughout the investigation.

Source	What was used
Glazer, arXiv:2311.13699v1	The precise published Presburger question, the distinction between basic-operation continuity and stronger arithmetic, and the separate reverse-mathematical problem
Two incoming Polish Presburger manuscripts	Existing split constructions, their proof interfaces, and their explicit unresolved questions
Incoming Hahn/omnific manuscript	Scope comparison; no support-complexity result from that manuscript is required by a new proof here
Omnific-normalization article	Existing algebraic splitting, residue-image, and unital-omnific-embedding criteria, including a stronger initial-image result
Presburger Lean syntax and decision source	The actual ordinary-integer semantic boundary of the existing formal development
Jeřábek and Ehrlich–Kaplan	Classical model theory and the ordered-group embedding input, respectively

The exact omnific embeddability criterion is *not* claimed as a new result relative to Provett. Its earlier article proves a stronger initial-realization version and explicitly states the equivalence even for noninjective unit-preserving homomorphisms. We include a direct normal-form proof to keep the application self-contained. The new use is the construction and analysis of perfect Polish models outside that split class. Likewise, the ordinary algebraic residue splitting facts are background; the automatic topological consequence is proved here.

The central further results are the exact continuity compiler, the target-sensitive Baire classification, the dimension and exact finite-pattern revision theorems, and the complete-metric obstruction. These were developed against the questions in the inspected manuscripts and audited independently at their delicate proof steps. No search can certify that an equivalent statement has never appeared elsewhere; global historical priority is not asserted.

The source and status metadata are also recorded in `provenance.json`. The bibliography provides permanent paper identifiers and repository links. A repository theorem is not treated as kernel-checked solely because it occurs in a research report.

[write] The “three incoming ZIPs” are batch 86’s manuscripts 03, 04 and 05, now [Parts I](#) and [II](#); the “omnific-normalization article” is the `discrete-initial-subgroups-and-omnific-normalization` report. `provenance.json` is shipped as `data/13-beyond-split-provenance.json`; its archive hashes and blob identifiers were verified at placement.

S Executable diagnostics and reproduction (source 13)

The accompanying `verify.py` is a standard-library Python program. It uses exact rational arithmetic and includes finite checks of the directional feasibility calculation, the nonsplit core’s Chinese remainder division formula, and stream examples attaining the stated revision bounds. Its output is recorded in `verification_results.json`.

The program implements finite algebraic diagnostics. It is not a complete implementation of arbitrary Presburger quantifier elimination, a proof of the infinite-stream lower bound, a proof that a space is Polish, or a machine-checked verification of the theorems. The exact algorithms in the text include transformations beyond the diagnostic program’s scope. This boundary is deliberate and explicit.

To reproduce the checks, run

`python3 verify.py`

from the unpacked package. To compile the standalone article, run

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

No external figures, bibliography database, network request, or repository checkout is needed to build the PDF. The source is a single file. The delivered PDF was checked by compilation, text extraction, and visual page inspection.

[write] In this report the program is `code/13-beyond-split-verify.py`; it writes `verification_results.json` beside itself, so it must be run on a copy (README). The recorded output is `data/13-beyond-split-verification_results.json`. The `article.tex` named above is source 13's delivered manuscript, merged into this report.

T An optional measure version of the final contradiction (source 14)

The category argument is sufficient and avoids a measure-theoretic dependency. For comparison with the original probabilistic mechanism, the multiplicative events in (8.5) also have fair-coin measure zero.

Fix n, k , and let $B = \{n+1, \dots, n+k\}$. No two distinct members of E_n agree outside B . To prove this, suppose r, s do, and let

$$p = \prod_{i \in r \setminus s} y_i, \quad q = \prod_{i \in s \setminus r} y_i.$$

Orient the pair so the largest differing index m belongs to $r \setminus s$. Repeated one-bit identities give

$$z_r = pu, \quad z_s = qu, \quad z_{r^*} = qv, \quad z_{s^*} = pv$$

for positive u, v . The estimate $Q_m^4 < y_m$, with $Q_m = \prod_{i < m} y_i$, gives

$$p^2 > y_m q^2 > y_n q^2.$$

Indeed $p \geq y_m$, $q \leq Q_m$, and $q^2 < y_m$. On the other hand, $r, s \in E_n$ implies

$$pu \leq y_n qv, \quad qu > pv.$$

Multiplying by q and p , respectively, and cancelling $v > 0$ gives $p^2 < y_n q^2$, a contradiction.

Thus the 2^k translates of E_n under flips in B are disjoint and have equal measure. The resulting bound is

$$\mu(E_n) \leq 2^{-k}.$$

Letting k increase gives $\mu(E_n) = 0$. Complement symmetry and the absence of ties imply $\mu(D) = 1/2$, contradicting the tail zero-one law. The non-strict finite-section bound is all that is needed.

Using direct inequalities with an inclusive upper endpoint, as in (8.5), avoids any need to identify an event using a floor of a ratio. This formulation also keeps every comparison inside the arithmetic semiring.

U Short checks on the elementary inequalities (source 14)

Three estimates support the whole construction.

Root scales. With $t_n \geq t_{n+1}^2$, $c \geq q_n t_n$, and $t_{n+1} \geq t_{n+2}^2$,

$$q_{n+1} \geq q_n t_{n+1} \geq q_n t_{n+2}^2.$$

Whenever $t_{n+2} > 8$, this exceeds $8q_n t_{n+2}$. The nonstandard construction guarantees this last hypothesis for every external n .

Finite-bit absorption. If $h_{i+1} > 8h_i$ and $k = \max(n, j) + 1$, then

$$h_k > h_n + 2h_j.$$

If $y_{i+1} > y_i^5$ and $y_i \geq 2$, the corresponding multiplicative estimate is

$$y_k > y_n y_j^2.$$

These are exactly the bounds needed for invariance under one flip.

Factorial obstruction. The identity $s_{n+1} = s_n + n!$ gives (10.3), and induction gives $n!b_n = b_1 + s_n$. The elementary bound

$$s_n \leq 2(n-1)! \quad (n \geq 3)$$

follows by induction. It guarantees that a fixed proposed value of b_1 eventually produces a residue strictly between consecutive multiples of $n!$, disproving a retraction without any limiting argument inside the group.

V Claim and source audit (source 15)

Claim	Evidence or proof supplied	Status
Research overlap	Glazer's topological arithmetic paper; inspected Presburger and omnific repository sources	Directly documented
Basic affirmative answer to Glazer's Question 2	Earlier repository manuscript; coefficient construction recalled in Section 118	Prior background
Full Presburger theory	Explicit residue division plus classical $\mathbb{Z}$ -group quantifier elimination	Classical semantic dependency
Borel multiplier barrier	Complete automatic-continuity and Cantor tail-event proof	Extension of Glazer's mechanism
Cardinality injection	Explicit map $a \mapsto ah + H$ and trivial-kernel argument	Proved here; priority not certified

Claim	Evidence or proof supplied	Status
Inherited locally compact classification	Cone reflection, classical LCA structure, measurable-order lemmas, rational-hull splitting	Full proof with named dependencies
Arithmetic boundary	Infinitesimal convex subgroup, endpoint identification, purity, and quantifier elimination	Proved here; every countable boundary realized
Continuum family	Finite condensation, Archimedean classes, and separable-space counting bound	Exact lower and upper bounds
Omnific realization	Finite normal-form map with sign and injectivity proofs	Set-sized mathematical construction
Formal implementation	Existing declarations inspected; roadmap given	No new Lean or Rocq certificate claimed
Historical breakthrough	Targeted comparison with retrieved primary sources and prior repository article	Not established or claimed

The source manifest accompanying this article records the pinned Git commit, the inspected repository paths and blob hashes, and the external reference URLs. It does not bundle copies of the external papers. The TeX file is self-contained and uses an inline bibliography. A `latexmk -pdf` invocation rebuilds the PDF with the indicated standard packages; alternatively, repeat `pdflatex` until the references are stable.

[write] “The source manifest accompanying this article” was delivered as `source_manifest.json` and is not shipped (it contains a third party’s personal profile address); [section E](#) summarizes its verified content. The “earlier repository manuscript” is [Part I](#).

W The precise quantifier-elimination input (source 16)

For completeness, this appendix records why the arithmetic argument is valid in every $\mathbb{Z}$ -group, rather than relying on calculations over the standard integers. The theorem itself is classical [\[3, 92\]](#).

Work in the group language with all congruence predicates. Put a quantifier-free formula into disjunctive normal form. Negated inequalities and equalities can be expressed using the linear order; negated congruences are finite disjunctions of the other standard residues. Thus it suffices to eliminate an existential variable from a conjunction of affine equalities, inequalities, and congruences.

Choose a positive common multiple L of the absolute values of all nonzero coefficients of the quantified variable x , taking $L = 1$ if there are none. Replace Lx by y and add $y \equiv_L 0$. After correcting signs, a constraint $ax < t$, $a > 0$, becomes $y < (L/a)t$. A congruence $m \mid (ax + s)$ becomes

$$(mL/a) \mid (y + (L/a)s).$$

This equivalence is valid in a torsion-free group: multiplying both the dividend and modulus by L/a preserves divisibility, with cancellation providing the reverse implication. Equalities

can be normalized in the same way; if one remains, substitute its specified value and check the other constraints.

In the equality-free case the conjunction has the form

$$\bigwedge_i \ell_i < y \quad \wedge \quad \bigwedge_j y < u_j \quad \wedge \quad P(y),$$

where P is a conjunction of congruences. Let N be a standard common multiple of their moduli, using $N = 1$ if there are none. Then $P(y)$ depends only on the complete congruence class of y modulo N . This is stronger than merely knowing $P(y) \leftrightarrow P(y + N)$ by a single standard shift.

If there is a lower bound, choose a maximal ℓ from the finite list. Any witness $y > \ell$ has the same residue as one of the finitely many candidates

$$\ell + 1, \ell + 2, \dots, \ell + N.$$

Choose the candidate strictly above ℓ with that residue. Standard division of the positive element $y - \ell$ shows the candidate is at most y . It therefore preserves every upper bound as well. The maximum can be selected in a finite disjunction of comparisons between the ℓ_i .

If there are upper bounds but no lower bounds, choose a minimal u and use the candidates $u - 1, \dots, u - N$. A witness can be replaced by the same-residue candidate at least as large as it and still strictly below u . If there are no bounds, standard division replaces any witness by one of $0, \dots, N - 1$. In every case existence is equivalent to a finite quantifier-free disjunction. Iterating eliminates all quantifiers.

Every closed quantifier-free sentence in the expanded language has the same truth value in every $\mathbb{Z}$ -group, because its terms are standard integer multiples of 1 and standard congruences have their prescribed residues. Thus the theory is complete and agrees with the standard integer group. Finally a formula over the positive cone is interpreted in the group by restricting every quantified variable to ≥ 0 . This proves the input used throughout the paper.

[write] This is the classical theorem printed as source 05's [theorem 4.9](#), source 04's [theorem 4.10](#) and source 06's [section I](#); it is printed again because it is source 16's, and its remark on invariance under complete congruence classes is the coset invariance of [lemma 4.7](#) and [theorem 4.8](#). "[92]" is source 05's delivered text, now [Part I](#).

X Claim ledger and source boundary (source 16)

Claim	Evidence and scope	Status
Polish Presburger cone construction	Section 128 ; existing ProveIt main and ordinal reports; classical $\mathbb{Z}$ -group theory	Prior construction, reproved
Unital homomorphisms are elementary embeddings	Theorem 129.1 ; explicit divisible-group and congruence argument	Proved here in general split form
Continuous matrix classification and closed image	Theorems 130.5 and 130.7 ; finite-dependence and finite-ancestry proofs	Proved specialization and extension
Closed-submodel normal forms, orbits, lattice	Theorems 131.1 , 131.3 , 132.1 and 132.4 ; finite-support dual separation	Proved for this family

Claim	Evidence and scope	Status
Automatic-continuity threshold	Theorems 133.1 and 133.3 ; rank argument and non-Borel shear	Exact theorem; related Hahn phenomena credited
Closed-image threshold	Theorems 134.1 and 134.2 ; terminal-tail quotient and shifted graph	Exact theorem for all countable ordinals
Self-copy core and clopen intersections	Theorems 135.1 and 135.2 ; fixed terminal ranks and finite coordinate approximations	Proved here
Omnific interpretation	Section 137 ; normal-form existence and fixed-support addition	Additive and ordered; topology transported
Finite example calculations	Included rational verification script and recorded output	Executed checks only
Proof-assistant verification	No new Lean or Rocq files supplied; modules proposed in section 137	Not claimed
Historical priority or major breakthrough	Bounded source and repository audit	Not certified

[write] For $\alpha = \beta = \omega$ the rows on the dictionary, the matrix classification, the closed image and automatic continuity were proved first by source 04 ([proposition 11.3](#), [theorem 11.4](#), and [corollary 11.5](#)), and the omnific interpretation is source 07’s [theorem 40.1](#); notes in [Part X](#) give the details.

Repository snapshot

The inspected repository snapshot is `0f084afa9be466593ffa6aff37749e78c0adc733`, with commit timestamp October 3, 2026, 19:50:54 Pacific time. The main relevant text is `Algebra/SurrealNumbers/docs/foundations-and-computation/polish-models-of-omnific-arithmetic/article.tex`. Its named research questions on coefficient endomorphisms and Polish elementary-submodel lattices determine the specific follow-up problem. The incoming ordinal-family source and other existing arithmetic archives were checked to avoid presenting their constructions as new. The archive’s source audit gives the exact paths and limits of this comparison.

[write] The source audit is shipped as `16-presburger-embeddings-SOURCE_AUDIT.md`. The archives it compared are now [Part III](#) (`polish_presburger_article`, source 06; `polish_presburger_research`, source 07), [Part IV](#) (`glazer_proveit_local_compactness`) and [Part V](#) (`Polish_Hahn_Puisseux_Levi_Civita_Research`), and the standard-cut manuscript `Glazer_ProveIt_Arithmetic_Research` is source 03 of `Algebra/SurrealNumbers/docs/surreal/discrete-initial-subgroups-and-omnific-normalization`; all were placed in `0bd0e5527`, after source 16’s pin.

Proof review

Independent mathematical reviews were used to challenge the statements before delivery. They particularly checked the distinction between a G_δ cone and a closed cone, inherited versus abstract Polishness, finite-support versus full-product codimension, the limit-coordinate inversion argument, and the different thresholds for map continuity and image closedness. This is ordinary mathematical review of an AI-assisted manuscript, not external refereeing or a formal proof certificate.

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- [85] Vladimir Reshetnikov and ProveIt contributors. `OmnificIntegers.lean` and `OmnificPurelyInfiniteModule.lean`. Same pinned commit as above. [Pinned surreal foundations source directory](#).
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